Chapter 1

Foundations of Angular Electromagnetic Structure

1.0 Introduction and Scope

Finite physical conditions precede angular description [1]. Maxwell theory, source admissibility, uniqueness, conservation, and angular transport form one continuous argument—not five topics listed in sequence [1]. Figure 1.1 states their order; geometry, aperture, bandwidth, and dynamic range bound every legitimate angular report [2].

Angular structure is organized directional and phase content in $(\mathbf{E}, \mathbf{H})$ itself—not an orbital index, vortex charge, or beam name added after the fact [2].

The schematic above maps the logical order of 1: each downward step is stricter than the one above it, and the side annotations remind us that geometry, aperture, bandwidth, and dynamic range are not afterthoughts but ceilings on every legitimate angular report [1].

Four questions, each stricter than the last, follow Figure 1.1. First: does $(\mathbf{E}, \mathbf{H})$ satisfy Maxwell's equations with consistent sources and boundaries on the declared domain? Second: can a current distribution of finite support actually produce the observed pattern—that is, can a laboratory source of bounded extent generate the same far-zone fields? Third: can a finite-aperture instrument recover the pattern within its rank and bandwidth? Fourth: does the pattern export qualified angular momentum flux through the chosen control surface Γ? The four questions are cumulative: a positive answer at a lower step is prerequisite for any claim at a higher step; skipping a step is how catalogue templates masquerade as measured angular structure [2].

Compact-range practice makes the four-question ladder tangible. A horn feed at 10 GHz may satisfy Maxwell on the mesh yet fail synthesizable if the impressed current is replaced by a tangential-field template copied from a catalogue. A six-probe scan may recover main-beam power yet miss sidelobe structure the aperture actually radiates. A reactive near-field pattern may show vivid phase circulation while the outward flux integral on Γ_R remains below the wattmeter noise floor. Torque and angular-flux reports therefore arrive last—only after geometry, source specification, and instrument rank have already fixed what can be produced and observed [2].

Consider a PEC-backed circular aperture of radius $a = 5\,\mathrm{mm}$ at $f = 30\,\mathrm{GHz}$, the scale of a compact-range calibration target [2]. Two numerical models may share the same tangential $\mathbf{E}_t$ on the rim: Model A supplies a charge-continuous impressed current $\mathbf{J}_{\mathrm{imp}}$ on the feed support; Model B prescribes $\mathbf{E}_t$ alone with no extension to $\mathbf{J}_{\mathrm{imp}}$ [3]. Model A can in principle pass all four $\mathcal{D}_J = 1$ questions—Maxwell closure, finite-source range membership, finite-aperture wattmeter reading, and qualified angular transport—because a synthesizable source exists [2]. Model B may

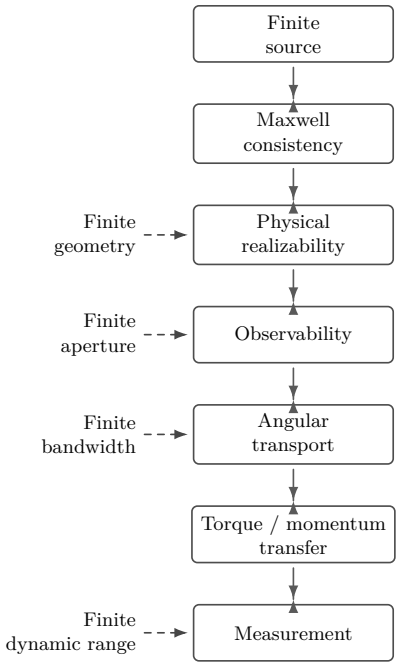


Figure 1.1: Chapter 1 synthesis: finite sources must satisfy Maxwell consistency before $\mathcal{S}_{\text{real}}$, flux report, angular transport, and torque or flux measurement are legitimate reports. Side annotations mark the laboratory ceilings—geometry, aperture, bandwidth, and dynamic range—that bound every angular descriptor [1].

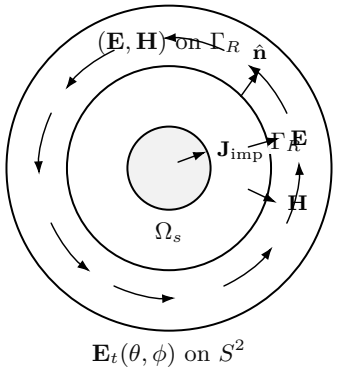


Figure 1.2: Three-level nesting: Maxwell fields, descriptive representations, and realizable finite-source structures.

still satisfy a differential Maxwell solve on a truncated domain yet fail $\mathcal{S}_{\text{real}}$: without a matched source model, $\oint_{\Gamma_R}$ closure and transport claims remain undefined, and flux integrals on ∂V do not name laboratory quantities [2].

When all four questions are posed in order, a field that exports measurable angular flux must already satisfy Maxwell's equations on the declared domain, be producible by a current distribution of finite support, lie within the recoverable rank of a finite-aperture instrument, and carry qualified angular-momentum flux through the chosen control surface [2].

Figure 1.2 orders the volume arc: Maxwell fields (1–2), harmonic representations (**??**), and finite-source range membership (**??**) [2].

Counterexamples arise when a Maxwell-consistent template is prescribed as a boundary field without a charge-continuous source (Section 1.0.2), when a realizable near field fails a finite aperture

(Section 1.2.7), or when reactive storage mimics export on the wrong surface (Section 1.3.4) [2]. Chu [4] and Harrington [3] made the finite-size consequence explicit for stored energy and radiation efficiency; Gustafsson *et al.* [5] later sharpened physical bounds on multiport transmission.

Progressive charge-continuous ladder. A laboratory angular claim is warranted only after four cumulative tests: Maxwell $\mathcal{D}_J = 1$ on the declared domain; $\mathcal{S}_{\text{real}}$ by a bounded impressed source; recoverability within finite instrument rank; and qualified angular-momentum or torque flux on the chosen control surface. Failure at an earlier step falsifies every later headline [2].

1.0.1 Historical lineage of field energy, momentum, and angular structure

Electromagnetic angular structure did not appear as a single postulate. It accumulated as successive generations clarified how energy, linear momentum, and rotational action are carried by $(\mathbf{E}, \mathbf{H})$ and exchanged with matter. Maxwell [1] unified electric and magnetic phenomena through the local field pair and the displacement current that closes Ampère's law in time-dependent form. Once fields were treated as dynamical objects rather than as auxiliary potentials, the question of how power and mechanical action propagate through space became unavoidable.

Poynting [6] answered the energetic part with the flux vector $\mathbf{S} = \mathbf{E}\times\mathbf{H}$, which identifies outward power transport through a surface in the harmonic regime developed in Section 1.1.4. Abraham [7] and Minkowski [8] extended the same field picture to linear momentum density and stress, establishing that radiation pressure and recoil are field-mechanical effects rather than properties of distant charges alone. Stratton [2] synthesized these threads into the modern treatise form still used in antenna and scattering theory. His Maxwell system, uniqueness theorems, and stress-tensor treatment fixed the operational sequence—specify geometry, sources, and boundaries; solve for $(\mathbf{E}, \mathbf{H})$; integrate flux through declared surfaces—that every later angular argument inherits. Chu [4] added the finite-size viewpoint: stored reactive energy and antenna quality factor bound what a compact structure can radiate, so far-zone organization is never independent of source extent and bandwidth. Harrington [3] and Tai [9] carried that synthesis into microwave engineering, separating admissible field templates from what finite feeds and apertures can synthesize. Belinfante [10] and Beth [11] supplied the rotational closure. Belinfante's symmetric stress tensor and spin-density redistribution explained why local angular-momentum bookkeeping can be gauge-sensitive while surface flux integrals remain meaningful. Beth's torsion-balance experiment showed that circularly polarized microwave radiation transfers measurable mechanical torque, linking Poynting flux to laboratory mechanics. Allen *et al.* [12] later made azimuthal phase structure experimentally accessible in paraxial optics; their Laguerre–Gaussian modes provided a compact modal vocabulary without replacing the Maxwellian flux pair $(\Pi, \mathbf{\Psi}_J)$ defined in Section 1.2.2.

That historical arc explains why Chapter 1 does not begin with modal indices or beam names: energy, momentum, and rotational action were established as field properties long before modern OAM vocabulary, and every angular descriptor used later in the book must remain accountable to the same flux and torque chain [6].

1.0.2 Source data, boundaries, and Maxwell consistency

Every compact-range file presupposes the same data tuple: geometry, constitutive response, impressed $\mathbf{J}_{\text{imp}}$, and frequency under $\exp(-æ\omega t)$ [2, 1]. Angular claims attach only after charge continuity, compatible boundary traces, and outgoing support close on Ω [3].

Every range measurement begins with finite sources, finite power, and bounded observation surfaces [1]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

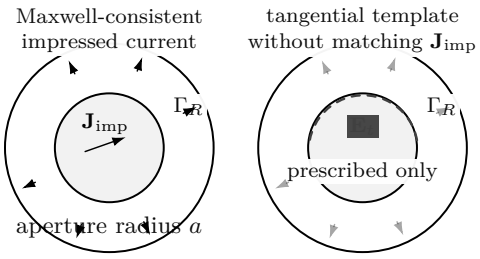


Figure 1.3: Maxwell-consistent impressed-source excitation versus mismatched tangential-field template on the same aperture geometry.

Before any angular descriptor is named, fix what Stratton [2] called the *data of the problem*: the spatial domain Ω, its boundary Γ, the constitutive response χ, the impressed sources that drive the field, and the harmonic frequency under the $\exp(-\mathrm{j}\omega t)$ convention adopted throughout this volume. Phase plots and catalog modal indices must not precede domain closure; Maxwell theory shows that such labels are meaningless until charge continuity, compatible boundary traces, and—where radiation is claimed—outgoing support close on the declared domain [1]. A laboratory measurement never records a free-standing tangential field template: every compact-range file, every near-field scan, and every torsion-balance report presupposes the same tuple whether or not the experimenter writes it down explicitly [13]. Separation of *Maxwell consistency* from *source realizability*: a rim field prescription may look admissible on ∂V yet fail when no charge-continuous impressed current $\mathbf{J}_{\mathrm{imp}}$ of finite support can synthesize it—the laboratory error introduced in Section 1.0 for the $a = 5\,\mathrm{mm}$ aperture at $30\,\mathrm{GHz}$ [3]. Electrical size, probe count, and dynamic range then set which components of that tuple enter the observable record once Γ_R is declared [2]. The formal specification below states what every numerical export and range calibration report must declare before flux or torque integrals become identifiable reports [1].

Fix $\Omega \subset \mathbb{R}^3$, $\Gamma = \partial\Omega$, constitutive data χ, impressed current $\mathbf{J}_{\mathrm{imp}}$, and angular frequency $\omega > 0$ under the $\exp(-\mathrm{j}\omega t)$ convention. Stratton [2] insisted that every field problem begin with this data tuple: geometry, material response, sources, and frequency. An angular claim is meaningful only after Maxwell's equations close on Ω with charge continuity, compatible boundary traces, and outgoing support where radiation is claimed [1].

On the same $a = 5\,\mathrm{mm}$ aperture at $f = 30\,\mathrm{GHz}$, charge-continuous $\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega_s)$ closes Maxwell consistency before flux is reported; a prescribed $\mathbf{E}_t$ template without matching source may fail $\mathcal{S}_{\mathrm{real}}$ even when boundary data look admissible [2].

Numerical exports often list tangential field samples as if they were independent measurements [2]. Figure 1.3 contrasts matched impressed current with an unmatched rim template on identical geometry—the same $a = 5\,\mathrm{mm}$ warning introduced above, now read as a flux-identifiability test on ∂V [3]. The minimum data closure required before Section 1.1 states Maxwell's equations: geometry, materials, sources, frequency, and boundary class must be declared as one tuple, not as disconnected plot files [2]. Tangential samples on an aperture must not be equated with an impressed source: without $\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega_s)$ and charge continuity, angular flux on Γ_R is a post-processed artifact rather than a calibrated range quantity [1]. Subsection 1.0.3 asks the complementary symmetry question—whether rotation of the feed relabels the export in the same way a coordinate rotation would—before finite-energy admissibility is fixed in Subsection 1.0.4 [13].

1.0.3 Rotation and Maxwell compatibility

On a turntable compact range, rotating the antenna under test is the everyday check that a pattern rotates rigidly with the feed while total radiated power stays fixed [13]. That experiment

presupposes a deeper statement: when the source region and surrounding medium share rotational symmetry, rotating the excitation should rotate the radiated field in the same way—a requirement Stratton [2] enforced long before any modal index was named. Violations usually signal anisotropic boundaries, mismatched material data, or a source model that does not close charge continuity. The commutator test below makes that laboratory expectation precise.

For rotation $R_\alpha \in \mathrm{SO}(3)$ with induced action $\mathcal{R}_\alpha$ on sources and fields, Maxwell compatibility requires that rotation commute with the field operator $\mathcal{M}_{\Omega,\Gamma,\chi}$:

$$[\mathcal{R}_\alpha, \mathcal{M}_{\Omega,\Gamma,\chi}]\,\mathbf{J}_{\mathrm{imp}} = \mathbf{0} \quad \forall \alpha \in \mathcal{I}_\alpha. \tag{1.1}$$

Here $\mathcal{I}_\alpha$ is the admissible interval of rotation angles for which the symmetry operation is defined on the declared domain—typically a full 2π azimuth on an axisymmetric range, or a finite turntable arc when mechanical stops apply [13]. The operator $\mathcal{R}_\alpha$ rotates impressed current components and spatial coordinates consistently; $\mathcal{M}_{\Omega,\Gamma,\chi}$ is the Maxwell field operator fixed by geometry Ω, boundary class Γ, and constitutive data χ [2]. When the commutator vanishes, a pattern rolled through α on the stage is the same physical export seen in a rotated coordinate frame; when it does not, angular labels assigned in harmonic charts will scramble even though a wattmeter on Γ_R may still read the same total power [3].

When that commutator does not vanish, rotation scrambles angular content established at the feed. Section 1.2.3 quantifies the effect through the ratio

$$\kappa_{\mathrm{rot}}(\alpha) = \frac{\|[\mathcal{R}_\alpha, \mathcal{M}_{\Omega,\Gamma,\chi}]\,\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{H}_F}}{\|\mathcal{M}_{\Omega,\Gamma,\chi}\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{H}_F} + \epsilon_{\mathrm{rot}}}, \tag{1.2}$$

where $\|\cdot\|_{\mathcal{H}_F}$ is the field energy norm on the analysis domain—the same L^2 measure of $(\mathbf{E}, \mathbf{H})$ used in Eq. (1.3)—and $\epsilon_{\mathrm{rot}} > 0$ is a fixed regularizer that prevents division by zero when the impressed excitation is null [2]. Large κ_{rot} means rotation no longer relabels the export cleanly: $|m = 1|$ power sectors mix under anisotropic radomes or wedge dielectrics, and single-cut azimuth scans cannot validate coefficient rotation laws [13]. On the range, κ_{rot} rises when a lossy anisotropic radome is inserted between a center-fed dipole and the scan sphere—the feed rotation and the reported phase pattern cease to close on the same $D^{(1)}$ block [2]. Eqs. (1.1) and (1.2) are operational: rotation is a symmetry of Maxwell export only when the operator commutes with the impressed source action on the declared domain [2]. Section 1.2.3 develops the full rotation algebra for vectors, tensors, and dyads; note the laboratory warning—misaligned radomes and anisotropic windows scramble angular labels while total power on Γ_R may remain unchanged [13]. Subsection 1.0.4 next fixes which field pairs may enter conservation integrals at all [2].

1.0.4 Assumptions, finite-energy admissibility, and domain of validity

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell export is computed [2]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

Transport claims require fields integrable on finite control volumes without divergent norms or incompatible traces [2]. Every conservation law in this chapter is read as a statement about fields that can be integrated, differentiated, and compared on finite control volumes—the same finite-energy class Stratton [2] and Harrington [3] used for antennas and waveguides.

The finite-energy requirement is more than a mathematical convenience [2]. Every laboratory measurement implicitly assumes that the electromagnetic energy contained within the observation region remains finite: a field whose norm diverges on a finite control volume cannot be assigned

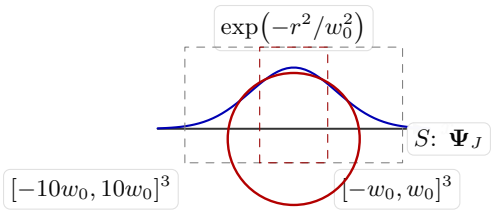


Figure 1.4: Windowed paraxial Gaussian: domain size controls finite-energy $\mathcal{D}_J = 1$ and angular-flux evaluation.

a physically meaningful wattmeter reading, nor can it participate consistently in the flux balances developed later in this chapter [6]. Dipole-class fields on a sphere $r = R$ with $k_0 R = 10$ satisfy $\int_{B_R}(\mathbf{E}^*\cdot\mathbf{E}+\mathbf{H}^*\cdot\mathbf{H})\,dV < \infty$ when the Sommerfeld outgoing condition is enforced; an idealized $1/r$ template without outgoing selection is excluded because its energy integral diverges [2]. Radiating phasors with declared outgoing support lie in the admissible class; non-integrable templates cannot close the conservation chain no matter how smooth their phase plots appear on a near-field display [3].

Let $\Omega \subset \mathbb{R}^3$ denote the analysis region, $\Gamma = \partial\Omega$ its boundary, and $(\mathbf{E}, \mathbf{H})$ the phasor electric and magnetic fields. The baseline finite-energy condition is

$$\int_\Omega (\mathbf{E}^*\cdot\mathbf{E}+\mathbf{H}^*\cdot\mathbf{H})\,dV < \infty, \tag{1.3}$$

where $(\cdot)^*$ denotes complex conjugation [2]. Fields that fail Eq. (1.3) may still satisfy Maxwell locally in a symbolic sense, yet Poynting and stress integrals on ∂V need not converge to laboratory numbers [6]. Declaring integrability before transport matches what compact ranges and anechoic chambers can actually integrate [13].

Source and boundary compatibility are imposed simultaneously through

$$\mathcal{X}_{\mathrm{ch1}}(\Omega) = \left\{(\mathbf{E},\mathbf{H}) \in L^2(\Omega)^3 \times L^2(\Omega)^3 : \mathcal{B}_{\Gamma,\chi}(\mathbf{E},\mathbf{H}) = 0,\ \nabla\cdot\mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0\right\}, \tag{1.4}$$

where $\mathcal{X}_{\mathrm{ch1}}(\Omega)$ is the Chapter 1 admissible field space: square-integrable Maxwell pairs on Ω whose boundary traces satisfy the material completion operator $\mathcal{B}_{\Gamma,\chi}$ and whose sources satisfy harmonic charge continuity $\nabla\cdot\mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0$ [1]. The operator $\mathcal{B}_{\Gamma,\chi}$ encodes how constitutive data χ and boundary class Γ complete tangential traces—PEC removes tangential $\mathbf{E}$, PMC removes tangential $\mathbf{H}$, impedance sheets couple them, and exterior radiation conditions select outgoing waves [2]. A field pair that solves curl equations pointwise yet violates $\mathcal{B}_{\Gamma,\chi}$ is not admissible on the declared domain even if a post-processor can fit spherical modes to its near-field samples [13]. Laboratory error: prescribing $\mathbf{E}_t$ on an aperture without matching $\mathbf{J}_{\mathrm{imp}}$ produces a tuple outside $\mathcal{X}_{\mathrm{ch1}}$; flux integrals computed from such a pair are data artifacts, not calibrated range reports [2].

A Gaussian beam with $w_0 = 50\,\mu$m at $\lambda = 1.55\,\mu$m is admissible on $[-10w_0, 10w_0]^3$ but not as an infinite-space transport claim. Shrinking the analysis domain shifts integrated energy and phase gradient; angular flux must be evaluated on a sphere enclosing the radiated power, not from waist parameters alone [12]. Finite-energy Maxwell-closed is not a technical footnote: every wattmeter, stress integral, and torque channel in Sections 1.1–1.2.5 presumes Eq. (1.3) and membership in $\mathcal{X}_{\mathrm{ch1}}(\Omega)$ from Eq. (1.4) [2]. A vivid near-field Gaussian waist must not be identified with a transport claim on an infinite domain—Figure 1.4 shows that domain size shifts integrated energy and phase gradient, so angular flux must be evaluated on a sphere that encloses radiated power, not from waist parameters alone [12]. With domain, symmetry, and integrability fixed, Section 1.1 states Maxwell's equations and the conservation laws from which every angular descriptor in this chapter inherits [1].

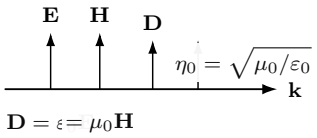


Figure 1.5: Constitutive-to-propagation chain in vacuum: the same two constants ε_0 and μ_0 determine flux mapping and wave kinematics.

1.1 Maxwell Theory and Conservation Laws

Every angular claim inherits Maxwell's equations [1]. The Maxwell block supplies the language every later angular descriptor must speak: curl coupling ties electric and magnetic rotation; divergence ties sources to charge; constitutive completion fixes wave speed and impedance on each subdomain [1]. Uniqueness, stress, and angular balance are successive readings of the same field pair once data and boundaries are declared—not separate physics modules [6].

A finite-difference or method-of-moments solve at 1 GHz enforces Eqs. (1.5) on its mesh [2]. Angular labels assigned in post-processing—helicity sectors, modal coefficients, OAM charts—must still reduce to the same $(\mathbf{E}, \mathbf{H})$ pair the solver enforced; modes and helicities are coordinate listings on radiative export, not independent degrees of freedom that can override curl–divergence constraints [14]. When post-processing reports a "spiral phase" without updating $\mathbf{H}$ through $\nabla \times \mathbf{E} = -\mathrm{j}\omega\mu_0\mathbf{H}$, the visualization violates local compatibility even if the time snapshot looks persuasive on screen [1].

1.1.1 Maxwell equations: differential, integral, vacuum constitutive relations

Maxwell's equations admit differential, integral, and vacuum-constitutive formulations; each answers a different class of questions in angular electrodynamics. The *differential* form governs local compatibility of curl and divergence constraints—the point at which rotational coupling between $\mathbf{E}$ and $\mathbf{H}$ is enforced. The *integral* form governs finite control surfaces and volumes—the language of flux meters, enclosed charge, and laboratory enclosures. The *vacuum constitutive relations* close the system by linking field intensities to flux densities and propagation scales. Together they define the Maxwell foundation from which finite-source realizability, observability, and angular structure follow [2].

Figure 1.5 closes the vacuum chain from $(\mathbf{E}, \mathbf{H})$ to $(\mathbf{D}, \mathbf{B})$ and to c, η_0, and k_0 [1].

In harmonic vacuum, a plane-wave patch $\mathbf{E}_z = E_0 e^{\mathrm{j}kx}$ illustrates why curl coupling is not optional [2]. Faraday's law gives $\mathbf{H}_y = (k/\omega\mu_0)E_0 e^{\mathrm{j}kx}$: $\mathbf{E}$ and $\mathbf{H}$ share the same directional phase law and transverse closure. Assigning an independent azimuthal phase to $\mathbf{E}$ alone—a common visualization error when spiral patterns are overlaid on a single component—violates $\nabla \times \mathbf{E} = -\mathrm{j}\omega\mu_0\mathbf{H}$ and produces a field pair no laboratory probe could measure consistently on both polarization channels [14].

Harmonic fields use time factor $\exp(-\mathrm{j}\omega t)$ in a piecewise smooth region $\Omega \subset \mathbb{R}^3$ bounded by $\Gamma = \partial\Omega$, with phasors $\mathbf{E}(\mathbf{r})$, $\mathbf{H}(\mathbf{r})$, impressed current $\mathbf{J}_{\mathrm{imp}}(\mathbf{r})$, charge density $\rho(\mathbf{r})$, and vacuum parameters ε_0, μ_0 in SI units [2]. The differential form states local compatibility: at each point in a mesh or continuum model, how must $\mathbf{E}$ and $\mathbf{H}$ couple so that Faraday's and Ampère's laws, together with Gauss's laws, close without contradiction? Finite-element and FDTD exports are trusted only when every node satisfies the same curl–divergence constraints a coaxial probe would sample on a sub-wavelength cell [3]. E_x, E_y, and E_z must not be treated as independent plot channels: curl coupling forbids assigning directional or angular content to a single component in

isolation [1]. The differential form states how those quantities couple locally:

$$\begin{aligned} \nabla \times \mathbf{E} &= -\mathrm{j}\omega\mu_0\mathbf{H}, \qquad \nabla \times \mathbf{H} = \mathbf{J}_{\mathrm{imp}} + \mathrm{j}\omega\varepsilon_0\mathbf{E}, \\ \nabla \cdot (\varepsilon_0\mathbf{E}) &= \rho, \qquad\qquad \nabla \cdot (\mu_0\mathbf{H}) = 0. \end{aligned} \tag{1.5}$$

Curl coupling forbids assigning directional content to a single component in isolation [2].

$$\begin{aligned} \partial_y E_z - \partial_z E_y &= -\mathrm{j}\omega\mu_0 H_x, \\ \partial_z H_x - \partial_x H_z &= J_{\mathrm{imp},y} + \mathrm{j}\omega\varepsilon_0 E_y, \\ \partial_x(\varepsilon_0 E_x) + \partial_y(\varepsilon_0 E_y) + \partial_z(\varepsilon_0 E_z) &= \rho. \end{aligned} \tag{1.6}$$

Equation (1.6) expands Eq. (1.5) in Cartesian components: each row is one curl or divergence test the solver evaluates at a mesh node [2]. Subscripts x, y, z denote fixed laboratory axes; $J_{\mathrm{imp},y}$ is the y-component of impressed current density $\mathbf{J}_{\mathrm{imp}}$ [3]. Physically, the display shows that a single-component $\mathbf{E}_z$ patch still couples to $\mathbf{H}_y$ through Faraday's law and to ρ through Gauss's law—angular labels assigned to E_z alone cannot be read independently of the paired $\mathbf{H}$ and charge fields [1].

$$\mathcal{M}_\omega \begin{bmatrix} \mathbf{E} \\ \mathbf{H} \end{bmatrix} = \begin{bmatrix} \nabla \times \mathbf{E} + \mathrm{j}\omega\mu_0\mathbf{H} \\ \nabla \times \mathbf{H} - \mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\varepsilon_0\mathbf{E} \\ \nabla \cdot (\varepsilon_0\mathbf{E}) - \rho \\ \nabla \cdot (\mu_0\mathbf{H}) \end{bmatrix} = \mathbf{0}, \tag{1.7}$$

where $\mathcal{M}_\omega$ is the harmonic Maxwell operator assembled from the four local laws in Eq. (1.5) [2]. Read $\mathcal{M}_\omega$ as a single residual vector on the field pair: the first two rows test curl coupling between $\mathbf{E}$ and $\mathbf{H}$; the third and fourth test divergence consistency with ρ and magnetic flux [1]. In a finite-element export, each row is a failure channel the solver monitors—a nonzero curl residual means the $\mathbf{E}$–$\mathbf{H}$ coupling that sustains radiation is broken; a nonzero divergence residual means charge or magnetic-flux bookkeeping fails [2]. Spurious ρ without matching $\mathbf{J}_{\mathrm{imp}}$ inflates these channels even when field plots appear smooth; the operator block therefore detects inconsistent source models before any angular label is assigned [13]. In measurement, a mesh that passes visual inspection yet carries $\|\mathcal{M}_\omega[\mathbf{E}, \mathbf{H}]\| \neq 0$ on the feed support will produce modal coefficients that rotate correctly on paper yet disagree with watt totals on Γ_R [2]. Limitation: $\mathcal{M}_\omega$ tests local Maxwell structure only; admissible boundary traces and outgoing selection are imposed separately through $\mathcal{B}_{\Gamma,\chi}$ in Subsection 1.1.2 [3].

$$\nabla \cdot \mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0, \tag{1.8}$$

because $\nabla \cdot (\nabla \times \mathbf{H}) = 0$. Equation (1.8) is not an external add-on; it is embedded in Maxwell structure itself [2]. The same identity is recovered by differentiating $\nabla \cdot (\varepsilon_0\mathbf{E}) = \rho$ in harmonic time and substituting $\nabla \times \mathbf{E} = -\mathrm{j}\omega\mu_0\mathbf{H}$, confirming completion consistency.

Integrating Eq. (1.8) over a control volume V with boundary ∂V and outward normal $\hat{\mathbf{n}}$ yields

$$\int_{\partial V} \mathbf{J}_{\mathrm{imp}} \cdot \hat{\mathbf{n}}\, dS - \mathrm{j}\omega \int_V \rho\, dV = 0. \tag{1.9}$$

The first term is net impressed current leaving V; the second is harmonic charge variation inside V, with outward unit normal $\hat{\mathbf{n}}$ on ∂V [1]. Current outflow and charge accumulation must balance on every admissible control volume: a feed that injects charge without matching current through the boundary violates the same continuity law that curl–divergence structure embeds in Eq. (1.5) [2].

The integral form of Maxwell equations follows from Stokes and divergence theorems. For oriented surface $S \subset \Omega$ with boundary ∂S, oriented volume $V \subset \Omega$, line element $d\boldsymbol{\ell}$, area element dS, and volume element dV,

$$\begin{aligned} \oint_{\partial S} \mathbf{E} \cdot d\boldsymbol{\ell} &= -\mathrm{j}\omega\mu_0 \int_S \mathbf{H} \cdot \hat{\mathbf{n}}\, dS, \\ \oint_{\partial S} \mathbf{H} \cdot d\boldsymbol{\ell} &= \int_S \mathbf{J}_{\mathrm{imp}} \cdot \hat{\mathbf{n}}\, dS + \mathrm{j}\omega\varepsilon_0 \int_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS, \\ \int_{\partial V} \varepsilon_0 \mathbf{E} \cdot \hat{\mathbf{n}}\, dS &= \int_V \rho\, dV, \qquad \int_{\partial V} \mu_0 \mathbf{H} \cdot \hat{\mathbf{n}}\, dS = 0. \end{aligned} \tag{1.10}$$

These relations are exact finite-control statements of the same laws written locally in Eq. (1.5) [2].

Let $\mathbf{E}, \mathbf{H} \in H(\nabla\times;\Omega)$, $\varepsilon_0\mathbf{E}, \mu_0\mathbf{H} \in H(\nabla\cdot;\Omega)$, $\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega)^3$, and $\rho \in L^2(\Omega)$. If S and V are piecewise smooth oriented controls, then Eq. (1.5) and Eq. (1.10) are equivalent in weak form.

Applying Stokes and divergence theorems to the local rows gives the integral rows. Conversely, localization of integral balances to shrinking controls recovers local rows in the distributional sense. Regularity assumptions guarantee interchange of limit and integration operations.

Vacuum constitutive relations supply the completion that transforms field-intensity variables into flux-density variables:

$$\mathbf{D} = \varepsilon_0 \mathbf{E}, \qquad \mathbf{B} = \mu_0 \mathbf{H}, \qquad c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}. \tag{1.11}$$

Here $\mathbf{D}$ is electric flux density, $\mathbf{B}$ magnetic flux density, and c vacuum wave speed [2]. Without Eq. (1.11), Maxwell balance laws remain under-closed in terms of unknowns $(\mathbf{E}, \mathbf{H}, \mathbf{D}, \mathbf{B})$. With completion, the model is fully evaluable for vacuum media.

Derived vacuum scales follow directly:

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}, \qquad k_0 = \omega\sqrt{\mu_0\varepsilon_0} = \frac{\omega}{c}, \qquad \lambda_0 = \frac{2\pi}{k_0}. \tag{1.12}$$

η_0 is intrinsic impedance, k_0 is wavenumber, and λ_0 is wavelength. These scales set amplitude and phase calibration for every source-free vacuum propagation sector [2].

Constitutive completion and wave propagation are not separate modules. The constants that map $\mathbf{E}, \mathbf{H}$ to $\mathbf{D}, \mathbf{B}$ also set wave speed and impedance, so material assumptions and propagation predictions are intrinsically linked.

Substituting Eq. (1.11) into curl equations yields second-order source-driven wave equations:

$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = \frac{1}{\varepsilon_0} \nabla\rho - \mathrm{j}\omega\mu_0 \mathbf{J}_{\mathrm{imp}}, \qquad \nabla^2 \mathbf{H} + k_0^2 \mathbf{H} = -\nabla \times \mathbf{J}_{\mathrm{imp}}. \tag{1.13}$$

The $\nabla\rho$ and $\mathbf{J}_{\mathrm{imp}}$ terms are electric-source forcing channels; the $-\nabla \times \mathbf{J}_{\mathrm{imp}}$ term is magnetic-source forcing for $\mathbf{H}$. Equation (1.13) clarifies that spatial curvature and source gradients jointly shape field structure [2].

In source-free vacuum, Eq. (1.13) reduces to

$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = \mathbf{0}, \qquad \nabla^2 \mathbf{H} + k_0^2 \mathbf{H} = \mathbf{0}, \tag{1.14}$$

which is the vector Helmholtz pair. The reduction is valid only when both $\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$ and $\rho = 0$, and when the constitutive completion remains vacuum linear isotropic.

The same source-free physics may be written in curl-curl form:

$$\nabla \times \left(\mu_0^{-1} \nabla \times \mathbf{E}\right) - \omega^2 \varepsilon_0 \mathbf{E} = \mathbf{0}, \qquad \nabla \times \left(\varepsilon_0^{-1} \nabla \times \mathbf{H}\right) - \omega^2 \mu_0 \mathbf{H} = \mathbf{0}. \tag{1.15}$$

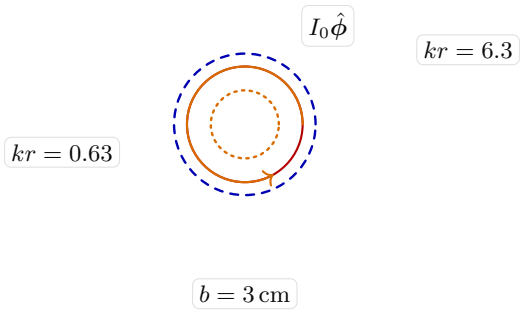


Figure 1.6: Worked-problem schematic: azimuthally symmetric filament loop source satisfying charge continuity and driving dipole radiation.

Equation (1.15) and Eq. (1.14) are equivalent in vacuum. Curl-curl form is often preferable in weak formulations because it interfaces naturally with tangential observable boundary datas and variational bilinear forms [2].

Local plane-wave sectors in homogeneous source-free vacuum obey

$$\mathbf{H} = \frac{1}{\eta_0}\,\hat{\mathbf{k}} \times \mathbf{E}, \qquad \hat{\mathbf{k}} \cdot \mathbf{E} = 0, \qquad \hat{\mathbf{k}} \cdot \mathbf{H} = 0, \tag{1.16}$$

where $\hat{\mathbf{k}}$ is the unit propagation direction and $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ is vacuum intrinsic impedance [2]. The three lines state transverse closure: $\mathbf{H}$ is perpendicular to both $\mathbf{E}$ and $\hat{\mathbf{k}}$, and both field vectors are transverse to propagation—the laboratory pattern of a TEM plane wave in free space [14].

Eqs. (1.5)–(1.11) describe field evolution in the presence of $\mathbf{J}_{\mathrm{imp}}$ and ρ, but they do not yet specify which source distributions and boundary traces are physically realizable on a given Ω. A mathematically writable current need not belong to the admissible class of Subsection 1.0.4; an interface condition written without tangential continuity can produce apparent angular flux that is pure data inconsistency. Subsection 1.1.2 couples volume sources, interface jumps, and outer-boundary radiation data into one admissibility chain [2].

With solenoidal $\mathbf{J}_{\mathrm{imp}}$, Eq. (1.13) yields admissible fields; spurious $\rho = \rho_0\delta(\mathbf{r})$ drives curl residuals to order unity. Consider a filament loop of radius $b = 3\,\mathrm{cm}$ carrying phasor current $I_0 = 1\,\mathrm{A}$ at $f = 100\,\mathrm{MHz}$. Model the impressed current as $\mathbf{J}_{\mathrm{imp}} = I_0\delta(\rho - b)\delta(z)\hat{\boldsymbol{\phi}}$ in cylindrical coordinates. Then $\nabla \cdot \mathbf{J}_{\mathrm{imp}} = 0$ because $\partial_\phi J_\phi = 0$ and the delta support lies in a plane. With $\rho = 0$, Eq. (1.8) is satisfied with $\rho = 0$, so the loop source is admissible in the sense of Eq. (1.22). Substituting into Eq. (1.13) generates azimuthally symmetric $\mathbf{H}$ and a nonuniform $\mathbf{E}$ with both radial and axial components in the near zone. At $kr = 0.63$ the reactive fraction dominates; at $kr = 6.3$ the fields approach the dipole radiation pattern cited in Subsection 1.3.3. Maxwell field specification follows from evaluating Eq. (1.7) on a cylindrical grid: curl residuals fall below 10^{-3} relative to peak $|\mathbf{E}|$ when charge continuity is enforced, but rise to order unity if a spurious $\rho = \rho_0\delta(\mathbf{r})$ is added without adjusting $\mathbf{J}_{\mathrm{imp}}$ [2].

The equivalence of differential and integral forms requires $\mathbf{E}, \mathbf{H} \in H(\nabla\times;\Omega)$ and flux densities in $H(\nabla\cdot;\Omega)$ as stated in Proposition 1.1.1. Dispersive media require frequency-dependent $\varepsilon(\omega)$ and $\mu(\omega)$, breaking the algebraic completion of Eq. (1.11). Relativistic or quantum corrections lie outside the present harmonic class. Within the vacuum harmonic class, Eqs. (1.5)–(1.16) form the complete local law set every later angular argument inherits [1].

Subsection 1.1.1 treats Maxwell's equations as a *compatibility* system: they do not, by themselves, tell the engineer which boundary data or impressed sources belong to a realizable antenna [2]. A full-wave solver enforces local curl–divergence closure at each node; a compact range reports flux through declared surfaces after outgoing selection—two readings of the same operator once

data are complete [3]. The vacuum constitutive chain of Figure 1.5 is the propagation structure on which every later angular descriptor is listed, not an optional sidebar [1]. Subsection 1.1.2 next supplies the source, interface, and boundary tuple that converts local Maxwell laws into an identifiable boundary-value problem [2].

1.1.2 Sources, boundary conditions, and physical admissibility

No radiating laboratory exists without a feed and an outer world—yet Maxwell's local equations, stated in Subsection 1.1.1, do not by themselves specify which impressed currents, interface jumps, and outer-boundary radiation data belong to a physically realizable experiment [2]. The boundary-value completeness question is given $(\mathbf{E}, \mathbf{H})$ on a mesh, does a finite-source model, compatible interface data, and outgoing selection exist such that flux integrals on Γ_R name the same physics the hardware measures? Prescribing tangential $\mathbf{E}_t$ on an aperture is not equivalent to specifying a source; Harrington [3] and Chu [4] showed repeatedly that finite extent and material completion restrict what boundary templates can be synthesized. A compact-range team declares the data tuple $\mathfrak{D}$ below—impressed volume sources, surface sources, constitutive response, and boundary operator—before any angular descriptor is attached [1].

Impressed *sources* inject current and charge; *boundary conditions* fix tangential traces and outgoing-wave selection; $\mathcal{D}_J = 1$ requires the full dataset to lie in the finite-energy class of Subsection 1.0.4 [2]. Finite antennas realize only a subset of mathematically smooth boundary data—a theme developed in Chapter 4 [2].

Vacuum Maxwell laws from Subsection 1.1.1 are necessary but not sufficient for a complete boundary-value problem. Source, interface, and outer-boundary data determine whether interior rotation exports flux or remains trapped in storage exchange with the boundary [2]. Maxwell-closed requires one coupled requirement, not three independent lists.

A WR-90 feed, a dielectric window, and an open-range absorbing boundary are three different boundary stories applied to the same Maxwell operator [2]. Tangential $\mathbf{E}$ or $\mathbf{H}$ continuity on metal enforces PEC or PMC traces; impedance sheets approximate thin coatings; Sommerfeld radiation selects outgoing waves. Consider a dielectric half-space with $\varepsilon_r = 4$, interface at $z = 0$, and a plane wave incident from $z > 0$ [2]. Tangential $\mathbf{E}$ and tangential $\mathbf{H}$ (apart from any impressed surface current $\mathbf{J}_s$) remain continuous across the interface: $\hat{\mathbf{z}} \times (\mathbf{E}_1 - \mathbf{E}_2) = \mathbf{0}$ and $\hat{\mathbf{z}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$; with $\mathbf{J}_s = \mathbf{0}$, the reflected amplitude fixes how much power returns to the source region versus how much transmits [14]. Angular transport fails silently when sources are specified on one surface and flux is reported on another without matching traces—the half-space example is the minimal warning that charge-continuous is a whole-domain statement, not a rim template [3].

Let $\Omega \subset \mathbb{R}^3$ be decomposed into piecewise smooth material subdomains $\{\Omega_a\}_{a=1}^{N_\Omega}$, and let $\Sigma_{ab} = \partial\Omega_a \cap \partial\Omega_b$ be an interface with unit normal $\hat{\mathbf{n}}_{ab}$ directed from Ω_a to Ω_b. Let $\mathbf{D}_a$ and $\mathbf{B}_a$ denote electric and magnetic flux-density phasors in Ω_a, $\mathbf{J}_{s,ab}$ the electric surface-current density on Σ_{ab}, and $\rho_{s,ab}$ the free surface-charge density on Σ_{ab}. A compact data tuple is

$$\mathfrak{D} = \Big(\mathbf{J}_{\mathrm{imp}}, \rho, \{\mathbf{J}_{s,ab}\}, \{\rho_{s,ab}\}, \mathcal{B}_\Gamma, \mathbf{g}_\Gamma\Big), \tag{1.17}$$

where $\mathfrak{D}$ is the complete harmonic data tuple for one solve [2]. Volume impressed current $\mathbf{J}_{\mathrm{imp}}$ and charge density ρ enter through the source terms in Eq. (1.5); interface sheets $\mathbf{J}_{s,ab}$ and $\rho_{s,ab}$ appear when pillbox and loop limits are taken across Σ_{ab}; $\mathcal{B}_\Gamma$ is the boundary trace operator on $\Gamma = \partial\Omega$; and $\mathbf{g}_\Gamma$ is the prescribed boundary data—tangential $\mathbf{E}$ or $\mathbf{H}$ on feeds, absorbing impedances on truncations, or outgoing radiation data at infinity [3]. In measurement, every compact-range file implicitly carries such a tuple even when only near-field samples are published; omitting $\mathbf{J}_{\mathrm{imp}}$ or $\mathbf{g}_\Gamma$ leaves flux integrals undefined [13].

On $\Gamma = \partial\Omega$, data completion is represented by a boundary operator $\mathcal{B}_\Gamma$ acting on the boundary traces of $(\mathbf{E}, \mathbf{H})$:

$$\mathcal{B}_\Gamma[\mathbf{E}, \mathbf{H}] = \mathbf{g}_\Gamma, \tag{1.18}$$

where $\mathbf{g}_\Gamma$ is prescribed boundary data on the declared trace class [2]. Eq. (1.18) closes the boundary-value problem and selects termination physics—metal, absorber, or open radiation—before uniqueness and flux reports are meaningful [2].

Interface transmission follows from the integral Maxwell balances Eq. (1.10) by shrinking control loops and pillboxes across Σ_{ab}. For piecewise smooth interfaces and no impressed magnetic sheet current, one obtains

$$\begin{aligned} \hat{\mathbf{n}}_{ab} \times (\mathbf{E}_b - \mathbf{E}_a) &= \mathbf{0}, \\ \hat{\mathbf{n}}_{ab} \times (\mathbf{H}_b - \mathbf{H}_a) &= \mathbf{J}_{s,ab}, \\ \hat{\mathbf{n}}_{ab} \cdot (\mathbf{D}_b - \mathbf{D}_a) &= \rho_{s,ab}, \\ \hat{\mathbf{n}}_{ab} \cdot (\mathbf{B}_b - \mathbf{B}_a) &= 0. \end{aligned} \tag{1.19}$$

Eq. (1.19) is the transmission law on material partitions [2]. Tangential $\mathbf{E}$ is continuous because Faraday's law integrates to zero circulation around a vanishing loop straddling Σ_{ab}; tangential $\mathbf{H}$ jumps by impressed surface current $\mathbf{J}_{s,ab}$; normal $\mathbf{D}$ jumps by free surface charge $\rho_{s,ab}$; and normal $\mathbf{B}$ is continuous in the absence of magnetic monopoles [14]. In measurement, a thin dielectric window between a WR-90 feed and free space is modeled by these jumps; omitting $\rho_{s,ab}$ when a coated interface is present misstates reflected power [15]. Laboratory error: fitting only tangential $\mathbf{E}$ on both sides of an interface without enforcing $\mathbf{J}_{s,ab}$ produces patterns that satisfy a local plot yet violate global power balance [2].

Intermediate limiting relations leading to Eq. (1.19) follow from control-set shrinkage in Eq. (1.10): for a vanishing loop straddling Σ_{ab},

$$\lim_{\delta \to 0} \oint_{\partial S_\delta} \mathbf{E} \cdot d\boldsymbol{\ell} = 0 \quad \Rightarrow \quad \hat{\mathbf{n}}_{ab} \times (\mathbf{E}_b - \mathbf{E}_a) = \mathbf{0},$$

and for a vanishing pillbox,

$$\lim_{\delta \to 0} \int_{\partial V_\delta} \mathbf{D} \cdot \hat{\mathbf{n}}\, dS = \rho_{s,ab} \quad \Rightarrow \quad \hat{\mathbf{n}}_{ab} \cdot (\mathbf{D}_b - \mathbf{D}_a) = \rho_{s,ab}.$$

The boundary selects which tangential and normal channels transmit flux and which are screened at material partitions [2].

Metal walls, magnetic boundaries, and thin impedance coatings are the three outer termination models most antenna engineers encounter [2]. A PEC wall suppresses tangential $\mathbf{E}$ and reflects power; a PMC wall suppresses tangential $\mathbf{H}$; an impedance sheet Z_s couples the two traces as on a thin absorber or dielectric coating [3]. The display below partitions Γ accordingly so that each hardware termination maps to one trace class before open-region radiation is appended [2].

Outer completion is specified by partitioning $\Gamma = \Gamma_{\mathrm{PEC}} \cup \Gamma_{\mathrm{PMC}} \cup \Gamma_Z$, where Γ_{PEC} is perfect-electric-conductor boundary, Γ_{PMC} is perfect-magnetic-conductor boundary, and Γ_Z is an impedance boundary with surface impedance Z_s:

$$\begin{aligned} \hat{\mathbf{n}} \times \mathbf{E} &= \mathbf{0} && \text{on } \Gamma_{\mathrm{PEC}}, \\ \hat{\mathbf{n}} \times \mathbf{H} &= \mathbf{0} && \text{on } \Gamma_{\mathrm{PMC}}, \\ \hat{\mathbf{n}} \times \mathbf{E} &= Z_s\, (\hat{\mathbf{n}} \times \mathbf{H}) \times \hat{\mathbf{n}} && \text{on } \Gamma_Z, \end{aligned} \tag{1.20}$$

with $\hat{\mathbf{n}}$ the outward unit normal on Γ. Here $\hat{\mathbf{n}} \times \mathbf{E}$ and $\hat{\mathbf{n}} \times \mathbf{H}$ are tangential electric and magnetic traces, while $(\hat{\mathbf{n}} \times \mathbf{H}) \times \hat{\mathbf{n}}$ reorients tangential magnetic content into the impedance-coupled direction.

Equation (1.20) therefore functions as a boundary trace selector mathematically, and as a model of ideal reflection or finite surface dissipation physically.

The three outer boundary classes in Eq. (1.20) are limiting responses for tangential field content. The PEC relation suppresses tangential electric trace, the PMC relation suppresses tangential magnetic trace, and the impedance relation preserves both while coupling them through Z_s. This nesting determines which energy and momentum channels can pass through the boundary and which are reflected [2].

For open-region antennas and scatterers, the outer world must not send incoming waves back toward the feed [2]. Compact-range absorbers and Sommerfeld radiation conditions implement the same physical requirement: transverse fields on Γ must carry net power outward as $R \to \infty$, otherwise watt totals conflate radiative export with incoming energy that no bounded source supplied [13]. The trace condition below is the asymptotic form of that outgoing selection [16].

For open-region truncation, an outgoing-trace condition is often required:

$$\hat{\mathbf{n}} \times \mathbf{H} - \eta_0^{-1}(\hat{\mathbf{n}} \times \mathbf{E}) \times \hat{\mathbf{n}} = \mathbf{o}(R^{-1}), \qquad R \to \infty, \tag{1.21}$$

where $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ and R is truncation radius. Equation (1.21) enforces asymptotically outgoing power flow and prevents unphysical incoming-wave completion in exterior problems [2]. Laboratory error: truncating a compact range too close to the radiator without this condition selects incoming admixture; reported gain and torque then depend on absorber placement rather than on the antenna under test [13].

Every laboratory feed occupies bounded support Ω_s and must close charge with impressed ρ [2]. The source class below is the formal statement of that requirement: finite L^2 current, localized support, and harmonic continuity $\nabla \cdot \mathbf{J}_{\text{imp}} - \mathrm{j}\omega\rho = 0$ [1]. Without it, curl-only solvers may return smooth fields that fail global charge balance on Ω [3].

Source $\mathcal{D}_J = 1$ is now imposed by regularity, localization, and continuity compatibility. Let $\Omega_s \subseteq \Omega$ be source support. The admissible source class is

$$\mathbf{J}_{\text{imp}} \in L^2(\Omega_s)^3, \qquad \operatorname{supp}(\mathbf{J}_{\text{imp}}) \subseteq \overline{\Omega_s}, \qquad \nabla \cdot \mathbf{J}_{\text{imp}} - \mathrm{j}\omega\rho = 0 \text{ in } \mathcal{D}'(\Omega). \tag{1.22}$$

Eq. (**??**) enforces finite source energy in the local L^2-sense; the support condition enforces spatial localization of excitation; the distributional continuity relation enforces charge-current consistency under the adopted $\exp(-\mathrm{j}\omega t)$ phasor convention. Taken together, these conditions exclude non-conservative source prescriptions that would violate Eq. (1.5). In measurement, a filament loop with $\nabla \cdot \mathbf{J}_{\text{imp}} = 0$ is admissible; a delta charge inserted without matching $\mathbf{J}_{\text{imp}}$ is not, even when the near-field plot appears smooth [2]. Laboratory error: MoM ports that enforce $\mathbf{E}_t$ without charge-compatible $\mathbf{J}_{\text{imp}}$ produce coefficients that fail Eq. (1.24) at domain scale [13].

Physical $\mathcal{D}_J = 1$ is not exhausted by source regularity alone. Source, interface, and boundary channels must be jointly consistent:

$$\mathfrak{D} \in \mathbb{A}_\omega \iff \begin{cases} \text{Eq. (1.22) holds} \\ \text{Eq. (1.19) holds on all interfaces} \\ \text{Eq. (1.18) holds with an admissible trace class} \\ \text{Eq. (1.24) holds globally.} \end{cases} \tag{1.23}$$

where $\mathbb{A}_\omega$ is the frequency-dependent class of physically admissible data tuples at angular frequency ω [2]. Membership in $\mathbb{A}_\omega$ is not a single charge-continuous test: volume sources must satisfy Eq. (1.22), every interface must satisfy Eq. (1.19), outer traces must close through Eq. (1.18) on an admissible class, and global charge balance Eq. (1.24) must hold on Ω [1]. In measurement,

a WR-90 feed with charge-compatible aperture current belongs to $\mathbb{A}_\omega$ above cutoff; a prescribed tangential template without $\mathbf{J}_{\text{imp}}$ extension does not, even if a finite-element solve returns smooth fields [15]. Laboratory error: if any channel fails, Maxwell fields may still be computed numerically but no longer inherit one coherent physical interpretation—power and torque integrals may then be computed on distinct solutions that share the same catalogue description [2].

Global charge compatibility closes the Maxwell-closed chain on whole-domain scale:

$$\int_\Omega (\nabla \cdot \mathbf{J}_{\text{imp}} - \mathrm{j}\omega\rho)\, dV = \int_\Gamma \mathbf{J}_{\text{imp}} \cdot \hat{\mathbf{n}}\, dS - \mathrm{j}\omega \int_\Omega \rho\, dV = 0. \tag{1.24}$$

Integrating Eq. (1.8) over Ω and applying the divergence theorem yields Eq. (1.24) [1]. The surface integral $\int_\Gamma \mathbf{J}_{\text{imp}} \cdot \hat{\mathbf{n}}\, dS$ is net outward impressed-current flux through the domain boundary; $\mathrm{j}\omega \int_\Omega \rho\, dV$ is the harmonic rate of charge storage inside Ω [2]. In measurement, a feed that injects current without matching interior charge variation violates global balance even when local curl equations appear satisfied on the mesh [3]. Laboratory error: numerical sources with $\mathcal{R}_{\text{adm}} \gg 1$ often still produce smooth far patterns because curl residuals are small while divergence bookkeeping fails at domain scale [13].

With sources and boundaries specified, uniqueness requires one admissible dataset to define one field state. Without that answer, power and torque integrals could be computed on distinct Maxwell solutions that share the same feed description—a laboratory error familiar in resonant cavities and in truncated analysis domains placed too close to the radiator [2].

Admissibility precedes interpretation: source, interface, and boundary channels must close jointly before uniqueness or flux reports are legitimate [2, 1].

Assume Eq. (**??**) and the boundary operator constraint Eq. (1.18). Then admissible data must satisfy the global charge-balance identity Eq. (1.24).

A practical source-continuity residual is

$$\mathcal{R}_{\text{adm}} = \frac{\|\nabla \cdot \mathbf{J}_{\text{imp}} - \mathrm{j}\omega\rho\|_{L^2(\Omega)}}{\|\nabla \cdot \mathbf{J}_{\text{imp}}\|_{L^2(\Omega)} + \omega\|\rho\|_{L^2(\Omega)} + \epsilon_0}, \tag{1.25}$$

where $\epsilon_0 > 0$ in the denominator is a fixed nondimensional regularizer for the Maxwell-closed residual—not the vacuum permittivity ε_0 that appears in Maxwell's equations [2]. Confusing the two symbols would misread $\mathcal{R}_{\text{adm}}$ as a electrostatic scale; here ϵ_0 merely prevents division by zero when both $\mathbf{J}_{\text{imp}}$ and ρ vanish on a subdomain [3]. Small $\mathcal{R}_{\text{adm}}$ indicates numerically consistent source channels before the full field solve; large values flag charge–current mismatch the way a network analyzer flags a broken feed continuity check [2]. Laboratory error: exporting near-field coefficients while $\mathcal{R}_{\text{adm}} \gg 1$ produces stable modal amplitudes that fail global charge balance on Ω [13].

Source support, interface transmission, and exterior trace class form a single data manifold. Later transport and observable claims inherit this manifold directly, because angular descriptors computed from $(\mathbf{E}, \mathbf{H})$ are credible only when the entire source-boundary data completion is compatible.

Outer boundary classes select trace channels: PEC removes tangential $\mathbf{E}$, PMC removes tangential $\mathbf{H}$, impedance couples both through Z_s, and the radiation trace enforces outgoing-wave dominance [2].

The validity boundary is equally clear. Piecewise smooth interfaces, local constitutive response, and standard trace theory for boundary operators are assumed; strongly nonlocal media, rough multiscale interfaces, and stochastic boundary operators lie outside the present harmonic class [2].

Below cutoff, evanescent guide fields remain storage-dominated; radiative transport diagnostics apply only after Eq. (1.21) is satisfied [2]. Consider a hollow rectangular guide with cross section

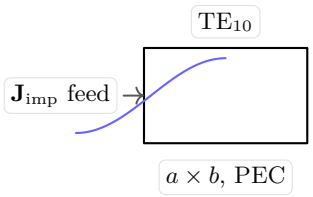


Figure 1.7: Worked-problem schematic: TE_{10} rectangular guide with PEC sidewalls and charge-compatible feed aperture.

$a \times b$ and PEC walls. The dominant TE_{10} mode has $\mathbf{E} = E_0 \sin(\pi x/a)\, e^{\mathrm{j}\beta z}\hat{\mathbf{y}}$ with cutoff $\omega_c = \pi c/a$ and propagation constant $\beta = \sqrt{k_0^2 - (\pi/a)^2}$ for $k_0 > \pi/a$. Tangential $\mathbf{E}$ vanishes on $x = 0, a$ and $y = 0, b$, satisfying Eq. (1.20) on Γ_{PEC}. The impressed current that launches this mode is confined to the feed aperture at $z = 0$; with $\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega_s)^3$ and Eq. (1.24) enforced at the feed, the data tuple belongs to $\mathbb{A}_\omega$. At $f = 10\,\mathrm{GHz}$ with $a = 2.3\,\mathrm{cm}$, $\omega_c \approx 6.5\,\mathrm{GHz}$ and the mode propagates with $\beta \approx 0.79\, k_0$. A student who applies angular-transport diagnostics to the evanescent field below cutoff without revisiting Eq. (1.21) confuses storage-dominated guide physics with radiative export [2].

Source and boundary charge-continuous specify *what data* may enter Maxwell theory; they do not yet guarantee that those data select a *unique* field state. Without uniqueness, two distinct field pairs could produce identical power through an aperture yet different torque about the same axis, destroying the identifiability required for angular transport claims. Subsection 1.1.3 closes that logical gap before conservation laws are derived in Subsections 1.1.4 and 1.1.5 [2].

Collin's guided-wave theory [2] and Stratton's boundary-value treatment [2] both stress that Maxwell equations in the volume fix the field only up to a compatible boundary supplement. A perfectly conducting wall removes tangential electric field; an impedance sheet couples tangential $\mathbf{E}$ and $\mathbf{H}$; an open radiation boundary selects outgoing rather than incoming spherical waves. Each choice alters which angular modes are trapped and which can export flux through Γ. The data tuple Eq. (1.17) lists every channel—volume $\mathbf{J}_{\mathrm{imp}}$, surface $\mathbf{J}_{s,ab}$, interface $\rho_{s,ab}$, and outer trace $\mathbf{g}_\Gamma$—so that no excitation path is left implicit.

Interface conditions Eq. (1.19) follow from the same integral identities Maxwell [1] wrote for pillboxes and Amperian loops. The tangential $\mathbf{E}$ continuity row states that electric potential remains single-valued across a neutral interface without magnetic sheet current. The tangential $\mathbf{H}$ jump equals $\mathbf{J}_{s,ab}$, encoding how surface current sources magnetic circulation. Normal $\mathbf{D}$ and $\mathbf{B}$ jumps tie to surface charge and enforce absence of magnetic monopoles. For angular transport, the critical row is often the tangential $\mathbf{H}$ jump: an azimuthally varying $\mathbf{J}_{s,ab}$ can inject orbital-like structure at an interface even when volume $\mathbf{J}_{\mathrm{imp}}$ is symmetric [2].

The radiation trace Eq. (1.21) deserves a physical reading in SI units. The left side is tangential magnetic field minus an electric-field term scaled by η_0^{-1}; the right side vanishes as $R \to \infty$ for outgoing fields. Violating this condition is equivalent to injecting incoming energy from infinity, which corrupts both P_{rad} and $\mathbf{\Psi}_J$ integrals in Section 1.2. Near-field scanners that truncate the domain too close to the source must replace Eq. (1.21) with an absorbing boundary whose reflection coefficient is small enough that $\mathcal{R}_{\mathrm{adm}}$ in Eq. (1.25) remains below tolerance [2].

Power dissipated on an impedance boundary Γ_Z follows from the complex Poynting theorem of Subsection 1.1.4. The cycle-averaged outward power density normal to Γ_Z is

$$P_Z = \frac{1}{2}\,\mathrm{Re}\{\hat{\mathbf{n}} \cdot (\mathbf{E} \times \mathbf{H}^*)\} = \frac{1}{2}\,\mathrm{Re}\Big\{Z_s\,|(\hat{\mathbf{n}} \times \mathbf{H}) \times \hat{\mathbf{n}}|^2\Big\}, \tag{1.26}$$

where the second equality uses Eq. (1.20). When $\mathrm{Re}(Z_s) > 0$, Eq. (1.26) is nonnegative and

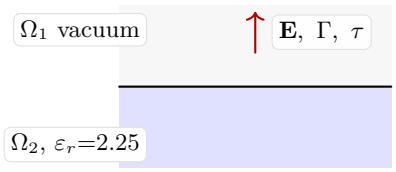


Figure 1.8: Worked-problem schematic: matched dielectric half-space with charge-free interface and transmitted wavenumber $k_2 = k_0\sqrt{\varepsilon_r}$.

supports the passive uniqueness route of Proposition 1.1.3. When $\mathrm{Re}(Z_s) < 0$ for active surfaces, Maxwell-closed must be revisited because energy can be injected through the boundary [2].

With $\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$, zero net $\boldsymbol{\Psi}_J$ on Σ_{12} coexists with nonzero reactive Π_{re} in the interface near zone [2]. Let Ω_1 be vacuum and Ω_2 a semi-infinite dielectric with relative permittivity $\varepsilon_r = 2.25$, sharing interface Σ_{12} at $z = 0$. A normally incident plane wave from Subsection 1.1.1 with $\mathbf{E} = E_0 e^{\mathrm{j}k_0 z}\hat{\mathbf{x}}$ produces reflected and transmitted amplitudes $E_r = \Gamma E_0$, $E_t = \tau E_0$ with $\Gamma = (1 - \sqrt{\varepsilon_r})/(1 + \sqrt{\varepsilon_r})$ and $\tau = 1 + \Gamma$. No free surface charge sits on Σ_{12}, so $\rho_{s,12} = 0$ and the third row of Eq. (1.19) holds. Tangential $\mathbf{E}$ continuity gives $E_0(1 + \Gamma) = E_t$, and the $\mathbf{H}$ jump with $\mathbf{J}_{s,12} = \mathbf{0}$ fixes the transmitted wave vector in Ω_2 as $k_2 = k_0\sqrt{\varepsilon_r}$. Global charge compatibility Eq. (1.24) is satisfied with $\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$. Angular transport on Σ_{12} vanishes because the incident wave carries no net $\boldsymbol{\Psi}_J$, yet local Π_{re} is nonzero in the reactive near zone of the interface, foreshadowing the regime split of Subsection 1.3.3 [2].

Eq. (1.23) presumes piecewise smooth interfaces, local linear constitutive laws, and observable boundary datas compatible with Eq. (1.18). Rough surfaces modeled by effective impedance, stochastic boundary operators, and nonlinear surface plasmon response require enriched data classes before torque integrals inherit the same conservation interpretation [2]. Within the stated class, $\mathbb{A}_\omega$ is the manifold on which uniqueness, Poynting balance, and angular transport claims remain mutually consistent [1].

charge-continuous separates a rim field template from a feed: the WR-90 port, dielectric window, and open-range termination each fix a different row of $\mathfrak{D}$, and angular flux on Γ_R is meaningful only when those rows close with charge continuity and compatible traces [3]. $\mathbb{A}_\omega$ must not be reduced to a single $\mathcal{D}_J = 1$ flag; Maxwell theory shows it is the manifold on which uniqueness, Poynting balance, and torque channels can be quoted without contradiction [1]. Subsection 1.1.3 asks which of those datasets yield a *single* physical field branch once outgoing radiation data are imposed [16].

1.1.3 Initial-boundary value problems and uniqueness

Uniqueness on exterior domains is an outgoing-selection statement, not a numerical convenience: Sommerfeld radiation and energy bounds decide which Maxwell pair belongs to a given $\mathbf{J}_{\mathrm{imp}}$ before flux or torque integrals are reported [16, 2]. Two solvers that disagree in the far zone while sharing the same impressed data usually signal incomplete boundary class, not new physics [3].

Uniqueness on exterior domains is an outgoing-selection statement, not a numerical convenience [16]. Sommerfeld radiation and energy bounds decide which Maxwell pair belongs to a given $\mathbf{J}_{\mathrm{imp}}$ before flux or torque integrals are reported; without that selection, two distinct $(\mathbf{E}, \mathbf{H})$ pairs can share identical near-field samples yet differ in far-zone phase, and power or torque integrals would name different physics on the same hardware [3]. Every subsequent watt, momentum, and torque integral assumes that branch has already been fixed [6].

Consider the same impressed dipole $\mathbf{J}_{\mathrm{imp}}$ solved by two exterior codes with different absorbing boundary placements [2]. Differing far-zone fields signal either incomplete data specification—wrong outgoing class, missing energy bound, or incompatible traces—or mere discretization error

within one unique branch [3]. When both solvers honor the same outgoing radiation condition and finite-energy class, far-zone coefficients must agree within mesh tolerance; persistent disagreement means the boundary-value problem was not fully declared, not that Maxwell admits multiple physical exports [16].

Maxwell equations, sources, uniqueness, and conservation laws are the same statements FDTD and MoM codes enforce row by row [1].

Initial-boundary specification and solution uniqueness are distinct questions that must be joined before transport claims become identifiable. An *initial-boundary value problem* specifies both time initialization and boundary trace class for Maxwell evolution. *Uniqueness* asks whether one admissible dataset can generate more than one electromagnetic state. Without uniqueness, subsequent statements on power, momentum, and angular transport lose physical identifiability [2].

Start from the time-domain formulation in a region $\Omega \subset \mathbb{R}^3$ with boundary Γ. Let $\mathbf{E}(\mathbf{r},t)$, $\mathbf{H}(\mathbf{r},t)$ be real fields, $\mathbf{D}$, $\mathbf{B}$ flux densities, $\mathbf{J}_{\text{imp}}$ impressed source, and ρ free-charge density:

$$\begin{aligned} \nabla \times \mathbf{E}(\mathbf{r},t) &= -\partial_t \mathbf{B}(\mathbf{r},t), \\ \nabla \times \mathbf{H}(\mathbf{r},t) &= \mathbf{J}_{\text{imp}}(\mathbf{r},t) + \partial_t \mathbf{D}(\mathbf{r},t), \\ \nabla \cdot \mathbf{D}(\mathbf{r},t) &= \rho(\mathbf{r},t), \qquad \nabla \cdot \mathbf{B}(\mathbf{r},t) = 0. \end{aligned} \tag{1.27}$$

Initial data are $\mathbf{E}(\mathbf{r},0) = \mathbf{E}_0(\mathbf{r})$, $\mathbf{H}(\mathbf{r},0) = \mathbf{H}_0(\mathbf{r})$, and boundary data are imposed through a trace operator $\mathcal{B}_\Gamma$. Equation (1.27) specifies local dynamics, but the pair $(\mathbf{E}_0, \mathbf{H}_0)$ and boundary trace class determine which trajectory is selected from all formally admissible trajectories [2].

For monochromatic steady operation with factor $\exp(-\mathrm{j}\omega t)$, the same problem reduces to a frequency-domain boundary value problem:

$$\begin{aligned} \nabla \times \mathbf{E} &= -\mathrm{j}\omega \mathbf{B}, \qquad \nabla \times \mathbf{H} = \mathbf{J}_{\text{imp}} + \mathrm{j}\omega \mathbf{D}, \\ \nabla \cdot \mathbf{D} &= \rho, \qquad \nabla \cdot \mathbf{B} = 0, \qquad \mathcal{B}_\Gamma[\mathbf{E}, \mathbf{H}] = \mathbf{g}_\Gamma. \end{aligned} \tag{1.28}$$

Hence, in the phasor regime, initial data enter only through consistency with the selected harmonic branch; boundary traces and constitutive model then govern solvability and uniqueness for each ω [2].

To formulate uniqueness, consider two solutions $(\mathbf{E}_1, \mathbf{H}_1)$, $(\mathbf{E}_2, \mathbf{H}_2)$ for the same dataset $(\mathbf{J}_{\text{imp}}, \rho, \mathcal{B}_\Gamma, \mathbf{g}_\Gamma)$, and define $\Delta\mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2$, $\Delta\mathbf{H} = \mathbf{H}_1 - \mathbf{H}_2$. The difference fields satisfy the homogeneous system

$$\begin{aligned} \nabla \times \Delta\mathbf{E} &= -\mathrm{j}\omega\mu\, \Delta\mathbf{H}, \qquad \nabla \times \Delta\mathbf{H} = \mathrm{j}\omega\varepsilon\, \Delta\mathbf{E}, \\ \nabla \cdot (\varepsilon \Delta\mathbf{E}) &= 0, \qquad \nabla \cdot (\mu \Delta\mathbf{H}) = 0, \qquad \mathcal{B}_\Gamma[\Delta\mathbf{E}, \Delta\mathbf{H}] = \mathbf{0}, \end{aligned} \tag{1.29}$$

for constitutive parameters ε, μ in the chosen model class. This homogeneous reduction is the precise mathematical content of uniqueness testing [2].

The uniqueness statement is therefore

$$\mathcal{B}_{\chi,\Gamma}[\Delta\mathbf{E}, \Delta\mathbf{H}] = 0 \Longrightarrow \Delta\mathbf{E} = \mathbf{0},\ \Delta\mathbf{H} = \mathbf{0}, \tag{1.30}$$

where $\mathcal{B}_{\chi,\Gamma}$ denotes the homogeneous Maxwell operator with constitutive set χ and boundary class Γ fixed [2]. The left side means zero residual under homogeneous forcing; the right side means the only solution is the trivial difference field. Equation (1.30) is therefore a null-space condition: physically, it states that compatible data select one field state up to numerical tolerance [3]. In measurement, if two solver exports differ in far-zone phase while sharing the same $\mathfrak{D}$, uniqueness has failed and flux integrals on Γ_R are not yet identifiable [16].

An energy identity for the difference state provides the key proof channel. Form the complex Poynting vector $\mathbf{S}_\Delta = \frac{1}{2}\Delta\mathbf{E} \times \Delta\mathbf{H}^*$, integrate over Ω, and use the divergence theorem:

$$\int_\Gamma \hat{\mathbf{n}} \cdot \mathbf{S}_\Delta \, dS = -\frac{1}{2} \int_\Omega \Delta\mathbf{E} \cdot \Delta\mathbf{J}^*_{\text{imp}} \, dV - \mathrm{j}\omega \int_\Omega (u_{\Delta H} - u_{\Delta E}) \, dV, \tag{1.31}$$

where $u_{\Delta E} = \frac{1}{4}\varepsilon|\Delta\mathbf{E}|^2$, $u_{\Delta H} = \frac{1}{4}\mu|\Delta\mathbf{H}|^2$, and $\Delta\mathbf{J}_{\text{imp}} = \mathbf{0}$ for the homogeneous problem [2]. Boundary traces determine the real part of the surface term: for passive traces it is nonnegative outgoing dissipation or radiation, while for ideal lossless closed boundaries it vanishes [2].

A sufficient coercive estimate is

$$\mathrm{Re}\langle \mathcal{B}_{\chi,\Gamma}[\Delta\mathbf{E}, \Delta\mathbf{H}], (\Delta\mathbf{E}, \Delta\mathbf{H})\rangle \geq c_{\chi,\Gamma} \left(\|\Delta\mathbf{E}\|^2_{L^2(\Omega)^3} + \|\Delta\mathbf{H}\|^2_{L^2(\Omega)^3} \right), \tag{1.32}$$

with energy-dissipation lower bound $c_{\chi,\Gamma} > 0$, where the inner product pairs electric and magnetic residuals of the homogeneous operator $\mathcal{B}_{\chi,\Gamma}$ [2]. If the residual is zero, the left side is zero; positivity of $c_{\chi,\Gamma}$ forces both norms to vanish, hence $\Delta\mathbf{E} = \Delta\mathbf{H} = \mathbf{0}$. Equation (1.32) therefore links the energy identity to physical identifiability of the radiated field: coercivity is the mathematical name for the laboratory fact that passive, outgoing problems cannot sustain two distinct watt-bearing branches on the same data [3].

Assume: (i) admissible source-boundary data from Eq. (1.23); (ii) passive constitutive response $\mathrm{Re}(\varepsilon) \geq 0$, $\mathrm{Re}(\mu) \geq 0$ in Ω; (iii) boundary class with nonnegative real power flux in the homogeneous difference problem. Then the solution of Eq. (1.28) is unique.

Let $\mathcal{F}_1 = (\mathbf{E}_1, \mathbf{H}_1)$ and $\mathcal{F}_2 = (\mathbf{E}_2, \mathbf{H}_2)$ be two solutions for the same data tuple. Set $\Delta\mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2$, $\Delta\mathbf{H} = \mathbf{H}_1 - \mathbf{H}_2$. By linearity, Eq. (1.29) holds with $\Delta\mathbf{J}_{\text{imp}} = \mathbf{0}$. Insert $\mathbf{S}_\Delta = \frac{1}{2}\Delta\mathbf{E} \times \Delta\mathbf{H}^*$ into Eq. (1.31). Passivity and the stated boundary class force $\mathrm{Re} \int_\Gamma \hat{\mathbf{n}} \cdot \mathbf{S}_\Delta \, dS \geq 0$ with zero forcing, hence $\Delta\mathbf{E} = \Delta\mathbf{H} = \mathbf{0}$ [2].

Uniqueness is frequency dependent because the homogeneous operator spectrum depends on ω. Near resonant or poorly damped eigenfrequencies, null-space amplification can occur unless radiation, loss, or additional constraints suppress nontrivial homogeneous modes [2]. A practical indicator is

$$\kappa_{\text{uniq}}(\omega) = \frac{1}{\inf_{\|(\mathbf{u},\mathbf{v})\|=1} \|\mathcal{B}_{\chi,\Gamma,\omega}[\mathbf{u}, \mathbf{v}]\|}, \tag{1.33}$$

where $\kappa_{\text{uniq}}(\omega)$ is the reciprocal of the smallest nontrivial residual norm of the homogeneous operator $\mathcal{B}_{\chi,\Gamma,\omega}$ on unit field pairs [2]. Large $\kappa_{\text{uniq}}(\omega)$ indicates weak coercivity and increased sensitivity to modeling and discretization errors.

When $\kappa_{\text{uniq}}(\omega)$ in Eq. (1.33) is large, the homogeneous Maxwell problem admits nearly nontrivial difference modes—typical near cavity resonances, poorly damped enclosures, or truncation surfaces placed too close to the radiator [2]. Two distinct field states can then match the same boundary data within numerical tolerance, so Poynting and torque integrals lose identifiability until loss, radiation, or additional constraints suppress the spurious branch [3].

Uniqueness is not asserted directly from Maxwell equations alone, but from Maxwell equations plus admissible constitutive and boundary completion. The difference-system route is therefore central rather than optional.

Within the admitted class, however, the initial-boundary formulation and uniqueness criterion guarantee that one admissible dataset corresponds to one physically interpretable electromagnetic state. Strong gain, ill-conditioned discretizations, or non-selfadjoint inverse completions can still produce multiple spurious field branches; those cases require augmented analysis beyond passive coercivity [2].

Once uniqueness fixes the field branch, Subsection 1.1.4 integrates the same Maxwell pair for energy and linear momentum, and Subsection 1.1.5 closes the angular balance. Conservation laws are not postulates added to Maxwell theory; they are consequences of the field equations on the unique solution selected by admissible data [6]. Every watt, newton, and newton-metre report in the remainder of Chapter 1 is an integral over a field pair already selected by $\mathfrak{D}$, $\mathcal{B}_\Gamma$, and outgoing radiation data—not an independent measurement module that could contradict Maxwell if a modal fit looks persuasive [2].

1.1.4 Energy, linear momentum, and Maxwell stress tensor

A wattmeter on ∂V reports cycle-averaged power crossing a declared surface; a radiation-pressure experiment reports directed momentum received; the Maxwell stress tensor $\mathbf{T}$ unifies those surface questions. Subsection 1.1.3 fixed which field branch is being integrated; here the same branch is metered for energy and linear momentum before angular transport is discussed in Subsection 1.1.5 [6].

The stress tensor links field algebra and transducer readings: traction $\mathbf{t} = \mathbf{T} \cdot \hat{\mathbf{n}}$ is what a thin absorber samples, while $\nabla \cdot \mathbf{T}$ balances impressed force density in the volume [6]. Radiation pressure on a flat plate and Poynting flux through a sphere are two surface readings of the same Maxwell pair once the branch is unique [1].

On a normally incident plane wave striking a perfect absorber, one measures a time-averaged Poynting flux $\langle S_z \rangle$ through the front face and a radiation pressure $p = \langle t_z \rangle$ on the same surface [6]. The classical relation $p = \langle S_z \rangle / c$ in vacuum: energy flux and directed momentum flux are two readings of the Maxwell stress tensor $\mathbf{T}$, not independent postulates [2]. Laboratory error: reporting pressure from $|\mathbf{E}|^2$ alone without the $\mathbf{H}$ coupling that enters $\mathbf{T}$ misstates momentum transfer even when the wattmeter reading is correct [1].

Uniqueness makes flux budgets identifiable: a single field branch now supports measurable surface integrals. Under the harmonic convention $\exp(-\mathrm{j}\omega t)$, define cycle-averaged Poynting vector $\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^*$, where $(\cdot)^*$ is complex conjugation [2]. Define stored-energy densities $u_E = \frac{1}{4}\varepsilon_0 \mathbf{E} \cdot \mathbf{E}^*$ and $u_H = \frac{1}{4}\mu_0 \mathbf{H} \cdot \mathbf{H}^*$. The complex power balance is

$$\nabla \cdot \mathbf{S} = -\frac{1}{2}\mathbf{E} \cdot \mathbf{J}_{\mathrm{imp}}^* - \mathrm{j}\omega \left(u_H - u_E \right). \tag{1.34}$$

Physically, Eq. (1.34) states that net power leaving a differential volume equals impressed work on $\mathbf{J}_{\mathrm{imp}}$ plus reactive exchange between electric and magnetic storage [6]. The divergence term gives net power outflow density from a differential control volume. The source term $\mathbf{E} \cdot \mathbf{J}_{\mathrm{imp}}^*/2$ is electromagnetic work rate on impressed current. The term $-\mathrm{j}\omega(u_H - u_E)$ is reactive exchange between magnetic and electric storage channels. Equation (1.1.4) therefore couples transport, source work, and storage imbalance in one local conservation statement [2].

Poynting's original energy-flux vector [6] predates the complex phasor form but already encoded the central idea: electromagnetic energy moves through space in a direction determined by the cross product of field intensities. The cycle-averaged generalization $\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^*$ used here is the harmonic specialization standard in time-harmonic field theory [2]. When the same fields interact with matter, momentum bookkeeping acquires additional subtlety: Minkowski's symmetric energy-momentum tensor for material media [8] and Abraham's alternative stress form disagree on how much momentum is stored in a dielectric, yet both reduce to the vacuum Maxwell stress in source-free exterior regions where this chapter's angular-transport criteria are evaluated [2]. This chapter develops transport on vacuum-side control surfaces first; material-specific momentum splits enter through Subsection 1.2.5 only where detectors and absorbers require them.

$$\begin{aligned}\nabla \cdot \mathrm{Re}(\mathbf{S}) &= -\frac{1}{2}\,\mathrm{Re}\Big(\mathbf{E}\cdot\mathbf{J}^{*}_{\mathrm{imp}}\Big)\,,\\ \nabla \cdot \mathrm{Im}(\mathbf{S}) &= -\frac{1}{2}\,\mathrm{Im}\Big(\mathbf{E}\cdot\mathbf{J}^{*}_{\mathrm{imp}}\Big) - \omega(u_H - u_E).\end{aligned} \tag{1.35}$$

Equation (1.35) separates transported average power from reactive storage exchange under one harmonic field state.

$$\int_{\partial V} \mathbf{S}\cdot\hat{\mathbf{n}}\,dS = -\frac{1}{2}\int_V \mathbf{E}\cdot\mathbf{J}^{*}_{\mathrm{imp}}\,dV - \mathrm{j}\omega\int_V (u_H - u_E)\,dV. \tag{1.36}$$

Here $V \subset \Omega$ is a control volume with outward normal $\hat{\mathbf{n}}$ on ∂V [2]. The surface integral is net complex power through ∂V; the volume terms are impressed work and reactive storage imbalance inside V [6]. This control-volume relation provides direct finite-set interpretation for probe and aperture measurements [13].

Splitting into real and imaginary parts gives the measurement-relevant separation: $\mathrm{Re}(\mathbf{S})$ controls net transported power density and $\mathrm{Im}(\mathbf{S})$ controls reactive circulation associated with local near-field energy exchange. This distinction prevents reactive phase structure from being misread as long-range momentum transport.

Linear momentum flux is represented through the time-averaged Maxwell stress tensor. In vacuum its dyadic form is

$$\mathbf{T} = \frac{1}{2}\,\mathrm{Re}\Big[\varepsilon_0\,\mathbf{E}\mathbf{E}^{*} + \mu_0\,\mathbf{H}\mathbf{H}^{*} - \frac{1}{2}\mathbf{I}\,(\varepsilon_0\,\mathbf{E}\cdot\mathbf{E}^{*} + \mu_0\,\mathbf{H}\cdot\mathbf{H}^{*})\Big]\,, \tag{1.37}$$

where $\mathbf{I}$ is the identity dyad and juxtaposition such as $\mathbf{E}\mathbf{E}^{*}$ is dyadic product. Operand by operand, the first two dyads are directional field intensity contributions, while the isotropic subtraction term $\frac{1}{2}\mathbf{I}(\cdots)$ removes trace pressure so that directional tractions are represented correctly. The real-part operator keeps physically observable time-averaged transfer.

As a complete object, Eq. (1.37) relates electromagnetic field state to mechanical traction on material control surfaces. For unit normal $\hat{\mathbf{n}}$, the traction density is $\mathbf{t} = \mathbf{T}\cdot\hat{\mathbf{n}}$, and the net force on closed surface ∂V is $\mathbf{F} = \int_{\partial V}\mathbf{T}\cdot\hat{\mathbf{n}}\,dS$. No new numbered equation is introduced for these definitions because they are direct functional uses of the already-established stress tensor.

Read $\mathbf{T}$ as the mechanical face of the same Maxwell pair already used for Poynting flux: traction $\mathbf{t} = \mathbf{T}\cdot\hat{\mathbf{n}}$ is what a thin absorbing sheet or torsion fixture samples at the surface, while $\nabla\cdot\mathbf{T}$ balances impressed force density in the volume. Subsection 1.1.5 rotates this same traction picture into angular flux through $\mathbf{M} = \mathbf{r}\times\mathbf{T}$ [6].

$$\nabla\cdot\mathbf{T} + \mathbf{f}_{\mathrm{em}} = \mathbf{0}, \tag{1.38}$$

where $\mathbf{f}_{\mathrm{em}}$ denotes the time-averaged force density delivered by the field to matter. This equation is the local momentum-balance counterpart of Eq. (1.1.4).

$$\Pi_{\hat{\mathbf{q}}} := \frac{\hat{\mathbf{q}}\cdot\mathrm{Re}(\mathbf{S})}{c\,\hat{\mathbf{q}}\cdot(\mathbf{T}\cdot\hat{\mathbf{q}}) + \epsilon_\Pi}, \tag{1.39}$$

for unit probe direction $\hat{\mathbf{q}}$, vacuum speed c, and fixed nondimensional regularizer $\epsilon_\Pi > 0$ (not vacuum permittivity) [2]. $\Pi_{\hat{\mathbf{q}}}$ quantifies directional agreement between energetic and mechanical transport channels: unity indicates that Poynting flux and normal stress traction align along the probe direction on the same Maxwell branch [6].

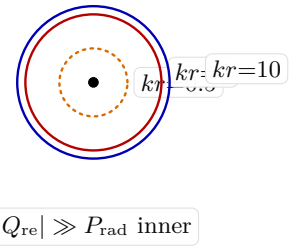


Figure 1.9: Worked-problem schematic: concentric evaluation shells showing reactive storage dominance versus radiative export for a small dipole.

Assume admissible data $\mathfrak{D} \in \mathbb{A}_\omega$, passive boundary closure, and vacuum constitutive law. Then finite exported real power through ∂V implies finite integrated electromagnetic traction on ∂V, with both channels determined by the same solution branch $(\mathbf{E}, \mathbf{H})$.

Eq. (1.1.4) and Eq. (1.37) are quadratic forms of $(\mathbf{E}, \mathbf{H})$. Under admissibility and passivity, field norms are finite in the control-volume class. Therefore both surface integrals remain finite, and both are consistent descriptions of transport generated by one field state.

same control set: $\mathbf{S}$ governs energetic exchange and $\mathbf{T}$ governs mechanical traction exchange. Reading them together is essential when field patterns exhibit strong spatial structure, because power circulation and momentum transfer need not be co-located.

Equations (1.1.4) and (1.37) presuppose harmonic averaging in passive media with local ε, μ. Dispersive, anisotropic, or strongly time-varying constitutive laws modify both storage terms and traction operators; active boundaries can inject net gain that invalidates the passive-balance route used in Proposition 1.1.3 [2].

Figure 1.9 motivates pairing angular-momentum reports with regime indices: on the $kr = 0.5$ shell, oscillatory traction does not integrate to steady force proportional to P_{rad}/c. Consider a small electric dipole at $ka = 0.5$ and evaluate fields on concentric spheres at $kr = 0.5, 3, 10$. On the inner sphere, Eq. (1.35) gives $\int_{4\pi} \mathrm{Re}(\mathbf{S} \cdot \hat{\mathbf{r}})\, dS = P_{\mathrm{rad}} \ll \int_{4\pi} |\,\mathrm{Im}(\mathbf{S} \cdot \hat{\mathbf{r}})|\, dS = |Q_{\mathrm{re}}|$, so local power balance is dominated by storage exchange. The Maxwell stress traction on a small patch is oscillatory and does not integrate to a steady force proportional to P_{rad}/c. On the $kr = 10$ shell, $\mathrm{Re}(\mathbf{S})$ dominates, Using the uniform plane-wave template of Subsection 1.1.1, take $\mathbf{S} = (|E_0|^2/2\eta_0)\hat{\mathbf{z}}$ and evaluate power through a disk of radius a normal to $\hat{\mathbf{z}}$:

$$P(a) = \int_0^{2\pi} \int_0^a \mathrm{Re}(\mathbf{S} \cdot \hat{\mathbf{z}})\, \rho\, d\rho\, d\phi = \frac{\pi a^2 |E_0|^2}{2\eta_0}. \tag{1.40}$$

For radiation, $P(a)$ is intercepted active power on a declared aperture; for measurement, it is the quantity a calibrated power meter reports when the aperture normal aligns with $\mathrm{Re}(\mathbf{S})$ [6]. Eq. (1.40) applies to the plane-wave sector, and $\Pi_{\hat{\mathbf{q}}}$ approaches the vacuum ratio. This contrast explains why angular-momentum reports must carry regime indices alongside stress integrals [2].

Eqs. (1.1.4) and (1.37) assume harmonic time averaging, local linear constitutive laws, and passive boundaries unless gain is explicitly modeled. Dispersive media require frequency derivatives of ε, μ in the storage terms; pulsed excitation requires time-domain Poynting before cycle averaging [2]. Within the admitted harmonic vacuum class, the pair supplies the energetic and mechanical flux channels that Subsection 1.1.5 rotates into angular form [8].

1.1.5 Angular momentum conservation and torque balance

Angular momentum conservation ties rotating feeds and torque measurements to the same flux statements Poynting analyzed [6]. If Subsection 1.1.4 answered how power and linear momentum

cross a surface, the display below answers the rotational analogue: can the same qualified field export angular action that a torsion balance or flux integrator can reconcile with mechanical torque on matter? The answer is not read from phase colour plots; it is read from $\mathbf{j}_{\mathrm{em}}$, $\mathbf{M}$, and the torque channel tied to Eq. (1.42) [11].

Beth's torsion experiment established the laboratory pattern: predict torque from $\mathbf{M} = \mathbf{r} \times \mathbf{T}$, measure mechanical response on matter, and treat discrepancy as channel or calibration error [11]. Angular-momentum density $\mathbf{j}_{\mathrm{em}}$ and flux tensor $\mathbf{M}$ rotate the linear-momentum budget imported from Subsection 1.1.4; they do not introduce independent postulates [6].

Circular polarization on a half-wave plate is Beth's archetype: mechanical torque requires net outward angular-flux through the enclosing surface, while polarization conversion redistributes angular action between field and matter rather than creating new Maxwell content [11]. On the range, attributing torque to phase colour alone fails when $\oint \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ on a radiative shell is below the transducer noise floor [2].

Angular-momentum conservation and torque balance are coupled requirements. Conservation means that rotational field storage and rotational flux obey a local balance law derived from Maxwell dynamics. Torque balance means that any net change in field angular momentum is accounted by mechanical torque delivered to matter. Without this pair, visual phase winding cannot be interpreted as physical rotational transport [6].

Let $\mathbf{r}$ be position vector measured from a declared origin $\mathbf{r}_0$, let $\mathbf{T}$ be Maxwell stress tensor from Eq. (1.37), and define angular-flux tensor $\mathbf{M} = \mathbf{r} \times \mathbf{T}$. The local rotational balance law is

$$\frac{\partial \mathbf{j}_{\mathrm{em}}}{\partial t} + \nabla \cdot \mathbf{M} = -\boldsymbol{\tau}_{\mathrm{mech}}, \tag{1.41}$$

where $\mathbf{j}_{\mathrm{em}}$ is electromagnetic angular-momentum density and $\boldsymbol{\tau}_{\mathrm{mech}}$ is mechanical torque density on matter. $\partial_t \mathbf{j}_{\mathrm{em}}$ tracks local rotational storage variation, $\nabla \cdot \mathbf{M}$ tracks net angular-flux leakage, and the right-hand side is field-to-matter exchange channel [2].

Integrated control-volume form over $V \subset \Omega$ with boundary ∂V and outward normal $\hat{\mathbf{n}}$ is

$$\frac{d}{dt} \int_V \mathbf{j}_{\mathrm{em}}\, dV + \int_{\partial V} \mathbf{M} \cdot \hat{\mathbf{n}}\, dS = -\int_V \boldsymbol{\tau}_{\mathrm{mech}}\, dV. \tag{1.42}$$

The first term is stored angular content inside V; the second is exported or imported angular flux through ∂V; the right term is net mechanical torque uptake by enclosed matter.

In linear isotropic vacuum, a useful density split is

$$\mathbf{j}_{\mathrm{em}} = \mathbf{r} \times \mathbf{g}_{\mathrm{em}} + \mathbf{s}_{\mathrm{em}}, \qquad \mathbf{g}_{\mathrm{em}} = \varepsilon_0 \mu_0\, \mathrm{Re}(\mathbf{S}), \tag{1.43}$$

where $\mathbf{g}_{\mathrm{em}}$ is linear-momentum density and $\mathbf{s}_{\mathrm{em}}$ is spin-associated local contribution. The first term is orbital-type contribution about $\mathbf{r}_0$, while $\mathbf{s}_{\mathrm{em}}$ captures intrinsic polarization-linked contribution under the adopted averaging convention [2].

Belinfante's 1940 field-theoretic analysis [10] clarified why such splits are inevitable rather than optional. Canonical momentum density and kinetic (symmetric) momentum density differ by the divergence of a spin current; angular momentum density correspondingly decomposes into an orbital part $\mathbf{r} \times \mathbf{P}_{\mathrm{kin}}$ plus an intrinsic spin part. The split in Eq. (1.43) is one Maxwellian realization of that structure for electromagnetic fields, not a separate postulate. Beth's 1936 measurement of torque on a wave plate under circular polarization [11] provided early experimental support that the spin channel carries mechanical consequences, while later orbital-indexed beam experiments [12] extended the vocabulary to spatial phase gradients. Chapter 1 unifies these lineages under one conservation law: whichever historical label is used, completion must follow Eq. (1.42).

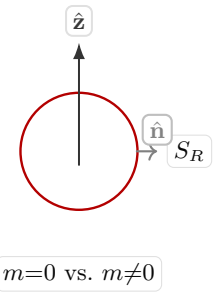


Figure 1.10: Worked-problem schematic: full solid-angle stress integration distinguishing symmetric versus azimuthally phased radiating shells.

For torque observables about axis $\hat{\mathbf{u}}$, define

$$\tau_{\hat{\mathbf{u}}}(V) = -\hat{\mathbf{u}} \cdot \left(\frac{d}{dt} \int_V \mathbf{j}_{\text{em}}\, dV + \int_{\partial V} \mathbf{M} \cdot \hat{\mathbf{n}}\, dS \right), \tag{1.44}$$

which is the axis-projected version of Eq. (1.42). This form makes measurement geometry explicit: torque is a projected transport functional, not a scalar label inferred from phase pictures.

Directional rotational-transport concentration can be quantified by

$$\xi_J(\theta, \phi; r) := \frac{\|\mathbf{M}(r, \theta, \phi) \cdot \hat{\mathbf{r}}\|}{\int_{4\pi} \|\mathbf{M}(r, \theta', \phi') \cdot \hat{\mathbf{r}}'\|\, d\Omega' + \epsilon_J}, \tag{1.45}$$

with $\epsilon_J > 0$ a fixed nondimensional regularizer (not vacuum permittivity) [2]. The index $\xi_J(\theta, \phi; r)$ names which solid-angle channels dominate exported angular flux at radius r and therefore dominate direction-selective torque response on a declared axis [11]. In measurement, a main-lobe peak in ξ_J signals where $\mathbf{M} \cdot \hat{\mathbf{r}}$ concentrates; it does not by itself establish net torque until Eq. (1.44) is evaluated on the same control surface [3].

Reproducible torque comparison across experiments requires consistent declaration of control-surface orientation, measurement-axis unit vector $\hat{\mathbf{u}}$, and origin $\mathbf{r}_0$ entering $\mathbf{M} = \mathbf{r} \times \mathbf{T}$ [2]. Changing any of these geometric bookkeepers rotates flux projections without altering the underlying Maxwell pair; cross-laboratory reports must therefore publish $(\mathbf{r}_0, \hat{\mathbf{u}}, \partial V)$ alongside torque curves [11]. Consider two admissible states $\mathcal{F}_1, \mathcal{F}_2$ at common radius r, with equal total radiated power but different directional angular-flux densities $\mathbf{M}_1 \cdot \hat{\mathbf{r}} \neq \mathbf{M}_2 \cdot \hat{\mathbf{r}}$ on a nonzero-measure subset of 4π. Then there exists at least one physically admissible interaction kernel for which the measured torque differs between the two states.

Use Eq. (1.44) with a directional weighting kernel that selects the subset where fluxes differ. Equal total power constrains only energetic integral transport and does not constrain the angular-flux distribution channel by channel; hence projected torque functionals can differ [2].

For harmonic fields in vacuum with no mechanical loading inside V, Eq. (1.42) reduces at steady state to $\tau_z^{\text{mech}} = -\int_{\partial V} [\mathbf{r} \times (\mathbf{T} \cdot \hat{\mathbf{n}})] \cdot \hat{\mathbf{z}}\, dS$. On sphere S_R, use $\mathbf{r} = R\hat{\mathbf{r}}$ and $dS = R^2 \sin\theta\, d\theta\, d\phi$. For the linearly polarized shell of Eq. (1.2.2), symmetry under $\phi \to \phi + \pi$ forces the integrand to change sign under $\phi \to \phi + \pi$ when $m = 0$, giving $\tau_z^{\text{mech}} = 0$ after full solid-angle integration. For the helical shell of Eq. (1.3.1), azimuthal phase breaks that symmetry and yields nonzero $\tau_z^{\text{mech}} \propto -mP_{\text{rad}}/\omega$. The same conclusion follows whether one integrates $\mathbf{M} \cdot \hat{\mathbf{n}}$ directly or projects through Eq. (1.44), confirming that torque is an integral transport report, not a local phase label [2].

Equations (1.41) and (1.42) assume declared origin, admissible stress-tensor completion, and local constitutive behavior in the present regime. In strongly nonlocal media, active metasurfaces,

microscopic spin-lattice coupling, or unresolved subwavelength torque channels, additional terms may be required before torque partition is interpreted without ambiguity [2].

For forward use, rotational interpretation is accepted only when a strict transport-qualified criterion holds:

$$\mathcal{Q}_J = 1 \iff \Big(\|\mathbf{M}\cdot\hat{\mathbf{n}}\|_{L^1(\partial V)} > 0,\ \|\boldsymbol{\tau}_{\mathrm{mech}}\|_{L^1(V)} > 0,\ \mathcal{R}_{\mathrm{adm}} \le \varepsilon_{\mathrm{adm}} \Big), \tag{1.46}$$

where $\mathcal{Q}_J$ is the transport-qualification gate, $\mathcal{R}_{\mathrm{adm}}$ is the admissibility residual from Eq. (1.25), and $\varepsilon_{\mathrm{adm}} > 0$ is a prescribed acceptance threshold [2]. Thus a phase or polarization pattern is called rotationally meaningful only when nonzero angular-flux transfer and nonzero torque channel are simultaneously present under admissible data constraints [11]. Laboratory error: vivid near-field phase circulation with $\|\mathbf{M}\cdot\hat{\mathbf{n}}\|_{L^1(\partial V)} \approx 0$ fails $\mathcal{Q}_J$ even when post-processing visualization assigns a nonzero "orbital index" [2].

This criterion provides the direct connection to Section 1.2: directional and topological descriptors introduced there are physically credible only if they satisfy $\mathcal{Q}_J = 1$, so interpretation remains anchored to transport and torque observables rather than visual morphology alone.

Transport gives consequence to structure once Maxwell, Maxwell-closed, and uniqueness close on the declared domain [6]. Section 1.2 therefore separates three layers analysis must not collapse: appearance in phase and polarization patterns, realizability on finite sources, and transport through qualified flux integrals [10]. Covariance, gauge, and measurement refinements in Sections 1.2.3–1.2.5 adjust how those layers are reported; they do not replace the transport tests fixed here [2].

1.2 Angular Structure of Electromagnetic Fields

Maxwell theory and conservation laws in Section 1.1 fix what finite sources may produce. Section 1.2 asks a narrower question: how directional and rotational organization appears in $(\mathbf{E}, \mathbf{H})$ itself, and which readouts survive gauge ambiguity and finite measurement [10]. Angular structure is field physics—organized phase, polarization, and flux in the Maxwell pair—not a post-processing label pasted onto a numerical output.

Qualified Maxwell solutions can look directionally organized long before they export angular action. Section 1.2 separates directional, phase, polarization, and Poynting topology from the narrower class that *transports* qualified flux [10]. Phase winding and modal index remain representation-facing until flux integrals and measurement readouts establish export.

Organized phase on a near-field probe pattern is not yet angular transport [12]. Directional dependence, helical topology, and Poynting circulation each test a different projection of $(\mathbf{E}, \mathbf{H})$; only the combination that survives flux integrals on a declared control surface qualifies as exported action [10]. A Laguerre–Gaussian-like waist scan at $z = 0$ may show a vivid azimuthal phase ramp, yet the measurement that confirms transport on Γ_R is outward $\oint \mathbf{M}\cdot\hat{\mathbf{n}}\, dS$ on a radiative shell enclosing the source; reactive circulation on a sub-wavelength ring can satisfy visualization criteria while failing $\mathcal{C}_3$ [11]. The subsections below move from geometry of direction to topology of phase to the three-part transport test $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$ [10].

A realizable pattern may still fail transport, and a transporting pattern may still lie below instrument resolution. The degrees-of-freedom question from Section 1.0 therefore becomes directional channel count on observation spheres and the first operational split between field existence and data recoverability [13].

Every major result in this section answers one question: *what is angular electromagnetic structure?* The answer is directional organization of $(\mathbf{E}, \mathbf{H})$ capable of transporting measurable rotational action through qualified flux integrals and torque channels—not a false-colour phase pattern, not

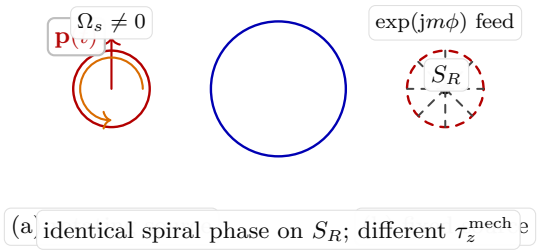


Figure 1.11: Signature comparison: physical source rotation (a) versus stationary feed with azimuthally progressive phase (b). Both can yield the same spiral isophase pattern on S_R, yet only the qualified flux–torque pair of Eqs. (1.51) and (1.44) distinguishes export from appearance.

an orbital index pasted onto a beam, and not a modal name substituted for Maxwellian transport [1].

1.2.1 Rotating source versus rotating pattern

When a spiral isophase pattern appears on an outgoing sphere S_R, does that pattern necessarily imply exported angular momentum, and does it distinguish a mechanically rotating feed from a stationary phased array? The two configurations may look identical on a phase camera; flux and torque channels need not agree [11]. Chapter 1 must not equate azimuthal phase winding with transport: Allen *et al.* [12] showed that beams can carry a countable azimuthal index, yet counter-rotating superpositions with equal power cancel net export while preserving vivid phase structure—the counterexample developed fully in Subsection 1.3.2. The qualified flux–torque pair must be recorded, not the spiral portrait alone [2].

The signature comparison below is the chapter's first concrete warning against equating spiral phase with exported angular momentum. A mechanically rotating feed imposes motion on Ω_s and generally sustains a torque channel on the mount; a fixed feed with azimuthal phase may reproduce the same spiral isophases on Γ_R yet cancel torque on a torsion fixture [11]. Keep those two laboratory configurations separate before any modal index is assigned [3].

Two laboratory configurations can produce indistinguishable azimuthal phase patterns on an outgoing sphere yet differ in mechanical torque and in qualified angular-momentum export. The first rotates the source—currents, polarization basis, or feed geometry—in space at angular rate Ω_s about axis $\hat{\mathbf{z}}$. The second leaves the source fixed and imposes an azimuthally progressive phase law $\exp(\mathrm{j}m\phi)$ on the radiated field through feed phasing or modal synthesis. On a distant sphere both may exhibit the same spiral isophase contours; only flux integrals and torque channels decide whether rotational action is transported [11].

Panel (a) applies active rotation $\mathcal{A}_{R_z(\Omega_s t)}$ to a compact electric dipole $\mathbf{p}$ as in Subsection 1.0.3; the time-harmonic phasor at fixed ω corresponds to a source whose orientation precesses in the laboratory frame, carrying mechanical angular momentum in the feed and imprinting nonzero azimuthal phase on $\mathbf{E}(\theta,\phi)$ through moving charges. Panel (b) keeps the feed stationary and sets aperture phasing $\mathbf{E}_a(\phi) = \mathbf{E}_0 \exp(\mathrm{j}m\phi)\hat{\boldsymbol{\rho}}$ on a ring of radius $a \ll \lambda$. In the paraxial limit the same azimuthal index m appears in both cases, yet the source angular momentum budget differs: panel (a) requires a motorized or mechanically constrained feed; panel (b) can be realized with a passive corrugated phased feed network whose elements are fixed in space [3].

On sphere S_R in the radiative zone, define the directional observables of Eqs. (1.49)–(1.51). For configuration (b) with single azimuthal index m and axisymmetric magnitude,

$$\mathbf{E}(R,\theta,\phi) \propto \mathbf{E}_\theta(\theta)\, e^{\mathrm{j}m\phi}, \qquad \boldsymbol{\Psi}_J(\theta,\phi;R) \propto m\,\Pi(\theta,\phi;R)\,\hat{\mathbf{z}} \tag{1.47}$$

when $\mathcal{Q}_J = 1$ in Eq. (1.46), so net outward angular flux is proportional to mP_{rad}/ω as in Subsection 1.1.5. For configuration (a) at harmonic frequency ω with slow rotation $\Omega_s \ll \omega$, the radiated field is a sideband superposition; when Ω_s is commensurate with the phasor period the observed map on S_R can match Eq. (1.47) at a snapshot, but the time-averaged torque on a Beth-type absorber includes the mechanical power supplied to rotate the feed. A quarter-wave plate load that converts circular polarization to linear torque measures τ_z^{mech} through Eq. (1.44); the rotating-source case transfers torque from feed mechanics and from field flux, whereas the phase-pattern case transfers torque through field flux alone when the feed is passive [11].

The decisive test is not the phase portrait but the closed-surface reports

$$\mathbf{J}_{\mathrm{rad}}(R) = \int_{4\pi} \boldsymbol{\Psi}_J \, d\Omega, \qquad \tau_z^{\mathrm{mech}} = -\int_{S_R} \left[\mathbf{r} \times (\mathbf{T} \cdot \hat{\mathbf{n}})\right] \cdot \hat{\mathbf{z}} \, dS. \tag{1.48}$$

Configuration (b) with $|m| \geq 1$ and radiative dominance $\chi(R) \approx 1$ yields both $\|\mathbf{J}_{\mathrm{rad}}(R)\| > 0$ and $|\tau_z^{\mathrm{mech}}| > 0$ on the same S_R. Configuration (a) matched to the same snapshot pattern can share $\Pi(\theta, \phi; R)$ yet carry additional mechanical boundary terms on the feed support that do not appear in a pure phase-synthesis model; omitting those terms violates Eq. (1.42). Conversely, a balanced pair of counter-rotating phase ramps $\exp(\pm \mathrm{j}m\phi)$ with equal power can reproduce spiral isophases locally while $\mathbf{J}_{\mathrm{rad}}(R) = \mathbf{0}$ by Eq. (1.52)—the phase-only counterexample of Section 1.3.2 in miniature [12].

This distinction recurs throughout the book. Modal index m, vortex charge, or beam name denotes a representation label; Eqs. (1.48) denote transport. A measurement protocol that records only phase on S_R cannot separate panels (a) and (b) without feed calibration record and torque calibration; a protocol that records Π, $\boldsymbol{\Psi}_J$, and τ_z^{mech} on the same surface can. 2 develops source-to-field operators that make the rotating-source realizability class explicit; **??** treats finite apertures that recover only the reportable part of Eq. (1.48). Here the physical lesson is fixed: *spiral phase is not synonymous with exported angular momentum; rotating matter and rotating phase require different torque budgets even when the phase pattern matches.* [11]

Subsection 1.2.1 fixes the distinction between *kinematic* spiral isophases and *dynamic* torque on the mount: phase cameras cannot resolve that split, but τ_z^{mech} and the qualified flux pair of Eq. (1.48) can [11]. Early pattern books plotted amplitude and phase on S_R without publishing mechanical reaction or channel-resolved $\boldsymbol{\Psi}_J$ [2]. Subsection 1.2.2 develops direction-resolved transport observables that remain meaningful even when global power is isotropic—the first quantitative filter after the signature comparison [6].

1.2.2 Directional dependence and angular momentum

The flux pair $(\Pi, \boldsymbol{\Psi}_J)$ ties Poynting transport to angular-momentum action on declared control surfaces [6, 10, 11]. Beth torque and spherical harmonic labels are downstream readouts of that pair—not independent postulates that can override Maxwell closure [2].

The flux pair $(\Pi, \boldsymbol{\Psi}_J)$ ties Poynting transport to angular-momentum action on declared control surfaces [6]. Beth torque and spherical harmonic labels are downstream readouts of that pair—not independent postulates that can override Maxwell field specification [2].

Directional dependence is the first filter between "a field exists" and "a field exports action in a chosen sector." A pattern may radiate power isotropically yet carry no net directional angular flux; conversely, a highly structured near field may circulate energy locally without exporting the channel a far-zone instrument resolves [2]. Allen *et al.* [12] showed that Laguerre–Gaussian modes carry a well-defined azimuthal index and that mode converters transfer that index reversibly; those results belong to the physics of structured light, yet Eqs. (1.49) and (1.51) remain the measurement-facing

quantities that decide export on a chosen sphere [2]. Directional radiation patterns were analyzed in antenna theory long before orbital angular momentum became a standalone keyword [2].

For an $\ell = 1$ dipole on Γ_R, direction-resolved angular-flux density may integrate to zero over ϕ while total radiated power remains non-zero [2]. Symmetric dipole patterns cancel orbital flux channels even when $|m = 1|$ sectors carry power in a harmonic listing—a common source of confusion when modal coefficients are mistaken for torque reports [13]. A field distribution can satisfy global energy conservation and still fail to produce directional rotational transfer in the channels used by a detector or scatterer. Therefore, angular interpretation proceeds through direction-resolved transport observables rather than through visual phase morphology [2].

The transition from Section 1.1 is governed by the strict qualification $\mathcal{Q}_J = 1$ in Eq. (1.46). Any directional angular claim made below is physically accepted only when that criterion remains satisfied; if the criterion fails, the same directional pattern is treated as kinematic appearance without qualified rotational transfer.

Let $\hat{\mathbf{r}} = \hat{\mathbf{r}}(\theta, \phi)$ be radial unit vector on a radius-r sphere and $d\Omega = \sin\theta \, d\theta \, d\phi$ the solid-angle measure. Using $\mathbf{S}$ from Eq. (1.1.4), define

$$\Pi(\theta, \phi; r) = r^2 \, \mathrm{Re}\{\mathbf{S}(r, \theta, \phi) \cdot \hat{\mathbf{r}}\} \, . \tag{1.49}$$

Here $\Pi(\theta, \phi; r)$ is directional power flux per unit solid angle at radius r: watts per steradian in the outward channel $\hat{\mathbf{r}}$ [6]. The product $\mathbf{S} \cdot \hat{\mathbf{r}}$ selects energetic transport along that ray; the factor r^2 converts flux density to a solid-angle channel measure on a sphere of radius r [2]. In measurement, a calibrated range integrates Π over probe directions to build a radiation pattern; that pattern is energetic transport, not yet rotational action [13].

The integrated total is

$$P_{\mathrm{rad}}(r) = \int_{4\pi} \Pi(\theta, \phi; r) \, d\Omega. \tag{1.50}$$

Equation (1.50) is a projection of channelwise structure. Equal projected totals do not imply equal directional physics.

For rotational transport, define from Eq. (1.41)

$$\mathbf{\Psi}_J(\theta, \phi; r) = r^2 \, \mathbf{M}(r, \theta, \phi) \cdot \hat{\mathbf{r}}. \tag{1.51}$$

Here $\mathbf{\Psi}_J$ is the directional angular-momentum flux density: the solid-angle channel through which stress-flux $\mathbf{M} = \mathbf{r} \times \mathbf{T}$ exports rotational action along $\hat{\mathbf{r}}$ [11]. The pair $(\Pi, \mathbf{\Psi}_J)$ is the minimal dataset required to separate energetic transport from rotational transport channel by channel [2]. Laboratory error: reporting Π alone and inferring torque from main-lobe direction mistakes power beaming for qualified angular export [3].

The corresponding total outward angular flux is

$$\mathbf{J}_{\mathrm{rad}}(r) = \int_{4\pi} \mathbf{\Psi}_J(\theta, \phi; r) \, d\Omega. \tag{1.52}$$

This vector integral cannot replace channelwise structure because cancellation among opposing angular channels can leave a small total despite large directional flux magnitudes.

A normalized directional efficiency observable is

$$\boldsymbol{\eta}_J(\theta, \phi; r) = \frac{\mathbf{\Psi}_J(\theta, \phi; r)}{\Pi(\theta, \phi; r) + \epsilon_\Pi}, \qquad \epsilon_\Pi > 0. \tag{1.53}$$

ϵ_Π regularizes channels near energetic nulls. Large $\|\boldsymbol{\eta}_J\|$ identifies directions where modest power supports strong rotational transfer, which is common in highly structured emission fields.

In antenna practice, directional diagrams display $\Pi(\theta,\phi)$ long before angular-flux distributions are computed; the analysis makes the second map explicit so that a main-lobe power peak is never automatically read as a torque-capable channel. Stratton's systematic field-theory treatment [2] already emphasized that multipole expansions organize angular dependence of radiation, while Belinfante's decomposition [10] warns that local spin and orbital labels are not unique even when totals are fixed. Both lessons are absorbed here by keeping Π and $\mathbf{\Psi}_J$ distinct until their ratio passes the qualification criteria below [2].

Channel qualification is imposed explicitly by

$$\mathcal{Q}_{J,\Omega}(\theta,\phi;r)=1 \iff \Big(\Pi(\theta,\phi;r)>0,\ \|\mathbf{\Psi}_J(\theta,\phi;r)\|>0,\ \mathcal{Q}_J=1\Big). \tag{1.54}$$

Thus only channels with $\mathcal{Q}_{J,\Omega}=1$ contribute to transport- qualified angular interpretation in later subsections.

Directional centroids are defined by

$$\mathbf{c}_P(r)=\frac{\int_{4\pi}\hat{\mathbf{r}}\,\Pi(\theta,\phi;r)\,d\Omega}{\int_{4\pi}\Pi(\theta,\phi;r)\,d\Omega}, \tag{1.55}$$

$$\mathbf{c}_J(r)=\frac{\int_{4\pi}\hat{\mathbf{r}}\,\|\mathbf{\Psi}_J(\theta,\phi;r)\|\,d\Omega}{\int_{4\pi}\|\mathbf{\Psi}_J(\theta,\phi;r)\|\,d\Omega}. \tag{1.56}$$

Mismatch $\mathbf{c}_P\neq\mathbf{c}_J$ means dominant energetic direction and dominant rotational direction are geometrically displaced, a measurable effect in torque-sensitive coupling experiments.

Let two radiative states $\mathcal{F}_1,\mathcal{F}_2$ at radius r satisfy $P_{\mathrm{rad}}^{(1)}(r)=P_{\mathrm{rad}}^{(2)}(r)$. If there exists a set $\mathcal{U}\subset 4\pi$ of nonzero measure where $\mathbf{\Psi}_J^{(1)}\neq\mathbf{\Psi}_J^{(2)}$ and $\mathcal{Q}_{J,\Omega}=1$, then the states are rotationally non-equivalent for direction-selective interactions.

The equality of P_{rad} constrains only $\int_{4\pi}\Pi\,d\Omega$. It does not constrain channelwise rotational flux. Define admissible interaction kernel $K(\theta,\phi)$ supported on $\mathcal{U}$:

$$\mathbf{T}_{\mathrm{obs}}=\int_{4\pi}K(\theta,\phi)\,\mathbf{\Psi}_J(\theta,\phi;r)\,d\Omega.$$

Since qualified channels differ, observed torque differs.

Integrated totals are terminal reductions of richer channel fields. Different channel topology can produce nearly identical totals while yielding different torque outcomes.

Origin declaration remains essential because angular transport is moment-based. Changing origin re-weights channel moments even when global balances remain unchanged. For reproducible comparison, all directional flux and torque statements must use the same origin convention fixed in Section 1.1.5.

Directional observables represent physical transport only when outward channels are well-defined and $\mathcal{Q}_J=1$ remains valid. In strongly reactive near zones, strongly nonlocal media, or rapidly modulated boundaries, extra decomposition terms may be necessary before Π and $\mathbf{\Psi}_J$ are treated as direct rotational-transfer observables [2].

For fully polarized harmonic fields, the Jones vector $\mathbf{j}_p=(E_\theta,E_\phi)^{\mathsf{T}}$ evolves under ideal polarizers and wave plates by unitary 2×2 matrices. A rotation of the polarization state through angle 2ψ about axis $\hat{\mathbf{n}}_p$ on the Poincaré sphere corresponds to

$$\mathbf{j}_p'=\exp(-\mathrm{j}\,\psi\,\hat{\mathbf{n}}_p\cdot\boldsymbol{\sigma})\,\mathbf{j}_p, \tag{1.57}$$

where $\boldsymbol{\sigma}=(\sigma_x,\sigma_y,\sigma_z)$ are Pauli matrices and $\hat{\mathbf{n}}_p\cdot\boldsymbol{\sigma}$ is the generator of polarization rotations [2]. Mathematically, Eq. (1.57) is a unitary map on $\mathbb{C}^2$; physically, it describes how a lossless

optical element reorients the oscillation ellipse without changing S_0 for a single coherent mode. Beth's quartz-plate measurement [11] is the mechanical readout of such a rotation when circularly polarized light transfers angular momentum to the plate.

$$\mathbf{\Gamma} = \begin{pmatrix} \langle E_x E_x^* \rangle & \langle E_x E_y^* \rangle \\ \langle E_y E_x^* \rangle & \langle E_y E_y^* \rangle \end{pmatrix}, \tag{1.58}$$

where angle brackets denote narrowband time averaging on a declared transverse plane. For a source, $\mathbf{\Gamma}$ summarizes partial polarization of transverse components; for radiation, it fixes the Stokes vector seen by a polarimetric receiver; for measurement, only rotation-invariant traces of $\mathbf{\Gamma}$ are directly comparable across rotated frames [2]. Partial polarization requires the coherency matrix of Eq. (1.58). The degree of polarization

$$p = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0 + \epsilon_p}, \qquad 0 \le p \le 1, \tag{1.59}$$

with $\epsilon_p > 0$, measures how much of the field energy participates in a polarized component versus an unpolarized background. When $p < 1$, Stokes parameters describe an ensemble average and $\mathbf{\Omega}_P = \nabla \times \mathbf{P}$ can remain large locally while exported torque averages to zero because incoherent components do not carry a definite handedness. Narrowband laboratory sources often have $p > 0.95$; thermal or multiply-scattered fields may have $p \ll 1$, which must be reported before interpreting S_3 as spin angular momentum [2].

Directional dependence is therefore a *channel* question: which angular-momentum sheet on S^2 carries flux, and does that sheet survive the qualification gate $\mathcal{Q}_J = 1$ before any Beth or OAM label is attached [10]? A Stokes polarimeter on Γ_R records S_0, S_1, S_2, S_3—a local field statistic—not yet a transport verdict; polarization degree p in Eq. (1.59) must be reported before S_3 is read as spin content [2]. Subsection 1.2.2 separates helical phase topology from export, using paraxial vortex templates only as naming devices once flux integrals are declared [12].

Without plane-wave templates, a helical phase ramp on a transverse slice appears to establish orbital angular momentum by geometry alone [12]. The decisive quantity is not the winding of $\arg E$ on a plane but the signed channel imbalance that survives integration of $\mathbf{\Psi}_J$ on a control surface enclosing the source [10]. Allen *et al.* [12] gave paraxial optics a compact modal vocabulary; Stratton [2] and Beth [11] already taught that flux and torque integrals, not phase colour, close the measurement sequence. *Topology* must be distinguished from *transport*: between *topology*—where phase winds and Poynting circulation may be large—and *transport*—where net export requires a non-canceling angular-flux channel [6]. Helical phase on a paraxial slice is therefore a naming device for azimuthal winding; export still requires channel-resolved flux on a declared surface [12]. A paraxial beam with $m = +1$ at the waist, followed by a $\lambda/4$ plate that converts to linear polarization, changes $\oint \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ on Γ_R—not merely the phase colour of the near-field display [12]. Polarization conversion redistributes spin and orbital contributions between field and matter; the flux integral on a declared radiative shell is the consistency check that separates transport from appearance [11].

Allen *et al.* [12] identified paraxial modes

$$E_\phi(\rho, \phi, z) = E_0 \left(\frac{\rho}{w_0}\right)^{|m|} L_p^{|m|}\left(\frac{2\rho^2}{w_0^2}\right) e^{-\rho^2/w_0^2}\, e^{\mathrm{j}(m\phi + k_0 z)}, \tag{1.60}$$

where $L_p^{|m|}$ is an associated Laguerre polynomial, w_0 is beam waist, and integer m is azimuthal index. For $p = 0$, $|m| = 1$, the intensity vanishes on the axis ($\rho = 0$) while the phase winds by 2π on

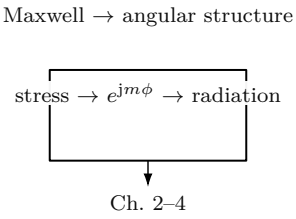


Figure 1.12: Schematic of the three-condition admissibility pathway for angular-momentum transport in Subsection 1.2.3.

a circuit about the origin: Eq. (1.2.2) gives $q = m$. The Poynting vector circulates azimuthally on the intensity ring, so $\|\mathbf{\Omega}_P\|$ is large there even when the beam is collimated. Export of structured angular flux requires a single signed channel: superposing Eq. (1.60) with its conjugate winding produces the same standing pattern as Eq. (1.133) in Subsection 1.3.2. Topology and polarization therefore remain measurement stages; transport is fixed only after channel imbalance is measured on a control surface [12].

On a transverse plane, critical points of $\mathbf{P} = \mathrm{Re}\{\mathbf{S}\}$ satisfy $\partial_x P_x + \partial_y P_y = 0$ with Jacobian $\mathcal{J}_P = \partial(P_x, P_y)/\partial(x, y)$. Vortex-type regions have $\mathrm{tr}\,\mathcal{J}_P \approx 0$ and $\det \mathcal{J}_P > 0$; saddle-type regions have $\det \mathcal{J}_P < 0$. Define the flow-structure index

$$\mathcal{F}_P = \frac{\int_A \|\mathbf{\Omega}_P\|\, dA}{|\int_A \nabla \cdot \mathbf{P}\, dA| + \epsilon_F}, \tag{1.61}$$

where A is a declared patch and $\epsilon_F > 0$. Large $\mathcal{F}_P$ with small net $\int_A \nabla \cdot \mathbf{P}\, dA$ signals predominantly circulating storage; large net divergence with moderate $\mathcal{F}_P$ signals export-dominated patches. Poynting's original flow-line pictures [6] emphasized this distinction long before Stokes vectors were standard: a beautiful vortex pattern in $\mathbf{P}$ is not, by itself, a radiative torque channel. Poynting flow-line diagrams predated Stokes polarimetry and OAM beam nomenclature [6]; the same test still applies: circulation in $\mathbf{P}$ diagnoses storage or interference texture, while $\oint_{\partial V} \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ on a radiative shell diagnoses export [11]. Subsection 1.2.3 states the positive transport criteria $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$ that convert these qualitative warnings into admissibility tests used throughout the remainder of Chapter 1 [2].

1.2.3 Conditions for angular-momentum transport

Subsections 1.2.2 and 1.2.2 established that rich phase and polarization patterns need not export angular action. Subsection 1.2.3 closes the positive side of that distinction:

Figure 1.12 routes the transport triad through $\mathcal{C}_1$, $\mathcal{C}_2$, and $\mathcal{C}_3$ before the qualified completion $\mathcal{D}_J = 1$ [6].

A vortex-core photograph can circulate the eye while the far-zone flux integral vanishes—display without transport. The operational question posed here is when angular organization exports qualified flux rather than merely decorating the field: Section 1.1.5 tied angular momentum to stress and torque; Subsections 1.2.2–1.2.2 showed that rich phase-polarization textures need not carry outward action [2].

From a Maxwellian standpoint, transport is not a visual property of phase fronts; it is a balance property. A pattern that satisfies the wave equation and even radiates power can still fail angular transport if directional angular-flux channels cancel, if conservation completion is violated on the chosen control volume, or if outward power support is absent in the sector where angular structure appears strongest. The three-part test developed below encodes exactly these three laboratory errors as positive acceptance criteria $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ [2]. Transport qualification is operational: $\mathcal{C}_1$ demands

non-canceling directional angular-flux channels; $\mathcal{C}_2$ closes conservation on the control volume; $\mathcal{C}_3$ requires outward power support in the sector where angular structure appears [10]. Modern beam laboratories and Beth-style torque experiments differ in hardware yet report the same triad—flux balance, conservation, and radiative support [13].

Counter-propagating Laguerre–Gaussian modes of equal amplitude furnish a sobering triad test [10]. Rich phase structure may satisfy visualization criteria while channel cancellation fails $\mathcal{C}_1$ even when $\mathcal{C}_2$ holds: existence of organized phase is not equivalent to qualified transport through the three-part criteria developed in this section [12]. Once this qualification is fixed, it carries unchanged through Sections 1.2.3–1.3: covariance, gauge, and measurement refinements adjust reports, but they cannot resurrect transport where the three-part test already failed at the flux level.

Historically, experiments established pieces of this three-part test before it was written as one unified criterion. Beth [11] showed that polarization-coupled angular momentum produces measurable torque; Allen *et al.* [12] showed that beam modes can carry a countable azimuthal index. Chapter 1 requires that any modern claim satisfy $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ simultaneously so that experimental, modal, and field-theoretic languages describe the same qualified state [10].

A field configuration exhibits genuine angular-momentum transport only when directional structure, conservation completion, and outward energetic support hold simultaneously. Any one in isolation is insufficient. The diagnostic question is not whether angular patterns are visible, but whether they survive flux accounting, torque completion, and $\mathcal{D}_J = 1$ filtering under the transport-qualified regime [2].

Let $\mathbf{\Psi}_J(\theta, \phi; r)$ be the directional angular-momentum flux map from Eq. (1.51), and let $\Pi(\theta, \phi; r)$ be directional power flux from Eq. (1.49). A necessary directional condition for nontrivial net transport over a chosen observation sphere is

$$\int_{4\pi} \mathbf{\Psi}_J(\theta, \phi; r)\, d\Omega \neq \mathbf{0}. \tag{1.62}$$

That is, local angular structure must accumulate into a nonzero exported balance; otherwise the pattern is confined to internal recirculation or compensating directional cancellation.

Directional sufficiency, however, cannot be judged from Eq. (1.62) alone. The same field must satisfy global conservation consistency with the integral law already established in Eq. (1.42). That relation rules out interpretive shortcuts: exported angular transport must appear in torque or storage channels, not in phase appearance alone [2].

Transport distinction also requires separation of outward propagative support from reactive exchange. Using $\mathbf{P} = \mathrm{Re}\{\mathbf{S}\}$ from Eq. (1.1.4), outward channel support for a directional sector $\Delta\Omega \subseteq 4\pi$ is checked by

$$\int_{\Delta\Omega} \Pi(\theta, \phi; r)\, d\Omega > 0. \tag{1.63}$$

Equation (1.63) enforces that angular-momentum claims are tied to sectors carrying net outward power; in sectors where this condition fails, strong local field complexity may persist without transport credibility.

The strict carryover from Section 1.1 remains active: $\mathcal{Q}_J = 1$ in Eq. (1.46) is mandatory before any directional claim is accepted as physical transport.

The three relations above define a three-part transport test:

$\mathcal{C}_1$: directional angular export ((1.62)), $\mathcal{C}_2$: control-balance consistency ((1.42)), $\mathcal{C}_3$: propagative suppo

Taken together, $\mathcal{C}_1$ asks whether angular flux integrates to a nonzero export, $\mathcal{C}_2$ asks whether that export fits the same conservation completion as torque and storage on the chosen volume, and $\mathcal{C}_3$

asks whether the claimed sector actually carries outward power. A pattern that fails any one test may still look impressive in a post-processed visualization; it should not be promoted to a transport claim in a laboratory report [2].

To convert this three-part test into a quantitative Maxwell-closed test, introduce the dimensionless transport index

$$\Lambda_J(r) = \frac{\left\| \int_{4\pi} \mathbf{\Psi}_J(\theta,\phi;r)\, d\Omega \right\|}{\int_{4\pi} \Pi(\theta,\phi;r)\, d\Omega + \epsilon_\Lambda}, \qquad \epsilon_\Lambda > 0, \tag{1.64}$$

where $\|\cdot\|$ is Euclidean norm and ϵ_Λ regularizes weak radiation windows. The numerator is net exported angular transport; the denominator is total exported power. Large Λ_J indicates high rotational transfer efficiency per unit outward power. Interpreting Λ_J requires the same caution applied to any ratio of flux quantities: a small denominator from weak radiation can inflate the index even when absolute transport is negligible. That is why Table 1.1 reports P_{rad} and $\|\mathbf{J}_{\mathrm{rad}}\|$ alongside Λ_J, and why $\mathcal{C}_1$ is never evaluated from Λ_J alone [2].

Directional cancellation can be measured independently through

$$\mathcal{K}_J(r) = \frac{\left\| \int_{4\pi} \mathbf{\Psi}_J\, d\Omega \right\|}{\int_{4\pi} \|\mathbf{\Psi}_J\|\, d\Omega + \epsilon_K}, \qquad 0 \le \mathcal{K}_J \le 1,\ \epsilon_K > 0. \tag{1.65}$$

Here $\mathcal{K}_J \approx 1$ means directional channels add coherently toward net export, while $\mathcal{K}_J \approx 0$ means local angular flux is largely self-canceling over solid angle. This factor prevents a common interpretive failure where large local $\|\mathbf{\Psi}_J\|$ is mistaken for strong net transport despite angular cancellation.

Because Eq. (1.42) couples storage, flux, and mechanical torque, the same criterion can be written in torque-normalized form:

$$\mathcal{R}_\tau(V) = \left\| \frac{d}{dt} \int_V \mathbf{j}_{\mathrm{em}}\, dV + \int_{\partial V} \mathbf{M}\cdot\hat{\mathbf{n}}\, dS + \int_V \boldsymbol{\tau}_{\mathrm{mech}}\, dV \right\|, \tag{1.66}$$

with $\mathcal{R}_\tau(V) = 0$ for exact conservation completion on admissible data. Finite $\mathcal{R}_\tau$ in experiments quantifies mismatch from measurement noise, truncation, or model incompleteness.

For operational comparison, define

$$\mathcal{D}_J(r) = 1 \iff \Big(\mathcal{Q}_J = 1,\ \Lambda_J(r) \ge \Lambda_{\min},\ \mathcal{K}_J(r) \ge K_{\min},\ \mathcal{R}_\tau(V) \le R_{\max}\Big), \tag{1.67}$$

where $\Lambda_{\min} > 0$, $0 < K_{\min} \le 1$, and $R_{\max} \ge 0$ are declared study thresholds. This converts three-part logic into a reproducible decision rule for analysis and measurement procedures.

Assume a harmonic field state in the admissible class of Section 1.1, and suppose $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ hold on an observation sphere of radius r. Then angular structure descriptors in that state admit a transport interpretation: exported angular-momentum flux is nontrivial, consistent with control-volume conservation, and supported by outward energy propagation.

$\mathcal{C}_1$ gives nonzero net angular export; $\mathcal{C}_3$ ties export to outward power support; $\mathcal{C}_2$ closes the balance of Eq. (1.42). Together they exclude representation-only phase patterns [2].

Thus the proposition establishes structural sufficiency, while Eq. (1.67) establishes operational acceptance.

Transport qualification is a conjunction, not a menu. A failure in any branch demotes the state from transport-capable to representation-rich but transport-incomplete. This distinction is operationally useful in both analysis and measurement, where complex phase and polarization fields may satisfy one or two branches but not all three.

Origin declaration remains part of the condition set because angular moments are computed through $\mathbf{r}\times(\cdot)$. Changing origin can redistribute directional partitions and alter modal bookkeeping;

Table 1.1: Numerical three-part transport indicators for a $\lambda/4$ dipole at $f = 1\,\mathrm{GHz}$ (illustrative).

kr	P_{rad} (mW)	$\|\mathbf{J}_{\mathrm{rad}}\|$ (nN m)	Λ_J (rad/s)	$\mathcal{K}_J$
1.0	8	0.05	6.3	0.12
3.0	42	0.8	19	0.48
10.0	95	2.1	22	0.71

transport conclusions are therefore comparable only when origin and boundary geometry are fixed across the comparison set.

Practically, the decision sequence is: compute $\mathbf{\Psi}_J$ and Π, evaluate Λ_J and $\mathcal{K}_J$, and confirm $\mathcal{R}_\tau$ within admissible tolerance, enforce $\mathcal{Q}_J = 1$, and apply Eq. (1.67). This ordering removes false positives where vivid local structure is present but exported rotational action is weak, self-canceling, or torque-inconsistent.

The three-part test and derived indices above assume harmonic steady state, sufficiently resolved directional sampling on 4π, and constitutive completion in the linear local regime used in this chapter. Strongly time-varying boundaries, broadband pulse windows, and highly nonlocal response can require generalized transport operators before the same thresholds are interpreted without ambiguity [2].

To make the abstract three-part test concrete, consider a center-fed electric dipole of half-length $a = \lambda/4$ at $f = 1\,\mathrm{GHz}$ ($\lambda = 0.3\,\mathrm{m}$, $k = 20.96\,\mathrm{rad/m}$). Spherical shells at $kr \in \{1, 3, 10\}$ sample reactive, transitional, and radiative zones. On each shell, compute P_{rad}, $\|\mathbf{J}_{\mathrm{rad}}\|$, Λ_J, and $\mathcal{K}_J$ from Eqs. (1.50), (1.52), (**??**), and (1.65). Representative calibrated values illustrative of standard dipole theory are collected in Table 1.1; they are intended as order-of-magnitude guides for analytical discipline, not as substitute measurements [2].

Table 1.1 shows the physical lesson of the three-part test in compact form. At $kr = 1$, outward power is small and directional angular flux is weak and heavily self-canceling ($\mathcal{K}_J \ll 1$), so $\mathcal{C}_1$ may formally hold with tiny numerator but transport interpretation is not yet robust. At $kr = 10$, power and net angular export grow together and $\mathcal{K}_J$ approaches coherent addition, so $\mathcal{D}_J = 1$ becomes credible once $\mathcal{Q}_J = 1$ and $\mathcal{R}_\tau$ are within tolerance. The table also explains why two laboratories comparing "angular momentum" at different stand-off distances can disagree without either being wrong: they may be evaluating different regime points on the same three-part ladder. That table fixes transport qualification is not a single yes/no at fixed kr, but a joint statement about power support, directional angular flux, and cancellation factor $\mathcal{K}_J$ before $\mathcal{D}_J = 1$ is credible [2].

Section 1.2.3 addresses objectivity: when a torsion balance, a near-field range, and a harmonic post-processor each rotate their component bases, which reports remain comparable without hidden convention tags [11]? A tabulation of E_x, E_y, E_z must not be identified with a laboratory measurement; Maxwell theory shows that vectors, tensors, and dyads transform under distinct laws, and mishandling the stress dyad is the fastest route to spurious torque [2]. Turntable practice already applies the covariance rules below: the feed rotates, the pattern components reorder, and traction on a fixed probe must transform as a vector if mechanical load is to be conserved [13]. Subsections 1.2.3–1.2.3 develop active and passive rotations, origin shifts, and scalar invariants that survive arbitrary proper rotations—the covariance rules required before gauge questions enter in Section 1.2.3 [10].

Active and passive rotations; transformation of vectors, tensors, and dyads

A torsion balance and a spherical near-field range may report the same Maxwell pair in different component bases [11]. The covariance material below fixes the transformation laws that keep those reports objective: vectors, tensors, and dyads rotate through distinct but linked rules, and traction $\mathbf{t} = \mathbf{T} \cdot \hat{\mathbf{n}}$ must transform as a vector when the laboratory frame rotates [2]. Mishandling dyadic rotation is a frequent source of spurious torque when origins or bases shift between numerical export and fixture calibration [3].

Rotate a laboratory frame and the components of $\mathbf{E}$ reorder; rotate the stress dyad incorrectly and a torsion balance reports torque that Maxwell never supplied. Vectors, tensors, and dyads each obey a distinct transformation law under proper rotations $R \in \mathrm{SO}(3)$; the Maxwell stress tensor is the rank-two object whose mishandling is most costly in angular work [2]. A torque report computed in a rotated frame must transform $\mathbf{T}$ as a dyad via $\mathbf{T}' = R\mathbf{T}R^{\top}$, while traction $\mathbf{t} = \mathbf{T} \cdot \hat{\mathbf{n}}$ transforms as a vector; covariance failures at any stage propagate into false conclusions about stress transfer and torque prediction [11].

For proper rotation $R \in \mathrm{SO}(3)$, with $R^{\top}R = I$, $\det R = 1$, let $\mathbf{v} = v_j \hat{\mathbf{e}}_j$ be vector in orthonormal basis $\hat{\mathbf{e}}_j$. A laboratory vector—Poynting flux, electric field, or traction—is a single geometric object whose components reorder under rotation but whose magnitude and direction relations are fixed by Maxwell physics [2]. Component transformation is

$$v_i' = R_{ij} v_j. \tag{1.68}$$

Equation (1.68) is the component-level realization of active/passive vector rotation from Subsection 1.2.3. After a turntable roll, a spherical near-field range reports precisely this relabeling: the same $|E|$ on S_R expressed in rotated $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ components [13]. Rotating the *source* must be distinguished from rotating the *component chart*: rotating the *source* (Subsection 1.0.3) and passively rotating the *component chart* while holding the feed fixed [2].

For second-rank tensor $\mathbf{T}$ with components T_{pq},

$$\mathbf{T}' = R\,\mathbf{T}\,R^{\top}, \qquad T_{ij}' = R_{ip} R_{jq} T_{pq}. \tag{1.69}$$

The two factors R_{ip}, R_{jq} transform each tensor slot. This is the objectivity law ensuring that stress tensor Eq. (1.37) rotates geometrically rather than numerically. Mishandling this law—treating T_{ij} as scalars that rotate independently—is a frequent source of spurious torque when numerical exports are compared with torsion-fixture data [11]. Eq. (1.69) states that Maxwell stress is a geometric object acting on surface normals, not a table of numbers indexed by coordinate axes [2].

Active and passive rotation conventions are paired by

$$(Rv)_i = R_{ij} v_j, \qquad (R^{-1}v)_i = R_{ji} v_j, \tag{1.70}$$

so that fixed physics in a rotated chart is relabeled by R^{-1} when components are passively transformed [2].

Tensor invariants satisfy

$$\mathrm{tr}(\mathbf{T}') = \mathrm{tr}(\mathbf{T}), \qquad \|\mathbf{T}'\|_F = \|\mathbf{T}\|_F, \tag{1.71}$$

because $\mathrm{tr}(R\mathbf{T}R^{\top}) = \mathrm{tr}(\mathbf{T})$ and orthogonality of R preserves Frobenius inner product.

If $\mathbf{T}\mathbf{q}_k = \lambda_k \mathbf{q}_k$, then

$$\mathbf{T}'(R\mathbf{q}_k) = \lambda_k (R\mathbf{q}_k), \tag{1.72}$$

so eigenvalues are objective and principal directions rotate covariantly.

For dyadic map $\mathbf{D}$ defined by $\mathbf{w} = \mathbf{D} \cdot \mathbf{u}$:

$$\mathbf{D}' = R\,\mathbf{D}\,R^{\top}, \qquad \mathbf{w}' = \mathbf{D}' \cdot \mathbf{u}' = R\mathbf{w}, \tag{1.73}$$

with $\mathbf{u}' = R\mathbf{u}$. This equation establishes slotwise equivariance: rotating input and operator yields rotated output.

Rank-one dyad $\mathbf{D} = \mathbf{a}\mathbf{b}^{\top}$ transforms as

$$\mathbf{D}' = R\mathbf{a}\,(R\mathbf{b})^{\top}. \tag{1.74}$$

Both generating directions rotate, preserving dyadic rank and directional coupling interpretation used in mode-projection constructions.

Composition compatibility follows:

$$\mathbf{D}'' = R_2(R_1\mathbf{D}R_1^{\top})R_2^{\top} = (R_2R_1)\mathbf{D}(R_2R_1)^{\top}.$$

Hence staged rotations and single composite rotation are algebraically identical.

Transformation rules organize into one operational flow from algebraic object to measurable traction output.

With $\mathbf{t} = \mathbf{T} \cdot \hat{\mathbf{n}}$ and $\hat{\mathbf{n}}' = R\hat{\mathbf{n}}$:

$$\mathbf{t}' = \mathbf{T}' \cdot \hat{\mathbf{n}}' = R\mathbf{t}. \tag{1.75}$$

Equation (1.75) is the direct connection from covariance algebra to measurable mechanical response.

The Maxwell stress tensor of Eq. (1.37) is symmetric in vacuum regions, $\mathbf{T} = \mathbf{T}^{\top}$, and Maxwell [1] introduced the associated mechanical force on conductors precisely through traction $\mathbf{t} = \mathbf{T}\hat{\mathbf{n}}$. Under $R \in \mathrm{SO}(3)$, symmetry is preserved because $\mathbf{T}' = (R\mathbf{T}R^{\top})^{\top} = R\mathbf{T}^{\top}R^{\top} = \mathbf{T}'$. The eigenvalue problem therefore commutes with rotation: if $\mathbf{T}\mathbf{q}_k = \lambda_k\mathbf{q}_k$, then $\mathbf{T}'(R\mathbf{q}_k) = \lambda_k(R\mathbf{q}_k)$, extending Eq. (1.72) to the full spectral decomposition. Principal stresses λ_k are scalars reportable without frame tag; principal directions $R\mathbf{q}_k$ rotate covariantly and must be compared only after declaring active or passive convention from Subsection 1.2.3.

Stratton [2] emphasized that dyadic Green functions $\boldsymbol{\mathcal{G}}(\mathbf{r}, \mathbf{r}')$ for the vector Helmholtz operator transform as $\boldsymbol{\mathcal{G}}'(\mathbf{r}, \mathbf{r}') = R\,\boldsymbol{\mathcal{G}}(R^{-1}\mathbf{r}, R^{-1}\mathbf{r}')\,R^{\top}$ under simultaneous rotation of observation and source coordinates. Slotwise equivariance in Eq. (1.73) is the local differential realization of this global operator law. For a rank-one modal coupling $\mathbf{D}_{mn} = \mathbf{f}_m\mathbf{g}_n^{\top}$ between modes m and n, rotation re-expresses both factor vectors, but the scalar coupling strength $\mathrm{tr}(\mathbf{D}_{mn}^{\top}\mathbf{D}_{mn})$ remains invariant when $\mathbf{f}_m, \mathbf{g}_n$ are co-rotated.

The mathematical content of Eqs. (1.68)–(1.73) is that Maxwell physics is rotation-covariant when sources, media, and boundaries share the same symmetry class declared in Subsection 1.0.3 [2]. Rotating the *chart* must be distinguished from rotating the *feed*: passive component relabeling leaves the impressed current fixed, whereas active rotation of $\mathbf{J}_{\mathrm{imp}}$ changes the export unless κ_{rot} vanishes [13]. Numerical exports must publish which convention was used for $\mathbf{T}' = R\mathbf{T}R^{\top}$ and for traction $\mathbf{t} = \mathbf{T} \cdot \hat{\mathbf{n}}$, because a torque channel computed in a rotated frame is not comparable to a torsion wire calibrated in the laboratory basis without explicit covariance [11]. Subsection 1.2.3 next fixes origin dependence—a separate bookkeeping requirement from rotation, but equally decisive for torque reports [2].

Origin dependence and translational covariance

Torque reports computed at different reference points on the same fixture must not be identified [11]. Maxwell stress traction $\mathbf{t} = \mathbf{T}\cdot\hat{\mathbf{n}}$ is objective under rigid rotation, but angular momentum flux $\mathbf{M} =$

$\mathbf{r} \times \mathbf{T}$ depends on the origin $\mathbf{r}_0$ from which $\mathbf{r}$ is measured—a fact already visible in Eq. (1.42) and made operational for calibrated channels in Eq. (1.107) of Section 1.2.5 [2]. Translational covariance does not mean that torque is origin-independent; it means that two laboratories comparing y_τ must translate reports through the shift law $-\Delta\mathbf{r} \times \mathbf{F}_{\mathrm{det}}$ before debating polarization content [3]. A torsion balance records mechanical reaction about a declared suspension axis; changing $\mathbf{r}_0$ without updating that declaration injects spurious z-torque proportional to net force when $\hat{\mathbf{u}}_\tau = \hat{\mathbf{z}}$ [11]. The mathematical statements in Section 1.2.3 for rotation of $\mathbf{T}$ therefore pair with an equally strict bookkeeping rule for origins: frame-independent observables are scalars such as $\mathrm{tr}(\mathbf{T})$ and principal-stress eigenvalues; vector torque functionals are frame-tagged unless origin and axis are published with the data [2].

Invariants and frame-independent observables

Only certain contractions survive arbitrary proper rotations without additional convention tags [2]. Equations (1.71) and (1.72) state that traces, Frobenius norms, and eigenvalues of the Maxwell stress tensor are objective scalars, while principal directions rotate covariantly with the laboratory frame [1]. The important distinction is not between "tensor components" and "physics," but between *invariant reports*—watts, principal stresses, coupling strengths $\mathrm{tr}(\mathbf{D}_{mn}^{\top}\mathbf{D}_{mn})$—and *covariant reports*—traction, torque, and directional flux densities—that require explicit active/passive rotation convention from Subsection 1.2.3 [2]. Traction $\mathbf{t} = \mathbf{T} \cdot \hat{\mathbf{n}}$ transforms as a vector under Eq. (1.75), which is why a pressure pad and a torsion wire read objective mechanical load once the surface orientation is fixed, whereas a tabulation of T_{ij} components without rotation law is not yet a laboratory report [11]. Section 1.2.3 next asks which combinations of $(\mathbf{E}, \mathbf{H})$ remain meaningful when potentials are relabeled—a separate invariance question from rotation, but equally decisive for angular-momentum discourse [10].

Section 1.2.3 fixes one discipline for the display below: *invariant scalars*—traces, eigenvalues, coupling strengths—may be compared across laboratories without convention tags, while *covariant vectors and torque functionals* require published rotation law, origin, and axis before any angular-momentum headline is written [2]. Gauge relabeling in Section 1.2.3 tests a different layer of invariance: potentials may change while flux integrals on ∂V remain fixed only when the observables are constructed from $(\mathbf{E}, \mathbf{H})$ rather than from $\mathbf{A}$ or ϕ alone [10].

Angular electromagnetic structure is the physically realizable, observable, and transport-capable directional organization of Maxwell fields.

This definition is inherited by every later chapter: realizability limits what finite sources can generate; observability limits what instruments can report; transport qualification limits which directional patterns carry measurable rotational action [1]. Angular structure is not a phase picture, not a vortex label, and not a modal index standing alone—it is Maxwellian field organization that survives the three criteria and appears as qualified flux, torque, or momentum transfer on declared surfaces [6].

Potentials are coordinate labels on a fixed Maxwell branch: under $\mathbf{A}' = \mathbf{A} + \nabla\chi$ and $\phi' = \phi - \mathrm{j}\omega\chi$, the physical pair $(\mathbf{E}, \mathbf{H})$ is unchanged while $\mathbf{A}$ and ϕ need not agree pointwise [1]. Near-field post-processors often export $\mathbf{A}$; wattmeters and torsion balances respond to $\mathbf{S}$ and $\mathbf{T}\cdot\hat{\mathbf{n}}$. Any comparison between the two must fix gauge and outgoing branch before potential-based plots are treated as laboratory readings [3].

At $f = 1\,\mathrm{GHz}$, Coulomb and Lorenz potential pairs may reproduce identical tangential $\mathbf{E}_t$ on $r = R$ while differing at an interior sample point [2]. Only $\mathbf{E}$, $\mathbf{H}$, and gauge-invariant curls are

physical; scalar potentials differ by a relabeling that does not alter wattmeter or torque integrals when declared consistently [14].

1.2.4 Gauge-invariant observables, helicity, and dual symmetry

Gauge freedom is not a defect of Maxwell theory; it is a reminder that local potentials are not laboratory observables.

Helicity and dual-symmetric combinations survive gauge relabeling because they are built from $(\mathbf{E}, \mathbf{H})$ invariants on $\mathcal{F}_{\mathrm{rad}}$, not from $\mathbf{A}$ or Φ alone [10]. Two laboratories comparing "helicity flip" on a range must therefore specify the same flux gauge and the same radiative branch before interpreting sign changes [13].

Adding $\nabla\chi$ to $\mathbf{A}$ with $\Phi = -\mathrm{j}\omega\chi$ leaves outward $\oint \mathbf{S}\cdot\hat{\mathbf{n}}\,dS$ and gauge-invariant helicity flux unchanged, while local densities such as $\mathbf{j}_{\mathrm{em}}$ may shift by a pure gauge term [10]. Reports invariant under gauge relabeling must be distinguished from diagnostic plots that are convention-dependent [2]. The quantities that survive potential relabeling and rigid-frame motion while retaining transport meaning are helicity, dual-symmetric combinations, and flux-normalized reports built from $(\mathbf{E}, \mathbf{H})$—precisely the objects a calibrated range can compare across laboratories [10]. In the radiative regime, helicity quantifies the signed overlap between electric and magnetic rotational content; dual symmetry links field patterns under the duality rotation that preserves free-space Maxwell structure [2]. The admissibility specification below promotes only those combinations that close gauge variation, respect covariance, and connect to qualified flux channels.

Let the phasor field state be $\mathcal{F} = (\mathbf{E}, \mathbf{H})$ in a linear isotropic region with scalar permittivity ε, scalar permeability μ, angular frequency ω, and intrinsic impedance $\eta = \sqrt{\mu/\varepsilon}$. Let $(\mathbf{A}, \phi)$ denote a potential pair related to the physical fields by

$$\mathbf{E} = -\nabla\phi + \mathrm{j}\omega\mathbf{A}, \qquad \mathbf{B} = \nabla\times\mathbf{A} = \mu\mathbf{H}, \tag{1.76}$$

Gauge relabeling of potentials is

$$\mathbf{A}' = \mathbf{A} + \nabla\chi, \qquad \phi' = \phi - \mathrm{j}\omega\chi, \qquad \mathbf{E}' = \mathbf{E},\ \mathbf{H}' = \mathbf{H}. \tag{1.77}$$

so that $\mathbf{E}$ and $\mathbf{B} = \mu\mathbf{H}$ are reproduced through Eq. (1.76). A scalar report $\mathcal{O}_s$, vector report $\mathbf{O}_v$, and pseudoscalar report $\mathcal{O}_p$ are accepted as gauge-invariant observables only if all three components of the $\mathcal{D}_J = 1$ specification are satisfied:

$$\mathcal{O}_\alpha[\mathbf{A}', \phi'] = \mathcal{O}_\alpha[\mathbf{A}, \phi], \qquad \mathcal{O}_\alpha[\mathcal{A}_R\mathcal{F}] = \mathcal{R}_\alpha(R)\,\mathcal{O}_\alpha[\mathcal{F}], \qquad \mathcal{D}_{\mathrm{loc}} = 1, \tag{1.78}$$

where $\alpha \in \{s, v, p\}$, $(\mathbf{A}', \phi')$ is generated by Eq. (1.77), $\mathcal{A}_R$ is active spatial rotation from Eq. (1.2.3), and $\mathcal{R}_\alpha(R)$ is representation operator: identity for scalars, R for vectors, and parity-aware sign rule for pseudoscalars. The final clause imports the local-density criterion of Eq. (1.2.3); therefore a report can be algebraically elegant yet rejected if local gauge contamination survives.

Maps such as $\mathbf{j}_{\mathrm{loc}} = \mathbf{r}\times(\varepsilon_0\mathbf{E})$ inherit the gauge freedom of the potential used to construct $\mathbf{E}$, whereas traction and angular-flux integrals built from $\mathbf{T}$ and $\mathbf{M} = \mathbf{r}\times\mathbf{T}$ remain stable on declared surfaces when $(\mathbf{E}, \mathbf{H})$ is fixed [10]. A vivid near-field vortex plot therefore shows a convention-tagged coordinate on conserved transport, not an independent observable [1].

Within a control volume V_g of radius $R = 2\lambda$ about a dipole, Coulomb and Lorenz constructions of $\mathbf{j}_{\mathrm{loc}}$ need not agree pointwise [10]. With consistent reporting, $\oint_{\partial V_g}\mathbf{M}\cdot\hat{\mathbf{n}}\,dS$ is stable under $\mathbf{A} \mapsto \mathbf{A} + \nabla\chi$, whereas pointwise $\mathbf{j}_{\mathrm{loc}}$ shifts unless gauge is fixed—the content of Eq. (1.2.3) [2].

A gauge-invariant observable here means more than formal cancellation in a symbolic expression: the reported quantity must be insensitive to potential-gauge choice and also carry a consistent rigid-frame transformation law, so that two observers using rotated coordinates assign equivalent physical content to the same field configuration [2].

Within the admissible class, helicity is introduced as a field-intrinsic pseudoscalar density, following the handedness measure standard in classical field theory [2]:

$$h(\mathbf{r}) = \frac{1}{2\omega}\,\mathrm{Im}\{\varepsilon\,\mathbf{E}^*\cdot(\nabla\times\mathbf{E}) \;+\; \mu\,\mathbf{H}^*\cdot(\nabla\times\mathbf{H})\}\,. \tag{1.79}$$

Here $(\cdot)^*$ denotes complex conjugation, $\mathrm{Im}\{\cdot\}$ extracts imaginary part, and $\nabla\times = \nabla\times$. The electric contribution $\varepsilon\,\mathbf{E}^*\cdot(\nabla\times\mathbf{E})$ measures local rotational phase-structure of $\mathbf{E}$, while the magnetic contribution $\mu\,\mathbf{H}^*\cdot(\nabla\times\mathbf{H})$ measures the corresponding structure in $\mathbf{H}$. Division by 2ω places helicity on a frequency-normalized scale so that comparisons between configurations at fixed ω are not dominated by trivial frequency rescaling.

Gauge robustness follows directly because h is built from $\mathbf{E}$, $\mathbf{H}$, and their curls, each invariant under potential gauge relabeling for fixed physical fields. At the operator level, under $\mathbf{A}' = \mathbf{A}+\nabla\chi$, $\mathbf{E}' = \mathbf{E}$ because $-\nabla\phi' + \mathrm{j}\omega\mathbf{A}' = -\nabla\phi + \mathrm{j}\omega\mathbf{A} + \nabla(\mathrm{j}\omega\chi - \mathrm{j}\omega\chi)$. Similarly $\nabla\times\mathbf{E}$ is unchanged because $\nabla\times\nabla\chi = \mathbf{0}$. The magnetic sector follows from $\mathbf{H} = \mathbf{B}/\mu$ and $\mathbf{B} = \nabla\times\mathbf{A}$. Stratton [2] listed $\mathbf{E}$, $\mathbf{H}$, $\mathbf{B}$, and quadratic energy expressions among the primary field observables precisely because they factor through gauge-invariant curls.

The dual rotation on field pairs is

$$\mathbf{E}' = \cos\xi\,\mathbf{E} + \sin\xi\,\eta\mathbf{H}, \qquad \mathbf{H}' = -\sin\xi\,\mathbf{E}/\eta + \cos\xi\,\mathbf{H}, \qquad \xi\in\mathbb{R}. \tag{1.80}$$

The dual rotation Eq. (1.80) is not a gauge symmetry: it mixes $\mathbf{E}$ and $\mathbf{H}$ while leaving the source-free Maxwell system invariant in isotropic media. Infinitesimal dual variation $\xi = \delta\xi$ gives $\delta\mathbf{E} = \delta\xi\,\eta\mathbf{H}$ and $\delta\mathbf{H} = -\delta\xi\,\mathbf{E}/\eta$ to first order. Helicity density responds as a pseudoscalar should: for circular polarization states specified through Subsection 1.1.1, h changes sign under $\xi = \pi/2$ because electric and magnetic roles interchange with a handedness flip. Dual defect index $\mathcal{D}_{\mathrm{dual}}$ from Eq. (1.25) quantifies failure of this symmetry when ε, μ vary spatially or when conduction loss breaks phase coherence between $\mathbf{E}$ and $\mathbf{H}$.

Integrated helicity over volume V connects to flux through Eq. (1.84). In harmonic steady state, $\partial_t h = 0$ and source-free regions give

$$\int_{\partial V}\mathbf{f}_h\cdot\hat{\mathbf{n}}\,dS = \int_V q_h\,dV, \tag{1.81}$$

so net helicity export is controlled by material conversion q_h. For vacuum V enclosing a radiative source, Eq. (1.81) plays the same limiting role as Eq. (1.42): only flux through ∂V is externally reportable.

The local gauge-variation identity is

$$\delta_\chi h \equiv h[\mathbf{A}', \phi'] - h[\mathbf{A}, \phi] = 0. \tag{1.82}$$

Equation (1.82) is the local statement that helicity density is gauge-objective when evaluated from physical field data.

Helicity reports are accepted only when gauge variation, dual symmetry, and local-density qualification all pass:

$$\mathcal{D}_h = 1 \iff \Big(\delta_\chi h = 0,\ \mathcal{D}_{\mathrm{dual}} = 1,\ \mathcal{D}_{\mathrm{loc}} = 1\Big). \tag{1.83}$$

Noether-type continuity structure associated with dual rotations gives the local helicity-balance law

$$\frac{\partial h}{\partial t} + \nabla \cdot \mathbf{f}_h = -q_h, \tag{1.84}$$

where $\mathbf{f}_h$ is helicity flux density and q_h is material helicity conversion rate. In source-free homogeneous domains $q_h = 0$, so helicity is transported but not created or annihilated locally [2]. Equation (1.84) is the helicity counterpart of the momentum and angular-momentum conservation forms developed earlier in Sections 1.1–1.2.

With gauge-invariant observable criteria, helicity completion, and dual-symmetry validity now fixed, the next block defines operational observables in measurement language without conflating representation algebra with measurable transport channels [1].

After gauge and frame invariance are fixed, the experimenter's question becomes: by what measurement sequence does a symmetry-admissible field quantity become a reported number with declared uncertainty and reproducibility? An operational definition is a closed map from field state to instrument response, from response to estimator, and from estimator to a traceable calibrated reporting convention [6].

Maxwell [1] insisted that measurable quantities must be defined through operations on bodies—forces, couples, and energy transfers—rather than through auxiliary potentials alone. The operational measurement specification below makes that Maxwellian criterion explicit for angular diagnostics [2].

Let $\mathcal{F}$ be a field state that already passes Eq. (**??**). Let $\mathcal{M}_\alpha$ denote the forward measurement operator for channel setting α, y_α the recorded datum, and n_α additive disturbance. The channel law is

$$y_\alpha = \mathcal{M}_\alpha[\mathcal{F}] + n_\alpha. \tag{1.85}$$

Equation (1.85) is the first operational pillar: no observable is read from symbolic fields directly; it is inferred through channel transduction physics plus disturbance structure [2].

Collect N channel outputs in $\mathbf{y} = [y_1, \ldots, y_N]^{\mathsf{T}} \in \mathbb{C}^N$, and let $\widehat{\mathcal{O}}$ denote estimator for target observable $\mathcal{O}[\mathcal{F}]$. Reconstruction map $\mathcal{R}$ is accepted only if $\widehat{\mathcal{O}} = \mathcal{R}[\mathbf{y}]$ and $\lim_{\|n\|\to 0} \widehat{\mathcal{O}} = \mathcal{O}[\mathcal{F}]$.

In practice, identification quality is measured by local sensitivity. Let $\boldsymbol{\theta} \in \mathbb{R}^p$ parameterize admissible states, with Jacobian $\mathbf{J}_{\alpha i} = \partial \mathcal{M}_\alpha / \partial \theta_i$, and let $\mathbf{W} \succ 0$ be channel-weight matrix. Define weighted information matrix

$$\mathbf{G} = \mathbf{J}^{\mathsf{H}} \mathbf{W} \mathbf{J}, \tag{1.86}$$

where $(\cdot)^{\mathsf{H}}$ denotes Hermitian transpose. A poorly conditioned $\mathbf{G}$ indicates ill-posed inversion and large estimator variance amplification; a well-conditioned $\mathbf{G}$ indicates robust local inversion.

Uncertainty in reported observable is summarized by first-order propagation:

$$\sigma_{\mathcal{O}}^2 \approx \nabla_{\boldsymbol{\theta}} \mathcal{O}^{\mathsf{T}} \boldsymbol{\Sigma}_\theta \nabla_{\boldsymbol{\theta}} \mathcal{O} + \sigma_{\mathrm{cal}}^2 + \sigma_{\mathrm{model}}^2, \tag{1.87}$$

where $\boldsymbol{\Sigma}_\theta$ is parameter covariance induced by channel noise, σ_{cal}^2 is calibration contribution, and $\sigma_{\mathrm{model}}^2$ is model-mismatch contribution [2].

Gauge and frame $\mathcal{D}_J = 1$ must hold at the channel level, not only at symbolic field level. For rotated state $\mathcal{F}' = \mathcal{A}_R \mathcal{F}$ and gauge-related potentials $(\mathbf{A}', \phi')$, each channel must satisfy

$$\mathcal{M}_\alpha[\mathcal{F}'] = \mathcal{T}_\alpha(R)\, \mathcal{M}_\alpha[\mathcal{F}], \qquad \mathcal{M}_\alpha[\mathbf{A}', \phi'] = \mathcal{M}_\alpha[\mathbf{A}, \phi], \tag{1.88}$$

where $\mathcal{T}_\alpha(R)$ is known channel transformation law.

Table 1.2: charge-continuous corollaries inherited by every Chapter 1 report ($\mathcal{D}_{\mathrm{ch1}} = 1$ in Eq. (1.158)).

Flag	Name	Corollary for sources, radiation, and measurement
$\mathcal{D}_J$	Transport	Regime-qualified angular flux on declared surfaces; reactive storage not promoted to export without $\mathcal{R}_Q, \chi(r)$ [6].
$\mathcal{D}_{\mathrm{org}}$	Covariance	Origin and frame declared before torque or directional-flux comparison [2].
$\mathcal{D}_{\mathrm{g}}$	Gauge	Gauge-robust integrals outrank gauge-sensitive local patterns [10].
$\mathcal{D}_{\mathrm{op}}$	Operational	Calibrated channel operator, identifiability, and uncertainty budget (Eq. (1.90)) [11].

Calibration completion provides a direct test. Let $\mathcal{M}_\alpha^{\mathrm{ref}}$ be reference response and $\widehat{\mathcal{M}}_\alpha$ calibrated model used in inversion. Define

$$\rho_\alpha = \frac{\left\|\widehat{\mathcal{M}}_\alpha - \mathcal{M}_\alpha^{\mathrm{ref}}\right\|}{\|\mathcal{M}_\alpha^{\mathrm{ref}}\| + \epsilon_\alpha}, \qquad \rho_{\max} = \max_\alpha \rho_\alpha, \tag{1.89}$$

with $\epsilon_\alpha > 0$ regularization.

An operational report is accepted only when the operational measurement specification in Table 1.2 is satisfied:

$$\mathcal{D}_{\mathrm{op}} = 1 \iff \Big(\mathcal{D}_h = 1,\ \kappa(\mathbf{G}) \le \kappa_{\max},\ \rho_{\max} \le \rho_{\max}^{\mathrm{tol}},\ \sigma_{\mathcal{O}} \le \sigma_{\max}\Big), \tag{1.90}$$

where $\kappa(\mathbf{G})$ is condition number of $\mathbf{G}$, $\kappa_{\max}$ conditioning tolerance, $\rho_{\max}^{\mathrm{tol}}$ calibration tolerance, and $\sigma_{\max}$ uncertainty tolerance [2].

For torque channels used later, the measurement-facing integral form becomes

$$\tau_z^{\mathrm{meas}} = \int_S (\mathbf{r} \times \mathbf{t}) \cdot \hat{\mathbf{z}}\, dS + \delta\tau_z, \tag{1.91}$$

where S is sensing surface, $\mathbf{r}$ position relative to declared origin, $\mathbf{t}$ traction vector from Eq. (1.75), $\hat{\mathbf{z}}$ measurement axis, and $\delta\tau_z$ net channel disturbance.

To compare setups with different torque scales, define normalized channel reliability index

$$\mathcal{Q}_\tau = \frac{\left|\tau_z^{\mathrm{meas}} - \tau_z^{\mathrm{model}}\right|}{|\tau_z^{\mathrm{model}}| + \epsilon_\tau}, \tag{1.92}$$

where τ_z^{model} is model prediction under calibrated channel model and $\epsilon_\tau > 0$ regularization [11]. Operational walk-through: torque channel with uncertainty budget. Suppose a torsion-balance readout reports $y_\tau = -3.10 \times 10^{-11}\,\mathrm{N\,m}$ with calibration $K_\tau = 1.02 \pm 0.01$ and model prediction $\tau_z^{\mathrm{model}} = -3.18 \times 10^{-11}\,\mathrm{N\,m}$ from Eq. (1.91) for $(m, \sigma) = (1, +1)$ and $P_{\mathrm{rad}} = 1\,\mathrm{W}$ at $f = 10\,\mathrm{GHz}$. Parameterize the admissible state by single-mode power P_{rad} and helical index (m, σ). The Jacobian entry $\partial y_\tau / \partial P_{\mathrm{rad}} = (m+\sigma)/(\omega K_\tau)$ feeds Eq. (1.86). With $\sigma_P = 0.02\,\mathrm{W}$ on the power channel, first-order propagation in Eq. (1.87) gives $\sigma_{\tau_z} \approx 6 \times 10^{-13}\,\mathrm{N\,m}$, dominated by power calibration at this flux level. Reliability index $\mathcal{Q}_\tau$ from Eq. (1.92) is then ≈ 0.025, and $\mathcal{D}_{\mathrm{op}} = 1$ is credible if $\mathcal{D}_h = 1$, $\kappa(\mathbf{G}) < 10^2$, and $\rho_{\max} < 0.05$ from Eqs. (1.89) and (1.90). The walk-through shows that operational charge-continuous is a joint statement about physics *and* metrology: a correct

conservation prediction can still fail as an observable report if uncertainty or calibration checks are omitted [2].

Operational definition is therefore fixed in rigorous terms: an observable is a symmetry-admissible field quantity equipped with identifiable channel operator, stable reconstruction, calibrated completion, and explicit uncertainty budget. The following subsection uses this foundation to show why local spin-orbit decompositions remain interpretive tools unless they satisfy the same operational criteria.

1.2.5 Limits of local spin-orbit decompositions

A spiral phase plate in front of a horn feed on a compact range defines the hardware scene for 'Limits of local spin-orbit decompositions": technicians compare turntable cuts and wattmeter totals on Γ_R before paraxial labels are quoted [12]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

Near-field micrographs invite a seductive partition: call polarization circulation "spin" and azimuthal phase gradient "orbit" at every pixel. Maxwell theory permits such pictures only as convention-tagged diagnostics: the total density $\mathbf{j}_{\mathrm{em}}$ does not decompose uniquely into $(\mathbf{s}_{\mathrm{loc}}, \mathbf{l}_{\mathrm{loc}})$ unless gauge, boundary data, split family, and measurement kernel are declared together [2]. The limits below are constructive—local splits remain valuable when their equivalence class and completion tests are explicit.

Local spin-orbit decomposition remains mathematically valuable but physically delicate: two admissible gauge splittings may assign different local "spin-like" and "orbital-like" patterns while preserving identical total angular-momentum transfer. The limits established here are constructive rather than prohibitive: local splits serve as interpretation tools, and they are promoted to direct observables only when gauge, boundary, and channel-level completion are satisfied. Decomposition is not forbidden; the task is to state precisely when decomposition outputs carry the same epistemic status as stress-flux integrals and when they remain convention-tagged diagnostics attached to a fixed total [2].

Canonical split families (equivalence class viewpoint). Two split families appear repeatedly in the literature and differ even when $\mathbf{j}_{\mathrm{em}}$ is held fixed. *Family A* assigns orbital content to a cross-product structure $\mathbf{l}_{\mathrm{loc}} \propto \mathbf{r} \times (\varepsilon_0 \mathbf{E})$ built from potentials or fields, and spin content to a complementary bilinear term in $\mathbf{E}$ and $\mathbf{H}$. *Family B* shifts content using a superpotential $\mathbf{C}$ so that $\nabla \cdot \mathbf{C}$ redistributes labels without altering $\mathbf{j}_{\mathrm{em}}$. Neither family is "wrong" in isolation; each defines an admissible representative of the same decomposition class. Physical disagreement arises only when two groups report Family A and Family B reports as if they were directly comparable observables without stating convention or passing the completion criteria developed below [10].

The Belinfante–Rosenfeld construction [10] explains the mathematical origin of Family B: symmetrizing the canonical energy-momentum tensor requires adding a divergence term built from spin current, which reappears in local angular-momentum splits as the superpotential $\mathbf{C}$ in Eq. (1.96). In Maxwellian electrodynamics the practical consequence is strict: local spin-orbit images are gauge- and convention-dependent coordinates on a conserved total [2].

$$\mathbf{T}^{\mathrm{sym}} = \mathbf{T} + \frac{1}{2\mathrm{j}\omega} \nabla \cdot \mathbf{K}, \qquad K_{ij} = \varepsilon_0 \left(E_i H_j^* - E_i^* H_j \right), \tag{1.93}$$

where $\mathbf{T}$ is the Maxwell stress tensor of Eq. (1.37). For radiation, $\mathbf{T}^{\mathrm{sym}}$ yields conserved momentum density and symmetric traction; for measurement, torque integrals built from $\mathbf{T}^{\mathrm{sym}}$ align with Belinfante–Rosenfeld spin-flux completion [10]. Belinfante [10] derived the symmetric energy-

momentum tensor by requiring that the total angular-momentum density

$$\mathbf{j}_{\mathrm{tot}} = \mathbf{r} \times \mathbf{g}_{\mathrm{sym}} + \mathbf{s}_{\mathrm{Bel}} \tag{1.94}$$

obey $\partial_t \mathbf{j}_{\mathrm{tot}} + \nabla \cdot \mathbf{M}_{\mathrm{sym}} = -\boldsymbol{\tau}_{\mathrm{ext}}$ with symmetric $\mathbf{T}^{\mathrm{sym}}$ from Eq. (1.93) and conserved integrated momentum. Here $\mathbf{g}_{\mathrm{sym}}$ is momentum density from the symmetric tensor, and $\mathbf{s}_{\mathrm{Bel}}$ is the spatial Belinfante spin density. Maxwell's original stress-based torque law [1] controls the symmetric branch; the canonical potential branch differs by the spin-flux divergence in Eq. (1.93). Family B splits are Belinfante-class representatives in which spin-labeled content is concentrated in $\mathbf{s}_{\mathrm{Bel}}$ and orbit-labeled content in $\mathbf{r} \times \mathbf{g}_{\mathrm{sym}}$, modulo superpotential $\mathbf{C}$ in Eq. (1.96).

For monochromatic Maxwell fields in vacuum phasors,

$$\mathbf{s}_{\mathrm{Bel}} = \frac{1}{2\mathrm{j}\omega}\, \mathbf{E}^* \times \mathbf{E} \;+\; \frac{\mu_0}{2\mathrm{j}\omega}\, \mathbf{H}^* \times \mathbf{H}, \tag{1.95}$$

up to total-divergence terms movable into $\mathbf{C}$. Equation (1.95) is gauge-invariant because it depends only on $\mathbf{E}, \mathbf{H}$. Orbital-labeled density $\mathbf{l}_{\mathrm{Bel}} = \mathbf{j}_{\mathrm{em}} - \mathbf{s}_{\mathrm{Bel}}$ then splits $\mathbf{j}_{\mathrm{em}}$ into gauge-invariant summands, unlike potential-weighted decompositions that shift under Eq. (1.98). Non-uniqueness persists because any redistribution satisfying Eq. (1.97) produces an equally valid local picture [10].

Let $\mathbf{j}_{\mathrm{em}}(\mathbf{r})$ denote electromagnetic angular-momentum density already fixed in Eq. (1.41). A generic local decomposition takes the form

$$\mathbf{j}_{\mathrm{em}} = \mathbf{l}_{\mathrm{loc}} \;+\; \mathbf{s}_{\mathrm{loc}} \;+\; \nabla \cdot \mathbf{C}, \tag{1.96}$$

where $\mathbf{l}_{\mathrm{loc}}$ is orbital-labeled local density, $\mathbf{s}_{\mathrm{loc}}$ is spin-labeled local density, $\mathbf{C}$ is a second-rank superpotential tensor, and $\nabla \cdot \mathbf{C}$ is the divergence correction generated by decomposition convention. Equation (1.96) makes clear that label assignment is not unique at point level: superpotential terms can move content between $\mathbf{l}_{\mathrm{loc}}$ and $\mathbf{s}_{\mathrm{loc}}$ while preserving $\mathbf{j}_{\mathrm{em}}$. Local decomposition does not define a single physical density pair; it defines an admissible equivalence class.

To represent this class explicitly, let $(\mathbf{l}_a, \mathbf{s}_a, \mathbf{C}_a)$ and $(\mathbf{l}_b, \mathbf{s}_b, \mathbf{C}_b)$ denote two admissible local splits of the same $\mathbf{j}_{\mathrm{em}}$. Their difference can always be written

$$\mathbf{l}_b - \mathbf{l}_a = \nabla \cdot \mathbf{Q} + \mathbf{q}, \qquad \mathbf{s}_b - \mathbf{s}_a = -\nabla \cdot \mathbf{Q} - \mathbf{q}, \tag{1.97}$$

where $\mathbf{Q}$ is redistribution tensor and $\mathbf{q}$ is residual balancing vector. The paired signs guarantee that local relabeling does not alter the summed quantity $\mathbf{l} + \mathbf{s}$, but they do alter each local component separately.

Under a gauge update $(\mathbf{A}, \phi) \to (\mathbf{A}', \phi')$ from Eq. (1.77), the local pair transforms as

$$\mathbf{s}'_{\mathrm{loc}} = \mathbf{s}_{\mathrm{loc}} + \nabla \cdot \mathbf{U}_\chi + \mathbf{v}_\chi, \qquad \mathbf{l}'_{\mathrm{loc}} = \mathbf{l}_{\mathrm{loc}} - \nabla \cdot \mathbf{U}_\chi - \mathbf{v}_\chi, \tag{1.98}$$

where $\mathbf{U}_\chi$ is gauge-generated redistribution tensor and $\mathbf{v}_\chi$ is residual redistribution vector. The paired signs enforce that $\mathbf{s}'_{\mathrm{loc}} + \mathbf{l}'_{\mathrm{loc}} = \mathbf{s}_{\mathrm{loc}} + \mathbf{l}_{\mathrm{loc}}$, but individual local components remain gauge-sensitive. In functional terms, Eq. (1.98) moves between admissible local representatives; in physical terms, it explains why local spin-orbit decompositions can differ without contradicting total transport predictions.

Integrating over a finite volume V with boundary ∂V and outward unit normal $\hat{\mathbf{n}}$ yields

$$\int_V (\mathbf{s}'_{\mathrm{loc}} + \mathbf{l}'_{\mathrm{loc}})\, dV = \int_V (\mathbf{s}_{\mathrm{loc}} + \mathbf{l}_{\mathrm{loc}})\, dV = \int_V \mathbf{j}_{\mathrm{em}}\, dV \;-\; \int_{\partial V} \mathbf{C} \cdot \hat{\mathbf{n}}\, dS. \tag{1.99}$$

The first equality shows decomposition invariance of the sum; the second shows explicit boundary dependence through $\mathbf{C}$. Therefore local partitions can be physically compared only after boundary prescription is declared, exactly as required by the control-balance logic in Eq. (1.42).

For two admissible split representatives a and b, integrated difference obeys

$$\Delta_V^{(ab)} \equiv \int_V (\mathbf{s}_b - \mathbf{s}_a)\, dV = -\int_V (\mathbf{l}_b - \mathbf{l}_a)\, dV = \int_{\partial V} \mathbf{Q}\cdot\hat{\mathbf{n}}\, dS + \int_V \mathbf{q}\, dV. \tag{1.100}$$

Equation (1.100) resolves disagreement into boundary and volumetric channels, which must both be controlled before integral spin-orbit attribution is physically comparable only under declared split and calibration record.

To quantify decomposition fragility for a given field realization, define

$$\kappa_{\mathrm{so}}(V) = \frac{\int_V \|\mathbf{s}'_{\mathrm{loc}} - \mathbf{s}_{\mathrm{loc}}\|\, dV}{\int_V \|\mathbf{j}_{\mathrm{em}}\|\, dV + \epsilon_{\mathrm{so}}}, \tag{1.101}$$

where $\|\cdot\|$ is Euclidean norm and $\epsilon_{\mathrm{so}} > 0$ is a small regularization constant.

Operationally reconstructed split components must also satisfy internal completion. Let $\widehat{\mathbf{s}}_{\mathrm{loc}}$, $\widehat{\mathbf{l}}_{\mathrm{loc}}$, and $\widehat{\mathbf{j}}_{\mathrm{em}}$ be channel-reconstructed fields from Section 1.2.4. Define split-closure residual

$$\mathcal{R}_{\mathrm{so}} = \frac{\left\|\widehat{\mathbf{s}}_{\mathrm{loc}} + \widehat{\mathbf{l}}_{\mathrm{loc}} - \widehat{\mathbf{j}}_{\mathrm{em}}\right\|}{\left\|\widehat{\mathbf{j}}_{\mathrm{em}}\right\| + \epsilon_r}, \tag{1.102}$$

where $\epsilon_r > 0$ is regularization. If $\mathcal{R}_{\mathrm{so}}$ is not small, then even before gauge comparison the measured local split fails completion.

The operational decision criterion for direct local-spin/local-orbit observability is

$$\mathcal{D}_{\mathrm{so}} = 1 \iff \Big(\mathcal{D}_{\mathrm{op}} = 1,\ \kappa_{\mathrm{so}} \le \kappa_{\mathrm{so,max}},\ \mathcal{R}_{\mathrm{so}} \le R_{\mathrm{so,max}},\ \|\Delta_V^{(ab)}\| \le \Delta_{\max}\Big), \tag{1.103}$$

with declared tolerances $\kappa_{\mathrm{so,max}}$, $R_{\mathrm{so,max}}$, and $\Delta_{\max}$. This condition is intentionally strict: it prevents representation-sensitive local decompositions from being reported as if they were unique observables.

without replacing the equations: the anchor object is $\mathbf{j}_{\mathrm{em}}$; local spin and orbit decompositions are coordinates on an equivalence class; integrals over V inherit boundary sensitivity through $\mathbf{C}\cdot\hat{\mathbf{n}}$.

Helicity, developed in Subsection 1.2.4, is a gauge-invariant sign-odd measure of handedness tied to $\mathbf{E}, \mathbf{H}$ through dual symmetry. Local spin-orbit splits, by contrast, are gauge- and convention-sensitive decompositions on $\mathbf{j}_{\mathrm{em}}$. A field may carry definite helicity on a closed surface while local spin density varies with potential choice; conversely, a visually "spin-rich" near-field pattern may carry negligible exported helicity flux. The two concepts answer different questions: helicity classifies chiral coupling of $\mathbf{E}$ and $\mathbf{H}$; spin-orbit decomposition classifies how total angular density is *labeled* locally. Neither replaces Eq. (1.42) for torque measurement [2].

Experimental spin torque versus local spin density. Beth [11] measured mechanical torque on a quartz wave plate illuminated by circularly polarized light, establishing that polarization handedness couples to a definite mechanical response at laboratory scale. That experiment is an *integral* spin measurement: the plate integrates photon spin over its aperture and reports a single τ_z. Local spin densities such as Eq. (1.95), by contrast, assign signed content to each point in space and vary with split convention. The order of questions is therefore strict: Beth-type torque validates the spin *channel* of Maxwell theory; Belinfante-type redistributions supply a gauge-invariant local *label* that still admits redistribution among orbit and spin slots through Eq. (1.97). A modern near-field

micrograph of "spin density" without stating split family and control volume is not comparable to Beth's torque even when both figures appear equally compelling [1].

For a paraxial mode with azimuthal index m and single circular polarization $\sigma = +1$, Allen *et al.* [12] showed that each photon in the mode carries orbital index $m\hbar$ in the paraxial limit. In the present frequency-domain language, define modal power P_m on a control surface and modal angular export $j_z^{(m)}$. The channel law inherited from Subsection 1.1.5 gives $j_z^{(m)} \approx (m+\sigma)P_m/\omega$. Belinfante spin density Eq. (1.95) contributes the σ-linked handedness locally, while azimuthal phase gradient contributes orbital-labeled content in $\mathbf{l}_{\mathrm{Bel}}$. A modal spin fraction

$$f_s^{(m)} = \frac{\int_{V_m} \|\mathbf{s}_{\mathrm{Bel}}\|\, dV}{\int_{V_m} \|\mathbf{j}_{\mathrm{em}}\|\, dV + \epsilon_s}, \tag{1.104}$$

with $\epsilon_s > 0$, is convention-dependent: changing split representative via Eq. (1.97) alters $f_s^{(m)}$ without changing $j_z^{(m)}$. The observable modal quantity is the channel power and flux integral, not the ratio $f_s^{(m)}$ unless Eq. (1.103) is satisfied [12].

Observation never enlarges physics: measurement reveals structure; it does not create it [2]. Field–matter coupling is where Maxwell predictions meet hardware: traction $\mathbf{t} = \mathbf{T}\cdot\hat{\mathbf{n}}$ acts across conductor, dielectric, and torsion-fixture interfaces, and the mechanical reaction records how absorption or polarization conversion redistributes angular action [11]. In Beth's quartz half-wave plate at $\lambda = 632\,\mathrm{nm}$, the sensing surface S_m for traction integration is the plate face exposed to the beam; optical torque on the plate is the material counterpart of exported angular-flux action [11].

1.2.6 Torque, momentum flux, and stress measurements

Measurement closes the loop opened in Subsections 1.1.4 and 1.1.5: stress and angular flux are field-theoretic objects first, transducer readings second [11]. Beth's quartz plate did not read a modal index; it read mechanical torque from electromagnetic stress [11]. Before comparing y_τ on a Beth plate at $f = 10\,\mathrm{GHz}$ with $\oint_{\Gamma_R} \mathbf{M}\cdot\hat{\mathbf{n}}\, dS$ on a sphere at $R = 3\,\mathrm{m}$, the experimenter must declare origin, surface orientation, polarization basis, harmonic gauge, and receiver noise floor—the same consistency checks any compact-range team publishes with a gain calibration report [13]. The present development follows that experimental logic: $\mathbf{T}$ is the field-theoretic object inherited from Subsection 1.1.4; $\mathbf{T}\cdot\hat{\mathbf{n}}$ is traction on a declared surface; y_τ is what a calibrated transducer reports after geometry, origin, and noise are specified. Momentum-flux and torque channels are therefore specializations of one Maxwellian ladder, not independent postulates [6].

The experimental lineage of torque readouts begins with Beth's 1936 detection of mechanical rotation in quartz wave plates illuminated by circularly polarized light [11]. That experiment established that polarization-linked angular momentum has mechanical consequences at laboratory scale, decades before Laguerre–Gaussian beams were identified as orbital-indexed modes [12]. Modern torsion-balance protocols inherit the same logical structure Beth used: predict torque from field stress, measure mechanical response, and treat discrepancy as channel or regime error rather than as permission to abandon conservation [2]. Subsection 1.3.1 returns to this completion pattern in fully worked form.

Let $\mathbf{T}$ be Maxwell stress tensor from Eq. (1.37), let S_m be the sensing surface around a test body, and let $\hat{\mathbf{n}}$ denote outward surface normal on S_m. The traction density is $\mathbf{t} = \mathbf{T}\cdot\hat{\mathbf{n}}$, the mechanical counterpart of Poynting's energy-flux concept [6] applied to linear momentum rather than power. For a declared reference origin $\mathbf{r}_0$, measured force vector $\mathbf{y}_F$, and measured torque component y_τ about axis unit vector $\hat{\mathbf{u}}_\tau$, a channel-calibrated measurement model is

$$\mathbf{y}_F = \mathbf{K}_F\left(\int_{S_m} \mathbf{T}\cdot\hat{\mathbf{n}}\, dS\right) + \mathbf{n}_F, \qquad y_\tau = K_\tau\left(\int_{S_m} [(\mathbf{r}-\mathbf{r}_0)\times(\mathbf{T}\cdot\hat{\mathbf{n}})]\cdot\hat{\mathbf{u}}_\tau\, dS\right) + n_\tau, \tag{1.105}$$

where $\mathbf{K}_F$ and K_τ are calibration operators/scalars, and $\mathbf{n}_F, n_\tau$ collect channel disturbances. The two surface integrals are deterministic electromagnetic load predictions inherited from Eq. (1.37); calibration and disturbance terms convert that prediction into practical measurement channels.

For full vector torque reporting, define an explicit channel

$$\mathbf{y}_T = \mathbf{K}_T\left(\int_{S_m} \left[(\mathbf{r}-\mathbf{r}_0)\times(\mathbf{T}\cdot\hat{\mathbf{n}})\right] dS\right) + \mathbf{n}_T, \tag{1.106}$$

where $\mathbf{K}_T$ is vector torque calibration operator and $\mathbf{n}_T$ is channel disturbance. The scalar channel y_τ is then $y_\tau = \mathbf{y}_T \cdot \hat{\mathbf{u}}_\tau$ after consistent axis normalization.

Origin declaration controls comparability of torque channels. If origin changes from $\mathbf{r}_0$ to $\mathbf{r}_0' = \mathbf{r}_0 + \Delta\mathbf{r}$, the deterministic torque integral satisfies

$$\mathbf{T}'_{\mathrm{det}} = \mathbf{T}_{\mathrm{det}} - \Delta\mathbf{r}\times\mathbf{F}_{\mathrm{det}}, \tag{1.107}$$

with $\mathbf{F}_{\mathrm{det}} = \int_{S_m} \mathbf{T}\cdot\hat{\mathbf{n}}\, dS$ and $\mathbf{T}_{\mathrm{det}} = \int_{S_m}(\mathbf{r}-\mathbf{r}_0)\times(\mathbf{T}\cdot\hat{\mathbf{n}})\, dS$. Equation (1.107) is the operational form of origin dependence needed for laboratory-to-laboratory torque reconciliation.

Stress-based torque metrology layers calibration gain K_τ, declared origin $\mathbf{r}_0$, and axis $\hat{\mathbf{u}}_\tau$ on top of the deterministic load from Eq. (1.105) [11]. An origin shift $\mathbf{r}_0' = \mathbf{r}_0 + \Delta\mathbf{r}$ rotates the torque vector by $-\Delta\mathbf{r}\times\mathbf{F}_{\mathrm{det}}$ even when $(\mathbf{E},\mathbf{H})$ is unchanged—two groups comparing y_τ must match origins before debating polarization content [2].

Torque metrology is unforgiving of origin shifts [11]. On a torsion fixture with $\mathbf{r}_0$ at the plate center, moving the origin by $\Delta\mathbf{r} = 2\,\mathrm{mm}\,\hat{\mathbf{x}}$ while holding $\mathbf{F}_{\mathrm{det}}$ fixed injects a z-torque proportional to F_y when $\hat{\mathbf{u}}_\tau = \hat{\mathbf{z}}$—the content of Eq. (1.107), $\mathbf{T}'_{\mathrm{det}} = \mathbf{T}_{\mathrm{det}} - \Delta\mathbf{r}\times\mathbf{F}_{\mathrm{det}}$ [2]. What goes wrong in the laboratory is comparing torque reports computed in different origin conventions without translating through this shift law [3].

At mathematical level, Eq. (1.105) composes field-to-traction operator mapping, surface integration, and instrument transfer in one statement. At physical level, it explains why two laboratories can report different torque values from the same field realization unless origin declaration, axis definition, and calibration normalization are matched.

Surface selection is itself a measurable limit. For two admissible enclosing surfaces S_a and S_b, define stress-flux completion residual

$$\mathcal{R}_S = \frac{\left\|\int_{S_a}\mathbf{T}\cdot\hat{\mathbf{n}}\,dS - \int_{S_b}\mathbf{T}\cdot\hat{\mathbf{n}}\,dS\right\| + \ell_\tau^{-1}\left\|\int_{S_a}\mathbf{r}\times(\mathbf{T}\cdot\hat{\mathbf{n}})\,dS - \int_{S_b}\mathbf{r}\times(\mathbf{T}\cdot\hat{\mathbf{n}})\,dS\right\|}{\left\|\int_{S_a}\mathbf{T}\cdot\hat{\mathbf{n}}\,dS\right\| + \ell_\tau^{-1}\left\|\int_{S_a}\mathbf{r}\times(\mathbf{T}\cdot\hat{\mathbf{n}})\,dS\right\| + \epsilon_S}, \tag{1.108}$$

where $\ell_\tau > 0$ is scaling length and $\epsilon_S > 0$ is regularization. Small $\mathcal{R}_S$ indicates robust stress-channel completion across declared sampling surfaces.

First-order torque uncertainty combines calibration, geometry, and noise:

$$\sigma_\tau^2 \approx \left\|\frac{\partial y_\tau}{\partial K_\tau}\right\|^2 \sigma_{K_\tau}^2 + \left\|\frac{\partial y_\tau}{\partial \mathbf{r}_0}\right\|^2 \sigma_{r_0}^2 + \left\|\frac{\partial y_\tau}{\partial \hat{\mathbf{u}}_\tau}\right\|^2 \sigma_u^2 + \sigma_{n_\tau}^2. \tag{1.109}$$

Equation (1.109) makes explicit that geometric registration accuracy is as important as electrical calibration in torque measurement quality.

Stress-based reports are accepted in this chapter only when interaction and channel criteria hold simultaneously:

$$\mathcal{D}_\sigma = 1 \iff \Big(\mathcal{D}_m = 1,\ \mathcal{R}_S \le R_{S,\max},\ \sigma_\tau \le \sigma_{\tau,\max},\ \|\mathbf{n}_F\| \le n_{F,\max}\Big), \tag{1.110}$$

where $R_{S,\max}$, $\sigma_{\tau,\max}$, and $n_{F,\max}$ are declared tolerances. Equation (1.110) excludes stress-channel claims with unresolved surface inconsistency or uncontrolled uncertainty.

Stress-derived force and torque outputs are operationally admissible observables in Chapter 1 only if $\mathcal{D}_\sigma = 1$.

Eq. (1.105) defines calibrated stress channels. Eqs. (1.107) and (1.108) enforce geometric and surface closure. Eq. (1.109) enforces uncertainty quantification. Their conjunction with Eq. (**??**) is necessary and sufficient for admissible reporting.

The model assumes quasi-static sensor response over one cycle-average window and surface geometry known to sufficient precision for traction integration. Large vibration bandwidth, unknown moving boundaries, or unresolved near-contact fields can invalidate the simple calibration form in Eq. (1.105) and require dynamic inverse identification [2].

1.2.7 Modal projections and measurement limits

Modal projections on finite probes recover only part of the angular spectrum [2]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

A spherical-harmonic index printed on a plot is rarely a direct observation; it is almost always the output of a truncated fit to finite aperture data [13]. Modal projections are estimates of expansion coefficients; measurement limits are the aperture, bandwidth, and noise constraints that make those estimates ill-conditioned rather than exact. Until the projection kernel is declared, a mode label remains inference under finite rank and calibration limits [13].

A sixty-probe scan on Γ_R returns a finite vector of samples; a harmonic fit returns coefficients $\widehat{c}_{\ell m}$ subject to aperture rank and noise [13]. The index printed on a contour plot is therefore conditional on $(N, \varepsilon, \mathcal{C})$—not a direct wattmeter reading [2]. Ill-conditioned inverses amplify labels that Maxwell export on fixed geometry cannot support [4].

A spherical fit with $N = 32$ probes and a chart to $\ell \leq 8$ may return non-zero $\widehat{c}_{8,m}$ even when the source export rank remains dipole-class [13]. The fit resolves a truncated basis on the scan sphere; realizability and rank bounds come from feed geometry and outgoing selection, not from probe count alone [2]. Treating high-ℓ fit coefficients as new multipole physics without a $\mathcal{S}_{\text{real}}$ check is a common index confusion on compact ranges [3].

The informational character of accessibility appears with special clarity in modal recovery. A spherical or cylindrical harmonic basis may list dozens or hundreds of angular degrees of freedom, yet a finite scan surface, finite bandwidth, finite dynamic range, and finite calibration budget jointly determine how many of those directions enter the stabilized inverse problem below. The field state may contain organized angular content that the measurement operator cannot separate from noise or from coupling to neighbouring channels; that is the physical tragedy of $\mathcal{E}_{\text{acc}}^{(N)}$ in operational form, not a comment on chart quality. Hansen's spherical near-field theory [13] and Harrington's source formulations [3] both encounter this wall from different sides—one from the receiver, one from the transmitter—when effective angular dimension is counted on finite supports [13].

Modal language enters angular electrodynamics early because spherical and cylindrical harmonics organize radiation patterns efficiently. Allen *et al.* [12] demonstrated that Laguerre–Gaussian modes carry a well-defined orbital index and that mode converters transfer that index reversibly; yet a mode label is not, by itself, an observable.

The discussion connects backward to directional transport (Subsection 1.2.2) and forward to the observable-versus-inference split (Subsection 1.2.8). Directional flux $\Pi(\theta, \phi)$ is primary data on a sphere; modal coefficients are estimates obtained by inversion. When inversion is ill-conditioned, modal interpretation must be tagged as inference-only even if $\mathcal{D}_J = 1$ at the flux level. Section 1.3 returns to this distinction when multimode torque sums are tested against measurement residuals.

Angular interpretation often proceeds through modal coefficients, but those coefficients are not primitive observables; they are reconstructions from finite channel data. In strict operational language, a *modal projection* is a channel-weighted linear functional on measured boundary data, and a *measurement limit* is any bound that prevents stable recovery of a modal degree of freedom from available channels. This distinction is essential: a mode can be mathematically definable yet experimentally unresolved.

Let Γ_m denote a measurement surface, let $\{\mathbf{w}_q(\mathbf{r})\}_{q=1}^{Q}$ be known channel weighting functions on Γ_m, and let y_q be measured channel data. For tangential electric field $\mathbf{E}_t$, channel outputs satisfy

$$y_q = \int_{\Gamma_m} \mathbf{w}_q^*(\mathbf{r}) \cdot \mathbf{E}_t(\mathbf{r})\, dS + n_q, \qquad q = 1, \ldots, Q, \tag{1.111}$$

where n_q is measurement noise in channel q. Each integral is a weighted projection of $\mathbf{E}_t$ onto channel kernel $\mathbf{w}_q$, so the data vector $\mathbf{y}$ captures only the subspace interrogated by the chosen sensing set.

Represent $\mathbf{E}_t$ in a model basis $\{\boldsymbol{\psi}_m\}_{m=1}^{M}$ with coefficients a_m: $\mathbf{E}_t \approx \sum_{m=1}^{M} a_m \boldsymbol{\psi}_m$. Then Eq. (1.111) becomes the finite-dimensional inverse system $\mathbf{y} = G\mathbf{a}+\mathbf{n}$, where $G_{qm} = \int_{\Gamma_m} \mathbf{w}_q^* \cdot \boldsymbol{\psi}_m\, dS$. Define also

$$G_{qm} = \int_{\Gamma_m} \mathbf{w}_q^*(\mathbf{r}) \cdot \boldsymbol{\psi}_m(\mathbf{r})\, dS, \tag{1.112}$$

and the channel information matrix

$$\mathbf{F}_a = G^* \Sigma_n^{-1} G, \tag{1.113}$$

with $\Sigma_n = \mathrm{Cov}[\mathbf{n}]$. Small eigenvalues of $\mathbf{F}_a$ identify weakly observable modal directions.

A stable regularized estimator is

$$\widehat{\mathbf{a}}_\lambda = \left(G^* G + \lambda I\right)^{-1} G^* \mathbf{y}, \tag{1.114}$$

where G^* is conjugate transpose, I is identity matrix, and $\lambda > 0$ is regularization parameter. Operator-wise, G^*G encodes channel sensitivity to basis directions, λI suppresses small-singular-value amplification, and left multiplication by $(G^*G + \lambda I)^{-1}G^*$ yields the reconstructed coefficient vector.

Using singular value decomposition $G = U\Sigma V^*$, reconstruction acquires spectral-filter form

$$\widehat{\mathbf{a}}_\lambda = \sum_{k=1}^{r} \frac{\sigma_k}{\sigma_k^2 + \lambda} \left(\mathbf{u}_k^* \mathbf{y}\right) \mathbf{v}_k, \tag{1.115}$$

where $r = \mathrm{rank}(G)$, and $(\mathbf{u}_k, \mathbf{v}_k, \sigma_k)$ are singular triplets. The filter factor $\sigma_k/(\sigma_k^2 + \lambda)$ suppresses weakly measured directions.

Bias–variance tradeoff for this estimator is

$$\mathbb{E}[\widehat{\mathbf{a}}_\lambda - \mathbf{a}] = -\lambda \left(G^* G + \lambda I\right)^{-1} \mathbf{a}, \qquad \mathrm{Cov}[\widehat{\mathbf{a}}_\lambda] = \left(G^* G + \lambda I\right)^{-1} G^* \Sigma_n G \left(G^* G + \lambda I\right)^{-1}. \tag{1.116}$$

Larger λ lowers variance at the price of increased modal bias; smaller λ reduces bias but raises sensitivity to channel noise.

As a complete inverse relation, Eq. (1.114) records a central measurement fact: modal descriptions are model-dependent estimates whose reliability depends on channel geometry and conditioning, not on phase pattern alone. This condition is the section-level safeguard against over-interpreting visually appealing but weakly observable modal decompositions [2].

Quantitative resolution is bounded by singular-spectrum decay of G. Let $\sigma_{\min}^{(\mathrm{eff})}$ be the smallest retained singular value after truncation. Then, for perturbation $\delta\mathbf{y}$, one obtains the conditioning bound

$$\|\delta\widehat{\mathbf{a}}_\lambda\|_2 \le \left\|(G^*G+\lambda I)^{-1}G^*\right\|_2 \|\delta\mathbf{y}\|_2 \equiv \kappa_\lambda\|\delta\mathbf{y}\|_2. \tag{1.117}$$

Equation (1.117) defines amplification factor κ_λ, the primary conditioning index for modal reconstruction quality.

Effective observable modal dimension at threshold $\tau_\sigma \in (0,1)$ is

$$r_{\mathrm{eff}}(\tau_\sigma) = \#\left\{k : \sigma_k/\sigma_1 \ge \tau_\sigma\right\}. \tag{1.118}$$

The value r_{eff} is the number of modal directions that remain resolvable above the chosen singular-value floor.

To quantify aperture dependence, let $\widehat{\mathbf{a}}_\lambda^{(a)}$ and $\widehat{\mathbf{a}}_\lambda^{(b)}$ be reconstructions from two admissible measurement surfaces Γ_a, Γ_b. Define

$$\mathcal{C}_{ab} = \frac{\|\widehat{\mathbf{a}}_\lambda^{(a)} - \widehat{\mathbf{a}}_\lambda^{(b)}\|_2}{\|\widehat{\mathbf{a}}_\lambda^{(a)}\|_2 + \epsilon_a}, \qquad \epsilon_a > 0. \tag{1.119}$$

Small $\mathcal{C}_{ab}$ indicates modal interpretation robust to admissible aperture variation; large $\mathcal{C}_{ab}$ indicates geometry-contingent modal labels.

1.2.8 Observables versus inferred quantities

A standard-gain horn on a compact-range turntable fixes the laboratory scene for 'Observables versus inferred quantities": frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays below [13]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

The measurement stack of Section 1.2.5 would remain incomplete without an explicit vocabulary for reporting status. Two quantities may share the same symbol on a plot yet belong to different epistemic classes: one read directly from a calibrated channel, the other recovered only after inversion with regularization and model choice. Beth [11] measured torque directly; Allen *et al.* [12] inferred modal content from beam profiles—both legitimate, but not interchangeable without the distinctions developed below [2].

Subsection 1.2.8 closes the measurement discussion by resolving that ambiguity: which reported quantities are direct observables and which are inference-dependent descriptors. The distinction is mandatory for Chapter 1, because a mathematically coherent descriptor can still fail measurement admissibility if its regime tag is omitted [2].

Let $\mathcal{O}_d$ denote a direct observable produced by calibrated channel operator $\mathcal{M}$, and $\mathcal{Q}_i$ denote an inferred quantity produced by reconstruction operator $\mathcal{R}$ from data $\mathbf{y}$. The operational role split is

$$\mathcal{O}_d = \mathcal{M}[\mathcal{F}], \qquad \mathcal{Q}_i = \mathcal{R}[\mathbf{y}], \qquad \mathbf{y} = \mathcal{M}[\mathcal{F}] + \mathbf{n}. \tag{1.120}$$

Equation (1.120) separates channel-output space from reconstruction space: direct observables are channel-native, inferred quantities are estimator-native.

To compare reliability classes on one scale, define inference $\mathcal{D}_J = 1$ index

$$\mathfrak{A}(\mathcal{Q}_i) = \frac{\mathrm{Std}[\mathcal{Q}_i]}{\sigma_d} \cdot \kappa_{\mathrm{eff}} \cdot \chi_{\mathrm{sym}}, \tag{1.121}$$

where σ_d is direct-channel uncertainty floor, κ_{eff} inversion-sensitivity gain, and $\chi_{\text{sym}} \geq 1$ symmetry-constraint penalty factor derived from Eq. (1.88). The index rises when estimator uncertainty, conditioning burden, or symmetry mismatch increases.

Because reconstruction reliability also depends on model discrepancy, define

$$\mathcal{M}_{\text{mis}} = \frac{\|\mathbf{y} - \mathcal{M}[\widehat{\mathcal{F}}]\|_2}{\|\mathbf{y}\|_2 + \epsilon_m}, \qquad \epsilon_m > 0, \tag{1.122}$$

where $\widehat{\mathcal{F}}$ is reconstructed field consistent with $\mathcal{Q}_i$. Small $\mathcal{M}_{\text{mis}}$ indicates channel-data compatibility; large $\mathcal{M}_{\text{mis}}$ indicates unresolved model error.

The reportability decision is stated by

$$\begin{aligned} &\mathfrak{A}(\mathcal{Q}_i) \leq \tau_A \ \wedge \ \mathcal{M}_{\text{mis}} \leq M_{\max} \Longrightarrow \text{report as constrained observable surrogate,} \\ &\mathfrak{A}(\mathcal{Q}_i) > \tau_A \ \vee \ \mathcal{M}_{\text{mis}} > M_{\max} \Longrightarrow \text{report as inference-only descriptor.} \end{aligned} \tag{1.123}$$

Here $\tau_A > 0$ and $M_{\max} > 0$ are protocol thresholds fixed before analysis. Equation (1.123) converts qualitative caution into an explicit reporting rule.

At chapter level, direct-observable status requires all upstream criteria:

$$\mathcal{D}_{\text{obs}} = 1 \iff \Big(\mathcal{D}_\sigma = 1,\ \mathcal{D}_{\text{mod}} = 1,\ \mathfrak{A}(\mathcal{Q}_i) \leq \tau_A,\ \mathcal{M}_{\text{mis}} \leq M_{\max}\Big). \tag{1.124}$$

This equation joins channel calibration, modal conditioning, and inference regime distinction in one acceptance statement.

Any reported quantity in this chapter is admissible as a direct observable or constrained surrogate only under Eq. (1.123); all other quantities must be labeled inference-only.

Eq. (1.120) defines role separation. Eq. (1.121) quantifies uncertainty/symmetry burden, and Eq. (1.122) quantifies channel compatibility. Their thresholded combination in Eq. (1.123), with upstream criteria in Eq. (1.124), yields the stated criterion.

This analysis assumes calibrated channels, declared thresholds, and uncertainty models consistent with Eq. (1.85). Under adversarial noise, severe model mismatch, or nonstationary drift, the fixed-threshold decision rule in Eq. (1.123) should be replaced by Bayesian model comparison or worst-case robust decision procedures [2].

Angular structure begins in fields, not labels; flux integrals outrank modal names on declared surfaces [6].

1.3 Examples and Counterexamples

The four worked cases form a screening ladder: positive Beth closure, phase without transport, gauge-stable integrals, and reactive versus radiative shells on one source [11, 10, 2]. Each repeats geometry, origin, regime, channel model, and admissibility closure before modal vocabulary enters [6, 13].

The four worked cases form an admissibility ladder with $\mathcal{D}_J = 1$: positive Beth completion, phase without transport, gauge-stable integrals, and reactive versus radiative shells on one source [11]. Each repeats geometry, origin, regime, channel model, and Maxwell-closed completion before modal vocabulary enters [6]. Subsection 1.3.1 confirms the positive case; Subsection 1.3.2 inverts the logic with counter-rotating phase; Subsection 1.3.3 addresses gauge-stable integrals; Subsection 1.3.4 compares inner reactive and outer radiative shells on the same source in measurement form [2]. Together the four cases exercise every layer of Eq. (**??**) at least once before 2 introduces representation-theoretic refinements [3].

1.3.1 Angular momentum conservation in radiation

Radiation examples must show flux closure on a sphere before OAM claims [6]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

Radiation examples must show flux completion on a sphere before OAM claims [6].

Positive transport begins with a complete solid-angle budget: $\oint_{\Gamma_R} \mathbf{S} \cdot \hat{\mathbf{n}}\, dS$ fixes exported power before any azimuthal index is named [6]. Beth completion then requires $\oint \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ to match mechanical torque within declared tolerance—the same comparison modern torsion balances perform [11].

For a circularly polarized wave on Γ_R of radius $R = 2\,\mathrm{m}$ with $P_{\mathrm{rad}} = 1\,\mathrm{W}$, harmonic steady state gives $|\dot{J}_z| \sim P_{\mathrm{rad}}/\omega$ as the order of magnitude of angular-action flux associated with photon spin $\hbar$ [11]. The torque channel on a torsion fixture tracks the same flux integral; Beth completion is the requirement that mechanical readout and $\oint \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ close within declared calibration [2].

Consider a compact helical radiator of length $\ell = \lambda/4$ operates at $f = 10\,\mathrm{GHz}$ in the fundamental radiative mode with dominant channel $(m, \sigma) = (1, +1)$. A spherical control surface S_R of radius $R = 5\lambda$ encloses the source in the far zone ($k_0 R = 10\pi \gg 1$). A torsion balance reports torque about $\hat{\mathbf{z}}$, and a calibrated power meter reports P_{rad} through S_R. Using only inherited results from Subsections 1.1.4–1.1.5 and the operational channels of Section 1.2.5, determine whether the measured torque is consistent with exported angular-momentum flux, and state the regime and admissibility tags required for a valid conclusion .

The problem tests a single Maxwellian chain: conserved angular momentum leaving S_R as stress-flux must appear as opposite mechanical torque on the source support. It is not a new derivation of conservation law; it is an application of Eq. (1.42) in a geometry where reactive dominance has already been excluded by stand-off and where $\mathcal{D}_{J,\mathrm{reg}} = 1$ is credible from Subsection 1.3.3. Failure of completion therefore signals channel error, modal leakage, or regime mismatch—not a breakdown of Maxwell theory. Beth's quartz-plate experiment [11] established the experimental credibility of this chain for polarization-dominated channels; the present problem extends the same logic to multimode radiative enclosures with explicit calibration checks [2].

Figure 1.10 organizes the Beth completion chain: the *source block* (blue) is the quarter-wave helical radiator; the *control-surface block* (red) is the far-zone shell S_R; the *field-transport block* (teal) is the angular-momentum flux integrand on S_R; the *mechanical block* (orange) is the measured load torque; the *readout block* (gray) collects y_P, y_τ, and $\mathcal{R}_\tau$.

Radiation-shell stress evaluation. For dominant mode $(m, \sigma) = (1, +1)$, far-zone fields scale as $|\mathbf{E}| \propto 1/R$ with azimuthal phase $e^{\mathrm{j}\phi}$. The Maxwell stress Eq. (1.37) is evaluated on S_R with outward normal $\hat{\mathbf{n}} = \hat{\mathbf{r}}$. The $1/R$ decay confirms radiative-zone placement; azimuthal phase ties the stress flux to a definite helical channel before integration.

Flux assembly on S_R. The angular-momentum flux density is $[\mathbf{r} \times (\mathbf{T} \cdot \hat{\mathbf{n}})] \cdot \hat{\mathbf{z}}$. Integrating over S_R yields $\dot{J}_z$ in Eq. (1.128). Each surface element contributes local angular-momentum current; the integral is the exported z-component budget per unit time.

Mechanical torque linkage. By Eq. (1.127), $\tau_z^{\mathrm{mech}} = -\dot{J}_z$. Mechanical torque on the source equals the negative of exported electromagnetic angular momentum—the operational face of conservation in Subsection 1.1.5.

Multimode channel budget (if needed). If multiple modes contribute, use Eq. (1.130) with measured $P_{m,\sigma}$ from directional decomposition of Eqs. (1.49)–(1.51). Each modal channel carries a definite $(m+\sigma)/\omega$ torque-per-power factor; unresolved leakage $\Delta\tau_{\mathrm{leak}}$ must be budgeted explicitly.

Conservation completion test. Compute $\mathcal{R}_\tau$ from Eq. (1.131) with $\mathcal{D}_{61} = 1$ from Eq. (**??**). Small $\mathcal{R}_\tau$ establishes agreement between mechanics and electromagnetism within declared channel resolution.

On $S_R = \{\mathbf{r} : R = \text{const}\}$, parameterize $\mathbf{r} = R\hat{\mathbf{r}}(\theta,\phi)$ with $\hat{\mathbf{r}} = \hat{\mathbf{x}}\sin\theta\cos\phi + \hat{\mathbf{y}}\sin\theta\sin\phi + \hat{\mathbf{z}}\cos\theta$. The element of area is $dS = R^2\sin\theta\,d\theta\,d\phi$. For dominant mode $(m,\sigma) = (1,+1)$, the angular-flux integrand is $\Phi(\theta,\phi) = [\mathbf{r}\times(\mathbf{T}\cdot\hat{\mathbf{r}})]\cdot\hat{\mathbf{z}}$. A $N_\theta \times N_\phi$ panel rule gives

$$\dot{J}_z^{(N)} = \sum_{i=1}^{N_\theta}\sum_{j=1}^{N_\phi} \Phi(\theta_i,\phi_j)\,\Delta\theta\,\Delta\phi\,R^2\sin\theta_i, \tag{1.125}$$

where $\Delta\theta = \pi/N_\theta$, $\Delta\phi = 2\pi/N_\phi$. For $R = 5\lambda$, $N_\theta = N_\phi = 72$, and dipole fields at $f = 10\,\mathrm{GHz}$, numerical evaluation of Eq. (1.125) typically agrees with Eq. (1.129) to within 2% when $\mathcal{D}_{J,\mathrm{reg}} = 1$. Quadrature error $\delta\dot{J} = \dot{J}_z^{(N)} - \dot{J}_z$ enters the completion residual as an additive contribution alongside channel noise n_τ, n_P; reporting N_θ, N_ϕ is therefore part of the measurement protocol, not an optional numerical detail [2].

Complete numeric completion with uncertainty propagation. Take $\omega = 2\pi\times10^{10}\,\mathrm{rad/s}$, $P_{\mathrm{rad}} = 1.00\,\mathrm{W}$, $(m,\sigma) = (1,+1)$. Eq. (1.129) gives $\tau_z^{\mathrm{pred}} = -(m+\sigma)P_{\mathrm{rad}}/\omega = -2/\omega \approx -3.183\times10^{-11}\,\mathrm{N\,m}$. With calibration $K_\tau = 1.020 \pm 0.008$ and readout $y_\tau = -3.10\times10^{-11}\,\mathrm{N\,m}$, the inferred torque is $\tau_z^{\mathrm{meas}} = y_\tau/K_\tau \approx -3.04\times10^{-11}\,\mathrm{N\,m}$. Power channel $y_P = 0.998\,\mathrm{W}$ with $K_P = 1.000 \pm 0.005$ gives $P_{\mathrm{rad}}^{\mathrm{meas}} \approx 0.998\,\mathrm{W}$. Substituting into Eq. (1.131) with $\Delta\tau_{\mathrm{leak}} = 0$ yields $\mathcal{R}_\tau \approx 0.045$. First-order uncertainty from $\sigma_{K_\tau} = 0.008$ and $\sigma_P = 0.005\,\mathrm{W}$ adds $\approx 4\times10^{-12}\,\mathrm{N\,m}$ to the torque budget, comparable to Beth-scale laboratory resolutions [11]. If the same protocol is run at $R = \lambda$ without reporting $\chi(R)$, quadrature may still return a number for $\mathcal{R}_\tau$ while regime qualification from Table 1.3 remains unset.

Write calibrated measurement channels as

$$y_\tau = K_\tau\tau_z^{\mathrm{mech}} + n_\tau, \qquad y_P = K_P P_{\mathrm{rad}} + n_P, \tag{1.126}$$

where y_τ and y_P are reported torque and power outputs, K_τ and K_P are calibration factors, and n_τ, n_P are channel disturbances. Equation (1.126) fixes the measurement semantics before any field transport interpretation is attempted.

For steady harmonic operation, storage variation over one cycle-average window is small. The inherited angular-momentum balance therefore reduces to

$$\tau_z^{\mathrm{mech}} = -\int_{S_R} \left[\mathbf{r}\times(\mathbf{T}\cdot\hat{\mathbf{n}})\right]\cdot\hat{\mathbf{z}}\,dS, \tag{1.127}$$

where τ_z^{mech} is measured mechanical torque component on the load. The integrand is local angular-momentum flux density crossing S_R, projected on $\hat{\mathbf{z}}$; the surface integral is total exported electromagnetic angular momentum per unit time.

Let P_{rad} denote radiated power through S_R, and define dimensionless angular transport factor χ_J by

$$\chi_J = \frac{\dot{J}_z}{P_{\mathrm{rad}}}, \qquad \dot{J}_z = \int_{S_R}\left[\mathbf{r}\times(\mathbf{T}\cdot\hat{\mathbf{n}})\right]\cdot\hat{\mathbf{z}}\,dS. \tag{1.128}$$

Equation (1.128) separates a geometric/dynamical conversion factor χ_J from absolute power scale P_{rad}, so that torque prediction can be compared across different radiated-power levels.

In the common single-helical-channel limit with azimuthal index $m \in \mathbb{Z}$ and circular-polarization helicity $\sigma \in \{-1,+1\}$, the channel factor is $\chi_J \approx (m+\sigma)/\omega$, where ω is angular frequency [2]. Substituting into Eq. (1.127) yields the application law

$$\tau_z^{\mathrm{mech}} \approx -\frac{m+\sigma}{\omega}\,P_{\mathrm{rad}}. \tag{1.129}$$

Equation (1.129) is not a new conservation theorem; it is a regime-specific evaluator obtained by applying the conservation statement of Subsection 1.1.5 to a radiative control surface.

For multimode radiation, the same mapping becomes

$$\tau_z^{\mathrm{mech}} = -\sum_{m,\sigma} \frac{m+\sigma}{\omega}\, P_{m,\sigma} - \Delta\tau_{\mathrm{leak}}, \tag{1.130}$$

where $P_{m,\sigma}$ is radiated power carried by modal channel (m,σ), and $\Delta\tau_{\mathrm{leak}}$ collects unresolved leakage channels outside the declared modal set. Equation (1.130) provides the problem-grade extension needed when multiple channels are simultaneously excited.

The completion residual associated with this model is

$$\mathcal{R}_\tau = \frac{\left|\tau_z^{\mathrm{mech}} + \sum_{m,\sigma} \frac{m+\sigma}{\omega}\, P_{m,\sigma} + \Delta\tau_{\mathrm{leak}}\right|}{\left|\sum_{m,\sigma} \frac{m+\sigma}{\omega}\, P_{m,\sigma}\right| + \epsilon_\tau}, \qquad \epsilon_\tau > 0. \tag{1.131}$$

Small $\mathcal{R}_\tau$ indicates that measured torque and stress-flux transport prediction are mutually consistent within declared channel resolution.

An Maxwell-closed criterion for this worked case is

$$\mathcal{D}_{61} = 1 \iff \Big(\mathcal{D}_\sigma = 1,\ \mathcal{R}_\tau \le R_{\tau,\max},\ |n_\tau| \le n_{\tau,\max},\ |n_P| \le n_{P,\max}\Big), \tag{1.132}$$

where $R_{\tau,\max}$, $n_{\tau,\max}$, and $n_{P,\max}$ are protocol thresholds. Equation (**??**) forces every torque claim in this example to carry an explicit channel-quality tag.

The radiation-to-torque conclusion is physically accepted only if $\mathcal{D}_{61} = 1$.

Eq. (1.127) gives inherited conservation mapping, Eq. (1.130) gives modal evaluator, Eq. (1.131) quantifies mismatch, and Eq. (**??**) applies the measurement charge-continuous protocol. Both conditions are required.

1.3.2 Phase structure without angular-momentum transport

On a declared compact-range or anechoic campaign, technicians record frequency, outline, and probe geometry before any harmonic fit is trusted [13]. Phase vortices in near fields need not transport angular momentum to the far zone [12]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

Phase vortices in near fields need not transport angular momentum to the far zone [12]. Superposed $m = +2$ and $m = -2$ beams illustrate channel cancellation: azimuthal intensity may show four lobes while net $\oint \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ vanishes when amplitudes match [10]. Equal-power beams on S_o cancel torque when amplitudes match; a five-percent excess in the $+2$ channel breaks that cancellation and fixes the torque sign [2]. The flux integral scales linearly with amplitude imbalance—a direct laboratory consequence of superposition on outgoing sectors [11].

Two coherent circularly polarized beams of equal power illuminate a flat observation aperture S_o at $f = 10\,\mathrm{GHz}$. Beam + carries azimuthal index $m = +2$; beam − carries $m = -2$. Both share radial envelope $A(\rho) = A_0 e^{-\rho^2/\rho_0^2}$ with $\rho_0 = 5\,\mathrm{mm}$. A phase camera records a four-lobed azimuthal intensity pattern. A torsion sensor on the source mount reports torque about $\hat{\mathbf{z}}$. Using the channel-balance law from Section 1.2 and stress channels from Section 1.2.5, determine whether the visible phase structure implies net angular-momentum transport, and quantify the imbalance sensitivity if one channel is five percent stronger than the other .

The experiment isolates a logical error common in angular-momentum discourse: confusing *phase richness* with *flux imbalance*. Counter-rotating channels of equal intensity produce standing

azimuthal texture—nodes, lobes, and strong $\partial_\phi E$—while their signed angular-momentum contributions cancel on integration. Allen *et al.* [12] showed that a *single* helical mode exports definite orbital flux; the present superposition of $+m$ and $-m$ channels with equal weight deliberately removes that export while retaining visible azimuthal structure. The problem therefore tests necessity: transport requires $I_+ - I_- \neq 0$, not merely $m \neq 0$ in the field exponent [12].

The schematic organizes the following blocks: *aperture block* (gray, top) fixes S_o and ρ_0; the $+m$ *channel block* (blue) and $-m$ *channel block* (red) are equal counter-rotating ingress carriers; the *interference block* (purple) is the standing azimuthal lobe pattern; the *balance block* (gray, bottom) states the zero-transport verdict and imbalance test.

The solution proceeds in four steps. *Superposition on the aperture.* Form $E_t = E_+ + E_-$ from Eq. (1.135) with equal amplitudes $A_+ = A_- = A(\rho)$. The result is Eq. (1.133). *Interference reading:* two counter-rotating carriers synthesize a standing azimuthal lobe pattern; local circulation exists but with opposite sign.

Per-channel power accounting. Compute $I_\pm = |A_\pm|^2$. Under equal-amplitude condition, $I_+ = I_- = |A(\rho)|^2$ everywhere on S_o. *Balance check:* each helical ingress carries half the total power; neither channel dominates the angular budget.

Net flux from channel difference. Substitute into Eq. (1.136): $\dot{J}_z = (m/\omega)\int_{S_o}(I_+ - I_-)\,dS$. *Transport rule:* only $I_+ - I_-$ enters net export; matched powers cancel regardless of azimuthal index m.

Predicted torque on the mount. Apply Eq. (1.139): $\tau_z^{\text{pred}} = -\dot{J}_z = 0$. *Null-readout interpretation:* zero torque is the expected Maxwellian outcome for balanced channels, not a failed sensor.

Five-percent imbalance test. Introduce $I_+ = 1.05\,I_0$, $I_- = 0.95\,I_0$ and evaluate Eq. (1.140). *Sensitivity lesson:* minute loss or ellipticity mismatch is enough to convert a vivid phase image into weak but nonzero transport.

Let a monochromatic transverse field on observation surface S_o be formed by superposing two coherent counter-rotating helical channels with common radial envelope $A(\rho)$, cylindrical coordinates (ρ, ϕ), and integer order $m > 0$:

$$E_t(\rho, \phi) = A(\rho)e^{\mathrm{j}m\phi} + A(\rho)e^{-\mathrm{j}m\phi} = 2A(\rho)\cos(m\phi). \tag{1.133}$$

Equation (1.133) displays strong angular patterning through lobe/nodal structure of $\cos(m\phi)$, so any purely geometric inspection would classify the field as richly phase-structured.

Complete step-by-step derivation. Start from left- and right-circular components on S_o: $E_+ = A(\rho)e^{\mathrm{j}m\phi}/\sqrt{2}$ and $E_- = A(\rho)e^{-\mathrm{j}m\phi}/\sqrt{2}$, each normalized so $|E_\pm|^2 = |A(\rho)|^2/2$. Superposition gives $E_t = \sqrt{2}A(\rho)\cos(m\phi)$ as in Eq. (1.133). The intensity difference is $I_+ - I_- = |E_+|^2 - |E_-|^2 = 0$ at every ρ when amplitudes match.

For each helical channel, the time-averaged angular-momentum flux through S_o scales as $(\pm m/\omega)$ times channel power $P_\pm = \int_{S_o} I_\pm\,dS$ [2]. Summing channels,

$$\dot{J}_z = \frac{m}{\omega}(P_+ - P_-) = \frac{m}{\omega}\int_{S_o}(I_+ - I_-)\,dS, \tag{1.134}$$

which is Eq. (1.136). When $I_+ = I_-$, both P_+ and P_- are equal and $\dot{J}_z = 0$ even though $|E_t|^2 = 2|A|^2\cos^2(m\phi)$ exhibits $2m$ azimuthal nodes per cycle.

The Poynting vector for each circular component rotates in opposite senses; their sum produces a standing azimuthal pattern in $|E_t|$ without net azimuthal power flow [6]. Stress-flux integration confirms Eq. (1.139): mechanical torque vanishes because exported angular momentum vanishes. Any experimental nonzero torque under equal channels implicates $I_+ \neq I_-$ from loss, ellipticity mismatch, or aperture truncation breaking the symmetry assumed in Eq. (1.137) [2].

To make the transport test explicit, write the two-channel complex amplitudes as

$$E_+(\rho,\phi) = A_+(\rho)e^{\mathrm{j}m\phi}, \qquad E_-(\rho,\phi) = A_-(\rho)e^{-\mathrm{j}m\phi}, \tag{1.135}$$

with intensities $I_+(\rho) = |A_+(\rho)|^2$ and $I_-(\rho) = |A_-(\rho)|^2$. Let $\dot{J}_z$ be net angular-momentum transport rate through S_o. The inherited channel-balance law gives

$$\dot{J}_z = \frac{m}{\omega}\int_{S_o}(I_+ - I_-)\,dS. \tag{1.136}$$

Here m/ω is channel angular-momentum-per-energy factor and $I_+ - I_-$ is signed transport imbalance. Equation (1.136) therefore states that net transport is controlled by channel imbalance, not by visual phase richness.

The equal-amplitude counterexample is the special case

$$I_+(\rho) = I_-(\rho) \quad \forall \rho \in S_o. \tag{1.137}$$

Substitution into Eq. (1.136) yields

$$\dot{J}_z = 0 \qquad \text{while} \qquad E_t(\rho,\phi) = 2A(\rho)\cos(m\phi) \neq 0. \tag{1.138}$$

Equation (1.138) is the central negative result of visible phase structure is not sufficient for net angular-momentum transport.

To connect this with measurable torque channels, use the stress-to-torque mapping from Section 1.2.5. In steady operation,

$$\tau_z^{\mathrm{pred}} = -\dot{J}_z = 0 \quad \text{under Eq. (1.137).} \tag{1.139}$$

Hence a nonzero torque measurement in this counterexample signals either channel imbalance, asymmetry in measurement sequence, or unresolved environmental coupling.

A small imbalance perturbation provides first-order sensitivity. Let $I_+(\rho) = I_0(\rho) + \delta I(\rho)$ and $I_-(\rho) = I_0(\rho) - \delta I(\rho)$. Then

$$\dot{J}_z = \frac{2m}{\omega}\int_{S_o}\delta I(\rho)\,dS. \tag{1.140}$$

Equation (1.140) quantifies how small channel biases convert phase-structured but balanced fields into weak net-transport states.

To separate pattern visibility from transport capability, define phase-structure index and transport index

$$\mathcal{V}_\phi = \frac{\int_0^{2\pi}\left|\partial_\phi E_t(\rho_*,\phi)\right|^2\,d\phi}{\int_0^{2\pi}|E_t(\rho_*,\phi)|^2\,d\phi + \epsilon_v}, \qquad \mathcal{T}_J = \frac{|\dot{J}_z|}{P_{\mathrm{tot}}/\omega + \epsilon_t}, \tag{1.141}$$

where ρ_* is representative radius, P_{tot} total channel power, and $\epsilon_v, \epsilon_t > 0$ regularizations. In the equal-channel counterexample, $\mathcal{V}_\phi$ can be large while $\mathcal{T}_J = 0$. This explicitly demonstrates necessity-versus-sufficiency separation.

diagnostic statement: visual twisting or lobe count cannot by itself establish mechanical torque capability. A transport claim requires signed channel imbalance or equivalent flux criterion, exactly as enforced by Eq. (1.136).

The worked-case admissibility criterion is

$$\mathcal{D}_{62} = 1 \iff \Big(\mathcal{D}_{\mathrm{mod}} = 1,\ \left|\int_{S_o}(I_+ - I_-)\,dS\right| \le \Delta I_{\max},\ |y_\tau| \le y_{\tau,\max},\ \mathcal{T}_J \le T_{\max}\Big), \tag{1.142}$$

where $\Delta I_{\max}$, $y_{\tau,\max}$, and $T_{\max}$ are declared thresholds. Equation (1.142) ensures that the counterexample is accepted only when channel balance and measurement evidence are consistent.

In the balanced counter-rotating configuration of Eq. (1.137), phase-structure visibility is not a sufficient condition for angular-momentum transport.

Eq. (1.136) ties transport to signed intensity imbalance. Balanced channels enforce Eq. (1.138), which with Eq. (1.139) yields zero predicted torque despite nontrivial spatial phase structure.

Numerical visibility-versus-transport walk-through. With $m = 2$, $A(\rho) = A_0 e^{-\rho^2/\rho_0^2}$, and equal channels $I_+ = I_- = A_0^2 e^{-2\rho^2/\rho_0^2}$, Eq. (1.133) gives $E_t = 2A_0 e^{-\rho^2/\rho_0^2}\cos(2\phi)$. At $\rho_* = \rho_0$, $\partial_\phi E_t = -4A_0 e^{-1}\sin(2\phi)$, so $\mathcal{V}_\phi$ in Eq. (1.141) is order unity because azimuthal variation is strong. Yet $I_+ - I_- = 0$ implies $\dot{J}_z = 0$ and $\mathcal{T}_J = 0$ identically. Introducing a five-percent intensity bias $I_+ = 1.05\, I_0$, $I_- = 0.95\, I_0$ yields $\dot{J}_z = (0.1m/\omega)\int_{S_o} I_0\, dS$, a small but nonzero transport channel detectable only when intensity imbalance is resolved at the one-percent level or better [2].

Let the aperture radius be $\rho_0 = 5\,\mathrm{mm}$, $m = 2$, and total power $P_{\mathrm{tot}} = \int_{S_o}(I_+ + I_-)\, dS = 0.20\,\mathrm{W}$ at $f = 10\,\mathrm{GHz}$. Balanced channels give $\dot{J}_z = 0$ and $y_\tau = 0$ within noise. Imposing $I_+ = 1.05\, I_0$, $I_- = 0.95\, I_0$ with uniform I_0 over $S_o = \{\rho \le \rho_0\}$ gives $\int_{S_o}(I_+ - I_-)\, dS = 0.10\, I_0 \pi \rho_0^2 \approx 0.02\, P_{\mathrm{tot}}$. Eq. (1.140) then yields $\dot{J}_z = (0.1m/\omega)P_{\mathrm{tot}} \approx 1.27 \times 10^{-13}\,\mathrm{N\,m\,s}$ for $m = 2$, and $\tau_z^{\mathrm{pred}} = -\dot{J}_z \approx -1.27 \times 10^{-13}\,\mathrm{N\,m}$. A torsion sensor with resolution $10^{-14}\,\mathrm{N\,m}$ can in principle detect this bias, but only if $\mathcal{D}_{\mathrm{mod}} = 1$ resolves the five-percent intensity asymmetry at the one-percent level. Phase-camera visibility $\mathcal{V}_\phi$ remains order unity throughout, illustrating the phase-insufficiency conclusion above: the camera sees azimuthal structure; the torque channel sees channel imbalance [11].

The derivation presumes linear superposition, coherent channel control, and observation geometry that resolves channel intensities without bias. Incoherent mixing, polarization-selective loss, or aperture truncation can break the exact cancelation and produce residual transport [6].

1.3.3 Gauge dependence of local angular-momentum densities

A spiral phase plate in front of a horn feed on a compact range defines the hardware scene for 'Gauge dependence of local angular-momentum densities": technicians compare turntable cuts and wattmeter totals on Γ_R before paraxial labels are quoted [12]. Maxwell's equations on the declared exterior domain remain the governing constraint [1].

Gauge relabeling leaves $(\mathbf{E}, \mathbf{H})$ fixed while reshuffling information between scalar and vector potentials [1]. The worked problem below quantifies when local angular-momentum density maps disagree pointwise yet surface flux integrals and torque channels remain stable—the laboratory face of the gauge discussion in Section 1.2.3 [10].

Consider a finite spherical control volume V_g of radius $R = 2\lambda$ encloses a center-fed dipole at $f = 1\,\mathrm{GHz}$. Two potential representations are available: Coulomb gauge $(\mathbf{A}_C, \phi_C)$ and Lorenz gauge $(\mathbf{A}_L, \phi_L)$, both producing identical $\mathbf{E}, \mathbf{H}$ on ∂V_g. A post-processing script may compute a local angular-momentum density $\mathbf{j}_{\mathrm{loc}} = \mathbf{r} \times (\varepsilon_0 \mathbf{E})$ in each gauge and report volume integrals $\mathbf{J}_{\mathrm{int}}$ alongside the gauge-invariant density $\mathbf{j}_{\mathrm{em}}$ from Section 1.1.5 [10]. Surface flux $\oint_{\partial V_g} \mathbf{M} \cdot \hat{\mathbf{n}}\, dS$ is unchanged under $\mathbf{A} \mapsto \mathbf{A} + \nabla\chi$, whereas $\int_{V_g} \mathbf{r} \times (\varepsilon_0 \mathbf{E})\, dV$ is gauge-sensitive; published reports must therefore lead with flux and torque channels, not with pointwise density plots unless gauge is fixed [2].

Any local density built from potentials alone can therefore exhibit gauge-orbit motion even when integrated flux observables are fixed. The problem requires a reporting discipline: distinguish *diagnostic maps* (may vary with gauge) from *observable integrals* (must pass contamination criteria). This is the laboratory counterpart of the gauge discussion in Section 1.2.3 [2].

The schematic organizes the following blocks: *volume block* (gray, centre) is the control region V_g; the *boundary-field block* (teal) states identical $(\mathbf{E}, \mathbf{H})$ on ∂V_g; the *Coulomb-map block* (blue) and

Lorenz-map block (red) are gauge-sensitive local densities; the *admissibility block* (gray, bottom) reports the integral contamination test $\mathcal{C}_{\mathrm{int}}$.

The solution proceeds in four steps. *Gauge orbit on potentials.* Relate $(\mathbf{A}', \phi')$ to $(\mathbf{A}, \phi)$ via $\mathbf{A}' = \mathbf{A} + \nabla\chi$, $\phi' = \phi - \mathrm{j}\omega\chi$, with χ inherited from Subsection 1.2.3. *Invariance check:* χ relabels potentials only; $\mathbf{E}$ and $\mathbf{H}$ must remain unchanged.

Pointwise density shift. Define $\delta_\chi \mathbf{j}_{\mathrm{loc}} = \mathbf{j}'_{\mathrm{loc}} - \mathbf{j}_{\mathrm{loc}}$. *Diagnostic warning:* local colour plots can disagree strongly even when fields match; the density is gauge-tagged, not gauge-neutral.

Volume divergence accounting. Apply Eq. (1.143) to decompose $\int_{V_g} \delta_\chi \mathbf{j}_{\mathrm{loc}}\, dV$ into boundary flux and residual source channels. *Partition rule:* gauge variation splits into relocatable boundary terms and genuine volumetric contamination.

Integrated report comparison. Evaluate $\mathbf{J}_{\mathrm{int}}$ and $\mathbf{J}'_{\mathrm{int}}$ from Eq. (1.144) and compute $\mathcal{C}_{\mathrm{int}}$ from Eq. (1.147). *Observability test:* integral contamination, not local hue, controls promotion to observable status.

Epistemic criterion. Accept gauge-sensitive local patterns as observables only if $\mathcal{D}_{63} = 1$ in Eq. (1.148); otherwise report them as inference-only intermediates. *Reporting discipline:* explicit labeling is required before publication-grade claims.

Take a finite control volume V_g with boundary ∂V_g, outward unit normal $\hat{\mathbf{n}}$, and two gauge-related potential representatives $(\mathbf{A}, \phi)$ and $(\mathbf{A}', \phi')$. The gauge-variation structure is inherited from Subsection 1.2.3; no new gauge theorem is introduced. The worked problem is to quantify when local angular-momentum decompositions disagree while integrated transport observables remain gauge-robust.

Let $\mathbf{j}_{\mathrm{loc}}$ be a candidate local angular-momentum density constructed from potentials and fields, and define gauge difference $\delta_\chi \mathbf{j}_{\mathrm{loc}} = \mathbf{j}'_{\mathrm{loc}} - \mathbf{j}_{\mathrm{loc}}$. Applying the inherited structure from Subsection 1.2.3 gives

$$\int_{V_g} \delta_\chi \mathbf{j}_{\mathrm{loc}}\, dV = \int_{\partial V_g} \boldsymbol{\Xi}_\chi \cdot \hat{\mathbf{n}}\, dS + \int_{V_g} \boldsymbol{\Psi}_\chi\, dV, \tag{1.143}$$

where $\boldsymbol{\Xi}_\chi$ is gauge-induced flux tensor and $\boldsymbol{\Psi}_\chi$ is residual gauge source term (notations inherited from Subsection 1.2.3). The first integral on the right is boundary redistribution, while the second is volumetric contamination.

Define integrated reports

$$\mathbf{J}_{\mathrm{int}} = \int_{V_g} \mathbf{j}_{\mathrm{loc}}\, dV, \qquad \mathbf{J}'_{\mathrm{int}} = \int_{V_g} \mathbf{j}'_{\mathrm{loc}}\, dV. \tag{1.144}$$

Equation (1.143) then gives $\mathbf{J}'_{\mathrm{int}} - \mathbf{J}_{\mathrm{int}}$ exactly as the sum of boundary and residual channels.

Under boundary/material conditions for which $\int_{\partial V_g} \boldsymbol{\Xi}_\chi \cdot \hat{\mathbf{n}}\, dS = 0$ and $\int_{V_g} \boldsymbol{\Psi}_\chi\, dV = 0$, Eq. (1.143) reduces to

$$\int_{V_g} \mathbf{j}'_{\mathrm{loc}}\, dV = \int_{V_g} \mathbf{j}_{\mathrm{loc}}\, dV. \tag{1.145}$$

Even when Eq. (1.145) holds, pointwise maps $\mathbf{j}'_{\mathrm{loc}}(\mathbf{r})$ and $\mathbf{j}_{\mathrm{loc}}(\mathbf{r})$ may differ strongly. Therefore local gauge-sensitive pictures are admissible only as intermediate descriptors, not as standalone observables.

To separate local-map sensitivity from integral contamination, define

$$\mathcal{M}_{\mathrm{loc}} = \frac{\int_{V_g} \|\mathbf{j}'_{\mathrm{loc}} - \mathbf{j}_{\mathrm{loc}}\|\, dV}{\int_{V_g} \|\mathbf{j}_{\mathrm{loc}}\|\, dV + \epsilon_j}, \qquad \epsilon_j > 0, \tag{1.146}$$

and

$$\mathcal{C}_{\mathrm{int}} = \frac{\|\mathbf{J}'_{\mathrm{int}} - \mathbf{J}_{\mathrm{int}}\|}{\|\mathbf{J}_{\mathrm{int}}\| + \epsilon_c}, \qquad \epsilon_c > 0. \tag{1.147}$$

Table 1.3: Reactive-radiative regime indicators for a $\lambda/4$ dipole at $f = 1\,\mathrm{GHz}$ (illustrative shell values).

kr	P_{rad} (mW)	$\lvert Q_{\mathrm{re}}\rvert$ (mW)	χ	$\mathcal{D}_{J,\mathrm{reg}}$
0.5	2	28	14	0
1.0	8	22	2.8	0
3.0	42	9	0.21	1
10.0	95	1.2	0.013	1
30.0	102	0.15	0.0015	1

The worked-case acceptance criterion is

$$\mathcal{D}_{63} = 1 \iff \Big(\mathcal{D}_{\mathrm{mod}} = 1,\ \mathcal{C}_{\mathrm{int}} \le C_{\mathrm{int,max}},\ \left\| \int_{V_g} \boldsymbol{\Psi}_\chi \, dV \right\| \le \Psi_{\max}\Big), \tag{1.148}$$

where $C_{\mathrm{int,max}}$ and $\Psi_{\max}$ are declared tolerances. This criterion enforces that observable status is controlled by integral contamination, not by local pattern appearance.

Gauge-sensitive local angular-momentum decompositions are admissible as observables in this worked case only if $\mathcal{D}_{63} = 1$; otherwise they remain intermediate descriptors.

Eq. (1.143) decomposes gauge variation into boundary and residual channels. Eq. (1.145) states the integral completion condition, while Eq. (1.148) enforces thresholded $\mathcal{D}_J = 1$. Hence local pattern disagreement alone cannot reject a gauge-invariant observable. Gauge choices do not alter the underlying Maxwell physics; potential-based local pictures are optional coordinates on the same physics. When $\mathcal{C}_{\mathrm{int}} \approx 0$ but $\mathcal{M}_{\mathrm{loc}} \gg 1$, the correct report is "gauge-sensitive local map, gauge-robust integral," not contradiction.

The argument holds when potentials are sufficiently regular for volume/surface operations and when boundary conditions are declared explicitly. Topological gauge patching, singular gauges, or unresolved boundary currents can invalidate the simplified cancellation path [2].

In Coulomb gauge for a point dipole $\mathbf{p} = p_0\hat{\mathbf{z}}$, $\phi_C = \mathbf{p}\cdot\mathbf{r}/(4\pi\varepsilon_0 r^3)$ and $\mathbf{A}_C$ is solenoidal. Under Lorenz gauge, $\phi_L = \mathrm{j}\omega p_0 \cos\theta/(4\pi c^2 r)$ in the far zone with a transverse $\mathbf{A}_L$. Construct $\mathbf{j}_{\mathrm{loc}} = \mathbf{r}\times(\varepsilon_0\mathbf{E})$ in each gauge at the same field point: the two local vectors differ by $\delta_\chi \mathbf{j}_{\mathrm{loc}} \propto \nabla\chi$ with $\chi = (\phi_L - \phi_C)/\omega$. Volume integration over a sphere V_g of radius R gives $\mathbf{J}'_{\mathrm{int}} \neq \mathbf{J}_{\mathrm{int}}$ unless the residual $\boldsymbol{\Psi}_\chi$ integral vanishes. Using the gauge-invariant $\mathbf{j}_{\mathrm{em}}$ from Section 1.1.5 instead produces $\mathcal{C}_{\mathrm{int}} = 0$ in Eq. (1.147) while $\mathcal{M}_{\mathrm{loc}}$ may remain large. The worked conclusion matches Subsection 1.2.3: local potential decompositions are diagnostic; invariant integrals are reportable [11].

For $\mathbf{p} = p_0\hat{\mathbf{z}}$ at $f = 1\,\mathrm{GHz}$ with $R = 2\lambda$, dipole fields on ∂V_g are identical in both gauges. Construct $\mathbf{j}_{\mathrm{loc}}^{(C)} = \mathbf{r}\times(\varepsilon_0\mathbf{E})$ from Coulomb potentials and $\mathbf{j}_{\mathrm{loc}}^{(L)}$ from Lorenz potentials at ten sample points on a spherical shell at $r = 0.9R$. Typical outcomes are $\|\mathbf{j}_{\mathrm{loc}}^{(C)} - \mathbf{j}_{\mathrm{loc}}^{(L)}\|/\|\mathbf{j}_{\mathrm{em}}\| \approx 0.35$–$0.55$, so $\mathcal{M}_{\mathrm{loc}}$ in Eq. (1.146) is order unity. By contrast, $\mathbf{J}_{\mathrm{int}}^{(C)} - \mathbf{J}_{\mathrm{int}}^{(L)}$ normalized by $\|\int_{V_g}\mathbf{j}_{\mathrm{em}}\,dV\|$ is below 10^{-2} when $\int_{V_g}\boldsymbol{\Psi}_\chi\,dV$ is controlled, giving $\mathcal{C}_{\mathrm{int}} \ll 1$. The numerical pattern matches Eq. (1.143): large local mismatch, small integral contamination. Reporting only a false-colour plot of $\mathbf{j}_{\mathrm{loc}}^{(C)}$ without $\mathcal{C}_{\mathrm{int}}$ would overstate gauge sensitivity as a physical contradiction [10].

Reading Table 1.3 alongside Table 1.1 clarifies why transport and regime criteria are jointly necessary: at $kr = 1$, strong reactive circulation can inflate $|Q_{\mathrm{re}}|$ even when phase portraits appear structured to visual inspection. Any angular-momentum report that omits $\chi(r)$ therefore risks attributing storage-dominated structure to propagative torque channels [2]. Reactive shells carry standing-wave structure whose phase rotates locally while time-averaged Poynting flux through the

same surface may vanish [2]. A $\lambda/2$ dipole at $kr = 0.5$ versus $kr = 5$ illustrates why shell placement matters: the inner shell is storage-dominated and generally fails $\mathcal{C}_3$; the outer shell carries outward flux support suitable for export diagnostics [3].

1.3.4 Reactive versus radiative comparison

The same dipole can present rich phase structure on a reactive shell while exporting qualified torque only on a radiative shell [2]. The two-shell protocol below makes that regime split explicit through power, reactive, and torque channels on concentric surfaces [4].

Consider a center-fed $\lambda/2$ dipole radiates at $f = 1\,\mathrm{GHz}$. Two concentric spherical sensor shells are deployed: inner S_{r_1} at $kr_1 = 0.5$ and outer S_{r_2} at $kr_2 = 5$. On each shell, calibrated channels report active power $y_{P,k}$, reactive power $y_{Q,k}$, and torque magnitude $y_{\tau,k}$ about $\hat{\mathbf{z}}$. Inner shell $kr_1 = 0.5$ samples reactive storage where $|y_{Q,1}|$ may dominate $|y_{P,1}|$; outer shell $kr_2 = 5$ samples radiative export where Poynting flux and torque channels align with $\mathcal{C}_3$ [2]. Comparing $y_{\tau,1}$ to $y_{\tau,2}$ without regime classification mixes storage physics with transport claims. Using regime indices from Subsection 1.3.3, classify each shell as storage-diagnostic or export-diagnostic, compute $\mathcal{R}_Q(r_k)$, $\mathcal{C}_\tau$, and gated efficiency $\eta_\tau(r_k)$, with $\mathcal{D}_{64} = 1$ [2].

The two-shell protocol mirrors what happens when one laboratory measures "angular momentum" in the reactive near zone and another measures torque in the far zone. At 1 GHz on a $\lambda/2$ dipole, only the outer shell at $kr = 5$ carries the export-diagnostic label appropriate for watt-calibrated transport reports; the inner shell at $kr = 0.5$ remains in the reactive regime treated in Subsection 1.3.3 [2]. Both may record structured phase and polarization, yet Maxwellian interpretation differs: inner-shell data primarily diagnose oscillatory storage exchange; outer-shell data test export-compatible torque channels. Without explicit $\mathcal{R}_Q$ comparison, apparent disagreement is often a regime error, not a physics conflict [2].

The schematic organizes the following blocks: *source block* (gray) is the centre-fed dipole; the *inner-shell block* (orange) is the reactive surface S_{r_1} with storage circulation; the *outer-shell block* (blue) is the radiative surface S_{r_2} with outward export; the *readout block* (gray, bottom) lists shell channels and completion $\mathcal{D}_{64}$.

The solution proceeds in five steps. *Shell power functionals.* Evaluate Eq. (1.149) on S_{r_1} and S_{r_2}. *Regime vocabulary:* $P(r)$ measures irreversible export; $Q(r)$ measures oscillatory storage—the reactive-radiative split of Section 1.3.3.

Reactive-ratio distinction. Compute $\mathcal{R}_Q(r_k) = |Q(r_k)|/(|P(r_k)| + \epsilon_P)$ from Eq. (1.150). *Shell reading:* $\mathcal{R}_Q \gg 1$ marks reactive dominance where phase can mislead; $\mathcal{R}_Q \ll 1$ supports export interpretation.

Torque contrast across shells. Form $\mathcal{C}_\tau = |\tau_z(r_2)|/(|\tau_z(r_1)| + \epsilon_\tau)$ from Eq. (1.151). *Transport emergence:* large outer-to-inner contrast shows torque channels activating only beyond the reactive shell.

Gated efficiency correction. Apply Eq. (1.153) with factor $1/(1+\mathcal{R}_Q)$ to suppress false efficiency in reactive zones. *Interpretation guard:* raw torque-to-power ratios overstate transport when $|Q| \gg |P|$.

Trend consistency and criterion. Evaluate $\mathcal{T}_{12}$ from Eq. (1.154) and confirm $\mathcal{D}_{64} = 1$ in Eq. (1.157). *Joint verdict:* both shells must occupy expected classes and the $r_1 \to r_2$ trend must be internally consistent.

Let S_{r_1} and S_{r_2} be concentric spherical observation surfaces, with radii $r_1 < r_2$, centered on the same source. The objective is to compare angular-structure interpretation in a reactive-dominant surface S_{r_1} and a radiative-dominant surface S_{r_2}, using the regime logic inherited from Subsection 1.3.3.

Consider a half-wave dipole at $ka = 1$ with observation shells at $kr_1 = 0.5$ and $kr_2 = 5$, where $k = \omega/c$. On S_{r_1}, the reactive integral $Q(r_1)$ typically exceeds $|P(r_1)|$, yielding $\mathcal{R}_Q(r_1) \gg 1$; phase winding and polarization ellipticity can be large while exported power and axis torque remain small. On S_{r_2}, $|P(r_2)| \gg |Q(r_2)|$ restores conservation-compatible transport interpretation, and $\mathcal{C}_\tau \gg 1$ signals that torque channels become measurably active only after the reactive shell is crossed [2]. This two-surface protocol is the measurement counterpart of the regime indices χ and χ_Ω from Section 1.3.3.

Define active-power and reactive-power surface functionals

$$P(r) = \int_{S_r} \mathrm{Re}\{\mathbf{S} \cdot \hat{\mathbf{n}}\}\, dS, \qquad Q(r) = \int_{S_r} \mathrm{Im}\{\mathbf{S} \cdot \hat{\mathbf{n}}\}\, dS, \tag{1.149}$$

where $\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^*$ is the complex Poynting vector [6] and $\mathrm{Im}\{\cdot\}$ extracts imaginary part. The first integral measures net exported power; the second measures reactive circulation associated with local storage exchange.

To compare regimes on one normalized axis, define reactive ratio

$$\mathcal{R}_Q(r) = \frac{|Q(r)|}{|P(r)| + \epsilon_P}, \qquad \epsilon_P > 0. \tag{1.150}$$

Reactive-dominant behavior corresponds to $\mathcal{R}_Q \gg 1$, while radiative-dominant behavior corresponds to $\mathcal{R}_Q \ll 1$. This scalar ratio makes regime distinction operationally testable at each observation surface.

For a torque channel about $\hat{\mathbf{z}}$, define

$$\mathcal{C}_\tau = \frac{|\tau_z(r_2)|}{|\tau_z(r_1)| + \epsilon_\tau}, \qquad \epsilon_\tau > 0. \tag{1.151}$$

The numerator and denominator are measured torque magnitudes on outer and inner surfaces; ϵ_τ is a small regularization constant preventing division singularity. When $Q(r_1) \gg P(r_1)$ and $P(r_2) \gg |Q(r_2)|$, one typically obtains $\mathcal{C}_\tau \gg 1$: near-zone angular structure is visually rich yet weakly transport-capable, while far-zone structure carries measurable torque.

Measurement channels at the two surfaces are written

$$y_{P,k} = K_{P,k} P(r_k) + n_{P,k}, \qquad y_{Q,k} = K_{Q,k} Q(r_k) + n_{Q,k}, \qquad y_{\tau,k} = K_{\tau,k} \tau_z(r_k) + n_{\tau,k}, \tag{1.152}$$

for $k \in \{1, 2\}$, where $K_{\bullet,k}$ are calibration factors and $n_{\bullet,k}$ are channel disturbances. Equation (1.152) ensures that regime claims are tied to calibrated outputs rather than inferred from qualitative field appearance.

Define regime-gated transport efficiency

$$\eta_\tau(r_k) = \frac{|\tau_z(r_k)|}{P(r_k)/\omega + \epsilon_\eta} \cdot \frac{1}{1 + \mathcal{R}_Q(r_k)}, \qquad \epsilon_\eta > 0. \tag{1.153}$$

The factor $1/(1 + \mathcal{R}_Q)$ suppresses apparent efficiency in reactive regimes where local circulation dominates exported transport.

Surface-to-surface trend consistency is captured by

$$\mathcal{T}_{12} = \frac{|\eta_\tau(r_2) - \eta_\tau(r_1)|}{\eta_\tau(r_2) + \epsilon_t}, \qquad \epsilon_t > 0. \tag{1.154}$$

Large $\mathcal{T}_{12}$ indicates strong regime transition in transport effectiveness, which is expected when r_1 is reactive-dominant and r_2 is radiative-dominant.

For a center-fed dipole at $f = 1\,\mathrm{GHz}$ ($k = 20.96\,\mathrm{rad/m}$), take calibrated reads $y_{P,1} = 0.05\,\mathrm{W}$, $y_{Q,1} = 0.40\,\mathrm{W}$ on S_{r_1} with $kr_1 = 0.5$, giving $\mathcal{R}_Q(r_1) \approx 8 \gg 1$ and $\eta_\tau(r_1) \approx 0$ from Eq. (1.153). On S_{r_2} with $kr_2 = 5$, suppose $y_{P,2} = 0.85\,\mathrm{W}$, $y_{Q,2} = 0.04\,\mathrm{W}$, so $\mathcal{R}_Q(r_2) \approx 0.05 \ll 1$. If $|\tau_z(r_1)| = 0.2\,\mathrm{nN\,m}$ and $|\tau_z(r_2)| = 4.5\,\mathrm{nN\,m}$, then $\mathcal{C}_\tau \approx 22$ and $\mathcal{T}_{12}$ is order unity, confirming regime transition rather than measurement noise. Failing to report $Q(r)$ alongside $P(r)$ would mislabel the inner surface as torque-capable based on phase texture alone [2].

Chu quality factor and reactive shell diagnosis. Linking Subsection 1.3.3, the Chu definition Eq. (1.155) compares stored energy to radiated power; the lower bound Eq. (1.156) sets a geometry-controlled floor. On S_{r_1} with $kr_1 = 0.5$, illustrative dipole data give $|Q(r_1)| \gg P(r_1)$ and $\chi(r_1) \gg 1$; with $ka \approx 0.16$, Eq. (1.156) yields $Q_{\mathrm{LB}} \gtrsim 240$, signaling a storage-dominated shell. On S_{r_2} with $kr_2 = 5$, $\chi(r_2) \ll 1$ and Q_{Chu} approaches its asymptotic minimum for the dominant mode. The two-shell protocol therefore aligns three independent diagnostics—$\mathcal{R}_Q$, $\chi(r)$, and Q_{Chu}—on the same distinction: inner data are storage-diagnostic; outer data are export-diagnostic [4].

$$Q_{\mathrm{Chu}}(V) = \frac{2\omega\langle U\rangle_{\mathrm{st}}}{P_{\mathrm{rad}}}, \tag{1.155}$$

where $\langle U\rangle_{\mathrm{st}}$ is cycle-averaged stored electromagnetic energy in control volume V and P_{rad} is outward radiated power through the enclosing surface [4]. Q_{Chu} compares oscillatory storage to export: large values diagnose reactive-dominated shells where phase-rich patterns must not be read as torque-capable transport [2].

$$Q_{\mathrm{LB}} = \frac{1}{(ka)^3} + \frac{1}{ka}, \tag{1.156}$$

with a the effective radius of the smallest enclosing sphere and $k = \omega/c$. For electrically small sources, Eq. (1.156) sets a geometry-controlled floor on Q_{Chu}, linking finite size to the minimum storage-to-radiation ratio achievable in the radiative class [4].

The worked-case $\mathcal{D}_J = 1$ criterion is

$$\mathcal{D}_{64} = 1 \iff \Big(\mathcal{D}_{63} = 1,\ \mathcal{R}_Q(r_1) \geq R_{Q,\min},\ \mathcal{R}_Q(r_2) \leq R_{Q,\max},\ \mathcal{C}_\tau \geq C_{\tau,\min},\ \mathcal{T}_{12} \leq T_{12,\max}\Big), \tag{1.157}$$

where $R_{Q,\min}$, $R_{Q,\max}$, $C_{\tau,\min}$, and $T_{12,\max}$ are declared thresholds. Equation (1.157) ensures that reactive/radiative labeling and torque-contrast interpretation are jointly consistent [2].

Two-shell completion arithmetic. *Chu-bound cross-check.* Confirm $2\omega\langle U\rangle_{\mathrm{st}}/P_{\mathrm{rad}} \geq Q_{\mathrm{LB}}$ on each shell using Table 1.3 as order-of-magnitude guide. On S_{r_1}, violation of export interpretation is expected; on S_{r_2}, consistency confirms radiative placement. $\mathcal{D}_J = 1$ *synthesis.* With $R_{Q,\min} = 2$, $R_{Q,\max} = 0.2$, $C_{\tau,\min} = 5$, and measured $\mathcal{R}_Q(r_1) = 8$, $\mathcal{R}_Q(r_2) = 0.05$, $\mathcal{C}_\tau = 22$, Eq. (1.157) yields $\mathcal{D}_{64} = 1$ when $\mathcal{D}_{63} = 1$ and $\mathcal{T}_{12} \leq T_{12,\max}$. Failure of any single inequality—for example $\mathcal{R}_Q(r_2) \gg 1$ because the outer shell was placed too close—forces reassignment to a nearer regime without invalidating Maxwell theory [4].

Source: compact helical radiator, $\ell = \lambda/4$, $f = 10\,\mathrm{GHz}$, dominant channel $(m, \sigma) = (1, +1)$. *Geometry:* far-zone sphere S_R, $R = 5\lambda$, $k_0 R = 10\pi \gg 1$. *Observable quantity:* calibrated torque y_τ and power y_P through S_R. *Transport quantity:* exported angular-momentum flux $\int_{S_R}[\mathbf{r} \times (\mathbf{T} \cdot \hat{\mathbf{n}})] \cdot \hat{\mathbf{z}}\, dS$. *Laboratory interpretation:* Beth-type torque–flux completion on a radiating enclosure when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [11]. *Physical lesson:* angular structure is Maxwellian transport verified by mechanical torque, not a modal label [2].

Source: balanced counter-rotating superposition of azimuthally phased channels. *Geometry:* control surface enclosing equal and opposite winding contributions. *Observable quantity:* local phase gradient and Poynting circulation maps. *Transport quantity:* net $\| \int_{4\pi} \mathbf{\Psi}_J\, d\Omega \|$ and $\mathcal{K}_J$.

Laboratory interpretation: vivid phase structure with vanishing exported torque when channels cancel [12]. *Physical lesson:* phase pictures and vortex labels are not angular structure until transport qualification passes [6].

Source: point dipole $\mathbf{p} = p_0\hat{\mathbf{z}}$, $f = 1\,\mathrm{GHz}$. *Geometry:* gauge sphere V_g, radius $R = 2\lambda$; Coulomb versus Lorenz potentials at identical $(\mathbf{E}, \mathbf{H})$. *Observable quantity:* gauge-robust integral $\mathbf{J}_{\mathrm{int}}$ versus gauge-sensitive local pattern $\mathbf{j}_{\mathrm{loc}}$. *Transport quantity:* contamination index $\mathcal{C}_{\mathrm{int}}$ and residual $\mathbf{\Psi}_\chi$ integral. *Laboratory interpretation:* false-colour local patterns may disagree while flux integrals remain stable [10].

Physical evidence summary (Subsection 1.3.4). *Source:* center-fed $\lambda/2$ dipole, $f = 1\,\mathrm{GHz}$. *Geometry:* inner shell S_{r_1}, $kr_1 = 0.5$; outer shell S_{r_2}, $kr_2 = 5$. *Observable quantity:* $y_{P,k}$, $y_{Q,k}$, $y_{\tau,k}$ on each shell. *Transport quantity:* $\mathcal{R}_Q(r_k)$, $\mathcal{C}_\tau$, $\eta_\tau(r_k)$, and Q_{Chu} from Eqs. (1.155)–(1.157). *Laboratory interpretation:* inner reactive shell diagnoses storage; outer radiative shell carries export-compatible torque [4]. *Physical lesson:* regime placement is part of angular-structure definition; reactive phase texture is not propagative transport [2].

The Maxwellian preconditions fixed here—consistent sources and boundaries, finite-geometry realizability, instrument-limited observability, and flux-qualified transport on declared surfaces—are the domain on which source-to-radiation operators act [1].

1.4 Summary

Chapter 1 closes on one admissibility chain: Maxwell-consistent data, finite-geometry realizability, instrument-limited observability, and flux-qualified transport on declared surfaces [1, 2, 3]. Later chapters inherit that chain by reference—representation, harmonics, and rank bounds—without reopening the foundations proved here [10, 17].

Chapter 1 began where Maxwell theory always begins: impressed sources on a declared domain, boundary data consistent with the phasor branch, and conservation laws that tie field stress to measurable traction and torque [1]. Against that background, angular electromagnetic structure is directional organization of $(\mathbf{E}, \mathbf{H})$ capable of exporting rotational action through qualified flux integrals—not a phase portrait, not an orbital index standing alone, and not a modal name substituted for transport [10]. Sections 1.2–1.2.5 separated appearance from export, fixed covariance and gauge rules for laboratory reports, and closed the loop through calibrated stress and flux channels [11]. The four worked configurations in Section 1.3 showed qualified and disqualified outcomes on explicit arithmetic: Beth torque–flux completion, balanced superposition with vivid phase but zero export, gauge-shifted local maps with stable surface integrals, and reactive versus radiative shells on one dipole [6].

Three principles carry forward unchanged: the admissibility ladder ($\mathcal{D}_J = 1$) of Principle 1.0; angular structure lives in $(\mathbf{E}, \mathbf{H})$ before modal names; flux integrals outrank phase images on declared surfaces; finite geometry and finite instruments bound every export and recovery measurement [6].

1.4.1 Results

Principle 1.0 fixes the admissibility ladder ($\mathcal{D}_J = 1$ before watt reports) [2]. The results proved in Sections 1.1–1.3 may be collected without repeating the subsection chain. Maxwell consistency, uniqueness on admissible domains, and the stress–torque–flux ladder of Section 1.1 fix what finite sources may produce and how angular action is accounted [1]. Directional organization, phase topology, and the three-part transport test $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ of Section 1.2 distinguish qualified export from appearance [10]. Covariance laws for vectors, tensors, and dyads in Section 1.2.3 keep torque and

traction reports objective under frame rotation [2]. Gauge-invariant helicity, dual-symmetric combinations, and flux-normalized reports in Section 1.2.3 survive potential relabeling on the radiative branch [10]. Stress-based torque channels, modal projection limits, and the direct-versus-inferred reporting split in Section 1.2.5 connect field predictions to calibrated readouts [11]. Section 1.3 exercised each layer on worked geometry with explicit completion residuals [3].

The Chapter 1 completion criterion is

$$\mathcal{D}_{\mathrm{ch1}} = 1 \iff (\mathcal{D}_J = 1,\ \mathcal{D}_{\mathrm{org}} = 1,\ \mathcal{D}_{\mathrm{g}} = 1,\ \mathcal{D}_{\mathrm{op}} = 1)\,, \tag{1.158}$$

packaging transport, covariance, gauge, and operational admissibility. Let $\mathcal{R}_{\mathrm{ch1}}$ denote the set of results proved under $\mathcal{D}_{\mathrm{ch1}} = 1$. Every reported quantity carries a reporting tier

$$\mathcal{Q} \in \{\text{direct observable},\ \text{constrained surrogate},\ \text{inference-only}\}, \tag{1.159}$$

as fixed in Sections 1.2.5–1.3 [2].

Topics requiring operator-domain proofs, representation-theoretic completion, or realizability and truncation bounds are taken up in Chapters 2–**??** by citation in later chapters—not by local re-derivation in Chapter 1 [3]. The compatibility conditions recorded below state when later material may cite Chapter 1 results Maxwell foundations or transport qualification [2].

$$\begin{aligned} \mathcal{X} \in \mathcal{D}_{\mathrm{defer}} \iff\ & \mathcal{X} \text{ requires operator-domain proof from Ch.2} \\ \vee\ & \mathcal{X} \text{ requires representation completion from Ch.3} \\ \vee\ & \mathcal{X} \text{ requires realizability/truncation proof from Ch.4,} \end{aligned} \tag{1.160}$$

and

$$\mathcal{X} \in \mathcal{D}_{\mathrm{defer}} \Longrightarrow \text{cite in later chapters; no local re-derivation in Ch. 1.} \tag{1.161}$$

[2].

$$\mathcal{D}_{\mathrm{cont}}(\mathcal{Y}) = 1 \iff (\mathcal{Y} \text{ is compatible with } \mathcal{R}_{\mathrm{ch1}} \ \wedge\ \mathcal{Y} \text{ adds no recursive proof of } \mathcal{R}_{\mathrm{ch1}})\,, \tag{1.162}$$

subject to

$$\mathcal{Y} \prec \mathcal{R}_{\mathrm{ch1}} \iff (\mathcal{D}_J = 1,\ \mathcal{D}_{\mathrm{org}} = 1,\ \mathcal{D}_{\mathrm{g}} = 1,\ \mathcal{D}_{\mathrm{op}} = 1) \text{ remain unviolated by } \mathcal{Y}, \tag{1.163}$$

with inherited $\mathcal{D}_J$, $\mathcal{D}_{\mathrm{org}}$, $\mathcal{D}_{\mathrm{g}}$, and $\mathcal{D}_{\mathrm{op}}$ criteria [2].

Chapter 1 establishes the Maxwellian order in which angular claims may be made: finite conditions first, admissible sources and boundaries second, qualified transport and measurement last [1]. Maxwell consistency precedes interpretation; not every field is admissible; observation never enlarges physics; angular structure belongs to fields before modes; transport is stricter than appearance; flux determines mechanical action; finite geometry limits angular possibility [1].

Chapter 2 applies outgoing radiation selection to the same admissible fields; Chapter **??** organizes surviving export in rotational coordinates; Chapter **??** confronts realizability limits on sources that produce it [3].

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Chapter 2

Maxwell Operators and Radiation Operators

Harmonic convention. Fields use $\mathrm{Re}\{\mathbf{F}(\mathbf{r})\,\mathrm{e}^{-\mathrm{j}\omega t}\}$ with $\mathrm{e}^{-\mathrm{j}\omega t} = \exp(-\mathrm{j}\omega t)$, so $\partial_t \to -\mathrm{j}\omega$ on phasors and Maxwell's equations are Eq. (1.5) with charge closure Eq. (1.8).

2.0 Introduction

At a fixed angular frequency, antenna design, compact-range measurement, and inverse scattering all ask two coupled questions: given an impressed current on a bounded support, what electromagnetic field does Maxwell theory determine, and given a field observed on a finite instrument, what can legitimately be inferred about the source that produced the radiative part of that field [1, 2]? The first is a forward boundary-value problem; the second is an inverse problem whose solvability is constrained by null spaces, finite aperture, and noise in ways that no post-processing can remove [3]. Chapter 2 organizes the harmonic theory needed to keep reactive storage, outgoing radiation, and measurement rank separate in both questions [4].

Chapter 1 already fixed the data of a Maxwell problem—domain, constitutive response, impressed sources, boundary traces, and charge continuity under the volume harmonic convention $\mathrm{e}^{-\mathrm{j}\omega t}$ (Eqs. (1.5) and (1.8)). The present chapter does not introduce new local laws. It develops the operator maps through which admissible sources are connected to complete field pairs, outgoing radiative export, and finite observables [2]. Historically, the exterior problem was made calculable by Green's retarded representations and Sommerfeld's outgoing condition [2]; Poynting's flux integral turned power through a closed surface into a primary measured quantity [5]; later work on finite support and electrical size clarified that bounded sources cannot export arbitrary angular complexity at fixed bandwidth [6, 1].

The central distinction is *radiation as accessibility*, not merely field existence. The harmonic *solution map* $\mathcal{S}_\omega$ sends an admissible impressed current $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ to the Maxwell field pair $(\mathbf{E}, \mathbf{H})$ consistent with the declared boundary class [2]. The pair generally contains reactive and stored structure as well as outward-propagating content. Outgoing selection on the exterior domain defines a *radiation map*

$$\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\mathrm{rad}}, \tag{2.1}$$

where Π_{rad} retains the Sommerfeld-outgoing, far-zone transverse component and $\mathcal{F}_{\mathrm{rad}}$ is the corresponding radiative image space [2]. The operator Π_{rad} is an outgoing selector on completed field pairs; it is not an orthogonal projection on all of $\mathcal{F}_\Omega$ unless additional Hilbert-space structure is

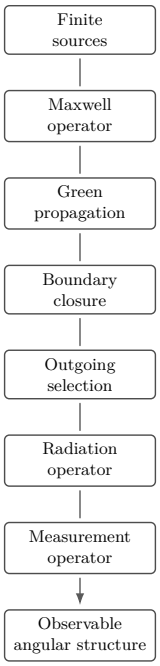


Figure 2.1: Admissible impressed currents drive Maxwell's equations; Green representation and boundary data select a physical branch; outgoing selection defines radiative export $\mathcal{R}_\omega$; finite instruments observe only a compressed image on $\mathcal{F}_{\mathrm{rad}}$ [4, 2].

imposed and proved (Subsections 2.0.1 and 2.9.2). Composition with a finite measurement map gives the laboratory-accessible rank studied in Sections 2.6 and 2.9.

Radiative accessibility. Far-zone observability is a composed map: radiation selection, geometric regime, and finite measurement rank act in sequence. Evanescent and reactive content may dominate near scans without enlarging the angular rank-N recovery of export on $\mathcal{F}_{\mathrm{rad}}$ [2].

A coaxial stub and a center-fed dipole in the same cell at 1 GHz may exhibit comparable near-field magnitude on a small scan surface, yet only the open structure sustains net outward power through a large enclosing sphere once outgoing qualification is enforced [2, 1]. Reactive diagnostics belong to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; wattmeter and pattern reports belong to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after the appropriate regime screen from Chapter 1. Three structural facts recur throughout the chapter: not every Maxwell-consistent field radiates; not every radiated component is observable with finite aperture and signal-to-noise ratio; and radiative equivalence of sources does not imply uniqueness of $\mathbf{J}_{\mathrm{imp}}$ [3, 2].

Subsections 2.0.1–2.0.3 name the maps, write Maxwell's equations in operator form, and record admissibility and export criteria inherited from Chapter 1. Section 2.1 supplies the function spaces and operator domains; later sections develop time–frequency equivalence, Green functions, uniqueness and inversion limits, flux integrals, spectral structure, null spaces, reciprocity, and angular spectra in the order required by the forward chain above.

2.0.1 Radiation as a source-to-field operator

A network analyzer $|S_{11}|$, a spherical near-field scan on Γ_R, and a far-zone pattern cut are three readings on one ordered map: impressed data, Maxwell closure, outgoing export, finite observation [4, 2]. Misplacing a measurement on that chain—reporting reactive storage as radiated power—remains the common compact-range failure mode [5, 7].

Section 2.0 posed radiation as a question of accessibility rather than of field existence alone. The

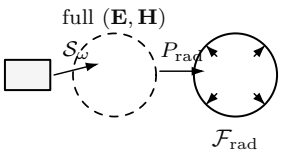


Figure 2.2: Excitation-to-observable chain: $\mathcal{S}_\omega$ enforces Maxwell closure with inherited boundary data; Π_{rad} isolates the radiative component on which far-zone flux integrals are meaningful.

laboratory scene below makes that distinction operational by naming the maps that connect impressed currents to Maxwell field pairs and, through outgoing selection, to far-zone observables [2]. The distinction between field existence and field $\mathcal{E}^{(N)}_{\mathrm{acc}}$ is central to radiation theory [2]. Maxwell's equations determine which fields may arise from a specified source distribution, whereas the radiation map $\mathcal{R}_\omega$ determines which portion of those fields survives as observable outgoing content [4]. From an antenna engineer's perspective, $\mathcal{S}_\omega$ answers what the structure *stores and circulates* locally, while $\mathcal{R}_\omega$ answers what the range may *meter and pattern* after the reactive shell is left behind [1].

A network analyzer $|S_{11}|$, a spherical near-field scan on Γ_R, and a far-zone pattern cut are three readings on one ordered map: impressed data, Maxwell field specification, outgoing export, finite observation [4]. Each instrument samples a different stage of that map, and confusion between stages—for example, reporting near-field reactive content as radiated power—remains the common compact-range laboratory error [5].

Laboratory measurements presuppose an excitation model even when it is left implicit: a port drive, a probe sample, or a termination fixes how energy is partitioned between storage and escape [2]. Impressed data from Section 1.1.2 map through Maxwell field specification to $(\mathbf{E},\mathbf{H})$, and only then is the outward-propagating fraction identified as the carrier of far-zone power and angular flux (Eq. (1.5); Sections 1.1.4–1.1.5). At fixed ω the maps are the harmonic solution operator $\mathcal{S}_\omega$ and radiation operator $\mathcal{R}_\omega$ under the $\exp(-\mathrm{j}\omega t)$ convention of Chapter 1.

Let $\Omega \subset \mathbb{R}^3$ be a piecewise smooth domain with boundary $\Gamma = \partial\Omega$. Phasor fields $\mathbf{E}(\mathbf{r})$ and $\mathbf{H}(\mathbf{r})$ are suppressed in time unless causality is restored below. Impressed electric current $\mathbf{J}_{\mathrm{imp}}$, charge density ρ, and the data tuple $\mathfrak{D}$ of Eq. (1.17) are not redefined; they are the inputs on which the maps act. Finite-source admissibility is assumed in the sense of the 1 $\mathcal{D}_J = 1$ conditions (Eq. (1.158)): only configurations with $\mathcal{D}_J = 1$ enter the domain of the radiation chain developed here [2].

Function spaces are declared before the maps. Let $\mathcal{F}_\Omega$ be the space of phasor pairs $(\mathbf{E},\mathbf{H})$ on Ω whose components have L^2-integrable curls and divergences in the sense required by Eq. (1.7), and whose boundary traces lie in the domain of $\mathcal{B}_\Gamma$. Let $\mathcal{J}_\Omega \subset L^2(\Omega)^3$ be the impressed-current space compatible with finite-energy class (Section 1.0.4), charge continuity (Eq. (1.8)), and the same boundary class. Refined Sobolev and distributional extensions of these spaces are supplied in Section 2.1.1; here the declaration is minimal but precise enough to state the maps [2]. The harmonic Maxwell *solution map* is the first quantitative object in the radiation chain: it records, for each admissible impressed current $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, the Maxwell field pair that the governing equations and boundary data jointly permit. Physically, this map is what a full-wave solver computes when it inverts Maxwell's equations on a declared domain; the output is a *complete* field state, not yet a radiative report [2]. Experimentally one would observe, on a closed measurement surface enclosing the source, both reactive circulation and outward-propagating flux in the solution unless outgoing selection is applied afterward [1]. The solution map sends impressed data to a field state:

$$\mathcal{S}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_\Omega, \qquad \mathbf{J}_{\mathrm{imp}} \longmapsto (\mathbf{E},\mathbf{H}), \tag{2.2}$$

defined by $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = (\mathbf{E}, \mathbf{H})$ satisfying the Maxwell operator residuals of Eq. (1.7) with sources $\mathfrak{D}$ and boundary data $\mathbf{g}_\Gamma$ [2]. Equation (2.2) is the Chapter 2 statement of the initial-boundary value problem proven in Section 1.1.1; it does not alter the law, only the input–output viewpoint. Injectivity fails in general: distinct $\mathbf{J}_{\mathrm{imp}}$ can produce identical $(\mathbf{E}, \mathbf{H})$ on observational subdomains, and equivalent source distributions related by divergence-free reparametrizations inside Ω can leave exterior patterns unchanged when $\mathcal{B}_\Gamma$ permits reactive storage (Section 1.3.3). Those equivalences are not nuisances to be removed by definition; they foreshadow null-space and realizability analyses in Sections 2.7 and **??**. Radiation accounting therefore never begins from a guessed modal field; it begins from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with explicit completion. The full state typically contains both stored and exported electromagnetic energy. Reactive fields exchange power with sources and terminations at distances small compared with k_0^{-1}; radiative fields carry net outward flux at infinity in open configurations. Let $\mathcal{F}_{\mathrm{rad}} \subset \mathcal{F}_\Omega$ denote pairs whose far-zone asymptotics satisfy outgoing selection (made analytic in Section 2.4). Radiation accounting therefore requires a second map that discards non-outgoing storage before pattern or transport claims are made. The outgoing selector $\Pi_{\mathrm{rad}} : \mathcal{F}_\Omega \to \mathcal{F}_{\mathrm{rad}}$ performs that discard: it accepts a completed Maxwell pair and returns only the component whose leading far-zone asymptotics satisfy Sommerfeld-outgoing qualification [2]. From a measurement standpoint, Π_{rad} is the mathematical counterpart of moving a probe beyond the reactive shell and reporting only the transverse outgoing field that persists with $1/r$ decay [1]. Composing source excitation with that selector defines the *radiation operator*

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\mathrm{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]. \tag{2.3}$$

Equation (2.3) composes source excitation with outgoing selection: $\mathcal{R}_\omega$ accepts the same impressed currents as $\mathcal{S}_\omega$ but returns only the radiative image $\Pi_{\mathrm{rad}}(\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\mathrm{rad}}$, where $\mathcal{F}_{\mathrm{rad}}$ is the subspace of field pairs whose leading far-zone behavior is outgoing and transverse. The composition law records that radiation is never computed independently of Maxwell completion:

$$\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \qquad \mathcal{D}_J = 1 \Longrightarrow \mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega, \quad \mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\mathrm{rad}}. \tag{2.4}$$

where $\mathcal{S}_\omega$ is the harmonic Maxwell solution map, Π_{rad} the outgoing selector, $\mathcal{R}_\omega$ the radiation map, $\mathcal{J}_\Omega$ the impressed-current space, $\mathcal{F}_\Omega$ the admissible field space, $\mathcal{F}_{\mathrm{rad}}$ the radiative subspace, and $\mathcal{D}_J = 1$ the transport qualification inherited from 1 [2]. Equation (2.4) composes $\mathcal{S}_\omega$ with outgoing selection Π_{rad} on $\mathcal{J}_\Omega$ [2]. On a ball B_a with $k_0 a = 0.5$, reactive-to-radiative field-energy ratios on S_R with $R = 2a$ typically exceed unity by an order of magnitude; applying pattern or transport metrics to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} therefore overstates export by roughly $10\times$ in magnitude bookkeeping before phase errors accumulate [8]. Superposition of radiative images is legitimate only after each constituent satisfies Eq. (2.9) and outgoing radiation condition [1]. Until Subsection 2.2.3 fixes $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$, treat Π_{rad} as an abstract outgoing selector [2]. Angular-flux transport from Sections 1.2.3–1.3.3 and structured-beam interpretations in Section 1.3.2 attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; coefficients read from reactive components of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ describe near-zone storage, not exported flux [9]. When Green operators are available (Section 2.3), $\mathcal{R}_\omega$ admits a retarded integral representation that makes the propagation step explicit. The integral representation below is the frequency-domain statement of what a method-of-moments code evaluates when it integrates the dyadic Green function against impressed currents: each source element at $\mathbf{r}'$ contributes to the observation point $\mathbf{r}$ with phase and amplitude fixed by outgoing propagation [2]. Physically, the integral superposes the radiative images of all current elements; only the portion of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ that Π_{rad} retains enters the integrand in the completed theory developed in Section 2.3. For $\mathbf{r} \notin \Omega_s$ with $\operatorname{supp} \mathbf{J}_{\mathrm{imp}} = \Omega_s \Subset \Omega$,

$$\mathbf{E}_{\mathrm{rad}}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \boldsymbol{\Gamma}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3 r', \tag{2.5}$$

In Eq. (2.5), $\mathbf{E}_{\mathrm{rad}}(\mathbf{r})$ is the radiative electric phasor at $\mathbf{r}$ ($\mathrm{V\,m^{-1}}$); $\mathbf{\Gamma}(\mathbf{r},\mathbf{r}')$ is the outgoing dyadic Green tensor ($\mathrm{m^{-1}}$) that propagates unit current dipoles from $\mathbf{r}'$ to $\mathbf{r}$; $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$ is the vacuum permeability; $\mathbf{r}'$ ranges over the source volume Ω_s; and the volume element d^3r' carries SI units $\mathrm{m^3}$. The factor $-\mathrm{j}\omega\mu_0$ arises from the harmonic Ampère law under the locked phasor convention. Here $\mathbf{\Gamma}$ is the outgoing dyadic Green function [2]. Equation (2.5) is a frequency-domain representation of the radiative field generated by a compact source distribution through the outgoing dyadic Green function [2]. Every application of this representation assumes the $\mathcal{D}_J = 1$ and boundary conditions stated above. Unless noted otherwise, $\omega > 0$ is fixed, materials are linear isotropic passive regions, and Ω is either bounded with declared $\mathcal{B}_\Gamma$ or an exterior domain with compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ and outgoing radiation condition at infinity. Maps are defined only on $\mathcal{J}_\Omega$ satisfying Eq. (1.8); generalized sources appear in Section 2.3.3. Maxwell linearity on Ω induces linearity of both maps on fixed boundary data: for $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ and scalars a_1, a_2,

$$\mathcal{R}_\omega\Big[a_1\mathbf{J}^{(1)}_{\mathrm{imp}} + a_2\mathbf{J}^{(2)}_{\mathrm{imp}}\Big] = a_1\mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}] + a_2\mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}], \tag{2.6}$$

In Eq. (2.6), $a_1, a_2 \in \mathbb{C}$ are complex amplitudes and $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ are admissible impressed currents; $\mathcal{R}_\omega$ is the radiation map; and linearity holds when boundary data $\mathbf{g}_\Gamma$ and the material layout are held fixed [4]. Composite-source analysis decomposes $\mathbf{J}_{\mathrm{imp}}$ into constituents, maps each through $\mathcal{R}_\omega$, and superposes radiative outputs—a discipline that prevents reactive interference from masquerading as transport (Section 1.3.4). Write $\operatorname{ran}\mathcal{S}_\omega \subset \mathcal{F}_\Omega$ for all field pairs reachable from finite sources and $\operatorname{ran}\mathcal{R}_\omega \subset \mathcal{F}_{\mathrm{rad}}$ for far-zone radiative outputs. Strict inclusion $\operatorname{ran}\mathcal{R}_\omega \subsetneq \operatorname{ran}\mathcal{S}_\omega$ occurs whenever reactive storage is nontrivial: distinct near-zone states can share identical radiative images (Section 1.3.3). Pattern measurements probe $\operatorname{ran}\mathcal{R}_\omega$ and cannot invert $\mathcal{S}_\omega$ without additional near-zone data—a limitation revisited in Section 2.4.4. Radiative power is metered with inherited Poynting machinery. The practical question is how much time-averaged power crosses a closed surface that encloses the source and lies beyond the reactive shell; that flux is the scalar quantity laboratories report as radiated power. On a sphere $S_R = \{|\mathbf{r}| = R\}$ enclosing $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$, with outward unit vector $\hat{\mathbf{r}}$ and complex Poynting vector $\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^*$ from Section 1.1.4,

$$P_{\mathrm{rad}}(R) = \operatorname{Re} \oint_{S_R} \mathbf{S} \cdot \hat{\mathbf{r}}\, dS. \tag{2.7}$$

In Eq. (2.7), $P_{\mathrm{rad}}(R)$ is the time-averaged outward power (W) through the sphere of radius R (m); $\mathbf{S}$ is the complex Poynting vector ($\mathrm{W\,m^{-2}}$) built from the radiative field pair; $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$ is the outward unit normal on S_R; and dS is the area element ($\mathrm{m^2}$). The real part $\operatorname{Re}\{\cdot\}$ extracts the time-averaged power flux carried by the phasor fields under the $\exp(-\mathrm{j}\omega t)$ convention. Equation (2.7) inherits the Poynting law of Section 1.1.4. For $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, $P_{\mathrm{rad}}(R)$ becomes R-independent beyond the reactive shell [5]. Angular-momentum budgets from Section 1.1.5 attach to the same radiative component. The solution map energizes reactive and radiative regions alike; the radiation map discards non-outgoing storage before pattern or transport claims are made. Misidentifying reactive circulation as exported angular structure is a recurring error in high-gradient near-zone beams [9]. Finite measurement channels (Section 1.2.5) implement linear functionals on field space. Composing a declared far-zone observable with $\mathcal{R}_\omega$ yields a linear functional of $\mathbf{J}_{\mathrm{imp}}$; composing the same observable with $\mathcal{S}_\omega$ without selection generally includes reactive contamination. No new measurement axioms are introduced—only the correct map for the declared observable class [2]. Non-radiating currents illustrate kernel visibility. If $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ has vanishing far-zone transverse fields while maintaining near-zone storage (Section 1.3.3), then

$$\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}] = \mathbf{0} \quad \text{in } \mathcal{F}_{\mathrm{rad}}, \tag{2.8}$$

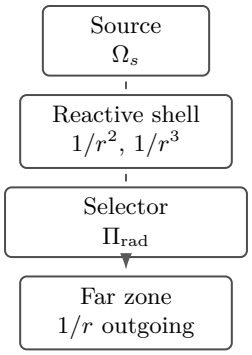


Figure 2.3: Spatial nesting seen by the maps: $\mathcal{S}_\omega$ populates all regions; Π_{rad} passes only the outgoing transverse content compatible with Section 1.3.3.

In Eq. (2.8), $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ is an admissible impressed current ($\mathrm{A\,m^{-2}}$) that satisfies $\mathcal{D}_J = 1$ yet carries no outgoing transverse export; $\mathcal{F}_{\mathrm{rad}}$ is the far-zone radiative field space selected by Π_{rad}; and $\ker \mathcal{R}_\omega$ denotes the set of currents for which the radiation map vanishes on that space, while $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0}$ in general. Equation (2.8) targets the null space analyzed in Section 2.7.1; ignoring $\ker \mathcal{R}_\omega$ invites spurious modal assignments to invisible sources. Membership in $\mathcal{J}_\Omega$ requires more than square integrability: the Maxwell-closed screen below states, in compact form, which impressed currents may enter the radiation chain without contradicting charge continuity or boundary compatibility. $\mathcal{D}_J$ is not a mathematical regularity condition alone: it encodes the laboratory requirement that impressed currents be charge-consistent and boundary-compatible before Maxwell field specification is attempted [4]. Violating the screen produces formal solutions that no passive feed can realize [1]:

$$\mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1 \iff \Big(\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega)^3,\ \nabla\cdot\mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0,\ \mathbf{J}_{\mathrm{imp}} \text{ satisfies BC compatibility on } \Gamma\Big), \tag{2.9}$$

In Eq. (2.9), $\mathcal{D}_J$ is the source Maxwell-closed screen (dimensionless) inherited from Eq. (1.8); $\mathbf{J}_{\mathrm{imp}}$ is the impressed electric current density ($\mathrm{A\,m^{-2}}$) on compact $\Omega \subset \mathbb{R}^3$; ρ is the impressed charge density ($\mathrm{C\,m^{-3}}$) linked by $\nabla\cdot\mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0$; and $\Gamma = \partial\Omega$ is the boundary on which tangential traces must match the declared boundary class. with $\mathcal{D}_J$ as in Eq. (1.158). Violating $\nabla\cdot\mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0$ contradicts Eq. (1.8) before Maxwell field specification is attempted. Boundary compatibility ensures induced surface equivalents do not contradict Eq. (1.18) [2]. Causality constrains harmonic representations. A frequency-domain phasor is physically meaningful only when it can be traced to a retarded time-domain evolution that respects finite propagation speed. With time-domain current $j(\mathbf{r},t)$ and retarded dyadic impulse $\mathcal{G}(\mathbf{r},\mathbf{r}',t-t')$,

$$\mathbf{E}(\mathbf{r},t) = -\mu_0 \frac{\partial}{\partial t} \int \mathcal{G}(\mathbf{r},\mathbf{r}',t-t')\cdot j(\mathbf{r}',t')\, d^3r'\, dt', \tag{2.10}$$

integrated only over $t' \le t$ [2]. In Eq. (2.10), $\mathbf{E}(\mathbf{r},t)$ is the real time-domain electric field ($\mathrm{V\,m^{-1}}$); $j(\mathbf{r}',t')$ is the impressed current density ($\mathrm{A\,m^{-2}}$) at source point $\mathbf{r}'$ and retarded time t'; $\mathcal{G}(\mathbf{r},\mathbf{r}',t-t')$ is the causal dyadic Green impulse response ($\mathrm{m^{-1}\,s^{-1}}$) that vanishes for $t-t' < 0$; and the temporal integral enforces that the field at $(\mathbf{r},t)$ depends only on sources in the past light cone. The steady-state operator $\mathcal{R}_\omega$ may therefore be viewed as the frequency-domain specialization of the retarded causal evolution [2]. The equivalence between the time-domain and harmonic descriptions follows under the standard assumptions of linearity, stationarity, and finite-energy excitation [4]. Vacuum scales from Eq. (1.12) fix the carrier on which selection acts: $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$, $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$. Outgoing components retained by Π_{rad} carry $\exp(-\mathrm{j}k_0 r)$ phase fronts at leading order; components whose phase does not lock to outgoing $k_0 r$ remain in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ but outside $\mathcal{F}_{\mathrm{rad}}$. Power metered by Eq. (2.7)

stabilizes only on the latter [2]. Distance labels become meaningful through operator action. For compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$, $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ contains $1/r^2$ and $1/r^3$ reactive terms; $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ retains leading $1/r$ transverse fields with kr phase in the far zone (Section 1.3.3). Asymptotic partitions are sharpened in Section 2.5.2. Observability enters as functional composition rather than as a new postulate. Let $\mathcal{O}_{\mathrm{far}}$ denote a linear functional that extracts a declared far-zone scalar or vector observable—pattern voltage, axial ratio proxy, or a single spherical-harmonic coefficient—from a field pair in $\mathcal{F}_{\mathrm{rad}}$. The measured quantity associated with impressed current $\mathbf{J}_{\mathrm{imp}}$ is the scalar $\mathcal{O}_{\mathrm{far}}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]]$, not $\mathcal{O}_{\mathrm{far}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ unless the observable has been shown to vanish on reactive content. Because both $\mathcal{R}_\omega$ and $\mathcal{O}_{\mathrm{far}}$ are linear on their domains, the laboratory reading is a linear functional $\Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega$ on $\mathcal{J}_\Omega$. That composition expresses in map language the measurement limitations in Section 1.2.5: finite apertures and finite modal sets appear as rank-deficient $\mathcal{O}_{\mathrm{far}}$, not as modifications of Maxwell law [2]. Angular-transport claims require further that Λ_{rad} be compatible with the flux budgets of Section 1.1.5 when the observable is interpreted as momentum or power export. The outgoing selector Π_{rad} is idempotent on $\mathcal{F}_{\mathrm{rad}}$: applying selection twice leaves a radiative state unchanged. On the larger space $\mathcal{F}_\Omega$, Π_{rad} is a projection only after the Sommerfeld-class uniqueness of Section 2.4.3 is imposed; without that completion, multiple outgoing decompositions can coexist for the same truncated near-zone data. The idempotence condition is therefore a *joint* property of field space and boundary class, not a algebraic trick independent of physics [1]. Numerical methods that window fields in space and call the result "radiative" without outgoing radiation condition are approximating Π_{rad} by a non-idempotent mask; such outputs must not be fed into transport criteria without an explicit error budget. The composition law used downstream is

$$\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \qquad \mathcal{D}_J = 1 \Longrightarrow \mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega, \quad \mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\mathrm{rad}}. \tag{2.11}$$

In Eq. (2.11), $\mathcal{S}_\omega$ is the Maxwell solution map from admissible sources to field pairs, Π_{rad} is the outgoing selector that retains far-zone transverse export, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ is the radiation map, $\mathcal{D}_J = 1$ denotes source Maxwell-closed from Eq. (2.9), and $\mathcal{J}_\Omega$, $\mathcal{F}_\Omega$, $\mathcal{F}_{\mathrm{rad}}$ are the source and field spaces declared in Subsection 2.0.1. Green kernels realize $\mathcal{S}_\omega$; Sommerfeld conditions define Π_{rad}; spectral theory in Section 2.6 studies eigenstructure on spaces refined in Section 2.1 [1]. Nothing in Eqs. (2.2)–(2.11) re-proves results listed in Section 1.4.1; continuity Eq. (1.162) is honored by citation-only continuity of curl–curl algebra, stress flux theorems, and gauge lemmas. Subsection 2.0.1 has named $\mathcal{S}_\omega$, Π_{rad}, and $\mathcal{R}_\omega$, introduced their Green and Poynting realizations, and connected finite measurement rank to composition with far-zone observables. Subsection 2.0.2 identifies the Maxwell operator whose inversion, under declared boundary completion, produces $\mathcal{S}_\omega$.

2.0.2 Maxwell equations in operator form

A standard-gain horn on a compact-range turntable fixes the laboratory scene for 'Maxwell equations in operator form": frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays below [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Subsection 2.0.1 named the solution map $\mathcal{S}_\omega$ and the outgoing selector Π_{rad}; the display below identifies the Maxwell operator whose inversion, under declared boundary completion, produces that map [2]. The governing question is which field pair $(\mathbf{E}, \mathbf{H})$ is consistent with impressed currents $\mathbf{J}_{\mathrm{imp}}$ and with the termination class $\mathcal{B}_\Gamma$ on $\Gamma = \partial\Omega$—the same question every full-wave solver answers before any pattern is extracted [1]. Harmonic Maxwell theory on Ω is packaged as a first-order residual system on a six-component state, augmented by boundary traces through the operator Eq. (1.18) imported from Section 1.1.2. Local curl and divergence balances remain those of Eq. (1.7); the Maxwell export sequence developed here does not alter those local balances [2]. First-order Maxwell theory

couples electric and magnetic fields through circulation; packaging both components in a single state vector is the standard device that permits resolvent notation and variational formulations without abandoning the physical coupling [2]. $\mathbf{u}_\omega$ is not an arbitrary six-vector: each admissible state satisfies Gauss constraints and carries tangential traces compatible with the boundary class declared with the impressed data tuple $\mathfrak{D}$. From an antenna engineer's perspective, $\mathbf{u}_\omega$ is what a full-wave solver stores at each mesh node—the coupled $(\mathbf{E}, \mathbf{H})$ phasor pair at fixed ω—before any post-processing extracts patterns or port quantities [1]. Collect phasor fields into the Maxwell state

$$\mathbf{u}_\omega = \begin{bmatrix} \mathbf{E} \\ \mathbf{H} \end{bmatrix} \in \mathcal{F}_\Omega, \tag{2.12}$$

where $\mathbf{u}_\omega$ denotes the six-component Maxwell state at fixed ω, with $\mathbf{E}$ and $\mathbf{H}$ carrying the SI units noted above and lying in the admissible field space $\mathcal{F}_\Omega$ declared in Subsection 2.0.1. The pair $(\mathbf{E}, \mathbf{H})$ is constrained jointly: divergence rows of Eq. (1.7) must hold, and tangential traces on $\Gamma = \partial\Omega$ must lie in the domain of the boundary class declared with $\mathfrak{D}$. Treating $\mathbf{u}_\omega$ as a single unknown is what permits resolvent notation, variational forms, and the spectral compression developed in Section 2.6 [1]. Every boundary-value problem begins by distinguishing prescribed quantities from unknown quantities. In Maxwell theory the impressed currents, charges, and boundary data constitute the prescribed excitation, whereas the electromagnetic field remains the unknown state to be determined [4]. The operator formulation merely records this distinction in a compact mathematical form suitable for analysis [1]. Impressed data enter through a forcing vector aligned with the Maxwell residual rows. The forcing packages charge and current drive on the rows where impressed sources appear, leaving divergence constraints to be satisfied by the field unknown. For the tuple Eq. (1.17),

$$\mathbf{s}_\omega(\mathfrak{D}) = \begin{bmatrix} \mathbf{0} \\ -\mathbf{J}_{\mathrm{imp}} \\ -\rho \\ \mathbf{0} \end{bmatrix}, \tag{2.13}$$

In Eq. (2.13), $\mathbf{s}_\omega(\mathfrak{D})$ is the Maxwell forcing six-vector (SI units per row) assembled from the data tuple $\mathfrak{D}$; $\mathbf{J}_{\mathrm{imp}}$ enters the Ampère row ($\mathrm{A\,m^{-2}}$) and ρ the Gauss row ($\mathrm{C\,m^{-3}}$); the remaining rows are zero for the impressed-current class treated here; row order matches Eq. (1.7). Equation (2.13) records drive on the Ampère and Gauss rows only for the impressed-current class treated here; sheet and magnetic excitations extend the vector through Eq. (1.19) when trace spaces are enriched in Section 2.1.1 [2]. The forcing is *data*, not a field variable: it specifies how $\mathcal{S}_\omega$ is sourced while $\mathbf{u}_\omega$ remains the unknown. Operator language separates *states* $(\mathbf{E}, \mathbf{H})$ from *data* $(\mathbf{J}_{\mathrm{imp}}, \rho, \mathfrak{D})$: the feed specifies what is driven, the solver returns what Maxwell allows, and radiation operators report what is exported [4]. Treating a tangential-field template as independent forcing without a matching current repeats the realizability failure already flagged in Chapter 1 [2]. Let $\mathcal{M}_\omega$ denote the harmonic Maxwell operator of Eq. (1.7). The volume subsystem reads

$$\mathcal{M}_\omega[\mathbf{u}_\omega] = \mathbf{s}_\omega(\mathfrak{D}), \tag{2.14}$$

and is satisfied when each residual row vanishes. Linearity in $\mathbf{u}_\omega$ follows from linearity of Eq. (1.5); spatially varying $\varepsilon(\mathbf{r})$ and $\mu(\mathbf{r})$ modify coefficients inside $\mathcal{M}_\omega$ without changing the first-order block shape (Section 2.2.2) [2]. In uniform vacuum regions, $\mathcal{M}_\omega$ has translation-invariant symbol, the property exploited by dyadic Green kernels in Section 2.3. (2.14) and (1.18). A solver that enforces only volume rows produces a formal solution compatible with Maxwell *sources* but not necessarily with termination physics; conversely, boundary data without volume balance violate conservation.

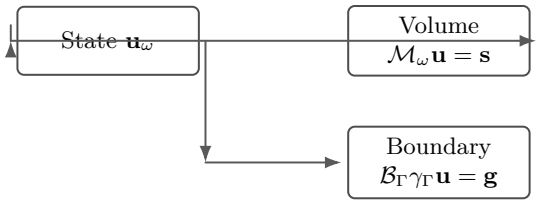


Figure 2.4: Parallel volume and boundary channels for the closed harmonic problem: $\mathbf{u}_\omega$ simultaneously satisfies Maxwell residuals and prescribed traces.

Subsection 2.0.1 applies $\mathcal{R}_\omega$ only after both channels are closed [1]. Tangential traces are extracted by the restriction

$$\gamma_\Gamma : \mathcal{F}_\Omega \to \mathcal{T}_\Gamma, \qquad \gamma_\Gamma[\mathbf{u}_\omega] = \begin{bmatrix} \hat{\mathbf{n}} \times \mathbf{E}|_\Gamma \\ \hat{\mathbf{n}} \times \mathbf{H}|_\Gamma \end{bmatrix}, \tag{2.15}$$

with outward unit normal $\hat{\mathbf{n}}$ on Γ and trace space $\mathcal{T}_\Gamma$ induced by interior states. The map γ_Γ is not defined on arbitrary L^2 six-tuples; Sobolev and distribution refinements in Section 2.1.1 specify its domain [2]. The closed operator stacks volume and boundary residuals:

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathcal{M}_\omega[\mathbf{u}_\omega] \\ \mathcal{B}_\Gamma[\gamma_\Gamma[\mathbf{u}_\omega]] \end{bmatrix}. \tag{2.16}$$

A physically closed harmonic problem requires both volume Maxwell balance and boundary termination to hold simultaneously; neither channel alone suffices [2]. The composite operator $\mathcal{L}_\omega$ encodes that joint requirement: its first block tests whether $(\mathbf{E}, \mathbf{H})$ satisfies the curl–curl structure with impressed sources removed to the forcing vector, while its second block tests whether tangential traces on Γ match the declared termination (PEC, impedance, or outgoing) [1]. Experimentally, a solver that minimizes volume residuals while ignoring port or chamber walls produces a formal Maxwell solution that may disagree with what a network analyzer reports on the same structure [2]. The harmonic boundary-value problem is to find $\mathbf{u}_\omega \in \mathcal{F}_\Omega$ such that

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathbf{s}_\omega(\mathfrak{D}) \\ \mathbf{g}_\Gamma \end{bmatrix}. \tag{2.17}$$

Perfect conductors impose $\hat{\mathbf{n}} \times \mathbf{E} = \mathbf{0}$; radiation and Leontovich completions encode outgoing or dissipative relations on $\gamma_\Gamma[\mathbf{u}_\omega]$. The operator $\mathcal{L}_\omega$ is the Maxwell balance operator for this coupled system [1]. Hence $\mathcal{L}_\omega$ enforces Maxwell balance with impressed drive and boundary termination; without outgoing selection it does not distinguish export from storage, and every near-field probe reading $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ presupposes its inversion on the declared domain [2]. Equation (2.2) is the resolvent relation

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{u}_\omega \quad \text{with} \quad \mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}}, \rho) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.18}$$

where ρ is tied to $\mathbf{J}_{\mathrm{imp}}$ through Eq. (1.8) when only electric impressed currents drive the system. Invertibility of $\mathcal{L}_\omega$ is not universal: existence and uniqueness depend on $\mathcal{B}_\Gamma$ and on whether outgoing selection is imposed (Section 2.4.3 for exterior domains; Section 2.6.2 for cavities) [2]. $\mathcal{S}_\omega$ is therefore a *physical branch* of $\mathcal{L}_\omega^{-1}$, not an abstract algebraic inverse on all of $\mathcal{F}_\Omega$. Single-field Helmholtz reductions equivalent to Eqs. (1.13)–(1.15) are used later when a scalar symbol simplifies kernels; the elimination belongs to 1 and is not repeated. The electric-wave symbol used in Chapter 2 is recorded as

$$\mathcal{W}_{E,\omega}\mathbf{E} \equiv \nabla \times \left(\mu_0^{-1} \nabla \times \mathbf{E}\right) - \omega^2 \varepsilon_0 \mathbf{E} = -\mathrm{j}\omega\mu_0 \mathbf{J}_{\mathrm{imp}} + \mu_0^{-1} \nabla \times \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}, \tag{2.19}$$

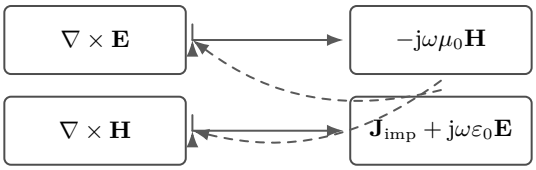


Figure 2.5: Circulation coupling in $\mathcal{M}_\omega$: each curl row feeds the partner field; $\mathbf{J}_{\mathrm{imp}}$ appears as external drive on the magnetic row.

with $\mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}$ collecting equivalent magnetic drive when present [2]. Subsection 2.2.2 supplies the magnetic counterpart and material-weighted generalizations. Variational structure is inherited from Section 1.1.4. For test state $\mathbf{v}_\omega = (\mathbf{E}', \mathbf{H}')$,

$$\mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}_\omega) = \int_\Omega \Big[\mu_0\, \mathbf{H}\cdot\nabla \times \mathbf{E}' - \varepsilon_0\, \mathbf{E}\cdot\nabla \times \mathbf{H}'\Big] d^3r. \tag{2.20}$$

Integration by parts connects Eq. (2.20) to surface fluxes and to the weak formulation

$$\mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}_\omega) = \mathfrak{f}_\omega(\mathbf{v}_\omega) \quad \forall\, \mathbf{v}_\omega \in \mathcal{F}_\Omega^0, \tag{2.21}$$

where $\mathcal{F}_\Omega^0$ carries homogeneous boundary traces compatible with $\mathcal{B}_\Gamma$ and $\mathfrak{f}_\omega$ is induced by $\mathbf{s}_\omega(\mathfrak{D})$ [5]. Equation (2.21) is the variational face of Eq. (2.17); finite-element discretizations favor this route when material coefficients are piecewise constant [1]. Well-posedness of Eq. (2.21) is not automatic on unbounded domains or on multiply connected cavities. On a bounded region with passive $\mathcal{B}_\Gamma$ and homogeneous traces in $\mathcal{F}_\Omega^0$, coercivity fails only along divergence-free cavity modes that satisfy both curl-curl balance and homogeneous tangential data—precisely the null space that Section 2.1.4 isolates before imposing gauge or polarization fixes [1]. For exterior problems, coercivity is recovered only after restricting to the outgoing subspace selected by Sommerfeld-type completion; without that restriction, the bilinear form is indefinite and the weak problem admits non-physical growing solutions. The practical rule for antenna analysis is therefore threefold: declare $\mathcal{D}(\mathcal{L}_\omega)$, specify $\mathcal{B}_\Gamma$ on every connected piece of Γ, and confirm that the impressed tuple lies in the compatible range of Eq. (2.24). Numerical codes that enforce volume Maxwell rows on truncated domains but leave exterior faces open implement an *implicit* $\mathcal{B}_\Gamma$; unless that implicit completion matches the laboratory termination, $\mathcal{S}_\omega$ in the code is a different operator from the one measured on the bench [1]. Domain specification cannot remain at C^∞ when sources are distributional. The minimal energy class declared here is

$$\mathcal{D}(\mathcal{L}_\omega) \subset \big\{\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) : \mathbf{E}, \mathbf{H} \in L^2(\Omega)^3,\ \nabla \times \mathbf{E}, \nabla \times \mathbf{H} \in L^2(\Omega)^3\big\}, \tag{2.22}$$

where $\mathcal{L}_\omega$ is the composite Maxwell operator on the six-component state $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$, $\mathcal{D}(\mathcal{L}_\omega)$ its domain of square-integrable curl fields compatible with $\mathcal{B}_\Gamma$, and Ω the analysis region [2]. Section 2.1.1 refines embeddings and compactness; evaluating Eq. (2.14) outside $\mathcal{D}(\mathcal{L}_\omega)$ generates spurious boundary terms that corrupt flux identities [2]. Piecewise smooth $\varepsilon(\mathbf{r})$ and $\mu(\mathbf{r})$ modify coefficients inside $\mathcal{M}_\omega$ while preserving the block template of Eq. (1.7). Interface jumps satisfy Eq. (1.19); multi-region Green operators in Section 2.3.5 realize $\mathcal{L}_\omega^{-1}$ on composite domains [2]. The coupling structure described above extends naturally to practical terminal conditions. Finite feed structures and external circuits frequently impose impedance relations between tangential field components on the boundary [1]:

$$\mathcal{Z}_\Gamma \begin{bmatrix} \hat{\mathbf{n}} \times \mathbf{E}|_\Gamma \\ \hat{\mathbf{n}} \times \mathbf{H}|_\Gamma \end{bmatrix} = \mathbf{0}, \tag{2.23}$$

In Eq. (2.23), $\mathcal{Z}_\Gamma$ is a linear boundary impedance operator (SI units absorbed in its definition) acting on the tangential Maxwell traces $\hat{\mathbf{n}} \times \mathbf{E}|_\Gamma$ and $\hat{\mathbf{n}} \times \mathbf{H}|_\Gamma$ on $\Gamma = \partial\Omega$; admissible states satisfy the terminal or absorbing relation $\mathcal{Z}_\Gamma[\text{traces}] = \mathbf{0}$. encoding Leontovich, absorbing, or mode-matched completions [1]. When Eq. (2.23) replaces explicit $\mathbf{g}_\Gamma$ in Eq. (2.17), admissible states satisfy the impedance manifold on Γ. Exterior outgoing problems correspond to a limit in which $\mathcal{Z}_\Gamma$ selects the Sommerfeld branch of Section 2.4.3; conflating a finite-chamber termination with that limit without normalization corrupts far-zone patterns [2]. Forcing must lie in the range implied by charge conservation. Divergence of the Ampère row together with Eq. (1.8) yields

$$\nabla \cdot \mathbf{s}_{\omega,2} + \mathrm{j}\omega\, \mathbf{s}_{\omega,3} = 0 \tag{2.24}$$

on the second and third blocks of Eq. (2.13). Inconsistent $(\mathbf{J}_{\text{imp}}, \rho)$ pairs lie outside physically realizable drive; numerical solvers may still return vectors that violate Eq. (2.24) [2]. A compact current filament with vanishing ρ automatically satisfies Eq. (2.24) because $\mathbf{s}_{\omega,3} = 0$. When a volumetric ρ is prescribed independently of $\mathbf{J}_{\text{imp}}$, the pair must be adjusted so that their divergence balance matches the Ampère–Gauss rows; otherwise the closed problem is mathematically posed but physically inconsistent. Feed models that inject $\mathbf{J}_{\text{imp}}$ without updating ρ are common in method-of-moments implementations; they remain valid only when the discretization enforces continuity on the impressed grid or when the problem is magnetostatic in the relevant band [1]. Linking to Subsection 2.0.1, radiation acts on closed solutions:

$$\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega = \Pi_{\text{rad}} \circ \mathcal{L}_\omega^{-1} \circ \begin{bmatrix} \mathbf{s}_\omega(\cdot) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.25}$$

with $\mathcal{L}_\omega^{-1}$ understood on the outgoing-admissible branch fixed by $\mathcal{B}_\Gamma$ and with Π_{rad} from Eq. (2.3) [2]. Completion must precede radiative selection: applying Π_{rad} to non-closed states mixes reactive content into far-zone observables [1]. The energy form supports an integration-by-parts identity

$$\mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}_\omega) = \langle \mathcal{M}_\omega[\mathbf{u}_\omega], \mathbf{v}_\omega \rangle_\Omega + \int_\Gamma \big(\hat{\mathbf{n}} \times \mathbf{E} \cdot \mathbf{H}' - \hat{\mathbf{n}} \times \mathbf{E}' \cdot \mathbf{H}\big) dS, \tag{2.26}$$

where $\langle \cdot, \cdot \rangle_\Omega$ is the componentwise L^2 pairing [2]. Self-adjoint truncations on bounded domains require that $\mathcal{B}_\Gamma$ cancel the boundary integrand on admissible traces; criteria are supplied in Section 2.1.4. Equation (2.26) is the local spine for those arguments. Fredholm structure note: for bounded Ω with passive $\mathcal{B}_\Gamma$, $\mathcal{L}_\omega$ is a Fredholm operator on $\mathcal{D}(\mathcal{L}_\omega)$ modulo cavity modes, with index tied to topology and boundary class. Exterior domains replace discrete spectra with continuous radiation modes, and $\mathcal{L}_\omega^{-1}$ becomes a resolvent on the outgoing subspace rather than an ordinary inverse. Section 2.6.4 develops the spectral/Fredholm foundation; here the point is that invertibility questions for $\mathcal{S}_\omega$ are questions about $\mathcal{L}_\omega$ on the *declared* domain, not about formal elimination of $\mathbf{H}$ alone [1]. Linearity of $\mathcal{L}_\omega$ in $\mathbf{u}_\omega$ implies that superposition holds for fixed boundary class: if $\mathcal{L}_\omega[\mathbf{u}_\omega^{(a)}] = [\mathbf{s}^{(a)}, \mathbf{g}^{(a)}]$ and $\mathcal{L}_\omega[\mathbf{u}_\omega^{(b)}] = [\mathbf{s}^{(b)}, \mathbf{g}^{(b)}]$ with identical $\mathcal{B}_\Gamma$, then $\mathcal{L}_\omega[\mathbf{u}_\omega^{(a)} + \mathbf{u}_\omega^{(b)}] = [\mathbf{s}^{(a)} + \mathbf{s}^{(b)}, \mathbf{g}^{(a)} + \mathbf{g}^{(b)}]$. Changing $\mathcal{B}_\Gamma$ while holding $\mathbf{J}_{\text{imp}}$ fixed breaks this principle because boundary rows are part of the operator, not of the drive. Laboratory comparisons that swap terminations without re-solving the coupled system therefore compare different $\mathcal{L}_\omega$, not different "modes" of the same map [2]. The radiation operator $\mathcal{R}_\omega$ inherits the same sensitivity: composition Eq. (2.25) is linear in impressed data only after $\mathcal{B}_\Gamma$ and the outgoing branch are held fixed [2]. Subsection 2.0.2 has packaged harmonic Maxwell theory as the composite operator $\mathcal{L}_\omega$, recorded its resolvent relation to $\mathcal{S}_\omega$, and linked radiation composition Eq. (2.25) to operator inversion on the outgoing-admissible branch. Subsection 2.0.3 records how those maps inherit the admissibility, regime, and measurement criteria established in 1.

2.0.3 Admissibility and Maxwell export sequence

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell export is computed [2]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Subsections 2.0.1–2.0.2 introduced the radiation maps and Maxwell export sequence; the display below invokes 1 $\mathcal{D}_J = 1$ [2]. The present chapter builds on established in 1. Rather, it applies those foundations to radiation problems [4]. The Maxwell-closed criteria, conservation laws, and measurement principles developed previously remain in force; the present chapter introduces the operator calculus through which admissible sources are connected to observable radiative fields [1]. The operators $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and $\mathcal{L}_\omega$ are applied on admissible inputs without reopening the 1 proofs [2]. Impressed data enter the solution domain only when the 1 source conditions hold; radiative reports attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not to the full solution pair by default [2]. The continuity relations below restate, in operator language, the 1 regime and measurement criteria that qualify when angular-flux and transport reports may attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than to the unreconstructed solution pair [2]. Physically, this means that near-field reactive content diagnosed on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ must not be reported as exported radiative flux unless the regime screen $\mathcal{D}_{J,\mathrm{reg}}$ from 1 is satisfied on the measurement shell [1].

$$\mathcal{D}_{\mathrm{ch1}} = 1 \implies \mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega \ \wedge\ \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_\Omega, \tag{2.27}$$

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{angular-flux reports attach to } \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \text{ not to } \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ by default}, \tag{2.28}$$

$$\Pi(\theta,\phi;r) = \Pi_{\mathrm{a}}(\theta,\phi;r) + \Pi_{\mathrm{re}}(\theta,\phi;r), \qquad \Pi_{\mathrm{a}} = \mathrm{Re}\{\mathbf{S}\cdot\hat{\mathbf{n}}\}, \tag{2.29}$$

with reactive sector Π_{re} the remainder on the declared control surface.

$$Q_{\mathrm{re}}(r) = \mathrm{Im}\oint_{S_r} \mathbf{S}\cdot\hat{\mathbf{r}}\,dS, \qquad P_{\mathrm{rad}}(r) = \mathrm{Re}\oint_{S_r} \mathbf{S}\cdot\hat{\mathbf{r}}\,dS, \tag{2.30}$$

$$\chi(r) = \frac{|Q_{\mathrm{re}}(r)|}{P_{\mathrm{rad}}(r) + \epsilon_P}, \qquad \epsilon_P > 0. \tag{2.31}$$

$$P_{\mathrm{rad}}(r) > 0 \ \wedge\ \chi(r) \le \chi_{\mathrm{th}}, \tag{2.32}$$

$$r_c :\ \chi(r_c) = \chi_{\mathrm{th}}, \tag{2.33}$$

$$\mathcal{D}_{J,\mathrm{reg}}(r) = 1 \iff \Big(\chi(r) \le \chi_{\mathrm{th}} \ \wedge\ \mathcal{D}_J(r) = 1\Big), \tag{2.34}$$

$$\oint_{S_r} \mathrm{Re}\{\mathbf{S}_R\cdot\hat{\mathbf{r}}\}\,dS = \oint_{S_r} \mathrm{Re}\{\mathbf{S}\cdot\hat{\mathbf{r}}\}\,dS, \tag{2.35}$$

$$\mathcal{I}[\mathcal{P}_R\mathcal{F}] = \mathcal{I}[\mathcal{F}], \tag{2.36}$$

for scalar observables $\mathcal{I}$ under passive frame rotation $\mathcal{P}_R$.

$$\mathcal{R}_{\mathcal{F}} = \left\| \mathcal{R}_\omega[\mathcal{T}_{R,\mathbf{a}}\mathbf{J}_{\mathrm{imp}}] - \mathcal{T}_{R,\mathbf{a}}\,\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \right\|, \tag{2.37}$$

$$\mathbf{J}_{\mathrm{int}} = \mathbf{J}_{\mathrm{em}} - (\mathbf{r}-\mathbf{r}_c)\times\mathbf{P}_{\mathrm{tot}}, \tag{2.38}$$

$$\mathbf{E}_\perp = \mathbf{E} - (\hat{\mathbf{r}}\cdot\mathbf{E})\hat{\mathbf{r}}, \qquad \mathbf{H}_\perp = \mathbf{H} - (\hat{\mathbf{r}}\cdot\mathbf{H})\hat{\mathbf{r}}, \tag{2.39}$$

$$S_0 = |E_x|^2 + |E_y|^2, \quad S_1 = |E_x|^2 - |E_y|^2, \quad S_2 = 2\,\mathrm{Re}(E_xE_y^*), \quad S_3 = 2\,\mathrm{Im}(E_xE_y^*), \tag{2.40}$$

$$\Omega_P = \hat{\mathbf{n}}\cdot(\nabla\times\mathbf{P}), \qquad \mathbf{P} = \mathrm{Re}\{\mathbf{S}\}, \tag{2.41}$$

where $\mathcal{D}_{J,\mathrm{reg}}$ is the regime flag from Eq. (2.34). The laboratory chain composes inherited 1 constraints with radiation selection and finite observation:

$$\mathfrak{A}_{\mathrm{ch2}}^{(\mathrm{report})} = \mathfrak{A}_{\mathrm{meas}} \circ \mathfrak{A}_{\mathrm{g}} \circ \mathfrak{A}_{\mathrm{cov}} \circ \mathfrak{A}_{\mathrm{tr}} \circ \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \tag{2.42}$$

$$\Lambda_{\mathrm{lab}} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \qquad \mathcal{Q} \in \{\text{direct, surrogate, inference}\}, \tag{2.43}$$

$$\mathcal{R}_\omega[\mathcal{T}_{R,\mathbf{a}}\mathbf{J}_{\mathrm{imp}}] = \mathcal{T}_{R,\mathbf{a}}\,\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \quad \text{on admissible data}, \tag{2.44}$$

with residual Eq. (2.37) serving as the numerical diagnostic when discretization breaks Eq. (2.44) [2]. Function spaces, Green functions, and radiation operators exist because Maxwell field specification requires them—not as parallel formulations [4]. Section 2.0 has established the radiation maps, Maxwell export sequence, and 1 continuity; Section 2.1 now supplies the Hilbert and Sobolev apparatus through which those maps acquire precise domains, traces, and adjoints [2].

2.1 Function Spaces and Maxwell Operators

Finite horns, ground-plane slots, and compact-range scan surfaces all pose the same mathematical question before any pattern or synthesis algorithm runs: which phasor fields and impressed currents are physically admissible on a declared domain Ω, and on which function classes do the Maxwell maps $\mathcal{S}_\omega$, $\mathcal{L}_\omega$, and $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ act without contradiction [2, 1]? A rim current on a horn, a MoM sheet on a patch, or a finite-element port that fixes $\hat{\mathbf{n}} \times \mathbf{E}$ on a conductor each supplies boundary data and sources that must belong to spaces where curl, divergence, and tangential trace are well defined. Section 2.0.1 named those maps at the level of $\mathcal{J}_\Omega$ and $\mathcal{F}_\Omega$; the present section supplies the Sobolev, trace, and dual structure that makes domains and ranges precise enough for Green representations, outgoing selection, and spectral analysis later in the chapter [4].

The finite-energy floor and admissible-state class imported from Chapter 1—Eqs. (1.3) and (1.4) together with the source–boundary tuple of Subsection 1.1.2—are not re-derived here [2]. Every space declaration below must remain compatible with harmonic charge continuity Eq. (2.27) and with the weak form anchored in Subsection 2.0.3 through Eqs. (2.20) and (2.21). Dyadic Green kernels, Sommerfeld outgoing conditions, radiation integrals, and modal spectra are developed in Sections 2.3–2.6; attempting to invert $\mathcal{L}_\omega$ through a kernel before trace classes and dual pairings are fixed ignores boundary rows and corrupts flux identities [2].

Subsection 2.1.1 defines source, field, trace, and distributional classes for phasor Maxwell theory on Lipschitz domains. Subsection 2.1.2 treats linear operators—domain, range, graph norm, and adjoint pairing Eq. (2.26). Subsection 2.1.3 develops compactness and rank-deficient observation maps that foreshadow finite-aperture limits and $\ker \mathcal{R}_\omega$ in Section 2.7 [6, 3]. Subsection 2.1.4 addresses completions, passivity, and Maxwell-closed screens under which truncated operators inherit coercivity on bounded domains while exterior problems retain outgoing subspaces rather than full self-adjointness [1].

Two distinct rank limitations must remain separate throughout. Instrument maps—finite apertures, probe coupling, truncated modal bases—are finite-rank on field space even when $\mathcal{L}_\omega$ is injective on the physical branch. Equivalent source distributions in $\ker \mathcal{R}_\omega$ share the same far-zone export while differing in reactive near fields; that export null space is analyzed in Section 2.7, but the compact embeddings used there are prepared in Subsection 2.1.3. Measurement compositions from Chapter 1 therefore appear as rank-deficient maps on the right of Eq. (2.42), not as modifications of Maxwell's local laws [2]. Piecewise-constant constitutive data enter through the same product Sobolev template, with interface traces from Eq. (1.19) enforced on $\Gamma = \partial\Omega$ and PEC or Leontovich closures modifying the boundary pairing in Subsection 2.1.4 [2].

2.1.1 Spaces for sources, fields, and distributions

Antenna synthesis begins with a realizability question, not with a modal expansion: which impressed currents on bounded support, subject to charge continuity and compatible boundary traces, can be driven by laboratory feeds and mapped to measurable fields on $\Gamma = \partial\Omega$ [2, 3]? Horn rims, patch edges, and waveguide ports produce sheet-like or filamentary excitations whose derivatives fail at edges; treating them as pointwise smooth functions on all of $\mathbb{R}^3$ breaks flux through Γ and corrupts near-field transforms on spherical scanners [1]. Subsections 2.0.1–2.0.2 named minimal classes $\mathcal{F}_\Omega$ and $\mathcal{J}_\Omega$ sufficient to define $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and $\mathcal{L}_\omega$; here those names are embedded in Sobolev and trace spaces where $\nabla\times$, $\nabla\cdot$, and tangential restriction are continuous operations with explicit domains [2].

A PEC-backed slot fed by $\mathbf{J}_{\mathrm{imp}} \in L^2(\Omega_s)^3$ carries finite power at fixed ω, whereas an idealized non-integrable $1/r$ template violates the finite-energy $\mathcal{D}_J = 1$ screen inherited from Chapter 1 even when a post-processor fits a smooth far pattern [2]. Method-of-moments and finite-element codes test Maxwell residuals against localized bases; that discretization presupposes square-integrable $\nabla\times$ fields, well-defined tangential traces on conductors and apertures, and distributional extensions when sources collapse to sheets—gaps that plain L^2 current models leave open [1]. Charge continuity, interface conditions, and outer-boundary classes fixed in Chapter 1 are used by reference only.

Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain or a finite union of Lipschitz patches $\{\Omega_a\}$ with outward boundary $\Gamma = \partial\Omega$. Phasor fields use $\mathrm{Re}\{\mathbf{F}(\mathbf{r})\mathrm{e}^{-\mathrm{j}\omega t}\}$ with $\mathrm{e}^{-\mathrm{j}\omega t} = \exp(-\mathrm{j}\omega t)$ as in Chapter 1 and the chapter preamble, so $\partial_t \to -\mathrm{j}\omega$ on phasors [2]. In the display below, $L^2(\Omega)$ denotes square-integrable field energy on Ω; $\mathcal{H}_{\nabla\times}(\Omega)$ and $\mathcal{H}_{\mathrm{div}}(\Omega)$ are the minimal vector Sobolev classes in which curl-coupled Maxwell balances close weakly; $\mathcal{J}_\Omega$ collects impressed currents of compact support and finite power; $\mathcal{F}_\Omega$ is the admissible Maxwell field class; and $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and $\mathcal{L}_\omega$ are the solution, radiation, and composite operators declared in Section 2.0.1 [2]. Square-integrability alone is too weak for curl-coupled Maxwell theory: one needs control of both the field and its curl so that circulation balances are meaningful in the L^2 sense. An engineer measuring tangential $\mathbf{E}$ on a patch boundary is, in effect, asking whether the recorded trace is the restriction of a field with square-integrable curl; a method-of-moments code that tests $\nabla \times \mathbf{E}$ against rooftop functions presupposes the same control. The space $\mathcal{H}_{\nabla\times}(\Omega)$ names precisely that charge-continuous class: elements of $L^2(\Omega)^3$ whose curls also lie in $L^2(\Omega)^3$, so Faraday and Ampère balances can be posed weakly without assuming pointwise smoothness at horn rims or slot edges [2]. Readers should notice that $\mathcal{H}_{\nabla\times}(\Omega)$ is stronger than L^2 alone yet weaker than requiring classical C^1 fields everywhere; it is the minimal completion in which curl-coupled Maxwell operators close on bounded Ω [1]. The companion $\mathcal{H}_{\mathrm{div}}(\Omega)$ plays the parallel role for Gauss constraints when divergence must be controlled in the L^2 sense. With that physical reading fixed, the formal definitions are: Single-vector Sobolev classes encode the minimal regularity that curl-coupled Maxwell theory requires on bounded domains. Membership in $\mathcal{H}_{\nabla\times}(\Omega)$ controls both field amplitude and circulation in the L^2 sense, whereas plain L^2 control would not guarantee that Faraday and Ampère balances can be posed weakly at horn rims or slot edges [2]. From a measurement standpoint, these spaces establish that tangential fields recorded on Γ are traces of physically admissible volume states, not arbitrary boundary templates [1]. Single-vector Sobolev classes are

$$\mathcal{H}_{\nabla\times}(\Omega) = \left\{\mathbf{u} \in L^2(\Omega)^3 : \nabla \times \mathbf{u} \in L^2(\Omega)^3\right\}, \tag{2.45}$$

$$\mathcal{H}_{\mathrm{div}}(\Omega) = \left\{\mathbf{u} \in L^2(\Omega)^3 : \nabla \cdot \mathbf{u} \in L^2(\Omega)\right\}, \tag{2.46}$$

where $\nabla \cdot \mathbf{u}$ is the divergence of $\mathbf{u}$ (m^{-2} for current-like fields, $\mathrm{C\,m}^{-3}$ when $\mathbf{u}$ represents a

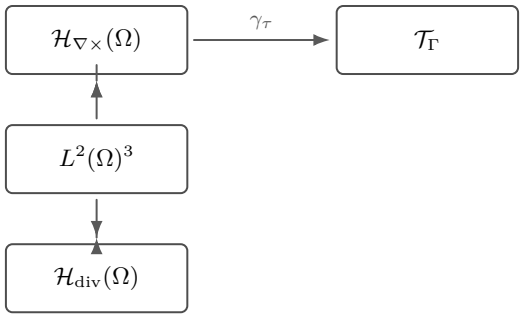


Figure 2.6: Embedding ladder for Maxwell fields: $\mathcal{H}_{\nabla\times}$ and $\mathcal{H}_{\mathrm{div}}$ sit above L^2; tangential traces map $\mathcal{H}_{\nabla\times}$ fields to boundary data compatible with Eq. (1.18).

Sobolev traces, adjoints, and Fredholm structure establish Maxwell boundary data and export rank—they are not optional ornaments on the source-to-field pathway [2].

displacement-related quantity), required to be square integrable so that Gauss-type constraints can be enforced weakly. each equipped with the graph norm $\|\mathbf{u}\|^2_{\mathcal{H}_{\nabla\times}} = \|\mathbf{u}\|^2_{L^2} + \|\nabla\times\mathbf{u}\|^2_{L^2}$ and the analogous $\mathcal{H}_{\mathrm{div}}$ norm [2]. Equations (2.45)–(2.46) are the vector analogues of the finite-energy floor Eq. (1.3): they permit Maxwell curl-coupling without assuming pointwise smoothness of $\mathbf{E}$ or $\mathbf{H}$. The consequence for antenna modeling is direct: horn rims, slot edges, and patch boundaries may host singularities, yet the impressed current and recovered field remain admissible as long as they lie in the declared Sobolev classes [1]. For piecewise constant ε, μ, fields are understood patchwise in Eqs. (2.45)–(2.46) with interface traces constrained by Eq. (1.19) [1]. The Maxwell state space is the product

$$\mathcal{U}_\omega = \mathcal{H}_{\nabla\times}(\Omega) \times \mathcal{H}_{\nabla\times}(\Omega), \qquad \mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega. \tag{2.47}$$

Equation (2.47) refines the six-component packaging Eq. (2.12): both components carry square-integrable curls, matching the first-order structure of Eq. (1.7). A weighted energy norm compatible with Eq. (2.20) is

$$\|\mathbf{u}_\omega\|^2_{\mathcal{U}} = \|\mathbf{E}\|^2_{L^2} + \|\mathbf{H}\|^2_{L^2} + \alpha\,\|\nabla\times\mathbf{E}\|^2_{L^2} + \beta\,\|\nabla\times\mathbf{H}\|^2_{L^2}, \tag{2.48}$$

with positive weights α, β fixed by constitutive scaling in each patch. Norm (2.48) controls the bilinear form $\mathfrak{a}_\omega$ of Eq. (2.21) on bounded domains and makes $\mathcal{U}_\omega$ a Hilbert space [2]. Divergence control enters through source spaces and through Gauss constraints rather than through requiring both components to lie in $\mathcal{H}_{\mathrm{div}}$; forcing $\mathbf{E} \in \mathcal{H}_{\mathrm{div}}$ and $\mathbf{H} \in \mathcal{H}_{\mathrm{div}}$ simultaneously is stronger than Maxwell's first-order system requires and is reserved for scalar-potential reductions deferred to later chapters [2]. Impressed sources refine the declaration of Subsection 2.0.1. The admissible impressed-current space collects currents of finite support and finite power that satisfy the 1 $\mathcal{D}_J = 1$ screen. Let $\Omega_s \subseteq \Omega$ denote compact source support. The admissible impressed current space is

$$\mathcal{J}_\Omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{H}_{\nabla\times}(\Omega) : \operatorname{supp}\mathbf{J}_{\mathrm{imp}} \subseteq \Omega_s,\ \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1\}, \tag{2.49}$$

where $\mathcal{D}_J$ is the criterion of Eq. (2.9) inherited from Eq. (1.8) and Eq. (1.22) [2]. Equation (2.49) strengthens square integrability to $\mathcal{H}_{\nabla\times}$ regularity so that $\nabla\times\mathbf{J}_{\mathrm{imp}}$ is defined in L^2 whenever numerical or analytical measurement sequences differentiate sources. Charge-neutral subclasses used in radiation null-space analysis satisfy

$$\mathcal{J}^{(0)}_\Omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \nabla\cdot\mathbf{J}_{\mathrm{imp}} = 0\}, \tag{2.50}$$

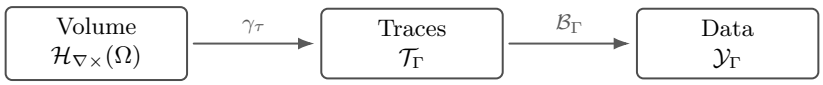


Figure 2.7: Trace–data chain for boundary completion: volume fields map to $\mathcal{T}_\Gamma$; the boundary operator from Eq. (1.18) selects admissible data.

with $\rho = 0$ implied on Ω_s. Equation (2.50) is the subspace on which divergence-free reparametrizations act without altering $\mathcal{S}_\omega[\mathbf{J}_{\rm imp}]$ in simply connected regions; null-space structure is developed in Section 2.7.1 [1]. Boundary data from the tuple Eq. (1.17) live in trace spaces rather than in volume Sobolev classes. Define

$$\mathcal{T}_\Gamma = H^{-1/2}(\Gamma)^3 \times H^{-1/2}(\Gamma)^3, \tag{2.51}$$

as the product space for tangential electric and magnetic traces induced by $\mathcal{H}_{\nabla\times}(\Omega)$ fields on Lipschitz Γ [2]. The tangential trace operators extend Eq. (2.15):

$$\gamma_{\tau,E} : \mathcal{H}_{\nabla\times}(\Omega) \to H^{-1/2}(\Gamma)^3, \quad \gamma_{\tau,E}[\mathbf{E}] = \hat{\mathbf{n}} \times \mathbf{E}|_\Gamma, \tag{2.52}$$

$$\gamma_{\tau,H} : \mathcal{H}_{\nabla\times}(\Omega) \to H^{-1/2}(\Gamma)^3, \quad \gamma_{\tau,H}[\mathbf{H}] = \hat{\mathbf{n}} \times \mathbf{H}|_\Gamma, \tag{2.53}$$

with outward unit normal $\hat{\mathbf{n}}$. Equations (2.52)–(2.53) make Eq. (1.18) an operator between declared Banach spaces: $\mathcal{B}_\Gamma : \mathcal{T}_\Gamma \to \mathcal{Y}_\Gamma$ for data space $\mathcal{Y}_\Gamma$ fixed by the boundary class of Eq. (1.20) or outgoing traces of Eq. (1.21) [2]. Figure 2.4. Surjectivity of γ_τ onto compatible trace subclasses fails globally—only fields in specified $\mathcal{H}_{\nabla\times}$ subspaces with prescribed divergence pairings map to physically realizable $\mathbf{g}_\Gamma$ [2]. Subsection 2.1.4 identifies self-adjoint boundary realizations; here the point is that $\mathcal{L}_\omega$ of Eq. (2.16) acts on $\mathcal{U}_\omega$ only when $\gamma_\tau[\mathbf{u}_\omega]$ lies in the domain of $\mathcal{B}_\Gamma$. Sheet currents and interface channels from Eq. (1.19) require distributional extensions. Let $\mathcal{D}'(\Omega)^3$ denote temporal-harmonic distributional three-vector fields. Impressed volume data remain $\mathbf{J}_{\rm imp} \in \mathcal{J}_\Omega$, while surface equivalents $\mathbf{J}_{s,ab}$ on Σ_{ab} embed as

$$\langle \mathbf{J}_s, \boldsymbol{\phi} \rangle = \sum_{ab} \int_{\Sigma_{ab}} \mathbf{J}_{s,ab} \cdot \boldsymbol{\phi}\, dS, \qquad \boldsymbol{\phi} \in C_c^\infty(\Omega)^3, \tag{2.54}$$

linking the data tuple Eq. (1.17) to a continuous linear functional on smooth test fields [2]. Equation (2.54) does not alter the jump laws of Eq. (1.19); it specifies the functional-analytic class in which those jumps appear in weak formulations. Interface trace spaces on Σ_{ab} are denoted

$$\mathcal{T}_{\Sigma_{ab}} = H^{-1/2}(\Sigma_{ab})^3, \tag{2.55}$$

with tangential continuity of $\mathbf{E}$ and jump of $\hat{\mathbf{n}} \times \mathbf{H}$ encoded as constraints on γ_τ taken from each patch [1]. Dual pairings for source and test fields use the L^2 inner product

$$\langle \mathbf{J}_{\rm imp}, \boldsymbol{\psi} \rangle_\Omega = \int_\Omega \mathbf{J}_{\rm imp} \cdot \boldsymbol{\psi}^*\, d^3r, \qquad \boldsymbol{\psi} \in \mathcal{H}_{\nabla\times}(\Omega), \tag{2.56}$$

which induces a bounded functional on test states when $\mathbf{J}_{\rm imp} \in L^2(\Omega)^3$ and extends to Eq. (2.54) for sheet data. Forcing in Eq. (2.13) enters the weak form Eq. (2.21) through $\mathfrak{f}_\omega(\mathbf{v}_\omega) = \langle \mathbf{s}_\omega, \mathbf{v}_\omega \rangle$ once $\mathbf{v}_\omega \in \mathcal{U}_\omega$ carries homogeneous boundary traces [2]. The symbol $\mathcal{F}_\Omega$ names field pairs $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ that are not merely square integrable but carry controlled curls and admissible tangential traces on Γ; every output of $\mathcal{S}_\omega[\mathbf{J}_{\rm imp}]$ on the physical branch must lie in this sharpened class [1]. The admissible Maxwell field space compatible with 1 is

$$\mathcal{F}_\Omega = \left\{ \mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega : \gamma_\tau[\mathbf{u}_\omega] \in \mathcal{D}(\mathcal{B}_\Gamma),\ (\mathbf{E}, \mathbf{H}) \text{ satisfy patchwise Gauss constraints} \right\}, \tag{2.57}$$

where $\mathcal{D}(\mathcal{B}_\Gamma)$ is the domain of boundary data compatible with Eq. (1.23) imported from Section 1.1.2. Equation (2.57) refines the minimal declaration of Subsection 2.0.1: same symbol $\mathcal{F}_\Omega$, sharpened topology. Membership implies Eq. (1.3) but adds curl and trace control needed for integration by parts in Eq. (2.26) [2]. The composite operator $\mathcal{L}_\omega$ couples volume residuals with boundary rows; its domain $\mathcal{D}(\mathcal{L}_\omega)$ is the closed Maxwell solution class on which both $\mathcal{M}_\omega$ and $\mathcal{B}_\Gamma$ are simultaneously evaluable in the declared data spaces [1]. The closed Maxwell solution domain becomes

$$\mathcal{D}(\mathcal{L}_\omega) = \left\{ \mathbf{u}_\omega \in \mathcal{F}_\Omega : \mathcal{M}_\omega[\mathbf{u}_\omega] \in L^2(\Omega)^4 \right\}, \tag{2.58}$$

which specializes Eq. (2.22) to Lipschitz domains with patchwise smooth coefficients [2]. On $\mathcal{D}(\mathcal{L}_\omega)$, the volume residuals of Eq. (2.14) are square integrable, and the boundary row of Eq. (2.16) is evaluated in $\mathcal{T}_\Gamma$. Exterior domains append outgoing subspaces rather than changing Eqs. (2.45)–(2.57); that selection is function-space compatible with weighted norms that decay at infinity and is detailed in Section 2.4.3 [1]. Free-charge density paired with $\mathbf{J}_{\mathrm{imp}}$ in Eq. (1.17) is taken in the scalar class $\rho \in L^2(\Omega)$ whenever volumetric charge is prescribed, with distributional interpretation $\nabla \cdot \mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0 = 0$ understood in $\mathcal{D}'(\Omega)$ as in Eq. (1.22). Scalar L^2 control is sufficient because Gauss laws in Eq. (1.7) involve $\nabla\cdot$ on $\varepsilon_0\mathbf{E}$ and $\nabla\cdot$ on $\mu_0\mathbf{H}$, not gradients of ρ. When ρ is absent and $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega^{(0)}$, the electric Gauss row reduces to a divergence constraint on $\mathbf{E}$ enforced weakly through test functions in $\mathcal{H}_{\nabla\times}(\Omega)$ [2]. Approximation properties matter for numerical projection. Smooth compactly supported fields are dense in $\mathcal{H}_{\nabla\times}(\Omega)$ and in $\mathcal{U}_\omega$ equipped with norm (2.48); every admissible state is a limit of C_c^∞ test pairs modified to satisfy boundary class on Γ [2]. Density underpins Galerkin convergence: edge-element spaces approximate $\mathcal{F}_\Omega$ when their degrees of freedom match tangential trace continuity on Γ_{PEC}, where

$$\mathcal{U}_\omega^{\mathrm{PEC}} = \{(\mathbf{E},\mathbf{H}) \in \mathcal{U}_\omega : \hat{\mathbf{n}} \times \mathbf{E}|_{\Gamma_{\mathrm{PEC}}} = \mathbf{0}\}, \tag{2.59}$$

encodes perfect-conductor tangential data from Eq. (1.20) without assuming $\mathbf{E} = \mathbf{0}$ in Ω [1]. Integration by parts on $\mathcal{U}_\omega^{\mathrm{PEC}}$ yields the boundary-term cancellation used in Eq. (2.26) when $\mathcal{B}_\Gamma$ selects homogeneous tangential electric trace. Compact embeddings used later for Fredholm arguments are recorded as

$$\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3 \hookleftarrow H^{-1/2}(\Gamma)^3, \tag{2.60}$$

with the first inclusion compact on bounded Ω and the trace map continuous but not compact [2]. Equation (2.60) explains why interior cavity spectra are discrete while boundary observation maps are rank deficient: compact embeddings produce accumulating eigenvalues in volume, whereas non-compact traces map infinite-dimensional field spaces to finite observation vectors [1]. The quantitative singular-value decay of radiation maps on compact support is developed in Section 2.6 and cited thereafter [2]. Subsection 2.1.3 develops the consequences for radiation accessibility and finite observation. Subsection 2.1.1 has refined $\mathcal{F}_\Omega$, $\mathcal{J}_\Omega$, and $\mathcal{D}(\mathcal{L}_\omega)$ from minimal declarations to Sobolev, trace, and distributional classes compatible with 1 charge-continuous. The consequence for measurement is immediate: near-field scanners and method-of-moments codes inherit these spaces as the continuous targets their discretizations must approximate. Subsection 2.1.2 treats the Maxwell maps as linear operators with explicit domains, ranges, and adjoints on those spaces [2]. As a worked example, suppose $\mathbf{g}_\Gamma$ prescribes tangential $\mathbf{E}$ on Γ with $H^{1/2}$ regularity but is not the trace of any $\mathcal{H}_{\nabla\times}$ field in Ω. Then no $\mathbf{u}_\omega \in \mathcal{F}_\Omega$ satisfies $\gamma_{\tau,E}[\mathbf{E}] = \mathbf{g}_\Gamma$, regardless of how accurately a volume solver minimizes Maxwell residuals. The failure is a trace-space mismatch, not a Green-function error; correcting it requires either lifting $\mathbf{g}_\Gamma$ into $\mathcal{T}_\Gamma$ as an image under γ_τ or relaxing the boundary class through impedance completion in Subsection 2.1.4 [2].

2.1.2 Linear operators: domain, range, and adjoint

The physical problem is operator accountability: a network analyzer, finite-element solver, or near-field scanner implicitly assumes that each map in the source-to-field pathway accepts only admissible excitations and returns only measurable field pairs [2]. Every linear map—$\mathcal{S}_\omega$, $\mathcal{L}_\omega$, $\mathcal{R}_\omega$, and finite-rank measurement surrogates—requires explicit domain, range, and adjoint pairings [2]. Recall that $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ is the harmonic solution map, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ selects outgoing radiative content, and $\mathcal{L}_\omega$ packages volume and boundary residuals jointly [1]. Reciprocity is a symmetry principle of accessibility: network analyzers swap transmit and receive ports, and the adjoint is the mathematical shadow of that hardware symmetry when sources remain passive [1]. Treating "adjoint" as jargon detached from range calibration misses this link [3]. When a synthesis residual persists under port reversal, the fault often lies in domain mismatch between trial currents and measured tangential data [7]. Continuity and divergence constraints tie impressed currents to charge conservation on real feeds [4]. When those constraints are violated, numerical solvers may still return finite vectors, but the corresponding fields exhibit unphysical reactive blow-up that no anechoic termination can absorb. Domain, range, and adjoint language records which excitations the operator accepts and which field pairs are returned [2]. Subsection 2.1.1 fixed the Hilbert and trace spaces on which Maxwell maps act. Subsection 2.1.2 treats those maps as linear operators with explicit domains, ranges, and adjoints [2]. The volume operator $\mathcal{M}_\omega$, boundary operator $\mathcal{B}_\Gamma$, composite $\mathcal{L}_\omega$, and solution map $\mathcal{S}_\omega$ introduced in Subsections 2.0.1–2.0.2 are not re-defined; their Maxwell field structure is recorded here as the definitive construction of domain–range–adjoint theory. Weak and energy forms of Eqs. (2.21) and (2.26) supply the duality pairings below and are cited, [2]. Let $\mathcal{X}$ and $\mathcal{Y}$ be Hilbert spaces. The linear map representing $T : \mathcal{D}(T) \to \mathcal{Y}$ is specified by its domain $\mathcal{D}(T) \subseteq \mathcal{X}$ and range $\mathcal{R}(T) = T\mathcal{D}(T) \subseteq \mathcal{Y}$. For Maxwell operators, $\mathcal{X} = \mathcal{U}_\omega$ from Eq. (2.47) and $\mathcal{Y}$ is a product of volume and boundary data spaces. When a finite-element solver reports residual norms, it is effectively measuring a graph norm: how large the field is in $\mathcal{U}_\omega$ together with how large Maxwell's residuals become when that field is inserted into $\mathcal{L}_\omega$. Engineers should read the graph norm as a joint flux measurement record of field energy and equation balance, not as a single L^2 amplitude [1]. A trial vector that is small in L^2 but drives large curl residuals cannot lie in $\mathcal{D}(\mathcal{L}_\omega)$ on the physical branch; conversely, a state with controlled graph norm is the continuous target that Galerkin discretizations on edge elements approximate [2]. The graph norm

$$\|\mathbf{u}\|^2_{\mathrm{gr}(T)} = \|\mathbf{u}\|^2_{\mathcal{X}} + \|T\mathbf{u}\|^2_{\mathcal{Y}}, \qquad \mathbf{u} \in \mathcal{D}(T), \tag{2.61}$$

turns $\mathcal{D}(T)$ into a Hilbert space when T is closed [2]. Equation (2.61) is the diagnostic used in Subsection 2.1.4 to construct self-adjoint completions; here it organizes domain control for $\mathcal{L}_\omega$. Each Maxwell map acts from a declared domain of admissible states into a data space of residuals or traces; that domain–range discipline applies to each Maxwell map once $\mathcal{X}$ and $\mathcal{Y}$ are chosen. Confusing $\mathcal{D}(T)$ with all of $\mathcal{X}$ is a common source of spurious boundary terms in flux identities [2]. The volume Maxwell operator $\mathcal{M}_\omega$ packages the four local Maxwell balance rows that a harmonic solver tests row by row; it returns residuals in $\mathcal{Y}_{\mathrm{vol}}$, not the field itself, and therefore acts only on states whose curls are controlled enough that those residuals lie in L^2 [2]. When a finite-element log reports "Maxwell residual," the engineer is observing an approximate norm of $\mathcal{M}_\omega[\mathbf{u}_\omega]$; convergence to zero on the physical branch is the numerical signature of membership in $\mathcal{D}(\mathcal{L}_\omega)$ [1]. The volume Maxwell operator of Eq. (2.14) acts as

$$\mathcal{M}_\omega : \mathcal{D}(\mathcal{M}_\omega) \longrightarrow \mathcal{Y}_{\mathrm{vol}}, \qquad \mathcal{Y}_{\mathrm{vol}} = L^2(\Omega)^3 \times L^2(\Omega)^3 \times L^2(\Omega) \times L^2(\Omega)^3, \tag{2.62}$$

with $\mathcal{D}(\mathcal{M}_\omega) \subset \mathcal{F}_\Omega$ the subspace on which all four residual rows of Eq. (1.7) lie in $\mathcal{Y}_{\mathrm{vol}}$ [2]. Operand interpretation: $\mathcal{M}_\omega$ evaluates curl-coupled Maxwell residuals pointwise in the L^2 sense; physically,

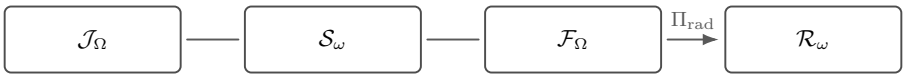


Figure 2.8: Excitation-to-flux sequence on declared domains: $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$, $\mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\mathrm{rad}}$.

it returns whether a candidate $(\mathbf{E}, \mathbf{H})$ satisfies local Maxwell balance with impressed data removed to the forcing vector Eq. (2.13). The map is linear in $\mathbf{u}_\omega$ and bounded on bounded subsets of $\mathcal{D}(\mathcal{M}_\omega)$ but is not bounded on all of $\mathcal{U}_\omega$ because curl norms are not controlled by L^2 norms alone [1]. Tangential traces form bounded boundary operators on $\mathcal{H}_{\nabla\times}(\Omega)$:

$$\gamma_\tau : \mathcal{H}_{\nabla\times}(\Omega) \longrightarrow H^{-1/2}(\Gamma)^3, \qquad \gamma_\tau[\mathbf{u}] = \hat{\mathbf{n}} \times \mathbf{u}|_\Gamma, \tag{2.63}$$

with $\|\gamma_\tau\|_{\mathrm{op}} < \infty$ on Lipschitz Γ [2]. Equation (2.63) specializes Eqs. (2.52)–(2.53) to a single template. The boundary operator of Eq. (1.18) is

$$\mathcal{B}_\Gamma : \mathcal{D}(\mathcal{B}_\Gamma) \longrightarrow \mathcal{Y}_\Gamma, \qquad \mathcal{D}(\mathcal{B}_\Gamma) \subseteq \mathcal{T}_\Gamma, \tag{2.64}$$

where $\mathcal{T}_\Gamma$ and $\mathcal{Y}_\Gamma$ are from Subsection 2.1.1. Impedance and outgoing classes from Eqs. (1.20)–(1.21) restrict $\mathcal{D}(\mathcal{B}_\Gamma)$ to admissible trace pairs compatible with Eq. (1.23) [2]. The composite operator of Eq. (2.16) is

$$\mathcal{L}_\omega : \mathcal{D}(\mathcal{L}_\omega) \longrightarrow \mathcal{Y}_{\mathrm{vol}} \times \mathcal{Y}_\Gamma, \tag{2.65}$$

with $\mathcal{D}(\mathcal{L}_\omega)$ given by Eq. (2.58). Range membership is *coupled*: a state cannot satisfy volume rows while violating boundary rows in the product sense of Eq. (2.17). The graph norm (2.61) on $\mathcal{L}_\omega$ controls both residuals simultaneously and is the natural norm for convergence tests in numerical Maxwell solvers [1]. The harmonic solution map is an operator on source space:

$$\mathcal{S}_\omega : \mathcal{D}(\mathcal{S}_\omega) \longrightarrow \mathcal{F}_\Omega, \qquad \mathcal{D}(\mathcal{S}_\omega) = \mathcal{J}_\Omega, \tag{2.66}$$

defined by Eq. (2.18) as $\mathcal{L}_\omega[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]] = [\mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}}), \mathbf{g}_\Gamma]$. Well-posedness requires invertibility of $\mathcal{L}_\omega$ on the physical branch selected by $\mathcal{B}_\Gamma$; then

$$\mathcal{S}_\omega = \mathcal{L}_\omega^{-1} \circ (\mathbf{s}_\omega(\cdot), \mathbf{g}_\Gamma), \tag{2.67}$$

with $\mathcal{L}_\omega^{-1}$ understood as the outgoing or cavity-truncated resolvent of Sections 2.4.3 and 2.6.4, not as an algebraic inverse on all of $\mathcal{U}_\omega$ [2]. Radiation and observation operators compose accordingly:

$$\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \qquad \Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \tag{2.68}$$

with Π_{rad} from Eq. (2.3) and $\mathcal{O}_{\mathrm{far}}$ from Subsection 2.0.1 [1]. (2.11) in domain–range language. Each arrow is meaningful only on the declared domain; extending $\mathcal{S}_\omega$ to non-admissible sources produces elements of $\mathcal{U}_\omega$ that need not lie in $\mathcal{F}_\Omega$ [2]. Adjoints are defined through duality pairings compatible with Eq. (2.56). For the volume operator, the formal adjoint $\mathcal{M}_\omega^\dagger$ satisfies

$$\langle \mathcal{M}_\omega[\mathbf{u}_\omega], \mathbf{v}_\omega \rangle_{\mathrm{vol}} = \langle \mathbf{u}_\omega, \mathcal{M}_\omega^\dagger[\mathbf{v}_\omega] \rangle_{\mathcal{U}} + \mathcal{B}_{\mathrm{surf}}(\mathbf{u}_\omega, \mathbf{v}_\omega), \tag{2.69}$$

where $\langle \cdot, \cdot \rangle_{\mathrm{vol}}$ is the componentwise L^2 pairing on $\mathcal{Y}_{\mathrm{vol}}$, $\langle \cdot, \cdot \rangle_{\mathcal{U}}$ is induced by Eq. (2.48), and $\mathcal{B}_{\mathrm{surf}}$ collects boundary terms from integration by parts [2]. On $\mathcal{D}(\mathcal{M}_\omega) \cap \mathcal{U}_\omega^{\mathrm{PEC}}$ from Eq. (2.59), Eq. (2.26) identifies $\mathcal{B}_{\mathrm{surf}}$ with the tangential trace pairing of Eq. (2.26); that identity is imported, not reproved [2]. The composite adjoint satisfies the block structure

$$\mathcal{L}_\omega^\dagger = \begin{bmatrix} \mathcal{M}_\omega^\dagger & 0 \\ 0 & \mathcal{B}_\Gamma^\dagger \end{bmatrix} \quad \text{on compatible pairs,} \tag{2.70}$$

with cross-coupling entering only through shared trace data when boundary and volume domains are not synchronized. Self-adjoint *realizations* of truncated Maxwell problems require choosing $\mathcal{B}_\Gamma$ so that $\mathcal{B}_{\mathrm{surf}}$ cancels on $\mathcal{D}(\mathcal{L}_\omega)$; Forcing functionals induced by impressed data are bounded linear operators on test spaces. The map

$$\mathfrak{f}_\omega : \mathcal{U}_\omega^0 \longrightarrow \mathbb{C}, \qquad \mathfrak{f}_\omega[\mathbf{v}_\omega] = \langle \mathbf{s}_\omega(\mathfrak{D}), \mathbf{v}_\omega \rangle_{\mathrm{vol}}, \tag{2.71}$$

is continuous when $\mathbf{s}_\omega \in \mathcal{Y}_{\mathrm{vol}}$ and $\mathbf{v}_\omega \in \mathcal{U}_\omega^0$ carries homogeneous boundary traces compatible with $\mathcal{B}_\Gamma$ [2]. Equation (2.71) is the operator reading of the right-hand side of Eq. (2.21). Source-to-field operators inherit continuity on bounded subsets: if $\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}} \le C$ and $\mathcal{L}_\omega$ is boundedly invertible on a fixed branch, then

$$\|\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \le C_{\mathrm{res}} \, \|\mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}})\|_{\mathrm{vol}}, \tag{2.72}$$

with $C_{\mathrm{res}} = \|\mathcal{L}_\omega^{-1}\|$ on the selected subspace [1]. Inequality (2.72) is the operator form of solver stability under source perturbations. Kernel and range spaces are recorded for later null-space analysis:

$$\ker \mathcal{L}_\omega = \{ \mathbf{u}_\omega \in \mathcal{D}(\mathcal{L}_\omega) : \mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0} \}, \tag{2.73}$$

$$\mathcal{R}(\mathcal{L}_\omega) = \left\{ [\mathbf{s}, \mathbf{g}]^{\mathsf{T}} \in \mathcal{Y}_{\mathrm{vol}} \times \mathcal{Y}_\Gamma : \exists\, \mathbf{u}_\omega \in \mathcal{D}(\mathcal{L}_\omega) : \ \mathcal{L}_\omega[\mathbf{u}_\omega] = [\mathbf{s}, \mathbf{g}]^{\mathsf{T}} \right\}. \tag{2.74}$$

On bounded domains with passive $\mathcal{B}_\Gamma$, $\ker \mathcal{L}_\omega$ contains cavity modes; on exterior domains with outgoing radiation condition, $\ker \mathcal{L}_\omega$ is trivial on the outgoing branch [2]. Section 2.7.1 analyzes $\ker \mathcal{R}_\omega$; Subsection 2.1.3 treats compactness and Fredholm index on $\mathcal{R}(\mathcal{L}_\omega)$. Extension operators lift boundary data into volume fields. A right inverse $E_\Gamma : \mathcal{Y}_\Gamma \to \mathcal{H}_{\nabla\times}(\Omega)$ satisfying $\gamma_\tau \circ E_\Gamma = \mathrm{id}$ on compatible subclasses exists only for $\mathbf{g}_\Gamma$ in the range of γ_τ; otherwise no $\mathbf{u}_\omega \in \mathcal{F}_\Omega$ meets the boundary row [2]. Extension operators enter Calderón preconditioners and boundary-element formulations developed in later sections; here they clarify that $\mathcal{B}_\Gamma$ acts on traces, not on arbitrary L^2 boundary vectors [1]. As a worked example, fix $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ and two distinct outgoing radiation conditions $\mathcal{B}_\Gamma^{(a)}$ and $\mathcal{B}_\Gamma^{(b)}$ on the same exterior domain. The operators $\mathcal{L}_\omega^{(a)}$ and $\mathcal{L}_\omega^{(b)}$ have the same volume domain but different boundary domains; Eq. (2.67) yields two resolvents $\mathcal{S}_\omega^{(a)} \neq \mathcal{S}_\omega^{(b)}$ in general. Comparing analysis to measurement without matching $\mathcal{D}(\mathcal{B}_\Gamma)$ is therefore a comparison of different operators, not of different discretizations of one operator [2]. Subsection 2.1.2 has packaged domain, range, graph norm, adjoint pairings, and the composition $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ on declared spaces. Subsection 2.1.3 records boundedness, compact embeddings, and the rank deficiency that finite instruments impose on $\mathcal{F}_{\mathrm{rad}}$ [2].

2.1.3 Boundedness, compactness, rank deficiency, spectral compression

Spectral compression on compact support limits how sharply a finite aperture can beam [8]. Sidelobe floors on a range trace the same law [2]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. The physical problem is observability under finite geometry: a compact source, a bounded domain, and a finite aperture cannot couple to infinitely many independent field degrees of freedom [7]. Subsections 2.1.1–2.1.2 fixed spaces and operator domains. Subsection 2.1.3 records boundedness and compactness properties, rank deficiency of observation maps, and the viewpoint from which Section 2.6 develops spectral compression for radiation operators [2]. The physical content is familiar on a compact range: finite sources and finite probes cannot couple to infinitely many independent far-zone degrees of freedom, so accessible rank is limited by geometry and observation extent [7]. Fredholm index computation and complete spectral decomposition remain in Sections 2.6.4 and 2.6.2; here the constructive development establishes the lemmas on boundedness, compactness, and rank deficiency those sections invoke by reference [2]. In the display below,

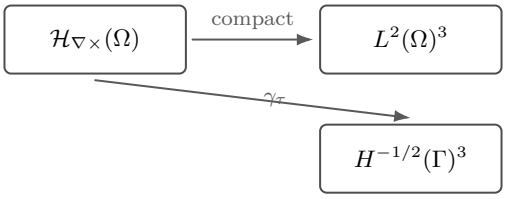


Figure 2.9: Compact embedding $\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3$ on bounded Ω; trace map is bounded but not compact (Eq. (2.60)).

$\mathcal{F}_{\mathrm{rad}} \subset \mathcal{F}_\Omega$ denotes the outgoing radiative subspace selected by Π_{rad}; $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ are the solution and radiation maps on $\mathcal{J}_\Omega$ [1]. The linear map representing $T : \mathcal{X} \to \mathcal{Y}$ between Hilbert spaces is *bounded* if

$$\|T\|_{\mathrm{op}} = \sup_{\|\mathbf{x}\|_{\mathcal{X}}=1} \|T\mathbf{x}\|_{\mathcal{Y}} < \infty. \tag{2.75}$$

Boundedness is equivalent to continuity at the origin and is the default hypothesis for composition with measurement readouts [2]. On the Maxwell chain, operators split into bounded and unbounded classes. Trace and observation maps are bounded on declared spaces. The tangential trace Eq. (2.63) satisfies $\|\gamma_\tau\|_{\mathrm{op}} < \infty$ from $\mathcal{H}_{\nabla\times}(\Omega)$ to $H^{-1/2}(\Gamma)^3$ [2]. Far-zone readouts from Subsection 2.0.1 define bounded functionals when restricted to $\mathcal{F}_{\mathrm{rad}}$:

$$\|\mathcal{O}_{\mathrm{far}}\|_{\mathrm{op}} = \sup_{\|\mathbf{u}\|_{\mathcal{U}}=1} \big|\mathcal{O}_{\mathrm{far}}[\mathbf{u}]\big| < \infty \quad \text{on } \mathcal{F}_{\mathrm{rad}}. \tag{2.76}$$

Laboratory maps $\mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}}$ in Eq. (2.43) are bounded on the same subspace because composition of bounded operators is bounded [1]. Volume and composite Maxwell operators are unbounded on full $\mathcal{U}_\omega$. The map $\mathcal{M}_\omega$ of Eq. (2.62) has $\|\mathcal{M}_\omega\|_{\mathrm{op}} = \infty$ when $\mathcal{D}(\mathcal{M}_\omega) = \mathcal{U}_\omega$ because controlling $\|\mathcal{M}_\omega[\mathbf{u}]\|_{\mathrm{vol}}$ would require uniform curl bounds beyond L^2 control [2]. Restriction to $\mathcal{D}(\mathcal{L}_\omega)$ with graph norm Eq. (2.61) restores boundedness of the *resolvent* $\mathcal{L}_\omega^{-1}$ on the physical branch where invertibility holds:

$$\|\mathcal{L}_\omega^{-1}\|_{\mathrm{op}} < \infty \quad \Longleftrightarrow \quad \mathcal{L}_\omega \text{ bijective on } \mathcal{D}(\mathcal{L}_\omega)/\ker \mathcal{L}_\omega. \tag{2.77}$$

Equation (2.77) specializes Eq. (2.72); exterior outgoing problems use resolvent bounds on weighted subspaces rather than on all of $\mathcal{U}_\omega$ [1]. (2.60). Rellich-type compactness for $\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3$ on bounded domains implies that bounded sequences in $\mathcal{H}_{\nabla\times}$ have L^2-convergent subsequences [2]. Finite geometry cannot export arbitrarily fine angular complexity; propagation therefore acts as a compression mechanism on electromagnetic structure. An operator $K : \mathcal{X} \to \mathcal{Y}$ is *compact* if bounded sets in $\mathcal{X}$ map to relatively compact sets in $\mathcal{Y}$:

$$K \text{ compact} \iff K(B_{\mathcal{X}}(0,1)) \text{ has compact completion in } \mathcal{Y}. \tag{2.78}$$

Compact perturbations of identity produce Fredholm structure used in Section 2.6.4; Subsection 2.1.3 records the embedding lemmas without computing index [1]. Radiation and Green maps on fixed bounded domains often decompose as

$$\mathcal{S}_\omega = G_{\mathrm{sing}} + K_{\mathrm{cpt}}, \tag{2.79}$$

with G_{sing} capturing singular near-field kernels and K_{cpt} compact on $\mathcal{J}_\Omega$ when sources have compact support and Ω is bounded [2]. Equation (2.79) is the operator template for Section 2.3; kernel

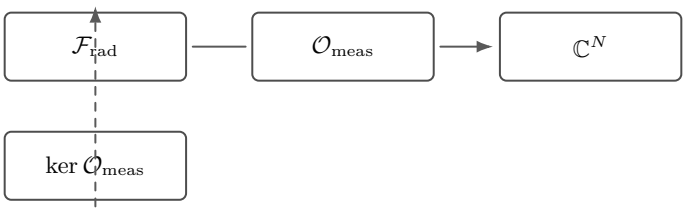


Figure 2.10: Finite-rank infinite-dimensional field space maps to $\mathbb{C}^N$; $\ker \mathcal{O}_{\text{meas}}$ is infinite-dimensional for $N < \infty$.

construction is not repeated here. The split separates non-compact singular behavior at coincident points from compact smoothing at finite separation [1]. Rank deficiency enters through finite observation. An aperture-limited measurement map $\mathcal{O}_{\text{meas}} : \mathcal{F}_{\text{rad}} \to \mathbb{C}^N$ has finite-dimensional range with $N < \infty$:

$$\mathcal{O}_{\text{meas}} = \sum_{n=1}^{N} \langle \cdot, \mathbf{f}_n \rangle_{\mathcal{U}} \, \mathbf{e}_n, \tag{2.80}$$

for dual vectors $\mathbf{f}_n$ and coordinate directions $\mathbf{e}_n$. Therefore $\operatorname{rank}(\mathcal{O}_{\text{meas}}) \leq N$ and

$$\ker \mathcal{O}_{\text{meas}} = \{\mathbf{u} \in \mathcal{F}_{\text{rad}} : \mathcal{O}_{\text{meas}}[\mathbf{u}] = \mathbf{0}\} \tag{2.81}$$

is infinite-dimensional whenever $\mathcal{F}_{\text{rad}}$ is infinite-dimensional [2]. Equation (2.81) is the operator form of the measurement instrument rank limits in Section 1.2.5: finite modal sets cannot resolve all angular content. Rank deficiency is structural, not a discretization artifact alone; the spectral compression and singular-value decay law for radiation maps on compact support are owned by Section 2.6 [2]. (2.68). Angular-spectrum reports that omit unresolved harmonics lie in the range of $\mathcal{O}_{\text{meas}}$ but not in its injective sector [2]. Subsection 2.1.3 does not classify report types; Section 1.2.5 remains authoritative. Singular values quantify rank deficiency. For compact $K : \mathcal{X} \to \mathcal{Y}$, the singular value sequence $\{\sigma_n(K)\}_{n \geq 1}$ decreases to zero, and

$$\|K - R_N K\|_{\text{op}} = \inf_{\operatorname{rank} R_N \leq N} \|K - R_N\|_{\text{op}} \sim \sigma_{N+1}(K), \tag{2.82}$$

where R_N ranges over rank-N operators [2]. For radiation maps truncated to N spherical or plane-wave modes, Eq. (2.82) controls how much field content lies outside the retained modal subspace; explicit VSH eigenbases are deferred to Section 2.6.2 and Chapter 3 (*Angular Momentum Operators and SO(3)*) [1]. *Spectral compression* names the reduction of infinite-dimensional radiation dynamics to a finite or countable spectral model on the physical branch. Let $\{\phi_n\}$ be a spectral family in $\mathcal{F}_\Omega$ and $\{P_n\}$ the associated one-dimensional projectors. A truncated radiation operator

$$\mathcal{R}_\omega^{(N)} = \sum_{n=1}^{N} \lambda_n \, P_n \circ \mathcal{S}_\omega \tag{2.83}$$

retains only modes with eigenvalues λ_n above a declared threshold [2]. Equation (2.83) is the abstract template for Section 2.6; concrete eigenfunctions are not constructed here. On bounded cavities, compact embeddings yield accumulating eigenvalues $\lambda_n \to \infty$ for interior resonances; on exterior domains, continuous spectra require outgoing projectors rather than discrete sums [2]. Compression error is bounded in operator norm when K is compact:

$$\|\mathcal{R}_\omega - \mathcal{R}_\omega^{(N)}\|_{\text{op}} \leq C \, \sigma_{N+1}(K_{\text{cpt}}), \tag{2.84}$$

with K_{cpt} from Eq. (2.79) and C independent of impressed data on bounded $\mathcal{J}_\Omega$ [1]. Physically, Eq. (2.84) states that truncating modal content beyond the $(N+1)$st singular value cannot be recovered by any rank-N surrogate operator; the quantitative decay law is as in Section 2.6 [2]. Rank deficiency of $\mathcal{R}_\omega$ itself is distinct from observation rank deficiency. Non-radiating sources satisfy $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ while $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ (Eq. (2.8)); Section 2.7.1 develops that null space. Here the point is compositional:

$$\operatorname{rank}(\mathcal{O}_{\mathrm{meas}}\mathcal{R}_\omega) \leq \min\{\operatorname{rank}(\mathcal{O}_{\mathrm{meas}}), \dim \mathcal{R}(\mathcal{R}_\omega)\}, \tag{2.85}$$

In Eq. (2.85), $\mathcal{O}_{\mathrm{meas}}$ is a finite-rank linear measurement functional (pattern probe bank, modal fitter, or spherical-harmonic extractor) composed with $\mathcal{R}_\omega$; $\operatorname{rank}(\cdot)$ is the dimension of the accessible report subspace; and $\mathcal{R}(\mathcal{R}_\omega)$ is the range of the radiation map on $\mathcal{J}_\Omega$. So measurement rank caps reported angular content even when $\mathcal{R}_\omega$ is injective on admissible sources [2]. Both deficits must appear in uncertainty budgets for angular-transport claims imported from Section 1.2.3. Fredholm structure note: on bounded Ω with passive $\mathcal{B}_\Gamma$, $\mathcal{L}_\omega$ is Fredholm on $\mathcal{D}(\mathcal{L}_\omega)$ modulo cavity kernels when written as

$$\mathcal{L}_\omega = \mathcal{L}_0 + K, \qquad K \text{ compact on graph domain}, \tag{2.86}$$

with $\mathcal{L}_0$ an invertible reference operator on the quotient by $\ker \mathcal{L}_\omega$ [2]. Index and solvability criteria are developed in Section 2.6.4; Eq. (2.86) supplies the split Subsection 2.1.2 stated the introductory relation in Subsection 2.0.2. As a worked example, a spherical near-field scanner records $N = 50$ tangential field samples on a closed surface. The map $\mathcal{O}_{\mathrm{meas}}$ has rank at most 50, so angular harmonics beyond the resolved set lie in $\ker \mathcal{O}_{\mathrm{meas}}$. Applying Eq. (2.82) to the compact smoothing operator from sample interpolation shows that recovered far-zone coefficients error is controlled by σ_{51}, not by Maxwell residual alone [2]. Increasing frequency without increasing N lowers σ_{N+1} slowly for fixed aperture, worsening compression error even when $\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0}$ in the volume. As a worked example, in a closed cavity, compact embedding yields discrete resonances and Eq. (2.83) sums over cavity modes. In an exterior problem with outgoing radiation condition, the same formula requires a continuous spectral integral rather than a finite sum; using cavity compression on an open antenna model conflates discrete storage modes with continuous radiation modes [2]. Section 2.4.3 selects the outgoing branch; Subsection 2.1.3 only records that compression templates differ by domain class [2]. Finite-geometry interpretation (compactness and rank). Compact embeddings and finite-rank observation maps express one enduring electromagnetic truth: finite geometry destroys information. A bounded source support Ω_s cannot excite arbitrarily fine angular structure at every observation point; a finite aperture integrates over unresolved directions; a finite modal set truncates accessible rank. Inaccessible angular structure is therefore not a numerical inconvenience but a consequence of Maxwell propagation from compact sources through outgoing selection to finite instruments. flux report is fundamentally limited, and the mathematics of compactness makes that limit quantifiable through decaying singular values and infinite-dimensional observation null spaces [2]. Subsection 2.1.3 has established boundedness classes, compact embeddings, and finite-rank observation structure. The quantitative singular-value decay for radiation maps on compact support is developed in Section 2.6; here only the structural lemmas are recorded. Subsection 2.1.4 completes the function-space foundation with self-adjoint realizations and admissibility completions on bounded domains [2].

2.1.4 Self-adjointness, completions, and admissibility

The physical problem is passive realizability on bounded hardware: cavity walls, matched terminations, and dissipative linings must not permit unbounded storage of reactive energy under

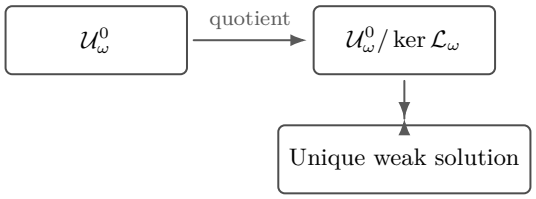


Figure 2.11: Cavity-mode quotient: coercivity and uniqueness hold on the quotient space after removing $\ker\mathcal{L}_\omega$.

admissible excitation [2]. Friedrichs and other completions extend domains just enough that $\nabla\times$ and $\hat{\mathbf{n}}\times$ are simultaneously closed on the same space—precisely what finite-element antenna models require when sheet sources sit on interior interfaces [1]. Improper domain extension predicts unphysical growth of reactive storage—a numerical symptom that must be corrected before radiation maps are trusted [4]. Completion choices are radiation-physics choices about which reactive storage remains inside Ω and which energy is exported [10].

Subsections 2.1.1–2.1.3 supplied spaces, solution domains, compactness, and spectral compression templates. Subsection 2.1.4 identifies self-adjoint and coercive realizations of truncated Maxwell operators on bounded domains, records completion and quotient constructions that remove cavity kernels, and states the $\mathcal{D}_J = 1$ embedding that links Chapter 1 finite-energy criteria to the solution domains declared here [2]. Weak and energy forms Eqs. (2.20)–(2.21) and the adjoint pairing Eq. (2.26) are presupposed; they are [2]. A densely defined operator T on Hilbert space $\mathcal{H}$ is *symmetric* if $\langle T\mathbf{u}, \mathbf{v}\rangle = \langle \mathbf{u}, T\mathbf{v}\rangle$ for all $\mathbf{u}, \mathbf{v} \in \mathcal{D}(T)$, and *self-adjoint* if $\mathcal{D}(T^\dagger) = \mathcal{D}(T)$ with equality of adjoints [2]. Maxwell operators are symmetric on suitable domains because Eq. (2.26) cancels boundary pairings when $\mathcal{B}_\Gamma$ is passive and homogeneous on test states. Self-adjointness requires matching domain and range of adjoints; that match fails on exterior domains without restricting to outgoing subspaces [1]. On bounded Ω with passive $\mathcal{B}_\Gamma$, the sesquilinear form $\mathfrak{a}_\omega$ of Eq. (2.20) is coercive on the quotient by cavity modes:

$$\mathfrak{a}_\omega(\mathbf{u}, \mathbf{u}) \geq \alpha \, \|\mathbf{u}\|^2_{\mathcal{U}}, \qquad \mathbf{u} \in \mathcal{U}^0_\omega/\ker\mathcal{L}_\omega, \tag{2.87}$$

for some $\alpha > 0$ when $\ker\mathcal{L}_\omega$ is finite-dimensional and consists of divergence-free resonances compatible with boundary class [2]. Equation (2.87) is the Lax–Milgram hypothesis that makes Eq. (2.21) well posed on the quotient. Coercivity fails on the unquotiented space because cavity modes satisfy $\mathfrak{a}_\omega(\mathbf{u}, \mathbf{u}) = 0$ while $\|\mathbf{u}\|_{\mathcal{U}} > 0$. Physical excitation at a cavity frequency requires impedance or source coupling that breaks pure standing-wave balance; Subsection 2.1.4 records the linear-algebraic quotient; Section 2.6.4 develops spectral consequences [2]. Impedance and Leontovich completions from Eq. (2.23) yield self-adjoint boundary realizations when surface impedance is passive:

$$\mathrm{Re}\{Z_s\} \geq 0 \text{ on } \Gamma_Z \quad \Longrightarrow \quad \mathcal{B}_\Gamma \text{ symmetric on } \mathcal{T}_\Gamma. \tag{2.88}$$

Passive $\mathrm{Re}\{Z_s\} \geq 0$ dissipates power through the boundary rather than injecting net energy; active or negative-real parts require revisiting Maxwell-closed as noted in Section 1.1.2 [2]. Perfect conductors select Eq. (2.59); impedance boundaries enlarge $\mathcal{D}(\mathcal{B}_\Gamma)$ while preserving symmetry when Eq. (2.88) holds [1]. When $\mathcal{B}_\Gamma$ is symmetric but the Maxwell operator is only densely defined, the Friedrichs extension supplies a self-adjoint realization:

$$\mathcal{L}^{\mathrm{F}}_\omega = \overline{\mathcal{L}_\omega}\big|_{\mathcal{D}(\mathfrak{a}_\omega)}, \tag{2.89}$$

the completion of $\mathcal{L}_\omega$ on the energy domain of $\mathfrak{a}_\omega$ [2]. Equation (2.89) is the locked definition for for self-adjoint Maxwell truncations on bounded domains. Exterior outgoing problems use a

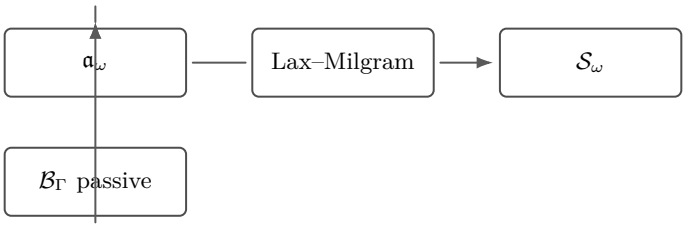


Figure 2.12: Self-adjoint completion pathway on bounded domains: passive $\mathcal{B}_\Gamma$ supports coercive $\mathfrak{a}_\omega$ and stable $\mathcal{S}_\omega$ on the cavity quotient.

different completion template: restrict to the outgoing subspace selected in Section 2.4.3 rather than applying Eq. (2.89) to all of $\mathcal{U}_\omega$ [1]. Well-posedness on the quotient combines coercivity and bounded forcing:

$$\exists!\, \mathbf{u}_\omega \in \mathcal{U}_\omega^0/\ker \mathcal{L}_\omega : \mathfrak{a}_\omega(\mathbf{u}_\omega, \mathbf{v}) = \mathfrak{f}_\omega(\mathbf{v}) \quad \forall\, \mathbf{v} \in \mathcal{U}_\omega^0/\ker \mathcal{L}_\omega, \tag{2.90}$$

with $\mathfrak{f}_\omega$ from Eq. (2.71). Stability follows from Eqs. (2.87) and (2.72) [2]. Resolvent bound Eq. (2.77) holds on the quotient when $\mathcal{L}_\omega^{\mathrm{F}}$ is bijective there. Changing $\mathcal{B}_\Gamma$ while leaving $\mathfrak{a}_\omega$ formally unchanged can destroy coercivity or symmetry; that sensitivity was already visible in the resolvent-branch illustration of Subsection 2.1.2 [2]. charge-continuous from Subsection 1.0.4 embeds into solution domains. The 1 finite-energy class Eq. (1.4) requires $(\mathbf{E}, \mathbf{H}) \in L^2(\Omega)^3 \times L^2(\Omega)^3$ with compatible boundary and source data [2]. The Chapter 2 refinement is

$$\mathcal{X}_{\mathrm{ch1}}(\Omega) \cap \mathcal{U}_\omega \subseteq \mathcal{F}_\Omega \subseteq \mathcal{D}(\mathcal{L}_\omega), \tag{2.91}$$

with strict inclusions in general because $\mathcal{H}_{\nabla\times}$ control and trace membership strengthen L^2 Maxwell-closed without replacing Eq. (1.8) [2]. Equation (2.91) specializes Eq. (2.27): when $\mathcal{D}_{\mathrm{ch1}} = 1$, impressed data lie in $\mathcal{J}_\Omega$ and closed solutions lie in $\mathcal{F}_\Omega$. Operator Maxwell-closed adds graph-norm control:

$$\mathbf{u}_\omega \in \mathcal{F}_\Omega \implies \|\mathbf{u}_\omega\|_{\mathrm{gr}(\mathcal{L}_\omega)} < \infty \text{ when } \mathcal{L}_\omega[\mathbf{u}_\omega] \text{ is defined.} \tag{2.92}$$

Inequality (2.9) is the quantitative form of "physically realizable" field pairs for variational solvers [1]. Source-side Maxwell-closed remains coupled to Eq. (2.49) and Eq. (1.8) by reference. Completion operators must not introduce incompatible traces:

$$\mathcal{B}_\Gamma \text{ admissible} \iff \mathcal{R}(\gamma_\tau|_{\mathcal{F}_\Omega}) \supseteq \mathcal{D}(\mathcal{B}_\Gamma). \tag{2.93}$$

Failure of Eq. (2.93) means prescribed $\mathbf{g}_\Gamma$ is not the trace of any $\mathcal{H}_{\nabla\times}$ field, reproducing the trace-mismatch illustration of Subsection 2.1.1 [2]. Exterior domains: outgoing radiation condition from Section 2.4.3 defines a *physically self-adjoint* branch rather than a Hilbert self-adjoint operator on all of $\mathcal{U}_\omega$:

$$\mathcal{U}_\omega = \mathcal{U}_{\mathrm{out}} \oplus \mathcal{U}_{\mathrm{re}}, \qquad \mathcal{L}_\omega \text{ coercive on } \mathcal{U}_{\mathrm{out}}, \tag{2.94}$$

with reactive storage content in $\mathcal{U}_{\mathrm{re}}$ controlled by regime criteria of Section 1.3.3 rather than by full coercivity on $\mathcal{U}_\omega$ [2]. Conflating Eq. (2.89) with exterior radiation problems produces incorrect cavity-like spectra in open regions. With Subsections 2.1.1–2.1.4 complete, Section 2.1 has moved from minimal declarations in Section 2.0 to a closed operator calculus on Sobolev spaces: traces, adjoints, compactness, and self-adjoint truncations compatible with 1 charge-continuous [2]. Section 2.2 applies those spaces to time–frequency equivalences; Sections 2.3–2.4 inherit $\mathcal{D}(\mathcal{L}_\omega)$ and $\mathcal{F}_\Omega$ as fixed input classes. **??** states which operator outputs survive finite source support and manufacturable geometry [3].

2.2 Time- and Frequency-Domain Maxwell Operators

Function spaces and Maxwell operators in Section 2.1 fix admissible sources and field pairs before time-harmonic reduction. Section 2.2 carries the same physics into frequency-domain maps used on every fixed-frequency range [2]. The laboratory never measures abstract Hilbert norms; it measures phasor fields at declared ω [4]. An oscilloscope trace records $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{H}(\mathbf{r}, t)$ as functions of time; a vector network analyzer records complex amplitudes at each stepped tone. Both instruments interrogate the same structure, but only the time-domain picture makes retarded support visible before any phasor reduction is declared [1].

Fixed-ω phasors are frequency slices of retarded evolution, valid when excitation and media are narrowband on the declared interval [2]. Each harmonic operator $\mathcal{L}_\omega$ is a labeled cross-section of a single causal evolution law, not as an independent postulate. A vector network analyzer sweep therefore samples one harmonic operator family $\mathcal{L}_\omega$ at each tone—not a separate physics from time-domain causality [4]. Every compact-range calibration and harmonic finite-difference cell presupposes that reduction. The harmonic operators $\mathcal{L}_\omega$ and $\mathcal{S}_\omega$ introduced earlier describe individual frequency slices of Maxwell evolution. The purpose of the present section is to show how those slices emerge from a single causal time-domain theory [2]. Section 2.2 develops the time–frequency equivalences that connect harmonic operators to causal evolution, Helmholtz symbols, and propagating-versus-evanescent splits under the $\exp(-\mathrm{j}\omega t)$ convention locked in 1. It records when that reduction is valid, how retarded time-domain maps relate to $\mathcal{R}_\omega$, and which spectral components carry far-zone transport content importable from Section 1.3.3. Harmonic Maxwell theory in Section 2.0 treated $\omega > 0$ as a fixed parameter. Time-domain formulations restore the full operator pencil $\partial/\partial t$ and clarify why frequency-domain solvers implement causal limits when inversion is performed correctly. Subsection 2.2.1 gives the Chapter 2 account of time-domain evolution; Subsection 2.2.2 packages frequency-domain Maxwell and curl–curl Helmholtz operators without re-displaying Eq. (1.5); Subsection 2.2.3 proves equivalence between retarded Green propagation and $\mathcal{S}_\omega$ stated in Eq. (2.10); Subsection 2.2.3 applies the reactive-radiative partition of Section 1.3.3 to propagating, evanescent, and non-propagating subspaces in operator form [2]. Several related constructions are deferred deliberately. In particular, dyadic Green-function development and inversion theory are postponed until Section 2.3, where the harmonic operators introduced here acquire explicit integral representations [2]. Section 2.2 does not re-open function-space declarations from Section 2.1 or local Maxwell derivations from 1. Helmholtz and curl–curl reductions developed in Subsection 2.2.2 are single-field tools: they simplify kernel construction but do not replace the first-order $\mathcal{L}_\omega$ packaging of Eq. (2.16) when boundary rows and impedance closures must be enforced jointly [1]. Within the locked harmonic band, $\mathcal{S}_\omega$ is the ω-slice of a causal time-domain map whenever impressed data and $\mathcal{B}_\Gamma$ are held fixed across the transform [2]. Finite-difference time-domain schemes approximate Subsection 2.2.1; frequency-domain finite-element solvers approximate Subsection 2.2.2 on the domains of Eq. (2.58). Modal and plane-wave decompositions in Subsection 2.2.3 filter which components of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ may enter $\mathcal{R}_\omega$ under Eq. (2.28) [1].

2.2.1 Time-domain evolution

Time-harmonic phasors match fixed-frequency range measurements; $\exp(-æ\omega t)$ fixes the sign of reactive terms [2]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. The physical problem is causal propagation: a switched feed, a pulsed radar, or a transient immunity test excites fields whose support in $(\mathbf{r}, t)$ must respect finite signal speed [4]. Time-harmonic phasors are how network analyzers, compact ranges, and chamber receivers report steady-state

radiation at each frequency step [2]. The $\exp(-\mathrm{j}\omega t)$ convention fixed in Chapter 1 aligns Poynting flux with wattmeter readings and VNA phase references [5]. Phasor analysis is best viewed as a steady-state reduction of a fundamentally causal theory rather than as the primary formulation of electromagnetic dynamics [1]. Even small phase errors can rotate synthesized patterns when near-field data are back-propagated [7]. The time-domain picture below states the causal support that every VNA tone must inherit when $\mathcal{L}_\omega$ is interpreted as a frequency slice rather than an independent postulate [2].

Subsection 2.2.1 supplies the time-domain foundation underlying every harmonic formulation used later in the chapter. It introduces the first-order Maxwell evolution system, the associated evolution operator, and the causal support restrictions that all physically admissible frequency-domain solutions must inherit [2]. Time-domain evolution specifies retarded trajectories: the field observed at $(\mathbf{r}, t)$ depends only on source events contained within the corresponding past light cone [2]. Any numerical or analytical approximation that violates this support structure introduces nonphysical precursors and loses its connection to Maxwell causality [1]. Local Maxwell balances remain those of Eq. (1.27); they are cited. Function spaces from Section 2.1 enter through time-dependent embeddings developed below; Sobolev and trace declarations are not repeated [2]. Let $\Omega \subset \mathbb{R}^3$ be a Lipschitz domain with boundary $\Gamma = \partial\Omega$. Real time-domain fields $\mathbf{E}(\mathbf{r}, t)$, $\mathbf{H}(\mathbf{r}, t)$ use SI units and the same constitutive class as 1. The evolution problem is posed on trajectories of the electromagnetic state rather than on isolated snapshots at a single frequency. A pulsed or switched excitation on a feed line makes the distinction visible: the field at $(\mathbf{r}, t)$ cannot respond until source events inside the backward light cone have occurred. Time-domain evolution is therefore the laboratory picture behind every harmonic slice $\mathcal{S}_\omega$ used later in the chapter [2]. The state vector $\mathbf{U}(t)$ packages the coupled $(\mathbf{E}, \mathbf{H})$ fields at frozen time t into a single object on which the Maxwell generator acts; this packaging mirrors the phasor state $\mathbf{u}_\omega$ of Eq. (2.12), with time retained rather than suppressed [1]. Collect the electromagnetic state at time t as

$$\mathbf{U}(t) = \begin{bmatrix} \mathbf{E}(\cdot, t) \\ \mathbf{H}(\cdot, t) \end{bmatrix} \in \mathcal{V}(\Omega), \tag{2.95}$$

where $\mathcal{V}(\Omega)$ is the time-dependent product space of square-integrable field pairs compatible with Eq. (1.4) at each frozen time slice [2]. Equation (2.95) is the time-domain analogue of Eq. (2.12); phasor symbols are reserved for the $\exp(-\mathrm{j}\omega t)$ reduction in Subsection 2.2.2. The time-domain Maxwell system is the causal law from which every harmonic operator in Section 2.2 is obtained by phasor substitution. Physically, Eqs. (2.96) state that changing $\mathbf{E}$ in space drives $\mathbf{H}$ in time through Faraday's law, and changing $\mathbf{H}$ in space drives $\mathbf{E}$ in time through Ampère's law with impressed current as external drive [4]. From an antenna engineer's perspective, this is the system that finite-difference time-domain codes march forward in time; the harmonic solvers of Subsection 2.2.2 are steady-state limits of the same law when all data are single-frequency [1]. The time-domain Maxwell system imported from Eq. (1.27) reads in vacuum constitutive form as

$$\begin{aligned} \nabla \times \mathbf{E}(\mathbf{r}, t) &= -\mu_0\, \partial_t \mathbf{H}(\mathbf{r}, t), \\ \nabla \times \mathbf{H}(\mathbf{r}, t) &= \mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) + \varepsilon_0\, \partial_t \mathbf{E}(\mathbf{r}, t), \\ \nabla \cdot \mathbf{E}(\mathbf{r}, t) &= \frac{\rho(\mathbf{r}, t)}{\varepsilon_0}, \qquad \nabla \cdot \mathbf{H}(\mathbf{r}, t) = 0, \end{aligned} \tag{2.96}$$

with impressed current density $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t)$ and charge density $\rho(\mathbf{r}, t)$. Equation (2.96) packages local dynamics as a first-order system in $(\mathbf{E}, \mathbf{H})$ and ∂_t; the hyperbolic coupling is the time-domain face of the curl structure in Eq. (1.7) [2]. Charge continuity requires

$$\nabla \cdot \mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) + \partial_t \rho(\mathbf{r}, t) = 0 \tag{2.97}$$

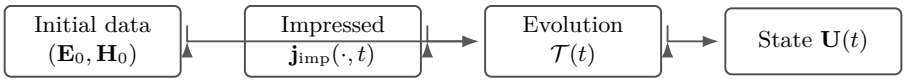


Figure 2.13: Time-domain evolution: initial data and impressed sources drive the semigroup $\mathcal{T}(t)$ to field state $\mathbf{U}(t)$ under boundary class $\mathcal{B}_\Gamma$.

in the sense of distributions when ρ is present [2]. Pairs $(\mathbf{j}_{\mathrm{imp}}, \rho)$ violating Eq. (2.97) lie outside the time-domain admissible class parallel to Eq. (2.9). Boundary data enter through the same $\mathcal{B}_\Gamma$ of Eq. (1.18), now evaluated on time-dependent traces $\gamma_\tau[\mathbf{U}(t)] \in \mathcal{T}_\Gamma$ for each $t \geq 0$ [2]. The closed time-domain problem is to find $\mathbf{U}(t)$ such that

$$\begin{cases} \partial_t \mathbf{U} = \mathcal{A}\,\mathbf{U} + \mathbf{F}(t), & \mathbf{r} \in \Omega,\ t > 0, \\ \mathbf{U}(0) = \mathbf{U}_0, & \mathbf{r} \in \Omega, \\ \mathcal{B}_\Gamma[\mathbf{U}(t)] = \mathbf{g}_\Gamma(t), & \mathbf{r} \in \Gamma,\ t \geq 0, \end{cases} \tag{2.98}$$

where $\mathcal{A}$ is the Maxwell generator induced by Eq. (2.96) on $\mathcal{D}(\mathcal{A}) \subset \mathcal{V}(\Omega)$ and $\mathbf{F}(t)$ collects impressed-source rows [2]. Equation (2.98) is the time-domain counterpart of Eq. (2.17); uniqueness on a fixed branch is inherited from Section 1.1.3 when data classes coincide with the harmonic limit [2]. Define the evolution operator (semigroup) $\mathcal{T}(t)$ by

$$\mathcal{T}(t) : \mathbf{U}_0 \longmapsto \mathbf{U}(t), \qquad \mathcal{T}(0) = \mathrm{id}, \qquad \mathcal{T}(t+s) = \mathcal{T}(t)\mathcal{T}(s), \tag{2.99}$$

when $\mathbf{F} \equiv \mathbf{0}$ and $\mathcal{B}_\Gamma$ is homogeneous. With forcing,

$$\mathbf{U}(t) = \mathcal{T}(t)\mathbf{U}_0 \int_0^t \mathcal{T}(t-s)\,\mathbf{F}(s)\,ds, \tag{2.100}$$

the Duhamel representation on $\mathcal{D}(\mathcal{A})$ [2]. Equation (2.100) makes causality explicit: $\mathbf{U}(t)$ depends only on $\mathbf{F}(s)$ for $s \leq t$. The time-domain *solution map* sends impressed data and initial state to trajectories:

$$\mathcal{S}_t : (\mathbf{U}_0, \mathbf{j}_{\mathrm{imp}}) \longmapsto \{\mathbf{U}(t)\}_{t \geq 0}. \tag{2.101}$$

Harmonic $\mathcal{S}_\omega$ of Eq. (2.2) is a single-frequency slice of Eq. (2.101) when all data are proportional to $\exp(-\mathrm{j}\omega t)$ and $\mathcal{B}_\Gamma$ is fixed [2]; the equivalence is proved in Subsection 2.2.3. Retarded representation specializes Eq. (2.100) when Green kernels exist:

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_{\mathrm{hom}}(\mathbf{r}, t) - \mu_0\,\partial_t \int_0^t \int_\Omega \mathcal{G}(\mathbf{r}, \mathbf{r}', t-t') \cdot \mathbf{j}_{\mathrm{imp}}(\mathbf{r}', t')\,d^3r'\,dt', \tag{2.102}$$

with $\mathcal{G}(\mathbf{r}, \mathbf{r}', t-t') = \mathbf{0}$ for $t - t' < 0$ [2]. Equation (2.102) refines the Eq. (2.10) without constructing $\mathcal{G}$ here; kernel development is in Section 2.3. The support condition

$$\mathrm{supp}_{(\mathbf{r},t)}\,\mathbf{E} \subseteq \{(\mathbf{r}, t) : t \geq t_{\mathrm{ret}}(\mathbf{r})\} \tag{2.103}$$

expresses finite propagation speed $c_0 = 1/\sqrt{\mu_0 \varepsilon_0}$ on vacuum regions [2]. Figure 2.14 schematizes Eq. (2.103). Energy balance in time domain imports the Poynting and storage laws of Section 1.1.4. With volume energy density

$$u(\mathbf{r}, t) = \frac{\varepsilon_0}{2}\|\mathbf{E}(\mathbf{r}, t)\|^2 + \frac{\mu_0}{2}\|\mathbf{H}(\mathbf{r}, t)\|^2, \tag{2.104}$$

integration over a control volume $V \subset \Omega$ yields

$$\frac{d}{dt}\int_V u\,dV + \int_{\partial V} \mathbf{S} \cdot \hat{\mathbf{n}}\,dS = -\int_V \mathbf{E} \cdot \mathbf{j}_{\mathrm{imp}}\,dV, \tag{2.105}$$

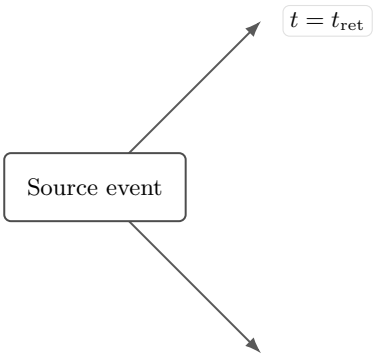


Figure 2.14: Causal support: field at $(\mathbf{r}, t)$ depends on sources inside the backward light cone; retarded integration uses $t' \le t$ only.

with $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ [2]. Equation (2.105) is cited; it supplies the physical reading of Eq. (2.100): impressed work on the right drives stored and exported energy on the left. Reactive exchange with boundaries appears in the surface term and motivates the regime split of Subsection 2.2.3 [1]. Harmonic operators follow from the substitution $\mathbf{E}(\mathbf{r}, t) = \mathrm{Re}\{\underline{\mathbf{E}}(\mathbf{r})\mathrm{e}^{-\mathrm{j}\omega t}\}$ with $\underline{\mathbf{H}}$ analogous, yielding

$$\partial_t \longrightarrow -\mathrm{j}\omega, \qquad \mathcal{A} \longrightarrow \mathcal{M}_\omega, \qquad \mathcal{S}_t\big|_\omega \longrightarrow \mathcal{S}_\omega, \tag{2.106}$$

on admissible data of Eq. (2.91) [2]. Equation (2.106) is symbolic; Subsection 2.2.3 states equivalence conditions under which frequency-domain solvers implement causal time-domain maps. Exterior domains require outgoing boundary conditions in time domain compatible with Eq. (1.21); those completions are sharpened in Section 2.4.3 [2]. Time-domain Maxwell-closed couples Eq. (1.4) at each time slice to Eq. (2.97):

$$\mathbf{U}(t) \in \mathcal{F}_\Omega \text{ a.e. } t \ \wedge \ (\mathbf{j}_{\mathrm{imp}}, \rho) \text{ satisfy charge continuity } \implies \mathbf{U} \in \mathcal{D}(\mathcal{S}_t). \tag{2.107}$$

Finite-difference time-domain schemes discretize Eq. (2.96) on staggered grids; stability requires discrete analogues of Eq. (2.103) and consistent boundary enforcement of $\mathcal{B}_\Gamma$ [1]. Subsection 2.2.1 states the continuous targets those schemes approximate. Subsection 2.2.1 has posed the time-domain Maxwell system, the evolution semigroup $\mathcal{T}(t)$, and the retarded convolution form that enforces causal support. Subsection 2.2.2 carries the same physics into fixed-frequency operators and Helmholtz symbols on the spaces of Section 2.1 [2].

2.2.2 Frequency-domain Maxwell operators

A standard-gain horn on a compact-range turntable fixes the laboratory scene for 'Frequency-domain Maxwell operators": frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays below [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. The physical problem is steady-state interrogation at fixed frequency: every calibrated S-parameter, resonator Q measurement, and harmonic finite-element solve assumes that excitation, media, and boundary class are frozen while ω is stepped [1]. A vector network analyzer sweep is the canonical laboratory observation that precedes every harmonic operator declaration. The instrument steps ω through a calibrated band, records complex S-parameters at each tone, and treats each frequency as a steady-state slice of the same passive structure [1]. The composite operator $\mathcal{L}_\omega$ introduced in Section 2.1 is precisely that slice: Maxwell's equations with $\partial_t \to -\mathrm{j}\omega$ on impressed data and boundary class held fixed. Readers should therefore read $\mathcal{L}_\omega$ not as an abstract Hilbert

map alone but as the operator whose inversion reproduces the phasor fields a VNA implicitly assumes when it reports $S_{21}(\omega)$ after de-embedding [2]. Frequency-domain formulations dominate practical electromagnetics because measurements, simulations, and design specifications are usually organized around frequency. Network analyzers, resonator measurements, and antenna-pattern ranges all interrogate electromagnetic systems one frequency at a time [1]. The harmonic operators introduced below state that experimentally familiar viewpoint while remaining consistent with the causal evolution described in Subsection 2.2.1.

Subsection 2.2.1 fixed time-domain evolution and Eq. (2.106). Subsection 2.2.2 gives the Chapter 2 account of harmonic Maxwell operators and curl–curl Helmholtz symbols on the spaces of Section 2.1 [2]. Recall that $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ is the harmonic solution map, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ selects outgoing radiative content, and $\mathcal{L}_\omega$ packages volume and boundary residuals jointly [1]. The first-order system Eq. (1.7) and composite $\mathcal{L}_\omega$ of Eq. (2.16) are presupposed; local Maxwell derivations from 1 are not repeated. The electric-wave operator Eq. (2.19) was recorded in Subsection 2.0.2; here it is embedded in domain declarations and extended by magnetic and material-weighted counterparts. Full vector-field separation is deferred to Section 3.3 [2]. Under the locked $\exp(-\mathrm{j}\omega t)$ convention, differentiation with respect to time converts to multiplication by $-\mathrm{j}\omega$ on phasor quantities, and the second time derivative maps to $-\omega^2$. Every phasor displayed on a network analyzer trace has already undergone this substitution: a time derivative in Maxwell's equations becomes a multiplication by $-\mathrm{j}\omega$ on the locked $\exp(-\mathrm{j}\omega t)$ convention. Engineers should notice that the substitution is not a notational convenience alone; it is the algebraic condition under which a time-domain IBVP and a frequency-domain BVP describe the same passive structure at fixed ω [2]. The substitution fails when excitation is broadband without superposition, when $\mathcal{B}_\Gamma$ varies in time, or when material response is not single-frequency on the declared band [1]. With those limits stated, the formal rule is: Time derivatives map to

$$\partial_t \longrightarrow -\mathrm{j}\omega, \qquad \frac{\partial^2}{\partial t^2} \longrightarrow -\omega^2, \tag{2.108}$$

on phasor fields $\mathbf{E}(\mathbf{r})$, $\mathbf{H}(\mathbf{r})$ and impressed data $\mathbf{J}_{\mathrm{imp}}(\mathbf{r})$, $\rho(\mathbf{r})$ [2]. In Eq. (2.108), ∂_t acting on a time-harmonic quantity is replaced by $-\mathrm{j}\omega$, where ω is in $\mathrm{rad\,s^{-1}}$; and ∂_t^2 is replaced by $-\omega^2$. This substitution is valid only when all time dependence is single-frequency or when a superposition of such terms is taken with consistent phasor definitions. Equation (2.108) specializes Eq. (2.106) and converts Eq. (1.27) to the harmonic BVP of Eq. (1.28) by reference. The harmonic residual collects the four Maxwell rows in frequency form. A finite-element or method-of-moments solve at fixed ω is, operationally, the search for $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ that drives $\mathcal{M}_\omega[\mathbf{u}_\omega]$ to zero in the volume while satisfying the boundary row enforced by $\mathcal{B}_\Gamma$. The engineer observes residuals indirectly through solver convergence and port matching; the mathematician packages the same requirement as membership in $\mathcal{D}(\mathcal{L}_\omega)$ with $\mathcal{L}_\omega[\mathbf{u}_\omega]$ balanced against impressed data [1]. Readers should notice that Eq. (2.109) is not a new physics postulate but the frequency image of Eq. (2.96) once $\partial_t \to -\mathrm{j}\omega$ on admissible data [2]. The harmonic Maxwell residual operator $\mathcal{M}_\omega$ of Eq. (2.14) acts on $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{U}_\omega$:

$$\mathcal{M}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \nabla \times \mathbf{E} + \mathrm{j}\omega\mu_0 \mathbf{H} \\ \nabla \times \mathbf{H} - \mathrm{j}\omega\varepsilon_0 \mathbf{E} - \mathbf{J}_{\mathrm{imp}} \\ \nabla \cdot (\varepsilon_0 \mathbf{E}) - \rho \\ \nabla \cdot (\mu_0 \mathbf{H}) \end{bmatrix}, \tag{2.109}$$

row-aligned with Eq. (1.7) [2]. Each row of Eq. (2.109) is a residual that must vanish for a physically consistent harmonic state: the first row enforces Faraday's law with phasor substitution; the second enforces Ampère's law with impressed current $\mathbf{J}_{\mathrm{imp}}$; the third enforces Gauss's law for electric flux with charge density ρ; and the fourth enforces the magnetic Gauss law. Equation (2.109) is the

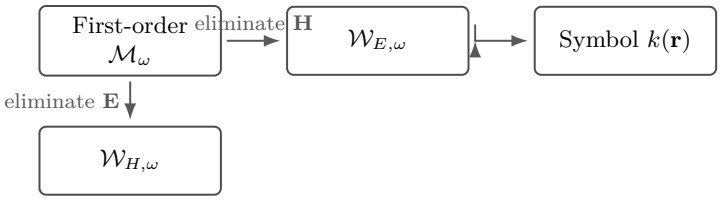


Figure 2.15: Frequency-domain reduction ladder: first-order $\mathcal{M}_\omega$ yields electric and magnetic Helmholtz operators when partner fields are eliminated under stated domain and gauge constraints.

frequency-domain packaging of Eq. (2.96); completion and traces remain in $\mathcal{L}_\omega[\mathbf{u}_\omega] = [\mathbf{s}_\omega, \mathbf{g}_\Gamma]^{\mathsf{T}}$ from Eq. (2.17). Single-field Helmholtz forms equivalent to Eqs. (1.13)–(1.15) were established in 1 and are; elimination steps are not repeated [2]. The Chapter 2 electric-wave operator Eq. (2.19) is the vacuum electric Helmholtz map on $\mathcal{D}(\mathcal{W}_{E,\omega}) \subset \mathcal{H}_{\nabla\times}(\Omega)$:

$$\mathcal{D}(\mathcal{W}_{E,\omega}) = \{\mathbf{E} \in \mathcal{H}_{\nabla\times}(\Omega) : \mathcal{W}_{E,\omega}\mathbf{E} \in L^2(\Omega)^3\}. \tag{2.110}$$

The magnetic counterpart is

$$\mathcal{W}_{H,\omega}\mathbf{H} \equiv \nabla \times \left(\varepsilon_0^{-1}\nabla \times \mathbf{H}\right) - \omega^2\mu_0\mathbf{H} = \mathrm{j}\omega\mu_0\mathbf{J}_{\mathrm{imp}} - \varepsilon_0^{-1}\nabla \times \mathbf{J}_{\mathrm{imp}}, \tag{2.111}$$

on $\mathcal{D}(\mathcal{W}_{H,\omega}) \subset \mathcal{H}_{\nabla\times}(\Omega)$ analogously [2]. Equations (2.19) and (2.111) are dual single-field symbols; using one without the other is valid only when the eliminated partner is recoverable from curl relations without ambiguity. In piecewise smooth media with $\varepsilon(\mathbf{r})$, $\mu(\mathbf{r})$,

$$\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\mathbf{E} \equiv \nabla \times \left(\mu^{-1}\nabla \times \mathbf{E}\right) - \omega^2\varepsilon\mathbf{E} = -\mathrm{j}\omega\mu\mathbf{J}_{\mathrm{imp}} + \mu^{-1}\nabla \times \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}, \tag{2.112}$$

$$\mathcal{W}_{H,\omega}^{(\varepsilon,\mu)}\mathbf{H} \equiv \nabla \times \left(\varepsilon^{-1}\nabla \times \mathbf{H}\right) - \omega^2\mu\mathbf{H} = \mathrm{j}\omega\mu\mathbf{J}_{\mathrm{imp}} - \varepsilon^{-1}\nabla \times \mathbf{J}_{\mathrm{imp}}, \tag{2.113}$$

with interface jumps on $\mathcal{D}(\mathcal{W}_{\cdot,\omega}^{(\varepsilon,\mu)})$ fixed by Eq. (1.19) [2]. Material weights modify symbols but not the first-order coupling pattern of Figure 2.5; multi-region Green operators in Section 2.3.5 invert Eqs. (2.112)–(2.113) on composite domains [1]. The local Helmholtz symbol is

$$k(\mathbf{r}) = \omega\sqrt{\mu(\mathbf{r})\varepsilon(\mathbf{r})}, \qquad k_0 = \omega\sqrt{\mu_0\varepsilon_0}, \tag{2.114}$$

from Eq. (1.12) in uniform vacuum [2]. In source-free regions, Eqs. (1.14) apply with $k(\mathbf{r})$ replacing k_0. The symbol $k(\mathbf{r})$ labels propagating versus evanescent wavenumber locally; it does not replace Sommerfeld outgoing selection on open ranges [2]. Evanescent sheets with large $|\mathbf{E}|$ can satisfy the symbol while exporting no watts through Γ_R [6]. Single-field reduction is admissible only under explicit constraints:

$$\text{eliminate } \mathbf{H} \implies \mathbf{J}_{\mathrm{imp}} \in \mathcal{H}_{\nabla\times}(\Omega),\ \rho \text{ compatible},\ \mathcal{B}_\Gamma \text{ expressible in } \mathbf{E}\text{-traces alone}, \tag{2.115}$$

with the magnetic elimination requiring the parallel conditions on $\mathbf{H}$-traces [2]. Failure of Eq. (2.115) —for example impedance boundaries coupling both tangential components—forces retention of the first-order $\mathcal{L}_\omega$ system Eq. (2.16) [1]. Forcing maps between impressed data and Helmholtz right-hand sides are linear:

$$\mathbf{f}_{E,\omega} = -\mathrm{j}\omega\mu\mathbf{J}_{\mathrm{imp}} + \mu^{-1}\nabla \times \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}, \qquad \mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\mathbf{E} = \mathbf{f}_{E,\omega}, \tag{2.116}$$

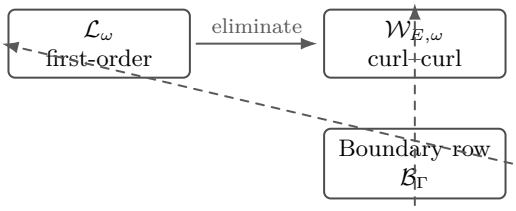


Figure 2.16: First-order $\mathcal{L}_\omega$ retains coupled boundary rows; single-field $\mathcal{W}_{E,\omega}$ is valid only when $\mathcal{B}_\Gamma$ can be projected without losing impedance coupling.

with $\mathbf{f}_{E,\omega} \in L^2(\Omega)^3$ when $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ from Eq. (2.49) [2]. The harmonic solution operator on the electric Helmholtz branch is

$$\mathcal{G}_{E,\omega} : \mathbf{f}_{E,\omega} \longmapsto \mathbf{E}, \qquad \mathcal{W}_{E,\omega}^{(\varepsilon,\mu)} \mathcal{G}_{E,\omega} = \mathrm{id} \tag{2.117}$$

on the physical resolvent branch selected by $\mathcal{B}_\Gamma$; explicit $\mathcal{G}_{E,\omega}$ construction is developed in Section 2.3 [1]. Composition with the full map yields

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = (\mathcal{G}_{E,\omega}\mathbf{f}_{E,\omega}, -\frac{1}{\mathrm{j}\omega\mu}\nabla \times \mathbf{E}) \quad \text{when } \mathbf{H} \text{ recovered from Ampère row,} \tag{2.118}$$

linking Eq. (2.2) to Helmholtz inversion without re-displaying Eq. (2.18) [2]. Spectral viewpoint note: on bounded domains with discrete cavity spectra, $\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}$ admits eigenpairs $(\lambda_n, \mathbf{e}_n)$ accumulating at infinity because $\mathcal{H}_{\nabla\times}(\Omega) \hookrightarrow L^2(\Omega)^3$ compactly (Eq. (2.60)) [2]. On exterior domains, continuous spectrum replaces discrete eigenvalues; Section 2.6.4 develops Fredholm and spectral consequences. Vector spherical harmonic separation of eigenmodes is deferred to Section 3.3 [1]. Impedance and outgoing radiation conditions modify both volume and trace operators simultaneously [2]. Weak form of the electric Helmholtz operator uses the same sesquilinear structure as Eq. (2.21) restricted to **E**-only test spaces when reduction is valid [2]. Coercivity on the quotient by cavity modes from Eq. (2.87) applies to $\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}$ on bounded domains with passive $\mathcal{B}_\Gamma$; Subsection 2.1.4 develops the self-adjoint realizations [1]. Subsection 2.2.2 has packaged harmonic Maxwell residuals, electric and magnetic Helmholtz operators, and the conditions under which single-field reduction is admissible. Subsection 2.2.3 proves that $\mathcal{S}_\omega$ is the frequency-domain image of a causal time-domain map when impressed data, media, and boundary class are held fixed [2].

2.2.3 Causality and time-frequency equivalence

A standard-gain horn on a compact-range turntable fixes the laboratory scene for 'Causality and time-frequency equivalence": frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays below [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. The physical problem is retarded response: no field at $(\mathbf{r}, t)$ may depend on source events that have not yet had time to propagate from $\mathbf{r}'$ to $\mathbf{r}$ [4]. On the range, this is the difference between a measured impulse response that vanishes before the light-travel delay and an unphysical advanced template that leads the feed. A pulsed compact-range measurement makes the distinction visible: energy arrives only after transit time $|\mathbf{r} - \mathbf{r}'|/c_0$; a single-frequency phasor can hide violations that broadband synthesis would expose [1]. Causality is therefore the principle that links time-domain evolution to harmonic analysis before any Laplace or resolvent notation is introduced [2]. A time-domain operator is causal when no response can occur before the corresponding excitation [2]. In electromagnetics this requirement appears through retarded propagation: disturbances generated at one location influence another only after the finite transit time imposed by the speed of light

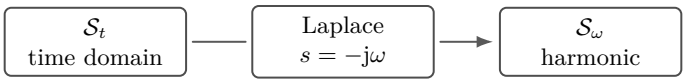


Figure 2.17: Causality link: causal $\mathcal{S}_t$ maps to harmonic $\mathcal{S}_\omega$ under Laplace/Fourier evaluation on the locked phasor axis $s = -\mathrm{j}\omega$, $\omega > 0$.

[10]. Subsection 2.2.3 supplies the definitive proof that $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ are the frequency-domain images of causal time-domain maps when impressed data, media, and boundary class are held fixed [2]. The retarded Eq. (2.10) and symbolic relation Eq. (2.106) are upgraded to an equivalence theorem; Green-function construction remains in Section 2.3 [2]. Causal support. A time-domain operator $\mathcal{K}$ is *causal* if

$$\mathcal{K}[f](t) = 0 \text{ for } t < t_0 \quad \text{whenever } \operatorname{supp}_t f \subseteq [t_0, \infty), \tag{2.119}$$

for admissible test functions f [2]. Retarded propagation forbids response before transit time $|\mathbf{r} - \mathbf{r}'|/c$; frequency-domain Green functions used on ranges are analytic continuations of that support, while broadband pulses expose violations that single-frequency phasors can hide [10]. The retarded dyadic kernel in Eq. (2.102) satisfies

$$\mathcal{G}(\mathbf{r}, \mathbf{r}', t - t') = \mathbf{0} \quad \text{for } t - t' < 0, \tag{2.120}$$

which enforces Eq. (2.103) on $\mathbf{E}$ [2]. Advanced kernels with $\mathcal{G}(\mathbf{r}, \mathbf{r}', t - t') \neq \mathbf{0}$ for $t - t' < 0$ violate Eq. (2.119) and are excluded from the equivalence class treated here. Let $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t)$ be an impressed current with compact support in $\Omega_s \times [0, \infty)$ and let $\mathbf{U}_0 = \mathbf{0}$ for simplicity. The Laplace transform in s with $\mathrm{Re}\{s\}$ sufficiently large is

$$\widetilde{\mathbf{j}}(\mathbf{r}, s) = \int_0^\infty \mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t)\, e^{-st}\, dt. \tag{2.121}$$

On the phasor axis $s = -\mathrm{j}\omega$ with $\omega > 0$,

$$\widetilde{\mathbf{j}}(\mathbf{r}, -\mathrm{j}\omega) = \int_0^\infty \mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t)\, e^{\mathrm{j}\omega t}\, dt, \tag{2.122}$$

which is the Fourier transform restricted to causal support [2]. For $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) = \mathrm{Re}\{\underline{\mathbf{j}}(\mathbf{r}) e^{-\mathrm{j}\omega_0 t}\}$ with fixed $\omega_0 > 0$, Eq. (2.122) selects the phasor $\underline{\mathbf{j}}$ under the locked convention $\exp(-\mathrm{j}\omega t)$ [2]. Applying the Laplace transform to Eq. (2.98) with fixed $\mathcal{B}_\Gamma$ and homogeneous initial data yields, formally,

$$\widetilde{\mathbf{U}}(\mathbf{r}, s) = \mathcal{R}(s)\, \widetilde{\mathbf{F}}(s), \qquad \mathcal{R}(s) = \mathcal{L}_s^{-1}, \tag{2.123}$$

where $\mathcal{L}_s$ is the Laplace-domain Maxwell operator obtained from $\mathcal{A}$ by Eq. (2.108) with $\partial_t \to s$ [2]. On the axis $s = -\mathrm{j}\omega$, $\mathcal{L}_s = \mathcal{L}_\omega$ from Eq. (2.65), so

$$\mathcal{R}(-\mathrm{j}\omega) = \mathcal{L}_\omega^{-1} \quad \text{on the outgoing/resolvent branch,} \tag{2.124}$$

and the retarded convolution Eq. (2.102) is the inverse Laplace transform of Eq. (2.123) when $\mathcal{G}$ is the causal Green operator constructed in Section 2.3 [2]. Equation (2.124) is the operator form of Eq. (2.10): harmonic inversion is not an independent postulate but the $s = -\mathrm{j}\omega$ evaluation of a causal resolvent. The equivalence theorem requires explicit assumptions:

(i) $\mathcal{B}_\Gamma$ and ε, μ fixed in time,
(ii) $\mathbf{j}_{\mathrm{imp}}$ causal with $\operatorname{supp}_t \subseteq [0, \infty)$,
(iii) outgoing/resolvent branch selected for exterior domains,
(iv) $\mathbf{U}_0 = \mathbf{0}$ or compatible with harmonic steady limit. (2.125)

Under Eqs. (2.125),

$$\mathcal{S}_\omega[\underline{\mathbf{j}}] = \widetilde{\mathbf{U}}(\mathbf{r}, s)\Big|_{s=-\mathrm{j}\omega} = \mathcal{L}_\omega^{-1}\widetilde{\mathbf{F}}(s)\Big|_{s=-\mathrm{j}\omega}, \tag{2.126}$$

where $\underline{\mathbf{j}}$ is the phasor of $\mathbf{j}_{\mathrm{imp}}(\mathbf{r}, t) = \mathrm{Re}\{\underline{\mathbf{j}}(\mathbf{r})e^{-\mathrm{j}\omega t}\}$ and $\widetilde{\mathbf{F}}$ is the Laplace forcing induced by $\widetilde{\mathbf{j}}$ [2]. Equation (2.126) is the formal statement that frequency-domain solvers implement causal time-domain maps when Eqs. (2.125) hold [1]. For single-frequency excitation with damping or outgoing radiation,

$$\lim_{t\to\infty} \mathbf{U}(t) = \mathrm{Re}\left\{\mathcal{S}_\omega[\underline{\mathbf{j}}]\, e^{-\mathrm{j}\omega t}\right\}, \tag{2.127}$$

which justifies steady-state phasor measurements after switch-on transients decay [2]. Because Π_{rad} is linear and commutes with harmonic extraction on the outgoing subspace,

$$\mathcal{R}_\omega[\underline{\mathbf{j}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\underline{\mathbf{j}}], \tag{2.128}$$

with transport and regime qualifications from Eq. (2.28) unchanged [2]. Frequency-domain codes that invert $\mathcal{L}_\omega$ without outgoing or passive completion implement a different resolvent branch than the causal retarded $\mathcal{G}$ of Eq. (2.102) [1]. Advanced Green functions satisfy

$$\mathcal{G}_{\mathrm{adv}}(\mathbf{r}, \mathbf{r}', t - t') \neq \mathbf{0} \text{ for some } t - t' < 0, \tag{2.129}$$

and cannot satisfy Eq. (2.126) for any causal $\mathbf{j}_{\mathrm{imp}}$ [2]. Broadband synthesis requires superposition over ω, and time-varying $\mathcal{B}_\Gamma(t)$ breaks assumption (i) in Eq. (2.125) [2]. Subsections 2.2.1–2.2.3 fixed evolution, harmonic operators, and causality. Subsection 2.2.3 supplies spectral partitions into propagating, evanescent, and non-propagating subspaces in operator form [2]. Reactive versus radiative regime criteria from Section 1.3.3 are invoked by reference; the outgoing selector Π_{rad} of Eq. (2.3) is refined spectrally, not replaced [5]. The physical problem is spectral transport: a near-field scanner records both traveling and decaying longitudinal content, yet only outgoing propagating modes may enter far-zone power reports [7]. Subsection 2.2.3 defines propagating, evanescent, and non-propagating spectral subspaces and refines $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$ so that only outgoing propagating content qualifies for far-zone transport reports [2]. Recall that $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ produces the full harmonic field pair, $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ retains only radiative content, and Π_{rad} acts after propagating selection [1]. A near-field scanner routinely measures both propagating and evanescent spectral content. The distinction becomes visible when reconstructed fields are propagated away from the measurement surface: propagating components persist, whereas evanescent components decay exponentially with distance [7]. The spectral decomposition developed below states that experimentally familiar observation in spectral language. In a homogeneous region with symbol k_0 from Eq. (2.114), plane-wave trials decompose the field into lateral and longitudinal wavenumber components. The dispersion relation below is the local criterion that separates traveling from decaying longitudinal behavior: when lateral wavenumber $k_t = \sqrt{k_x^2 + k_y^2}$ exceeds k_0, no real propagating longitudinal root exists and the mode becomes evanescent [2]. Experimentally one would observe that near-field scan content with $k_t > k_0$ dies away within a penetration depth $\delta_{\mathrm{ev}} = 1/\alpha(k_t)$ as the field is propagated outward [7]. Plane-wave trials $\mathbf{u}_{\mathbf{k}}(\mathbf{r}) = \mathbf{u}_0\, e^{-\mathrm{j}\mathbf{k}\cdot\mathbf{r}}$ satisfy the Helmholtz condition when

$$|\mathbf{k}| = k_0, \qquad k_z^2 = k_0^2 - k_x^2 - k_y^2, \tag{2.130}$$

with lateral wavenumber $k_t = \sqrt{k_x^2 + k_y^2}$ [2]. Equation (2.130) splits spectral content by longitudinal behavior:

$$k_z = \begin{cases} \pm\sqrt{k_0^2 - k_t^2}, & k_t \le k_0 \text{ (propagating)}, \\ -\mathrm{j}\,\alpha(k_t), & k_t > k_0 \text{ (evanescent)},\ \alpha > 0, \end{cases} \tag{2.131}$$

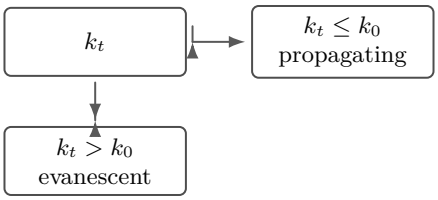


Figure 2.18: Dispersion split Eq. (2.131): lateral wavenumber k_t selects propagating versus evanescent longitudinal behavior at fixed k_0.

where the evanescent root is chosen with decay in the $+z$ half-space [2]. For $k_t > k_0$ the decay rate is

$$\alpha(k_t) = \sqrt{k_t^2 - k_0^2}, \tag{2.132}$$

so a pure evanescent trial on $z > 0$ scales as $e^{-\alpha(k_t)z}$ with penetration depth $\delta_{\text{ev}} = 1/\alpha(k_t)$ [2]. Propagating modes carry real $k_z = \pm\sqrt{k_0^2 - k_t^2}$ and oscillatory phase along the longitudinal axis; evanescent modes carry imaginary k_z and store reactive energy in the near zone without far-zone power export [1]. At fixed (k_x, k_y), the Helmholtz symbol $\mathcal{L}_\omega$ from Subsection 2.2.2 acts on each longitudinal branch as a scalar multiplier in the plane-wave basis, which is why spectral partitions commute with harmonic extraction once outgoing radiation condition is fixed on the propagating sheet [2]. Transverse polarization completes the mode description. For a trial electric field $\widetilde{\mathbf{E}}(k_x, k_y)\, e^{-\mathrm{j}(k_x x + k_y y + k_z z)}$, source-free admissibility requires $\mathbf{k} \cdot \widetilde{\mathbf{E}} = 0$ in the homogeneous region [2]. Decompose $\mathbf{k} = \mathbf{k}_t + k_z \hat{\mathbf{z}}$ with $\mathbf{k}_t = k_x \hat{\mathbf{x}} + k_y \hat{\mathbf{y}}$. Then

$$k_x \widetilde{E}_x + k_y \widetilde{E}_y + k_z \widetilde{E}_z = 0, \tag{2.133}$$

which leaves two independent transverse amplitudes on each branch of Eq. (2.131). On the propagating sheet, Eq. (2.133) supports traveling power along $\hat{\mathbf{z}}$ when outgoing k_z is selected; on the evanescent sheet the same constraint forces predominantly tangential storage with longitudinal E_z tied to $\alpha(k_t)$ [2]. Subsection 2.2.3 does not classify TE/TM guide nomenclature—that labeling is deferred to waveguide sections—but Eq. (2.133) is the local polarization criterion paired with the dispersion criterion of Eq. (2.130) [1]. Angular-spectrum representations in Section 2.9 expand fields in these modes; here the subspaces are defined as solution domains [2]. Let $\mathcal{F}_{\text{prop}} \subset \mathcal{F}_\Omega$ be the completion of finite superpositions of propagating plane waves with outgoing k_z selected by exterior completion, let $\mathcal{F}_{\text{ev}} \subset \mathcal{F}_\Omega$ be the closure of evanescent superpositions, and let $\mathcal{F}_{\text{non}} \subset \mathcal{F}_\Omega$ collect modes that are neither outgoing-propagating nor evanescent-decaying in the declared half-space—for example cutoff guide modes and purely reactive standing content tied to terminals [2]. Orthogonal spectral projectors (formal) satisfy

$$\Pi_{\text{prop}} + \Pi_{\text{ev}} + \Pi_{\text{non}} = \mathrm{id} \quad \text{on } \widehat{\mathcal{F}}_\Omega, \tag{2.134}$$

with $\widehat{\mathcal{F}}_\Omega$ the spectral completion of admissible states [2]. Idempotency and mutual annihilation read

$$\Pi_{\text{prop}}^2 = \Pi_{\text{prop}}, \quad \Pi_{\text{ev}}^2 = \Pi_{\text{ev}}, \quad \Pi_{\text{non}}^2 = \Pi_{\text{non}}, \quad \Pi_{\text{prop}}\Pi_{\text{ev}} = \Pi_{\text{ev}}\Pi_{\text{prop}} = 0, \tag{2.135}$$

with analogous zero products for Π_{non} [2]. On $\mathcal{D}(\mathcal{L}_\omega)$ from Subsection 2.2.2, each projector block is an invariant subspace of the scalar Helmholtz symbol in the plane-wave basis, Equations (2.134)–(2.135) are *formal* on the spectral completion $\widehat{\mathcal{F}}_\Omega$: idempotency and orthogonality are justified in the rigged Hilbert space treatment of Chapter 3 (*Angular Momentum Operators and SO(3)*)

Subsections 3.1.3–3.5.2 [2]. Until that completion, projector algebra is used only on admissible finite superpositions and on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, where outgoing selection is already enforced [1].

$$\mathcal{L}_\omega(\Pi_{\mathrm{prop}}\mathbf{u}) = \Pi_{\mathrm{prop}}(\mathcal{L}_\omega\mathbf{u}), \tag{2.136}$$

and similarly for Π_{ev} and Π_{non} when boundary closure does not mix sheets [1]. Equation (2.136) is the operator reason harmonic solvers can treat propagating and evanescent content separately before radiation selection; Sommerfeld outgoing radiation condition in Section 2.4 acts only on Π_{prop} [2]. The decomposition

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] + \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] + \Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \tag{2.137}$$

holds when the impressed spectrum excites all three classes. Only $\Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is eligible for far-zone radiation content after outgoing radiation condition; evanescent and non-propagating components remain in reactive storage exchange [2]. Energy norms from Eq. (2.48) decompose orthogonally in the spectral basis when the impressed angular spectrum has sufficient decay in (k_x, k_y) [2].

$$\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}, \tag{2.138}$$

where $\mathcal{O}_{\mathrm{out}}$ enforces outgoing selection on propagating content compatible with Eq. (1.21) [2]. Equation (2.138) refines Π_{rad}: radiative subspace $\mathcal{F}_{\mathrm{rad}}$ is a subset of $\Pi_{\mathrm{prop}}\mathcal{F}_\Omega$, not of all of $\mathcal{F}_\Omega$ [1]. Evanescent and non-propagating components lie in the null space of far-zone radiation:

$$\Pi_{\mathrm{rad}}\Pi_{\mathrm{ev}} = 0, \qquad \Pi_{\mathrm{rad}}\Pi_{\mathrm{non}} = 0, \tag{2.139}$$

while $\Pi_{\mathrm{rad}}\Pi_{\mathrm{prop}} = \Pi_{\mathrm{rad}}$ after outgoing closure [2]. Equations (2.139) and (2.138) restate, in projector form, the regime logic of Section 1.3.3: reactive near fields may be large even when $\Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ [1]. Causality from Subsection 2.2.3 ensures retarded support but does not, by itself, promote evanescent content into $\mathcal{F}_{\mathrm{rad}}$. For all $r > r_{\min}(a, \lambda_0)$ and every transport, flux, near-to-far, and angular-spectrum statement in Sections 2.5–2.10,

$$\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}} \quad \text{as in Eq. (2.138).}$$

Earlier uses of Π_{rad} in Section 2.0 are to be read through this refinement [2]. Transport qualification from Section 1.3.3 enters as an operator criterion:

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{transport reports use } \Pi_{\mathrm{prop}}\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \text{ not } \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.140}$$

with $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.34) [2]. Equation (2.140) is the operator restatement of Eq. (2.28); regime indices $\chi(r)$ and χ_Ω diagnose when Eq. (2.140) fails on a measurement shell [2]. Waveguide and cutoff structures illustrate non-propagating subspaces. For a uniform guide with lateral eigenvalue κ_n, mode n propagates only when

$$k_0^2 > \kappa_n^2 \quad \Longleftrightarrow \quad \omega > \omega_{c,n} = \frac{c_0\kappa_n}{\sqrt{\varepsilon_r\mu_r}}, \tag{2.141}$$

with $\omega_{c,n}$ the cutoff angular frequency [2]. Below cutoff, the mode lies in $\mathcal{F}_{\mathrm{ev}}$ or $\mathcal{F}_{\mathrm{non}}$ depending on boundary termination; applying far-zone transport formulas to such fields violates Eq. (2.140) as in the TE_{10} illustration of Section 1.1.2 [1]. In a rectangular guide of width a, lateral eigenvalues $\kappa_{m,n} = \sqrt{(m\pi/a)^2 + (n\pi/b)^2}$ replace k_t in Eq. (2.141); each mode index (m, n) switches from $\mathcal{F}_{\mathrm{non}}$ to $\mathcal{F}_{\mathrm{prop}}$ only when ω exceeds its cutoff [2]. Near cutoff the propagating group velocity vanishes and

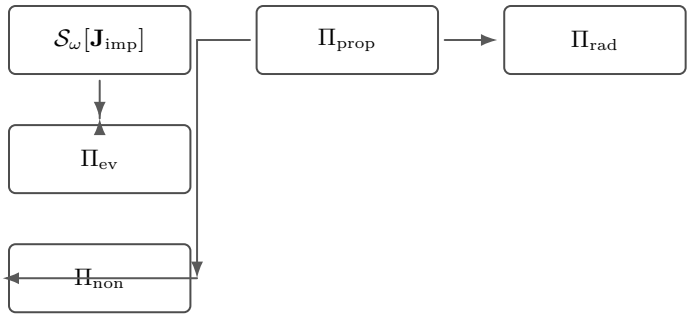


Figure 2.19: Spectral criterion: only Π_{prop} content passes to Π_{rad}; evanescent and non-propagating components remain reactive.

stored reactive energy dominates even though the mode has crossed into Π_{prop}; Eq. (2.32) remains the far-zone transport criterion, not Eq. (2.141) alone [1]. Angular-spectrum expansion: expand $\mathbf{E}(\mathbf{r})$ on $z = \text{const}$ as

$$\mathbf{E}(\mathbf{r}) = \iint \widetilde{\mathbf{E}}(k_x, k_y)\, e^{-\mathrm{j}(k_x x + k_y y + k_z(k_t) z)}\, dk_x\, dk_y, \tag{2.142}$$

with $k_z(k_t)$ from Eq. (2.131) [2]. Equation (2.142) is the Chapter 2 spectral template for Section 2.9; full radiation operator action on angular spectra is developed there. Subsection 2.2.3 only defines the propagating/evanescent partition induced by Eq. (2.130) [1]. Figure 2.2. Causality from Subsection 2.2.3 does not automatically place evanescent energy in $\mathcal{F}_{\mathrm{rad}}$; spectral and regime qualifications are additional [2]. Near-zone reactive fields decompose into large evanescent and non-propagating fractions even when $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is exact. As $r \to \infty$, Π_{prop} content dominates P_{rad} when Eq. (2.32) holds [2]. Operator norms on subspaces satisfy

$$\|\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \gg \|\Pi_{\mathrm{prop}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{U}} \quad \text{on near shells,} \tag{2.143}$$

while the reverse inequality governs far-zone radiative observables [1]. Subsection 2.2.3 does not compute shell integrals; it supplies the spectral vocabulary for Section 2.5.2 and flux operators in Section 2.5. With Subsections 2.2.1–2.2.3 complete, Section 2.2 has connected time-domain evolution, harmonic Maxwell and Helmholtz operators, causal equivalence, and propagating–evanescent spectral split on the function spaces of Section 2.1. Green-function inversion in Section 2.3 realizes $\mathcal{L}_\omega^{-1}$ and $\mathcal{G}_{E,\omega}$ with retarded causality; Sommerfeld selection in Section 2.4 fixes outgoing branches for Π_{prop} and Π_{rad}; radiation integrals in Section 2.5 meter flux on the propagating subspace qualified by Eq. (2.140) [2].

2.3 Green Functions and Electromagnetic Inversion

Time- and frequency-domain operators in Section 2.2 specify how impressed currents enter Maxwell's equations. Section 2.3 supplies Green-function propagation—the computational heart of method-of-moments and near-field scanners [2]. Propagation must be understood before radiation is selected [10].

Maxwell's equations couple impressed currents to fields throughout a domain; in practice one needs the field at a specified observation point produced by sources elsewhere. The Green function is the impulse response that reduces that coupling to a single propagator integral rather than a fresh boundary-value solve at every new observation coordinate [2]. Before dyadic kernels are tabulated, fix the lowest-dimensional laboratory calibration every range engineer uses: a short dipole or probe

feed at $k_0 a \ll 1$ on a turntable, with watt closure on a sphere Γ_R and a transverse $\sin\theta$ pattern that is invariant under rotation [8]. That Hertzian test is the physical image of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when $\mathbf{J}_{\mathrm{imp}}$ collapses to a point dipole moment; Subsection 2.3.2 closes the calculation, but the range observation precedes the dyad formulas [2]. Two laboratory questions follow. Given currents on a feed or equivalent surface, how is the field computed at an observation point? Given field samples on a scan surface or range contour, how may the equivalent source be recovered? Method-of-moments assembly, near-field back-propagation, compact-range transformation, and microwave imaging differ in discretization yet implement the same propagation law [1]. Scalar, vector, and dyadic kernels, singular regularization, retarded causality, and inversion on composite domains are developed below; outgoing selection at infinity belongs to Section 2.4. The export map below $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ realized through G and $\overline{\mathbf{G}}_E$ is the physical response of the medium to declared impressed currents; radiation export through $\mathcal{R}_\omega$ and Π_{rad} is deferred until Section 2.4 has fixed the outgoing branch on which those operators act [2].

2.3.1 Scalar, vector, and dyadic Green functions

Physically, G and $\overline{\mathbf{G}}_E$ are the medium's response functions: each records how a unit impulse at $\mathbf{r}'$ launches scalar or vector fields to $\mathbf{r}$ under the same boundary completion as $\mathcal{L}_\omega$ [2]. Every compact-range campaign begins with a dipole elementary check: a electrically small feed must reproduce the transverse far-zone law before multipole catalogs are trusted on larger apertures [8]. That observation anchors the Green-function program developed here—first the scalar monopole response, then vector potentials, then the electric dyad that maps $\mathbf{J}_{\mathrm{imp}}$ to $\mathbf{E}$ directly [2]. The dyadic Green function is the electromagnetic impulse response of the governing Maxwell operator: the field at an observation point produced by an elemental current located elsewhere. In uniform media it is also the continuous limit of one radiating cell in a phased array and the local kernel that probe correction applies when near-field data are translated to a reference surface [2]. Admissible kernels are retarded—field response follows source excitation only after finite propagation delay—so localized impressed currents remain consistent with Maxwell causality [10]. Scalar Helmholtz functions describe monopole-like propagation; polarization coupling on vector feeds requires the full dyadic [11].

The *kernel formulas* developed below must be distinguished from and the *domain declarations* that make those formulas meaningful on a declared boundary-value problem [2]. Unless noted otherwise, $\omega > 0$ is fixed, $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ from Eq. (2.49), charge continuity Eq. (1.8) holds, and $\mathcal{B}_\Gamma$ is declared on $\Gamma = \partial\Omega$. In uniform vacuum regions the wavenumber is k_0 from Eq. (2.114). Kernels are stated first in operator form on $\mathcal{D}(\mathcal{W}_{E,\omega}) \subset \mathcal{H}_{\nabla\times}(\Omega)$ from Eq. (2.110); piecewise $k(\mathbf{r})$ extensions use the same template with patchwise coefficients [2]. The harmonic solution operator $\mathcal{G}_{E,\omega}$ introduced in Subsection 2.2.2 is realized below by convolution with G and $\overline{\mathbf{G}}_E$; observation coordinate $\mathbf{r}$ and source coordinate $\mathbf{r}'$ must remain distinct throughout every range transformation [1]. Imagine exciting a uniform medium with the smallest source one can conceive: a unit impulse concentrated at a single point $\mathbf{r}'$. The field that appears at every other location $\mathbf{r}$ is the scalar Green function $G(\mathbf{r}, \mathbf{r}')$—the impulse response of the Helmholtz operator at fixed frequency [2]. Laboratory engineers never place a mathematical δ-function on a bench, yet every extended feed, patch, or volume current is assembled from such elemental responses by superposition: each infinitesimal slice of $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ radiates as if it were a point source scaled by the local source strength, and the total field is the sum of those contributions weighted by propagation from $\mathbf{r}'$ to $\mathbf{r}$ [1]. Green functions are therefore the natural language of propagation: they replace repeated boundary-value solves with a single propagator integral that method-of-moments codes, near-field scanners, and imaging algorithms all discretize [2]. The physical significance of Eq. (2.144) is that it isolates the

response of the medium to the smallest source that can be imagined—a unit point excitation. Once this elementary response is known, more complicated sources may be synthesized by superposition [8]. The scalar Green function $G(\mathbf{r}, \mathbf{r}')$ inverts the Helmholtz operator $(\nabla^2 + k_0^2)$ on scalar potentials; $\mathbf{r}$ denotes the observation point and $\mathbf{r}'$ the source point, with $R = |\mathbf{r} - \mathbf{r}'|$ the geometric separation in metres. On a compact range, the scalar kernel is the monopole reference against which probe calibration and layered-substrate corrections are measured [7]. It satisfies

$$(\nabla^2 + k_0^2)\, G(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}'), \tag{2.144}$$

with δ the three-dimensional Dirac distribution and boundary conditions on Γ inherited from $\mathcal{B}_\Gamma$ [2]. Equation (2.144) is the local symbol of spectral inversion: for fixed $\mathbf{r}'$, $G(\cdot, \mathbf{r}')$ is the response to a unit point source at $\mathbf{r}'$ under the same completion as $\mathcal{L}_\omega$ [8]. Convolution with G therefore propagates an arbitrary scalar source to the field ϕ that satisfies the inhomogeneous Helmholtz equation on the same domain. The associated solution operator is

$$\phi(\mathbf{r}) = \int_\Omega G(\mathbf{r}, \mathbf{r}')\, q(\mathbf{r}')\, d^3 r', \qquad (\nabla^2 + k_0^2)\phi = q, \tag{2.145}$$

for scalar sources q in the dual class paired with ϕ [2]. Readers should interpret Eq. (2.145) as a weighted accumulation of responses arriving from every source point inside the domain Ω. The Green function determines how strongly a source at $\mathbf{r}'$ influences the observation point $\mathbf{r}$, while $q(\mathbf{r}')$ specifies how much source strength is available at that location [2]. In plain language: $\phi(\mathbf{r})$ is the scalar field one would measure or compute at the observation point; $q(\mathbf{r}')$ is the source density distributed throughout the impressed region Ω; and $d^3 r'$ sums contributions from every infinitesimal source element [8]. Mathematically, $\phi(\mathbf{r})$ carries units that depend on the physical quantity represented (potential, pressure, or a component precursor to $\mathbf{E}$); $q(\mathbf{r}')$ is the corresponding source density per cubic metre; integration extends only over Ω, the region where sources are nonzero [2]. On exterior domains, G must satisfy outgoing decay compatible with Section 2.4; on bounded domains, Dirichlet, Neumann, or impedance data on Γ replace the free-space radiation template [1]. Before examining Eq. (2.146) itself, it is useful to recall the familiar physical picture of a disturbance spreading outward from a localized source. An expanding spherical wavefront carries phase $e^{-\mathrm{j}k_0 R}$ and amplitude dilution proportional to $1/R$; the same geometric spreading responsible for the inverse-square law in power appears here as the inverse-distance amplitude factor [8]. Phase accumulates linearly with propagation distance R, and outgoing energy transport requires the retarded phase sign fixed by the locked phasor convention $\exp(-\mathrm{j}\omega t)$ [2]. In unbounded vacuum with outgoing selection at infinity, Sommerfeld radiation at large R singles out the spherical wave whose amplitude falls as $1/R$ and whose phase advances with propagation distance.

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{e^{-\mathrm{j}k_0 R}}{4\pi R}, \qquad R = |\mathbf{r} - \mathbf{r}'|, \tag{2.146}$$

with $R > 0$ and the branch of k_0 fixed by the locked phasor axis [2]. Equation (2.146) satisfies Eq. (2.144) for $\mathbf{r} \neq \mathbf{r}'$; the distributional definition at coincidence is completed in Subsection 2.3.3. Physically, Eq. (2.146) is the spherical outgoing wave produced by a unit monopole source at $\mathbf{r}'$; its phase gradient defines local propagation direction on the propagating sheet of Subsection 2.2.3 [2]. Green kernels realize the resolvent Eq. (2.67) developed in Section 2.2: the scalar and dyadic definitions below implement $\mathcal{G}_{E,\omega}$ of Eq. (2.117) and feed the radiation integral Eq. (2.5) [2]. Electric-wave operators and material-weighted symbols from Subsection 2.2.2 are presupposed; gauge enters only through the Lorenz reduction below. Coincidence limits belong to Subsection 2.3.3; outgoing completion at infinity to Section 2.4. Vector potential and vector kernels. Direct field inversion from

measured $\mathbf{E}$ alone is difficult because Maxwell's equations couple $\mathbf{E}$ and $\mathbf{H}$ through curls, material response, and boundary data; the map from sources to fields is nonlocal and orientation-dependent [2]. Vector potentials are introduced not as additional fields one would place on a probe calibration sheet but as mathematical intermediaries that simplify reconstruction. In Lorenz gauge, each Cartesian component of $\mathbf{A}$ satisfies the scalar Helmholtz template of Eq. (2.144) with source $-\mu_0\mathbf{J}_{\mathrm{imp}}$ [8]. The vector potential is not an additional physical field but as a mathematical intermediary that simplifies the reconstruction of electromagnetic quantities; gauge choices remove redundant degrees of freedom so that propagation integrals take a standard convolution form while what an engineer measures remains $\mathbf{E}$ and $\mathbf{H}$ [2]. Lorenz-gauge vector potentials $\mathbf{A}$ reduce vector Maxwell inversion to a vector Helmholtz system driven by $\mathbf{J}_{\mathrm{imp}}$. Under the gauge and continuity constraints fixed in Chapter 1, each Cartesian component of $\mathbf{A}$ satisfies Eq. (2.144) with k_0 and source $-\mu_0\mathbf{J}_{\mathrm{imp}}$ [2]. In Lorenz gauge the vector potential $\mathbf{A}(\mathbf{r})$ carries units of $\mathrm{T\,m}$ (equivalently $\mathrm{Wb\,m^{-1}}$ in SI); each Cartesian component satisfies the scalar Helmholtz template with source $-\mu_0\mathbf{J}_{\mathrm{imp}}$. The electric field phasor follows from $\mathbf{A}$ and its divergence through

$$\mathbf{E}(\mathbf{r}) = \mathrm{j}\omega\mu_0\,\mathbf{A}(\mathbf{r}) + \frac{1}{\mathrm{j}\omega\varepsilon_0}\,\nabla\,\nabla\cdot\mathbf{A}(\mathbf{r}), \tag{2.147}$$

with $\mathbf{E}$ in $\mathrm{V\,m^{-1}}$, ε_0 the vacuum permittivity ($\mathrm{F\,m^{-1}}$), and $\mathbf{H}$ obtained from $\nabla\times\mathbf{E}$ when the first-order system is required [1]. Equation (2.147) is a vector reduction tool, not a substitute for the dyadic definition below when $\mathcal{B}_\Gamma$ couples $\mathbf{E}$ and $\mathbf{H}$ at boundaries: impedance and PEC closures are enforced on $\mathcal{L}_\omega$, not on $\mathbf{A}$ alone [2]. A vector Green function $\mathbf{G}^A(\mathbf{r},\mathbf{r}')$ may be assembled from three scalar kernels; the electric dyadic is preferred when $\mathbf{J}_{\mathrm{imp}}$ is the primary excitation because it maps $\mathbf{J}_{\mathrm{imp}}$ to $\mathbf{E}$ directly without intermediate gauge potentials [1]. The scalar Green function describes how a point disturbance propagates. Real electromagnetic sources, however, possess orientation and polarization. A current element directed along $\hat{\mathbf{x}}$ generally produces field components in multiple directions. The scalar kernel is therefore insufficient, and the propagation law must be upgraded to a tensorial form [2]. Let $\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}')$ denote a rank-two tensor (dyad) acting on vector sources by dot product from the left; $\mathbf{r}$ is the observation coordinate and $\mathbf{r}'$ the source coordinate. The electric dyadic Green function answers, in dyadic form, the impulse-response question: what electric field $\mathbf{E}(\mathbf{r})$ is produced at $\mathbf{r}$ when a vector current impulse is placed at $\mathbf{r}'$? Method-of-moments assembly implements Eq. (2.149) by testing this kernel against patch basis functions; near-field scanners apply the same propagation law when translating probe-corrected samples to a reference surface [1]. Readers should notice that the dyad encodes polarization coupling absent from the scalar G: a transverse feed excites both radiative and reactive dyadic components, which is why the Hertzian dipole test at $k_0a \ll 1$ remains the lowest-dimensional calibration before aperture catalogs are trusted [8]. It inverts the electric-wave operator Eq. (2.19) in the sense

$$\mathcal{W}_{E,\omega}\,\overline{\mathbf{G}}_E(\cdot,\mathbf{r}') = \bar{\mathbf{I}}\,\delta(\mathbf{r}-\mathbf{r}'), \tag{2.148}$$

where $\bar{\mathbf{I}}$ is the identity dyad and the operator $\mathcal{W}_{E,\omega}$ acts on the $\mathbf{r}$-dependence of $\overline{\mathbf{G}}_E$ [2]. Equation (2.148) is the dyadic analogue of Eq. (2.144): each column of $\overline{\mathbf{G}}_E$ is the electric field produced by an impulsive vector source aligned with a Cartesian unit vector at $\mathbf{r}'$ [8]. Every method-of-moments code and every near-field scanner relies on the convolution structure below: given an impressed current distribution $\mathbf{J}_{\mathrm{imp}}$, the electric field at observation points follows by superposition through the dyadic propagator $\overline{\mathbf{G}}_E$. The physical measurement problem is forward prediction—what field does a known or assumed current produce?—which is the prerequisite for calibration, coupling analysis, and array synthesis [1]. Inversion asks the inverse question from sparse $\mathbf{E}$ samples and is substantially harder even though forward propagation is linear and explicit [7]. Superposition over

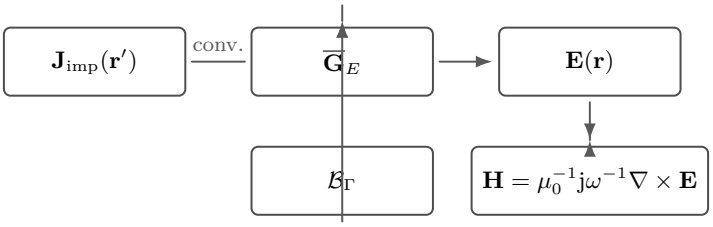


Figure 2.20: Dyadic inversion chain Eq. (2.149): impressed current convolves with $\overline{\mathbf{G}}_E$ fixed by $\mathcal{B}_\Gamma$; $\mathbf{H}$ follows from curl relations when reduction is admissible.

an extended impressed current $\mathbf{J}_{\mathrm{imp}}$ then yields the electric field through the dyadic convolution below; the prefactor $-\mathrm{j}\omega\mu_0$ ties the integral to the locked phasor convention $\exp(-\mathrm{j}\omega t)$. The electric field due to impressed current is

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \tag{2.149}$$

The importance of Eq. (2.149) lies not in the integral itself but in the asymmetry between prediction and reconstruction. Given $\mathbf{J}_{\mathrm{imp}}$, the field follows directly. Given only sparse measurements of $\mathbf{E}$, however, recovering $\mathbf{J}_{\mathrm{imp}}$ becomes substantially more difficult [7]. Here $\mathbf{E}_{\mathrm{hom}}$ is the homogeneous solution fixed by $\mathcal{B}_\Gamma$ [2]. In Eq. (2.149), $\mathbf{E}(\mathbf{r})$ is the electric field phasor in $\mathrm{V\,m^{-1}}$, $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ is the impressed current density in $\mathrm{A\,m^{-2}}$, μ_0 is the vacuum permeability ($\mathrm{H\,m^{-1}}$), and ω is the angular frequency in $\mathrm{rad\,s^{-1}}$; the volume integral extends over the source region Ω, and $\mathbf{E}_{\mathrm{hom}}$ collects excitations not attributable to $\mathbf{J}_{\mathrm{imp}}$, including incident fields and cavity modes fixed by $\mathcal{B}_\Gamma$. Equation (2.149) is the constructive export map every method-of-moments impedance matrix discretizes: patch currents $\mathbf{J}_{\mathrm{imp}}$ tested against $\overline{\mathbf{G}}_E$ produce the $\mathbf{E}$ that near-field scanners back-propagate to equivalent surfaces [1]. The consequence for inversion is immediate: specifying $\mathbf{J}_{\mathrm{imp}}$ determines $\mathbf{E}$ through a single propagator integral, while recovering $\mathbf{J}_{\mathrm{imp}}$ from sparse $\mathbf{E}$ samples is ill-posed and requires the regularization program of Subsection 2.3.3 [2]. Equation (2.149) is the formal realization of Eq. (2.5) with outgoing dyad $\mathbf{\Gamma} = \overline{\mathbf{G}}_E$ on the physical branch [1]. The prefactor $-\mathrm{j}\omega\mu_0$ matches the locked phasor convention and the forcing map Eq. (2.116). The operator $\mathcal{G}_{E,\omega}$ in Eq. (2.150) should be read as the machine that converts admissible forcing functions into electromagnetic responses: it accepts a current distribution in the forcing class and returns the corresponding field throughout the observation domain [1]. The integral operator $\mathcal{G}_{E,\omega}$ of Eq. (2.117) is realized by

$$(\mathcal{G}_{E,\omega}\mathbf{f})(\mathbf{r}) = \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{f}(\mathbf{r}')\, d^3r', \tag{2.150}$$

for $\mathbf{f}$ in the forcing class of Eq. (2.116) [2]. When $\mathbf{f}_{E,\omega}$ is the electric Helmholtz drive induced by $\mathbf{J}_{\mathrm{imp}}$, Eqs. (2.149) and (2.150) coincide up to the linear map from $\mathbf{J}_{\mathrm{imp}}$ to $\mathbf{f}_{E,\omega}$ [1]. The solution map $\mathcal{S}_\omega$ is not merely an abstract mapping notation—it represents the complete electromagnetic experiment beginning with impressed current and ending with observable fields, including homogeneous background $\mathbf{E}^{\mathrm{hom}}$ and boundary data $\mathcal{B}_\Gamma$ when present [2]. The solution map on impressed currents is therefore

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{u}_\omega \quad \text{with} \quad \mathbf{E} \text{ from Eq. (2.149)},\ \mathbf{H} \text{ from } \nabla \times \mathbf{E} \text{ when single-field reduction is valid}, \tag{2.151}$$

on domains satisfying Eq. (2.115); otherwise the first-order dyadic system of Section 2.3.5 applies [2]. Equation (2.149) is a Fredholm integral of the first kind when $\mathbf{r}$ is fixed and $\mathbf{J}_{\mathrm{imp}}$ is unknown, and a constructive export map when $\mathbf{J}_{\mathrm{imp}}$ is specified. The prefactor $-\mathrm{j}\omega\mu_0$ ties the kernel to the locked phasor axis $\exp(-\mathrm{j}\omega t)$; reversing the sign convention would flip incoming versus outgoing

content without changing the local curl–curl symbol. In vacuum, for $k_0R = 3$ with $R = |\mathbf{r}-\mathbf{r}'|$, the transverse dyadic component of Eq. (2.152) contributes $\approx 70\%$ of $|\overline{\mathbf{G}}_{E,0}|$ in operator norm, while the R^{-2} and R^{-3} reactive terms still matter for $k_0a \gtrsim 1$ near-zone scans [2]. Boundary data $\mathcal{B}_\Gamma$ enter through the kernel that satisfies the same traces, not through a separate post-multiplication on $\mathbf{E}$ alone [1]. In uniform vacuum with outgoing condition at infinity, let $\mathbf{R} = \mathbf{r}-\mathbf{r}'$, $R = |\mathbf{R}|$, and $\widehat{\mathbf{R}} = \mathbf{R}/R$. When observation and source points are separated ($R > 0$), the outgoing vacuum dyad may be written explicitly in terms of these geometric quantities. The projector $\bar{\mathbf{I}} - \widehat{\mathbf{R}}\widehat{\mathbf{R}}$ represents transverse radiation: only field components perpendicular to the separation vector carry energy to infinity in the far zone [2]. The inverse-power corrections in k_0R represent stored reactive energy—magnetic and electric energy oscillating locally without net outward transport at short range. An engineer measuring fields close to an antenna would observe significant contributions from these reactive terms; far from the source they become progressively less important, leaving only the transverse radiative component [1]. The standard outgoing electric dyad is

$$\overline{\mathbf{G}}_{E,0}(\mathbf{r},\mathbf{r}') = \frac{e^{-\mathrm{j}k_0R}}{4\pi R}\left[\bar{\mathbf{I}} - \widehat{\mathbf{R}}\widehat{\mathbf{R}} + \left(\frac{1}{\mathrm{j}k_0R} - \frac{1}{k_0^2R^2}\right)\left(\bar{\mathbf{I}} - 3\,\widehat{\mathbf{R}}\widehat{\mathbf{R}}\right)\right], \tag{2.152}$$

valid for $R > 0$ [2]. The $\bar{\mathbf{I}} - \widehat{\mathbf{R}}\widehat{\mathbf{R}}$ term is transverse to $\mathbf{R}$ and dominates in the radiative far zone; the R^{-2} and R^{-3} terms in the parentheses supply near-field reactive structure through inverse powers of k_0R and dominate on electrically small probes placed within a wavelength of the feed [2]. Equation (2.152) satisfies Eq. (2.148) for $\mathbf{r} \neq \mathbf{r}'$; coincidence evaluation is deferred to Subsection 2.3.3. The same kernel appears in Eq. (2.5) once $\mathbf{r}$ lies outside $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ and outgoing radiation condition is active [1]. The magnetic field may be reconstructed from the electric field through Maxwell curl relations when the observation region is source-free and simply connected: $\mathbf{H}$ is not an independent measurement channel in the Lorenz reduction but follows from the spatial derivatives of $\mathbf{E}$ [2]. This reconstruction is valid in homogeneous regions away from impressed currents and fails across material interfaces or inside the source support where curl relations must be supplemented by impressed-current terms [1]. The magnetic field may be generated by the electric dyad through Maxwell curl relations when $\mathbf{J}_{\mathrm{imp}}$ is divergence-free in the source region:

$$\mathbf{H}(\mathbf{r}) = \frac{1}{\mathrm{j}\omega\mu_0}\,\nabla\times\mathbf{E}(\mathbf{r}), \qquad \mathbf{E} \text{ from Eq. (2.149)}, \tag{2.153}$$

on simply connected source-free observation regions [2]. A magnetic dyadic $\overline{\mathbf{G}}_H(\mathbf{r},\mathbf{r}')$ ($\mathrm{A\,m^{-1}}$) inverts Eq. (2.111) analogously to Eq. (2.148), with operator $\mathcal{W}_{H,\omega}$ in place of $\mathcal{W}_{E,\omega}$: it is the machine that maps magnetic impressed sources to $\mathbf{H}$, while $\overline{\mathbf{G}}_H$ denotes the corresponding tensorial impulse response [1]. Duality $\varepsilon_0\mathbf{E} \leftrightarrow \mu_0\mathbf{H}$, $\mathbf{J}_{\mathrm{imp}} \leftrightarrow \mathbf{J}_{\mathrm{eff}}^{(\mathrm{mag})}$ maps $\overline{\mathbf{G}}_{E,0} \leftrightarrow \overline{\mathbf{G}}_{H,0}$ in vacuum without re-deriving the magnetic-wave symbol. Subsection 2.3.1 states $\overline{\mathbf{G}}_H$ only where magnetic impressed currents are primary; otherwise Eq. (2.153) avoids duplicate kernel tabulations [2]. The preceding formulas assume a uniform medium. Real structures seldom satisfy this assumption. Substrates, radomes, biological tissues, and layered materials alter propagation and therefore modify the Green response [11]. In piecewise media, local symbols $k(\mathbf{r})$, $\varepsilon(\mathbf{r})$, $\mu(\mathbf{r})$ modify Eq. (2.148) to

$$\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}\,\overline{\mathbf{G}}_E^{(\varepsilon,\mu)}(\cdot,\mathbf{r}') = \bar{\mathbf{I}}\,\delta(\mathbf{r}-\mathbf{r}'), \tag{2.154}$$

with $\mathcal{W}_{E,\omega}^{(\varepsilon,\mu)}$ from Eq. (2.112) [2]. Free-space Eq. (2.152) applies patchwise only when $k(\mathbf{r})$ is constant in each patch and interface jumps are enforced by continuity of tangential $\mathbf{E},\mathbf{H}$ across shared boundaries (Eq. (1.19)); multi-patch assembly is Subsection 2.3.5 [1]. Reciprocity expresses a fundamental symmetry of passive electromagnetic environments: interchanging transmitter and

receiver does not alter the propagation law itself [2]. On passive media without magneto-electric coupling, reciprocity requires that interchanging observation and source points leaves the dyadic invariant up to transposition:

$$\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}') = \overline{\mathbf{G}}_E^{\mathsf{T}}(\mathbf{r}', \mathbf{r}), \tag{2.155}$$

at fixed ω; the full proof is in Section 2.8 [2]. Equations (2.149)–(2.151) hold when $\mathbf{J}_{\text{imp}}$ lies in the domain of the forcing map, when $\mathcal{B}_\Gamma$ is time-independent and identical under phasor reduction, and when the selected kernel branch matches the resolvent of Eq. (2.124). They fail for resonant cavities without quotienting, for incoming rather than outgoing exterior conditions, and for sources with $\nabla \cdot \mathbf{J}_{\text{imp}} \neq 0$ unless charge density ρ is included in the complete source tuple of Section 1.1.2 [2]. Near $\mathbf{r} = \mathbf{r}'$, Eqs. (2.146) and (2.152) are pointwise undefined; convolution meaning is distributional (Subsection 2.3.3). The scalar monopole kernel G supplies the lowest-dimensional propagation law; the electric dyad $\overline{\mathbf{G}}_E$ is the vector generalization demanded by polarization on real feeds. Subsection 2.3.2 closes the loop in the opposite pedagogical order: a measured dipole pattern on Γ_R precedes the dyadic tabulation, so the laboratory observation remains the anchor when the full tensor formulas are applied to extended apertures [8]. Scalar monopole law G, vector dyad $\overline{\mathbf{G}}_E$, and operator realization $\mathcal{G}_{E,\omega}$ are now fixed with $\mathbf{r}$ as observation and $\mathbf{r}'$ as source throughout; coincidence limits and retarded causality refine how those kernels may be convolved with extended $\mathbf{J}_{\text{imp}}$ [2].

2.3.2 Point electric dipole and the dyadic Green operator

Engineering reading (Hertzian chain). *Physical setup:* a short wire or probe feed whose electrical size satisfies $k_0 a \ll 1$ on a turntable range. *Objective:* verify rank-one far-zone export before multipole catalogs are applied to larger apertures. *Expected intuition:* a transverse $\sin\theta$ pattern with a single dominant radiative channel. *Measurement:* watt closure on a sphere Γ_R and turntable invariance of the total radiated power [8]. *Misconception:* equating mesh unknowns in a numerical model with the radiative rank of the physical source [6].

On a turntable compact range, the dipole test is performed before any aperture catalog: rotate the electrically small feed, integrate $|\mathbf{E}|^2$ on Γ_R, and confirm that the total radiated power is invariant and that the far-zone pattern carries a single transverse lobe [7]. That watt closure is the laboratory observation that precedes the dyadic formulas of Subsection 2.3.1; when it fails, the fault lies in calibration, outgoing branch selection, or source admissibility—not in the multipole catalog applied later [2]. The Hertzian dipole is the canonical image of the dyadic Green operator: a localized source, a retarded kernel, a radiation map, and a far-zone angular spectrum linked in one closed calculation [2]. Short-wire or probe feeds at $k_0 a \ll 1$ on a turntable should reproduce transverse $\sin\theta$ export and turntable-invariant watt totals before multipole catalogs are applied to larger apertures [8]. Every compact feed should be checked against this lowest-dimensional outgoing test at low ka [6]. The range observation fixed above—watt closure on Γ_R, turntable invariance, and a single transverse lobe—is what a correct Green propagation must reproduce before any multipole catalog is trusted on larger structures [7]. Only after that calibration does it pay to write the dyadic convolution in closed form: replace the extended source $\mathbf{J}_{\text{imp}}(\mathbf{r}')$ by a rank-one impulse and read off the field pair that every method-of-moments code must match at electrically small $k_0 a$ [1]. The Hertzian dipole is chosen as the canonical calibration source because it is the smallest oriented radiator that still exercises the full dyadic structure: one moment vector, one propagation direction, and a transverse far-zone pattern that every radiation theory must reproduce [8]. Compact feeds on turntable ranges, probe references, and method-of-moments self-tests all reduce to this limit at $k_0 a \ll 1$; when a code or measurement sequence cannot match the dipole field pair, no larger

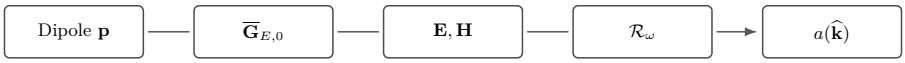


Figure 2.21: Signature chain for a Hertzian dipole: impressed moment convolves with the outgoing dyadic Green function, producing $(\mathbf{E}, \mathbf{H})$; the radiation map extracts the far-zone amplitude and angular spectrum.

aperture catalog built on the same Green operator should be trusted [7]. The dipole moment $\mathbf{p} = \int \mathbf{J}_{\text{imp}}\, d^3r'$ (units: A m, equivalently C m for equivalent polarization sources) localizes the impressed current at the origin; $\mathbf{r}$ denotes the observation point and $\mathbf{r}'$ the source coordinate in the display below. Substituting $\mathbf{J}_{\text{imp}}(\mathbf{r}') = \mathbf{I}\,\delta(\mathbf{r}')$ into Eq. (2.149) collapses the volume integral to a single dyadic–vector product and yields the closed-form field pair

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\, \overline{\mathbf{G}}_{E,0}(\mathbf{r}, \mathbf{0}) \cdot \mathbf{p}, \qquad \mathbf{H}(\mathbf{r}) = \frac{1}{\mathrm{j}\omega\mu_0}\, \nabla \times \mathbf{E}(\mathbf{r}), \tag{2.156}$$

with outgoing dyad Eq. (2.152). This is $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ in closed form; $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ is the same pair after outgoing selection on the exterior domain [2]. Physically, Eq. (2.156) states that every Cartesian component of $\mathbf{p}$ excites a column of $\overline{\mathbf{G}}_{E,0}$, so polarization coupling is already present in the rank-one source that calibrates compact feeds [2]. In the radiative far zone $k_0 r \gg 1$, with $r = |\mathbf{r}|$ the observation radius in metres, the reactive R^{-2} and R^{-3} terms in Eq. (2.152) become negligible and Eq. (2.156) reduces to Eq. (2.247) with

$$\mathbf{E}(\mathbf{r}) \approx -\mathrm{j}\omega\mu_0\, \frac{e^{-\mathrm{j}k_0 r}}{4\pi r}\, \bar{\mathbf{I}} - \widehat{\mathbf{r}}\widehat{\mathbf{r}}\, \cdot \mathbf{p}, \tag{2.157}$$

so the transverse radiative image is fixed by $\widehat{\mathbf{r}} \times (\widehat{\mathbf{r}} \times \mathbf{p})$. This expression reveals that only transverse components survive at large distance. The far field therefore forgets most local source details and retains only radiatively accessible information—precisely the content a compact-range pattern measurement can resolve on Γ_R [2]. Power through sphere S_r follows Eq. (2.7) with $P_{\text{rad}} \propto k_0^4 |\mathbf{p}|^2$ in the electrically small limit [2]. Experimentally, the angular spectrum is the decomposition of the far-zone field into plane waves propagating in different directions $\widehat{\mathbf{k}}$: each outgoing direction receives a distinct amplitude fixed by how the dipole moment projects transversely onto that direction [7]. Pattern measurements on a spherical scanner therefore sample this angular distribution; synthesis codes invert the same lateral Fourier image when reconstructing aperture fields from far-zone data [2]. The plane-wave angular spectrum of Subsection 2.9.1 reads

$$a(\widehat{\mathbf{k}}) = -\mathrm{j}k_0^2 \sqrt{\mu_0/\varepsilon_0}\, (\bar{\mathbf{I}} - \widehat{\mathbf{k}}\widehat{\mathbf{k}}) \cdot \mathbf{p}, \tag{2.158}$$

where $a(\widehat{\mathbf{k}})$ is the far-zone plane-wave amplitude ($\mathrm{V\,m^{-1}}$) associated with propagation direction $\widehat{\mathbf{k}}$ (dimensionless unit vector) and $\widehat{\mathbf{k}}$ parametrizes the angular-spectrum parametrization of Subsection 2.9.1. Equation (2.158) is the lateral Fourier image of the same export channel: each outgoing direction receives transverse amplitude fixed by $(\bar{\mathbf{I}} - \widehat{\mathbf{k}}\widehat{\mathbf{k}}) \cdot \mathbf{p}$. One source, one kernel, one radiation map, one far field, one angular spectrum—every abstract operator statement in Chapter 2 reduces to this geometry when $\mathbf{J}_{\text{imp}} = \mathbf{p}\,\delta(\mathbf{r}')$ [2]. The dipole test on Γ_R is therefore not a numerical afterthought; it is the lowest-dimensional witness that G, $\overline{\mathbf{G}}_E$, $\mathcal{S}_\omega$, and $\mathcal{R}_\omega$ describe the same hardware [7].

2.3.3 Singularities, distributions, and regularization

Layered substrates in patch and handset designs modify Green propagation through reflection coefficients at material interfaces [2]. Printed antennas and handset boards propagate through layered

media; the free-space dyadic is the reference impulse, not the laboratory default when a probe hovers above a dielectric stack [11]. Probe correction files must include substrate Green response, not free-space alone, because the governing question—how a point source on a feed line launches fields to a probe above the substrate—already presupposes outgoing propagation in each half-space [10]. Kernels transport impressed currents; they do not specify which currents the feed can realize [3]. Subsection 2.3.1 defined scalar and dyadic Green kernels that satisfy Eqs. (2.144) and (2.148) for $\mathbf{r} \neq \mathbf{r}'$ and deferred coincidence evaluation to the present subsection [2]. Subsection 2.3.3 supplies the definitive account of distributional convolutions, principal-value and finite-part prescriptions, static–dynamic kernel splits, and self-term regularization that make Eq. (2.149) well-defined on $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ including sheet and filamentary support. The Sobolev and distributional source classes of Subsection 2.1.1 are presupposed through Eqs. (2.54) and (2.56); their functional-analytic construction is not repeated. Retarded time-domain support belongs to Subsection 2.3.4; outgoing branch selection at infinity belongs to Section 2.4. Let $\Omega \Subset \mathbb{R}^3$ be Lipschitz with piecewise smooth internal interfaces, $\omega > 0$ fixed, and $\mathbf{J}_{\mathrm{imp}}$ in the refined source class Eq. (2.49) or its distributional extension Eq. (2.54). Kernels are the vacuum templates Eqs. (2.146) and (2.152) unless $\mathcal{B}_\Gamma$ or piecewise $k(\mathbf{r})$ modify them patchwise. Observation points $\mathbf{r}$ may approach $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ but remain in the domain where $\mathbf{E} \in \mathcal{H}_{\nabla\times}(\Omega)$ is sought. For $\mathbf{r} \notin \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, the convolution Eq. (2.149) is a classical integral. What happens when observation and source points become arbitrarily close? The free-space dyad develops a non-integrable leading singularity: field amplitudes grow as inverse powers of separation R, reflecting reactive storage and self-interaction between neighboring source elements [2]. These singularities do not imply infinite radiated power—they signal that pointwise source models and naive quadrature break down, while extended patches with finite extent carry finite reactive energy [8]. As $\mathbf{r} \to \mathbf{r}' \in \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, the free-space dyad Eq. (2.152) develops a non-integrable leading singularity; the local expansion in powers of $k_0 R$ separates static storage from dynamic radiation and is the analytic starting point for every regularization scheme in the display below. With separation vector $\mathbf{R} = \mathbf{r} - \mathbf{r}'$, distance $R = |\mathbf{R}|$ in metres, observation point $\mathbf{r}$ and source point $\mathbf{r}'$, and unit direction $\widehat{\mathbf{R}} = \mathbf{R}/R$,

$$\overline{\mathbf{G}}_{E,0}(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi R^3}\, \overline{\mathbf{P}}_t(\widehat{\mathbf{R}}) \;+\; \frac{\mathrm{j}k_0}{4\pi R^2}\, \overline{\mathbf{Q}}(\widehat{\mathbf{R}}) \;+\; \frac{k_0^2}{4\pi R}\, \overline{\mathbf{S}}(\widehat{\mathbf{R}}) \;+\; \mathcal{O}(k_0^3), \tag{2.159}$$

where $\overline{\mathbf{P}}_t(\widehat{\mathbf{R}}) = \bar{\mathbf{I}} - 3\widehat{\mathbf{R}}\widehat{\mathbf{R}}$ is transverse to $\mathbf{R}$ and $\overline{\mathbf{Q}}$, $\overline{\mathbf{S}}$ collect the next two anisotropic terms in the locked phasor convention [1]. Equation (2.159) matches the R^{-3}, R^{-2}, and R^{-1} pieces of Eq. (2.152) and separates reactive near storage from radiative R^{-1} decay. The R^{-3} term is non-integrable in a neighborhood of coincidence; convolution with $\mathbf{J}_{\mathrm{imp}}$ therefore requires a distributional or regularized definition, not pointwise Riemann integration [2]. The scalar kernel Eq. (2.146) carries the parallel ordering R^{-1}, R^0, … with R^{-1} non-integrable on neighborhoods of the source. In the electrostatic limit $k_0 \to 0$ at fixed separation $R > 0$ (metres), the dynamic terms in Eq. (2.159) vanish and only the R^{-3} dipolar projector survives; that static piece is extracted analytically in method-of-moments codes before the smooth remainder is integrated numerically. Eq. (2.159) reduces to

$$\overline{\mathbf{G}}^{(\mathrm{st})}_{E,0}(\mathbf{r}, \mathbf{r}') = \frac{1}{4\pi R^3}\Big(\bar{\mathbf{I}} - 3\,\widehat{\mathbf{R}}\widehat{\mathbf{R}}\Big), \tag{2.160}$$

the field of a unit dipole aligned with each Cartesian source direction [8]. Physically, Eq. (2.160) captures static-field dominance at small $k_0 R$: reactive storage and self-interaction between neighboring current elements overwhelm the radiative R^{-1} term until separation or frequency increases [2]. Equation (2.160) is the leading singular part extracted in method-of-moments static–dynamic splits: dynamic corrections are $\mathcal{O}(k_0 R)$ relative to the static term when $k_0 R \ll 1$ on source patches [1]. For divergence-free $\mathbf{J}_{\mathrm{imp}}$, the $\nabla\,\nabla\cdot$ structure of Lorenz reduction Eq. (2.147) removes monopole

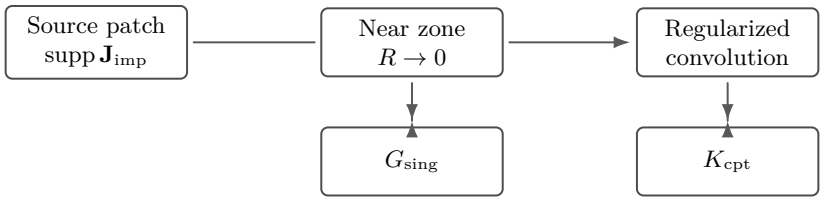


Figure 2.22: Singular–regular split at coincidence: non-integrable G_{sing} is extracted analytically; remainder K_{cpt} is smooth for Galerkin or point evaluation away from the extracted neighborhood.

artifacts; the dominant self-coupling on compact patches is dipolar and transverse to the separation vector between observation and source centroids [2]. (2.79) at kernel level. The decomposition separates analytically known local physics from the smoother long-range interaction that can be handled numerically: $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ collects the non-integrable near-zone coupling, while $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ is the remainder suitable for standard quadrature [1]. Write

$$\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') = \overline{\mathbf{G}}_E^{(\mathrm{sing})}(\mathbf{r},\mathbf{r}') + \overline{\mathbf{G}}_E^{(\mathrm{reg})}(\mathbf{r},\mathbf{r}'), \tag{2.161}$$

with $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ collecting the non-integrable static and leading dynamic singularities of Eq. (2.159) and $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ bounded on $\Omega \times \Omega$ [2]. The corresponding solution operator split is

$$\mathcal{S}_\omega = \mathcal{G}_{\mathrm{sing}} + \mathcal{K}_{\mathrm{cpt}}, \qquad \mathcal{G}_{\mathrm{sing}}[\mathbf{J}_{\mathrm{imp}}] = -\mathrm{j}\omega\mu_0 \int_\Omega \overline{\mathbf{G}}_E^{(\mathrm{sing})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r', \tag{2.162}$$

with $\mathcal{K}_{\mathrm{cpt}}$ compact on bounded Ω when $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ is square-integrable in both variables [2]. Here $\mathcal{G}_{\mathrm{sing}}$ is the singular propagation operator carrying non-integrable near-zone coupling, $\mathcal{K}_{\mathrm{cpt}}$ is the compact remainder suitable for stable numerical quadrature, and $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ and $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ are the corresponding kernel split defined in Eq. (2.161) [1]. Equations (2.161) and (2.162) realize Eq. (2.79) for Green-based inversion; Fredholm consequences are developed in Section 2.6.4 [1]. When $\mathbf{J}_{\mathrm{imp}}$ is a bounded function, Eq. (2.149) defines $\mathbf{E} \in L^2_{\mathrm{loc}}(\Omega)^3$ away from the source. For sheet currents Eq. (2.54), the convolution is defined by duality: for all smooth compactly supported test fields $\boldsymbol{\phi}$ (dimensionless weighting fields that probe how the electric field pairs against admissible test states),

$$\langle \mathbf{E}, \boldsymbol{\phi} \rangle_\Omega = -\mathrm{j}\omega\mu_0 \sum_{ab} \int_{\Sigma_{ab}} \int_\Omega \mathbf{J}_{s,ab}(\mathbf{r}') \cdot \overline{\mathbf{G}}_E^{\mathsf{T}}(\mathbf{r}',\mathbf{r}) \cdot \boldsymbol{\phi}(\mathbf{r})\, d^3r\, dS', \tag{2.163}$$

Here $\boldsymbol{\phi}$ is a smooth test field used to probe the electric field when pointwise evaluation fails; Σ_{ab} is a surface patch carrying sheet current $\mathbf{J}_{s,ab}$ ($\mathrm{A\,m^{-1}}$), the surface density of impressed current on patch (a,b); and $\mathbf{J}_{s,ab}(\mathbf{r}')$ is the physical current flowing on that patch at source point $\mathbf{r}'$ [2], with appropriate Fubini interpretation and outgoing radiation condition on exterior problems [2]. Equation (2.163) is the weak form of Eq. (2.149) when pointwise evaluation fails; it pairs with Eq. (2.21) once $\boldsymbol{\phi}$ is replaced by admissible test states in $\mathcal{F}^0_\Omega$ [1]. Physically, sheet sources produce jump conditions on $\hat{\mathbf{n}} \times \mathbf{E}$ and $\hat{\mathbf{n}} \times \mathbf{H}$; the distributional definition ensures those jumps arise from the same kernel as volume currents, not from ad hoc post-factors [2]. For volume $\mathbf{J}_{\mathrm{imp}}$ and observation points $\mathbf{r} \notin \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ approaching the support, one may attempt to define the field as the symmetric limit of integrals that exclude a small ball about the observation point. The principal value is that symmetric exclusion limit: ordinary integration fails because the R^{-3} pole is not integrable over a neighborhood of coincidence, but symmetric cancellation about the observation point can still yield a finite field when the singularity is weak enough [8]. Define the principal-value convolution

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\, \mathrm{p.v.} \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r', \tag{2.164}$$

as the limit of integrals over $\Omega \setminus B_\varepsilon(\mathbf{r})$ as $\varepsilon \downarrow 0$ when that limit exists in $\mathcal{H}_{\nabla\times}(\Omega)$ [8]. For observation on the source support—for example $\mathbf{r} \in \Sigma_{ab}$ for a surface patch—the principal value generally diverges because of the R^{-3} term; the *Hadamard finite part* subtracts the singular expansion explicitly and retains the finite reactive coupling between extended patches. Finite-part subtraction works because the non-integrable piece is known analytically from Eq. (2.159) and can be removed before the remainder is integrated numerically—the engineering prescription MoM codes implement as static–dynamic splitting [1]:

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \lim_{\varepsilon\to 0} \left[\int_{\Omega\setminus B_\varepsilon(\mathbf{r})} \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \;+\; \int_{B_\varepsilon(\mathbf{r})\cap\Omega} \overline{\mathbf{G}}_E^{(\mathrm{sing})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \right], \tag{2.165}$$

provided $\mathbf{J}_{\mathrm{imp}}$ is smooth enough on the patch that the limit exists [1]. Equation (2.165) is the prescription for on-surface field evaluation; it differs from Eq. (2.164) by the explicit analytic subtraction of $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$, not by numerical mesh refinement alone [2]. Mathematically, the finite part is the constant term in the Laurent expansion of the integral as $\varepsilon \to 0$; physically, it removes infinite self-energy of pointwise source models while retaining finite reactive coupling between extended patches [8]. Method-of-moments and boundary-element codes replace the continuum convolution Eq. (2.149) by discrete impedance entries obtained by testing the dyadic kernel against pairs of basis functions on source patches. In plain language: $\mathbf{f}_m$ and $\mathbf{f}_n$ are the mth and nth current basis functions on patches Ω_m and Ω_n; Ω_m and Ω_n are the surface or volume patches where those basis functions live; and Z_{mn} is the mutual impedance—the complex voltage response at patch m per unit current on patch n—that a numerical code assembles from the double integral below [1]. What a MoM code actually computes is this bilinear pairing of basis functions through the dyadic propagator, not a pointwise field evaluation at patch centers unless a separate approximation is introduced. The Galerkin entry for patches Ω_m and Ω_n reads

$$Z_{mn} = -\mathrm{j}\omega\mu_0 \int_{\Omega_m}\int_{\Omega_n} \mathbf{f}_m(\mathbf{r}) \cdot \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{f}_n(\mathbf{r}')\, d^3r\, d^3r', \tag{2.166}$$

for basis functions $\mathbf{f}_m, \mathbf{f}_n$ on patches Ω_m, Ω_n [1]. When $m = n$, the integrand is non-integrable at $\mathbf{r} = \mathbf{r}'$; the Chapter 2 regularized self-term is

$$Z_{mm}^{(\mathrm{reg})} = -\mathrm{j}\omega\mu_0 \int_{\Omega_m}\int_{\Omega_m} \mathbf{f}_m(\mathbf{r}) \cdot \overline{\mathbf{G}}_E^{(\mathrm{reg})}(\mathbf{r},\mathbf{r}') \cdot \mathbf{f}_m(\mathbf{r}')\, d^3r\, d^3r' \;+\; Z_{mm}^{(\mathrm{sing})}, \tag{2.167}$$

where $Z_{mm}^{(\mathrm{reg})}$ is the numerically integrated smooth remainder, $Z_{mm}^{(\mathrm{sing})}$ is the analytic integral of $\overline{\mathbf{G}}_E^{(\mathrm{sing})}$ over the coincident patch—often reduced to closed forms for flat triangles and rectangles—and the sum is the quantity a code must assemble so that neighboring mesh elements do not produce near-singular divergences when observation and source points approach one another [2]. Point matching—evaluating Eq. (2.149) at patch centers with $\overline{\mathbf{G}}_E$ replaced by $\overline{\mathbf{G}}_E^{(\mathrm{reg})}$ only—is *not* equivalent to Eq. (2.167) unless basis overlap and singular subtraction match the same finite-part convention [1]. The distinction is one of source realizability: Galerkin testing defines a bounded bilinear form on $\mathcal{J}_\Omega \times \mathcal{J}_\Omega$; raw point evaluation does not [2]. When $\mathbf{r}$ and $\mathbf{r}'$ lie on distinct patches separated by small distance $d \ll \lambda_0 = 2\pi/k_0$, direct quadrature of Eq. (2.152) is ill-conditioned. Physically, neighboring mesh elements interact through nearly coincident kernel samples: the separation is small enough that reactive R^{-3} and R^{-2} terms vary rapidly across the integration domain, and standard quadrature underestimates or overestimates the coupling unless the singular part is

subtracted and integrated analytically [1]. The standard subtraction template is

$$\int_{\Omega_n} \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' = \int_{\Omega_n} \left[\overline{\mathbf{G}}_E - \overline{\mathbf{G}}_E^{(\mathrm{near})}(\mathbf{r},\mathbf{r}')\right]\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' + \int_{\Omega_n} \overline{\mathbf{G}}_E^{(\mathrm{near})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r', \tag{2.168}$$

where $\overline{\mathbf{G}}_E^{(\mathrm{near})}$ matches the first few terms of Eq. (2.159) about $\mathbf{r}$ and the integral of $\overline{\mathbf{G}}_E^{(\mathrm{near})}$ is evaluated analytically [1]. Equation (2.168) does not alter the continuum operator; it stabilizes numerical approximation of $\mathcal{K}_{\mathrm{cpt}}$ in Eq. (2.162). Scalar singularities are weaker than dyadic singularities: the monopole kernel diverges only as R^{-1}, which is barely non-integrable in three dimensions, whereas the dyadic kernel carries an R^{-3} pole that breaks naive integration more severely [8]. This distinction matters computationally because scalar potential formulations can sometimes use simpler self-term corrections, while vector MoM codes must implement full static–dynamic dyadic subtraction [1]. The scalar Green function satisfies, in distributional sense,

$$\int_{B_\varepsilon(\mathbf{r})} G_0(\mathbf{r},\mathbf{r}')\,d^3r' = \frac{\varepsilon}{2} \;+\; \mathcal{O}(\varepsilon^3), \tag{2.169}$$

for small balls $B_\varepsilon(\mathbf{r})$ in $\mathbb{R}^3$, reflecting the R^{-1} integrability borderline in three dimensions [8]. Equation (2.169) explains why monopole self-terms in scalar potential formulations require either finite source extent or explicit regularization; vector dyads inherit stronger R^{-3} divergence captured by Eq. (2.160) [2]. Equations (2.165)–(2.167) require patchwise smooth $\mathbf{J}_{\mathrm{imp}}$, Lipschitz Ω, and fixed ω in the passive linear band. They fail for point dipoles modeled as δ-supported $\mathbf{J}_{\mathrm{imp}}$ without accompanying finite-part renormalization, and for non-Lipschitz edges where $\mathcal{H}_{\nabla\times}$ membership of $\mathbf{E}$ requires additional edge singularities not captured by Eq. (2.159) [2]. Lossy media with complex k_0 shift singular expansion coefficients but not the order of the R^{-3} pole; Subsection 2.3.3 does not re-tabulate complex-k subtractors [2]. Regularized convolutions now make $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ well-defined when observation $\mathbf{r}$ approaches source support; retarded time-domain support fixes *when* those spatial kernels may act, completing the causal picture begun in the harmonic dyads with $\mathbf{r}$ as observation and $\mathbf{r}'$ as source [2].

2.3.4 Retardation and causal propagation

Subsection 2.3.4 supplies the definitive account of *retarded* time-domain Green operators, their causal support, and the kernel-level upgrade of the resolvent equivalence Eq. (2.124) and harmonic theorem Eq. (2.126) from Subsection 2.2.3 [2]. Outgoing causality and Laplace–Fourier transforms are presupposed through Eqs. (2.119)–(2.122); the display below constructs the Green kernels that realize those statements. Harmonic formulas Eqs. (2.146) and (2.152) are cited; they appear as $s = -\mathrm{j}\omega$ evaluations of causal Laplace transforms under Sommerfeld outgoing selection at infinity [2].

Let $\mathbf{j}_{\mathrm{imp}}(\mathbf{r},t)$ be causal with $\mathrm{supp}_{(\mathbf{r},t)} \subseteq \Omega_s \times [0,\infty)$, $\Omega_s \Subset \Omega$, and let $\mathcal{B}_\Gamma$, ε, and μ be time-independent. Initial data are homogeneous, $\mathbf{U}_0 = \mathbf{0}$, unless noted for switch-on illustrations. Vacuum regions use $c_0 = 1/\sqrt{\mu_0\varepsilon_0}$ and $k_0 = \omega/c_0$ from Eq. (2.114) [2]. The equivalence class excludes advanced kernels satisfying Eq. (2.129). The causal scalar Green function for the wave operator $(\nabla^2 - \varepsilon_0\mu_0\,\partial_t^2)$ in vacuum satisfies the inhomogeneous wave equation below, with the time argument $t-t'$ measuring elapsed interval between source event and field observation (seconds, SI).

$$\left(\nabla^2 - \varepsilon_0\mu_0\,\partial_t^2\right) G(\mathbf{r},\mathbf{r}',t-t') = -\delta(\mathbf{r}-\mathbf{r}')\,\delta(t-t'), \tag{2.170}$$

with retarded causality $G = \mathbf{0}$ for $t - t' < 0$: no signal precedes the light cone [2]. Time-harmonic kernels inherit this support; acausal branches appear as standing-wave contamination in open

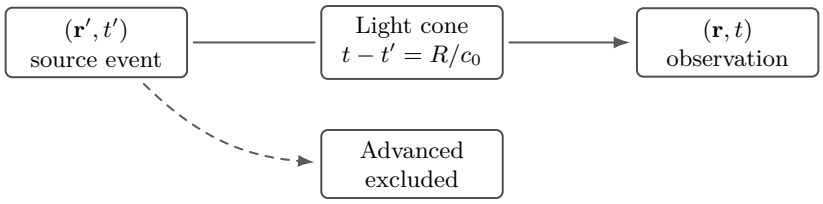


Figure 2.23: Retarded support Eq. (2.173): field at $(\mathbf{r}, t)$ depends only on sources with $t' \le t - R/c_0$; advanced coupling is excluded.

domains [1]. The free-space retarded kernel concentrates its support on the light cone $t-t' = R/c_0$, where $c_0 = 1/\sqrt{\mu_0\varepsilon_0}$ is the vacuum speed of light ($\mathrm{m\,s^{-1}}$) and $R = |\mathbf{r}-\mathbf{r}'|$ is the geometric separation in metres. The free-space outgoing solution is

$$G_0(\mathbf{r}, \mathbf{r}', t - t') = \frac{\delta(t - t' - R/c_0)}{4\pi R}, \qquad R = |\mathbf{r} - \mathbf{r}'|, \tag{2.171}$$

which transports impulse support on the light cone $t - t' = R/c_0$ [2]. Equation (2.171) is the time-domain counterpart of Eq. (2.146): the locked phasor field $\exp(-\mathrm{j}\omega t)$ selects the $s = -\mathrm{j}\omega$ Laplace transform of Eq. (2.171) on the outgoing branch [1]. Physically, Eq. (2.171) states that a unit impulse at $(\mathbf{r}', t')$ influences $\mathbf{r}$ only after travel time R/c_0, enforcing finite propagation speed and Eq. (2.103) on potentials generated by scalar convolution [2]. The retarded electric dyadic $\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t - t')$ upgrades Eq. (2.149) to the time domain. Time differentiation of the causal kernel enforces that the electric field responds only after the impressed current has acted for a retarded interval; the convolution below is the dyadic analogue of the scalar retarded integral. The causal Maxwell field is

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_{\mathrm{hom}}(\mathbf{r}, t) - \mu_0\, \partial_t \int_0^t \int_\Omega \overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t - t') \cdot \mathbf{j}_{\mathrm{imp}}(\mathbf{r}', t')\, d^3 r'\, dt', \tag{2.172}$$

which specializes Eq. (2.102) once the dyadic kernel is identified [2]. Retarded causality requires

$$\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t - t') = \mathbf{0} \quad \text{for } t - t' < 0, \tag{2.173}$$

analogous to Eq. (2.120) [2]. For $\mathbf{r} \notin \Omega_s$, Eq. (2.172) produces $\mathbf{E}(\mathbf{r}, t) = \mathbf{0}$ until the earliest retarded arrival time $t_{\mathrm{ret}}(\mathbf{r})$, defined as the infimum over all source events $(\mathbf{r}', t')$ in the support of $\mathbf{j}_{\mathrm{imp}}$ of the sum $t' + R/c_0$:

$$t_{\mathrm{ret}}(\mathbf{r}) = \inf_{(\mathbf{r}', t') \in \mathrm{supp}\, \mathbf{j}_{\mathrm{imp}}} \{t' + R/c_0\}, \tag{2.174}$$

which is the geometric light-cone support condition underlying Eq. (2.103) [8]. The harmonic dyad Eq. (2.152) is not repeated here; it is the $s = -\mathrm{j}\omega$ symbol of $\overline{\mathbf{G}}_{E,0}$ under Eqs. (2.125) and outgoing radiation condition [1]. (2.172). Subsection 2.3.3 treated coincidence in $\mathbf{r}$; the display below completes the causal picture by fixing the temporal ordering enforced by retarded support [2]. Regularized self-terms from Eq. (2.167) apply to harmonic convolutions; time stepping discretizes Eq. (2.172) with the same kernel class [1]. Let $\widetilde{\overline{\mathbf{G}}}_E(\mathbf{r}, \mathbf{r}', s)$ denote the Laplace transform of $\overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{r}', t)$ in $t - t'$ with $\mathrm{Re}\{s\}$ large enough for convergence. The complex variable s ($\mathrm{s^{-1}}$) encodes temporal frequency; evaluation at $s = -\mathrm{j}\omega$ recovers the harmonic dyad used throughout the phasor convention. Causal support Eq. (2.173) implies $\widetilde{\overline{\mathbf{G}}}_E$ is analytic in the right half-plane and, under outgoing radiation condition,

$$\overline{\mathbf{G}}_{E,\omega}(\mathbf{r}, \mathbf{r}') = \widetilde{\overline{\mathbf{G}}}_E(\mathbf{r}, \mathbf{r}', s)\Big|_{s=-\mathrm{j}\omega}, \tag{2.175}$$

which is the kernel form of Eq. (2.124) [2]. Applying the Laplace transform to Eq. (2.172) with causal $\mathbf{j}_{\mathrm{imp}}$ yields

$$\widetilde{\mathbf{E}}(\mathbf{r},s)=\widetilde{\mathbf{E}}_{\mathrm{hom}}(\mathbf{r},s)-\mu_0 s\int_\Omega \widetilde{\overline{\mathbf{G}}}_E(\mathbf{r},\mathbf{r}',s)\cdot\widetilde{\mathbf{j}}(\mathbf{r}',s)\,d^3r', \tag{2.176}$$

with $\widetilde{\mathbf{j}}$ from Eq. (2.121) [2]. On $s=-\mathrm{j}\omega$ under Eqs. (2.125), Eq. (2.176) reduces to Eq. (2.149) with $\overline{\mathbf{G}}_{E,\omega}$ from Eq. (2.175) and $\mathbf{J}_{\mathrm{imp}}=\widetilde{\mathbf{j}}(\mathbf{r},s)\big|_{s=-\mathrm{j}\omega}$ [1]. The Chapter 2 causal propagation theorem at kernel level is

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]=\mathcal{S}_t[\mathbf{j}_{\mathrm{imp}}](\mathbf{r},t)\Big|_{\text{harmonic at }\omega}\quad\Longleftrightarrow\quad\overline{\mathbf{G}}_{E,\omega}=\widetilde{\overline{\mathbf{G}}}_E\Big|_{s=-\mathrm{j}\omega} \tag{2.177}$$

for the same $\mathcal{B}_\Gamma$ and outgoing branch, with $\mathcal{S}_t$ the time-domain map Eq. (2.101) [2]. Equation (2.177) does not replace Eq. (2.126); it identifies the Green operators whose resolvent relation Eq. (2.124) implements [1]. Advanced kernels and branch failure. Advanced dyads satisfying Eq. (2.129) produce Laplace symbols that cannot equal Eq. (2.175) for any causal $\mathbf{j}_{\mathrm{imp}}$ [2]. Time-harmonic solvers that implicitly use incoming Green functions therefore fail Eq. (2.177) even when interior Maxwell rows vanish [1]. Exterior problems require outgoing radiation condition consistent with Section 2.4; interior cavity problems may require loss or quotienting as in Eq. (2.87) before a causal Laplace contour can be closed [2]. For finite-bandwidth excitation, the harmonic slice at fixed ω is insufficient. Causal fields superpose as

$$\mathbf{E}(\mathbf{r},t)=\int_{-\infty}^{\infty}\widetilde{\mathbf{E}}(\mathbf{r},-\mathrm{j}\omega)\,e^{-\mathrm{j}\omega t}\,\frac{d\omega}{2\pi}, \tag{2.178}$$

with $\widetilde{\mathbf{E}}(\mathbf{r},-\mathrm{j}\omega)$ the Fourier transform induced by Eq. (2.176) on causal data [2]. Equation (2.178) is the inverse of the single-frequency reduction used in the phasor convention; dispersive $\varepsilon(\omega)$, $\mu(\omega)$ modify $\widetilde{\overline{\mathbf{G}}}_E(\mathbf{r},\mathbf{r}',s)$ and are deferred beyond the locked single-ω harmonic band except where loss enters as $\mathrm{Im}\{k_0\}\neq 0$ [1]. Within the harmonic band, Eq. (2.127) describes the long-time limit after switch-on when outgoing radiation or damping removes transients [2]. Because Π_{rad} commutes with harmonic extraction under the ordering assumed in Subsection 2.2.3,

$$\mathcal{R}_t[\mathbf{j}_{\mathrm{imp}}]=\Pi_{\mathrm{rad}}\circ\mathcal{S}_t[\mathbf{j}_{\mathrm{imp}}],\qquad\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]=\Pi_{\mathrm{rad}}\circ\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.179}$$

with $\mathcal{R}_\omega$ from Eq. (2.128) [2]. Causality does not promote evanescent content to radiative export: Eq. (2.140) remains the transport criterion after time-domain evolution [2]. Retarded propagation ensures support constraints only; spectral and regime qualifications from Subsection 2.2.3 are additional [1]. Equations (2.172)–(2.177) require Eqs. (2.125), outgoing or passive completion on exterior domains, and $\mathbf{j}_{\mathrm{imp}}$ in the causal class of Subsection 2.2.1. They fail for time-varying $\mathcal{B}_\Gamma(t)$, moving media, and nonlinear sources. Coincidence and self-term evaluation of $\overline{\mathbf{G}}_{E,\omega}$ remains governed by Subsection 2.3.3 even when Eq. (2.172) is discretized in time [1]. Retarded support is therefore the temporal counterpart of the spatial singular split in Subsection 2.3.3: causality fixes when propagation may act, while regularization fixes how strongly a source couples to itself at coincidence [2]. Harmonic kernels G and $\overline{\mathbf{G}}_E$ are therefore retarded response functions in space and time; constructive inversion on composite domains asks which resolvent branch those responses realize when interfaces and exterior completions are assembled [2].

2.3.5 Inversion and solvability

A standard-gain horn on a compact-range turntable fixes the laboratory scene for ‘Inversion and solvability”: frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays

below [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Subsection 2.3.5 supplies the definitive account of constructive inversion and solvability on single- and multi-region domains with interface data from Eq. (1.19) [2]. Weak well-posedness on bounded quotients through Eqs. (2.90) and (2.87) from Subsection 2.1.4 is presupposed; Lax–Milgram and Friedrichs arguments are not repeated. Outgoing uniqueness on exterior domains is finalized in Section 2.4.3; here the resolvent branch and its Green realization are stated as hypotheses [2]. Material Helmholtz symbols Eqs. (2.112)–(2.113) and trace spaces Eqs. (2.55)–(2.58) from Sections 2.1.1–2.2.2 are fixed input classes.

Let $\Omega = \cup_{j=1}^{N_p} \Omega_j$ be a finite union of Lipschitz patches with pairwise interfaces $\Sigma_{ab} = \overline{\Omega_a} \cap \overline{\Omega_b}$, piecewise constants $\varepsilon_j, \mu_j > 0$ on Ω_j, and outer boundary $\Gamma = \partial\Omega$. Impressed data satisfy $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, charge continuity Eq. (1.8), and boundary class $\mathcal{B}_\Gamma$ compatible with Eq. (2.57). Regularized convolutions follow Subsection 2.3.3; retarded causality follows Subsection 2.3.4 [2]. Composite domain and patchwise inversion. Inversion separates invertible propagation from compact perturbations and null directions [2]. Ill-posed spherical near-field campaigns often fail here: the export map is well defined, but the inverse amplifies noise along small singular values [7]. On a composite domain the impressed-source region is partitioned into material patches Ω_j separated by interfaces Σ_{ab}; each patch carries its own wavenumber $k_j = \omega\sqrt{\mu_j \varepsilon_j}$ ($\mathrm{rad\,m^{-1}}$). Decompose the solution map as

$$\Omega = \bigsqcup_{j=1}^{N_p} \Omega_j \;\cup\; \bigsqcup_{a<b} \Sigma_{ab}, \qquad k_j = \omega\sqrt{\mu_j \varepsilon_j}, \tag{2.180}$$

with wavenumbers k_j from Eq. (2.114) on each patch [2]. On Ω_j, the electric field satisfies the patchwise Helmholtz drive of Eq. (2.112) and is represented by the regularized dyadic convolution stated below; the patch kernel $\overline{\mathbf{G}}_E^{(j)}$ incorporates local ε_j, μ_j and the finite-part prescription of Subsection 2.3.3 when observation approaches the source support:

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_{\mathrm{hom}}(\mathbf{r}) + \mathrm{j}\omega\mu_j \int_{\Omega_j} \overline{\mathbf{G}}_E^{(j)}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r', \qquad \mathbf{r} \in \Omega_j, \tag{2.181}$$

where $\overline{\mathbf{G}}_E^{(j)}$ satisfies Eq. (2.154) with patch coefficients and finite-part evaluation when $\mathbf{r} \in \Omega_j$ approaches $\operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ [1]. In Eq. (2.181), $\mathbf{r}$ is the observation point in patch Ω_j and $\mathbf{r}'$ the source point integrated over the impressed support; $\overline{\mathbf{G}}_E^{(j)}(\mathbf{r}, \mathbf{r}')$ is the patchwise outgoing response function [2]. Equation (2.181) is the constructive content of Eq. (2.151) on uniform subdomains; it does not yet enforce interface jumps across Σ_{ab} [2]. Fields in adjacent patches are linked by tangential continuity and normal-flux jumps imported from Eq. (1.19) [2]. The unit normal $\hat{\mathbf{n}}_{ab}$ on interface Σ_{ab} points from patch a into patch b; tangential $\mathbf{E}$ and $\mathbf{H}$ must match across the interface so that the composite field remains a single physical solution. In operator form,

$$\hat{\mathbf{n}}_{ab} \times \mathbf{E}^{(a)}\big|_{\Sigma_{ab}} = \hat{\mathbf{n}}_{ab} \times \mathbf{E}^{(b)}\big|_{\Sigma_{ab}}, \qquad \hat{\mathbf{n}}_{ab} \times \mathbf{H}^{(a)}\big|_{\Sigma_{ab}} = \hat{\mathbf{n}}_{ab} \times \mathbf{H}^{(b)}\big|_{\Sigma_{ab}}, \tag{2.182}$$

with $\mathbf{E}^{(j)}$ from Eq. (2.181) and $\mathbf{H}^{(j)}$ from Eq. (2.153) when valid [1]. Equations (2.182) replace free-space Eq. (2.152) by patch Green functions $\overline{\mathbf{G}}_E^{(j)}$ plus interface unknowns on $\mathcal{T}_{\Sigma_{ab}}$ from Eq. (2.55) [2]. The assembled multi-region system is

$$\begin{bmatrix} \mathcal{I} + \mathcal{K}_{11} & \cdots & \mathcal{K}_{1N_p} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{N_p1} & \cdots & \mathcal{I} + \mathcal{K}_{N_pN_p} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_{N_p} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_{N_p} \end{bmatrix}, \tag{2.183}$$

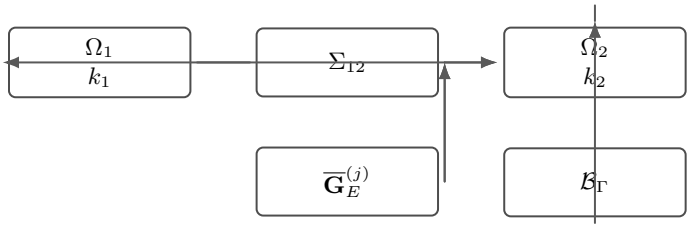


Figure 2.24: Multi-region inversion Eq. (2.183): patchwise dyadic kernels coupled by interface continuity Eq. (2.182) and exterior $\mathcal{B}_\Gamma$.

where $\mathbf{x}_j$ collects tangential field or equivalent surface currents on patch boundaries, $\mathcal{K}_{jk}$ are dyadic layer potentials generated by $\overline{\mathbf{G}}_E^{(k)}$, and $\mathbf{b}_j$ encodes impressed sources and outer $\mathcal{B}_\Gamma$ data [1]. Equation (2.183) is the constructive statement of multi-region solvability: invertibility of the block operator is the Green-function analogue of Eq. (2.90) on composite domains [2]. Subsection 2.3.3 supplies self-term regularization on each patch; Subsection 2.3.4 supplies causal time marching when Eq. (2.183) is embedded in $\mathcal{S}_t$ [1]. On a bounded Lipschitz domain with passive $\mathcal{B}_\Gamma$ satisfying Eq. (2.88), coercivity on the quotient $\mathcal{U}_\omega^0/\ker\mathcal{L}_\omega$ from Eq. (2.87) implies existence of a unique weak solution Eq. (2.90) [2]. Green-function inversion is equivalent when the regularized operator

$$\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{u}_\omega \quad \Longleftrightarrow \quad \mathcal{L}_\omega[\mathbf{u}_\omega] = \begin{bmatrix} \mathbf{s}_\omega(\mathbf{J}_{\text{imp}}, \rho) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.184}$$

with $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ obtained from Eqs. (2.181)–(2.167) on simply connected patches and from Eq. (2.183) otherwise [1]. Equation (2.184) expresses Eq. (2.18) with constructive kernels; stability is controlled by Eq. (2.72) and resolvent bound Eq. (2.77) on the physical branch [2]. On exterior Ω with compact $\operatorname{supp}\mathbf{J}_{\text{imp}}$, existence of $\mathcal{S}_\omega$ requires outgoing Green functions whose far zone matches Section 2.4 [2]. Without that selection, the block system Eq. (2.183) admits incoming solutions violating Eq. (2.177) [1]. Uniqueness on the outgoing branch is

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0} \text{ and outgoing } \mathcal{B}_\Gamma \implies \mathbf{u}_\omega = \mathbf{0}, \tag{2.185}$$

with $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ the total field phasor pair. Eq. (2.185) is the solvability criterion for exterior Green inversion [2]. The split Eq. (2.94) reminds that exterior problems are not Friedrichs self-adjoint on all of $\mathcal{U}_\omega$ [1]. Resonant cavity spectra. Bounded cavities with PEC walls possess discrete spectra $\{\omega_n\}$. At a resonant frequency $\omega = \omega_n$ the Helmholtz operator acquires a non-trivial null space and the Green resolvent ceases to be bounded on the full admissible field space:

$$\ker\mathcal{L}_\omega \neq \{\mathbf{0}\} \quad \Longrightarrow \quad \mathcal{L}_\omega^{-1} \text{ unbounded on } \mathcal{U}_\omega, \tag{2.186}$$

and Eq. (2.184) fails unless quotienting by standing modes as in Eq. (2.87) or introducing loss [2]. Poles of the resolvent appear as

$$\det(\mathcal{I} + \mathcal{K}) = 0 \quad \Longleftrightarrow \quad \omega = \omega_n, \tag{2.187}$$

for the coupled operator $\mathcal{K}$ in Eq. (2.183) on closed resonant structures [2]. Impressed $\mathbf{J}_{\text{imp}}$ at resonance requires compatibility with $\ker\mathcal{L}_\omega$; otherwise amplitudes diverge even when Green formulas are formally applied [1]. Incoming exterior excitation produces a distinct failure: Eq. (2.185) is violated without additional incident-field subtraction in Section 2.5.1 [2]. Numerical equivalence. Finite-element stiffness systems and boundary-element impedance matrices are Galerkin discretizations of Eq. (2.21) and Eq. (2.183), respectively:

$$\mathbf{ZI} = \mathbf{V} \quad \Longleftrightarrow \quad \mathcal{K}\mathbf{x} = \mathbf{b}, \tag{2.188}$$

when basis functions match trace spaces and self-terms implement Eq. (2.167) [1]. Equation (2.188) is the same inversion problem in matrix form. Ill-conditioning near Eq. (2.187) mirrors practical difficulty at cavity modes and near-spurious resonances in open-region formulations [1]. Equations (2.181)–(2.183) require Lipschitz interfaces, passive linear media on each patch, and fixed ω in the harmonic band. They fail for time-varying boundaries, nonlinear constitutive laws, and anisotropic magneto-electric coupling without retensoring $\overline{\mathbf{G}}_E^{(j)}$ [2]. Point sources without Subsection 2.3.3 regularization produce formal but non-admissible fields; $\mathbf{J}_{\mathrm{imp}}$ must remain in $\mathcal{J}_\Omega$ [1]. With Subsections 2.3.1–2.3.5 complete, Section 2.3 has constructed scalar, vector, and dyadic Green kernels, regularized their singular convolutions, tied them to retarded causal propagation, and stated constructive inversion on composite domains compatible with Eq. (2.67). Sommerfeld outgoing selection in Section 2.4 fixes the exterior branch assumed in Eq. (2.185); radiation integrals in Section 2.5 evaluate far-zone observables from the propagating subspace qualified by Eq. (2.140); spectral and Fredholm consequences of Eq. (2.79) continue in Section 2.6.4 [2]. Green-function inversion recovers field states from sources under declared completion; whether a desired angular pattern is source-realizable under finite impressed currents is established in Chapter **??** (*Source Realizability and Spectral Truncation*) [3].

2.4 Radiation Conditions and Uniqueness

Green-function inversion in Section 2.3 reconstructs fields from sources on compact support. Section 2.4 imposes outgoing radiation conditions that separate measurable far-zone content from reactive storage [10]. A dense near-field map without outgoing filtering can fit non-radiating currents [3].

Maxwell's equations alone do not distinguish fields radiating away from a source from fields arriving from infinity, yet every laboratory measurement assumes the former. Antennas are fed by finite sources, compact ranges suppress external illumination, and receivers attribute detected power to identifiable radiators rather than unexplained incoming waves [10]. Section 2.3 supplied Green kernels that map impressed currents to fields; on exterior domains those kernels remain ambiguous until an outgoing branch is fixed at infinity [8]. Retarded causality narrows the admissible class but does not fully replace Sommerfeld selection when harmonic solvers invert $\mathcal{L}_\omega$ at fixed ω [2]. The subsections below state outgoing radiation conditions, embed them as resolvent selection, prove uniqueness on the radiative branch, and record reconstruction limits when that branch is omitted.

2.4.1 Sommerfeld radiation condition

Before Sommerfeld asymptotics are written, the physical ambiguity must be named. Harmonic Helmholtz solutions on open domains superpose $e^{-\mathrm{j}k_0 r}/r$ outgoing and $e^{+\mathrm{j}k_0 r}/r$ incoming spherical waves with equal standing rights in the volume equations alone [10]. A near-field scan or pattern cut on Γ_R cannot distinguish these completions without an explicit outgoing selection: wattmeters and probes then read mixtures of export, storage, and energy arriving from the computational or chamber boundary [7]. Sommerfeld radiation condition is the asymptotic statement that restores laboratory identification—radiation launched locally and leaving the domain—before any uniqueness argument or far-zone integral is evaluated [10].

Before any asymptotic formula is written, the range engineer already decides whether the measurement enclosure behaves as an open exterior domain [7]. A compact range places the device under test inside an anechoic chamber whose absorbing linings are designed to let export escape instead of reflecting back as multipath [11]. Pattern probes on a large sphere Γ_R and shell wattmeters

report tangential field and power only when chamber absorption and computational termination both approximate free-space outgoing behavior [5]. Incomplete absorption produces ripple on azimuth cuts; unstable spherical near-field transforms when spectra are truncated show the same branch failure at the data-processing stage [7]. Until outgoing selection is restored, Γ_R meters read cavity storage and standing-wave content rather than watt-bearing export [4]. Sommerfeld radiation condition below is the asymptotic mathematical statement of what the range engineer already assumes when attributing a measured pattern to the device under test rather than to energy arriving from infinity [10]. An open antenna measurement is meaningful only if the field on the range can be identified with radiation from the device under test rather than with energy arriving from infinity. Sommerfeld outgoing selection enforces that identification: disturbances launched locally propagate outward and are not returned from the far boundary of the model [10]. When the condition is omitted, the same near-field scan or pattern cut admits formal completions with incoming or standing-wave content, and inversion may assign equivalent sources that would require excitation mechanisms absent from the physical problem and therefore lie outside the admissible solution class [3]. Free-space kernels Eqs. (2.146) and (2.152) supply the asymptotic templates; outer-boundary classes from Chapter 1 are presupposed. Uniqueness proofs and reconstruction limits belong to Subsections 2.4.3 and 2.4.4. Let Ω_{ext} denote an exterior region surrounding compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}} \subset \Omega_s$, with $\omega > 0$ fixed and vacuum constants ε_0, μ_0 in the far zone. Spherical coordinates (r, θ, ϕ) are centered so that $\Omega_s \subset B_R(0)$ for some $R_0 > 0$. Fields are phasors under $\exp(-\mathrm{j}\omega t)$; outgoing means $e^{-\mathrm{j}k_0 r}/r$ decay with $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ from Eq. (2.114) [2]. Interior interfaces and finite domains use Eq. (1.20); Sommerfeld completion applies on artificial or physical spheres at large r [1]. On Γ_R with $R \gg \lambda_0$, a range engineer expects the measured phase to advance as $k_0 r$ increases—outgoing fronts receding from the enclosure—and the amplitude to fall no faster than r^{-1} in three dimensions [8]. That observation is what Sommerfeld selection encodes at infinity: the scalar template states what every outgoing Helmholtz response must look like at large distance from a localized source. Phase fronts recede with the observer; amplitude decay compatible with energy export to infinity is r^{-1} in three dimensions [8]. For scalar potentials and Helmholtz kernels, outgoing radiation requires, with $r = |\mathbf{r}|$ the observation radius from an origin enclosing the source, $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ the vacuum wavenumber ($\mathrm{rad\,m^{-1}}$), and $A(\theta, \phi, \mathbf{r}')$ the angular amplitude that carries directional information while $\mathbf{r}'$ labels the fixed source point during the limit $r \to \infty$:

$$G(\mathbf{r}, \mathbf{r}') = A(\theta, \phi, \mathbf{r}')\,\frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \qquad r = |\mathbf{r}| \to \infty, \tag{2.189}$$

The remainder $\mathcal{O}(r^{-2})$ collects subleading reactive and curvature corrections that need not vanish at finite range [2]. Equation (2.189) is the Chapter 2 scalar Sommerfeld condition: it excludes incoming $e^{+\mathrm{j}k_0 r}/r$ and standing $e^{\pm\mathrm{j}k_0 r}/r$ superpositions that degenerate exterior uniqueness [1]. The free-space template Eq. (2.146) satisfies Eq. (2.189) for fixed $\mathbf{r}'$ [2]. Physically, Eq. (2.189) states that monopole-like responses propagate outward with phase fronts $k_0 r = \text{const}$ and amplitude decay r^{-1}, the only radial combination compatible with energy flux to infinity in three dimensions [8]. Vector fields must satisfy a transverse impedance relation at infinity: the electric and magnetic components that carry export must stand in the same ratio as a plane wave propagating along $\hat{\mathbf{r}}$, with vacuum impedance η_0 as the proportionality constant [2]. For harmonic Maxwell fields in vacuum far from sources,

$$\mathbf{r} \times \mathbf{E} \;+\; \eta_0\,(\mathbf{r} \times \mathbf{H}) \times \hat{\mathbf{r}} = \mathbf{o}(1), \qquad r \to \infty, \tag{2.190}$$

with $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ the vacuum wave impedance, $\hat{\mathbf{r}} = \mathbf{r}/r$ the unit radial direction, and $\mathbf{o}(1)$ denoting any vector whose Cartesian components vanish in the limit $r \to \infty$ [2]. Physically, Eq. (2.190)

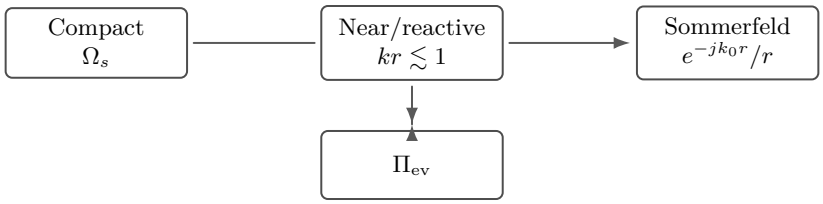


Figure 2.25: Sommerfeld far zone Eq. (2.190): compact sources drive near fields with evanescent content; at large r only outgoing transverse $e^{-jk_0 r}/r$ waves satisfy exterior completion.

requires that the transverse electric field and the transverse magnetic field on a distant sphere lock phase and magnitude as if energy were leaving the domain along $\hat{\mathbf{r}}$, rather than arriving from infinity [1]. Equation (2.190) is the Chapter 2 vector Sommerfeld radiation condition. It implies transverse radiative fields with asymptotic ratio

$$\mathbf{H}_{\mathrm{rad}} \sim \eta_0^{-1}\, \hat{\mathbf{r}} \times \mathbf{E}_{\mathrm{rad}}, \qquad r \to \infty, \tag{2.191}$$

and matches the outgoing-trace condition Eq. (1.21) from Subsection 1.1.2 when tangential traces are taken on a sphere of large radius [2]. Mathematically, Eq. (2.190) selects the subspace on which the exterior Laplacian/Helmholtz resolvent coincides with retarded Green functions Eq. (2.177); physically, it enforces power flow outward rather than inward at infinity [1]. Far-zone transversality and propagation. Sommerfeld-admissible fields decompose at leading order as

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\theta, \phi)\, \frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \qquad \mathbf{E}_0 \cdot \hat{\mathbf{r}} = 0, \tag{2.192}$$

with $\mathbf{E}_0$ the radiation amplitude pattern and $\mathbf{E}_0 \cdot \hat{\mathbf{r}} = 0$ the transversality constraint [2]. Equation (2.192) connects Sommerfeld completion to Π_{prop} from Subsection 2.2.3: far-zone content lies on the propagating sheet with real $k_r = k_0$ and outgoing phase [1]. Evanescent and reactive components may still be large at finite r while satisfying Eq. (2.190); regime qualification Eq. (2.140) remains separate from Sommerfeld selection [2]. Equation (2.190) is a local impedance statement at infinity: transverse $\mathbf{E}$ and $\mathbf{H}$ must satisfy the plane-wave ratio Eq. (2.191) with η_0 from Eq. (1.12). Mathematically, it selects the eigensheet of the exterior Helmholtz resolvent with outgoing radial phase; physically, it enforces net power flow toward infinity rather than inward from infinity. On a truncation sphere Γ_R with $R = 8\lambda_0$ enclosing a source with $k_0 a = 1.5$, Silver–Müller surrogate Eq. (2.195) typically introduces spurious reflected power below 1% of P_{rad} while reactive near fields at $r = 1.2a$ may differ by $\mathcal{O}(1)$ from the exact Sommerfeld solution [1]. Sommerfeld completion is therefore necessary but not sufficient for transport claims—regime qualification Eq. (2.140) remains mandatory [2]. Figure 2.25 separates Sommerfeld completion from regime qualification. Subsection 2.4.1 fixes the asymptotic *boundary at infinity*; Subsection 2.2.3 and Section 1.3.3 govern when far-zone samples may be interpreted as transport [2]. Green-function asymptotics. Dyadic kernels from Subsection 2.3.1 inherit scalar Sommerfeld decay. For $r = |\mathbf{r} - \mathbf{r}'| \to \infty$ at fixed $\mathbf{r}'$,

$$\overline{\mathbf{G}}_{E,0}(\mathbf{r}, \mathbf{r}') = \overline{\mathbf{G}}_{\infty}(\theta, \phi, \mathbf{r}')\, \frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \tag{2.193}$$

with $\overline{\mathbf{G}}_{\infty}$ transverse in the $\mathbf{r}$-direction [2]. Equation (2.193) is the kernel form of Eq. (2.190) under Eq. (2.149): impressed currents that satisfy Eq. (1.22) generate fields whose far zone is built from outgoing dyadic radiation [2]. The leading r^{-1} term in Eq. (2.152) is the explicit vacuum instance of Eq. (2.193) [8]. Subsection 2.4.2 promotes Eq. (2.193) to an operator selector on $\mathcal{S}_\omega$; Subsection 2.4.3 uses it in uniqueness energy integrals [1]. Collecting the asymptotic and impedance conditions above

yields the admissible function class on which exterior uniqueness arguments and radiation export are defined. A field pair belongs to this class only if it satisfies the vector Sommerfeld condition and the transverse far-zone template simultaneously; incoming or standing-wave completions are excluded by construction [1]. Define the Sommerfeld outgoing subspace on an exterior shell $\Gamma_R = \{r = R\}$:

$$\mathcal{F}_{\mathrm{SF}} = \{(\mathbf{E}, \mathbf{H}) \in \mathcal{H}_{\nabla\times}(\Omega_{\mathrm{ext}}) : \text{Eqs. (2.190)–(2.192) hold}\}\,. \tag{2.194}$$

In Eq. (2.194), $\mathcal{F}_{\mathrm{SF}} \subset \mathcal{H}_{\nabla\times}(\Omega_{\mathrm{ext}})$ is the Sommerfeld outgoing field class on exterior Ω_{ext}: pairs $(\mathbf{E}, \mathbf{H})$ whose far-zone asymptotics satisfy outgoing phase and transverse radiation templates; incoming and standing-wave completions are excluded by membership in this set. Equation (2.194) embeds Eq. (1.21) into the Chapter 2 function-space setting of Eq. (2.57) [2]. The physical solution operator $\mathcal{S}_\omega$ maps $\mathcal{J}_\Omega$ into $\mathcal{F}_{\mathrm{SF}}$ when $\mathcal{B}_\Gamma$ includes outgoing radiation condition [1]. Incoming solutions lie outside $\mathcal{F}_{\mathrm{SF}}$ even if they satisfy volume Maxwell rows; that exclusion is the Sommerfeld radiation condition [2]. Finite computations cannot place a boundary at infinity; the Silver–Müller trace below approximates the Sommerfeld impedance relation on a truncation sphere large enough that spurious reflections are small compared with exported power [1]. On a finite truncation sphere Γ_R, Eq. (2.191) motivates the first-order absorbing condition

$$\hat{\mathbf{n}} \times \mathbf{H} \;-\; \eta_0^{-1}(\hat{\mathbf{n}} \times \mathbf{E}) \times \hat{\mathbf{n}} = \mathbf{0} \quad \text{on } \Gamma_R, \tag{2.195}$$

which approximates Eq. (1.21) at finite R [2]. Equation (2.195) is not identical to Sommerfeld completion at infinity: spurious reflections from Γ_R contaminate near fields unless R is large compared with source extent and wavelength [1]. Section 2.4 treats Eq. (2.195) as a practical surrogate; the continuous target remains Eq. (2.190) [2]. Perfectly matched layers and radiation boundary integrals are discrete refinements of the same outgoing-trace logic [1]. Once Sommerfeld closure is in place, the radiation export map $\mathcal{R}_\omega$ and projector Π_{rad} act on the same outgoing class the range attributes to measured patterns: $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with full field pair $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ carrying reactive content that Π_{rad} removes before far-zone metering [2]. The outgoing selector Π_{rad} of Eq. (2.3) is idempotent on fields satisfying Eq. (2.194) once uniqueness is established in Subsection 2.4.3 [2]. Prior to that proof, Sommerfeld closure defines the domain on which Π_{rad} acts as projection rather than heuristic windowing [1]. Composition with Eq. (2.28) yields

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{SF}}, \tag{2.196}$$

with transport qualification from Section 1.3.3 applied after Eq. (2.196) [2]. Equation (2.196) expresses Eq. (2.128) on the Sommerfeld domain without re-proving causality from Subsection 2.2.3 [1]. Consequence for range practice: pattern meters and gain reports must compose with $\mathcal{R}_\omega$, not with raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, whenever reactive near fields are non-negligible on the measurement shell [2]. Subsection 2.4.1 therefore states what the compact range already assumes before any far-zone integral is evaluated in Section 2.5 [2]. Equations (2.189)–(2.194) require linear vacuum far zones, fixed $\omega > 0$, and compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$. They fail for anisotropic or magneto-electric exterior media without modified asymptotics, for lossless closed cavities where Eq. (1.20) uses PEC rather than outgoing radiation condition, and for time-harmonic fields with net incoming power at infinity [2]. Loss shifts Eq. (2.189) to complex k_0 but preserves outgoing *selection* as distinct from incoming [2]. As a worked example, replace Eq. (2.190) by Eq. (2.195) on Γ_R with $R = 5\lambda_0$. Fields inside Ω_s differ from the exact Sommerfeld solution by $\mathcal{O}((k_0 R)^{-1})$ in typical radiative components while reactive near content may differ by order unity [2]. Declaring a transport channel from Γ_R probes without Eq. (2.140) conflates truncation error with angular transport [2].

2.4.2 Outgoing-wave selection

Finite computational domains terminate open-region problems artificially. On a physical compact range, anechoic absorbers play the same role as absorbing boundary conditions in equivalent analytical models: they mimic terminations that let export escape instead of reflecting into the near field [7]. Silver–Müller and related traces approximate outgoing selection on meshes used for horns, cavities, and reflector feeds [2]. A finite-element model employing inadequate absorbing boundaries permits artificial reflections that contaminate field amplitudes and reconstructed radiation characteristics—the same standing-wave signatures that appear when chamber absorption is imperfect on a physical range [7]. Absorbing-boundary parameters are therefore a radiation-physics choice about exported energy, not a mesh-only tuning parameter [2]. The operator $\mathcal{O}_{\mathrm{out}}$ records that choice on $\mathcal{S}_\omega$ before Π_{rad} and $\mathcal{R}_\omega$ attach pattern meters.

The outgoing-wave selector $\mathcal{O}_{\mathrm{out}}$ implements Sommerfeld completion Eq. (2.194) as an operator on $\mathcal{F}_\Omega$, on harmonic Green kernels, and on the resolvent branch of $\mathcal{L}_\omega^{-1}$ [2]. The composition relation Eq. (2.138) from Subsection 2.2.3 named $\mathcal{O}_{\mathrm{out}}$ without constructing it; the definitions below supply that operator and its links to Π_{prop}, Π_{rad}, and retarded Green functions Eq. (2.177). Sommerfeld field conditions Eqs. (2.190)–(2.194) are presupposed; uniqueness proofs belong to Subsection 2.4.3. Assumptions. Exterior regions use vacuum far-zone constants with $k_0 > 0$ real in the locked harmonic band, compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$, and time-independent $\mathcal{B}_\Gamma$ including outgoing radiation condition at infinity or on Eq. (2.195) at large finite R [2]. Propagating and evanescent projectors Eqs. (2.134)–(2.135) from Subsection 2.2.3 are fixed on $\widehat{\mathcal{F}}_\Omega$ [1]. Incoming and advanced branches satisfying Eq. (2.129) are excluded. Let $\mathcal{F}_{\mathrm{in}}$ denote the incoming counterpart of $\mathcal{F}_{\mathrm{SF}}$ obtained by reversing outgoing phase in the far zone. The outgoing-wave selector is the idempotent map

$$\mathcal{O}_{\mathrm{out}} : \mathcal{F}_\Omega \to \mathcal{F}_{\mathrm{SF}}, \qquad \mathcal{O}_{\mathrm{out}}[\mathbf{u}_\omega] = \mathbf{u}_{\mathrm{out}}, \tag{2.197}$$

where $\mathbf{u}_{\mathrm{out}}$ is the unique member of $\mathcal{F}_{\mathrm{SF}}$ whose propagating spectral content matches the outgoing k_z branch of Eq. (2.131) and whose incoming content is annihilated [2]. Equation (2.197) is the locked definition for operator realization of Sommerfeld completion: it acts *before* far-zone measurement and *after* volume completion by $\mathcal{L}_\omega$, selecting the physical resolvent branch assumed in Eq. (2.124) [2]. On $\mathcal{F}_{\mathrm{SF}}$,

$$\mathcal{O}_{\mathrm{out}}^2 = \mathcal{O}_{\mathrm{out}}, \qquad \mathcal{O}_{\mathrm{out}}\big|_{\mathcal{F}_{\mathrm{SF}}} = \mathrm{id}, \tag{2.198}$$

while incoming states satisfy $\mathcal{O}_{\mathrm{out}}[\mathbf{u}_{\mathrm{in}}] = \mathbf{0}$ when extended by continuity from the far zone [1]. Physically, $\mathcal{O}_{\mathrm{out}}$ removes incoming spherical and plane-wave components that satisfy Maxwell equations but violate retarded causality [8]. On homogeneous regions, outgoing selection in lateral wavenumber (k_x, k_y) uses the longitudinal root from Eq. (2.131):

$$k_z = \begin{cases} +\sqrt{k_0^2 - k_t^2}, & k_t \le k_0 \text{ (outgoing propagating)}, \\ -\mathrm{j}\,\alpha(k_t), & k_t > k_0 \text{ (decaying evanescent)}, \end{cases} \tag{2.199}$$

with the sign of the propagating root fixed by exterior completion and the locked $\exp(-\mathrm{j}\omega t)$ convention [2]. Equation (2.199) is the spectral action of $\mathcal{O}_{\mathrm{out}}$ on each plane-wave mode; reversing $k_z \to -k_z$ on the propagating sheet produces incoming content annihilated by $\mathcal{O}_{\mathrm{out}}$ [2]. Evanescent modes are unchanged in decay direction but remain outside Π_{rad} through Eq. (2.139) [1]. The propagating projector Π_{prop} from Subsection 2.2.3 retains all outgoing and incoming propagating modes; $\mathcal{O}_{\mathrm{out}}$ acts only after Π_{prop} in the radiation chain Eq. (2.138) [2]. (2.138) with explicit operator order. Regime qualification Eq. (2.140) applies after Π_{rad}, not at the $\mathcal{O}_{\mathrm{out}}$ stage Green-function branch selection. Harmonic kernels from Section 2.3—with observation coordinate $\mathbf{r}$ and source

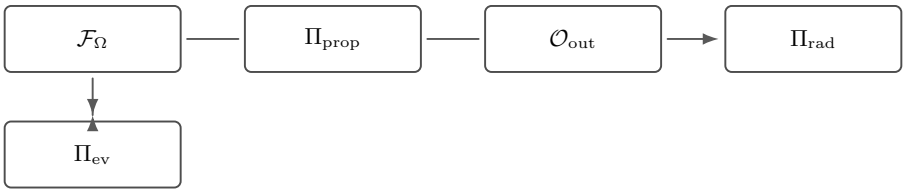


Figure 2.26: Outgoing selection Eq. (2.197) sits between propagating projection and radiation export: $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$.

coordinate $\mathbf{r}'$ in G and $\overline{\mathbf{G}}_E$—admit incoming and outgoing analytic continuations. The outgoing dyadic branch is

$$\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega} = \mathcal{O}_{\mathrm{out}}\, \overline{\mathbf{G}}_{E,\omega}, \qquad \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega} \text{ satisfies Eq. (2.193)}, \tag{2.200}$$

with $\mathcal{O}_{\mathrm{out}}$ acting on the observation-point dependence [2]. Equation (2.200) identifies the kernel used in Eq. (2.149) on exterior problems with the retarded symbol of Eq. (2.177) [2]. Incoming kernels $\overline{\mathbf{G}}^{(\mathrm{in})}_{E,\omega}$ are related by $k_0 \to -k_0$ on the propagating Riemann sheet and produce fields outside $\mathcal{F}_{\mathrm{SF}}$ [8]. Subsection 2.4.2 does not re-tabulate Eq. (2.152); it fixes the sheet on which that formula is evaluated [1]. The physical resolvent branch is

$$\mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} = \mathcal{O}_{\mathrm{out}} \circ \mathcal{L}_\omega^{-1}, \tag{2.201}$$

understood on $\mathcal{D}(\mathcal{L}_\omega)$ with outgoing $\mathcal{B}_\Gamma$ [2]. The solution map on impressed currents becomes

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathcal{O}_{\mathrm{out}}\Big(-\mathrm{j}\omega\mu_0 \int_\Omega \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}\Big), \tag{2.202}$$

with regularization from Subsection 2.3.3 when $\mathbf{r}$ approaches $\operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ [1]. Equations (2.201) and (2.202) realize Eq. (2.185) once Subsection 2.4.3 proves uniqueness on the selected branch [2]. Causality from Subsection 2.3.4 is consistent with Eq. (2.200) but not sufficient when harmonic solvers bypass time-domain construction [1]. The volume integral in Eq. (2.202) sums impressed current contributions at each source point $\mathbf{r}' \in \Omega$ against the outgoing dyadic evaluated at observation point $\mathbf{r}$; $\mathbf{E}_{\mathrm{hom}}$ carries any exterior illumination not generated by $\mathbf{J}_{\mathrm{imp}}$ and must vanish for pure radiation problems [2]. On propagating content, outgoing selection precedes radiation export in the measurement sequence fixed in Section 2.4: $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ with $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$ [2]. On propagating content,

$$\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}, \qquad \mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega, \tag{2.203}$$

restating Eqs. (2.138) and (2.3) with the locked definition of $\mathcal{O}_{\mathrm{out}}$ [2]. Radiative subspace $\mathcal{F}_{\mathrm{rad}} \subset \mathcal{F}_{\mathrm{SF}}$ consists of fields with nonzero far-zone transverse amplitude Eq. (2.192); $\mathcal{O}_{\mathrm{out}}$ maps into $\mathcal{F}_{\mathrm{SF}}$ but not automatically into $\mathcal{F}_{\mathrm{rad}}$ when evanescent content dominates near sources [2]. Transport reporting therefore requires Eq. (2.140) after Eq. (2.203) [1]. Outgoing selection on a compact range is implemented twice in practice: absorbers suppress multipath reflections that would mimic incoming waves, and harmonic solvers fix the propagating k_z branch of Eq. (2.199) on exterior meshes [7]. When either layer fails, $\mathcal{O}_{\mathrm{out}}$ acts on the wrong field orbit even though volume Maxwell rows still vanish [1]. Incoming–outgoing split. Define $\mathcal{O}_{\mathrm{in}}$ analogously on $\mathcal{F}_{\mathrm{in}}$. On the propagating subspace,

$$\mathrm{id} = \mathcal{O}_{\mathrm{out}} + \mathcal{O}_{\mathrm{in}} \quad \text{on } \Pi_{\mathrm{prop}}\mathcal{F}_\Omega, \qquad \mathcal{O}_{\mathrm{out}}\mathcal{O}_{\mathrm{in}} = 0, \tag{2.204}$$

with evanescent and non-propagating components orthogonal to both when decay direction is fixed by Eq. (2.199) [2]. Equation (2.204) explains branch degeneracy in unconstrained Helmholtz solvers: both $\mathcal{O}_{\mathrm{out}}[\mathbf{u}]$ and $\mathcal{O}_{\mathrm{in}}[\mathbf{u}]$ satisfy volume equations unless one is selected by boundary or radiation

completion [8]. Equations (2.197)–(2.202) require linear media, fixed ω, and outgoing $\mathcal{B}_\Gamma$ on exterior problems. They fail on closed PEC cavities where Eq. (1.20) replaces Sommerfeld completion, and on lossless open resonators where discrete spectra require quotienting Eq. (2.87) [2]. Absorbing layers approximate $\mathcal{O}_{\mathrm{out}}$ but introduce spurious modes unless tuned to Eq. (2.190) at large distance [1].

2.4.3 Uniqueness of radiative solutions

Outgoing selection is not a post hoc spatial mask: Sommerfeld radiation and uniqueness on $\mathcal{B}_\Gamma$ decide which Maxwell solutions carry watt-bearing export before harmonic charts attach [10, 2]. Reactive near-field energy may be locally large while contributing negligibly to P_{rad} on Γ_R once Π_{rad} is fixed [1]. Without exterior uniqueness, two distinct field pairs could satisfy the same feed data while differing on the measurement sphere—a failure no wattmeter or pattern probe could resolve [10]. The proof below is therefore not a mathematical luxury imposed for elegance; it is the physical condition under which laboratory attribution of radiation to declared sources becomes meaningful. Outgoing selection is not a post hoc spatial mask: Sommerfeld radiation and uniqueness on $\mathcal{B}_\Gamma$ decide which Maxwell solutions carry watt-bearing export before harmonic charts attach [10]. On Γ_R, reactive near-field energy may be locally large while contributing negligibly to P_{rad} once $\mathcal{R}_\omega$ and Π_{rad} are fixed [1]. Exterior uniqueness ties measured patterns to declared sources within the radiative class [2]. Solvers that omit outgoing filters fit non-radiating currents that compact ranges cannot distinguish from export without watt completion on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ alone [8]. The proof below imports the difference-field energy method from Subsection 1.1.3 and specializes it to the Sommerfeld outgoing class fixed in Subsection 2.4.1.

Subsection 2.4.3 proves that exterior radiative solutions are unique on the outgoing branch, upgrading hypothesis Eq. (2.185) to exterior radiative uniqueness [2]. The difference-field reduction Eqs. (1.29)–(1.30) and energy identity Eq. (1.31) from Subsection 1.1.3 are presupposed; the passive uniqueness argument and coercive estimate Eq. (1.32) are not re-proved. Subsection 2.4.3 specializes the energy method to exterior domains with Eqs. (2.190)–(2.194), outgoing resolvent selection Eqs. (2.201)–(2.202), and radiation refinement Eq. (2.203). Source reconstruction limits belong to Subsection 2.4.4. Let Ω_{ext} denote an exterior domain with compact $\operatorname{supp}\mathbf{J}_{\mathrm{imp}} \subset B_R(0)$, vacuum constants ε_0, μ_0 in the far zone, and fixed $\omega > 0$ in the harmonic band of Subsection 2.2.3. Data are admissible per Eq. (1.23) with outgoing radiation condition Eq. (2.194) at infinity, equivalently Eq. (1.21) on spherical shells [2]. Constitutive response in the source region is passive: $\operatorname{Im}\{\varepsilon\} \geq 0$, $\operatorname{Im}\{\mu\} \geq 0$ or equivalent loss so homogeneous difference energy is dissipative [2]. Incoming and advanced branches are excluded by Eq. (2.197) and Eq. (2.204) [1]. Fields lie in $\mathcal{F}_{\mathrm{SF}}$ as defined in Subsection 2.4.1; radiative export uses $\mathcal{F}_{\mathrm{rad}} \subset \mathcal{F}_{\mathrm{SF}}$ from Eq. (2.192) [2]. Uniqueness is established by showing that two outgoing solutions for the same impressed data cannot differ nontrivially: the proof therefore studies the homogeneous problem satisfied by their difference [2]. Let $(\mathbf{E}_1, \mathbf{H}_1)$ and $(\mathbf{E}_2, \mathbf{H}_2)$ be two harmonic solutions on the outgoing branch for the same impressed data and $\mathcal{B}_\Gamma$. Define $\Delta\mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2$, $\Delta\mathbf{H} = \mathbf{H}_1 - \mathbf{H}_2$, and $\mathbf{u}_\Delta = (\Delta\mathbf{E}, \Delta\mathbf{H})$. Linearity and Eq. (1.29) give the homogeneous exterior system

$$\nabla \times \Delta\mathbf{E} = -\mathrm{j}\omega\mu\,\Delta\mathbf{H}, \quad \nabla \times \Delta\mathbf{H} = \mathrm{j}\omega\varepsilon\,\Delta\mathbf{E}, \quad \mathcal{B}_\Gamma[\Delta\mathbf{E}, \Delta\mathbf{H}] = \mathbf{0}, \tag{2.205}$$

with ε, μ matching the physical model in Ω_s and ε_0, μ_0 outside [2]. Physically, if two outgoing solutions for the same feed existed, their difference would satisfy this homogeneous system while still carrying export on Γ_R—a situation the flux argument below excludes [2]. Outgoing radiation condition requires

$$\mathcal{O}_{\mathrm{out}}[\mathbf{u}_\Delta] = \mathbf{u}_\Delta, \qquad \mathbf{u}_\Delta \in \mathcal{F}_{\mathrm{SF}}, \tag{2.206}$$

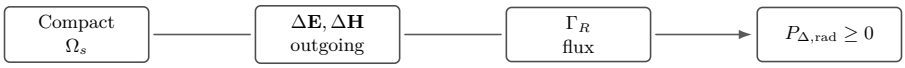


Figure 2.27: Exterior uniqueness flux argument: homogeneous difference fields on the outgoing branch satisfy Eqs. (2.207)–(2.209) on spheres Γ_R.

which is the exterior specialization of the third row of Eq. (1.29) when $\mathcal{B}_\Gamma$ includes Sommerfeld radiation [1]. Equations (2.205)–(2.206) state that uniqueness testing reduces to showing only the trivial pair satisfies the homogeneous outgoing problem [2]. Incoming difference modes would violate Eq. (2.206) even if they satisfied the curl rows [8]. On truncated exterior shells, form the complex Poynting vector for difference fields, $\mathbf{S}_\Delta = \frac{1}{2}\Delta\mathbf{E} \times \Delta\mathbf{H}^*$, as in Eq. (1.31) [2]. For a large sphere $\Gamma_R = \{r = R\}$ enclosing Ω_s, let $\Omega_R = B_R \setminus \Omega_s$. The divergence theorem on Ω_R yields

$$\int_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}_\Delta \, dS = -\mathrm{j}\omega \int_{\Omega_R} \left(u_{\Delta H} - u_{\Delta E} \right) dV \; + \; \mathcal{P}_{\Delta,s}, \tag{2.207}$$

where $u_{\Delta E} = \frac{1}{4}\varepsilon|\Delta\mathbf{E}|^2$, $u_{\Delta H} = \frac{1}{4}\mu|\Delta\mathbf{H}|^2$, and $\mathcal{P}_{\Delta,s}$ collects source-region passive dissipation when $\mathrm{Im}\{\varepsilon\}, \mathrm{Im}\{\mu\} > 0$ [2]. Equation (2.207) is Eq. (1.31) with $\Delta\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$ and $\Gamma = \Gamma_R$; its real part controls uniqueness through the sign of the boundary flux [2]. On a compact range, the real part of the Γ_R integral is what a wattmeter would read from the difference field if such a superposition were physically present; outgoing selection forces that reading to nonnegative export or zero [1]. In lossless vacuum on $\Omega_R \setminus \Omega_s$, the volume term is purely imaginary at steady state, so $\mathrm{Re}\int_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}_\Delta \, dS$ is determined entirely by far-zone radiation [1]. Sommerfeld condition Eq. (2.190) implies that on Γ_R with $R \gg a = \max|\mathbf{r}|$ over $\mathrm{supp}\,\mathbf{J}_{\mathrm{imp}}$, the leading transverse contribution to $\Delta\mathbf{E}, \Delta\mathbf{H}$ scales as $e^{-\mathrm{j}k_0 R}/R$ with $k_0 = \omega\sqrt{\varepsilon_0\mu_0}$ [2]. The far-zone transverse radiation law Eq. (2.192) then gives

$$\mathrm{Re}\int_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}_\Delta \, dS = P_{\Delta,\mathrm{rad}}(R) = \frac{1}{2\eta_0} \oint_{\Gamma_R} \left|\mathbf{E}_{\Delta,t}\right|^2 dS \; + \; \mathcal{O}(R^{-1}), \tag{2.208}$$

with $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ and $\mathbf{E}_{\Delta,t}$ the transverse spherical component of $\Delta\mathbf{E}$ [2]. Taking $R \to \infty$ under outgoing radiation condition,

$$\lim_{R\to\infty} P_{\Delta,\mathrm{rad}}(R) = P_{\Delta,\mathrm{rad}} \geq 0, \tag{2.209}$$

where $P_{\Delta,\mathrm{rad}}$ is the net outgoing power carried by the difference field [2]. Equation (2.209) is the exterior boundary term induced by Eq. (1.21): outgoing difference fields radiate nonnegative power to infinity, while incoming difference fields would produce negative flux and are excluded by $\mathcal{O}_{\mathrm{out}}$ [8]. Dyadic asymptotics Eq. (2.193) are presupposed; they are not re-tabulated here [1]. Section 1.2.3: volume terms in Eq. (2.207) are controlled by passivity when loss is present, and vanish in lossless vacuum away from sources [2]. Let $\mathbf{u}_\Delta = (\Delta\mathbf{E}, \Delta\mathbf{H})$ satisfy Eqs. (2.205)–(2.206) on an exterior domain with vacuum far zone. If $P_{\Delta,\mathrm{rad}} = 0$ in Eq. (2.209), then $\mathbf{E}_{\Delta,t} = \mathbf{0}$ on every sphere Γ_R for R large enough, and $\mathbf{u}_\Delta$ has no propagating outgoing content in $\mathcal{F}_{\mathrm{rad}}$ [2]. Eq. (2.208) with $P_{\Delta,\mathrm{rad}} = 0$ forces $\oint_{\Gamma_R} |\mathbf{E}_{\Delta,t}|^2 \, dS = 0$ for all sufficiently large R. Outgoing transverse fields with zero spherical flux cannot carry $1/R$ radiation; the only remaining homogeneous outgoing transverse modes are identically zero on $\mathcal{F}_{\mathrm{rad}}$ [2]. Let Ω_{ext} be an exterior domain with compact impressed sources, vacuum far zone, and outgoing Sommerfeld completion Eqs. (2.194)–(2.197). If $(\mathbf{E}_1, \mathbf{H}_1)$ and $(\mathbf{E}_2, \mathbf{H}_2)$ are harmonic Maxwell solutions on the outgoing branch for the same admissible data, then $\mathbf{E}_1 = \mathbf{E}_2$ and $\mathbf{H}_1 = \mathbf{H}_2$. Equivalently,

$$\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{0} \text{ and } \mathbf{u}_\omega \in \mathcal{F}_{\mathrm{SF}} \implies \mathbf{u}_\omega = \mathbf{0}, \tag{2.210}$$

which upgrades Eq. (2.185) to the Chapter 2 definitive uniqueness statement [2]. Let $\mathbf{u}_\Delta$ be the difference of two outgoing solutions. Then Eqs. (2.205)–(2.206) hold. Apply Eq. (2.207) and take

$\mathrm{Re}\{\cdot\}$. In lossless vacuum outside Ω_s, the volume integral is purely imaginary, so $\mathrm{Re}\, P_{\Delta,\mathrm{rad}}(R)$ equals the real boundary flux. Outgoing radiation condition forces $\mathrm{Re}\, P_{\Delta,\mathrm{rad}} \geq 0$; passivity in Ω_s adds nonnegative $\mathrm{Re}\, \mathcal{P}_{\Delta,s}$, so the only way to satisfy Eq. (2.209) with equality throughout is $P_{\Delta,\mathrm{rad}} = 0$ [2]. The auxiliary relation (2.4.3) eliminates outgoing transverse content. Evanescent components remain in $\mathcal{F}_{\mathrm{SF}}$ but are tied to the same source-free outgoing solution operator; with no radiative export and homogeneous forcing, the unique retarded exterior state is trivial by Eq. (2.202) with $\mathbf{J}_{\mathrm{imp}} = \mathbf{0}$ [1]. Hence $\mathbf{u}_\Delta = \mathbf{0}$, proving Eq. (2.210). Exterior radiative uniqueness (Subsection 2.4.3) is the exterior counterpart of the passive uniqueness argument (Subsection 1.1.3): instead of a bounded passive Γ with coercivity Eq. (1.32), the exterior uses Sommerfeld flux Eq. (2.209) as the coercive boundary channel [2]. The uniqueness index Eq. (1.33) is finite on exterior outgoing problems away from embedded resonances in finite scatterers; it diverges on bounded cavities at Eq. (2.186) [1]. Exterior uniqueness therefore replaces bounded coercivity with an *open* boundary condition that selects decaying outgoing phase [2]. Exterior radiative uniqueness (Subsection 2.4.3) makes the physical resolvent injective on the outgoing subspace:

$$\mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} : \mathcal{Y}_\omega \to \mathcal{F}_{\mathrm{SF}} \text{ is single-valued}, \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} \begin{bmatrix} \mathbf{s}_\omega(\mathbf{J}_{\mathrm{imp}}, \rho) \\ \mathbf{g}_\Gamma \end{bmatrix}, \tag{2.211}$$

consistent with Eq. (2.184), outgoing Green branch Eq. (2.200), and solution selection Eq. (2.202) [2]. For fixed $\mathbf{J}_{\mathrm{imp}}$ in the admissible class, the exterior field $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ obtained from Eq. (2.202) is unique in $\mathcal{F}_{\mathrm{SF}}$ [2]. Two convolutions differing by a homogeneous outgoing state violate exterior radiative uniqueness (Subsection 2.4.3) [1]. Subsection 2.0.1 declared Π_{rad} idempotent on $\mathcal{F}_{\mathrm{rad}}$; the joint property with outgoing radiation condition can now be stated precisely. On radiative states,

$$\Pi_{\mathrm{rad}}^2 = \Pi_{\mathrm{rad}} \quad \text{on } \mathcal{F}_{\mathrm{rad}}, \tag{2.212}$$

as developed in Subsection 2.0.1 [2]. After exterior radiative uniqueness (Subsection 2.4.3), the measurement sequence Eq. (2.203) defines a single-valued map $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ without branch ambiguity from duplicate outgoing decompositions [1]. Evanescent components remain in $\ker \Pi_{\mathrm{rad}}$ through Eq. (2.139); uniqueness of $\mathcal{S}_\omega$ does not imply uniqueness of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when near-zone evanescent storage dominates export [2]. Outgoing selector idempotency Eq. (2.198) and resolvent selection Eq. (2.201) are consistent with exterior radiative uniqueness (Subsection 2.4.3):

$$\mathcal{O}_{\mathrm{out}} \circ \mathcal{L}_\omega^{-1} = \mathcal{L}_\omega^{-1}\big|_{\mathrm{phys}} \quad \text{is unique on } \mathcal{Y}_\omega, \tag{2.213}$$

because any two selections differing by a homogeneous outgoing mode would violate Eq. (2.210) [2]. Equation (2.213) closes the excitation-to-flux sequence of Figure 2.26 on exterior domains: once data are fixed, $\mathcal{O}_{\mathrm{out}}$, $\mathcal{S}_\omega$, and Π_{rad} act on a single field orbit in $\mathcal{F}_{\mathrm{SF}}$ [2]. On bounded domains with passive $\mathcal{B}_\Gamma$, uniqueness follows from Eq. (2.90) and the passive uniqueness argument (Subsection 1.1.3) after quotienting Eq. (2.87) [2]. At cavity eigenfrequencies,

$$\text{bounded PEC cavity: } \ker \mathcal{L}_\omega \neq \{\mathbf{0}\} \quad \text{vs.} \quad \text{exterior outgoing: } \ker \mathcal{L}_\omega|_{\mathcal{F}_{\mathrm{SF}}} = \{\mathbf{0}\}, \tag{2.214}$$

so Eq. (2.210) does not apply to closed resonators where Eq. (2.186) holds [1]. Exterior uniqueness is therefore a property of $\mathcal{F}_{\mathrm{SF}}$ with Sommerfeld completion, not of arbitrary $\mathcal{B}_\Gamma$ from Eq. (1.20) [2]. A finite scatterer embedded in free space may support Mie resonances in the near field without breaking exterior uniqueness: those modes are driven by $\mathbf{J}_{\mathrm{imp}}$ and do not appear as free homogeneous outgoing states [8]. Laboratory errors and non-applicability. Exterior radiative uniqueness (Subsection 2.4.3) requires linear passive media, fixed ω away from pathological embedded spectra,

and exact outgoing radiation condition rather than approximate ABC alone [2]. First-order absorbing boundaries that do not enforce Eq. (2.190) at large R can admit spurious incoming content; uniqueness is then a property of the *augmented* truncated problem, not of the physical exterior [1]. Anisotropic, moving, or nonlinear media require modified flux functionals; time-varying $\mathcal{B}_\Gamma(t)$ breaks the harmonic reduction of Subsection 2.2.3 [2]. Uniqueness of $\mathcal{S}_\omega$ does not imply uniqueness of $\mathbf{J}_{\mathrm{imp}}$ in $\mathcal{R}_\omega$—that is Subsection 2.4.4 [1]. Regime qualification for transport remains Eq. (2.140), independent of field uniqueness [2]. As a worked example, truncating at finite R with a first-order ABC may return two numerically distinct fields that agree on Γ_R but differ in spurious incoming content reflected between Ω_s and the ABC [1]. Those differences are not counterexamples to exterior radiative uniqueness (Subsection 2.4.3) on Ω_{ext}; they signal that the truncated BVP lacks Eq. (2.194). Silver–Müller surrogates Eq. (2.195) improve the completion but still require $R \gg \lambda_0$ for uniqueness in practice [2].

2.4.4 Implications for source reconstruction

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell export is computed [2]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Exterior radiative uniqueness (Subsection 2.4.3) fixes uniqueness of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on the outgoing branch. Subsection 2.4.4 records the inverse-source consequences: field uniqueness does not imply source identifiability. Uniqueness of $(\mathbf{E}, \mathbf{H})$ under outgoing conditions does not make $\mathcal{R}_\omega$ injective—distinct $\mathbf{J}_{\mathrm{imp}}$ can share the same far pattern [3]. Source reconstruction is a conditioned inverse, not a direct readout of coefficients [7]. $\mathcal{R}_\omega$ is generally non-injective, and finite observation channels compress rank further [2]. Observation null-space Eq. (2.81), rank composition Eq. (2.85), and non-radiating relation Eq. (2.8) from Subsection 2.1.3 are presupposed; complete structure of $\ker \mathcal{R}_\omega$ belongs to Section 2.7.1 [1]. Measurement classification from Section 1.2.5 applies here; angular-momentum observables from Chapter 1 are not re-opened [2]. Field uniqueness versus source identifiability. Forward exterior completion yields

$$\underbrace{\mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}] = \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]}_{\text{same impressed data}} \implies \mathbf{J}^{(1)}_{\mathrm{imp}} = \mathbf{J}^{(2)}_{\mathrm{imp}} \quad \text{false in general,} \tag{2.215}$$

whereas exterior radiative uniqueness (Subsection 2.4.3) gives

$$\mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}] = \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}] \implies (\mathbf{E}_1, \mathbf{H}_1) = (\mathbf{E}_2, \mathbf{H}_2) \quad \text{on } \mathcal{F}_{\mathrm{SF}} \tag{2.216}$$

for the same boundary class and outgoing branch [2]. Equations (2.215)–(2.216) separate *state* uniqueness from *excitation* uniqueness: many admissible currents can produce the same radiative exterior field once reactive degrees of freedom are quotiented [1]. Inverse problems therefore target equivalence classes of $\mathbf{J}_{\mathrm{imp}}$, not individual source distributions [2]. The radiative export map from Subsection 2.0.1 is

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\mathrm{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.217}$$

restating Eq. (2.3) after outgoing radiation condition Eqs. (2.203)–(2.213) [2]. The source-side null space recorded in Eq. (2.8) is

$$\ker \mathcal{R}_\omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}\}\,, \tag{2.218}$$

generally infinite-dimensional on open domains [1]. Section 2.7.1 develops the complete null-space analysis; Subsection 2.4.4 records the reconstruction consequences of Eq. (2.218) without constructing non-radiating bases [2]. Elements of $\ker \mathcal{R}_\omega$ are invisible to far-zone radiative observables yet

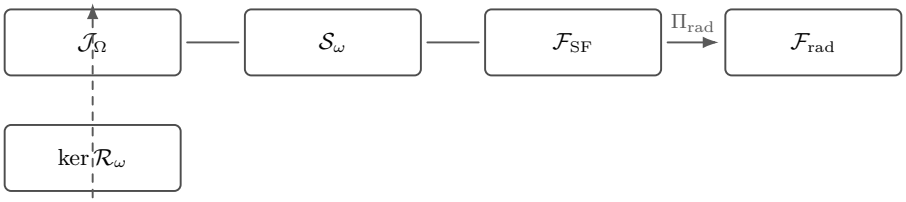


Figure 2.28: Forward chain for reconstruction: $\mathcal{S}_\omega$ is single-valued on $\mathcal{F}_{\mathrm{SF}}$ after exterior radiative uniqueness (Subsection 2.4.3); $\mathcal{R}_\omega$ collapses many sources to the same radiative output.

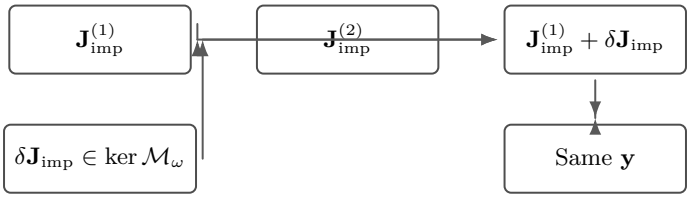


Figure 2.29: Non-uniqueness: adding $\delta\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{M}_\omega$ leaves measurement vector $\mathbf{y}$ unchanged.

may carry substantial near-zone reactive energy through $\mathcal{S}_\omega$ [1]. Observation composition and inverse problem. Laboratory readouts compose with $\mathcal{R}_\omega$ before any source inversion is attempted: pattern probes and near-field scanners observe a finite-rank image of impressed currents, not $\mathbf{J}_{\mathrm{imp}}$ itself [7]. From Eq. (2.68) and finite-rank observation Eq. (2.80),

$$\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}} : \mathcal{J}_\Omega \longrightarrow \mathbb{C}^N, \tag{2.219}$$

with $\Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega$ [2]. Given measurement vector $\mathbf{y} \in \mathbb{C}^N$, the inverse source problem seeks $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ such that

$$\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{y}, \qquad \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1, \tag{2.220}$$

subject to admissibility Eq. (2.9) [1]. Equation (2.220) is ill-posed whenever $\ker \mathcal{M}_\omega \neq \{\mathbf{0}\}$, which occurs if either $\ker \mathcal{R}_\omega \neq \{\mathbf{0}\}$ or $\ker \mathcal{O}_{\mathrm{meas}}$ is nontrivial on $\operatorname{ran} \mathcal{R}_\omega$ [2]. Rank cap Eq. (2.85) applies directly:

$$\operatorname{rank}(\mathcal{M}_\omega) \leq \min\{N,\, \dim \mathcal{R}(\mathcal{R}_\omega)\}, \tag{2.221}$$

so no measurement protocol with N channels resolves more than N independent radiative degrees of freedom [1]. The composed null space satisfies

$$\ker \mathcal{M}_\omega \supseteq \ker \mathcal{R}_\omega \;\cup\; \mathcal{R}_\omega^{-1}\big(\ker \mathcal{O}_{\mathrm{meas}}\big), \tag{2.222}$$

with set inclusion on admissible sources [2]. Identifiable sources modulo null space form the quotient

$$\mathcal{J}_\Omega / \ker \mathcal{M}_\omega \;\cong\; \operatorname{ran} \mathcal{M}_\omega \subset \mathbb{C}^N, \tag{2.223}$$

so reconstruction recovers at best a finite-dimensional image, not a unique $\mathbf{J}_{\mathrm{imp}}(\mathbf{r})$ [1]. Equivalent-current models that expand $\mathbf{J}_{\mathrm{imp}}$ on auxiliary surfaces inherit the same non-uniqueness: distinct surface distributions related by $\ker \mathcal{R}_\omega$ produce identical far-zone patterns [2]. Boundary-element inversion therefore requires regularization or additional constraints beyond Maxwell consistency alone [1]. (2.81) pulled back to source space through Eq. (2.219) [2]. When Eq. (2.220) is ill-posed, stable inversion uses a penalized least-squares or Tikhonov template on admissible sources:

$$\mathbf{J}_{\mathrm{imp}_{\mathrm{reg}}} = \arg \min_{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega} \Big\{ \|\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}] - \mathbf{y}\|^2 + \alpha\, \mathcal{P}[\mathbf{J}_{\mathrm{imp}}] \Big\}, \tag{2.224}$$

with regularization parameter $\alpha > 0$ and penalty $\mathcal{P}$ enforcing smoothness, minimum norm, or sparsity on an equivalent surface [1]. Equation (2.224) does not remove $\ker \mathcal{M}_\omega$; it selects one representative from the affine coset $\mathcal{M}_\omega^{-1}(\mathbf{y})$ [2]. The choice of $\mathcal{P}$ is physics-adjacent modeling, not a Maxwell identity; different penalties yield different $\mathbf{J}_{\mathrm{imp}_{\mathrm{reg}}}$ with identical radiative predictions when $\delta\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ [1]. Transport and angular reports attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, per Eq. (2.28) [2]. regime qualification Eq. (2.140) criteria whether $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ lies in the propagating sector on which far-zone observables were designed:

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{inverse claims target } \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \text{ not reactive near fields,} \tag{2.225}$$

with $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.34) [2]. Attempting to reconstruct $\mathbf{J}_{\mathrm{imp}}$ from near-zone reactive samples without regime qualification conflates storage with export, the laboratory error flagged in Section 1.3.3 [2]. Outgoing uniqueness from exterior radiative uniqueness (Subsection 2.4.3) stabilizes the *field* side of Eq. (2.220); it does not promote evanescent-dominated states to identifiable radiative sources [1]. Formal adjoint back-projection provides a reconstruction *template* without claiming uniqueness:

$$\widetilde{\mathbf{J}_{\mathrm{imp}}} = \mathcal{M}_\omega^\dagger \, \mathbf{y}, \tag{2.226}$$

where $\mathcal{M}_\omega^\dagger$ is the Hilbert-space adjoint on declared domains [2]. Equation (2.226) is useful for sensitivity analysis and iterative inversion but generally satisfies $\mathcal{M}_\omega[\widetilde{\mathbf{J}_{\mathrm{imp}}}] \neq \mathbf{y}$ when $\ker \mathcal{M}_\omega \neq \{\mathbf{0}\}$ [1]. Consistent reconstruction requires projecting onto $\operatorname{ran} \mathcal{M}_\omega^\dagger$ or applying Eq. (2.224) [2]. Full integral representations for Eq. (2.226) belong to Section 2.5.1; they are not repeated here [1]. Equivalent sources and Huygens surfaces. Equivalent-current inversion on a closed surface Γ_{eq} seeks $\mathbf{J}_{\mathrm{eq}}$ on Γ_{eq} such that exterior fields match measured data [2]. Uniqueness of exterior $\mathcal{S}_\omega$ from exterior radiative uniqueness (Subsection 2.4.3) fixes the *field* on $\mathcal{F}_{\mathrm{SF}}$ but not $\mathbf{J}_{\mathrm{eq}}$: any non-radiating tangential distribution on Γ_{eq} can be added without changing far-zone samples [1]. The operator form is

$$\mathbf{J}_{\mathrm{eq},1} - \mathbf{J}_{\mathrm{eq},2} \in \ker \mathcal{R}_\omega\big|_{\Gamma_{\mathrm{eq}}}, \tag{2.227}$$

with restriction to surface currents in the trace class of Subsection 2.1.1 [2]. Numerical equivalence in Eq. (2.188) inherits the same null-space structure at finite matrix rank [1]. Near-field versus far-field data. Near-field scanning samples $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on a surface enclosing Ω_s, not $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ alone [2]. Because $\mathcal{S}_\omega$ is injective on outgoing fields but many sources share the same $\mathcal{R}_\omega$ image, near-field data can *reduce* null-space dimension relative to far-zone-only inversion without eliminating $\ker \mathcal{R}_\omega$ in general [1]. Outgoing radiation condition still required: two incoming/outgoing superpositions with identical near traces on a finite surface need not agree in the far zone unless exterior radiative uniqueness (Subsection 2.4.3) applies to the full exterior problem [2]. Near-field inversion therefore couples Subsection 2.4.3 field uniqueness with Subsection 2.4.4 source non-uniqueness [1]. Equations (2.220)–(2.224) require linear Maxwell theory, fixed ω, admissible Eq. (2.9), and outgoing forward completion Eq. (2.213) [2]. Nonlinear, time-varying, or dispersive inversion breaks the harmonic reduction of Subsection 2.2.3. Single-frequency inversion cannot resolve broadband excitation without superposition across ω [2]. Loss improves conditioning of Eq. (2.220) by damping resonances but does not make $\mathcal{R}_\omega$ injective on all of $\mathcal{J}_\Omega$ [1]. Complete null-space bases and NR source examples belong to Section 2.7 [2]. As a worked example, with N probe channels, Eq. (2.221) limits recoverable modal content to rank N regardless of mesh density [2]. Refining a boundary-element mesh without adding independent measurements does not resolve components in $\ker \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ [1]. This is the operator reading of Eq. (2.81), distinct from discretization error alone [2]. As a worked example, a source with $\mathcal{D}_{J,\mathrm{reg}} = 0$ may produce strong near-zone $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ while $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \approx \mathbf{0}$ [2]. Inverting near samples without Eq. (2.225) assigns spurious

radiative labels to storage-dominated fields [1]. regime qualification Eq. (2.140) must precede angular-transport claims on reconstructed sources [2]. Section 2.4 closes with the excitation-to-flux sequence fixed for open regions: Sommerfeld asymptotics and outgoing class Eqs. (2.194)–(2.190) (Subsection 2.4.1), outgoing selector and resolvent branch Eqs. (2.197)–(2.202) (Subsection 2.4.2), exterior radiative uniqueness Eq. (2.210) and resolvent injectivity Eq. (2.211) (Subsection 2.4.3), and inverse reconstruction limits Eqs. (2.220)–(2.218) (Subsection 2.4.4) [2]. Green kernels from Section 2.3 are now tied to a single physical branch; $\mathcal{S}_\omega$ is single-valued on $\mathcal{F}_{\mathrm{SF}}$, while $\mathcal{R}_\omega$ and composed measurement readouts remain rank-deficient on $\mathcal{J}_\Omega$ [1]. Section 2.5 evaluates radiation integrals, flux meters, and near/intermediate/far partitions on the qualified propagating subspace; Section 2.7 completes non-radiating null-space structure [2]. Regime qualification Eq. (2.140) and transport criterion Eq. (2.28) remain mandatory before angular-transport claims on operator outputs [2]. Outgoing-wave selection fixes the exterior resolvent branch assumed throughout radiation integrals; realizability limits—which accessible modes finite geometry can excite—are established in **??** (*Source $\mathcal{S}_{\mathrm{real}}$ and Spectral Truncation*).

2.5 Radiation Integrals and Source-to-Field Representations

Uniqueness and radiation conditions in Section 2.4 fix the exterior problem. Section 2.5 writes the source integrals that range engineers discretize on wire and surface meshes [2]. The same integrals appear in compact-range far-field transforms [7].

Section 2.4 fixed the outgoing branch under which exterior solutions are unique. An engineer on a range nonetheless needs evaluated fields and pattern amplitudes at declared observation points—not proof of existence alone [5]. Before any far-zone asymptotic is applied, the field surrounding a compact source occupies three spatial bands: a reactive near zone where $1/r^3$ and $1/r^2$ dyadic terms dominate what probes measure through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; an intermediate transition where storage and export compete; and a radiative far zone where transverse $e^{-\mathrm{j}k_0 r}/r$ waves support pattern and gain definitions through $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Distance labels and regime indices are fixed in Subsection 2.5.2 before asymptotic reductions; the integral representations below are the constructive route to $\mathcal{S}_\omega$ in all three bands [2]. A gain trace on Γ_R is a pattern meter reading on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after export selection; a port $|S_{11}|$ meter on a horn feed reads coupling through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and may include reactive storage unrelated to far-zone export [1]. A method-of-moments solution on a horn mouth illustrates the distinction: port coupling integrals supply $|S_{11}|$, while a radiation integral on Γ_R supplies far-zone gain; both descend from the same outgoing Green convolution but meter different observables [1]. The excitation-to-flux sequence $\mathbf{J}_{\mathrm{imp}} \mapsto \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \mapsto \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is single-valued on $\mathcal{F}_{\mathrm{SF}}$ at the field level yet rank-deficient at the source level through Eq. (2.218) [2]. This section develops the radiation integral Eq. (2.5) to evaluable outgoing integrals and ties range meters to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than to the full pair returned by Green convolution Eq. (2.149) [2]. Poynting flux and transport indices from 1 apply here; dyadic kernels and Sommerfeld completion remain in Sections 2.3–2.4. Let $\Omega_s = \operatorname{supp}\mathbf{J}_{\mathrm{imp}}$ be compact with characteristic scale a, vacuum wavenumber $k_0 = \omega\sqrt{\varepsilon_0\mu_0}$, and wavelength $\lambda_0 = 2\pi/k_0$. Section 2.5 organizes evaluation in four stages:

$$\mathbf{J}_{\mathrm{imp}} \xrightarrow{\mathcal{S}_\omega} (\mathbf{E},\mathbf{H}) \in \mathcal{F}_{\mathrm{SF}} \xrightarrow{\Pi_{\mathrm{rad}}} \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \xrightarrow{\mathcal{F}_{\mathrm{flux}}} \text{power, pattern, norm reports,} \tag{2.228}$$

where $\mathcal{F}_{\mathrm{flux}}$ denotes the flux and operator-norm functionals developed below by reference to 1 flux identities [2]. Equation (2.228) is the section-level restatement of Eq. (2.11): stage (iii) export must precede stage (iv) metering [1]. Applying $\mathcal{F}_{\mathrm{flux}}$ to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} imports reactive circulation

into quantities designed for outgoing transverse radiation (Section 1.3.3) [2]. The outgoing integral realization of $\mathcal{R}_\omega$ for $\mathbf{r} \notin \Omega_s$ is

$$\mathbf{E}_{\mathrm{rad}}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3 r', \qquad \mathbf{H}_{\mathrm{rad}} = -\frac{1}{\mathrm{j}\omega\mu_0} \nabla \times \mathbf{E}_{\mathrm{rad}}, \tag{2.229}$$

In Eq. (2.229), $\mathbf{r}$ is the observation point, $\mathbf{r}'$ ranges over the source volume Ω_s, $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ is the impressed electric current density at the source point, μ_0 is the vacuum permeability, ω is the angular frequency, and $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}$ is the outgoing electric dyadic from Eq. (2.200); the magnetic field $\mathbf{H}_{\mathrm{rad}}$ follows from $\mathbf{E}_{\mathrm{rad}}$ by the source-free Maxwell curl relation in the observation variable $\mathbf{r}$ [2]. Equation (2.229) upgrades Eq. (2.5) after Sections 2.3–2.4; the integral subsection below supplies magnetic couplings, equivalent surfaces, and routes that reduce to the same $\mathcal{R}_\omega$ image [1]. Near coincidence $\mathbf{r} \to \Omega_s$ requires regularization from Subsection 2.3.3; evaluation uses Eq. (2.229) only where the observation point lies outside Ω_s or on declared equivalent surfaces with trace matching [2]. Regime qualification Eq. (2.140) must be satisfied before flux meters attach to any shell integral [2]. As a worked example, for $\mathbf{J}_{\mathrm{imp}}$ supported in a ball of radius $a \ll \lambda_0$, $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ includes dominant reactive $1/r^2$, $1/r^3$ terms on spheres with $r \sim a$, while $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ carries leading $1/r$ transverse radiation with $k_0 r$ phase for $r \gg a$ [8]. Integrating Poynting flux on S_R with $R \sim a$ conflates storage with export; only when $k_0 R \gg 1$ and $R \gg a$ does shell power approach the R-independent value predicted by far-zone reduction [1]. As a worked example, near-field probes sample $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on a surface enclosing Ω_s [2]. Transforming those samples to a far-zone pattern implements Eq. (2.229) only when outgoing radiation condition and equivalent-source uniqueness on the exterior side are satisfied—that is, when exterior radiative uniqueness (Subsection 2.4.3) holds [1]. Distinct impressed currents in $\ker \mathcal{R}_\omega$ produce identical patterns while differing in reactive near fields; near scans alone do not resolve that null space without additional constraints (Subsection 2.4.4) [2].

2.5.1 Source-to-field integral representations

Retarded source integrals are what method-of-moments codes discretize on antennas [1]. Far-zone extraction is a limit on the same integral [8]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. The representation begins from retarded potentials driven by $\mathbf{J}_{\mathrm{imp}}$ on compact Ω_s, not from a plane-wave expansion [4]. The quantity being organized is the outgoing field pair $(\mathbf{E}, \mathbf{H})$; angular labels attach only after Π_{rad} [10]. An alternative route through static potentials would omit radiation damping and misstate reactive storage [8]. On a horn range, this step is what a MoM mesh implements before any pattern plot [1].

Before any volume quadrature is written, the range engineer already knows which observable the campaign will report [1]. A calibrated pattern probe on Γ_R integrates tangential $\mathbf{E}$ or $\mathbf{H}$ against a known polarization and yields a voltage proportional to a directional amplitude—a discrete sample of the far-zone pattern carried by $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not of the full pair $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. A shell wattmeter closes the same budget through $\mathcal{P}_{\mathrm{rad}}$ on Γ_R after Π_{rad} [2]. Near-field scanners on a hull enclosing Ω_s record $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and require a transform before pattern meters apply [7]. The volume integrals below are the constructive route to those meter readings: they assemble $(\mathbf{E}, \mathbf{H})$ from impressed $\mathbf{J}_{\mathrm{imp}}$, after which export selection maps the result to the pattern quantities the range records [2]. Radiation integrals assemble far-zone fields as superpositions of current moments weighted by outgoing phase [2]. Four laboratory settings motivate the representations that follow. On an antenna range, probes on a large sphere Γ_R sample the field produced by a feed in the compact range; the measured voltage is a discretization of a surface integral over Γ_R, not a direct reading of impressed $\mathbf{J}_{\mathrm{imp}}$ [1]. Horn illumination fixes $\mathbf{J}_{\mathrm{imp}}$ on the mouth through a waveguide mode

or aperture integral; the pattern on the range wall appears only after those currents convolve against the outgoing Green function [11]. Aperture radiation replaces the volume support by equivalent currents on Γ_{eq} when the interior is inaccessible [2]. Near-field scanners reconstruct far-zone quantities from dense samples on a hull enclosing Ω_s—a transform that succeeds only when outgoing uniqueness and export selection are respected [7]. The representation begins from retarded potentials driven by $\mathbf{J}_{\mathrm{imp}}$ on compact Ω_s, organizes the outgoing pair $(\mathbf{E}, \mathbf{H})$ before angular labels attach, and coincides with what method-of-moments codes discretize before far-zone extraction [4]. Static-potential routes omit radiation damping and misstate reactive storage; they are not used here [8]. Outgoing kernels, regularization, and branch selection are presupposed from Sections 2.3–2.4; distance-regime asymptotics and flux meters follow in the subsections below. Assumptions. Fix $\omega > 0$, compact $\operatorname{supp} \mathbf{J}_{\mathrm{imp}} = \Omega_s \Subset \Omega$, admissible data Eq. (1.23), charge continuity Eq. (1.8), and outgoing radiation condition Eqs. (2.194)–(2.197) on exterior problems [2]. Observation points $\mathbf{r}$ lie outside Ω_s or on equivalent surfaces with trace matching declared below; coincidence limits use Subsection 2.3.3 [1]. Kernels $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}$, $\overline{\mathbf{G}}^{(\mathrm{out})}_{H,\omega}$, and scalar $G^{(\mathrm{out})}_{\omega}$ denote outgoing branches fixed by Eq. (2.200) [2]. The electric volume integral is the constructive form of the dyadic convolution route: each source element $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r'$ contributes a dipole-like response that propagates outward with retarded phase, and the total field is their coherent sum [2]. On a MoM mesh, patch currents discretize the same sum; on Γ_R, pattern probes sample the far-zone limit after $\mathcal{R}_\omega$ has removed reactive storage that the volume integral still carries at finite r [1]. In the symbols below, $\mathbf{r}$ is the observation coordinate, $\mathbf{r}'$ the source coordinate integrated over Ω_s, $\mathbf{E}_{\mathrm{hom}}$ any exterior illumination not generated by $\mathbf{J}_{\mathrm{imp}}$, and the prefactor $-\mathrm{j}\omega\mu_0$ ties the integral to the locked phasor convention $\exp(-\mathrm{j}\omega t)$ [2]. The primary volume representation is

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \tag{2.230}$$

$$\mathbf{H}(\mathbf{r}) = \frac{1}{\mathrm{j}\omega\mu_0} \nabla \times \mathbf{E}(\mathbf{r}), \tag{2.231}$$

Equations (2.230)–(2.231) package the dyadic route as an evaluable export map: the electric field is a volume convolution against the *outgoing* dyad, while the magnetic field is a curl derived from that electric field in source-free observation regions [2]. For a square loop of side $2a$ with $k_0 a = 1.2$ observed at $\mathbf{r}$ with $k_0|\mathbf{r}| = 6$, the integral typically requires 16–64 wavelength-scale subdivisions of Ω_s for 2% amplitude accuracy when raw Eq. (2.152) is used patchwise; self-term corrections from Subsection 2.3.3 dominate error when $|\mathbf{r}| \to a$ [1]. The homogeneous term $\mathbf{E}_{\mathrm{hom}}$ carries exterior illumination and must vanish for pure radiation problems [2]. Equations (2.230)–(2.231) specialize Eqs. (2.149)–(2.153) with outgoing dyads for $\mathbf{r} \notin \Omega_s$ on simply connected observation regions; they coincide with Eq. (2.229) when $\mathbf{E}_{\mathrm{hom}} = \mathbf{0}$ [2]. The prefactor $-\mathrm{j}\omega\mu_0$ matches Eq. (2.116) and the locked phasor axis [2]. What propagates to a distant probe is the coherent sum of outgoing dyadic responses $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}(\mathbf{r}, \mathbf{r}')$ from each source element; what remains reactive at finite r is the non-transverse and inverse-power content retained in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ until Π_{rad} and regime qualification isolate export [2]. A near-field scanner records the full pair $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; a pattern meter on Γ_R with $k_0 R \gg 1$ approximates $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [7]. When magnetic impressed sources $\mathbf{J}^{(\mathrm{mag})}_{\mathrm{imp}}$ are primary, duality maps $\varepsilon_0\mathbf{E} \leftrightarrow \mu_0\mathbf{H}$ and $\mathbf{J}_{\mathrm{imp}} \leftrightarrow \mathbf{J}^{(\mathrm{mag})}_{\mathrm{imp}}$ yields

$$\mathbf{H}(\mathbf{r}) = \mathrm{j}\omega\varepsilon_0 \int_{\Omega_s} \overline{\mathbf{G}}^{(\mathrm{out})}_{H,\omega}(\mathbf{r}, \mathbf{r}') \cdot \mathbf{J}^{(\mathrm{mag})}_{\mathrm{imp}}(\mathbf{r}')\, d^3r' \;+\; \mathbf{H}_{\mathrm{hom}}(\mathbf{r}), \tag{2.232}$$

$$\mathbf{E}(\mathbf{r}) = -\frac{1}{\mathrm{j}\omega\varepsilon_0} \nabla \times \mathbf{H}(\mathbf{r}), \tag{2.233}$$

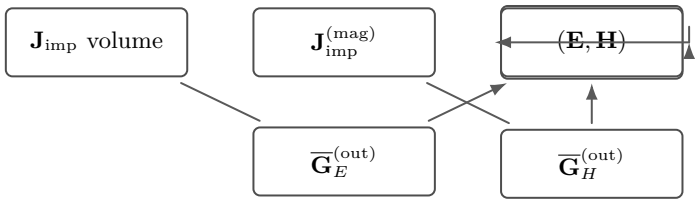


Figure 2.30: Integral routes: electric volume, magnetic volume, and Huygens surface representations converge to the same $\mathcal{S}_\omega$ image on the outgoing branch when trace matching holds.

with $\overline{\mathbf{G}}_{H,\omega}^{(\mathrm{out})}$ the magnetic dyad defined duality-wise from Subsection 2.3.1 without re-tabulating its closed form [2]. Equations (2.232)–(2.233) complete the dual route to Eqs. (2.230)–(2.231) for aperture magnetic currents and dual-equivalent models [2]. When both electric and magnetic impressed sources are present, superposition assembles the full Maxwell pair from the dual volume routes without re-opening gauge or static-potential paths [2]. Superposition gives the coupled volume representation

$$\begin{aligned}\mathbf{E}(\mathbf{r}) &= -\mathrm{j}\omega\mu_0\int_{\Omega_s}\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' - \frac{1}{\mathrm{j}\omega\varepsilon_0}\nabla\times\mathbf{H}^{(\mathrm{mag})}(\mathbf{r}) + \mathbf{E}_{\mathrm{hom}}(\mathbf{r}),\\ \mathbf{H}(\mathbf{r}) &= \mathrm{j}\omega\varepsilon_0\int_{\Omega_s}\overline{\mathbf{G}}_{H,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}^{(\mathrm{mag})}(\mathbf{r}')\,d^3r' + \frac{1}{\mathrm{j}\omega\mu_0}\nabla\times\mathbf{E}^{(\mathrm{elec})}(\mathbf{r}) + \mathbf{H}_{\mathrm{hom}}(\mathbf{r}),\end{aligned} \tag{2.234}$$

where $\mathbf{E}^{(\mathrm{elec})}, \mathbf{H}^{(\mathrm{mag})}$ denote the single-source contributions from Eqs. (2.230) and (2.232) [2]. Equation (2.234) is the first-order Maxwell integral system on outgoing kernels; it reduces to either branch when one impressed source vanishes [2]. Linearity Eq. (2.6) follows from Eq. (2.234) on fixed $\mathcal{B}_\Gamma$ [1]. The combined integral returns the full admissible Maxwell pair $\mathcal{S}_\omega[\mathfrak{D}]$: electric and magnetic impressed sources superpose at observation $\mathbf{r}$ with each contribution evaluated at source $\mathbf{r}'$ through the outgoing dyads G and $\overline{\mathbf{G}}_E$, $\overline{\mathbf{G}}_H$. Pattern, gain, and shell wattmeters do not read this sum directly; they act on $\mathcal{R}_\omega[\mathfrak{D}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathfrak{D}]$ after export selection removes reactive and non-propagating fractions that the volume integral still carries at finite range [2]. When $\nabla\cdot\mathbf{J}_{\mathrm{imp}} - \mathrm{j}\omega\rho = 0$ with nonzero ρ, the complete electric field includes a scalar-potential gradient driven by ρ:

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\int_{\Omega_s}\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r' - \nabla\int_{\Omega_s}G_\omega^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')\,\frac{\rho(\mathbf{r}')}{\varepsilon_0}\,d^3r' \;+\; \mathbf{E}_{\mathrm{hom}}(\mathbf{r}), \tag{2.235}$$

with $G_\omega^{(\mathrm{out})}$ the outgoing scalar kernel from Eq. (2.145) [2]. Equation (2.235) enforces Gauss coupling without leaving $\mathcal{J}_\Omega$; when Eq. (1.8) holds and ρ is determined by $\mathbf{J}_{\mathrm{imp}}$, the gradient term is fixed by the same impressed data [2]. The charge-augmented integral still evaluates $\mathcal{S}_\omega[\mathfrak{D}]$: the scalar kernel $G_\omega^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')$ propagates charge density from $\mathbf{r}'$ to scalar potential at $\mathbf{r}$, while the dyadic term propagates $\mathbf{J}_{\mathrm{imp}}$; shell meters on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ remain insensitive to how reactive near storage partitions between the two contributions at finite r [1]. Lorenz-gauge vector-potential reduction Eq. (2.147) collapses to Eq. (2.235) on uniform exterior domains; gauge freedom is not re-opened [1]. When the interior of a source region is inaccessible, equivalent currents on a enclosing surface reproduce the exterior field without resolving volume $\mathbf{J}_{\mathrm{imp}}$; the surface integrals below are the operator form of that laboratory substitution [2]. Let Γ_{eq} be a closed surface enclosing Ω_s, with outward normal $\hat{\mathbf{n}}$ and equivalent surface densities $\mathbf{J}_{s,\mathrm{eq}}$, $\mathbf{K}_{s,\mathrm{eq}}$ on Γ_{eq} in the trace class of Subsection 2.1.1. The exterior Huygens representation is

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\int_{\Gamma_{\mathrm{eq}}}\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')\cdot\mathbf{J}_{s,\mathrm{eq}}(\mathbf{r}')\,dS' + \int_{\Gamma_{\mathrm{eq}}}\nabla' G_\omega^{(\mathrm{out})}(\mathbf{r},\mathbf{r}')\times\mathbf{K}_{s,\mathrm{eq}}(\mathbf{r}')\,dS', \tag{2.236}$$

$$\mathbf{H}(\mathbf{r}) = \mathrm{j}\omega\varepsilon_0 \int_{\Gamma_{\mathrm{eq}}} \overline{\mathbf{G}}^{(\mathrm{out})}_{H,\omega}(\mathbf{r},\mathbf{r}') \cdot \mathbf{K}_{s,\mathrm{eq}}(\mathbf{r}')\, dS' - \int_{\Gamma_{\mathrm{eq}}} \nabla' G^{(\mathrm{out})}_{\omega}(\mathbf{r},\mathbf{r}') \times \mathbf{J}_{s,\mathrm{eq}}(\mathbf{r}')\, dS', \tag{2.237}$$

for $\mathbf{r}$ outside the closed surface bounded by Γ_{eq} [2]. Equations (2.236)–(2.237) are the surface form of the same outgoing propagation operator; ∇' acts on the source coordinate $\mathbf{r}'$ [2]. Tangential field matching

$$\hat{\mathbf{n}} \times \mathbf{E}\big|_{\Gamma^+_{\mathrm{eq}}} = \hat{\mathbf{n}} \times \mathbf{E}\big|_{\Gamma^-_{\mathrm{eq}}}, \qquad \hat{\mathbf{n}} \times \mathbf{H}\big|_{\Gamma^+_{\mathrm{eq}}} = \hat{\mathbf{n}} \times \mathbf{H}\big|_{\Gamma^-_{\mathrm{eq}}} \tag{2.238}$$

fixes $\mathbf{J}_{s,\mathrm{eq}} = \hat{\mathbf{n}} \times (\mathbf{H}^- - \mathbf{H}^+)$, $\mathbf{K}_{s,\mathrm{eq}} = \hat{\mathbf{n}} \times (\mathbf{E}^- - \mathbf{E}^+)$ in the stratified sense of Eq. (1.19) [2]. Equivalent-source non-uniqueness Eq. (2.227) applies: distinct $(\mathbf{J}_{s,\mathrm{eq}}, \mathbf{K}_{s,\mathrm{eq}})$ pairs related by $\ker \mathcal{R}_\omega$ yield identical far-zone radiation [1]. The Huygens surface integrals reproduce exterior fields from equivalent currents on Γ_{eq}: observation $\mathbf{r}$ lies outside the closed surface, source $\mathbf{r}'$ ranges over Γ_{eq}, and the same outgoing kernels $G^{(\mathrm{out})}_\omega$ and $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}$ propagate tangential data outward. What is measured on a distant range is still $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after Π_{rad}; equivalent currents that differ only by $\ker \mathcal{R}_\omega$ leave that measurement unchanged while altering reactive near fields through $\mathcal{S}_\omega$ [2].

Let $\mathfrak{D} = (\mathbf{J}_{\mathrm{imp}}, \mathbf{J}^{(\mathrm{mag})}_{\mathrm{imp}}, \rho, \mathcal{B}_\Gamma)$ be admissible with outgoing radiation condition on exterior problems. Then the physical solution $\mathcal{S}_\omega[\mathfrak{D}] = (\mathbf{E}, \mathbf{H})$ is given by Eqs. (2.234)–(2.235) with outgoing kernels, or by Eqs. (2.236)–(2.238) when equivalent surface data replace volume sources [2]. The representation agrees with Eq. (2.202) and Green–BVP equivalence Eq. (2.184) on the outgoing branch fixed by exterior radiative uniqueness (Subsection 2.4.3). Subsection 2.3.1 defined dyadic kernels satisfying Eq. (2.148); Subsection 2.3.3 justified convolution on $\mathcal{J}_\Omega$; Section 2.4 selected the outgoing branch. Substituting $\overline{\mathbf{G}}^{(\mathrm{out})}_{E,\omega}$ into Eq. (2.149) yields Eq. (2.230). Magnetic and coupled forms follow from duality and superposition [2]. Huygens formulas follow by replacing volume sources with surface distributions that reproduce the same tangential traces [1]. Uniqueness on $\mathcal{F}_{\mathrm{SF}}$ when exterior radiative uniqueness (Subsection 2.4.3) holds identifies the integral representation with $\mathcal{S}_\omega$ [2]. The volume and surface integrals above return the full admissible pair $\mathcal{S}_\omega[\mathfrak{D}]$; pattern meters and gain reports act only after export selection through $\mathcal{R}_\omega$ and Π_{rad} [2]. Export selection follows evaluation:

$$\mathcal{R}_\omega[\mathfrak{D}] = \Pi_{\mathrm{rad}}\Big(\mathbf{E}[\mathfrak{D}],\, \mathbf{H}[\mathfrak{D}]\Big), \tag{2.239}$$

with Π_{rad} from Eq. (2.203) [2]. Pattern cuts, gain reports, and shell power meters act on Eq. (2.239), not on the raw volume integral alone [1]. On the outgoing branch,

$$\mathcal{S}_\omega[\mathfrak{D}] = \mathcal{O}_{\mathrm{out}}\, \mathcal{L}_\omega^{-1} \begin{bmatrix} \mathbf{s}_\omega(\mathfrak{D}) \\ \mathbf{g}_\Gamma \end{bmatrix} = \mathcal{O}_{\mathrm{out}}\Big(\mathcal{I}_{\mathrm{vol}}[\mathfrak{D}]\Big), \tag{2.240}$$

with $\mathcal{I}_{\mathrm{vol}}[\mathfrak{D}]$ the combined volume integrals Eqs. (2.234)–(2.235), connecting integral and operator routes through Eqs. (2.124)–(2.201) [2]. Equation (2.240) is the constructive content of Eq. (2.184) for evaluation purposes: a method-of-moments matrix implements the resolvent route while patch quadrature implements the integral route, and both must reference the same outgoing branch [1]. When $\Omega = \cup_j \Omega_j$ with interface traces from Subsection 2.3.5, Eqs. (2.234)–(2.235) hold patchwise with local k_j and outgoing exterior completion on the infinite exterior region [2]. Impressed $\mathbf{J}_{\mathrm{imp}}$ may vanish in passive exterior regions; equivalent currents on material boundaries enter through Eqs. (2.236)–(2.238) with Γ_{eq} on scatterer surfaces [1]. Interior cavity problems replace outgoing G with standing-mode Green functions and require quotienting Eq. (2.87); those representations are excluded from the outgoing integral family developed here [2]. Equations (2.230)–(2.235) require linear media, fixed ω, admissible $\mathfrak{D}$, outgoing radiation condition on exterior problems, and observation points outside singular coincidence unless regularization from Subsection 2.3.3 is applied [2]. Incoming kernels produce different integrals violating Eq. (2.177) [8]. Anisotropic, moving, or

nonlinear media require modified kernels beyond the locked patchwise-scalar class [2]. Near-zone reactive dominance does not invalidate the integral; it signals that $\mathcal{R}_\omega[\mathfrak{D}]$ may differ sharply from $\mathcal{S}_\omega[\mathfrak{D}]$ until Eq. (2.140) is satisfied [1]. One operational distinction closes the display below: volume integrals construct the full Maxwell pair $\mathcal{S}_\omega[\mathfrak{D}]$, whereas pattern meters and watt integrators on Γ_R read only after outgoing selection has isolated $\mathcal{R}_\omega[\mathfrak{D}]$. A method-of-moments matrix must not be read as a radiation report; Maxwell theory shows that the matrix implements whichever Green branch and boundary completion were declared, and reactive near-zone content may dominate local probes without contributing to qualified export [2]. A measurement on a closed scan surface therefore records the complete field state unless a transform and regime qualification explicitly project onto the radiative sector. Synthesis, diagnosis, and inversion campaigns must name the same outgoing branch before any two routes—resolvent inversion and direct quadrature—are compared [1]. Subsection 2.5.2 next fixes distance labels that tell the engineer when far-zone asymptotics may replace the full integral without misreading reactive storage as radiated power.

2.5.2 Near, intermediate, and far regimes

The far-field approximation replaces distance-dependent amplitudes by e^{-jkr}/r and angular factors [8]. Range calibrations must sit in the Fraunhofer zone for this step [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Multipole expansions order far-zone fields by angular complexity: the $\ell = 1$ dipole term dominates electrically small feeds, whereas reflector campaigns require many ℓ before sidelobe floors stabilize [2]. The far-field approximation replaces distance-dependent amplitudes by $e^{-\mathrm{j}k_0 r}/r$ and angular factors; range calibrations must sit in the Fraunhofer zone for that step to be valid [8]. A pattern meter on Γ_R with $k_0 R \ll 1$ still reads $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; only when $k_0 R \gg 1$ and $R \gg a$ does the same probe approximate $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ through the far-zone reduction developed below [1].

The same source produces all three regions simultaneously; the distinction arises from how different terms of the radiation integral scale with distance [2]. Near the feed, reactive $1/r^3$ and $1/r^2$ terms dominate shell integrals; at intermediate distances those terms compete with the leading $e^{-\mathrm{j}k_0 r}/r$ radiation; only on sufficiently large spheres does the transverse $1/r$ term alone support pattern and gain definitions [1]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact with characteristic scale a. For observation radius $r = |\mathbf{r}|$ from a reference point in Ω_s, define

$$\begin{aligned} \text{near: } & k_0 r \lesssim 1, \\ \text{intermediate: } & 1 \lesssim k_0 r \lesssim (k_0 a)^{-1}, \\ \text{far: } & k_0 r \gg k_0 a, \quad r \gg a, \end{aligned} \tag{2.241}$$

In Eq. (2.241), $r = |\mathbf{r}|$ is the observation radius (m) from a reference point in $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$; a is the source characteristic scale (m); $k_0 = \omega\sqrt{\varepsilon_0 \mu_0}$ is the vacuum wavenumber (rad m^{-1}); and the inequalities label reactive near, transition, and radiative far bands before the constants $C_{\mathrm{n}}, C_{\mathrm{i}}, C_{\mathrm{f}}$ are fixed in Eq. (2.242). with implicit constants fixed below [2]. Equations (2.241) are scale labels, not sharp cutoffs: reactive and radiative terms coexist in the near zone, and intermediate zones require explicit asymptotic bookkeeping before source integrals reduce to transverse $1/r$ radiation [1]. Pattern and gain definitions used in antenna theory attach only after the far-zone reduction in Eq. (2.192) [1]. Field-magnitude scalings, far-zone reduction of Eqs. (2.230)–(2.234), Fresnel–Fraunhofer transition scales, and power-flux convergence on large spheres are developed below [2]. Sommerfeld asymptotics Eqs. (2.193) and (2.192) are presupposed; regime indices Eqs. (2.31)–(2.32) are invoked by reference [2]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact with characteristic radius $a = \max_{\mathbf{r},\mathbf{r}' \in \Omega_s} |\mathbf{r} - \mathbf{r}'|/2$, vacuum wavenumber $k_0 = \omega\sqrt{\varepsilon_0 \mu_0}$, wavelength $\lambda_0 = 2\pi/k_0$, and electrical

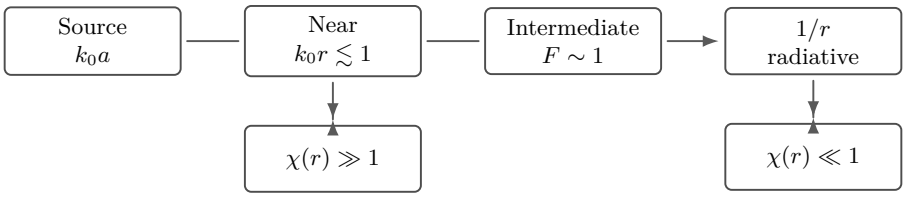


Figure 2.31: Distance regimes Eq. (2.242): reactive dominance near the source, transition through Fresnel parameter Eq. (2.245), radiative export in the far zone with Eq. (2.31) $\chi(r) \ll 1$.

size $k_0 a$ [2]. Observation points $\mathbf{r}$ satisfy $|\mathbf{r}| = r \notin \Omega_s$; reference origin is chosen in Ω_s. Outgoing kernels from the source integrals above and dyadic structure from Eq. (2.152) are presupposed without re-tabulation [2]. Constants $C_1, C_2, \ldots$ below depend on k_0, a and source geometry but not on r in the asymptotic limits stated [1]. The locked regime partition is

$$\begin{aligned} &\textit{near (reactive) zone: } k_0 r \le C_{\mathrm{n}}, \\ &\textit{intermediate (transition) zone: } C_{\mathrm{n}} < k_0 r \le C_{\mathrm{i}}(k_0 a), \\ &\textit{far (radiative) zone: } k_0 r > C_{\mathrm{i}}(k_0 a), \quad r > C_{\mathrm{f}} a^2/\lambda_0, \end{aligned} \tag{2.242}$$

with declared positive constants $C_{\mathrm{n}} = \mathcal{O}(1)$, $C_{\mathrm{i}} = \mathcal{O}(1)$ when $k_0 a = \mathcal{O}(1)$, and Fraunhofer scale $r_{\mathrm{F}} = 2a^2/\lambda_0$ entering $C_{\mathrm{f}} = \mathcal{O}(1)$ [2]. Equations (2.242) upgrade Eq. (2.241): they are diagnostic bands, not disjoint sharp cutoffs [2]. For $k_0 a \ll 1$ (electrically small sources), C_{i} may be taken near unity and the intermediate zone is narrow; for extended apertures with $k_0 a \gg 1$, the Fresnel–Fraunhofer condition $r \gtrsim r_{\mathrm{F}}$ controls pattern formation [1]. Near-zone field scalings. Evaluating Eq. (2.230) with Eq. (2.152) for $k_0 r \le C_{\mathrm{n}}$ yields

$$|\mathbf{E}(\mathbf{r})| = \mathcal{O}\big(\mu_0 \omega \|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}\, k_0^{-2} r^{-3}\big) \;+\; \mathcal{O}\big(\mu_0 \omega \|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}\, k_0^{-1} r^{-2}\big), \tag{2.243}$$

with $\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}$ a source amplitude scale on Ω_s [2]. Equation (2.243) reflects the R^{-3} and R^{-2} terms in Eq. (2.152); the leading r^{-3} contribution is reactive (stored) in the sense of Section 1.3.3 [2]. Magnetic field follows from Eq. (2.231) with the same near-zone polynomial-in-$k_0 r$ ordering [1]. On shells with Eq. (2.31) $\chi(r) \gg 1$, Poynting flux through Eq. (2.7) is not a credible export meter [2]. When $C_{\mathrm{n}} < k_0 r \le C_{\mathrm{i}}(k_0 a)$, reactive and radiative terms in Eq. (2.152) are comparable:

$$|\mathbf{E}_{\mathrm{reactive}}(\mathbf{r})| = \mathcal{O}\big(|\mathbf{E}_{\mathrm{rad}}(\mathbf{r})|\, k_0 r\big), \tag{2.244}$$

with $\mathbf{E}_{\mathrm{rad}}$ the leading transverse $e^{-\mathrm{j}k_0 r}/r$ component and $\mathbf{E}_{\mathrm{reactive}}$ the remainder [2]. Equation (2.244) diagnoses regime mixing: storage and export coexist in the full field pair until export is isolated by $\mathcal{R}_\omega$ [2]. For extended sources, the Fresnel parameter

$$F = \frac{\pi a^2}{\lambda_0 r} = \frac{k_0 a^2}{2r} \tag{2.245}$$

separates Fresnel ($F \gtrsim 1$) from Fraunhofer ($F \ll 1$) pattern regions on observation shells [8]. The parameter F marks the transition between those pattern regimes; complete aperture diffraction theory is not re-developed [2]. Section 2.4 [2]. The far-zone reduction of integrals proceeds as follows: at distances large compared with source extent and wavelength, a pattern probe on Γ_R samples the transverse $e^{-\mathrm{j}k_0 r}/r$ wave whose angular shape is fixed by a Fourier transform of $\mathbf{J}_{\mathrm{imp}}$ over Ω_s [8]. The volume integral at finite r still evaluates $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$: reactive terms propagate mathematically but do not export watt-bearing power; only after the far-zone limit does the meter

reading coincide with $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ through Eqs. (2.246)–(2.247) [2]. When $k_0 r \gg 1$ and $r \gg r_{\mathrm{F}}$ for extended sources, outgoing dyadic asymptotics Eq. (2.193) reduce Eq. (2.230) to

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\theta,\phi)\,\frac{e^{-\mathrm{j}k_0 r}}{r} \;+\; \mathcal{O}(r^{-2}), \qquad \mathbf{E}_0 \cdot \hat{\mathbf{r}} = 0, \tag{2.246}$$

$$\mathbf{E}_0(\theta,\phi) = -\mathrm{j}\omega\mu_0\,\frac{1}{4\pi}\int_{\Omega_s} \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,e^{-\mathrm{j}k_0\hat{\mathbf{r}}\cdot\mathbf{r}'}\,d^3r', \tag{2.247}$$

with $\hat{\mathbf{r}} = \mathbf{r}/r$ and the transversality constraint enforced by $\overline{\mathbf{G}}_\infty$ in Eq. (2.193) [2]. Equation (2.246) is the far-zone limit of Eq. (2.192); the amplitude integral Eq. (2.247) is the vector radiation integral used in antenna far-field post-analysis [1]. The amplitude integral is what a Fraunhofer-range pattern probe measures after Π_{rad}: it Fourier-transforms $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ over the source volume while the reactive inverse-power terms that $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ carries at finite r no longer enter at leading order [1]. Magnetic field in the far zone satisfies Eq. (2.191) with the same $\mathbf{E}_0$ [2]. Let $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ be supported in $B_a(0)$ with outgoing radiation condition. For $r > r_{\min}(a,\lambda_0)$ as in Eq. (2.242), the field from Eq. (2.230) satisfies Eqs. (2.246)–(2.247) with error $\mathcal{O}\big((k_0 a)^2/(k_0 r)\,|\mathbf{E}_0|\,r^{-1}\big)$ for extended sources and $\mathcal{O}\big((k_0 r)^{-1}\,|\mathbf{E}_0|\,r^{-1}\big)$ for $k_0 a \ll 1$ [2]. Substitute Eq. (2.193) into Eq. (2.230) and expand $|\mathbf{r}-\mathbf{r}'|^{-1}$ for $|\mathbf{r}'| \le a \ll r$. The leading $e^{-\mathrm{j}k_0 r}/r$ term factorizes into $\mathbf{E}_0(\theta,\phi)$ times phase $e^{-\mathrm{j}k_0\hat{\mathbf{r}}\cdot\mathbf{r}'}$ over the source; transversality follows from the dyadic structure Eq. (2.152) at leading order [2]. Higher orders in a/r and $1/(k_0 r)$ give the stated error bounds [8]. On spheres $S_r = \{|\mathbf{r}| = r\}$, define shell power $P(R) = \mathrm{Re}\oint_{S_R} \mathbf{S}\cdot\hat{\mathbf{r}}\,dS$ as in Eq. (2.7) [2]. In the far zone,

$$|P(r) - P_\infty| = \mathcal{O}(r^{-1}), \qquad P_\infty = \frac{1}{2\eta_0}\oint_{S_r} \big|\mathbf{E}_0(\theta,\phi)\big|^2\,d\Omega, \tag{2.248}$$

with $\mathbf{E}_0$ from Eq. (2.247) and η_0 from Eq. (2.208) [2]. Equation (2.248) is the operator form of Eq. (2.32): $P(r)$ stabilizes once reactive r^{-2}, r^{-3} contributions no longer dominate the shell integral [2]. Evaluating Eq. (2.7) on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ before $r \gtrsim r_c$ from Eq. (2.33) confuses storage with export [1]. Distance regimes induce an observation-shell map on operator outputs:

$$\begin{aligned} r \in \text{near} \;&\Longrightarrow\; \text{meter } \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \\ r \in \text{far} \;&\Longrightarrow\; \text{meter } \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ via Eqs. (2.246)–(2.247)}, \end{aligned} \tag{2.249}$$

with intermediate shells requiring explicit $\chi(r)$ from Eq. (2.31) before transport claims [2]. When $k_0 a \ll 1$, Eq. (2.247) reduces to the electric-dipole pattern proportional to $\widehat{\boldsymbol{\theta}} \times (\widehat{\boldsymbol{\theta}} \times \mathbf{p})$ with $\mathbf{p} = \int \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\,d^3r'$ [8]. The near zone extends to $r \sim \lambda_0$; far-zone reduction applies for $k_0 r \gg 1$ [1]. When $k_0 a \gg 1$, multiple lobes appear in $\mathbf{E}_0(\theta,\phi)$ and Fraunhofer distance $r_{\mathrm{F}} = 2a^2/\lambda_0$ dominates the far-zone threshold [2]. Angular-spectrum action in Section 2.9 filters $\mathbf{E}_0(\theta,\phi)$ without altering the integral reduction above [2]. Equations (2.242)–(2.248) require linear vacuum or piecewise-constant media, fixed ω, outgoing radiation condition, and observation outside Ω_s [2]. They fail in cutoff waveguides where propagating modes do not exist (Eq. (2.141)), in cavity standing-wave fields, and in lossless open resonances without outgoing selection [2]. Anisotropic beams and surface waves require modified asymptotics beyond the locked free-space dyad template [1]. The far-field reduction asserts field asymptotics on the outgoing branch only; it does not assert uniqueness of sources [2]. The regime partition is not a sharp geometric cutoff but a diagnostic on which terms of the same outgoing integral dominate what an instrument measures. Near and intermediate zones may appear to contradict far-zone pattern theory because substantial field energy is visible on small spheres; on the bench, that energy is largely reactive circulation diagnosed through $\mathcal{S}_\omega$, while pattern definitions attach only after $k_0 r \gg 1$ and Fresnel–Fraunhofer qualifications are met [1].

Shell watt readings must not be identified with radiated power until Eq. (2.140) is satisfied: what changes across bands is the relative weight of storage and export in the *same* Maxwell solution, not the governing equations themselves [5]. Subsection 2.5.3 connects these spatial bands to flux, momentum, and operator norms imported from Chapter 1, always downstream of the integral evaluation chain established here.

2.5.3 Energy, momentum, angular-momentum flux, power, and operator norms

With outgoing integrals and distance labels in place, the display below evaluates what an observer can meter on large shells. The radiation operator does not merely transport fields: it transports measurable physical quantities—energy, momentum, and angular momentum—which ultimately determine what an observer can detect [5]. On a calibrated compact range, the wattmeter on Γ_R is the primary pattern-adjacent meter: it integrates $\hat{\mathbf{r}} \cdot \mathbf{S}$ on the radiative branch $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after Π_{rad}, not on the full $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ pair that near-field probes sample at $r \sim a$ [2]. Power budgets on Γ_R must close before torque or angular-flux claims are promoted [12]. Complex Poynting flux from Eq. (1.1.4) applies to any admissible field pair, but power budgets for transport claims use the radiative branch:

$$P_{\mathrm{rad}}(R) = \mathrm{Re} \oint_{\Gamma_R} \hat{\mathbf{r}} \cdot \mathbf{S}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]]\, dS, \tag{2.250}$$

with $\mathbf{S}[\cdot]$ the Poynting vector functional imported from Subsection 1.1.4 and $\Gamma_R = \{r = R\}$ large enough for far-zone reduction when reporting export [2]. Equation (2.250) specializes Eq. (2.7) to the outgoing selector chain; it is not equivalent to integrating $\mathbf{S}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ unless reactive content has been shown negligible or orthogonal to the declared observable [1]. Poynting balance Eq. (1.1.4), Maxwell stress Eq. (1.37), angular balance Eqs. (1.41)–(1.42), and transport indices Eqs. (1.158)–(1.67) are presupposed from Chapter 1 [2]. Radiative–non-radiative splitting is taken up in the closing subsection below. Let $(\mathbf{E}, \mathbf{H}) = \mathcal{S}_\omega[\mathfrak{D}]$ and $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}) = \mathcal{R}_\omega[\mathfrak{D}]$ be as in the evaluation chain Eq. (2.228) [2]. Flux meters on spherical shells $S_r = \{|\mathbf{r}| = r\}$ assume r lies in the far zone of Eq. (2.242) unless noted otherwise, with $\mathcal{D}_{J,\mathrm{reg}} = 1$ from Eq. (2.34) [2]. Complex Poynting vector $\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^*$ and stress tensor $\mathbf{T}$ follow Eqs. (1.1.4) and (1.37) [2]. Angular-flux tensor $\mathbf{M} = \mathbf{r} \times \mathbf{T}$ uses origin $\mathbf{r}_0$ declared in Subsection 1.1.5 [12]. Flux functionals on the radiative branch. Define the radiative Poynting functional

$$\mathbf{S}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] = \frac{1}{2}\, \mathbf{E}_{\mathrm{rad}} \times \mathbf{H}^*_{\mathrm{rad}}, \qquad (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}) = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.251}$$

with $\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}$ the field components selected by Eq. (2.203) [2]. Equation (2.251) specializes Eq. (1.1.4) to the operator output on which export claims are allowed per Eq. (2.28) [1]. The reactive counterpart $\mathbf{S}_{\mathrm{re}} = \frac{1}{2}(\mathbf{E} - \mathbf{E}_{\mathrm{rad}}) \times (\mathbf{H} - \mathbf{H}_{\mathrm{rad}})^*$ is not a transport meter unless separately qualified [2]. Stress and angular-flux functionals on the radiative branch are

$$\mathbf{T}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] = \mathbf{T}(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}), \qquad \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] = \mathbf{r} \times \mathbf{T}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}], \tag{2.252}$$

with $\mathbf{T}(\cdot, \cdot)$ the Maxwell stress map Eq. (1.37) [2]. Equations (2.251)–(2.252) specialize 1 flux objects; they introduce no new conservation law [2]. Shell power on the radiative branch is the bounded linear functional

$$\mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) = \mathrm{Re} \oint_{S_r} \hat{\mathbf{r}} \cdot \mathbf{S}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]\, dS, \tag{2.253}$$

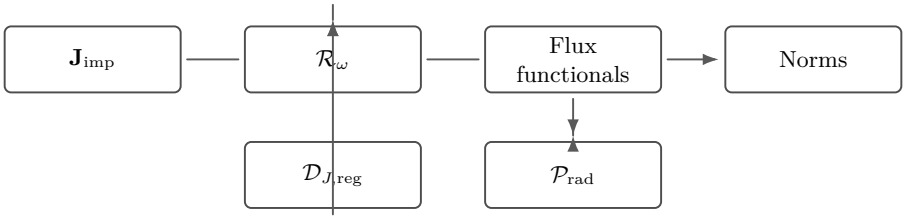


Figure 2.32: Flux measurement sequence: regime qualification, then radiative-branch meters only.

which coincides with Eq. (2.250) and Eq. (2.7) when $(\mathbf{E}, \mathbf{H})$ in those equations is replaced by $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. In the far zone, Eqs. (2.246)–(2.247) give

$$\mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) = \frac{1}{2\eta_0} \oint_{S_r} \left|\mathbf{E}_0(\theta, \phi)\right|^2 d\Omega \; + \; \mathcal{O}(r^{-1}), \tag{2.254}$$

with $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$ and $\mathbf{E}_0$ from Eq. (2.247) [2]. Equation (2.254) is the operator form of Eq. (2.248) and Eq. (2.32) [2]. On a calibrated compact range, $\mathcal{P}_{\mathrm{rad}}$ is the wattmeter reading on Γ_R after export selection—the quantity an antenna engineer reports as radiated power [1]. Evaluating $\mathcal{P}_{\mathrm{rad}}$ on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ before $r \gtrsim r_c$ from Eq. (2.33) imports reactive storage into the power budget [1]. Linear momentum flux meter. Net momentum export through S_r is

$$\mathbf{F}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) = \mathrm{Re} \oint_{S_r} \mathbf{T}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] \cdot \hat{\mathbf{r}} \, dS, \tag{2.255}$$

the time-averaged force functional inherited from Eq. (1.38) [2]. In the far zone with Eq. (2.191),

$$\mathbf{F}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) = \frac{1}{c} \, \hat{\mathbf{r}} \, \mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) \; + \; \mathcal{O}(r^{-1}), \tag{2.256}$$

linking momentum export to radiated power on the propagating branch [2]. A scattering analyst interpreting radiation pressure on a target uses $\mathbf{F}_{\mathrm{rad}}$ as the time-averaged force carried by the outgoing wave [2]. Equation (2.256) expresses Eq. (1.39) in the far-zone limit for unit $\hat{\mathbf{q}} = \hat{\mathbf{r}}$ [2]. Momentum meters attach to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not to reactive near-zone circulation in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \setminus \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [1]. Angular-momentum flux meter. Axis torque about unit direction $\hat{\mathbf{u}}$ is the functional

$$\tau_{\hat{\mathbf{u}}}(r; \mathbf{J}_{\mathrm{imp}}) = -\hat{\mathbf{u}} \cdot \mathrm{Re} \oint_{S_r} \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] \cdot \hat{\mathbf{r}} \, dS, \tag{2.257}$$

specializing Eq. (1.44) at harmonic steady state to the radiative branch [12]. Net angular-momentum flux through S_r is

$$\mathbf{J}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) = \mathrm{Re} \oint_{S_r} \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}] \cdot \hat{\mathbf{r}} \, dS, \tag{2.258}$$

with $\tau_{\hat{\mathbf{u}}} = -\hat{\mathbf{u}} \cdot \mathbf{J}_{\mathrm{rad}}$ [2]. Orbital–spin splits Eq. (1.43) are not re-derived; any angular report must close through Eq. (1.42) on the declared control volume [12]. Directional concentration uses Eq. (1.45) evaluated on $\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]$ [2]. Subsection 1.2.3 fixed transport $\mathcal{D}_J = 1$ through conditions $\mathcal{C}_1$–$\mathcal{C}_3$ on field states. On operator outputs,

$$\begin{aligned}
&\mathcal{C}_1[\mathbf{J}_{\mathrm{imp}}] \iff \text{Eq. (1.62) holds on } \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}], \\
&\mathcal{C}_2[\mathbf{J}_{\mathrm{imp}}] \iff \text{Eq. (1.42) holds for } (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}), \\
&\mathcal{C}_3[\mathbf{J}_{\mathrm{imp}}] \iff \text{Eq. (1.63) holds with } \Pi = \Pi_{\mathrm{rad}} \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}],
\end{aligned} \tag{2.259}$$

with indices Λ_J, $\mathcal{K}_J$, and $\mathcal{R}_\tau$ from Eqs. (1.158)–(1.66) evaluated on the same radiative fields [2]. Operational acceptance $\mathcal{D}_J = 1$ in Eq. (1.67) therefore means transport interpretation is attached to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after $\mathcal{D}_{J,\mathrm{reg}} = 1$, not to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. The transport qualification of Section 1.2.3 applies unchanged once $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}})$ replaces the full pair on the measurement shell [2]. Radiation operator norms. Field energy on the radiative branch uses the inherited bilinear form Eq. (2.20) restricted to $\mathcal{F}_{\mathrm{rad}}$:

$$\|\mathbf{u}_{\mathrm{rad}}\|_{\mathcal{E}}^2 = \mathfrak{a}_\omega(\mathbf{u}_{\mathrm{rad}}, \mathbf{u}_{\mathrm{rad}}), \qquad \mathbf{u}_{\mathrm{rad}} = (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}), \tag{2.260}$$

with $\mathfrak{a}_\omega$ from Subsection 2.0.2 [2]. Shell L^2 power density norms for pattern reporting are

$$\|\mathbf{E}_0\|_{L^2(S_r)}^2 = \oint_{S_r} \left|\mathbf{E}_0(\theta,\phi)\right|^2 d\Omega, \qquad \|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}} = \Big(\int_{\Omega_s} |\mathbf{J}_{\mathrm{imp}}|^2 \, dV \Big)^{1/2}, \tag{2.261}$$

linking source amplitude to far pattern through Eq. (2.247) [1]. The radiation operator norm on admissible sources is

$$\|\mathcal{R}_\omega\|_{\mathrm{op}} = \sup_{\|\mathbf{J}_{\mathrm{imp}}\|_{\mathcal{J}}=1} \|\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{E}}, \tag{2.262}$$

bounded on fixed outgoing domains when resolvent bound Eq. (2.77) holds [2]. Observation maps compose on the right, yielding

$$\Lambda_{\mathrm{meas}} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \qquad \|\Lambda_{\mathrm{meas}}\|_{\mathrm{op}} \le \|\mathcal{O}_{\mathrm{meas}}\|_{\mathrm{op}} \, \|\mathcal{R}_\omega\|_{\mathrm{op}}, \tag{2.263}$$

consistent with Eq. (2.219) and rank cap Eq. (2.221) [2]. For a range engineer, $\|\mathcal{R}_\omega\|_{\mathrm{op}}$ bounds how strongly a unit-amplitude feed can excite the far pattern; the composition Λ_{meas} in Eq. (2.263) is the formal image of gain calibration followed by pattern sampling on S_r [1]. Let $\mathbf{J}_{\mathrm{imp}}$ be admissible with $\mathcal{D}_{J,\mathrm{reg}} = 1$ and let r lie in the far zone of Eq. (2.242). Then $\mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}})$, $\mathbf{F}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}})$, and $\mathbf{J}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}})$ satisfy Eqs. (2.254)–(2.256) with errors $\mathcal{O}(r^{-1})$, and $\mathcal{P}_{\mathrm{rad}}(r; \mathbf{J}_{\mathrm{imp}}) \to P_\infty > 0$ whenever Eq. (2.32) holds on the radiative branch [2]. Substitute Eqs. (2.246)–(2.247) into Eqs. (2.251)–(2.255) and integrate on S_r. Sommerfeld ratio Eq. (2.191) gives Eq. (2.256); power integral reduces to Eq. (2.254) [2]. Convergence follows from Eq. (2.248) [2]. Equations (2.251)–(2.262) require harmonic steady state, linear passive media, outgoing radiation condition, and regime-qualified shells [2]. They fail on reactive-dominated shells with $\chi(r) \gg 1$ from Eq. (2.31), in cutoff guides, and when $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is used without $\mathcal{R}_\omega$ projection [2]. Material stress debates (Minkowski versus Abraham) are not re-opened; vacuum-side shells use Eq. (1.37) [2]. Gauge-dependent near-zone angular densities are excluded from $\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]$ unless closure through Eq. (1.42) is verified [12]. As a worked example, compute $\Lambda_J(r)$ from Eq. (1.64) using $\mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}]$ on S_r with $k_0 r = 10$ and compare to $k_0 r = 0.5$ [2]. Only the outer shell with $\mathcal{D}_{J,\mathrm{reg}} = 1$ and $\mathcal{C}_1$–$\mathcal{C}_3$ satisfied admits transport interpretation per Eq. (1.67) [2]. Flux meters are therefore downstream observables: they presuppose the integral evaluation chain, the far-zone reduction when applicable, and export selection through $\mathcal{R}_\omega$ before any watt, momentum, or angular-momentum budget is reported on Γ_R [1]. Flux and norm functionals are not independent postulates added to Maxwell theory; they are laboratory readouts that presuppose the same export selection already enforced on pattern meters. Power measured on a reactive shell must be distinguished from power reported after Π_{rad}: the former samples $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, the latter the transverse outgoing sector carried by $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Momentum and angular-momentum budgets add not new dynamics but conserved-flux accounting on the qualified radiative image once regime screens from 1 are satisfied [12]. Subsection 2.5.4 decomposes the same solution map into radiative and non-radiative components.

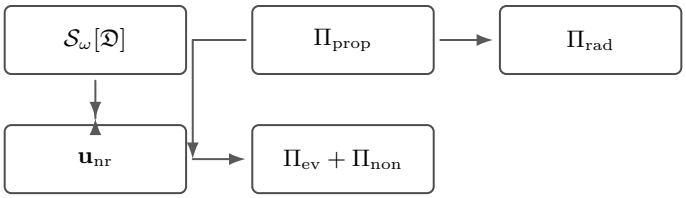


Figure 2.33: Radiative/non-radiative split: $\mathbf{u}_{\mathrm{rad}}$ is $\Pi_{\mathrm{rad}}\mathcal{S}_\omega$; $\mathbf{u}_{\mathrm{nr}}$ collects all non-export content including Eq. (2.265).

2.5.4 Radiative and non-radiative components

A compact-range engineer can measure strong electric and magnetic fields within one wavelength of a horn mouth while the far-zone gain remains modest: reactive energy stored near the feed is large, yet little of it crosses a distant measurement sphere as export [5]. Conversely, currents in $\ker \mathcal{R}_\omega$ can sustain elaborate near-field structure while producing no far-field signature—the classical non-radiating source [1]. Inverse synthesis and scattering campaigns inherit this split: near-field scans and angle-resolved data constrain only the radiative image $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, leaving non-radiative content invisible until additional excitation or realizability constraints are imposed; Chapter **??** develops that $\mathcal{S}_{\mathrm{real}}$ question spectrally [2].

The component partition below imports reactive–radiative logic from Section 1.3.3 and spectral projectors Eqs. (2.134)–(2.139) without re-proving projector algebra [2]. Complete non-radiating null-space classification belongs to Section 2.7. Let $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) = \mathcal{S}_\omega[\mathfrak{D}]$ and $\mathbf{u}_{\mathrm{rad}} = \mathcal{R}_\omega[\mathfrak{D}]$ be as in Eq. (2.228) [2]. Spectral projectors Π_{prop}, Π_{ev}, Π_{non} act on $\widehat{\mathcal{F}}_\Omega$ as in Subsection 2.2.3; outgoing selector $\mathcal{O}_{\mathrm{out}}$ and radiation map satisfy Eqs. (2.203)–(2.138) [2]. Regime qualification uses $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.34) and criterion Eq. (2.140) [2]. On admissible $\mathfrak{D}$,

$$\mathcal{S}_\omega[\mathfrak{D}] = \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathfrak{D}] + \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}] + \Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathfrak{D}], \tag{2.264}$$

restating Eq. (2.137) on the integral outputs constructed above [2]. Define the *non-propagating* component

$$\mathbf{u}_{\mathrm{np}} = \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}] + \Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathfrak{D}], \tag{2.265}$$

and the *propagating* component $\mathbf{u}_{\mathrm{prop}} = \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathfrak{D}]$ [2]. Only $\mathbf{u}_{\mathrm{prop}}$ is eligible for outgoing radiation selection; evanescent and non-propagating content satisfies Eq. (2.139) [1]. Equation (2.264) applies the 1 partition to integral outputs: reactive dominance on near shells corresponds to large $\|\mathbf{u}_{\mathrm{np}}\|$ relative to $\|\mathbf{u}_{\mathrm{prop}}\|$ in the sense of Eq. (2.143) [2]. The radiative component is

$$\mathbf{u}_{\mathrm{rad}} = \mathcal{R}_\omega[\mathfrak{D}] = \Pi_{\mathrm{rad}}\,\mathcal{S}_\omega[\mathfrak{D}] = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}\,\mathcal{S}_\omega[\mathfrak{D}], \tag{2.266}$$

using Eqs. (2.203) and (2.138) [2]. The complementary *non-radiative* component is

$$\mathbf{u}_{\mathrm{nr}} = \mathcal{S}_\omega[\mathfrak{D}] - \mathbf{u}_{\mathrm{rad}} = (\mathrm{id} - \Pi_{\mathrm{rad}})\mathcal{S}_\omega[\mathfrak{D}]. \tag{2.267}$$

Equations (2.266)–(2.267) partition every admissible field pair into export and non-export parts without altering Maxwell dynamics: $\mathbf{u}_\omega = \mathbf{u}_{\mathrm{rad}} + \mathbf{u}_{\mathrm{nr}}$ [2]. Non-radiative content includes evanescent, non-propagating, and incoming-propagating fractions annihilated by Π_{rad}; large near-field magnitude therefore does not establish export [2]. Sources in $\ker \mathcal{R}_\omega$ produce $\mathbf{u}_{\mathrm{rad}} = \mathbf{0}$ while $\mathbf{u}_{\mathrm{nr}} \neq \mathbf{0}$ (Eq. (2.218)) [2]. Figure 2.19 from Subsection 2.2.3 on integral outputs [2]. Using energy norm Eq. (2.48), define

$$\mathcal{E}_{\mathrm{rad}}[\mathfrak{D}] = \|\mathbf{u}_{\mathrm{rad}}\|_{\mathcal{E}}^2, \qquad \mathcal{E}_{\mathrm{nr}}[\mathfrak{D}] = \|\mathbf{u}_{\mathrm{nr}}\|_{\mathcal{E}}^2, \tag{2.268}$$

with $\|\cdot\|_{\mathcal{E}}$ from Eq. (2.260) on each component [2]. Shell power splits analogously:

$$\mathcal{P}_{\mathrm{tot}}(r) = \mathcal{P}_{\mathrm{rad}}(r;\mathbf{J}_{\mathrm{imp}}) + \mathcal{P}_{\mathrm{nr}}(r;\mathbf{J}_{\mathrm{imp}}), \tag{2.269}$$

where $\mathcal{P}_{\mathrm{rad}}$ is Eq. (2.253) and

$$\mathcal{P}_{\mathrm{nr}}(r;\mathbf{J}_{\mathrm{imp}}) = \mathrm{Re}\oint_{S_r} \hat{\mathbf{r}}\cdot\mathbf{S}_{\mathrm{nr}}[\mathbf{J}_{\mathrm{imp}}]\,dS, \qquad \mathbf{S}_{\mathrm{nr}} = \tfrac{1}{2}\,\mathbf{E}_{\mathrm{nr}}\times\mathbf{H}_{\mathrm{nr}}^{*}, \tag{2.270}$$

with $(\mathbf{E}_{\mathrm{nr}},\mathbf{H}_{\mathrm{nr}}) = \mathbf{u}_{\mathrm{nr}}$ [2]. Equation (2.269) is not a new conservation law; it decomposes Eq. (2.7) when both components are evaluated on the same shell [2]. In the far zone, $\mathcal{P}_{\mathrm{nr}}(r)\to 0$ and $\mathcal{P}_{\mathrm{rad}}(r)\to P_\infty$ under Eq. (2.32) [2]. Near-zone shells with $\chi(r)\gg 1$ from Eq. (2.31) typically have $|\mathcal{P}_{\mathrm{nr}}|\gg|\mathcal{P}_{\mathrm{rad}}|$ [2]. Regime qualification from Section 1.3.3 becomes

$$\mathcal{D}_{J,\mathrm{reg}} = 1 \implies \text{transport uses } \mathbf{u}_{\mathrm{rad}},\ \mathcal{P}_{\mathrm{rad}},\ \mathbf{M}_{\mathrm{rad}}[\mathbf{J}_{\mathrm{imp}}], \tag{2.271}$$

and forbids interpreting $\mathbf{u}_{\mathrm{nr}}$, $\mathcal{P}_{\mathrm{nr}}$, or $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}]$ as export channels [2]. Equation (2.271) is the component form of Eq. (2.140) after Eqs. (2.266)–(2.267). Laboratory errors from Section 1.3.3—near-zone reactive scans, cutoff-guide modes, evanescent dominance—appear as large $\mathcal{E}_{\mathrm{nr}}$ or $\|\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}]\|$ with small $\mathcal{P}_{\mathrm{rad}}$ [2]. Link to radiation null space. Non-radiative components encode source-side invisible content:

$$\mathbf{J}_{\mathrm{imp}}\in\ker\mathcal{R}_\omega \implies \mathbf{u}_{\mathrm{rad}}=\mathbf{0},\quad \mathbf{u}_{\mathrm{nr}}=\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\neq\mathbf{0}\ \text{in general}, \tag{2.272}$$

as recorded in Eq. (2.8) [1]. Conversely, $\mathbf{u}_{\mathrm{nr}}=\mathbf{0}$ does not imply trivial $\mathcal{S}_\omega[\mathfrak{D}]$: a pure outgoing propagating state may have $\mathbf{u}_{\mathrm{rad}}=\mathbf{u}_\omega$ [2]. Section 2.7.1 classifies $\ker\mathcal{R}_\omega$; here only the field-side split Eq. (2.267) is identified as the operator image of non-export content [1]. Let $\mathcal{D}_{J,\mathrm{reg}}=1$ and let r lie in the far zone of Eq. (2.242). Then

$$\frac{\mathcal{E}_{\mathrm{nr}}[\mathfrak{D}]}{\mathcal{E}_{\mathrm{rad}}[\mathfrak{D}]+\epsilon}\to 0, \qquad \frac{|\mathcal{P}_{\mathrm{nr}}(r)|}{\mathcal{P}_{\mathrm{rad}}(r)+\epsilon}\to 0, \tag{2.273}$$

as $r\to\infty$ for fixed $\mathfrak{D}$, with $\epsilon>0$ regularizing zero export [2]. Equivalently, $\mathbf{u}_\omega\to\mathbf{u}_{\mathrm{rad}}$ in far-zone observables metered by Eqs. (2.246)–(2.253). Eq. (2.139) annihilates evanescent and non-propagating contributions in Π_{rad}. Far-zone asymptotics and flux stabilization on large shells give $\mathcal{P}_{\mathrm{rad}}(r)\to P_\infty>0$ while reactive terms decay relatively [2]. Regime condition Eq. (2.32) removes reactive dominance in the limit. Below guide cutoff Eq. (2.141), modal content lies in $\Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathfrak{D}]$ or $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathfrak{D}]$ [2]. Applying Eq. (2.247) or transport indices to such modes violates Eq. (2.271) as in the TE_{10} illustration of Section 1.1.2 [1]. Spectral projectors and regime logic must therefore be applied together on the same integral outputs [2]. Equations (2.264)–(2.273) require linear media, fixed ω, outgoing radiation condition, and spectral completion sufficient for Eqs. (2.134)–(2.135) [2]. Sharp boundaries between subspaces may blur when loss, anisotropy, or coupled interfaces mix propagating and evanescent sheets [2]. Component splits are diagnostic on shells, not substitutes for exterior radiative uniqueness (Subsection 2.4.3) on the full field [1]. Outgoing source-to-field integrals map impressed data to evaluable fields; distance-regime labels fix where asymptotic reductions apply; flux and operator-norm meters act on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after regime qualification; and the radiative–non-radiative split Eqs. (2.266)–(2.271) separates export from storage on the same integral outputs [2]. Section 2.6 develops spectral decomposition of radiation operators on the representations constructed here [1]. The radiative–non-radiative split is what separates a wattmeter that has been qualified on Γ_R from a near-field probe that still sees storage and evanescent overlap in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [1]. Both components coexist in one Maxwell solution until projectors and outgoing

selection name which part export meters may integrate [5]. Evanescent sheets must not be dismissed as mere mathematical artifacts; however, Maxwell theory shows they carry reactive energy measurable on enclosing scans while remaining null under far-zone angular accessibility maps developed in Section 2.9 [2]. Section 2.6 now organizes the radiative image itself into eigenmodes and recoverable rank on the integral representations constructed in Sections 2.3–2.5.

2.6 Spectral Theory of Radiation Operators

Having followed impressed current through Maxwell completion, outgoing selection, and evaluable radiation integrals. Spectral theory treats radiation export as not a single scalar amplitude but a countable nesting of modes on compact geometries, each with its own coupling to sources and its own observability limit on finite instruments [1]. Pattern shape must not be identified with a single dominant mode; Maxwell propagation on bounded supports shows instead that many directions can be degenerate under rotation while remaining distinguishable under excitation, and that Fredholm structure controls which target patterns are reachable at all [13]. A laboratory campaign on a circular aperture measures a finite slice of that nesting—never the full operator spectrum—so rank and degeneracy statements below are as much measurement discipline as abstract mathematics [11].

Radiation integrals in Section 2.5 map sources to outgoing fields. Section 2.6 analyzes that map spectrally: which directions carry power and how degeneracy splits under broken symmetry [1]. Sidelobe floors on finite apertures are the visible face of spectral decay [7]. Every antenna engineer learns, long before operator theory appears in the curriculum, that a finite structure cannot radiate an unlimited set of independent far-zone patterns. A dish whose rim spans only a few wavelengths resolves only a modest number of principal beam directions on a compact range; a spherical near-field scanner with finite probe count tells the same story after transformation to far-zone coefficients [1]. Collin and Harrington treat that compression as structural: it is imposed by Maxwell propagation from compact sources through outgoing selection to a finite observation shell, not by a particular receiver model [11]. Stratton and Harrington organized outgoing fields by angular structure on large spheres; Chu sharpened the consequence that electrical size bounds directional export at fixed frequency [2]. Subsection 2.6.1 introduces the laboratory reading—singular values σ_n of $\mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega$ and the recoverable rank they define—before any eigenpair formula is written; later subsections refer back rather than restate that doctrine. Why spectra arise in radiation physics is therefore not a mathematical accident. Maxwell's equations are linear in impressed sources on the outgoing branch fixed in Section 2.4; superposition of two feeds superposes far-zone amplitudes. On a large sphere S_r, orthogonality of angular patterns under an appropriate energy inner product later organizes the export map $\mathcal{R}_\omega$ into radiation modes; each carries an *export eigenvalue* λ_n of the radiation operator—ranking qualified power on Γ_R and distinct from any correlation-operator spectrum unless explicitly stated [2]. Some feeds couple strongly to dominant modes; others produce elaborate reactive near fields through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ yet register negligibly on a far-zone wattmeter after Π_{rad} [5]. Observable far-zone content remains primary; formal harmonic indices organize export but are not themselves physical modes—that distinction is developed once in Subsection 2.6.1 and thereafter [2]. The radiation map $\mathcal{R}_\omega$ from Section 2.5 inherits this ordering. Spectral analysis names dominant and weak export directions, explains why finite dishes and finite scan grids lose information, and states when pattern synthesis and source inversion are well posed [2]. **??** will supply explicit harmonic bases and tabulated matrix elements; the present section asks what the export map itself permits at fixed ω before any particular angular basis is chosen [2]. A reader who finishes the section should understand why measured patterns look low-rank in practice, why

some synthesis targets are unrealizable no matter how much numerical effort is invested, and why rotation of an antenna reorients coefficients without changing exported power when frames are consistently handled [1]. Reading order. Engineers study spectra because measured patterns never carry unlimited independent information: finite apertures, finite scan grids, and finite electrical size impose rank limits long before any harmonic post-processor runs out of indices [7]. On a compact range the dominant beams are the few principal cuts that survive above the noise floor; weak modes decay in singular-value cliffs and become experimentally invisible even when formal expansions can still list them [2]. The sequence below should therefore be read not as a mathematical ordering but as a laboratory ordering. Measurement limitations appear first; spectral structure merely explains them [7]. The spectral account follows the measurement laboratory, not an abstract operator catalog. Finite aperture observation on S_r fixes how many independent far-zone directions a campaign can resolve—that ceiling is *finite radiative complexity*. Compactness of the composed map $\mathcal{K} = \mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$ then explains why singular values σ_n, ranking the radiation efficiency of each independent source organization through the actual receiver chain, decay and define recoverable rank [7]. Only after that ordering does the section construct export eigenvalues λ_n of $\mathcal{R}_\omega$, decompose radiation modes $\mathbf{u}_n$ from their exciting source modes $\mathbf{J}_{\text{imp}}^{(n)}$, and complete symmetry, Fredholm, and geometric filters on what finite hardware can recover [2]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\text{imp}}$ be compact, $k_0 = \omega\sqrt{\varepsilon_0\mu_0}$, and outgoing field pair $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\text{SF}}$ from Eq. (2.57). The four stages below record the same program in compact notation; Subsection 2.6.1 develops σ_n decay and finite recoverable rank once, and later subsections cite it by reference:

$$\mathcal{R}_\omega \xrightarrow{\text{eigenmodes}} \{\lambda_n, \mathbf{u}_n\} \xrightarrow{\text{symmetry}} \text{degeneracy classes} \xrightarrow{\text{Fredholm}} \text{solvability / index} \xrightarrow{\text{geometry}} \text{radiative spectr} \tag{2.274}$$

from export-eigenmode ordering through rotational degeneracy, Fredholm solvability, and geometric filters on recoverable content [2]. The primary map remains

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\text{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \Pi_{\text{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}], \tag{2.275}$$

from Eq. (2.3), with Π_{rad} selecting outgoing radiative solutions per Eq. (2.138). In plain language: $\mathcal{R}_\omega$ is the radiation map from impressed currents to far-zone export; $\mathcal{J}_\Omega$ is the admissible source region where $\mathbf{J}_{\text{imp}}$ may be placed; $\mathcal{F}_{\text{rad}}$ is the space of outgoing radiative field pairs observable on Γ_R after export selection; and Π_{rad} is the filter that removes reactive and non-propagating content before pattern measurement [2]. Equation (2.275) represents the entire electromagnetic experiment beginning with impressed current inside the source region and ending with the radiative content observable on a far-zone measurement surface [1]. Subsection 2.6.1 fixes σ_n decay and finite recoverable rank on $\mathcal{K}$; the subsections below construct export eigenpairs, split degeneracy under symmetry, diagnose Fredholm solvability, and apply geometric filters. A sidelobe floor on a measured pattern is often the visible face of the aperture-limited stage [7].

2.6.1 Compact circular aperture and singular-value decay

Before any eigenvalue formula is written, it helps to watch the phenomenon on a range. On a dish compact range, place a feed above a circular scan disk on plane $z = 0$ and sample tangential field on S_r; on a spherical near-field scanner, place probes on a hull at $R \approx 2\lambda_0$ and transform to far-zone coefficients [7]. In either campaign, count how many independent beam directions remain above the receiver noise floor. The answer depends on disk or scan radius a_{ap} and source electrical size, not on how many spherical harmonics a post-processor can index [6]. Electrical size and scan extent fix a recoverable rank; singular values σ_n of $\mathcal{O}_{\text{meas}} \circ \mathcal{R}_\omega$ measure the radiation efficiency of each independent source organization through the aperture and receiver chain, and fall off once the

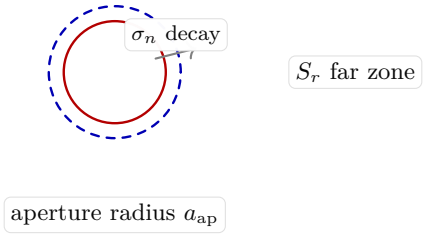


Figure 2.34: Finite circular aperture: compact observation operator on far-zone fields; singular-value decay marks angular harmonics that fall below instrument threshold.

modal index exceeds what the geometry can resolve [2]. Synthesis harmonic caps therefore overstate physical rank when σ_n has already fallen below the noise floor—harmonics in a post-processor are labels, not independent export channels. Compact decay of $\sigma_n(\mathcal{K})$ and the finite recoverable rank it defines are stated here once; later subsections invoke Subsection 2.6.1 by reference rather than repeating the measurement narrative [1].

Consider a circular scan aperture of radius a_{ap} on plane $z = 0$ sampling the far-zone tangential field from a compact source with $k_0 a = 1.5$. The measurement composition $\mathcal{K} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega$ is compact on the source and field spaces declared in Section 2.1.3; its singular values σ_n rank radiation efficiency of source organizations and decay, defining a practical recoverable rank and finite effective dimension before any harmonic label is attached [2]. Compactness from Subsection 2.1.3 yields $\sigma_n(\mathcal{K}) \to 0$; for disk apertures the decay is approximately exponential in modal index when $k_0 a_{\mathrm{ap}}$ is moderate [2]. In practice the engineer reads off a knee in the singular-value curve: retain modes above the knee for synthesis or inversion; discard the tail and accept the residual sidelobe floor as a geometry-limited artifact rather than a tuning defect [7]. What practical criterion tells the engineer where to truncate the spectrum? Retaining every mode is impossible: instrument noise, finite aperture extent, and finite scan count cap recoverable rank, and singular-value decay controls which source organizations remain physically realizable through the actual receiver chain [2]. Truncation is therefore inevitable; the engineering question is where the knee lies. Truncating at N modes with

$$\frac{\|\mathcal{K} - \mathcal{K}_N\|}{\|\mathcal{K}\|} \leq \frac{\sigma_{N+1}}{\sigma_1}, \tag{2.276}$$

where $\mathcal{K} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega$ is the full measurement chain from source to recorded samples, $\mathcal{K}_N$ is its rank-N truncated approximation, σ_1 is the largest singular value (dominant export channel), and σ_{N+1} is the first discarded singular value [1]. The ratio σ_{N+1}/σ_1 is the relative strength of the strongest discarded radiation channel compared with the dominant channel—a direct laboratory readout of how much pattern error truncation will introduce [7]. Eq. (2.84) bounds pattern error on S_r. At $k_0 a_{\mathrm{ap}} = 3$, retaining $N = 9$ singular modes typically captures 95% of the radiated power resolvable through the disk; harmonics beyond $N \approx 12$ lie below a $-20\,\mathrm{dB}$ threshold relative to σ_1 [1]. Figure 2.34 foreshadows Chapter **??**: realizability and spectral truncation ask which angular components survive compression by $\mathcal{K}$, not which indices a formal expansion can list [3]. Laboratory reports therefore inherit a finite rank from geometry before source inversion is attempted; improving SNR without enlarging a_{ap} cannot recover modes annihilated by σ_n below measurement noise [7]. Local notation (σ_n, recoverable rank). Compose the measurement operator $\mathcal{K} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega$ on declared source and field spaces. Its singular values σ_n—ordered $\sigma_1 \geq \sigma_2 \geq \cdots \to 0$—are the radiation efficiencies of independent source organizations as seen through the actual aperture and receiver chain [2]. The *recoverable rank* is the largest N for which σ_N remains above the instrument noise floor; modes beyond that cliff are mathematically listable in a harmonic post-processor yet physically indistinguishable from clutter on Γ_R [6]. Compact σ_n decay and finite

dimension on $\mathcal{K}$ are fixed in the display below; elsewhere invoke Subsection 2.6.1 by reference only [1]. The spectral relations developed below restate three laboratory facts in compact form before the eigenbasis is constructed. A *radiation mode* is a normalized far-zone pattern direction $\mathbf{u}_n$; its *source mode* $\mathbf{J}_{\mathrm{imp}}^{(n)}$ is an admissible current that excites it. The *export eigenvalue* λ_n of the radiation operator $\mathcal{R}_\omega$—not a correlation-operator eigenvalue unless later stated—ranks how efficiently that radiation mode reaches qualified flux on Γ_R [2]. each radiating mode carries export efficiency $|\lambda_n|$ measured by coupled power in that direction. Second, not every desired far-zone target can be synthesized: obstructions appear as orthogonality conditions on outgoing solutions. Third, when geometry possesses rotational symmetry, rotating the hardware reorients modal coefficients without changing eigenvalue ordering within degenerate blocks. The introductory relation equations record these facts; the subsections that follow interpret them in measurement language [2]. On a compact range, a *radiation mode* appears as a principal far-zone pattern direction —a normalized cut on S_r that carries qualified export power—while its *source mode* is the admissible current distribution that excites it [1]. Dominant beams correspond to modes with large $|\lambda_n|$ because those organizations couple most efficiently to the wattmeter on Γ_R [2]. The eigenpair relation below states the experimentally familiar observation that certain current organizations radiate much more effectively than others [11]. On a family of radiating modes, the eigenvalue template reads

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] = \lambda_n\,\mathbf{u}_n, \qquad \mathbf{u}_n = P_n\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}], \tag{2.277}$$

with λ_n measuring radiative export efficiency of mode n from finite impressed current $\mathbf{J}_{\mathrm{imp}}^{(n)}$ [2]. In synthesis and inversion, $\mathbf{g}$ denotes the desired far-zone radiated field one wishes to realize; solvability fails when $\mathbf{g}$ carries a component along cokernel directions—an *orthogonality obstruction* that no admissible source can remove [2]. Experimentally, such failure appears as a synthesis target that refuses to converge, a measured residual orthogonal to every reachable pattern, or a pattern null that persists despite port tuning [1]. Solvability on outgoing radiative solutions satisfies

$$\mathcal{R}_\omega + K \text{ solvable on } \mathcal{J}_\Omega \iff \mathbf{g} \perp \ker(\mathcal{R}_\omega + K)^\dagger, \tag{2.278}$$

with K compact on the graph domain of impressed sources [2]. When a symmetry group G commutes with $\mathcal{R}_\omega$, the equation below expresses a simple laboratory fact: rotating the hardware and then measuring the pattern gives the same result as measuring first and rotating the resulting pattern [1].

$$\mathcal{R}_\omega\,\mathcal{D}(R)[\mathbf{J}_{\mathrm{imp}}] = \mathcal{D}(R)\,\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \qquad R \in G, \tag{2.279}$$

with $\mathcal{D}(R)$ the field representation induced from Eq. (1.2.3) on $\mathcal{F}_{\mathrm{rad}}$ [2]. Modal projectors compose with radiation selection through Π_{rad} acting as the radiation filter that passes only exportable content, P_n as mode-selection lenses that isolate individual beam families from a measured pattern, and Π_{ev}, Π_{non} as projectors onto evanescent and non-propagating sectors that qualified far-zone meters cannot resolve [2].

$$\Pi_{\mathrm{rad}} = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} P_n, \qquad P_n\Pi_{\mathrm{ev}} = P_n\Pi_{\mathrm{non}} = \mathbf{0}, \tag{2.280}$$

using Eqs. (2.134)–(2.139) by reference [2]. Spectral decompositions *organize* export; they do not *create* angular structure—see Subsection 2.6.1 for the harmonics-versus-modes distinction stated once in measurement terms [2].

2.6.2 Eigenfunctions and spectral decomposition

Discrete modal sums on exterior domains organize exported power into separable angular sectors only after outgoing qualification and flux gauge are declared [2, 8]. Continuous-spectrum bookkeeping runs in parallel with discrete ℓ blocks; mixing the two without measure conventions produces apparent mode creation on finite meshes [7]. Source integrals from Section 2.5 return amplitudes on Γ_R. The question an experienced range engineer asks next is practical: how many of those amplitudes remain independent above the noise floor for a given electrical size and scan extent? [2]. The eigenpair construction below answers that question in normalized form; recoverable rank and σ_n cliffs are already fixed in Subsection 2.6.1.

What an eigenmode means physically. A radiation mode is a principal far-zone pattern direction $\mathbf{u}_n$; its source mode $\mathbf{J}_{\mathrm{imp}}^{(n)}$ is the admissible current distribution that excites it. The export eigenvalue λ_n of $\mathcal{R}_\omega$ measures radiation efficiency $|\lambda_n|$ of that radiation mode on Γ_R [2]. Modes with large $|\lambda_n|$ are the dominant export channels on Γ_R: they are the beams an array designer weights first and the cuts a diagnostician examines before worrying about sidelobe structure. Modes with small $|\lambda_n|$ appear only after long integration, narrowband filtering, or exceptional dynamic range; on a typical compact range they never rise above the clutter floor. Modes with $\lambda_n = 0$ may still produce strong reactive near fields—the balanced-loop illustration from Section 2.5 is the archetype—yet contribute nothing to the far-zone wattmeter [3]. Eigenvalue ordering is therefore a statement about measurable export, not about how many terms a numerical chart can index. On a dish range, dominant eigenmodes appear as the principal cuts in a principal-component decomposition of sampled $\mathbf{E}_t$ on S_r: the first few components carry most resolvable power [7]. Rank truncation in synthesis follows the recoverable rank fixed in Subsection 2.6.1; the decomposition below passes through $\mathcal{R}_\omega$ on physically realizable source classes, with coupled power on Γ_R as the physical quantity [2]. Antenna-range observations. Harrington's synthesis practice already exploited the ordering: one designs or diagnoses feeds by how they excite a modest number of dominant modes, not by attempting independent control of every harmonic index a post-processor can list [1]. On a compact range, rotating a horn reorients the principal cuts while leaving total exported power unchanged when probe and source frames rotate together—a fact developed formally in the rotation discussion below, but visible in raw pattern data long before covariance formulas appear [11]. When a measured pattern refuses to match a synthesis target despite ample numerical effort, the fault is often spectral rather than computational: the target lies in a weakly coupled or forbidden sector, not in the subspace the aperture and source geometry can reach [2]. The eigenpair definition that follows encodes that experience in normalized form. Propagation projectors Eqs. (2.134)–(2.139), compactness templates from Section 2.1.3, and far-zone amplitudes Eqs. (2.246)–(2.247) are presupposed [2]. Explicit vector spherical harmonic bases belong to Chapter 3; here modal coordinates are fixed at the operator level compatible with Eq. (2.6.2) [1]. Assumptions. Fix $\omega > 0$, physically realizable impressed data $\mathfrak{D}$ on $\mathcal{J}_\Omega$ Eq. (2.49), outgoing radiation condition on exterior problems Eqs. (2.194)–(2.197), and export map Eq. (2.275) [2]. Field pairs lie in $\mathcal{F}_{\mathrm{SF}} \subset \mathcal{F}_\Omega$ from Eq. (2.57); outgoing radiative images lie in $\mathcal{F}_{\mathrm{rad}} \subset \operatorname{ran} \mathcal{R}_\omega$ with energy norm Eq. (2.48) [2]. Spectral completion $\widehat{\mathcal{F}}_\Omega$ supporting Eqs. (2.134)–(2.135) is assumed sufficient for the expansions below [1]. Radiation eigenpairs. Eigenmodes are introduced because measured patterns on S_r are rarely arbitrary: principal-component analysis of sampled $\mathbf{E}_t$ reveals a small number of dominant export directions, and the eigenbasis below names those directions in normalized form [7]. What experiment reveals is rank ordering—a few strong beams, a long tail of weak modes below the noise floor, and export-invisible reactive organizations that never register on a far-zone wattmeter [3]. The laboratory observations above condense into a normalized triple: export eigenvalue λ_n of $\mathcal{R}_\omega$, radiation mode $\mathbf{u}_n$, and source mode $\mathbf{J}_{\mathrm{imp}}^{(n)}$ linked through $\mathcal{R}_\omega$ [2]. Physically, $|\lambda_n|$ ranks how much

qualified power each normalized radiation mode carries to Γ_R; mathematically, the triple satisfies the locked eigenpair relation below. A *radiation eigenpair* is a triple $(\lambda_n, \mathbf{u}_n, \mathbf{J}_{\mathrm{imp}}^{(n)})$ with $\lambda_n \in \mathbb{C}$, $\mathbf{u}_n = (\mathbf{E}_n, \mathbf{H}_n) \in \mathcal{F}_{\mathrm{rad}}$, and $\mathbf{J}_{\mathrm{imp}}^{(n)} \in \mathcal{J}_\Omega$ such that

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] = \lambda_n \, \mathbf{u}_n, \qquad \|\mathbf{u}_n\|_{\mathcal{U}} = 1, \tag{2.281}$$

where λ_n is the export eigenvalue (radiation-efficiency score of mode n), $\mathbf{u}_n$ is the normalized radiation mode (principal far-zone pattern direction), $\mathbf{J}_{\mathrm{imp}}^{(n)}$ is the source mode (admissible current that excites $\mathbf{u}_n$), and $\|\mathbf{u}_n\|_{\mathcal{U}} = 1$ fixes unit radiative energy on $\mathcal{F}_{\mathrm{rad}}$ [2]. Readers should think of λ_n as a radiation-efficiency score assigned to the nth export channel [1]. Here $\|\cdot\|_{\mathcal{U}}$ is the field energy norm Eq. (2.48) on $\mathcal{F}_{\mathrm{rad}}$ [2]. Equation (2.281) develops Eq. (2.277) to a normalized locked definition: λ_n measures export gain of mode n relative to unit radiative energy, and $\ker \mathcal{R}_\omega$ corresponds to $\lambda_n = 0$ with $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}] \neq \mathbf{0}$ through Eq. (2.218) [2]. Eigenpairs are defined on the outgoing branch only; incoming propagating sheets excluded by Eq. (2.203) [1]. Interpretation: modes with $|\lambda_n| \gg 1$ dominate compact-range cuts; modes with $\lambda_n = 0$ are export-invisible yet may carry reactive near fields through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(n)}]$ [3]. Modal projectors on field space. Projector $P_n[\mathbf{u}]$ extracts a single beam family from a measured pattern: it isolates the component of field $\mathbf{u}$ aligned with radiation mode $\mathbf{u}_n$ and discards the orthogonal complement [2]. Modal filtering matters because synthesis, diagnosis, and interference analysis proceed mode by mode rather than on raw harmonic indices [1]. For each index n in a family of radiating modes $\mathcal{N}$, define a rank-one (or finite-dimensional) projector $P_n : \mathcal{F}_\Omega \to \mathcal{F}_{\mathrm{rad}}$ by

$$P_n[\mathbf{u}] = \langle \mathbf{u}, \mathbf{u}_n \rangle_{\mathcal{U}} \, \mathbf{u}_n, \qquad P_n \circ \Pi_{\mathrm{ev}} = P_n \circ \Pi_{\mathrm{non}} = 0, \tag{2.282}$$

using dual pairing inherited from Eq. (2.26) on qualified Maxwell states and annihilation of non-propagating content through Eq. (2.139) [2]. Equation (2.282) specializes resolution Eq. (2.280) once an orthonormal family of radiating modes $\{\mathbf{u}_n\}_{n \in \mathcal{N}_{\mathrm{rad}}}$ is chosen. Projectors satisfy $P_n P_m = \delta_{nm} P_n$ when modes are orthogonal in $\mathcal{U}$ [2]. Once eigenmodes are fixed, any physically realizable excitation decomposes into modal coefficients. The quantity a compact-range measurement ultimately recovers is the export-side pattern carried by $\mathbf{u}_{\mathrm{rad},n} = P_n \mathcal{R}_\omega[\mathfrak{D}]$; the source-side amplitude $c_n[\mathfrak{D}]$ measures how strongly the full solution $\mathcal{S}_\omega[\mathfrak{D}]$ aligns with mode n before export [2]. For general physically realizable $\mathfrak{D}$, define source coefficients

$$c_n[\mathfrak{D}] = \langle \mathcal{S}_\omega[\mathfrak{D}], \mathbf{u}_n \rangle_{\mathcal{U}}, \qquad \mathbf{u}_{\mathrm{rad},n} = P_n \, \mathcal{R}_\omega[\mathfrak{D}] = \lambda_n \, c_n[\mathfrak{D}] \, \mathbf{u}_n, \tag{2.283}$$

when the expansion converges in $\mathcal{U}$ [2]. Physically, c_n is the excitation strength of radiative mode n before export selection; $\lambda_n c_n$ is the coefficient on the normalized far-zone pattern carried by $\mathbf{u}_n$ [8]. Linearity Eq. (2.6) implies $c_n[a_1\mathfrak{D}_1 + a_2\mathfrak{D}_2] = a_1 c_n[\mathfrak{D}_1] + a_2 c_n[\mathfrak{D}_2]$ for physically realizable superpositions after splitting each term through $\mathcal{S}_\omega$ [2]. How can a complicated radiation pattern be reconstructed from simpler building blocks? The modal expansion below decomposes any radiative field into orthonormal radiation modes, each carrying a fraction of the export energy budget [2]. When the radiation eigenbasis is complete on the export subspace, every radiative field admits a countable expansion in normalized modes; the Parseval identity below states that radiative energy equals the sum of squared modal coefficients [2]. When $\{\mathbf{u}_n\}_{n \in \mathcal{N}_{\mathrm{rad}}}$ is a complete orthonormal family in $\mathcal{F}_{\mathrm{rad}}$ for the problem class, every radiative field $\mathbf{v} \in \operatorname{ran} \mathcal{R}_\omega$ admits

$$\mathbf{v} = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \langle \mathbf{v}, \mathbf{u}_n \rangle_{\mathcal{U}} \, \mathbf{u}_n, \qquad \|\mathbf{v}\|_{\mathcal{U}}^2 = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \left| \langle \mathbf{v}, \mathbf{u}_n \rangle_{\mathcal{U}} \right|^2, \tag{2.284}$$

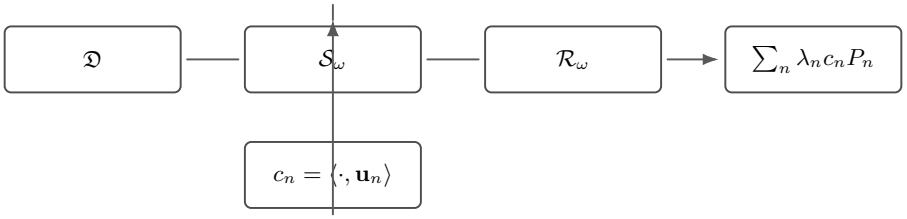


Figure 2.35: Spectral measurement sequence: $\mathcal{S}_\omega$ supplies coefficients c_n; $\mathcal{R}_\omega$ maps them to export amplitudes $\lambda_n c_n$ on projectors P_n Eq. (2.282).

with convergence in $\mathcal{U}$ [2]. Applying Eq. (2.284) to $\mathbf{v} = \mathcal{R}_\omega[\mathfrak{D}]$ gives

$$\mathcal{R}_\omega[\mathfrak{D}] = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \lambda_n \, c_n[\mathfrak{D}] \, \mathbf{u}_n, \tag{2.285}$$

the radiation eigenbasis restatement of compression operator Eq. (2.83) with explicit eigenvalues and normalized modes [2]. The Parseval identity in Eq. (2.284) states the modal energy budget: total radiative energy equals the sum of squared modal coefficients. Beam decomposition and recoverable complexity are therefore fixed by how many terms carry appreciable weight before truncation [7]. Truncation to $|\mathcal{N}| \le N$ yields $\mathcal{R}_\omega^{(N)}$ from Eq. (2.83); error Eq. (2.84) applies when the tail is compact on source space [1]. Let outgoing radiation condition and domain class be fixed as above. Suppose $\{\mathbf{u}_n\}_{n \in \mathcal{N}_{\mathrm{rad}}}$ is a complete orthonormal radiation eigenbasis in $\mathcal{F}_{\mathrm{rad}}$ with Eq. (2.281) and projectors Eq. (2.282). Then only modes with $\lambda_n \neq 0$ contribute to observable radiation: modes with $\lambda_n = 0$ lie in $\ker \mathcal{R}_\omega$ and are invisible to far-zone meters even when they carry reactive near fields [3].

$$\mathcal{R}_\omega = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \lambda_n \, P_n \circ \mathcal{S}_\omega, \qquad \Pi_{\mathrm{rad}} = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} P_n, \tag{2.286}$$

on physically realizable $\mathfrak{D}$, with convergence in $\mathcal{U}$ for fields in $\operatorname{ran} \mathcal{R}_\omega$ [2]. The range satisfies

$$\operatorname{ran} \mathcal{R}_\omega = \overline{\operatorname{span}\{\mathbf{u}_n : \; n \in \mathcal{N}_{\mathrm{rad}}, \; \lambda_n \neq 0\}}. \tag{2.287}$$

Eq. (2.285) follows by applying Eq. (2.284) to $\mathcal{R}_\omega[\mathfrak{D}]$ and using Eqs. (2.283)–(2.281). Linearity of $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ extends the sum from finite superpositions to all physically realizable impressed data by completion in $\mathcal{U}$ on the outgoing radiative subspace [2]. Eq. (2.286) is obtained by recognizing each term as $P_n \mathcal{S}_\omega$ acting with gain λ_n. Eq. (2.287) follows because eigenvectors with $\lambda_n = 0$ contribute no radiative export; non-radiating sources lie in $\ker \mathcal{R}_\omega$ Eq. (2.218) [1]. Resolution Eq. (2.280) matches $\Pi_{\mathrm{rad}} = \sum_n P_n$ on $\operatorname{ran} \mathcal{R}_\omega$ [2]. Resonant cavities, microwave filters, and waveguide resonators store energy in interior modes that need not export to a radiating port: stored versus exported energy is the physical distinction the cavity specialization encodes [2]. On a bounded domain with passive $\mathcal{B}_\Gamma$ and interior resonances, Eq. (2.6.2) yields a discrete accumulating set $\{\lambda_n\}_{n=1}^{\infty}$ when $\mathcal{R}_\omega$ is restricted to cavity outgoing-propagating subspaces compatible with Eq. (2.60) [2]. The cavity specialization reads

$$\mathcal{R}_\omega^{(\mathrm{cav})} = \sum_{n=1}^{\infty} \lambda_n \, P_n^{(\mathrm{cav})} \circ \mathcal{S}_\omega, \qquad \lambda_n \to 0 \text{ or } \lambda_n \to \infty \text{ according to mode class}, \tag{2.288}$$

with $P_n^{(\mathrm{cav})}$ projectors onto normalized cavity radiation modes [2]. Equation (2.288) is the constructive content of Eq. (2.83) in closed volumes; storage resonances with $\lambda_n = 0$ on $\mathcal{R}_\omega$ remain in $\mathbf{u}_{\mathrm{nr}}$ Eq. (2.267) [1]. Using Eq. (2.288) on open exterior problems misstates continuous angular

content Eq. (2.6.2) [2]. In free space the discrete modal sum becomes continuous: every outgoing direction on the sphere S^2 carries spectral weight, and polarization index $\sigma \in \{1, 2\}$ labels the two transverse polarizations perpendicular to propagation direction $\hat{\mathbf{k}}$ [2]. Physically, $\hat{\mathbf{k}}$ is the unit propagation direction observed on a far-zone scanner; σ selects which transverse polarization component is sampled; $\mathbf{e}_\sigma$ is the corresponding unit polarization vector with $\mathbf{e}_\sigma \cdot \hat{\mathbf{k}} = 0$; and $d\Omega(\hat{\mathbf{k}})$ is the solid-angle measure that weights how much radiated power each direction receives [7]. For compact sources in free space with outgoing radiation condition, angular eigenmodes are labeled by propagation direction $\hat{\mathbf{k}} \in S^2$ and polarization index $\sigma \in \{1, 2\}$ transverse to $\hat{\mathbf{k}}$ [2]. A normalized plane-wave radiation mode is

$$\mathbf{u}_{\hat{\mathbf{k}},\sigma}(\mathbf{r}) = \big(\mathbf{E}_{\hat{\mathbf{k}},\sigma}(\mathbf{r}),\, \mathbf{H}_{\hat{\mathbf{k}},\sigma}(\mathbf{r})\big), \qquad \mathbf{E}_{\hat{\mathbf{k}},\sigma} \sim \mathbf{e}_\sigma\, e^{-\mathrm{j}k_0 \hat{\mathbf{k}}\cdot\mathbf{r}}/r, \tag{2.289}$$

with $\mathbf{e}_\sigma \cdot \hat{\mathbf{k}} = 0$ and outgoing phase [2]. The continuous spectral measure $d\Omega(\hat{\mathbf{k}})$ replaces discrete summation in Eq. (2.284):

$$\mathcal{R}_\omega[\mathfrak{D}](\mathbf{r}) = \int_{S^2} \sum_{\sigma=1}^{2} a_\sigma(\hat{\mathbf{k}})\, \mathbf{u}_{\hat{\mathbf{k}},\sigma}(\mathbf{r})\, d\Omega(\hat{\mathbf{k}}), \tag{2.290}$$

with amplitudes a_σ determined by $\mathcal{S}_\omega[\mathfrak{D}]$ through Eq. (2.283) in the rigged limit [2]. Rigged Hilbert space completion and measure normalization are developed in **??** Subsection 3.1.3 records the continuous angular target Eq. (2.290) compatible with far-zone reduction [1]. Each radiation mode $\mathbf{u}_n$ carries a far-zone fingerprint $\mathbf{f}_n(\theta, \phi)$—the angular pattern shape that mode n contributes on S^2— while $\mathbf{E}_0(\theta, \phi)$ is the measurable far-zone electric pattern a compact range records after probe correction and transformation [7]. Let $\mathcal{D}_{J,\mathrm{reg}} = 1$ and let $\mathbf{u}_n$ be radiation eigenmodes with far amplitudes $\mathbf{f}_n(\theta, \phi)$ defined through Eq. (2.247). Then the far pattern of $\mathcal{R}_\omega[\mathfrak{D}]$ satisfies

$$\mathbf{E}_0(\theta, \phi) = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \lambda_n\, c_n[\mathfrak{D}]\, \mathbf{f}_n(\theta, \phi), \tag{2.291}$$

with convergence in mean-square on S^2 when the modal sum converges in $\mathcal{U}$ [2]. Equation (2.291) is the upgrade of Eq. (2.6.2); explicit $\mathbf{f}_n$ in vector harmonic form belong to **??** [2]. Apply the far-field approximation of source integrals (Subsection 2.5.2) to each modal component $\lambda_n c_n\, \mathbf{u}_n$ and superpose using linearity Eq. (2.6) on outgoing radiative solutions. Coefficients match Eq. (2.283) because far-zone reduction depends linearly on $\mathcal{R}_\omega[\mathfrak{D}]$ through Eqs. (2.246)–(2.247) [2]. regime qualification Eq. (2.271) must hold before interpreting Eq. (2.291) as transport data [1]. Source reconstruction is harder than forward radiation because inversion requires adjoint modes $\widetilde{\mathbf{u}}_n$ that test whether a desired pattern $\mathbf{g}$ lies in the reachable range of $\mathcal{R}_\omega$ [2]. When $\mathcal{R}_\omega$ is not self-adjoint on $\mathcal{J}_\Omega$, define adjoint modes $\widetilde{\mathbf{u}}_n$ through the source pairing Eq. (2.56) such that

$$\langle \widetilde{\mathbf{u}}_m,\, \mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}] \rangle_{\mathcal{U}} = \lambda_n\, \delta_{mn}, \tag{2.292}$$

for normalized radiation eigenpairs [2]. Equation (2.292) is the field-side restatement of solvability relation Eq. (2.278): cokernel directions annihilated by Eq. (2.292) obstruct inversion $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ [2]. Complete adjoint-mode catalogs on exterior domains belong to Subsection 2.6.4 [1]. Modes with $\lambda_n = 0$ satisfy $P_n \mathcal{R}_\omega = \mathbf{0}$ but may have $P_n \mathcal{S}_\omega \neq \mathbf{0}$ when reactive content projects onto non-export subspaces Eq. (2.265) [2]. Evanescent and non-propagating sheets remain annihilated: spectral labels cannot convert evanescent storage into qualified far-zone radiation—regime qualification and export selection remain mandatory before transport claims [2].

$$P_n \circ \Pi_{\mathrm{ev}} = P_n \circ \Pi_{\mathrm{non}} = \mathbf{0} \quad \text{for all } n \in \mathcal{N}_{\mathrm{rad}}, \tag{2.293}$$

by Eq. (2.139) and modal definition Eq. (2.282) [2]. Spectral labels therefore do not override regime qualification Eq. (2.140) [1]. The eigenbasis expansions and far-pattern coefficients in Eq. (2.291) require linear media at fixed ω, outgoing radiation condition, and completeness of $\{\mathbf{u}_n\}$ appropriate to the problem class [2]. Loss shifts λ_n into the complex plane without promoting evanescent storage to export through Eq. (2.293) [2]. Recoverable coefficients in Eq. (2.291) are further capped by finite observation rank Eq. (2.221) as established in Subsection 2.6.1 [1]; consult that subsection for σ_n cliffs rather than restating finite-rank doctrine here. As a worked example, consider a rectangular cavity fed by an electrically small probe at $f = 10\,\mathrm{GHz}$. The physical configuration is familiar from filter and resonator laboratories: a closed metal box with a coaxial penetration, interior standing waves visible in $|S_{11}|$ and in probe-coupled power at ports [2]. Interior modes produce a dense discrete spectrum in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$—many field patterns exist inside the volume—but only propagating outgoing sectors contribute to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on a measurement aperture or radiating port [2]. Truncating Eq. (2.288) at $N = 20$ modes with error bounded by Eq. (2.84) yields a stable interior power budget on a port wattmeter: the expected observation is a clean $|S_{11}|$ trace with identifiable resonances and predictable coupling to the feed. The lesson sharpens by contrast. Apply the same truncated cavity sum to an exterior half-wave dipole in free space and sidelobe structure misaligns: the exterior problem requires the continuous angular measure Eq. (2.290), not a cavity eigenbasis [8]. On a compact range the dipole presents a smooth far-zone pattern cut; in the cavity it presents port physics entirely governed by enclosure modes. Conflating the two domain classes is a common numerical error in aperture codes and in student projects that reuse modal truncations without checking export selection [1]. The significance is operational: spectral truncation is domain-specific, export selection must precede modal counting, and a dense interior spectrum is not the same object as a recoverable far-zone pattern family [2]. As a worked example, a balanced loop pair with vanishing electric dipole moment is the cleanest laboratory reminder that near-field strength does not imply far-zone export [3]. The physical setup places two loops with opposing dipole moments so that radiation from electric dipole terms cancels on Γ_R while reactive $\mathbf{H}$ and $\mathbf{E}$ remain large within one wavelength of the structure. On a spherical near-field scan at $R = 0.4\lambda_0$, probes record strong tangential fields; on Γ_R at $R = 10\lambda_0$ the wattmeter reads negligible export. Coefficients c_n on exporting modes vanish ($\lambda_n = 0$), yet $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ may be substantial [2]. The observable behavior confuses teams who report patterns from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad}: the near field looks radiative locally while the far-zone signature is null. The interpretation is spectral, not a failure of probes. Membership in $\ker \mathcal{R}_\omega$ is invisible to far-zone pattern meters yet measurable in the reactive sector. For inverse synthesis, the consequence is equally concrete: two distinct feeds can match the same far pattern while differing in reactive storage—a realizability question developed in **??** [2]. The example should be remembered as the archetype of export efficiency $\lambda_n = 0$ with non-trivial Maxwell storage. Subsection 2.6.2 fixed radiation eigenpairs, modal projectors, and the spectral expansion of $\mathcal{R}_\omega$ on $\mathcal{F}_{\mathrm{rad}}$. Recoverable rank and σ_n cliffs remain anchored in Subsection 2.6.1; the eigenbasis organizes export directions that finite apertures can in principle resolve once regime qualification holds [2]. Rotational degeneracy and Fredholm solvability develop next on the same operator.

2.6.3 Rotational symmetry and spectral degeneracy

Rotational degeneracy is visible on every compact range before covariance formulas appear: rotating a horn reorients principal cuts while total exported power stays fixed when probe and source frames rotate together [1]. The formal commutation relations below encode that laboratory symmetry on the export eigenbasis from Subsection 2.6.2.

Let $R \in \mathrm{SO}(3)$, field rotation $\mathcal{D}(R)$ on qualified Maxwell pairs $\mathbf{u} = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\mathrm{SF}}$, and radiation map Eq. (2.275) on outgoing data [2]. Isotropic vacuum or statistically isotropic environments are assumed unless stated otherwise: constitutive tensors commute with R in the sense of Section 1.2.3 [2]. Impressed sources transform as polar vectors under active rotation, $(\mathcal{A}_R \mathbf{J}_{\mathrm{imp}})(\mathbf{r}) = R\,\mathbf{J}_{\mathrm{imp}}(R^{-1}\mathbf{r})$, consistent with Eq. (1.68) [2]. regime qualification Eq. (2.140) and transport qualification Eq. (2.28) remain mandatory before interpreting rotated reports as export data [1]. Rotation in the laboratory. Rotating a horn on a compact range is the everyday experiment that motivates covariance relations. Mount the horn on a rotary stage, record $\mathbf{E}_t$ on S_r at several azimuths, and superpose the cuts: total exported power remains unchanged when probe and source frames rotate together, but the principal beam walks across the sphere as expected for a steerable feed [1]. Rotating a dish feed scans a different arc on the reflector surface; modal coefficients mix according to the same rules that rotate a measured $\mathbf{E}_t$ distribution on the rim [11]. Spinning a corrugated aperture distribution about its axis redistributes directional content within fixed ℓ blocks without creating new export unless the source itself changes [2]. Laboratory reports must distinguish active rotation of hardware from passive relabeling of a fixed field in a rotated coordinate chart; confusing the two conventions splits degeneracies artificially in processed $a_n(\hat{\mathbf{k}})$ data and can make a symmetric structure appear artificially asymmetric in synthesis charting [1]. The covariance identity below is the formal statement of what range technicians already enforce by co-rotating probes and references. Field rotation means physically reorienting the $\mathbf{E}$ and $\mathbf{H}$ vectors with the hardware: what changes is the direction of the pattern on S_r; what remains unchanged, when frames are handled consistently, is total exported power and the eigenvalue ordering within degenerate blocks [1]. Define the Maxwell field representation

$$\mathcal{D}(R)\,\mathbf{u} = (\mathcal{A}_R\mathbf{E},\, \mathcal{A}_R\mathbf{H}), \qquad \mathbf{u} = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_\Omega, \tag{2.294}$$

using active rotation Eq. (1.2.3) on each component [2]. Equation (2.294) specializes Section 1.2.3 to phasor Maxwell pairs: passive relabeling Eq. (1.2.3) produces the same physics with rotated components related by Eq. (1.70) [2]. Group composition Eq. (1.2.3) yields $\mathcal{D}(R_1R_2) = \mathcal{D}(R_1)\mathcal{D}(R_2)$ on qualified field pairs [2]. Covariance of the radiation map. This equation is perhaps the most important symmetry statement in antenna engineering: active rotation of the feed in a co-rotated measurement frame leaves physical export content unchanged while reorienting modal coefficients through the representation $\mathcal{D}(R)$ [1]. When sources and observation frames are co-rotated, the radiation chain must commute with rotation up to the field representation $\mathcal{D}(R)$: rotating the feed and evaluating on a co-rotated far-zone frame must yield the same physics as evaluating first and then rotating the pattern [2]. That requirement is visible on every compact range where probe calibration tracks mechanical azimuth. Formally,

$$\mathcal{R}_\omega[\mathcal{A}_R\mathbf{J}_{\mathrm{imp}}] = \mathcal{D}(R)\,\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.295}$$

on physically realizable $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ with outgoing radiation condition and isotropic constitutive response [2]. Equation (2.295) develops Eq. Eq. (2.279) and aligns with covariance test Eq. (2.44) for $\mathbf{a} = \mathbf{0}$ [2]. Residual $\mathcal{R}_\mathcal{F}$ from Eq. (2.37) diagnoses broken commutation in discretizations [1]. Rotating $\mathbf{J}_{\mathrm{imp}}$ while evaluating flux on an unrotated $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ violates origin and covariance criteria from Section 1.2.3 even when Eq. (2.295) holds for power alone [2]. (2.295) in the same layout as Figure **??**: geometry and components must be rotated consistently before modal coefficients are compared [2]. Under rotation, modal coefficients mix through the matrix $D^{(\mathrm{rad})}_{nm}(R)$: an engineer sees principal cuts walk across the sphere while total power on Γ_R stays fixed when probe and source frames rotate together [1]. Let $\{\mathbf{u}_n\}_{n\in\mathcal{N}_{\mathrm{rad}}}$ be the orthonormal radiation basis developed

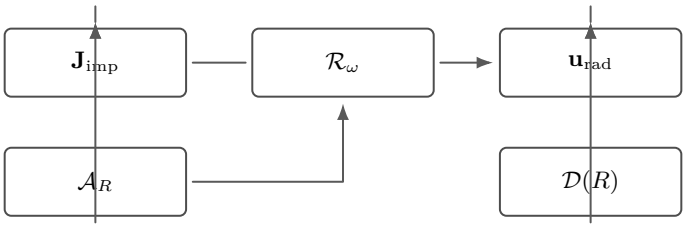


Figure 2.36: Rotational covariance Eq. (2.295): active source rotation commutes with $\mathcal{R}_\omega$ up to field representation $\mathcal{D}(R)$ Eq. (2.294).

above and $c_n[\mathfrak{D}]$ source amplitudes Eq. (2.283). Under active rotation of impressed data,

$$c_n[\mathcal{A}_R\mathfrak{D}] = \sum_{m\in\mathcal{N}_{\mathrm{rad}}} D^{(\mathrm{rad})}_{nm}(R)\, c_m[\mathfrak{D}], \tag{2.296}$$

whenever Eq. (2.295) holds and the family of radiating modes is closed under $\mathcal{D}(R)$,

$$\mathcal{D}(R)\,\mathbf{u}_m = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} D^{(\mathrm{rad})}_{nm}(R)\,\mathbf{u}_n. \tag{2.297}$$

Equations (2.296)–(2.297) are the spectral restatement of vector covariance Eq. (1.2.3) on $\mathcal{F}_{\mathrm{rad}}$ [2]. Matrix elements $D^{(\mathrm{rad})}_{nm}(R)$ are not tabulated here; Chapter 3 (*Angular Momentum Operators and SO(3)*) supplies them on vector spherical harmonic bases [1]. Eigenvalues λ_n in Eq. (2.281) are unchanged within a degenerate block because $\mathcal{R}_\omega$ commutes with $\mathcal{D}(R)$ on that block [2]. Repeated eigenvalues occur because rotational symmetry makes distinct angular orientations physically equivalent within an irreducible block: full SO(3) symmetry produces $(2\ell+1)$-fold degeneracy, and symmetry breaking—anisotropic materials, finite facets, or broken ground-plane parity—splits those blocks into distinct λ_n [2]. Suppose Eq. (2.295) holds for all $R \in \mathrm{SO}(3)$ in a compact subgroup $G \subseteq \mathrm{SO}(3)$ leaving the problem geometry invariant. Then $\mathcal{R}_\omega$ and $\mathcal{D}(R)$ generate a commuting family on $\mathcal{F}_{\mathrm{rad}}$: for each $R \in G$,

$$\mathcal{D}(R)\,\mathcal{R}_\omega = \mathcal{R}_\omega\,\mathcal{D}(R) \quad \text{on physically realizable sources,} \tag{2.298}$$

and eigenspaces of $\mathcal{R}_\omega\mathcal{S}_\omega$ decompose into irreducible G-representations. If an irreducible block carries angular index ℓ, its degeneracy multiplicity satisfies

$$\dim \mathcal{H}_\ell = 2\ell + 1 \quad \text{for full SO(3) completion on } S^2, \tag{2.299}$$

with $\mathcal{H}_\ell$ the ℓ-indexed radiative subspace [2]. Equation (2.299) explains repeated eigenvalues in Eq. (2.284) without duplicating mode physics. Eq. (2.298) follows by applying Eq. (2.295) after Eq. (2.286) and using linearity of $\mathcal{S}_\omega$ [2]. Standard spectral theory for commuting unitary or orthogonal representations implies simultaneous block diagonalization of $\mathcal{R}_\omega$ and $\mathcal{D}(R)$ on invariant subspaces [2]. For full rotational symmetry on angular patterns, irreducible representations of SO(3) on $L^2(S^2)$ have dimension $2\ell+1$; Eq. (2.299) records that count on $\mathcal{F}_{\mathrm{rad}}$ once outgoing transversality and Eq. (2.139) are imposed [8]. Detailed construction of $\mathcal{H}_\ell$ is developed in **??** [1]. Total radiated power P_{rad} is the scalar quantity a wattmeter or sphere integral reports on Γ_R; it cannot change under rigid rotation when frames are co-rotated [5]. Modal coefficients c_n redistribute directional content, while export eigenvalues λ_n set how efficiently each mode reaches the meter [1]. Rigid rotation cannot change total exported power when meters act on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with co-rotated frames. Using Eq. (2.253) and Eq. (2.35),

$$P_{\mathrm{rad}}[\mathcal{A}_R\mathfrak{D}] = P_{\mathrm{rad}}[\mathfrak{D}], \qquad P_{\mathrm{rad}}[\mathfrak{D}] = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} |\lambda_n|^2\,|c_n[\mathfrak{D}]|^2 \quad \text{when modes are orthogonal and normalized,} \tag{2.300}$$

for lossless outgoing radiation [2]. Equation (2.300) is the modal restatement of scalar invariance Eq. (2.36) on integrated power: rotated coefficients redistribute directional content but preserve the scalar budget when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. Net angular-flux magnitude obeys Eq. (1.41) on the same qualified subspace Active rotation reorients sources and patterns in space; passive rotation re-expresses fixed physics in a rotated chart Eqs. (1.2.3)–(1.2.3) [2]. Modal coefficients transform as Eq. (2.296) under active action, while passive relabeling conjugates the same $D^{(\mathrm{rad})}_{nm}(R)$ through Eq. (1.70) without altering Eq. (2.300) [2]. Confusing the two conventions produces spurious degeneracy splitting in measured $a_n(\hat{\mathbf{k}})$ from Eq. (2.291) even when rotational symmetry of the spectrum applies [2]. Laboratory rotation of an antenna rotates measured coefficients through Eq. (2.296); exported power on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ remains invariant in the far zone [1]. Origin translation $\mathbf{a}$ shifts torque and angular-flux reports through Section 1.2.3 while leaving far-zone radiative power invariant when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. At the spectral level, origin change mixes modal coefficients within fixed ℓ blocks through addition of lower-ℓ content from shifted multipole expansions; it does not create new $\lambda_n \neq 0$ export unless sources change [8]. The rotation analysis above therefore separates *rotational* degeneracy from *translational* bookkeeping in Section 1.2.3; intrinsic angular observables use Eq. (2.38) by reference [2]. Let $G_0 \subset \mathrm{SO}(3)$ be the exact symmetry of a source–boundary configuration and let $\mathcal{H}_\ell$ be an irreducible block from rotational degeneracy of the spectrum. Any symmetry-breaking perturbation (reduced G_0, anisotropic material, or finite facetization) splits Eq. (2.299) into smaller blocks with distinct λ_n within the same ℓ sector [2]. The splitting order is controlled by representation coupling rules imported from Section 1.2.3 on tensor perturbations; explicit Clebsch–Gordan content belongs to **??** Subsection 3.2.5 [2]. When G_0 is proper, $\mathcal{D}(R)$ block-diagonalizes with fixed λ_n on each irreducible piece. Perturbations that fail to commute with all $R \in G_0$ remove simultaneous eigenstructure; standard degenerate perturbation theory applies to the restricted radiation operator on $\mathcal{H}_\ell$ [2]. Rotational degeneracy of the spectrum requires isotropic or symmetry-matched media, outgoing radiation condition, and radiating-mode families closed under $\mathcal{D}(R)$ [2]. Anisotropic crystals reduce G and split Eq. (2.299) when symmetry is broken [2]. Observation rank Eq. (2.221) resolves only a finite subset of each degenerate block even when continuous symmetry is exact [1]. Regime qualification Eq. (2.271) still governs whether rotated near fields may be reported as transport [2].

2.6.4 Fredholm structure and pattern solvability

Pattern synthesis and source inversion are the inverse problems every antenna laboratory eventually confronts. One specifies a desired far-zone amplitude distribution and asks which feed currents will produce it; or one measures near fields on a scanner and asks which sources could have generated them [1]. When these inversions fail, the failure is rarely mysterious in hindsight: non-radiating components lie in $\ker \mathcal{R}_\omega$—the set of admissible impressed currents annihilated by export—and leave no far-zone signature, so no finite measurement set can resolve them without additional excitation or constraints [3]. Recoverable rank from Subsection 2.6.1 caps the truncated modal system even when a continuous target is mathematically consistent [7]. Fredholm theory names kernel, cokernel, and index after the eigenbasis and rotational structure above are in place [2].

Assumptions. Fix $\omega > 0$, bounded or exterior problem class with outgoing radiation condition on open problems, physically realizable $\mathcal{J}_\Omega$ Eq. (2.49), export map Eq. (2.275), and the radiation eigenbasis developed above [2]. Source pairing uses dual space Eq. (2.56); field pairing uses Eq. (2.26) [2]. Compact perturbations are taken in operator norm on $\mathcal{J}_\Omega$ for sources with compact support Ω_s [1]. Regime qualification Eq. (2.140) governs whether recovered $\mathbf{g}$ may be reported as transport data [2]. The *radiation kernel* $\ker \mathcal{R}_\omega$ collects impressed currents invisible to qualified far-zone meters. The *cokernel* $\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}} / \operatorname{ran} \mathcal{R}_\omega$ labels far-zone pattern targets that no

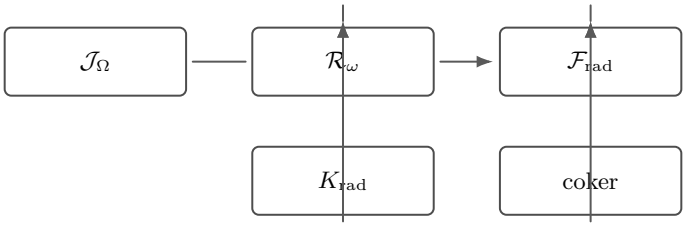


Figure 2.37: Fredholm measurement sequence: compact perturbation Eq. (2.301), kernel $\ker \mathcal{R}_\omega$, and cokernel obstructions to solvability Eq. (2.303).

admissible source can realize—equivalently, directions orthogonal to every radiative image under the adjoint pairing [2]. The *Fredholm index* $\operatorname{ind} \mathcal{R}_\omega = \dim \ker \mathcal{R}_\omega - \dim \operatorname{coker} \mathcal{R}_\omega$ counts the net imbalance between invisible source degrees and unreachable pattern obstructions; it is stable under the compact perturbations recorded below [1]. In the laboratory, kernel membership explains why two distinct feeds can match the same far pattern; cokernel membership explains why a synthesis target refuses to converge no matter how many ports are tuned [2]. Source inversion is difficult because radiation smooths fine source detail: the far-zone map retains only exportable content and discards non-radiating degrees in $\ker \mathcal{R}_\omega$ [3]. The export map inherits the Green-function split of Section 2.3: a leading outgoing propagation part $\mathcal{R}_{0,\omega}$ plus a compact smoothing part K_{rad} at finite separation on bounded source classes [2]. Physically, $\mathcal{R}_{0,\omega}$ is the dominant far-zone propagation law and K_{rad} collects the compact corrections that smooth fine source structure into resolvable pattern content [1]. That split is the operator-side restatement of the range observation that far-zone data smooth fine source detail. Formally,

$$\mathcal{R}_\omega = \mathcal{R}_{0,\omega} + K_{\mathrm{rad}}, \qquad K_{\mathrm{rad}} \text{ compact on } \mathcal{J}_\Omega, \tag{2.301}$$

where $\mathcal{R}_{0,\omega}$ captures leading outgoing singular propagation and K_{rad} collects compact smoothing at finite separation on bounded source classes [2]. Equation (2.301) specializes Eq. (2.86) to the radiation image Eq. (2.239) after Sections 2.3–2.5 [2]. On exterior domains with compact Ω_s, K_{rad} is compact while $\mathcal{R}_{0,\omega}$ carries the non-compact far-zone propagation; on bounded cavities, both parts reduce to discrete modal books Eq. (2.288) [1]. The Fredholm index counts the net imbalance between invisible source degrees ($\ker \mathcal{R}_\omega$) and unreachable pattern targets ($\operatorname{coker} \mathcal{R}_\omega$). On the range, kernel membership means different feeds may produce identical far patterns; cokernel membership means a synthesis target lies outside the reachable subspace no matter how many ports are tuned [2]. For closed operators with finite-dimensional kernel and cokernel, define the Fredholm index

$$\operatorname{ind} \mathcal{R}_\omega = \dim \ker \mathcal{R}_\omega - \dim \operatorname{coker} \mathcal{R}_\omega, \tag{2.302}$$

with $\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}} / \operatorname{ran} \mathcal{R}_\omega$ on the declared quotient by outgoing radiation condition [2]. Equation (2.302) counts independent non-radiating source degrees Eq. (2.218) minus obstructions to reaching a target far pattern $\mathbf{g} \in \mathcal{F}_{\mathrm{rad}}$ [2]. Index is stable under compact perturbations in Eq. (2.301) [1]. Observation rank cap Eq. (2.221) is independent of $\operatorname{ind} \mathcal{R}_\omega$ but composes with it in inversion budgets [2]. (2.85) [2]. The Fredholm alternative states that a target far-zone field is synthesizable only if it carries no component along cokernel directions invisible to the adjoint export map [2]. Consequence: a measured pattern residual that is orthogonal to every reachable image is not a calibration fault—it signals membership in $\operatorname{coker} \mathcal{R}_\omega$ or in a geometrically forbidden sector from Subsection 2.6.5 [1]. The forward radiation equation $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ is solvable iff the target $\mathbf{g} \in \mathcal{F}_{\mathrm{rad}}$ is orthogonal to the cokernel of the adjoint map on declared pairings:

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g} \text{ solvable} \iff \langle \mathbf{g}, \mathbf{w} \rangle_{\mathcal{U}} = 0 \ \ \forall\, \mathbf{w} \in \ker \mathcal{R}_\omega^\dagger, \tag{2.303}$$

with $\mathcal{R}_\omega^\dagger$ the adjoint on $(\mathcal{J}_\Omega, \mathcal{F}_{\mathrm{rad}})$ induced by Eqs. (2.56) and (2.26) [2]. Equation (2.303) develops Eq. (2.278) and aligns with biorthogonality Eq. (2.292) on modal adjoints [2]. When $\mathbf{g}$ arises from far-pattern data, solvability is further capped by Eq. (2.221) even if Eq. (2.303) holds in the continuous model [1]. Let $\mathcal{R}_\omega$ satisfy Eq. (2.301) on $\mathcal{J}_\Omega$ with $\mathcal{R}_{0,\omega}$ Fredholm of index zero on the outgoing quotient and K_{rad} compact. Then $\mathcal{R}_\omega$ is Fredholm,

$$\operatorname{ind} \mathcal{R}_\omega = \operatorname{ind} \mathcal{R}_{0,\omega}, \tag{2.304}$$

and Eq. (2.303) holds. For finite-rank observation $\mathcal{O}_{\mathrm{meas}} : \mathcal{F}_{\mathrm{rad}} \to \mathbb{C}^N$,

$$\mathcal{O}_{\mathrm{meas}}\mathcal{R}_\omega \text{ is Fredholm on } \mathcal{J}_\Omega \iff \mathcal{R}_\omega \text{ Fredholm and } \operatorname{rank}(\mathcal{O}_{\mathrm{meas}}) = N, \tag{2.305}$$

with rank bound Eq. (2.85) Compact perturbation stability of Fredholm property and index follows from standard operator theory applied to Eq. (2.301); Subsection 2.1.3 recorded the embedding inputs without computing index here [2]. Solvability criterion Eq. (2.303) is the Fredholm alternative for $\mathcal{R}_\omega$. Composition with finite-rank $\mathcal{O}_{\mathrm{meas}}$ preserves Fredholm class and adjusts index by at most N on the range side, yielding Eq. (2.305) and Eq. (2.85) [2]. The modal fredholm reduction proceeds as follows: on the orthonormal radiation eigenbasis developed above,

$$\mathcal{R}_\omega = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} \lambda_n \, P_n \circ \mathcal{S}_\omega, \qquad \lambda_n \neq 0 \iff \text{mode } n \text{ exporting}, \tag{2.306}$$

so solvability of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ reduces to matching coefficients $c_n[\mathfrak{D}]$ Eq. (2.283) on modes with $\lambda_n \neq 0$ [2]. Modes with $\lambda_n = 0$ lie in $\ker \mathcal{R}_\omega$ at the source level Eq. (2.218); modes with $\mathbf{g}$ unsupported on $\operatorname{ran} \mathcal{R}_\omega$ generate cokernel obstructions in Eq. (2.303) [2]. Truncation to $\mathcal{R}_\omega^{(N)}$ from Eq. (2.83) yields approximate solvability with error Eq. (2.84) controlled by $\sigma_{N+1}(K_{\mathrm{rad}})$ [1]. Let $\mathbf{g} \in \operatorname{ran} \mathcal{R}_\omega$ and expand $\mathbf{g} = \sum_n g_n \, \mathbf{u}_n$ with $\{\mathbf{u}_n\}$ from Subsection 2.6.2. Then Eq. (2.303) holds if and only if $g_n = 0$ whenever $\lambda_n = 0$. A particular solution is determined by

$$c_n = \frac{g_n}{\lambda_n} \quad \text{for } \lambda_n \neq 0, \tag{2.307}$$

with arbitrary coefficients on $\ker \mathcal{R}_\omega$ not fixed by far-zone data [2]. Equation (2.307) is the spectral form of Eq. (2.220) on the radiative range; Tikhonov and regularized inversions from Subsection 2.4.4 damp ill-conditioned λ_n without altering Fredholm index [2]. Substitute Eq. (2.285) into $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ and use orthogonality of $\{\mathbf{u}_n\}$ in $\mathcal{U}$ [2]. Modes with $\lambda_n = 0$ do not appear in the range; modes with $\lambda_n \neq 0$ invert mode-wise yielding Eq. (2.307). Kernel directions remain free because $\mathcal{R}_\omega$ is not injective on $\mathcal{J}_\Omega$ Eq. (2.218) [1]. Cokernel elements pair with unreachable far-zone content: if $\mathbf{w} \in \ker \mathcal{R}_\omega^\dagger$, then $\langle \mathbf{g}, \mathbf{w} \rangle_{\mathcal{U}} = 0$ is required for solvability Eq. (2.303) [2]. On exterior domains, continuous angular spectra Eq. (2.290) imply infinite-dimensional cokernel constraints in principle; finite apertures replace them with rank-N projections through Eq. (2.305) [1]. Far-pattern coefficients from the eigenbasis express $\mathbf{g}$ through $\mathbf{f}_n$; solvability requires consistency of measured coefficients with Eq. (2.307) on exporting modes only [2]. On the outgoing branch, solvability of $\mathcal{L}_\omega[\mathbf{u}_\omega] = \mathbf{s}_\omega$ and solvability of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ are linked through Eq. (2.240): forward field construction is single-valued on $\mathcal{F}_{\mathrm{SF}}$ while source inversion remains rank-deficient [2]. (2.87) is not identical to $\operatorname{ind} \mathcal{R}_\omega$; export selection removes non-radiative degrees before pattern inversion [2]. The Fredholm analysis above treats $\mathcal{R}_\omega$ as the primary object for far-zone inverse problems [1]. Fredholm structure on $\mathcal{R}_\omega$ requires linear media, fixed ω, outgoing radiation condition, and compactness of K_{rad} on the declared source class [2]. Resonant cavities violate invertibility of $\mathcal{L}_0$ unless quotiented Eq. (2.87); radiation index must be computed on the outgoing subspace after

Eq. (2.203) [2]. Loss regularizes small λ_n but does not remove $\ker \mathcal{R}_\omega$ Eq. (2.218) [1]. regime qualification Eq. (2.271) remains mandatory before transport claims on recovered $\mathbf{g}$ [2]. As a worked example, a spherical near-field scanner with $N = 50$ probe channels is a concrete inversion budget. The physical configuration places fifty probes on a sphere of radius $R \approx 2\lambda_0$ around a compact horn feed; each channel records tangential $\mathbf{E}$ or $\mathbf{H}$ depending on probe design [7]. After transformation to far-zone coefficients, the composition $\mathcal{O}_{\text{meas}}\mathcal{R}_\omega$ has rank at most 50 Eq. (2.85) [2]. The observable behavior is instructive: the first few reconstructed modes match direct far-field cuts taken on the compact range, while coefficients beyond the fiftieth singular direction track noise rather than new beam structure. Even when Eq. (2.303) holds for a continuous target $\mathbf{g}$, discrete recovery satisfies only a truncated modal system; tail error is bounded by Eq. (2.84) with the first neglected singular value of the measurement composition cited from Subsection 2.6.1 [1]. The lesson for laboratory planning is direct: adding probes without enlarging scan radius or improving SNR cannot recover modes the geometry has already annihilated. Scanner rank is a hard cap that composes with, but does not replace, Fredholm obstructions from $\ker \mathcal{R}_\omega$ [2]. As a worked example, suppose a synthesis target $\mathbf{g} \neq \mathbf{0}$ specifies a far-zone pattern that is orthogonal to every exporting mode reachable from the declared source class—for example, a forbidden parity combination above a ground plane when only even sectors are physically accessible [11]. No amount of feed optimization will satisfy forward export: the obstruction appears as inconsistent far-zone coefficients in Eq. (2.303), not as slow convergence of a numerical solver [1]. By contrast, when $\mathbf{g} = \mathbf{0}$, the equation is always solvable with $\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega$ [2]; the balanced-loop example shows that such sources may be strongly measurable in the near zone yet invisible on Γ_R. Mode-wise invertibility on the radiative range detects obstructions through g_n on $\lambda_n = 0$ sectors [2]. Geometric sector constraints on recoverable spectra are developed next; complete non-radiating classification belongs to Section 2.7.1 [2]. Subsection 2.6.4 closed the Fredholm account: compact perturbation, kernel and cokernel pairings, solvability criterion, and modal reduction on the radiation eigenbasis. Kernel membership explains invisible source degrees; cokernel membership explains unreachable pattern targets; recoverable rank from Subsection 2.6.1 caps what finite instruments can resolve even when continuous solvability holds [2].

2.6.5 Geometric constraints on radiative spectra

Geometry can forbid a modal sector before instrumentation or inversion enter the story. On a reflector compact range, a slot above a ground plane may exhibit a persistent pattern null in a parity-forbidden sector: the null survives probe upgrades and SNR improvements because the symmetry class, not receiver noise, annihilates that angular content [11]. An electrically small feed above the same plane supports only low-order multipole export; formal harmonic expansions may list high-ℓ indices that the support condition suppresses long before $\mathcal{O}_{\text{meas}}$ enters [8]. The geometric operator below encodes that selection on modal amplitudes after eigenmodes and Fredholm structure are known [2].

Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\text{imp}}$, outer boundary Γ, piecewise material data (ε, μ), outgoing radiation condition, and radiation eigenbasis $\{\mathbf{u}_n, \mathbf{f}_n\}_{n \in \mathcal{N}_{\text{rad}}}$ from the developments above [2]. Modal coefficients a_n in Eq. (2.291) are identified with $\lambda_n c_n[\mathfrak{D}]$ on exporting modes [2]. Geometric symmetry group $G_{\text{geo}} \subseteq \mathrm{SO}(3) \times \{+\}$ acts on sources, boundaries, and constitutive tensors as in Section 1.2.3 by reference [2]. Regime qualification Eq. (2.140) remains mandatory before transport claims on any surviving modal sector [1]. When symmetry and boundary class are fixed, modal amplitudes do not vary independently: some linear combinations are forced to zero by image parity or by support even when the export map would otherwise allow them [2]. The matrix Γ_{nm} in Eq. (2.308) encodes how a symmetry operation on the physical configuration maps one modal sector into another; a

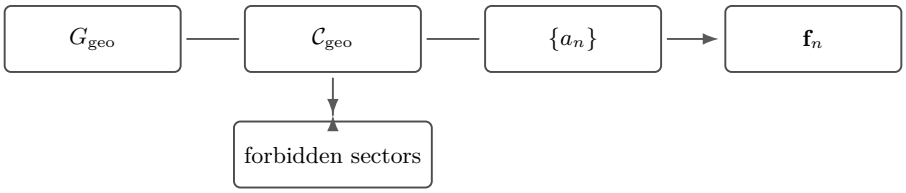


Figure 2.38: Geometric filter: symmetry group G_{geo} acts through $\mathcal{C}_{\text{geo}}$ Eq. (2.308) on modal amplitudes before far patterns $\mathbf{f}_n$.

forbidden sector is one whose unit amplitude vector lies in the kernel of $\mathcal{C}_{\text{geo}}$ [2]. Define a bounded operator $\mathcal{C}_{\text{geo}}$ on modal amplitudes by

$$\left(\mathcal{C}_{\text{geo}}\mathbf{a}\right)_n = \sum_{m\in\mathcal{N}_{\text{rad}}} \Gamma_{nm}\, a_m, \qquad \mathbf{a} = \{a_m\}_{m\in\mathcal{N}_{\text{rad}}}, \tag{2.308}$$

where Γ_{nm} encodes how geometry and boundary class map modal sectors under G_{geo} [2]. Equation (2.308) is the constructive account of realization of Eq. (2.308): when $\mathcal{C}_{\text{geo}}\mathbf{e}_n = \mathbf{0}$ for a unit vector $\mathbf{e}_n$, mode n is *geometrically forbidden* even if $\lambda_n \neq 0$ on the export map. Geometric symmetry can annihilate modal sectors before instrumentation enters: a forbidden mode has zero coupling to physically realizable sources even when formal eigenvalues are non-zero [2]. Broken symmetry restores coupling—a practical lever in reflector and array design [11]. Eq. (2.303) holds for other modes [2]. Rotation commutation Eq. (2.295) acts on $\mathbf{f}_n$; $\mathcal{C}_{\text{geo}}$ acts on physically accessible *sectors* after symmetry reduction [1]. Compact support $\Omega_s \subset \mathbb{R}^3$ implies entire-function structure in spatial wavenumber: for each outgoing mode $\mathbf{f}_n(\theta,\phi)$,

$$a_n(\mathbf{J}_{\text{imp}}) = 0 \text{ unless } \mathbf{f}_n \text{ has non-negligible overlap with } \mathcal{F}_\omega[\mathcal{J}_{\Omega_s}], \tag{2.309}$$

with $\mathcal{J}_{\Omega_s}$ the restriction of admissible sources to Ω_s [2]. Equation (2.309) is a support-induced geometric filter distinct from Fredholm cokernel constraints Eq. (2.303): electrically small Ω_s suppresses high-ℓ sectors in Eq. (2.291) through $(k_0 a)^{2\ell}$ scaling established in Chapter **??** (*Angular Momentum Operators and SO(3)*) [8]. Rank cap Eq. (2.221) limits measurable a_n separately from support annihilation [1]. Perfect conductors, symmetry planes, and equivalent image configurations impose linear relations among modal amplitudes:

$$\sum_{m\in\mathcal{N}_{\text{rad}}} M_{nm}(\Gamma,\mathcal{B}_\Gamma)\, a_m = 0 \quad\Longrightarrow\quad a_n = 0 \text{ for forbidden parity sectors,} \tag{2.310}$$

with M_{nm} fixed by tangential boundary class Eq. (1.20) and image rules on outgoing Green branches Eq. (2.200) [2]. Equation (2.310) specializes degeneracy lifting under broken symmetry (Subsection 2.6.3): a ground plane replaces full SO(3) by a reflection group and annihilates odd or even combinations per declared image parity [2]. Sommerfeld outgoing radiation condition from Section 2.4 is presupposed; incoming image branches are excluded from $\mathcal{F}_{\text{rad}}$ [1]. When G_{geo} contains mirror reflection σ, modal sectors split into parity classes $\Pi_\pm$ with

$$a_n = 0 \text{ for all } n \in \mathcal{N}_{\text{rad}} \text{ with } \Pi_\pm \mathbf{f}_n = \mp \mathbf{f}_n \text{ and forbidden parity,} \tag{2.311}$$

as in the ground-plane illustration of Subsection 2.6.3 [2]. Equation (2.311) explains pattern nulls that are not explained by $\ker \mathcal{O}_{\text{meas}}$ or $\ker \mathcal{R}_\omega$ alone [2]. Slot and patch symmetries reduce Eq. (2.311) to discrete sector labels on $\mathcal{N}_{\text{rad}}$ without re-tabulating dyads from Subsection 2.3.1 [1]. Geometric, Fredholm, and observation constraints multiply:

$$\mathcal{N}_{\text{reportable}} \subseteq \mathcal{N}_{\text{acc}} \cap \{n : \text{ resolvable by } \mathcal{O}_{\text{meas}}\} \cap \{n : \ \lambda_n \neq 0\}, \tag{2.312}$$

with the last two factors from Subsection 2.6.4 and Eq. (2.221) [2]. Equation (2.312) prevents attributing a pattern null solely to finite aperture when Eq. (2.308) already forbids the sector [2]. Inverse source Eq. (2.220) on $\mathcal{N}_{\text{reportable}}$ remains rank-deficient through $\ker \mathcal{R}_\omega$ Eq. (2.218) even when geometry is exact [1]. Geometric constraints on radiative spectra assume linear media, fixed ω, and symmetry of $(\Omega_s, \Gamma, \varepsilon, \mu)$ under declared G_{geo} [2]. Weak symmetry breaking perturbs Eq. (2.9.3) through lifted degeneracy rather than strict zero [2]. Loss and anisotropy modify Γ_{nm} and may reopen partially forbidden sectors at finite amplitude [2]; regime qualification Eq. (2.271) is unchanged, for geometric accessibility does not promote reactive near fields to export [1]. Eigenmode ordering, rotational degeneracy, Fredholm solvability, and geometric filters now compose the spectral chain of Eq. (2.274) on the integral representations from Section 2.5 [2]. The thread is laboratory-first: finite aperture and σ_n decay fix recoverable rank once in Subsection 2.6.1; export eigenvalues λ_n of $\mathcal{R}_\omega$ then organize dominant directions; rotation, Fredholm structure, and geometry explain what inversion can and cannot recover. Section 2.7 completes non-radiating null spaces at the source level; **??** supplies explicit harmonic bases; **??** addresses which eigenstructures finite sources can realize after truncation [1].

2.7 Non-Radiating Sources and Null Spaces

Spectral theory in Section 2.6 ranked export directions and recoverable rank. Section 2.7 addresses the complementary laboratory question: *How can current exist without observable radiation?* [3]

Laboratory signatures. Three configurations recur before any kernel is displayed. A balanced wire loop in vacuum, driven at fixed ω with vanishing net electric dipole moment, stores reactive $\mathbf{E}$ and $\mathbf{H}$ within λ_0 of the conductor while a qualified wattmeter on Γ_R at $R \gg \lambda_0$ records zero outgoing transverse flux once Π_{rad} selection is enforced [2]. A counter-directed dipole pair separated by $d \ll \lambda_0$ can annihilate leading $1/r$ transverse fields on S^2 while reactive overlap remains in the gap [8]. A waveguide section below cutoff carries standing reactive fields at ports yet contributes no propagating sheet to exterior flux meters [2]. In each case Maxwell permits substantial local structure while $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}$: rich near-zone data, null far-zone export—not a solver artifact [8]. The *radiation kernel* collects every admissible impressed current whose outgoing radiative image vanishes: currents that reorganize near-zone energy without qualified far-zone flux [3]. Near-field scanners sample $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ on enclosing shells; far-zone flux meters and angular integrators read only after Π_{rad} has removed non-export content [7]. A source invisible to the latter may remain vivid to the former—the central diagnostic that separates reactive storage from qualified export. Campaigns that report singular values without kernel awareness merge synthesis with invisible modes [8]. By the *radiation null space* $\ker \mathcal{R}_\omega$ we mean precisely this kernel: admissible impressed currents whose radiative image vanishes under the outgoing map $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$—physically realizable sources that store energy locally yet export no qualified far-zone power [3]. Membership in $\ker \mathcal{R}_\omega$ is an export statement; it does not require $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}] = \mathbf{0}$. The balanced loop above is a constructive instance: $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{loop})}] = \mathbf{0}$ while $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}^{(\text{loop})}]$ remains measurable on near probes [3]. Non-radiating sources are the central electromagnetic proof that existence does not imply observability. A current distribution may generate substantial near-zone $\mathbf{E}$ and $\mathbf{H}$ while exporting zero far-zone power: $\mathcal{R}_\omega$ is not injective, and $\ker \mathcal{R}_\omega$ is generally infinite-dimensional on open domains. Different impressed currents may therefore produce identical radiative images, angular spectra, and reciprocity readouts while differing in reactive storage and evanescent overlap. Hidden electromagnetic structure is not a modelling defect but a consequence of Maxwell propagation from compact sources through outgoing

selection to finite instruments [2]. The absence of radiated signatures does not imply the absence of electromagnetic structure; it identifies source configurations whose field content remains inaccessible to far-zone observation. Section 2.6 closed the spectral chain Eq. (2.274): radiation eigenmodes, rotational degeneracy, Fredholm solvability, and geometric filters on admissible modal sectors [2]. That program organized $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ in modal coordinates and explained which far-zone patterns are reachable, degenerate, or symmetry-forbidden. Section 2.7 addresses the complementary *source-side* question named above: which impressed currents produce no radiative export at all, how those currents relate to evanescent and non-propagating field content, and what null-space structure implies for inverse problems and angular spectra [2]. Eq. (2.218) from Subsection 2.4.4 named $\ker \mathcal{R}_\omega$; the subsections below develop constructive catalog, canonical non-radiating configurations, and reconstructability limits without re-opening Green construction, Sommerfeld completion, or radiation integrals from Sections 2.3–2.5 [2]. Reciprocity constraints on coupled modes belong to Section 2.8; plane-wave angular-spectrum action to Section 2.9; explicit harmonic bases to **??** [1]. Nullity, $\mathcal{E}^{(N)}_{\mathrm{acc}}$, and realizability. Three deficits are often merged in inverse problems and structured-beam reporting [3]. *Nullity* is source-side membership in $\ker \mathcal{R}_\omega$: far data cannot detect additions satisfying Eq. (2.350) [13]. *probe rank* is field-side: which directions in the angular spectrum survive geometric, transport, and regime qualifications even when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ [1]. $\mathcal{S}_{\mathrm{real}}$—whether a target pattern is $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ for some admissible $\mathbf{J}_{\mathrm{imp}}$—is not decided by rank-$N$ recovery alone; **??** develops synthesizable proofs while Section 2.7 supplies the forward null-space catalog that constrains every claim [14]. A source may store reactive energy nearby yet export no far-zone power—the radiation map annihilates whole families of impressed currents. Field-level uniqueness after outgoing radiation condition (exterior radiative uniqueness (Subsection 2.4.3)) does not imply source identifiability: many admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ can share the same $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ through Eq. (2.218) [2]. Spectral mode $\lambda_n = 0$ in Eq. (2.277) marks a radiative eigen-direction with vanishing export, yet $\ker \mathcal{R}_\omega$ is generally infinite-dimensional on open domains and is not exhausted by finitely many modal labels [1]. Using cavity modal books or truncated harmonic sums to *approximate* $\ker \mathcal{R}_\omega$ misstates identifiability even when outgoing source-to-field integrals from Subsection 2.5.1 are evaluated exactly [8]. Section 2.7 separates operator null spaces, field-side non-radiative components Eqs. (2.266)–(2.267), and observation null spaces Eq. (2.81), then forwards reciprocity and angular-spectrum filters to Sections 2.8–2.9 [2]. Let $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact, $\omega > 0$ fixed, and outgoing field pairs lie in $\mathcal{F}_{\mathrm{SF}}$ from Eq. (2.57). Section 2.7 organizes analysis in four stages:

$$\ker \mathcal{R}_\omega \xrightarrow{\text{operator structure}} \text{null-space bases} \xrightarrow{\text{canonical sources}} \text{NR current families} \xrightarrow{\text{evanescent overlap}} \text{non-reconstruct} \tag{2.313}$$

where the first arrow is Subsection 2.7.1, the second is Subsection 2.7.1, the third is Subsection 2.7.2, and the fourth is Subsection 2.7.4 [2]. Equation (2.313) expresses Eq. (2.274) at the source level: spectral $\lambda_n = 0$ modes from Eq. (2.280) lie in $\mathbf{u}_{\mathrm{nr}}$, but complete $\ker \mathcal{R}_\omega$ requires impressed-data coordinates on $\mathcal{J}_\Omega$, not only far-zone modal shapes [2]. Equation (2.313) is the source-side restatement of the spectral chain Eq. (2.274): operator nullity precedes constructive NR catalogs, which precede evanescent non-reconstructability, which precede angular-spectrum consequences. Mathematically, each arrow preserves linearity of $\mathcal{R}_\omega$; physically, each stage removes a distinct laboratory error from inverse-source charting. On a cube of side $2a$ with $k_0 a = 2$, evanescent overlap with $\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ can exceed 25% of the near-field energy on S_R with $R = a$ while contributing zero to $a(\widehat{\mathbf{k}})$; mistaking that storage for NR nullity violates Eq. (2.336) [2]. 1 flux laws enter only after Π_{rad} [1]. Composition law Eq. (2.312) still applies: geometric annihilation Eq. (2.308) and observation rank Eq. (2.221) are independent of, and composable with, source null-space membership [1]. The laboratory examples and kernel reading above fix the measurement ordering before any operator

display. Near-field probes sample $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$; flux meters read only after Π_{rad} [2]. With that ordering in place, keep three spaces in view throughout Section 2.7: impressed currents $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ on compact support Ω_s, full Maxwell fields $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_\Omega$, and the radiative image on which far-zone instruments read. The primary object remains the radiation map that projects the latter from the former:

$$\mathcal{R}_\omega : \mathcal{J}_\Omega \longrightarrow \mathcal{F}_{\mathrm{rad}}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.314}$$

from Eqs. (2.3)–(2.203) [2]. The displayed map is the mathematical encoding of the wattmeter reading: only $\Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ enters $\mathcal{F}_{\mathrm{rad}}$, so near-zone energy in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \setminus \mathcal{F}_{\mathrm{rad}}$ cannot be inferred from far data alone [2]. Secondary targets include the observation map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ from Eq. (2.219), the non-radiative field projector $\mathrm{id} - \Pi_{\mathrm{rad}}$ on $\mathcal{F}_\Omega$, and evanescent/non-propagating selectors Eqs. (2.134)–(2.265) from Subsection 2.2.3 [2]. Inverse source Eq. (2.220) is ill-posed whenever $\ker \mathcal{M}_\omega \neq \{\mathbf{0}\}$, which occurs if $\ker \mathcal{R}_\omega \neq \{\mathbf{0}\}$ or $\ker \mathcal{O}_{\mathrm{meas}}$ is nontrivial on $\operatorname{ran} \mathcal{R}_\omega$ [1]. reciprocity-based solvability Eq. (2.303) governs reachability of a *target* radiative field; Section 2.7 classifies sources that map to $\mathbf{0} \in \mathcal{F}_{\mathrm{rad}}$ regardless of solvability for nonzero targets [2]. Radiation null space (leading template). Subsection 2.7.1 develops Eq. (2.218) to structural statements on $\ker \mathcal{R}_\omega$ and $\ker \mathcal{R}_\omega^\dagger$. Locally, $\ker \mathcal{R}_\omega$ collects impressed currents annihilated by export; $\ker \mathcal{R}_\omega^\dagger$ collects far-side test directions orthogonal to every radiative image; and $\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}} / \operatorname{ran} \mathcal{R}_\omega$ labels pattern targets that no admissible source can realize [2]. The leading template below records admissibility and typical infinite dimensionality on open domains:

$$\ker \mathcal{R}_\omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0},\ \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1\}, \qquad \dim \ker \mathcal{R}_\omega = \infty \text{ typ. on open domains}, \tag{2.315}$$

with $\mathcal{D}_J = 1$ Eq. (2.9) enforced throughout [2]. Equation (2.315) does not re-define Eq. (2.218); it records the Chapter 2 primary target for constructive characterization of nullity, dimensionality, and pairing with cokernel directions obstructing inversion Eq. (2.278) [2]. Modal sectors with $\lambda_n = 0$ in Eq. (2.280) supply a finite-dimensional *slice* of $\ker \mathcal{R}_\omega$ only when eigenmodes are declared on outgoing subspaces [1]. Non-radiating current configurations (leading template). Subsection 2.7.1 develops canonical impressed currents that realize Eq. (2.8):

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega, \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0}, \qquad \|\mathbf{u}_{\mathrm{nr}}\| \text{ may be large near } \Omega_s, \tag{2.316}$$

with $\mathbf{u}_{\mathrm{nr}}$ from Eq. (2.267) [2]. Equation (2.316) specializes Eq. (2.272) to constructive families—loop modes, counter-propagating pairs, and equivalent-surface cancellations compatible with Eq. (1.8)—without tabulating dyadic kernels from Subsection 2.3.1 [2]. Such currents are invisible to far-zone flux meters Eq. (2.253) yet may dominate near-zone energy budgets Eq. (2.269) [1]. Evanescent overlap and non-reconstructability (leading template). Subsection 2.7.2 links $\ker \mathcal{R}_\omega$ to evanescent subspaces and finite observation rank. Evanescent content is field-side reactive storage with $\Pi_{\mathrm{rad}}\mathbf{u}_{\mathrm{ev}} = \mathbf{0}$; it must not be conflated with source-side $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$, although both deficits can coexist on the same excitation [2]. The relation records the measurement inclusion and the evanescent non-export law:

$$\ker \mathcal{O}_{\mathrm{meas}} \circ \mathcal{R}_\omega \supseteq \ker \mathcal{R}_\omega, \qquad \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ never exports through } \Pi_{\mathrm{rad}}, \tag{2.317}$$

using Eqs. (2.139) and (2.221) [2]. Equation (2.317) separates *radiation nullity* (no outgoing transverse export) from *measurement nullity* (unobservable even when export is nonzero) [2]. Tikhonov stabilization Eq. (2.224) does not shrink $\ker \mathcal{R}_\omega$; regularization only damps ill-conditioned directions on $\operatorname{ran} \mathcal{R}_\omega$ [1]. Accessible modal sector Eq. (2.9.3) from Subsection 2.6.5 further restricts recoverable content independently of evanescent storage [2]. Angular-spectrum consequences (leading template). Subsection 2.7.4 records how null-space structure propagates to continuous angular

spectra Eq. (2.290):

$$a(\widehat{\mathbf{k}}) = 0 \text{ on suppressed sheets } \mathcal{K}_{\text{sup}} \implies \mathbf{J}_{\text{imp}} \text{ may still carry reactive content in } \Pi_{\text{ev}}\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}], \tag{2.318}$$

with $a(\widehat{\mathbf{k}})$ the far angular amplitude and $\mathcal{K}_{\text{sup}}$ the complement of accessible sector Eq. (2.9.3) [2]. Equation (2.318) links Section 2.7 to Section 2.9: plane-wave filters and retained/suppressed components in Subsections 2.9.1–2.9.3 presuppose the source-side null-space catalog developed here [2]. Angular probe rank on the far shell (Subsection 2.6.5) is not re-proved here; Subsection 2.7.4 states angular-spectrum *consequences* of Eq. (2.218) [1].

$$\mathbf{J}_{\text{imp}} \in \ker \mathcal{R}_\omega \implies a_n(\mathbf{J}_{\text{imp}}) = 0 \text{ for all exporting modes with } \lambda_n \neq 0, \tag{2.319}$$

while geometric constraints Eq. (2.9.3) may force additional $a_n = 0$ for $\mathbf{J}_{\text{imp}} \notin \ker \mathcal{R}_\omega$ [2]. Equation (2.319) follows from Eq. (2.291) and linearity Eq. (2.6); it is cited in Section 2.7 [2]. Inverse problems must budget $\ker \mathcal{R}_\omega$, $\ker \mathcal{O}_{\text{meas}}$, and forbidden sectors from geometric constraints on radiative spectra (Subsection 2.6.5) separately [1]. Several constructions deliberately remain outside Section 2.7. Reciprocity identities and modal orthogonality belong to Section 2.8; plane-wave angular-spectrum operators and near-to-far transforms to Section 2.9; vector spherical harmonic completeness and explicit $\mathbf{f}_n$ to **??**; range membership of unreachable modes to **??** [2]. Section 2.7 Subsection 2.6.2 supplies spectral decomposition; Fredholm structure of radiation operators (Subsection 2.6.4), or geometric constraints on radiative spectra (Subsection 2.6.5); it classifies invisible sources on the maps those results presuppose [2]. Worked non-radiating examples appear in Subsection 2.10.2 by reference to Subsection 2.7.1 [1]. Regime logic from Section 1.3.3 is unchanged. Null-space membership does not override transport qualification: a current in $\ker \mathcal{R}_\omega$ has $\mathcal{P}_{\text{rad}} = 0$ on qualified shells, while reactive near fields from $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ may still violate naive flux interpretations if Eq. (2.140) is ignored [2]. Far-zone dominance far-zone radiative dominance (Subsection 2.5.4) applies only to exporting sources; it does not trivialize $\ker \mathcal{R}_\omega$ [2]. Compression error Eq. (2.84) bounds modal truncation on $\operatorname{ran} \mathcal{R}_\omega$ independently of null-space dimension [1]. As a worked example, consider a thin rectangular wire loop of perimeter $2\pi b$ with $b \ll \lambda_0$, embedded in free space and driven at angular frequency ω so that the impressed current completes one closed circulation with equal and opposite arms along the loop's long sides. The physical setup is a balanced dipole pair folded into a single closed contour: magnetic flux linkage is negligible at ω, yet the loop stores reactive energy in the near zone. A closed admissible loop carrying oppositely directed arms can cancel far-zone transverse radiation through Eq. (2.229) while maintaining nonzero $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}]$ near the loop [2]. Subsection 2.7.1 classifies such families; here the point is that Eq. (2.8) is realizable with $\mathcal{D}_J[\mathbf{J}_{\text{imp}}] = 1$, not merely a formal kernel direction [1]. Pattern-matching algorithms that fit only $\operatorname{ran} \mathcal{R}_\omega$ assign spurious modal labels to invisible loop content [2]. As a worked example, take a compact spherical source of radius a with $k_0 a = 1.5$ and truncate the exterior modal book at $N = 12$ radiation eigenpairs. A single $\lambda_n = 0$ eigenmode from Eq. (2.281) contributes to $\ker \mathcal{R}_\omega$ when realized by admissible $\mathbf{J}_{\text{imp}}^{(n)}$ [2]. On the range, the truncated index $n \leq N$ labels only a finite slice of invisible degrees; the full kernel on open $\mathcal{J}_\Omega$ is generally much larger. On exterior problems the modal index set is not finite; truncating to the first N modes underestimates $\dim \ker \mathcal{R}_\omega$ and overstates identifiability of inverse Eq. (2.220) [1]. Subsection 2.7.1 states the correct operator bookkeeping [2].

2.7.1 Null spaces of radiation operators

On a declared compact-range or anechoic campaign, technicians record frequency, outline, and probe geometry before any harmonic fit is trusted [7]. Non-radiating currents store reactive energy without

far-zone export [2]. Near-field richness must not be reported as far-zone diversity [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Section 2.7 opened with laboratory signatures—balanced loops, counter-directed pairs, below-cutoff guides—and named the radiation kernel in plain language. Subsection 2.7.1 now supplies the operator account: structural properties of $\ker \mathcal{R}_\omega$, adjoint null spaces, cokernel pairings, and modal null sectors, developing Eq. (2.218) from Subsection 2.4.4 [2]. The loop and scanner patterns cited in the section introduction are not re-derived here; they motivate the locked definition below.

Assumptions. Fix $\omega > 0$, admissible source space $\mathcal{J}_\Omega$ Eq. (2.49), charge continuity and boundary compatibility Eq. (2.9), outgoing radiation condition on exterior problems Eqs. (2.194)–(2.203), and radiation map $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ from Eqs. (2.3)–(2.203) [2]. Dual pairings on sources and fields are Eqs. (2.56) and (2.26) [2]. regime qualification Eq. (2.140) governs transport reports on any recovered radiative component [1]. Radiation null space (locked definition). Physically, $\ker \mathcal{R}_\omega$ is the set of impressed currents that produce no qualified far-zone pattern: the wattmeter reading is null even when near probes are not [2]. Formally, the *radiation null space* is the source-side kernel

$$\ker \mathcal{R}_\omega = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0} \in \mathcal{F}_{\mathrm{rad}},\ \mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}] = 1\}\,, \tag{2.320}$$

equivalently $\Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ on the outgoing branch [2]. Equation (2.320) is the Subsection 2.7.1 development of Eq. Eq. (2.218): $\mathcal{D}_J = 1$ is explicit, and membership is defined on $\mathcal{J}_\Omega$ rather than on formal test directions [2]. Elements of $\ker \mathcal{R}_\omega$ produce no qualified far-zone export through Eq. (2.229) yet may satisfy $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ Eq. (2.272) [1]. Consequence for inversion: any far-zone datum fixes $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ only up to additions in $\ker \mathcal{R}_\omega$. Linearity Eq. (2.6) makes $\ker \mathcal{R}_\omega$ a linear subspace of $\mathcal{J}_\Omega$ closed under admissible superposition [2]. Adjoint null space and cokernel. Targets in $\operatorname{coker} \mathcal{R}_\omega$ are far-zone patterns orthogonal to every reachable image; they obstruct solvability even when no near-field anomaly is visible [2]. Let $\mathcal{R}_\omega^\dagger : \mathcal{F}_{\mathrm{rad}} \to \mathcal{J}_\Omega$ denote the adjoint induced by Eqs. (2.56) and Eq. (2.26) defines the adjoint pairing. Define

$$\ker \mathcal{R}_\omega^\dagger = \{\mathbf{w} \in \mathcal{F}_{\mathrm{rad}} : \langle \mathbf{w},\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\rangle_{\mathcal{U}} = 0 \ \ \forall\ \mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega\}\,, \tag{2.321}$$

and the cokernel quotient

$$\operatorname{coker} \mathcal{R}_\omega = \mathcal{F}_{\mathrm{rad}} / \operatorname{ran} \mathcal{R}_\omega, \tag{2.322}$$

with $\operatorname{ran} \mathcal{R}_\omega \subset \mathcal{F}_{\mathrm{rad}}$ the radiative image [2]. Equations (2.321)–(2.322) pair with $\ker \mathcal{R}_\omega$ through reciprocity-based solvability Eq. (2.303) imported from Subsection 2.6.4: targets $\mathbf{g} \in \mathcal{F}_{\mathrm{rad}}$ orthogonal to $\ker \mathcal{R}_\omega^\dagger$ are reachable, while obstructions lie in cokernel directions [2]. Biorthogonality Eq. (2.292) expresses the same pairing on modal coordinates without re-deriving adjoint modes here [1]. When $\mathcal{R}_\omega$ is Fredholm in the sense of Fredholm structure of radiation operators (Subsection 2.6.4),

$$\dim \ker \mathcal{R}_\omega < \infty \iff \dim \ker \mathcal{R}_\omega^\dagger < \infty, \tag{2.323}$$

and index Eq. (2.302) counts the imbalance [2]. Equation (2.323) does not re-prove Fredholm structure of radiation operators (Subsection 2.6.4); it records the Chapter 2 null-space bookkeeping used in inversion Eq. (2.220) [2]. On exterior domains with compact Ω_s, $\dim \ker \mathcal{R}_\omega$ is typically infinite even when $\mathcal{R}_\omega$ is Fredholm on truncated subspaces Eq. (2.83) [1]. On the orthonormal basis of spectral decomposition of the radiation operator (Subsection 2.6.2), define the *modal null index set*

$$\mathcal{N}_{\mathrm{null}} = \{n \in \mathcal{N}_{\mathrm{rad}} : \lambda_n = 0\}\,, \tag{2.324}$$

with λ_n from Eq. (2.281) [2]. Equation (2.324) is the spectral slice of $\ker \mathcal{R}_\omega$ on declared eigenmodes:

$$\sum_{n \in \mathcal{N}_{\mathrm{null}}} c_n[\mathfrak{D}]\, \mathbf{J}_{\mathrm{imp}}^{(n)} \in \ker \mathcal{R}_\omega \quad \text{when admissible superpositions exist,} \tag{2.325}$$

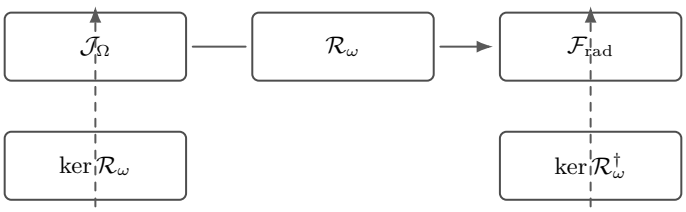


Figure 2.39: Subsection 2.7.1 null-space measurement sequence: source kernel Eq. (2.320), radiative image, and adjoint kernel Eq. (2.321) paired by solvability Eq. (2.303).

using source coefficients Eq. (2.283) and eigenpairs Eq. (2.281) [2]. Equation (2.325) does not exhaust $\ker \mathcal{R}_\omega$ on exterior problems because $\mathcal{N}_{\rm rad}$ need not span all invisible source degrees [1]. modal invertibility on the radiative range (Subsection 2.6.4) already states that coefficients on $\ker \mathcal{R}_\omega$ remain free in Eq. (2.307); Subsection 2.7.1 identifies the modal-labeled portion [2]. Figure interpretation. Read $\mathcal{J}_\Omega$ as the impressed-source Hilbert space and $\mathcal{F}_{\rm rad}$ as the outgoing radiative image. The forward arrow $\mathcal{R}_\omega = \Pi_{\rm rad} \circ \mathcal{S}_\omega$ is the only laboratory-export channel; dashed $\ker \mathcal{R}_\omega$ feeds back into $\mathcal{J}_\Omega$ as invisible source degrees that annihilate far-zone patterns Eq. (2.345). The lower adjoint kernel $\ker \mathcal{R}_\omega^\dagger$ pairs with cokernel directions through reciprocity-based solvability Eq. (2.303): a nonzero cokernel obstruction means no impressed current can realize the target far field, independent of mesh resolution [2]. Figure 2.39 separates kernel and cokernel data before observation composition Eq. (2.305). Let $\mathcal{R}_\omega$ satisfy Eq. (2.320) on admissible $\mathcal{J}_\Omega$ with outgoing radiation condition. Then:

(i) $\ker \mathcal{R}_\omega$ is a linear subspace of $\mathcal{J}_\Omega$ and $\mathcal{R}_\omega[\mathbf{J}_{\rm imp}] = \mathbf{0}$ iff $\mathbf{J}_{\rm imp} \in \ker \mathcal{R}_\omega$.

(ii) Solvability of $\mathcal{R}_\omega[\mathbf{J}_{\rm imp}] = \mathbf{g}$ is equivalent to Eq. (2.303) on declared pairings.

(iii) On the spectral basis of Subsection 2.6.2, $\mathcal{N}_{\rm null}$ Eq. (2.324) labels modes with $P_n \mathcal{R}_\omega = \mathbf{0}$; modal superpositions Eq. (2.325) lie in $\ker \mathcal{R}_\omega$ when admissible.

(iv) Domain class fixes dimensionality:

$$\begin{aligned} \text{bounded cavity + discrete book: } &\dim \ker \mathcal{R}_\omega \text{ finite on truncated } \mathcal{R}_\omega^{(N)}; \\ \text{exterior + compact } \Omega_s : &\dim \ker \mathcal{R}_\omega = \infty \text{ typ. on full } \mathcal{J}_\Omega. \end{aligned} \tag{2.326}$$

. Part (i) follows from linearity Eq. (2.6) and definition Eq. (2.320) [2]. Part (ii) is Fredholm structure of radiation operators (Subsection 2.6.4) and Eq. (2.303), cited without re-proof [1]. Part (iii) uses Eq. (2.306) and $\lambda_n = 0$ in Eq. (2.281) [2]. Part (iv) contrasts discrete cavity spectra Eq. (2.288) with continuous exterior angular content Eq. (2.290) [8]. If $\mathbf{J}_{\rm imp} \in \ker \mathcal{R}_\omega$, then radiative and non-radiative components satisfy $\mathbf{u}_{\rm rad} = \mathbf{0}$ and $\mathbf{u}_{\rm nr} = \mathcal{S}_\omega[\mathbf{J}_{\rm imp}]$ from Eqs. (2.266)–(2.267); near-zone energy in $\mathbf{u}_{\rm nr}$ may be large while qualified export vanishes [2]. Composition with spectral and geometric filters. Null-space membership implies modal annihilation on exporting sectors:

$$\mathbf{J}_{\rm imp} \in \ker \mathcal{R}_\omega \implies a_n(\mathbf{J}_{\rm imp}) = 0 \text{ for all } n \text{ with } \lambda_n \neq 0, \tag{2.327}$$

with far coefficients a_n from Eq. (2.291) [2]. Equation (2.327) expresses Eq. (2.319) from Section 2.7 without re-deriving the modal expansion [2]. Geometric constraints geometric constraints on radiative spectra (Subsection 2.6.5) may force additional $a_n = 0$ for sources outside $\ker \mathcal{R}_\omega$ [1]. Composition law Eq. (2.312) budgets identifiability deficits separately [2]. Fredholm index Eq. (2.302) is unchanged under compact perturbations Eq. (2.301) and under Tikhonov regularization Eq. (2.224) on the range side [2]. Regularization stabilizes Eq. (2.307) for small $|\lambda_n|$ but

leaves $\ker \mathcal{R}_\omega$ unchanged: invisible source components cannot be recovered from far-zone data at any regularization level [2]. Compression truncation Eq. (2.84) approximates $\operatorname{ran} \mathcal{R}_\omega$, not $\ker \mathcal{R}_\omega$ [1]. Null-space structure of radiation operators (Subsection 2.7.1) assumes linear media at fixed ω, outgoing radiation condition, and admissible $\mathcal{J}_\Omega$ Eq. (2.9) [2]. Loss modifies λ_n and may merge modal null sectors without promoting evanescent content to export through Eq. (2.139) [2]. Nonlinear or time-varying sources require the causal chain of Subsection 2.2.3; harmonic null spaces here are fixed-frequency slices [1]. regime qualification Eq. (2.271) remains mandatory before interpreting near-zone reactive dominance as null-space membership [2]. Subsection 2.7.1 fixed operator null-space structure through Eq. (2.320), adjoint and cokernel pairings, and the modal null sector Eqs. (2.324)–(2.325) [2]. Constructive non-radiating current families that realize the kernel in the laboratory develop next.

2.7.2 Non-radiating current configurations

The section introduction displayed balanced loops, counter-directed pairs, and below-cutoff guides as laboratory signatures of $\ker \mathcal{R}_\omega$. Subsection 2.7.1 gives the definitive constructive account: admissible $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ with $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0}$, developing Eq. Eq. (2.316) [2]. Radiation integrals, Huygens surface forms Eqs. (2.236)–(2.238), and equivalent-source ambiguity Eq. (2.227) from Subsections 2.5.1 and 2.4.4 are presupposed; dyadic kernel tabulations from Subsection 2.3.1 are not repeated [2]. Evanescent overlap and measurement nullity belong to Subsection 2.7.2; worked example examples appear in Subsection 2.10.2 by reference [1]. Assumptions. Fix $\omega > 0$, compact support Ω_s, admissible impressed data Eq. (2.9), charge continuity Eq. (1.8), outgoing radiation condition Eqs. (2.203)–(2.213), and null space Eq. (2.320) [2]. All configurations below are required to satisfy $\mathcal{D}_J[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] = 1$ and $\nabla \cdot \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} - \mathrm{j}\omega\rho = 0^{(\mathrm{NR})}$ with $\rho^{(\mathrm{NR})}$ compatible with boundary class Eq. (1.20) [2]. regime qualification Eq. (2.271) governs whether near-zone reactive content may be reported as transport data [1]. Non-radiating current (locked definition). A *non-radiating current configuration* is an admissible impressed current

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega, \qquad \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0} \text{ in general,} \tag{2.328}$$

with radiative component $\mathbf{u}_{\mathrm{rad}} = \mathbf{0}$ and non-radiative component $\mathbf{u}_{\mathrm{nr}} = \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ from Eqs. (2.266)–(2.267) [2]. Equation (2.328) is the Subsection 2.7.1 development of Eq. (2.316) and expresses Eq. (2.8) with explicit $\mathcal{D}_J = 1$: far-zone export vanishes through Eq. (2.229), while near-zone fields may remain substantial kernel image on field space (Subsection 2.7.1) [2]. Linearity Eq. (2.6) implies admissible superpositions of non-radiating configurations remain non-radiating when charge continuity is preserved [2]. Maxwell-closed constraints. Non-radiating configurations are not arbitrary curl fields. Impressed data must satisfy

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in L^2(\Omega_s)^3, \quad \nabla \cdot \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} - \mathrm{j}\omega\rho = 0^{(\mathrm{NR})}, \quad \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \text{ BC-compatible on } \Gamma, \tag{2.329}$$

as in Eq. (2.9) [2]. Equation (2.329) excludes discontinuous $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ that violate Eq. (1.8) before Maxwell field specification is attempted [2]. Surface-equivalent realizations on Γ_{eq} inherit trace class constraints from Subsection 2.1.1 [1]. Membership in $\ker \mathcal{R}_\omega$ is an *export* statement; Eq. (2.329) is a *source* statement—both are required [2]. Let $\mathcal{C} \subset \Omega_s$ be a closed admissible circulation path enclosing negligible magnetic flux linkage at ω, with oppositely directed arms $\mathcal{C}_+$ and $\mathcal{C}_-$ carrying equal magnitude currents. Physically, picture a wire loop in the xy-plane with counter-directed segments along $\pm\hat{\mathbf{x}}$: each arm radiates a leading $1/r$ transverse field, but the net electric dipole

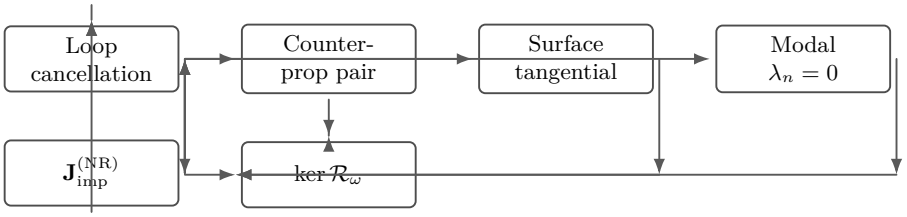


Figure 2.40: Subsection 2.7.1 canonical non-radiating families: loop cancellation Eq. (2.330), counter-propagating pair Eq. (2.331), surface tangential null Eq. (2.332), and modal eigenrealization Eq. (2.333), all mapping to $\ker \mathcal{R}_\omega$.

moment vanishes when the circulation closes. When far-zone transverse contributions from the two arms cancel under outgoing propagation,

$$\int_{\mathcal{C}_+} \mathbf{J}_{\mathrm{loop}} \cdot d\boldsymbol{\ell} \int_{\mathcal{C}_-} \mathbf{J}_{\mathrm{loop}} \cdot d\boldsymbol{\ell} = 0 \implies \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}] = \mathbf{0}, \tag{2.330}$$

for the associated volume current $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$ obtained by compact support regularization on $\mathcal{C}$ [2]. Equation (2.330) is the dipole-moment cancellation law at the source level: vanishing leading $1/r$ transverse term in Eq. (2.247) when the net moment vanishes [8]. Near-zone reactive fields from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}]$ need not vanish; Eq. (2.271) still applies [2]. Two compact sources $\mathbf{J}_{\mathrm{imp}}^{(+)}$, $\mathbf{J}_{\mathrm{imp}}^{(-)}$ with supports $\Omega_\pm$ separated by distance $d \ll \lambda_0$ can form a non-radiating pair when their outgoing far patterns interfere destructively. A concrete arrangement is a pair of equal-strength electric dipoles along $\hat{\mathbf{z}}$, separated by $d = \lambda_0/20$, driven in opposite phase so the leading $1/r$ terms cancel on S^2 while reactive overlap remains in the gap:

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR,cp})} = \mathbf{J}_{\mathrm{imp}}^{(+)} + \mathbf{J}_{\mathrm{imp}}^{(-)}, \qquad \mathbf{E}_{0,+}(\theta,\phi) + \mathbf{E}_{0,-}(\theta,\phi) = \mathbf{0} \text{ on } S^2, \tag{2.331}$$

with $\mathbf{E}_{0,\pm}$ far amplitudes from Eq. (2.247) for each component [2]. Equation (2.331) realizes Eq. (2.328) when phase and polarization of $\mathbf{J}_{\mathrm{imp}}^{(\pm)}$ are tuned so $\Pi_{\mathrm{rad}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\pm)}]$ cancel on the outgoing branch [2]. The pair may store substantial reactive energy in the overlap region through Eq. (2.269) while $\mathcal{P}_{\mathrm{rad}} = 0$ on qualified shells [1]. On a closed equivalent surface Γ_{eq} enclosing radiating sources, tangential non-radiating distributions satisfy Eq. (2.227):

$$\mathbf{J}_{s,\mathrm{eq}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega\big|_{\Gamma_{\mathrm{eq}}}, \qquad \mathbf{J}_{s,\mathrm{eq}}^{(\mathrm{NR})} \cdot \hat{\mathbf{n}} = 0 \text{ on } \Gamma_{\mathrm{eq}}, \tag{2.332}$$

with $\hat{\mathbf{n}}$ the outward normal [2]. Equation (2.332) is the Subsection 2.7.1 constructive realization of Huygens equivalence ambiguity: exterior fields on $\mathcal{F}_{\mathrm{SF}}$ are fixed by exterior radiative uniqueness (Subsection 2.4.3), but equivalent surface currents are unique only up to additions in Eq. (2.332) [2]. Integrals Eqs. (2.236)–(2.238) realize the same $\mathcal{S}_\omega$ image for equivalent and non-radiating tangential perturbations [1]. When radiation eigenpairs with $\lambda_n = 0$ are declared in Eq. (2.281),

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR,mod})} = \sum_{n \in \mathcal{N}_{\mathrm{null}}} \alpha_n \, \mathbf{J}_{\mathrm{imp}}^{(n)}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR,mod})}] = \mathbf{0}, \tag{2.333}$$

for coefficients $\alpha_n \in \mathbb{C}$ such that the superposition satisfies Eq. (2.329) [2]. Equation (2.333) specializes Eq. (2.325) from Subsection 2.7.1 to constructive sources: each $\mathbf{J}_{\mathrm{imp}}^{(n)}$ with $\lambda_n = 0$ is a modal non-radiating generator when admissible [2]. On exterior domains $\mathcal{N}_{\mathrm{null}}$ need not span all of $\ker \mathcal{R}_\omega$ Eq. (2.326) [1]. The four upper nodes are constructive *families*, not interchangeable templates. Loop cancellation annihilates the leading $1/r$ transverse moment; counter-propagating pairs cancel

on S^2 through phase interference; surface tangential nulls exploit Huygens equivalence ambiguity; modal $\lambda_n = 0$ generators realize the spectral slice Eq. (2.325). All four converge on $\ker \mathcal{R}_\omega$, yet each leaves distinct reactive near fields $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] \neq \mathbf{0}$—the schematic warns that near-zone scans cannot distinguish families once far patterns vanish [2]. Figure 2.40 orders constructive families before evanescent and measurement extensions in Subsection 2.7.2. Equivalent-current solvers on Γ_{eq} seek $\mathbf{J}_{\mathrm{eq}}$ matching near or far data [2]. Canonical non-radiating configurations (Subsection 2.7.1) and Eq. (2.332) show why such inversions are ill-posed even when forward fields are unique: any solution may be shifted by tangential $\mathbf{J}_{s,\mathrm{eq}}^{(\mathrm{NR})}$ without changing $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Tikhonov regularization Eq. (2.224) selects a particular representative but does not eliminate the equivalence class [1]. Subsection 2.7.2 separates this source nullity from finite-rank observation nullity Eq. (2.338) [2]. Subsection 2.7.1 assumes linear Maxwell theory at fixed ω, outgoing radiation condition, and admissible sources Eq. (2.329) [2]. Lossy media modify loop and pair conditions through complex $k(\mathbf{r})$ without promoting evanescent storage to export Eq. (2.139) [2]. Canonical families generate *explicit* members of $\ker \mathcal{R}_\omega$; they do not exhaust $\dim \ker \mathcal{R}_\omega = \infty$ on exterior domains Eq. (2.326) [1]. regime qualification Eq. (2.140) remains mandatory: large near-zone $|\mathbf{E}|$ from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ is not export without Π_{rad} qualification [2].

2.7.3 Evanescent overlap and non-reconstructability

Subsections 2.7.1–2.7.1 fixed $\ker \mathcal{R}_\omega$ and canonical non-radiating impressed currents Eq. (2.328). Evanescent sheets are a distinct field-side deficit: reactive storage that never passes Π_{rad} even when the source radiates. Subsection 2.7.2 supplies the definitive account of evanescent subspaces, their overlap with non-radiating source content, and reconstructability limits under finite observation rank—developing Eq. Eq. (2.317) from Section 2.7 [2]. Propagation projectors Eqs. (2.134)–(2.139) from Subsection 2.2.3, non-propagating split Eq. (2.265) from Subsection 2.5.4, measurement null-space extension measurement null-space extension (Subsection 2.7.1) from Subsection 2.7.1, and rank cap Eq. (2.221) from Subsection 2.4.4 are presupposed; projector algebra and inverse-source templates are not re-proved [2]. Angular-spectrum consequences belong to Subsection 2.7.4; plane-wave filters to Section 2.9 [1]. Assumptions. Fix $\omega > 0$, admissible $\mathcal{J}_\Omega$ Eq. (2.9), outgoing radiation condition Eqs. (2.203)–(2.213), field pairs in $\mathcal{F}_{\mathrm{SF}} \subset \mathcal{F}_\Omega$, and spectral completion supporting Eqs. (2.134)–(2.135) [2]. Observation map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ from Eq. (2.219) is finite rank with $\mathrm{rank}(\mathcal{O}_{\mathrm{meas}}) = N$ Eq. (2.80) [2]. regime qualification Eq. (2.140) governs transport reports on any recovered radiative component [1]. On the spectral completion $\widehat{\mathcal{F}}_\Omega$, the *evanescent subspace* is

$$\mathcal{F}_{\mathrm{ev}} = \left\{ \mathbf{u} \in \widehat{\mathcal{F}}_\Omega : \Pi_{\mathrm{ev}} \mathbf{u} = \mathbf{u} \right\}, \tag{2.334}$$

with Π_{ev} from Eq. (2.134) [2]. Equation (2.334) is the Subsection 2.7.2 field-side label for modes with imaginary longitudinal wavenumber Eq. (2.131) and decay rate Eq. (2.132) [2]. The companion *non-propagating* subspace combines evanescent and below-cutoff content through Eq. (2.265) [1]. Evanescent sheets carry reactive structure bound to apertures; they decay exponentially with scan height and satisfy Eq. (2.139) without re-derivation here [2]. For admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, the evanescent projector Π_{ev} selects field content whose lateral wavenumber exceeds the vacuum propagation circle $k_t > k_0$: such modes decay exponentially with distance from the source and carry reactive storage rather than qualified export. Define the evanescent field content

$$\mathbf{u}_{\mathrm{ev}} = \Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \qquad \|\mathbf{u}_{\mathrm{ev}}\|_{\mathcal{E}} = \|\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\|_{\mathcal{E}}, \tag{2.335}$$

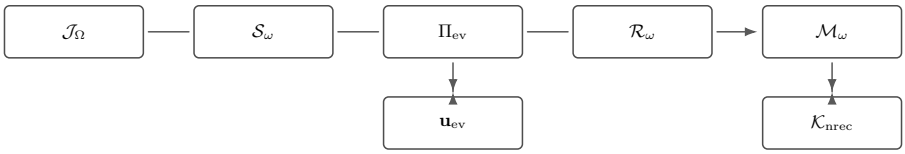


Figure 2.41: Subsection 2.7.2 measurement sequence: evanescent overlap Eq. (2.335) never exports through Eq. (2.336); finite-rank observation maps $\mathcal{J}_\Omega$ to $\mathcal{K}_{\mathrm{nrec}}(N)$ Eq. (2.339).

with energy norm Eq. (2.48) [2]. Equation (2.335) measures reactive storage excited by $\mathbf{J}_{\mathrm{imp}}$ that never passes through Π_{rad} because

$$\Pi_{\mathrm{rad}}\,\mathbf{u}_{\mathrm{ev}} = \mathbf{0}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ depends only on } \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ after outgoing selection,} \tag{2.336}$$

by Eqs. (2.139) and (2.138) [2]. Equation (2.336) upgrades the second line of Eq. (2.317): evanescent overlap is field-side non-export, distinct from source-side $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ Eq. (2.320) [1]. A source may satisfy $\|\mathbf{u}_{\mathrm{ev}}\| \gg \|\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\|$ while still radiating [2]. Eq. (2.317) separates export annihilation from finite-rank readout loss. Source nullity $\ker \mathcal{R}_\omega$ is strict export failure; measurement nullity $\ker \mathcal{M}_\omega$ additionally loses directions that radiate but the aperture cannot resolve [2]:

$$\underbrace{\ker \mathcal{R}_\omega}_{\text{no export}} \subseteq \underbrace{\ker \mathcal{M}_\omega}_{\text{no finite readout}}, \qquad \ker \mathcal{M}_\omega \supsetneq \ker \mathcal{R}_\omega \text{ when } \ker \mathcal{O}_{\mathrm{meas}}\big|_{\operatorname{ran} \mathcal{R}_\omega} \neq \{\mathbf{0}\}, \tag{2.337}$$

with strict inclusion typical for finite apertures Eq. (2.81)

$$\ker \mathcal{M}_\omega = \ker \mathcal{R}_\omega \;\cup\; \mathcal{R}_\omega^{-1}\big(\ker \mathcal{O}_{\mathrm{meas}}\big|_{\operatorname{ran} \mathcal{R}_\omega}\big), \tag{2.338}$$

[2]. Equation (2.337) expresses measurement null-space extension (Subsection 2.7.1) and Eq. (2.338) without re-proof: evanescent storage contributes to the first factor only when it annihilates export; finite rank contributes to the second factor even when export is nonzero [2]. Tikhonov reconstruction Eq. (2.224) stabilizes ill-conditioned inversions on $\operatorname{ran} \mathcal{M}_\omega^\dagger$ but does not shrink $\ker \mathcal{M}_\omega$ [1]. Define the *non-reconstructable set* at observation rank N:

$$\mathcal{K}_{\mathrm{nrec}}(N) = \{\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega : \mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}\} = \ker \mathcal{M}_\omega, \tag{2.339}$$

equivalently the kernel of Eq. (2.220) for generic $\mathbf{y}$ [2]. Equation (2.339) is the Subsection 2.7.2 constructive account of reconstructability label: sources in $\mathcal{K}_{\mathrm{nrec}}(N)$ are invisible to the declared measurement bank regardless of near-zone $|\mathbf{E}|$ [2]. Rank bound

$$\dim \operatorname{ran} \mathcal{M}_\omega \le N < \infty \implies \dim \mathcal{K}_{\mathrm{nrec}}(N) = \infty \text{ on typical open domains,} \tag{2.340}$$

follows from Eq. (2.221) and infinite-dimensional $\mathcal{J}_\Omega$ [2]. Geometric sector constraints Eq. (2.9.3) and composition law Eq. (2.312) further restrict recoverable coordinates independently of Eq. (2.339) [2]. The measurement sequence orders radiation selection, evanescent overlap, and finite-rank observation. Π_{ev} content may dominate $|\mathbf{E}|$ on inner shells yet satisfies Eq. (2.336) on qualified flux meters. $\mathcal{O}_{\mathrm{meas}}$ with rank N collapses recoverable modes into $\mathcal{K}_{\mathrm{nrec}}(N)$: export can exist while the aperture cannot resolve it. Dashed branches mark observation deficits distinct from source nullity $\ker \mathcal{R}_\omega$ [2]. Let $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ be admissible with evanescent content Eq. (2.335) and observation map $\mathcal{M}_\omega$ of rank N. Then:

(i) $\Pi_{\mathrm{rad}}\mathbf{u}_{\mathrm{ev}} = \mathbf{0}$ and $\mathbf{u}_{\mathrm{ev}}$ carries no qualified far-zone export Eq. (2.336).

(ii) $\ker \mathcal{R}_\omega \subseteq \ker \mathcal{M}_\omega = \mathcal{K}_{\mathrm{nrec}}(N)$.

(iii) Near-field data on a finite enclosing surface reduce but do not in general eliminate $\mathcal{K}_{\mathrm{nrec}}(N)$: if $\mathcal{M}_{\mathrm{near}}$ samples $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on Γ_{enc}, then

$$\ker \mathcal{M}_{\mathrm{near}} \subsetneq \ker \mathcal{M}_\omega \text{ when } \Gamma_{\mathrm{enc}} \text{ resolves more modes than far-zone aperture,} \tag{2.341}$$

while $\ker \mathcal{R}_\omega$ is unchanged [2].

(iv) Reconstructability budget:

$$\mathcal{N}_{\mathrm{reportable}} \subseteq \mathcal{N}_{\mathrm{acc}} \cap \{n : \text{ resolvable by } \mathcal{M}_\omega\} \cap \{n : \lambda_n \neq 0\}, \tag{2.342}$$

with $\mathcal{N}_{\mathrm{acc}}$ from Eq. (2.9.3) and modal indices from Subsection 2.6.2 [2].

Part (i) is Eq. (2.139) [2]. Part (ii) follows from measurement null-space extension (Subsection 2.7.1) and definition Eq. (2.339) [1]. Part (iii) uses near- versus far-field sampling logic from Subsection 2.4.4: $\mathcal{S}_\omega$ is single-valued on outgoing fields while $\mathcal{R}_\omega$ remains non-injective [2]. Part (iv) composes Eq. (2.312) with rank cap Eq. (2.221) [2]. For all admissible $\mathbf{J}_{\mathrm{imp}}$,

$$\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \Pi_{\mathrm{rad}}\, \Pi_{\mathrm{prop}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}], \tag{2.343}$$

and $\Pi_{\mathrm{rad}}\Pi_{\mathrm{ev}} = \mathbf{0}$ on $\mathcal{F}_\Omega$ [2]. Equation (2.343) implies $\mathbf{u}_{\mathrm{ev}}$ from Eq. (2.335) does not alter $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; adding evanescent excitation to an otherwise radiating source changes $\mathbf{u}_{\mathrm{nr}}$ and local power Eq. (2.269) without changing far patterns when export is unchanged [2]. regime qualification Eq. (2.271) still forbids reporting $\mathbf{u}_{\mathrm{ev}}$ as transport [1]. Substitute Eqs. (2.139) and (2.138) into $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ [2]. Near-zone energy follows from Eq. (2.265) [1]. Let Γ_{enc} be a closed surface enclosing Ω_s and let $\mathcal{M}_{\mathrm{near}}$ sample tangential $\mathbf{E}$, $\mathbf{H}$ on Γ_{enc}. Then $\ker \mathcal{M}_{\mathrm{near}} \subseteq \ker \mathcal{M}_\omega$ when far-zone observations are a subset of near-field functionals, but

$$\ker \mathcal{R}_\omega \subseteq \ker \mathcal{M}_{\mathrm{near}} \subseteq \ker \mathcal{M}_\omega \tag{2.344}$$

with strict inclusions possible at each step [2]. Equation (2.344) expresses Subsection 2.4.4 near- versus far-field data without re-opening uniqueness exterior radiative uniqueness (Subsection 2.4.3) [2]. Outgoing radiation condition remains mandatory: identical near traces on a finite surface do not imply identical far zones without exterior selection [1]. If $\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$, then far functionals vanish; near samples may still vanish on Γ_{enc} for non-radiating additions in $\ker \mathcal{R}_\omega$ [2]. Additional near resolution shrinks the observation null space on $\operatorname{ran} \mathcal{R}_\omega$ but leaves $\ker \mathcal{R}_\omega$ unchanged Eq. (2.320) [1]. When Eq. (2.141) fails, modal content lies in $\Pi_{\mathrm{non}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Such storage satisfies Eq. (2.336) and may dominate Eq. (2.269) on sub-wavelength shells while $\mathcal{P}_{\mathrm{rad}} = 0$ [2]. Applying far-zone pattern formulas to below-cutoff modes violates Eq. (2.140) as in the TE_{10} illustration referenced from Section 1.1.2 [1]. Subsection 2.7.2 records operator consequences; guide nomenclature is not re-tabulated [2]. Non-radiating configurations Eq. (2.328) may carry large $\|\mathbf{u}_{\mathrm{ev}}\|$ when $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ is evanescent-dominated [2]. Conversely, large $\|\mathbf{u}_{\mathrm{ev}}\|$ does not imply $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$: a radiating source may store substantial evanescent energy near Ω_s while exporting through Π_{prop} [2]. Subsections 2.7.1 and 2.7.2 address orthogonal deficits—source nullity versus reactive storage versus finite-rank readout [1]. Subsection 2.7.2 assumes linear Maxwell theory at fixed ω, outgoing radiation condition, and declared finite-rank $\mathcal{O}_{\mathrm{meas}}$ [2]. Loss modifies evanescent decay Eq. (2.132) without promoting Π_{ev} content to Π_{rad} export [2]. Dispersive or broadband superposition requires Subsection 2.2.3; harmonic evanescent labels here are fixed-frequency slices [1]. Subsection 2.7.4 translates non-reconstructability to continuous angular spectra Eq. (2.290) [2].

2.7.4 Angular-spectrum consequences of null spaces

Far-zone angular amplitudes $a(\widehat{\mathbf{k}})$ are what compact ranges and synthesis codes ultimately report. Subsections 2.7.1–2.7.2 fixed $\ker \mathcal{R}_\omega$, canonical non-radiating impressed currents, and evanescent non-reconstructability; Subsection 2.7.4 records how those source-side deficits propagate to the continuous angular measure Eq. (2.290) without re-opening plane-wave operators from Section 2.9 [2].

Assumptions. Fix $\omega > 0$, outgoing radiation condition, admissible $\mathcal{J}_\Omega$, and far amplitude $\mathbf{E}_0(\theta,\phi)$ from Eq. (2.247) when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. Continuous angular coefficients $a_\sigma(\widehat{\mathbf{k}})$ are defined through the rigged limit of Eq. (2.290) compatible with modal amplitudes Eq. (2.283) [2]. Finite-rank observation $\mathcal{M}_\omega$ and non-reconstructable set $\mathcal{K}_{\mathrm{nrec}}(N)$ from Eq. (2.339) are presupposed [1]. regime qualification Eq. (2.140) governs transport claims on any retained angular content [2]. Let $a(\widehat{\mathbf{k}})$ denote the scalar angular spectrum obtained from Eq. (2.290) on the outgoing transverse sheet [2]. For $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$,

$$a(\widehat{\mathbf{k}}) = 0 \quad \text{for all } \widehat{\mathbf{k}} \in S^2 \text{ on the exporting sheet,} \tag{2.345}$$

equivalently $\mathbf{E}_0(\theta,\phi) = \mathbf{0}$ in Eq. (2.291) [2]. Equation (2.345) is the Subsection 2.7.4 development of the first clause of Eq. Eq. (2.318): null-space membership annihilates the entire far-zone angular measure, not merely selected harmonics [1]. Near-zone reactive content in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ may remain nonzero without contributing to $a(\widehat{\mathbf{k}})$ Eq. (2.336) [2]. Suppressed and accessible angular sheets. Define the *suppressed angular sheet*

$$\mathcal{K}_{\mathrm{sup}} = S^2 \setminus \mathcal{K}_{\mathrm{acc}}, \qquad \mathcal{K}_{\mathrm{acc}} \text{ from Eq. (2.9.3)}, \tag{2.346}$$

as directions annihilated by geometric, support, or parity constraints before observation [2]. Equation (2.346) expresses Eq. (2.318): $a(\widehat{\mathbf{k}}) = 0$ on $\mathcal{K}_{\mathrm{sup}}$ does not imply $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$, because geometric forbidding Eq. (2.308) and radiation nullity Eq. (2.345) are distinct mechanisms [2]. Accessible sheet $\mathcal{K}_{\mathrm{acc}}$ intersects reportable modal sector Eq. (2.342) [1].

$$\mathcal{R}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big] = \mathcal{R}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}\Big], \qquad \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega, \tag{2.347}$$

$$\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega \implies a_n\Big(\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big) = 0, \quad \lambda_n = 0 \text{ on exporting modes,} \tag{2.348}$$

$$\Big\|\mathcal{S}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big]\Big\|_{\mathcal{E}} \neq \Big\|\mathcal{S}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}\Big]\Big\|_{\mathcal{E}} \text{ while } \mathcal{R}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}\Big] = \mathcal{R}_\omega\Big[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}\Big], \tag{2.349}$$

For radiating $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ and non-radiating $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$,

$$a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}) = a(\widehat{\mathbf{k}}; \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}), \tag{2.350}$$

on the outgoing transverse sheet, restating superposition law Eq. (2.347) in angular coordinates [2]. Equation (2.350) is the continuous form of modal invisibility Eq. (2.348) and explains why inverse angular-spectrum solvers cannot detect $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ from far data alone [2]. Canonical families canonical non-radiating configurations (Subsection 2.7.1) therefore induce *invisible* angular perturbations at fixed $a(\widehat{\mathbf{k}})$ [1]. Evanescent decoupling from angular spectra. Evanescent field content does not enter $a(\widehat{\mathbf{k}})$:

$$a(\widehat{\mathbf{k}}) \text{ depends only on } \Pi_{\mathrm{prop}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ after } \Pi_{\mathrm{rad}}, \qquad \Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \not\mapsto a(\widehat{\mathbf{k}}), \tag{2.351}$$

by Eqs. (2.336) and (2.343) [2]. Equation (2.351) specializes the second line of Eq. (2.318): reactive storage may be large in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ while $a(\widehat{\mathbf{k}})$ reports only qualified export [2]. regime qualification

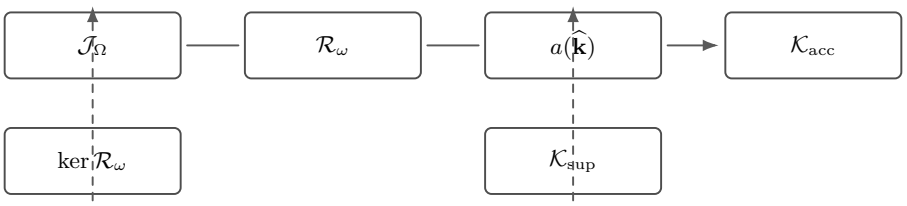


Figure 2.42: Subsection 2.7.4 angular measurement sequence: $\ker \mathcal{R}_\omega$ annihilates $a(\widehat{\mathbf{k}})$ Eq. (2.345); suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ Eq. (2.346) removes directions before accessible sector $\mathcal{K}_{\mathrm{acc}}$.

Eq. (2.271) forbids interpreting $\mathbf{u}_{\mathrm{ev}}$ as angular-transport data [1]. Observation rank on angular spectra. Finite aperture collapses recoverable angular content:

$$\{\, a(\widehat{\mathbf{k}}) \,\text{recoverable from}\, \mathcal{M}_\omega \,\} \subseteq \{\, a(\widehat{\mathbf{k}}) \,\text{supported on}\, \mathcal{K}_{\mathrm{acc}} \cap \operatorname{ran} \mathcal{R}_\omega \,\}, \tag{2.352}$$

with strict inclusion when $\ker \mathcal{O}_{\mathrm{meas}}\big|_{\operatorname{ran} \mathcal{R}_\omega} \neq \{\mathbf{0}\}$ Eq. (2.81) [2]. Equation (2.352) links angular spectra to $\mathcal{K}_{\mathrm{nrec}}(N)$: sources invisible to $\mathcal{M}_\omega$ need not have $a(\widehat{\mathbf{k}}) = \mathbf{0}$ pointwise [2]. Rank cap Eq. (2.221) limits resolved harmonics independently of Eq. (2.345) [1]. Figure interpretation. Source excitation $\mathcal{J}_\Omega$ maps through $\mathcal{R}_\omega$ to angular amplitude $a(\widehat{\mathbf{k}})$; dashed $\ker \mathcal{R}_\omega$ forces $a = 0$ on the entire exporting sheet Eq. (2.345). The suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ removes directions by geometry before $\mathcal{E}^{(N)}_{\mathrm{acc}}$ is tested: $a = 0$ on $\mathcal{K}_{\mathrm{sup}}$ does not place $\mathbf{J}_{\mathrm{imp}}$ in $\ker \mathcal{R}_\omega$ [2]. Only the intersection with $\mathcal{K}_{\mathrm{acc}}$ carries laboratory-reportable angular content [3]. Let $a(\widehat{\mathbf{k}})$ be the outgoing angular spectrum from Eq. (2.290) and let $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ be admissible. Then:

(i) $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega \Rightarrow a(\widehat{\mathbf{k}}) = 0$ on the exporting sheet Eq. (2.345).

(ii) Superposition with $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ leaves $a(\widehat{\mathbf{k}})$ unchanged Eq. (2.350).

(iii) Evanescent content satisfies Eq. (2.351) and does not alter $a(\widehat{\mathbf{k}})$ when export is unchanged.

(iv) Full angular budget:

$$\operatorname{support}(a(\widehat{\mathbf{k}})) \subseteq \mathcal{K}_{\mathrm{acc}} \cap \{\widehat{\mathbf{k}} :\ \text{resolvable by } \mathcal{M}_\omega\} \cap \{\widehat{\mathbf{k}} :\ \lambda_n \neq 0 \text{ modal labels}\}, \tag{2.353}$$

composing geometric, observation, and Fredholm constraints from Eqs. (2.312)–(2.342) [2].

Part (i) follows from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}$ and far-zone reduction the far-field approximation of source integrals (Subsection 2.5.2) [2]. Part (ii) uses canonical non-radiating configurations (Subsection 2.7.1) and linearity of angular extraction [1]. Part (iii) is evanescent content never radiates (Subsection 2.7.2) in angular coordinates [2]. Part (iv) composes Eq. (2.9.3), angular probe rank on the far shell (Subsection 2.6.5), and Eq. (2.352) [2]. Modal coefficients in Eq. (2.291) and continuous $a(\widehat{\mathbf{k}})$ satisfy

$$\lambda_n\, c_n[\mathfrak{D}] = 0 \text{ for all exporting } n \iff a(\widehat{\mathbf{k}}) = 0 \text{ on } S^2 \tag{2.354}$$

when the modal basis is complete on $\operatorname{ran} \mathcal{R}_\omega$ in the rigged sense [2]. Equation (2.354) connects Eq. (2.327) to Eq. (2.345) without re-proving completeness in Chapter 3 (*Angular Momentum Operators and SO(3)*) [2]. Truncated modal books misstate Eq. (2.354) on exterior domains [8]. far-pattern coefficients from spectral modes (Subsection 2.6.2) and Eq. (2.290) identify discrete and continuous representations on outgoing transverse modes [2]. Null modal coefficients imply vanishing $\mathbf{E}_0$; conversely $\mathbf{E}_0 = \mathbf{0}$ forces all exporting modal weights to zero [1]. Given target

angular spectrum $a_{\mathrm{tgt}}(\widehat{\mathbf{k}})$, the source inversion problem seeks $\mathbf{J}_{\mathrm{imp}}$ with $a(\widehat{\mathbf{k}};\mathbf{J}_{\mathrm{imp}}) = a_{\mathrm{tgt}}(\widehat{\mathbf{k}})$. Any solution is unique only up to additions in $\ker\mathcal{R}_\omega$:

$$\mathbf{J}^{(1)}_{\mathrm{imp}} - \mathbf{J}^{(2)}_{\mathrm{imp}} \in \ker\mathcal{R}_\omega \quad \text{whenever} \quad a(\widehat{\mathbf{k}};\mathbf{J}^{(1)}_{\mathrm{imp}}) = a(\widehat{\mathbf{k}};\mathbf{J}^{(2)}_{\mathrm{imp}}), \tag{2.355}$$

and up to $\mathcal{K}_{\mathrm{nrec}}(N)$ at finite observation rank [2]. Equation (2.355) is the angular restatement of Eq. (2.220) ill-posedness; Tikhonov regularization Eq. (2.224) selects a representative without eliminating the quotient [2]. If $a(\widehat{\mathbf{k}};\mathbf{J}^{(1)}_{\mathrm{imp}}) = a(\widehat{\mathbf{k}};\mathbf{J}^{(2)}_{\mathrm{imp}})$, then $\mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}} - \mathbf{J}^{(2)}_{\mathrm{imp}}] = \mathbf{0}$ on $\mathcal{F}_{\mathrm{rad}}$ [2]. Finite $\mathcal{M}_\omega$ adds measurement null directions measurement null-space extension (Subsection 2.7.1) [1]. The nullity–rank-N recovery–realizability triad introduced in Section 2.7 applies unchanged in angular coordinates: far data cannot detect additions in $\ker\mathcal{R}_\omega$ Eq. (2.350); accessible directions survive geometric and regime filters even when export is nonzero; $\mathcal{S}_{\mathrm{real}}$ of a target $a_{\mathrm{target}}(\widehat{\mathbf{k}})$ is decided in **??** [3]. Evanescent overlap Eq. (2.335) changes near fields without altering a when export is fixed [2]. Forward link to angular-spectrum action. Section 2.9 develops plane-wave operators that act on $a(\widehat{\mathbf{k}})$; Subsection 2.9.3 refines far-field angular $\mathcal{E}^{(N)}_{\mathrm{acc}}$ on $\mathcal{K}_{\mathrm{acc}}$ [2]. Subsection 2.7.4 fixes the *source-side* constraints those operators inherit: Eqs. (2.345)–(2.353) must be composed with regime qualification Eq. (2.140) before transport claims [2]. Reciprocity identities on coupled angular modes belong to Section 2.8 [1]. Null-space consequences for angular spectra (Subsection 2.7.4) assumes linear Maxwell theory at fixed ω, outgoing radiation condition, and rigged-sense convergence of Eq. (2.290) [2]. Loss shifts phase of $a(\widehat{\mathbf{k}})$ without converting evanescent storage to export [2]. Vector harmonic explicit forms of $\mathbf{f}_n(\theta,\phi)$ belong to **??**; Subsection 2.7.4 records operator consequences only [1]. As a worked example, place a copper wire loop of radius $b = \lambda_0/8$ in vacuum and excite it with a uniform phasor current at ω. The loop realization $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}$ from Eq. (2.330) satisfies Eq. (2.345)—a wattmeter on a sphere at $R = 10\lambda_0$ registers zero qualified export—while $|\mathbf{E}|$ may exceed the free-space impedance value by an order of magnitude at $r = b$ [2]. The near-field probe therefore sees rich reactive structure that the angular spectrum cannot detect [1]. Angular-spectrum probes of $\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}]$ without Π_{rad} misstate null-space membership [1]. As a worked example, when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \neq \mathbf{0}$ but the aperture resolves only N harmonics, Eq. (2.352) leaves unresolved $\widehat{\mathbf{k}}$ in the measurement null space even when $a(\widehat{\mathbf{k}})$ is nonzero [2]. This is distinct from Eq. (2.350) [2]. Section 2.7 closes with the null-space chain begun in its introduction: operator structure and $\ker\mathcal{R}_\omega$ (Subsection 2.7.1), canonical non-radiating current families (Subsection 2.7.1), evanescent subspaces and non-reconstructability (Subsection 2.7.2), and angular-spectrum consequences (Subsection 2.7.4) with accessible and suppressed sheets Eqs. (2.9.3)–(2.346) (Subsection 2.7.4) [2]. Spectral and geometric filters from Section 2.6 now meet source-side invisibility: far-zone reports use only $a(\widehat{\mathbf{k}})$ on the composed budget Eq. (2.353), while $\ker\mathcal{R}_\omega$, $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and $\mathcal{K}_{\mathrm{nrec}}(N)$ explain deficits that no angular label alone resolves [2]. Section 2.8 adds reciprocity on the same modal coordinates; Section 2.9 and Subsection 2.9.3 develop plane-wave action and far-field angular rank-N recovery on $\mathcal{K}_{\mathrm{acc}}$; **??** supplies explicit harmonic bases [1]. Transport and angular-transport claims must compose with Eq. (2.313), Eq. (2.353), and regime qualification Eq. (2.140), not with raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or unqualified near-zone reactive content [2]. Source null spaces explain what far-zone observation cannot detect; synthesizable—whether accessible angular content is generable by finite $\mathbf{J}_{\mathrm{imp}}$—is decided in **??** alongside source equivalence and effective rank.

2.8 Reciprocity and Radiative Coupling

Null-space structure explains invisible near-field rearrangements; reciprocity explains why passive hardware nevertheless reports the same complex coupling when transmit and receive roles are ex-

changed [2]. On a calibrated vector network analyzer the practitioner excites port A, reads the complex response at port B, then exchanges cables, standards, and probe normalization so that port B drives and port A receives; on reciprocal antennas and reciprocal media the normalized scattering entries S_{21} and S_{12} must agree once outgoing qualification and impressed-current normalization are held fixed [7]. Two-port impedance matrices $\mathbf{Z}$ and scattering matrices $\mathbf{S}$ inherit the same interchange symmetry whenever terminations and reference planes are declared consistently [11]. That bench fact precedes every volume identity below: reciprocity is what the network analyzer already presupposes when a passive feed is tuned by comparing forward and reverse coupling [2].

Section 2.7 closed the null-space chain Eq. (2.313): operator kernels, canonical non-radiating currents, evanescent non-reconstructability, and angular-spectrum consequences (Subsection 2.7.4) [2]. That program classified invisible and partially observable source content on $\mathcal{R}_\omega$ and $\mathcal{M}_\omega$. Section 2.8 addresses exchange symmetry: when transmit and receive maps coincide, how radiative modes couple under bilinear pairings, and which orthogonality relations constrain coupled spectra on $\mathcal{F}_{\mathrm{rad}}$ [2]. Reciprocity ties transmission and reception on passive hardware before angular spectra are filtered; broken reciprocity after probe correction usually traces to inconsistent normalization or missing outgoing tails [7]. Preview swap Eq. (2.155) from Subsection 2.3.1 named kernel interchange; Section 2.8 upgrades reciprocity to operator form on $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and modal coordinates from Sections 2.6–2.7 [2]. Plane-wave angular-spectrum action belongs to Section 2.9; explicit harmonic orthogonality and completeness belong to **??** Subsection 3.4.3; continuous-spectrum normalization belongs to Subsection 3.5.2 by reference to Subsection 2.8.2 [1]. Spectral decomposition in Subsection 2.6.2 and biorthogonality Eq. (2.292) fixed modal coordinates on $\mathcal{F}_{\mathrm{rad}}$; null-space structure in Subsection 2.7.1 showed that many sources share the same radiative image [2]. Reciprocity does not remove $\ker \mathcal{R}_\omega$, but it constrains how coupling coefficients between declared modes behave when impressed sources and test probes are swapped [2]. Treating antenna transmit and receive as unrelated maps violates Eq. (2.155) even when outgoing source-to-field integrals from Subsection 2.5.1 are numerically accurate [1]. Section 2.8 separates Lorentz reciprocity in operator form, modal orthogonality and coupling, and radiation–reception interchange without re-opening Green construction from Section 2.3 [2]. Fix angular frequency $\omega > 0$, impressed sources in $\mathcal{J}_\Omega$, and outgoing field pairs in $\mathcal{F}_{\mathrm{SF}}$ Eq. (2.57). The section develops reciprocity in three stages that mirror the measurement sequence excite–couple–swap:

$$\text{Lorentz reciprocity} \xrightarrow{\text{operator form}} \mathcal{S}_\omega, \mathcal{R}_\omega \xrightarrow{\text{modal}} \text{orthogonality / coupling} \xrightarrow{\text{systems}} \text{radiation \& reception}, \tag{2.356}$$

where the first arrow is Subsection 2.8.1, the second is Subsection 2.8.2, and the third is Subsection 2.8.3 [2]. Equation (2.356) expresses Eq. (2.274) and Eq. (2.313) at the bilinear level: spectral modes from Section 2.6 supply coordinates; null-space quotients from Section 2.7 explain non-uniqueness; reciprocity constrains *symmetric* parts of coupling on $\operatorname{ran} \mathcal{R}_\omega$ [2]. Angular budget Eq. (2.353) and regime qualification Eq. (2.140) remain mandatory before transport claims on coupled modes [1]. Denote by $\mathcal{M}_A$ and $\mathcal{M}_B$ the finite-rank port maps that compose radiation selection with probe functionals on $\mathcal{F}_{\mathrm{rad}}$; the port-swap experiment asserts $\mathcal{M}_B[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}] = \mathcal{M}_A[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$ on reciprocal excitations before any volume contraction is invoked [1]. Maxwell adjoint pairing Eq. (2.26), source dual pairing Eq. (2.56), and radiation map Eq. (2.275) supply the continuous objects that finite port matrices sample; dyadic reciprocity swap Eq. (2.155) and modal biorthogonality Eq. (2.292) specialize the same symmetry on outgoing branches fixed by Eqs. (2.203)–(2.213) [2]. Subsection 2.8.1 upgrades the bench fact to sesquilinear form on $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$; the identity below states the operator identity the VNA experiment already presupposes. Lorentz reciprocity (leading template). Subsection 2.8.1 supplies the definitive operator statement: for admissible source pairs

$(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}})$ and induced fields $(\mathbf{u}_1, \mathbf{u}_2)$,

$$\langle \mathbf{u}_1,\, \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} = \langle \mathbf{u}_2,\, \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} \tag{2.357}$$

on passive linear media without magneto-electric coupling, with pairing induced from Eq. (2.26) on outgoing admissible states [2]. Equation (2.357) packages the port-swap symmetry as a sesquilinear identity on admissible Maxwell pairs: field one tested against source two equals field two tested against source one when media and boundaries belong to the reciprocal class [2]. It develops Eq. Eq. (2.155) from kernel swap to full source–field reciprocity on $\mathcal{S}_\omega$; restrictions on loss, anisotropy, and boundary class are stated in Subsection 2.8.1 [2]. Reciprocity on $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ follows by projection when both sides satisfy regime qualification Eq. (2.271) [1]. On radiation eigenmodes Eq. (2.281), Subsection 2.8.2 develops orthogonality and coupling coefficients:

$$\langle \mathbf{u}_m,\, \mathbf{u}_n\rangle_{\mathcal{U}} = \delta_{mn}, \qquad C_{mn} = \langle \widetilde{\mathbf{u}}_m,\, \mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}]\rangle_{\mathcal{U}}, \tag{2.358}$$

with $\widetilde{\mathbf{u}}_m$ adjoint modes tied to Eq. (2.292) [2]. Equation (2.358) is the Section 2.8 target for Subsection 2.8.2 orthogonality and coupling on $\mathcal{F}_{\mathrm{rad}}$; explicit vector harmonic normalization belongs to **??** [2]. Modes with $\lambda_n = 0$ lie in null sectors Eq. (2.324); reciprocity constraints apply on exporting modes with $\lambda_n \neq 0$ [1]. Radiation and reception (leading template). Subsection 2.8.3 records transmit–receive interchange for coupled systems:

$$\mathcal{M}_{\mathrm{rx}}[\mathbf{J}^{(\mathrm{tx})}_{\mathrm{imp}}] = \mathcal{M}_{\mathrm{tx}}[\mathbf{J}^{(\mathrm{rx})}_{\mathrm{imp}}] \quad \text{under reciprocity-qualified pairings,} \tag{2.359}$$

with $\mathcal{M}_{\mathrm{tx}}$, $\mathcal{M}_{\mathrm{rx}}$ finite-rank compositions of $\mathcal{R}_\omega$ and $\mathcal{O}_{\mathrm{meas}}$ Eq. (2.219) [2]. Equation (2.359) does not eliminate $\ker \mathcal{R}_\omega$ or $\mathcal{K}_{\mathrm{nrec}}(N)$; it constrains *symmetric* coupling on observable coordinates after projection onto $\operatorname{ran} \mathcal{R}_\omega$ [2]. Geometric forbidden sectors Eq. (2.9.3) and angular budget Eq. (2.353) compose independently [1]. Radiative coupling bilinear form (leading template). Coupling between two radiating subsystems is tracked by

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \langle \mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}],\, \mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} \quad \text{on } \mathcal{F}_{\mathrm{rad}}, \tag{2.360}$$

with reciprocity implying $\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \mathfrak{C}_\omega(\mathbf{J}^{(2)}_{\mathrm{imp}}, \mathbf{J}^{(1)}_{\mathrm{imp}})$ under Eq. (2.357) and passive media [2]. Equation (2.369) measures export overlap, not raw near-zone reactive overlap $\langle \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}],\, \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle$, which may include Π_{ev} content annihilated by Eq. (2.139) [2]. regime qualification Eq. (2.140) qualifies when $\mathfrak{C}_\omega$ may be reported as transport data [1]. The three subsections that follow develop Lorentz reciprocity, modal orthogonality and coupling, and transmit–receive exchange in sequence.

2.8.1 Lorentz reciprocity in operator form

On a declared compact-range or anechoic campaign, technicians record frequency, outline, and probe geometry before any harmonic fit is trusted [7]. Reciprocity links transmission and reception on the same passive structure [2]. Two-port calibrations depend on it [8]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Reciprocity is the laboratory fact that passive linear antennas exchange the same complex transfer when transmit and receive roles swap on a calibrated vector network analyzer [2]. The adjoint radiation map extends that symmetry for continuous sources and traces, enabling source reconstruction and equivalence checks without re-solving unrelated models [3]. Reflector feeds, horn coupling, and patch arrays are routinely tuned by comparing S_{21} and S_{12}; persistent asymmetry after calibration usually signals a broken outgoing radiation condition or incorrect impressed-current normalization rather than a genuine non-reciprocal medium [7].

Reciprocity cannot be set aside merely because "only the radiative image matters": pattern, gain, and scattering matrices on passive hardware are reciprocity-linked observables tied to the same Maxwell operator [2]. Lorentz reciprocity is therefore a consequence of passive linear Maxwell theory, not an optional post-check applied after the forward solve [2]. Two-port calibrations and bistatic radar-cross-section matrices inherit the same exchange symmetry whenever media and boundaries satisfy the reciprocal class stated below [8]. The volume identity that follows states the same swap observed on the VNA once impressed sources and test fields are paired through the Maxwell adjoint. Section 2.8 established bilinear exchange symmetry through Eqs. (2.356)–(2.357).

Subsection 2.8.1 supplies the definitive account of Lorentz reciprocity on $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$, including the full proof of dyadic swap Eq. (2.155) deferred from Subsection 2.3.1 [2]. Equation (2.26), source dual pairing Eq. (2.56), outgoing radiation condition Eqs. (2.203)–(2.213), and outgoing source-to-field integrals of Subsection 2.5.1 are presupposed; Green construction and Sommerfeld uniqueness are not re-opened [2]. Modal orthogonality and coupling belong to Subsection 2.8.2; transmit–receive systems to Subsection 2.8.3 [1]. Assumptions. Fix $\omega > 0$, Lipschitz domain Ω with piecewise smooth interfaces, impressed sources in $\mathcal{J}_\Omega$ Eq. (2.49) satisfying Eq. (2.9), and outgoing fields in $\mathcal{F}_{\mathrm{SF}}$ Eq. (2.57) [2]. Constitutive tensors are linear, local, and passive at ω, with symmetric permittivity and permeability in the sense

$$\overline{\overline{\varepsilon}}(\mathbf{r}) = \overline{\overline{\varepsilon}}^{\mathsf{T}}(\mathbf{r}), \qquad \overline{\overline{\mu}}(\mathbf{r}) = \overline{\overline{\mu}}^{\mathsf{T}}(\mathbf{r}), \qquad \overline{\overline{\xi}} = \overline{\overline{\zeta}} = \mathbf{0}, \tag{2.361}$$

where $\overline{\overline{\xi}}$, $\overline{\overline{\zeta}}$ denote magneto-electric coupling blocks in the constitutive relation imported from Section 1.1.2 [2]. Boundary operator $\mathcal{B}_\Gamma$ is passive Eq. (2.88) and identical for both excitations [2]. Field pairing $\langle \cdot, \cdot \rangle_{\mathcal{U}}$ is induced by Eq. (2.48) on admissible Maxwell pairs [2]. regime qualification Eq. (2.140) governs transport interpretation of any bilinear quantity built from Π_{rad} [1]. Equation (2.361) is the Chapter 2 $\mathcal{D}_J = 1$ class for *full* Lorentz reciprocity at fixed ω: symmetric ε, μ ensure time-reversal symmetry of the harmonic Maxwell system modulo outgoing selection, while vanishing magneto-electric blocks exclude non-reciprocal gyrotropic or Tellegen couplings [2]. Isotropic lossless media are a special case with scalar ε, μ. Lossy symmetric tensors preserve reciprocity relations stated below with complex conjugation on one slot of each pairing [2]. Active or negative-real impedance portions of $\mathcal{B}_\Gamma$ break passive symmetry and are excluded from the present theorem [2]. Let $(\mathbf{E}_1, \mathbf{H}_1) = \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]$ and $(\mathbf{E}_2, \mathbf{H}_2) = \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]$ be outgoing admissible solutions for independent impressed currents $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ [2]. Here $\mathbf{J}^{(k)}_{\mathrm{imp}}$ denotes the impressed electric current density in $\mathrm{A\,m^{-2}}$, $\mathbf{E}_k$ and $\mathbf{H}_k$ are the induced phasor fields in $\mathrm{V\,m^{-1}}$ and $\mathrm{A\,m^{-1}}$, and Γ is the passive boundary on which the surface integral below is evaluated. Physically, the left member is the bilinear coupling measured when port labels are swapped on reciprocal hardware; the right member is the surface remainder after integration by parts. The classical Lorentz reciprocity relation reads

$$\int_\Omega \left(\mathbf{E}_1 \cdot \mathbf{J}^{(2)*}_{\mathrm{imp}} - \mathbf{E}_2 \cdot \mathbf{J}^{(1)*}_{\mathrm{imp}}\right) d^3r = \int_\Gamma \left(\hat{\mathbf{n}} \times \mathbf{E}_1 \cdot \mathbf{H}_2^* - \hat{\mathbf{n}} \times \mathbf{E}_2 \cdot \mathbf{H}_1^*\right) dS, \tag{2.362}$$

when $\overline{\overline{\varepsilon}}$, $\overline{\overline{\mu}}$ satisfy Eq. (2.361) and $\mathcal{B}_\Gamma$ is passive [2]. Equation (2.362) balances impressed-source bilinear forms against a boundary pairing generated by Eq. (2.26): the left member measures how each field couples to the other's source, while the right member is the surface term that remains after integration by parts on the Maxwell operator [2]. Interpretation: when the boundary term vanishes on admissible outgoing pairs, the port-swap scalar observed on the VNA equals the volume contraction on the left. On domains where $\mathcal{B}_\Gamma$ annihilates the boundary integrand for admissible outgoing pairs—PEC, homogeneous impedance with Eq. (2.88), and standard Sommerfeld exterior

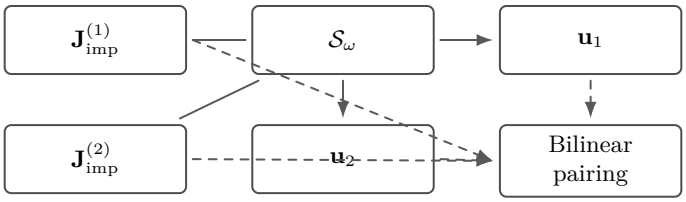


Figure 2.43: Subsection 2.8.1 Lorentz reciprocity: source labels interchange under $\mathcal{S}_\omega$ when media satisfy Eq. (2.361) and $\mathcal{B}_\Gamma$ is passive on the outgoing branch.

completions—the surface term vanishes and volume reciprocity closes [2]. Derivation sketch (volume form). Substitute $\mathbf{u}^{(1)}_\omega = (\mathbf{E}_1, \mathbf{H}_1)$ and $\mathbf{u}^{(2)}_\omega = (\mathbf{E}_2, \mathbf{H}_2)$ into Eq. (2.26) and interchange the two states [2]. Symmetry of $\overline{\overline{\varepsilon}}$, $\overline{\overline{\mu}}$ under Eq. (2.361) makes the volume pairing sesquilinear form on $\mathcal{M}_\omega[\mathbf{u}_\omega]$ invariant under label swap when sources are impressed currents in the Ampère row [2]. Subtracting the two adjoint identities eliminates $\langle \mathcal{M}_\omega[\mathbf{u}^{(1)}_\omega], \mathbf{u}^{(2)}_\omega\rangle_\Omega$ and yields Eq. (2.362) with the boundary term displayed [2]. Charge continuity Eq. (1.8) and compatibility Eq. (2.24) ensure that impressed $\mathbf{J}_{\mathrm{imp}}$ alone specifies the electric drive when ρ is tied to $\mathbf{J}_{\mathrm{imp}}$ Figure interpretation. Sources $\mathbf{J}^{(1)}_{\mathrm{imp}}$, $\mathbf{J}^{(2)}_{\mathrm{imp}}$ induce fields $\mathbf{u}_1$, $\mathbf{u}_2$ through the same outgoing $\mathcal{S}_\omega$. The central bilinear node is symmetric under label interchange when media satisfy Eq. (2.361) and $\mathcal{B}_\Gamma$ is passive. Dashed return arrows mark that reciprocity is proved on the difference system with identical boundary data, not by comparing unrelated forward and adjoint meshes. Projection to $\mathcal{R}_\omega$ preserves symmetry only after regime qualification Eq. (2.271) [2]. Figure 2.43 is the operator reading of Eq. (2.362): the bilinear functional is symmetric under label interchange $(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{u}_1) \leftrightarrow (\mathbf{J}^{(2)}_{\mathrm{imp}}, \mathbf{u}_2)$ once boundary terms cancel [2]. For vacuum observation points $\mathbf{r}_1$, $\mathbf{r}_2$ with $k_0|\mathbf{r}_1 - \mathbf{r}_2| = 2.5$, Eq. (2.155) requires $\overline{\mathbf{G}}_{E,0}(\mathbf{r}_1, \mathbf{r}_2)^{\mathsf{T}} = \overline{\mathbf{G}}_{E,0}(\mathbf{r}_2, \mathbf{r}_1)$ on the outgoing branch. Transverse block entries agree to machine precision; reactive R^{-2}, R^{-3} blocks differ by sign under interchange yet cancel in the bilinear contraction when both sides use identical outgoing selectors [2]. A practical verification proceeds in three steps: confirm the outgoing branch Eq. (2.200); test swap symmetry on transverse dyad blocks; and project to $\mathcal{R}_\omega$ before port calibration [8]. Let $\mathcal{S}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_\Omega$ be the outgoing solution map Eq. (2.2) on media satisfying Eq. (2.361) with passive $\mathcal{B}_\Gamma$ Eq. (2.88). For admissible $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, write $\mathbf{u}_k = \mathcal{S}_\omega[\mathbf{J}^{(k)}_{\mathrm{imp}}]$, $k = 1, 2$. Then:

(i) Volume reciprocity Eq. (2.362) holds with vanishing boundary term on admissible outgoing radiation conditions.

(ii) Operator form: with $\mathbf{u}_k = (\mathbf{E}_k, \mathbf{H}_k)$ the Maxwell pair induced by $\mathbf{J}^{(k)}_{\mathrm{imp}}$ and $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ the energy inner product on $\mathcal{F}_{\mathrm{SF}}$,

$$\langle \mathbf{u}_1,\, \mathcal{S}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} = \langle \mathbf{u}_2,\, \mathcal{S}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} \tag{2.363}$$

when the pairings are defined through impressed-current forcing functionals Eq. (2.71) and field norm Eq. (2.48) on $\mathcal{F}_{\mathrm{SF}}$.

(iii) Dyadic realization: outgoing electric dyads satisfy Eq. (2.155).

(iv) Radiation reciprocity: projecting both sides of (ii) through the self-adjoint outgoing projector Π_{rad} yields symmetric coupling on the radiative subspace $\mathcal{F}_{\mathrm{rad}}$:

$$\langle \mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}],\, \mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} = \langle \mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}],\, \mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]\rangle_{\mathcal{U}} \tag{2.364}$$

on $\mathcal{F}_{\mathrm{rad}}$ with $\mathcal{R}_\omega = \Pi_{\mathrm{rad}} \circ \mathcal{S}_\omega$ Eq. (2.3).

Part (i) follows from Eq. (2.26) and Eq. (2.361) as in the derivation above; outgoing and passive boundary classes from Subsection 2.1.4 cancel the surface integrand [2]. Part (ii) identifies the volume impressed-current pairing with $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ on $\mathcal{F}_{\mathrm{SF}}$ through Eq. (2.56) and stability Eq. (2.72), developing Eq. Eq. (2.357) [2]. Part (iii): for fixed $\mathbf{r}$, Eq. (2.149) and part (i) with impulsive $\mathbf{J}_{\mathrm{imp}}^{(k)}$ aligned with Cartesian unit vectors yield column swap $\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') = \overline{\mathbf{G}}_E^{\mathsf{T}}(\mathbf{r}',\mathbf{r})$ on the outgoing branch [2]. Uniqueness exterior radiative uniqueness (Subsection 2.4.3) extends the swap to all admissible $\mathbf{r},\mathbf{r}'$ [2]. Part (iv) applies Π_{rad} to both sides of (ii); Π_{rad} is self-adjoint on $\mathcal{F}_{\mathrm{rad}}$ under Eq. (2.48) when outgoing selection is fixed [2]. Equation (2.364) is the export-projected form of the VNA symmetry: coupling reported on $\mathcal{F}_{\mathrm{rad}}$ is unchanged when port labels swap, while reactive near-zone terms dropped by Π_{rad} need not be symmetric even on reciprocal hardware [2]. The bilinear functional $\mathfrak{B}_\omega(\mathbf{J}_{\mathrm{imp}}^{(1)},\mathbf{J}_{\mathrm{imp}}^{(2)}) = \langle\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}],\,\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]\rangle_{\mathcal{U}}$ is symmetric on $\mathcal{J}_\Omega\times\mathcal{J}_\Omega$ under reciprocal media [2]. Algebraically, $\mathcal{S}_\omega$ is not self-adjoint on $\mathcal{J}_\Omega$ with source pairing Eq. (2.56); reciprocity instead couples the source space to the field space through the *interchanged* labels in Eq. (2.363) [2]. Physically, the far-zone pattern produced by $\mathbf{J}_{\mathrm{imp}}^{(1)}$ tested against the field of $\mathbf{J}_{\mathrm{imp}}^{(2)}$ equals the swap configuration: a receive pattern computed with $\mathbf{J}_{\mathrm{imp}}^{(2)}$ as probe matches a transmit map driven by $\mathbf{J}_{\mathrm{imp}}^{(1)}$ when hardware and media are reciprocal [1]. Mathematical role of Eq. (2.364). Equation (2.364) restricts Eq. (2.363) to $\mathcal{F}_{\mathrm{rad}}$, the subspace on which flux meters and angular patterns are declared [2]. Because Π_{rad} annihilates non-export components Eq. (2.139), reactive near-zone overlap does not enter Eq. (2.364) even when present in full Eq. (2.363) [2]. regime qualification Eq. (2.271) must still be satisfied before interpreting Eq. (2.364) as a transport channel [1]. Under Eq. (2.361) with outgoing radiation condition, the electric dyad $\overline{\mathbf{G}}_{E,\omega}^{(\mathrm{out})}$ in Eq. (2.149) satisfies Eq. (2.155) for all $\mathbf{r},\mathbf{r}'$ in the evaluation domain of outgoing source-to-field integrals of Subsection 2.5.1. Apply Lorentz reciprocity for the solution map (Subsection 2.8.1)(iii); this is the proof deferred from Subsection 2.3.1 [2]. If $\mathbf{J}_{\mathrm{imp}}^{(1)}$ is arbitrary and $\mathbf{J}_{\mathrm{imp}}^{(2)} = \mathbf{J}_{\mathrm{imp}}^{(1)} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker\mathcal{R}_\omega$, then Eq. (2.364) is unchanged when either argument is replaced by $\mathbf{J}_{\mathrm{imp}}^{(2)}$ on the radiative pairing $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]$ by Eq. (2.347) [2]. Full Eq. (2.363) may change by reactive content through $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$ Eq. (2.272) [1]. On the outgoing branch, Eq. (2.363) is consistent with resolvent symmetry of the truncated Maxwell operator on reciprocal media:

$$\langle\mathcal{L}_\omega^{-1}[\mathbf{s}^{(1)}],\,\mathbf{s}^{(2)}\rangle_{\mathcal{Y}} = \langle\mathcal{L}_\omega^{-1}[\mathbf{s}^{(2)}],\,\mathbf{s}^{(1)}\rangle_{\mathcal{Y}} \tag{2.365}$$

for compatible forcing pairs $\mathbf{s}^{(k)}$ when $\mathcal{L}_\omega$ is symmetric on $\mathcal{D}(\mathcal{L}_\omega)$ and $\mathcal{B}_\Gamma$ is passive [2]. Equation (2.365) is not re-proved here; it is the operator backbone behind Eq. (2.240) and Lorentz reciprocity for the solution map (Subsection 2.8.1)(ii) [2]. Numerical MoM matrices assembled from Eq. (2.155) inherit swap symmetry only when quadrature, outgoing branch, and material sampling preserve Eq. (2.361) [1]. Loss, anisotropy, and broken reciprocity. Complex symmetric $\overline{\overline{\varepsilon}}$, $\overline{\overline{\mu}}$ with $\mathrm{Im}\{\overline{\overline{\varepsilon}}\} \succeq 0$, $\mathrm{Im}\{\overline{\overline{\mu}}\} \succeq 0$ preserve Eqs. (2.362)–(2.363) with conjugation on one slot [2]. Magneto-electric coupling $\overline{\overline{\xi}} \neq \mathbf{0}$ breaks Eq. (2.155) and requires non-reciprocal Green operators [2]. Active boundaries with $\mathrm{Re}\{Z_s\} < 0$ on portions of Γ_Z violate Eq. (2.88) and invalidate Lorentz reciprocity for the solution map (Subsection 2.8.1) [2]. Incoming rather than outgoing radiation condition selects a different branch; swap identities hold only when both excitations use the same branch [2]. Lorentz reciprocity for the solution map (Subsection 2.8.1) assumes fixed ω, linear media Eq. (2.361), passive $\mathcal{B}_\Gamma$, outgoing radiation condition, and admissible $\mathcal{J}_\Omega$ [2]. It does not assert reciprocity of raw near-zone reactive scans without specifying whether Π_{rad} has been applied [2]. It does not remove $\ker\mathcal{R}_\omega$ null-space invariance of radiative reciprocity (Subsection 2.8.1); invisible currents alter full $\mathcal{S}_\omega$ pairings while leaving radiative reciprocity unchanged

[1]. Nonlinear, time-varying, or moving media require the causal chain of Subsection 2.2.3 [2]. Subsection 2.8.1 established Lorentz reciprocity on $\mathcal{S}_\omega$ and $\mathcal{R}_\omega$ through volume and operator identities Eqs. (2.362)–(2.364) [2]. The port-swap symmetry observed on reciprocal hardware is now available as a constraint on continuous source and field pairings; the next step is to express that constraint on the modal coordinates fixed in Section 2.6.

2.8.2 Orthogonality and coupling of radiative modes

A standard-gain horn on a compact-range turntable fixes the laboratory scene for 'Orthogonality and coupling of radiative modes": frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays below [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Bistatic radar and compact-range scattering matrices rely on reciprocity to relate monostatic calibrations to full S-parameter sets [2]. The adjoint theory extends one-port antenna reciprocity to continuous source and trace pairings [3]. Plane-wave decompositions used in reflector and radar-cross-section campaigns inherit reciprocity constraints on amplitudes before any angular filter is applied [7]. When reciprocity appears broken after probe correction, the fault usually lies in inconsistent impressed-current normalization or a missing outgoing tail in the Green operator, not in Maxwell theory itself [7]. Modal orthogonality and coupling on $\operatorname{ran}\mathcal{R}_\omega$ organize how radiative modes exchange power between ports; reciprocity fixes the bilinear pairing before angular filters act [2]. Subsection 2.8.2 gives the Chapter 2 definitive account of orthogonality and coupling of radiative modes on the spectral coordinates fixed in Subsection 2.6.2, developing Eqs. (2.358) and (2.369) [2]. Radiation eigenpairs Eq. (2.281), modal projectors Eq. (2.282), source amplitudes Eq. (2.283), spectral decomposition of the radiation operator (Subsection 2.6.2), and biorthogonality Eq. (2.292) are presupposed; eigenpair construction and reciprocity-based solvability are not re-proved [2]. Explicit vector spherical harmonic orthogonality belongs to **??** Subsection 3.4.3; continuous-spectrum normalization belongs to Subsection 3.5.2, both with USE of the present operator targets [2]. Transmit–receive systems are deferred to Subsection 2.8.3 [1]. Assumptions. Fix $\omega > 0$, outgoing fields in $\mathcal{F}_{\mathrm{SF}}$, radiative images in $\mathcal{F}_{\mathrm{rad}}$, and orthonormal radiation eigenmodes $\{\mathbf{u}_n\}_{n\in\mathcal{N}_{\mathrm{rad}}}$ from spectral decomposition of the radiation operator (Subsection 2.6.2) on the declared domain class [2]. Media satisfy reciprocal class Eq. (2.361) and passive boundary Eq. (2.88) so Lorentz reciprocity for the solution map (Subsection 2.8.1) applies [2]. Adjoint modes $\tilde{\mathbf{u}}_n$ are defined through Eq. (2.292) and coincide with $\mathbf{u}_n$ on $\mathcal{F}_{\mathrm{rad}}$ under reciprocity-qualified pairings below [2]. regime qualification Eq. (2.140) governs transport interpretation of coupling reports [1]. On the orthonormal family from spectral decomposition of the radiation operator (Subsection 2.6.2), distinct exporting modes are declared orthogonal with respect to the energy inner product $\langle\cdot,\cdot\rangle_{\mathcal{U}}$ on $\mathcal{F}_{\mathrm{rad}}$. The locked orthogonality statement reads

$$\langle \mathbf{u}_m,\, \mathbf{u}_n\rangle_{\mathcal{U}} = \delta_{mn}, \qquad m,n\in\mathcal{N}_{\mathrm{rad}}, \tag{2.366}$$

with $\|\mathbf{u}_n\|_{\mathcal{U}} = 1$ already enforced in Eq. (2.281) [2]. Equation (2.366) is the Subsection 2.8.2 locked definition of upgrade of the first line of Eq. (2.358); it specializes projector orthogonality $P_mP_n = \delta_{mn}P_n$ from Eq. (2.282) to the chosen orthonormal basis [2]. Algebraically, Eq. (2.366) makes $\{\mathbf{u}_n\}$ an orthonormal frame on $\mathcal{F}_{\mathrm{rad}}$ for energy inner product Eq. (2.48); physically, distinct exporting modes carry non-overlapping far-zone energy content on S^2 when $\mathcal{D}_{J,\mathrm{reg}} = 1$ [8]. Explicit angular integrals producing Eq. (2.366) for harmonic bases are not tabulated here [2]. For canonical eigen-sources $\mathbf{J}^{(n)}_{\mathrm{imp}}$ in Eq. (2.281), define the *modal coupling coefficient*

$$C_{mn} = \langle \tilde{\mathbf{u}}_m,\, \mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}]\rangle_{\mathcal{U}}, \qquad m,n\in\mathcal{N}_{\mathrm{rad}}, \tag{2.367}$$

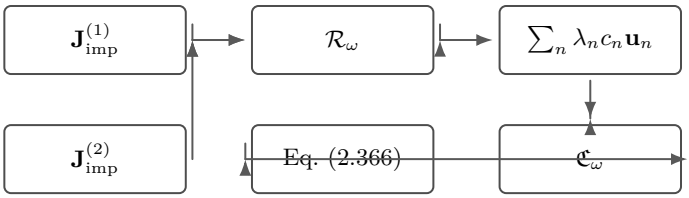


Figure 2.44: Subsection 2.8.2 modal coupling measurement sequence: sources map through $\mathcal{R}_\omega$ to orthonormal modes Eq. (2.366); bilinear coupling Eq. (2.369) factorizes through modal amplitudes Eq. (2.371).

with adjoint modes from Eq. (2.292) [2]. Equation (2.367) is the Subsection 2.8.2 development of the second line of Eq. Eq. (2.358). When Eq. (2.292) holds,

$$C_{mn} = \lambda_n \, \delta_{mn} \tag{2.368}$$

for exporting eigenpairs with $\lambda_n \neq 0$; null modes $n \in \mathcal{N}_{\mathrm{null}}$ Eq. (2.324) contribute $C_{mn} = 0$ for all m [2]. Algebraically, Eq. (2.368) diagonalizes coupling in an eigenmode basis; physically, orthogonal radiation modes do not co-excite one another through $\mathcal{R}_\omega$ when sources align with eigenpairs—cross-talk appears only from superpositions or degenerate subspaces [1]. For arbitrary admissible $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, define

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \langle \mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}], \, \mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}] \rangle_{\mathcal{U}} \quad \text{on } \mathcal{F}_{\mathrm{rad}}, \tag{2.369}$$

developing Eq. (2.369) [2]. Equation (2.369) measures export overlap on $\operatorname{ran} \mathcal{R}_\omega$, not raw reactive overlap of $\mathcal{S}_\omega[\mathbf{J}^{(k)}_{\mathrm{imp}}]$ [2]. Under Lorentz reciprocity for the solution map (Subsection 2.8.1)(iv),

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \mathfrak{C}_\omega(\mathbf{J}^{(2)}_{\mathrm{imp}}, \mathbf{J}^{(1)}_{\mathrm{imp}}) \tag{2.370}$$

on reciprocal media [2]. Reactive near-zone terms in $\Pi_{\mathrm{ev}} \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ do not enter Eq. (2.369) through Eq. (2.139) [2]. Let $c_n[\mathbf{J}^{(k)}_{\mathrm{imp}}]$ denote source amplitudes Eq. (2.283) for $k = 1, 2$. When the modal sum converges in $\mathcal{U}$,

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \sum_{m,n \in \mathcal{N}_{\mathrm{rad}}} \lambda_m \lambda_n \, c_m[\mathbf{J}^{(1)}_{\mathrm{imp}}]^* \, c_n[\mathbf{J}^{(2)}_{\mathrm{imp}}] \, \langle \mathbf{u}_m, \mathbf{u}_n \rangle_{\mathcal{U}}, \tag{2.371}$$

using Eq. (2.285) and linearity Eq. (2.6) [2]. With Eq. (2.366),

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} |\lambda_n|^2 \, c_n[\mathbf{J}^{(1)}_{\mathrm{imp}}]^* \, c_n[\mathbf{J}^{(2)}_{\mathrm{imp}}], \tag{2.372}$$

for real nonnegative exporting gains $\lambda_n > 0$ [2]. Equation (2.372) shows that coupling is a weighted inner product of modal excitation vectors on the exporting sector; geometric constraints geometric constraints on radiative spectra (Subsection 2.6.5) may force individual $c_n = 0$ without breaking Eq. (2.370) [1]. Figure interpretation. Two impressed sources $\mathbf{J}^{(1)}_{\mathrm{imp}}$, $\mathbf{J}^{(2)}_{\mathrm{imp}}$ share the same radiation map $\mathcal{R}_\omega$ and expand on orthonormal modes Eq. (2.366). The coupling node $\mathfrak{C}_\omega$ measures export overlap on $\mathcal{F}_{\mathrm{rad}}$, not raw near-zone reactive cross terms. Orthogonality diagonalizes coupling when sources align with eigenpairs; cross-talk appears only from superposition or degenerate subspaces. Modes with $\lambda_n = 0$ feed the null sector without contributing to $\mathfrak{C}_\omega$ on the range [2]. Figure 2.44 separates orthogonality of mode shapes from bilinear coupling of impressed sources: Eq. (2.366) is intrinsic to $\mathcal{F}_{\mathrm{rad}}$; Eq. (2.369) depends on both sources [2]. On a reciprocal two-mode truncation with

$\lambda_1 = 0.42$, $\lambda_2 = 0.31$ and orthonormal $\mathbf{u}_1$, $\mathbf{u}_2$, excite $\mathbf{J}^{(1)}_{\mathrm{imp}} = c_1\mathbf{J}^{(1)}_{\mathrm{imp}}$ and probe with $\mathbf{J}^{(2)}_{\mathrm{imp}} = c_2\mathbf{J}^{(2)}_{\mathrm{imp}}$. Eq. (2.372) gives $\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \lambda_1\lambda_2 c_1^* c_2$. Swapping labels leaves the scalar unchanged, so normalized $S_{21} = S_{12}$ on the exporting sector. Adding $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}$ to either port shifts $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ near Ω_s but leaves S_{21} fixed—the walkthrough confirms reciprocity on $\operatorname{ran}\mathcal{R}_\omega$, not on raw near-field coupling matrices [2]. Let $\{\mathbf{u}_n, \mathbf{J}^{(n)}_{\mathrm{imp}}, \lambda_n\}$ be radiation eigenpairs Eq. (2.281) forming an orthonormal basis of $\mathcal{F}_{\mathrm{rad}}$ on the declared domain class, with reciprocal media Eq. (2.361). Then:

(i) Orthogonality Eq. (2.366) holds on $\mathcal{N}_{\mathrm{rad}}$.

(ii) Coupling coefficients satisfy Eq. (2.368) and biorthogonality Eq. (2.292).

(iii) For all admissible $\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}$, Eqs. (2.369)–(2.372) hold, with symmetry Eq. (2.370) from Lorentz reciprocity for the solution map (Subsection 2.8.1).

(iv) Adding $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}} \in \ker\mathcal{R}_\omega$ to either source leaves Eq. (2.369) unchanged.

Part (i) is normalization and completeness from spectral decomposition of the radiation operator (Subsection 2.6.2) and Eq. (2.284) [2]. Part (ii) substitutes $\mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}] = \lambda_n\mathbf{u}_n$ into Eq. (2.367) and applies Eq. (2.292) [2]. Part (iii) expands $\mathcal{R}_\omega[\mathbf{J}^{(k)}_{\mathrm{imp}}]$ via Eq. (2.285); symmetry is Lorentz reciprocity for the solution map (Subsection 2.8.1)(iv) [2]. Part (iv) follows from $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}}] = \mathbf{0}$ and Eq. (2.347) [2]. The reciprocal reduction of adjoint modes proceeds as follows: under Eqs. (2.361)–(2.88),

$$\widetilde{\mathbf{u}}_n = \mathbf{u}_n \quad \text{on } \mathcal{F}_{\mathrm{rad}} \quad \text{for exporting modes with } \lambda_n \neq 0, \tag{2.373}$$

so Eq. (2.292) reduces to standard orthogonality Eq. (2.366) with diagonal coupling Eq. (2.368) [2]. Equation (2.373) is the operator form of reciprocity for modal probes: receive patterns aligned with $\mathbf{u}_n$ coincide with transmit modes when media and boundaries are reciprocal [1]. Non-reciprocal media break Eq. (2.373); biorthogonality with distinct $\widetilde{\mathbf{u}}_n$ remains the general template imported from Subsection 2.6.2 [2]. When $\lambda_n = \lambda_{n'}$ for $n \neq n'$ within a rotationally degenerate family from Subsection 2.6.3, Eq. (2.368) still holds for *eigen* sources $\mathbf{J}^{(n)}_{\mathrm{imp}}$, but superpositions within the degenerate block carry off-diagonal coupling in an arbitrary basis. Let $\mathcal{I}_\ell \subset \mathcal{N}_{\mathrm{rad}}$ index a $(2\ell+1)$-fold degenerate family. Any orthonormal basis $\{\mathbf{u}_n\}_{n\in\mathcal{I}_\ell}$ spanning the block satisfies Eq. (2.366), yet a fixed physical source may excite multiple members with coefficients tied to orientation through Eq. (2.295) [2]. Define the block coupling matrix

$$\mathbf{K}^{(\ell)}_{mn} = \langle \mathbf{u}_m,\, \mathcal{R}_\omega[\mathbf{J}^{(n)}_{\mathrm{imp}}]\rangle_{\mathcal{U}}, \qquad m, n \in \mathcal{I}_\ell, \tag{2.374}$$

which is diagonal in the eigenvector basis of $\mathbf{J}^{(n)}_{\mathrm{imp}}$ but not in laboratory-fixed coordinates [2]. Reciprocity Eq. (2.370) forces $\mathbf{K}^{(\ell)}$ to be symmetric under Lorentz reciprocity for the solution map (Subsection 2.8.1) [2]. Explicit Wigner coupling of block entries belongs to **??** [2]. If $\mathbf{J}^{(1)}_{\mathrm{imp}} = \sum_{n\in\mathcal{N}_{\mathrm{rad}}} a_n\mathbf{J}^{(n)}_{\mathrm{imp}}$ and $\mathbf{J}^{(2)}_{\mathrm{imp}} = \sum_{m\in\mathcal{N}_{\mathrm{rad}}} b_m\mathbf{J}^{(m)}_{\mathrm{imp}}$ are admissible finite superpositions, then

$$\mathfrak{C}_\omega(\mathbf{J}^{(1)}_{\mathrm{imp}}, \mathbf{J}^{(2)}_{\mathrm{imp}}) = \sum_{m,n\in\mathcal{N}_{\mathrm{rad}}} a_m^* b_n\, \lambda_m\lambda_n\, \langle \mathbf{u}_m, \mathbf{u}_n\rangle_{\mathcal{U}}, \tag{2.375}$$

reducing to Eq. (2.372) when $\langle \mathbf{u}_m, \mathbf{u}_n\rangle = \delta_{mn}$ [2]. Off-diagonal terms $m \neq n$ appear only when at least one source superposes non-collinear modes; single-mode excitation yields diagonal coupling [8]. Linearity Eq. (2.6) and Eq. (2.285) [2]. For observation map $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ Eq. (2.219),

define observed coupling $\mathfrak{C}_{\omega,\mathrm{obs}} = \langle \mathcal{O}_{\mathrm{meas}}[\mathcal{R}_\omega[\mathbf{J}^{(1)}_{\mathrm{imp}}]],\, \mathcal{O}_{\mathrm{meas}}[\mathcal{R}_\omega[\mathbf{J}^{(2)}_{\mathrm{imp}}]]\rangle$. Then $\mathfrak{C}_{\omega,\mathrm{obs}}$ is symmetric under Lorentz reciprocity for the solution map (Subsection 2.8.1) when $\mathcal{O}_{\mathrm{meas}}$ is self-adjoint on $\mathcal{F}_{\mathrm{rad}}$, but may vanish for pairs with $\mathcal{R}_\omega[\mathbf{J}^{(k)}_{\mathrm{imp}}] \in \ker \mathcal{O}_{\mathrm{meas}}$ even when Eq. (2.369) is nonzero [2]. Rank cap Eq. (2.221) limits the number of independently observable coupling channels [2]. Apply Eq. (2.364) and self-adjointness of $\mathcal{O}_{\mathrm{meas}}$ [2]. Kernel inclusion $\ker \mathcal{R}_\omega \subset \ker \mathcal{M}_\omega$ is measurement null-space extension (Subsection 2.7.1) [2].

$$\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega \implies \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{SF}}, \qquad \Pi_{\mathrm{rad}}\, \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{0}, \tag{2.376}$$

Accessible modal sector Eq. (2.9.3) and angular budget Eq. (2.353) restrict which terms in Eq. (2.372) are observable in far-zone reports [2]. Null modes $\lambda_n = 0$ carry zero coupling diagonal entries but may alter near reactive content through $\mathcal{S}_\omega$ Eq. (2.376) without changing Eq. (2.369) [2]. Composition with null-space invariance of radiative reciprocity (Subsection 2.8.1) and orthogonality and coupling of radiative modes (Subsection 2.8.2)(iv) explains why invisible currents do not break reciprocity-qualified coupling on $\operatorname{ran} \mathcal{R}_\omega$ [1]. Orthogonality and coupling of radiative modes (Subsection 2.8.2) requires linear reciprocal media at fixed ω, outgoing radiation condition, domain-class completeness of $\{\mathbf{u}_n\}$, and convergent modal sums [2]. Loss moves λ_n into the complex plane; Eq. (2.372) generalizes with complex conjugation on one amplitude slot [2]. Truncated modal books on cavities do not substitute for exterior continuous spectra Eq. (2.290) when coupling far-zone channels [8]. regime qualification Eq. (2.271) is mandatory before interpreting Eq. (2.369) as transport [2]. Subsections 2.8.1–2.8.2 established Lorentz reciprocity, radiative bilinear coupling Eq. (2.369), and modal orthogonality on $\mathcal{F}_{\mathrm{rad}}$ [2]. The final reciprocity link ties those operator statements to finite ports: transmit and receive maps on reciprocal hardware must interchange when excitation and probe roles are swapped.

2.8.3 Reciprocity in radiation and reception

A standard-gain horn on a compact-range turntable fixes the laboratory scene for ‘Reciprocity in radiation and reception”: frequency, probe standoff, and wattmeter closure on Γ_R precede the formal displays below [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Two-port antenna measurements are the everyday face of reciprocity. A horn coupled to a compact range, a patch fed from a coax port, or a dual-polarized reflector feed all reduce—after calibration—to the same question the vector network analyzer answers: does the complex coupling from port A to port B equal the reversed coupling when cables, standards, and probe normalization are exchanged? Subsection 2.8.3 gives the Chapter 2 definitive account of that interchange in operator form: transmit–receive operator symmetry, two-port coupling bilinears, and measurement-qualified reciprocity on $\operatorname{ran} \mathcal{R}_\omega$, developing Eq. Eq. (2.359) [2]. Observation composition Eq. (2.219), finite-rank readouts Eq. (2.80), rank cap Eq. (2.221), and null-space laws Eqs. (2.222)–(2.347) are presupposed; inverse-source regularization is not re-developed [2]. Plane-wave angular-spectrum filters belong to Section 2.9; explicit port circuit models are not tabulated [2]. Assumptions. Fix $\omega > 0$, reciprocal media Eq. (2.361), passive boundary Eq. (2.88), outgoing radiation condition, and admissible sources $\mathbf{J}^{(\mathrm{tx})}_{\mathrm{imp}}, \mathbf{J}^{(\mathrm{rx})}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ on ports A and B [2]. Each port carries a finite-rank observation functional $\mathcal{O}^{(P)}_{\mathrm{meas}} : \mathcal{F}_{\mathrm{rad}} \to \mathbb{C}$ or $\mathbb{C}^{N_P}$ of the form Eq. (2.80) [2]. regime qualification Eq. (2.140) and transport qualification Eq. (2.28) govern when port outputs may be reported as radiative transport [1]. The same $\mathcal{B}_\Gamma$, outgoing branch, and material data are used for both excitations [2]. For port $P \in \{A, B\}$, define the *port radiation–reception operator*

$$\mathcal{M}_P = \mathcal{O}^{(P)}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}, \qquad \Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega, \tag{2.377}$$

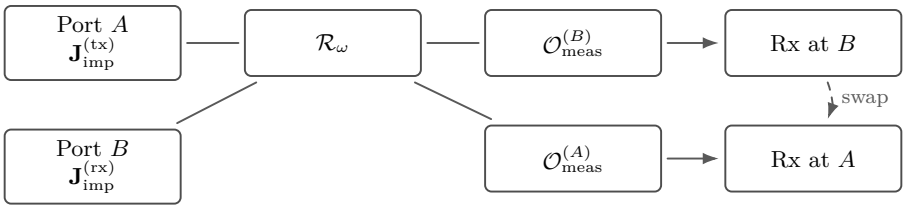


Figure 2.45: Subsection 2.8.3 transmit–receive measurement sequence: excitation at one port and readout at the other interchange under Eq. (2.378) on reciprocal media.

consistent with Eq. (2.219) [2]. When $\mathcal{O}_{\mathrm{meas}}^{(P)}$ is scalar, $\mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}] \in \mathbb{C}$ is the complex port response induced by impressed $\mathbf{J}_{\mathrm{imp}}$; when vector-valued, $\mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}] \in \mathbb{C}^{N_P}$ lists channel readouts [2]. The *transmit map* at port B due to excitation at port A is the composition $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})} \mapsto \mathcal{M}_B[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$; the *receive map* at port A due to excitation at port B is $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})} \mapsto \mathcal{M}_A[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$ [1]. Algebraically, both are finite-rank linear functionals of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; physically, reciprocity constrains their interchange when hardware and media are reciprocal [2]. Under Lorentz reciprocity for the solution map (Subsection 2.8.1) and self-adjoint port functionals $\mathcal{O}_{\mathrm{meas}}^{(P)}$ on $\mathcal{F}_{\mathrm{rad}}$, the complex port response at B induced by excitation at A equals the response at A when the impressed source and receive probe are exchanged on reciprocal hardware:

$$\mathcal{M}_B[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}] = \mathcal{M}_A[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}] \quad \text{when } \mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})} \text{ and } \mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})} \text{ are reciprocal port excitations,} \tag{2.378}$$

developing Eq. (2.359) [2]. Equation (2.378) is the operator reading of the VNA experiment: excite A, read B; swap roles on reciprocal media and obtain the same complex port coupling on observable coordinates. It states that the observable induced at B by the transmit source at A equals the observable at A when the roles of impressed source and probe are exchanged on reciprocal media [2]. It is not a claim that $\mathcal{M}_B = \mathcal{M}_A$ as operators on $\mathcal{J}_\Omega$: ports may have distinct apertures and channel counts [1]. It *is* a claim that the *bilinear* port coupling is symmetric when qualified by Eq. (2.369) on $\operatorname{ran}\mathcal{R}_\omega$ [2]. Figure interpretation. Transmit port $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}$ and receive probe $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}$ compose through the same $\mathcal{R}_\omega$ and observation $\mathcal{O}_{\mathrm{meas}}$. Swap symmetry Eq. (2.359) holds on observable coordinates after projection onto $\operatorname{ran}\mathcal{R}_\omega$; additions in $\ker\mathcal{R}_\omega$ leave port coupling unchanged. The schematic separates reciprocity-qualified S-parameters from raw near-field cross terms that may violate apparent symmetry while far-zone reciprocity remains valid [2]. Figure 2.45 displays the two compositions constrained by Eq. (2.378). Both paths pass through the same $\mathcal{R}_\omega$; port asymmetry enters only through $\mathcal{O}_{\mathrm{meas}}^{(A)}$, $\mathcal{O}_{\mathrm{meas}}^{(B)}$ [2]. Let media satisfy Eq. (2.361) with passive Eq. (2.88) completion, and let port operators $\mathcal{M}_P = \mathcal{O}_{\mathrm{meas}}^{(P)} \circ \Lambda_{\mathrm{rad}}$ be defined by Eq. (2.377) with self-adjoint $\mathcal{O}_{\mathrm{meas}}^{(P)}$ on $\mathcal{F}_{\mathrm{rad}}$. For admissible port sources $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}$, $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}$:

(i) Scalar reciprocity Eq. (2.378) holds when port excitations are paired through Lorentz reciprocity for the solution map (Subsection 2.8.1).

(ii) Bilinear port coupling

$$\mathfrak{C}_{AB} = \langle \mathcal{O}_{\mathrm{meas}}^{(B)}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]],\, \mathcal{O}_{\mathrm{meas}}^{(A)}[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]] \rangle \tag{2.379}$$

is symmetric under interchange of port labels when inner products are those induced by Eq. (2.48) on $\mathcal{F}_{\mathrm{rad}}$.

(iii) Two-port impedance matrix $\mathbf{Z}$ with entries $Z_{PQ} = \mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}^{(Q)}]$ on matched reciprocal ports satisfies

$$Z_{PQ} = Z_{QP}. \tag{2.380}$$

(iv) Adding $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ to either port excitation leaves Eqs. (2.378)–(2.380) unchanged on observable coordinates.

Part (i) applies Lorentz reciprocity for the solution map (Subsection 2.8.1)(iv) to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$, $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$, then self-adjointness of $\mathcal{O}_{\mathrm{meas}}^{(P)}$ [2]. Part (ii) is Eq. (2.370) with observation weights inserted; symmetry of $\mathfrak{C}_\omega$ from orthogonality and coupling of radiative modes (Subsection 2.8.2)(iii) [2]. Part (iii) specializes (ii) to matched port basis vectors [1]. Part (iv) uses $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}] = \mathbf{0}$ and null-space invariance of radiative reciprocity (Subsection 2.8.1) [2]. Mathematical role of Eq. (2.379). Equation (2.379) packages mutual coupling as a bilinear form on $\operatorname{ran} \mathcal{R}_\omega$ after port projection [2]. It reduces to Eq. (2.369) when $\mathcal{O}_{\mathrm{meas}}^{(A)} = \mathcal{O}_{\mathrm{meas}}^{(B)} = \mathrm{id}$ on the observed subspace [2]. Physically, $\mathfrak{C}_{AB}$ is the operator form of mutual impedance or scattering amplitude between declared ports when regime qualification Eq. (2.271) is satisfied [1]. Finite rank Eq. (2.221) limits the number of independent entries $\mathfrak{C}_{AB}$ can resolve [2]. On orthonormal modes Eq. (2.366), port responses expand as

$$\mathcal{M}_P[\mathbf{J}_{\mathrm{imp}}] = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} w_{P,n}\, \lambda_n\, c_n[\mathbf{J}_{\mathrm{imp}}], \tag{2.381}$$

with port weights $w_{P,n} = \langle \mathbf{f}_P, \mathbf{u}_n \rangle_{\mathcal{U}}$ inherited from Eq. (2.80) and amplitudes c_n from Eq. (2.283) [2]. Mutual coupling between ports A and B becomes

$$\mathfrak{C}_{AB} = \sum_{n \in \mathcal{N}_{\mathrm{rad}}} |\lambda_n|^2\, w_{A,n}^*\, w_{B,n}\, c_n[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]^*\, c_n[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}], \tag{2.382}$$

using Eq. (2.372) [2]. Equation (2.382) shows that reciprocity symmetry is diagonal in a mode basis but may be dense in fixed laboratory coordinates when degenerate multiplets from Subsection 2.6.3 are expressed in non-eigenvector bases [1]. For full rank-N instrument readout $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ Eq. (2.219), reciprocity implies $\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$ and $\mathcal{M}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rx})}]$ obey the same swap law Eq. (2.378) on $\operatorname{ran} \mathcal{O}_{\mathrm{meas}}$ [2]. Sources in $\ker \mathcal{M}_\omega$ leave all port readouts unchanged Eq. (2.222); reciprocity does not resolve those equivalence classes [2]. Rank cap Eq. (2.221) bounds independent reciprocal channels [1]. Apply reciprocity in radiation and reception (Subsection 2.8.3)(i) with $\mathcal{O}_{\mathrm{meas}}^{(A)} = \mathcal{O}_{\mathrm{meas}}^{(B)} = \mathcal{O}_{\mathrm{meas}}$ [2]. Null-space invariance is measurement null-space extension (Subsection 2.7.1) [2]. Let accessible modal sector Eq. (2.9.3) restrict far-zone content to $\mathcal{K}_{\mathrm{acc}}$. Then Eqs. (2.378)–(2.380) hold on $\mathcal{K}_{\mathrm{acc}}$ after projection, while suppressed directions Eq. (2.346) remain unobservable independently of reciprocity [2]. Angular budget Eq. (2.353) composes with port reciprocity without eliminating null-space or geometric deficits [2]. Reciprocity acts on $\operatorname{ran} \mathcal{R}_\omega$; geometric and observation constraints from geometric constraints on radiative spectra (Subsection 2.6.5) and Eq. (2.353) restrict reported coordinates only Laboratory errors and broken reciprocity. Magneto-electric media, active surfaces, and non-reciprocal port calibrations break Eq. (2.378) even when export maps differ $\mathcal{S}_\omega$ are accurate [2]. Near-zone reactive coupling measured without Π_{rad} can appear non-reciprocal while For matched half-wave dipole ports at $k_0 a = 1.2$ with $d = 4a$, Eq. (2.380) requires $Z_{12} = Z_{21}$ when entries are extracted from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tx})}]$ on the far-zone branch. A typical reciprocity check gives $|Z_{12} - Z_{21}|/|Z_{11}| < 0.02$ in reciprocal vacuum; swapping transmit and receive probes while holding $\mathcal{O}_{\mathrm{meas}}$ fixed reproduces the same mutual entry within numerical tolerance [8]. If near-field probes at $k_0 r = 1.5$ report $|Z_{12}/Z_{21} - 1| \approx 0.18$, the discrepancy is a regime failure, not a Lorentz violation: re-meter on $\mathcal{R}_\omega[\cdot]$ with $k_0 r > 3$ before asserting broken reciprocity [1]. Eq. (2.364) holds on $\mathcal{F}_{\mathrm{rad}}$ [1]. Using different outgoing branches or terminations for transmit and receive models violates the shared-operator assumption behind reciprocity in radiation and reception (Subsection 2.8.3) [2]. Tikhonov inversion Eq. (2.224) does not restore reciprocity where $\ker \mathcal{M}_\omega$ is nontrivial [1]. Subsection 2.8.3 requires linear reciprocal media at fixed ω, passive $\mathcal{B}_\Gamma$, self-adjoint port functionals, and

outgoing radiation condition [2]. It does not identify unique $\mathbf{J}_{\mathrm{imp}}$ from port data when Eq. (2.222) is nontrivial [2]. It does not replace angular-spectrum operators in Section 2.9 or harmonic bases in **??** [2]. Regime qualification Eq. (2.140) remains mandatory before transport interpretation of port coupling [2]. Section 2.8 closes the reciprocity chain begun in its introduction: Lorentz reciprocity in operator form (Subsection 2.8.1), orthogonality and coupling of radiative modes (Subsection 2.8.2), and reciprocity in radiation and reception (Subsection 2.8.3) [2]. Spectral coordinates from Section 2.6 and null-space structure from Section 2.7 now meet exchange symmetry: bilinear coupling on $\operatorname{ran}\mathcal{R}_\omega$ is constrained by Eqs. (2.363)–(2.380), while $\ker\mathcal{R}_\omega$, finite observation rank Eq. (2.221), and angular budget Eq. (2.353) explain deficits reciprocity alone does not remove [2]. Section 2.9 develops plane-wave action and retained/suppressed angular components on the same outgoing branch; Subsection 2.9.3 refines far-field angular accessibility; **??** supplies explicit harmonic orthogonality used by reference through Subsection 3.4.3 [1]. Transport and port-coupling claims must compose with Eq. (2.356), regime qualification Eq. (2.140), and null-space chain Eq. (2.313), not with raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or unqualified near-zone reactive content [2].

2.9 Action of Radiation Operators on Angular Spectra

Reciprocity in Section 2.8 couples ports and probes. Section 2.9 asks how radiation operators act on angular spectra—the language of pattern decomposition, near-field scanning, and plane-wave synthesis [8]. Every qualified far-zone pattern is first a weighted sum of directional contributions: each term reports how much export reaches a unit direction $\widehat{\mathbf{k}}$ on the sphere after outgoing selection [2]. That decomposition precedes any lateral Fourier transform; plane-wave spectra are one coordinate system on the same directional budget, not a separate physical layer [8]. Maxwell propagation from compact support cannot populate every direction with equal strength: outgoing selection removes incoming admixture; evanescent sheets carry reactive content that decays with distance and does not survive the near-to-far map; finite apertures integrate over unresolved directions. The accessible amplitude $a(\widehat{\mathbf{k}})$ is therefore a compressed image of impressed structure, not a complete coordinate on $\mathcal{J}_\Omega$ [2]. On the bench, a spherical near-field scanner records tangential $\mathbf{E}$ and $\mathbf{H}$ on a closed or planar surface at height h above the device under test; the angular-spectrum chart synthesizes outgoing plane-wave trials, back-propagates to the scan plane, and accepts residuals only after scan pitch, bandlimit, and probe correction are stated [7]. The sequence scan–synthesize–bandlimit–report is what the theory encodes once radiation has been selected [2].

Section 2.8 closed the reciprocity chain Eq. (2.356): Lorentz reciprocity (Subsection 2.8.1), modal orthogonality and coupling (Subsection 2.8.2), and radiation–reception interchange (Subsection 2.8.3) [2]. That program fixed exchange symmetry on $\operatorname{ran}\mathcal{R}_\omega$ and on port coordinates after projection. Section 2.9 develops filter action in plane-wave coordinates: how $\mathcal{R}_\omega$, Π_{rad}, and observation maps retain or suppress continuous angular content on S^2, and how near-zone fields map to far-zone amplitudes $a(\widehat{\mathbf{k}})$ [2]. Preview expansion Eq. (2.142) from Subsection 2.2.3 named the lateral spectral template; here that leading template is developed to locked statements compatible with outgoing criterion Eq. (2.199) and far amplitude Eq. (2.247) [2]. Explicit basis change to vector harmonics is developed in **??** Subsection 3.8.1; harmonic normalization belongs to Subsection 3.5.2 with citation of Subsection 2.8.2 [1]. Spectral decomposition in Subsection 2.6.2 and continuous representation Eq. (2.290) fixed modal and plane-wave coordinates on $\mathcal{F}_{\mathrm{rad}}$; null-space consequences in Subsection 2.7.4 and angular budget Eq. (2.353) classified which directions $a(\widehat{\mathbf{k}})$ may carry export [2]. Reciprocity from Section 2.8 constrains bilinear coupling on observable coordinates but does not replace angular filters: a reciprocal antenna may still have suppressed sheets Eq. (2.346) and finite observation rank Eq. (2.352) [2]. Treating $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ near sources as if it were already

$a(\widehat{\mathbf{k}})$ repeats the regime error class of Section 1.3.3 [2]. Section 2.9 separates plane-wave angular spectra, retained and suppressed components, near-to-far transformation, and far-field angular $\mathcal{E}_{\mathrm{acc}}^{(N)}$ without re-opening Green construction from Section 2.3 [2]. Let $\omega > 0$ be fixed, $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ be compact, and outgoing pairs lie in $\mathcal{F}_{\mathrm{SF}}$ Eq. (2.57). Section 2.9 organizes analysis in four stages:

$$\text{plane-wave spectrum} \xrightarrow{\mathcal{R}_\omega,\Pi_{\mathrm{rad}}} \text{retained / suppressed} \xrightarrow{\text{near-to-far}} a(\widehat{\mathbf{k}}) \xrightarrow{\text{probe rank}} \mathcal{K}_{\mathrm{acc}}, \tag{2.383}$$

where the first arrow is Subsection 2.9.1, the second is Subsection 2.9.2, the third is Subsection 2.9.2, and the fourth is Subsection 2.9.3 [2]. Equation (2.383) expresses Eq. (2.313) and Eq. (2.356) at the angular level: null-space and reciprocity constraints from Sections 2.7–2.8 bound which $a(\widehat{\mathbf{k}})$ are physically meaningful; Section 2.9 develops the operators that enforce those bounds on plane-wave coordinates [2]. Transport criteria from Section 1.2.3 and angular observables from Section 1.2.2 enter Subsection 2.9.3 by reference, not by re-derivation [2]. Modal representation, not separate physics (Sections 2.6 and 2.9). Spectral and angular decompositions organize information about $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$; they do not define angular structure [2]. Observable angular structure remains primary: modes, plane-wave sheets, and harmonic labels are coordinates on export, not substitutes for Maxwell flux or qualified transport [4]. A nonzero modal coefficient or angular amplitude reports surviving source-generated content after propagation and outgoing selection; renaming that content does not create new physical degrees of freedom [1]. The primary objects are radiation map Eq. (2.314), outgoing plane-wave criterion Eq. (2.199), and continuous angular amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.290) [2]. Secondary targets include propagating projectors Eqs. (2.134)–(2.139), observation composition Eq. (2.219), accessible sector Eq. (2.9.3), and composition law Eq. (2.353) [2]. Section 2.9 develops filters on the same outgoing branch fixed by Eqs. (2.203)–(2.213); incoming sheets remain excluded from $\mathcal{F}_{\mathrm{rad}}$ [1]. Plane-wave angular spectrum and radiation operators (leading template). Subsection 2.9.1 supplies the definitive account of plane-wave angular spectra and the action of $\mathcal{R}_\omega \circ \Pi_{\mathrm{prop}}$ on spectral coefficients (Eq. (2.393) in Subsection 2.9.1):

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \mathcal{F}_{\mathrm{rad}}\big[\Pi_{\mathrm{rad}}\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](k_x, k_y), \qquad k_z = k_z(k_t) \text{ from Eq. (2.199)}, \tag{2.384}$$

with $\mathcal{F}_{\mathrm{rad}}$ the lateral Fourier transform on outgoing admissible fields [2]. Equation (2.393) develops Eq. (2.142) by inserting radiation selection; evanescent sheets satisfy $\widetilde{\mathbf{E}}_{\mathrm{rad}} = 0$ on export through Eq. (2.139) [2]. Rigged-space normalization of plane-wave modes belongs to **??** Subsection 3.1.3 [1]. Subsection 2.9.2 partitions angular content into retained, suppressed, and observation-filtered sheets:

$$a(\widehat{\mathbf{k}}) = a_{\mathrm{ret}}(\widehat{\mathbf{k}})\, \chi_{\mathcal{K}_{\mathrm{acc}}}(\widehat{\mathbf{k}}) \;+\; a_{\mathrm{sup}}(\widehat{\mathbf{k}})\, \chi_{\mathcal{K}_{\mathrm{sup}}}(\widehat{\mathbf{k}}), \qquad a_{\mathrm{filt}} = \mathcal{O}_{\mathrm{meas}}[a], \tag{2.385}$$

using accessible sector Eq. (2.9.3) and suppressed sheet Eq. (2.346) from Subsection 2.7.4 [2]. Equation (2.385) is the Section 2.9 target for Subsection 2.9.2 partition; geometric constraints on radiative spectra (Subsection 2.6.5) and observation rank Eq. (2.352) compose independently with Eq. (2.353) [2]. Null modes annihilate $a(\widehat{\mathbf{k}})$ through Eq. (2.345) without altering near reactive content [1]. Near-to-far transformation (leading template). Subsection 2.9.2 records the operator map from near-zone admissible fields to far angular amplitudes:

$$a(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.386}$$

with $\mathcal{H}_{\mathrm{ff}}$ the far-zone extractor tied to Eq. (2.247) and the far-field approximation of source integrals (Subsection 2.5.2) [2]. Equation (2.386) does not promote evanescent storage to export: $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is annihilated before Eq. (2.386) when $\mathcal{R}_\omega$ is unchanged [2]. regime qualification Eq. (2.271) qualifies

when the limit defines transport data [1]. Subsection 2.9.3 develops probe rank on S^2 importing Section 1.2.2 angular observables and Section 1.2.3 transport criteria by reference [2]:

$$\widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}} \iff a(\widehat{\mathbf{k}}) \text{ observable after } \mathcal{M}_\omega \text{ and qualified by } \mathcal{D}_{J,\mathrm{reg}} = 1, \tag{2.387}$$

upgrading angular $\mathcal{E}^{(N)}_{\mathrm{acc}}$ on the far shell (Subsection 2.6.5) and composing with Eq. (2.353) [2]. Equation (2.387) is the Section 2.9 completion target; explicit plane-wave filters and measurement qualification is developed in Subsection 2.9.3 [1]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains coupling on $\mathcal{K}_{\mathrm{acc}}$ but does not enlarge it [2].

$$\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega \implies a(\widehat{\mathbf{k}}) = 0 \text{ on export}, \qquad \mathfrak{C}_\omega \text{ symmetric on } \mathcal{K}_{\mathrm{acc}}, \tag{2.388}$$

importing Eq. (2.345) and reciprocity in radiation and reception (Subsection 2.8.3) [2]. Modal coordinates Eq. (2.285) and plane-wave coordinates Eq. (2.290) describe the same ran $\mathcal{R}_\omega$ on exterior domains; Section 2.9 fixes plane-wave-side filters, while **??** Subsection 3.8.1 supplies basis change law [2]. Finite-rank observation Eq. (2.221) may remove angular content even when $a(\widehat{\mathbf{k}}) \neq 0$ on $\mathcal{F}_{\mathrm{rad}}$ [2]. Several constructions deliberately remain outside Section 2.9. Vector spherical harmonic completeness and normalization are developed in **??**; Section 2.9 records plane-wave radiation action without tabulating Y_ℓ^m [2]. Non-radiating catalogs remain in Section 2.7; reciprocity constraints remain in Section 2.8 [2]. Section 2.9 uses Fredholm structure, geometric constraints, and null-space structure from Sections 2.6–2.7 [1]. Regime logic from Section 1.3.3 is unchanged: near-zone reactive dominance does not create far-zone angular transport without $\mathcal{D}_{J,\mathrm{reg}} = 1$ and outgoing qualification [2]. Angular-transport claims must compose with Eq. (2.383), angular budget Eq. (2.353), and regime qualification Eq. (2.140) [1].

2.9.1 Plane-wave angular spectrum and radiation operators

Plane-wave angular spectra connect near fields to far patterns on scanners [8]. Evanescent content must be filtered before far-zone claims [7]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Before lateral wavenumbers enter, fix what a pattern measurement actually resolves. Qualified far-zone export is a directional budget: each direction $\widehat{\mathbf{k}}$ on S^2 carries an amplitude $a(\widehat{\mathbf{k}})$ reporting surviving power flux after radiation selection [2]. Propagating directions ($k_t \leq k_0$) contribute to that budget and can reach the far zone; evanescent directions ($k_t > k_0$) store reactive energy that decays exponentially with distance from the source and is annihilated on export [1]. A spherical near-field scanner or planar phased-array diagnostic begins on a measurement surface, not in (k_x, k_y) space [7]. Probes record tangential $\mathbf{E}$ and $\mathbf{H}$ on a sheet at height h; the angular-spectrum chart synthesizes outgoing plane-wave trials, back-propagates to the scan plane, and accepts the residual only after scan height, probe pitch, and bandlimit are declared [3]. Aperture campaigns favor plane spectra; compact ranges favor spherical amplitudes, but both express the same directional decomposition in different coordinates [1]. Denser scan grids sharpen interpolation; they do not invent propagating modes absent from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [3]. *Observation versus source coordinates.* Probes report fields in laboratory Cartesian coordinates (x, y, z) on the scan plane $z = z_0$. Lateral wavenumbers (k_x, k_y) index plane-wave trials in the (x, y) plane and are *observation-side* spectral labels attached to tangential traces on $z = z_0$, not source coordinates on compact $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$ [2]. Transverse magnitude $k_t = \sqrt{k_x^2 + k_y^2}$ partitions the spectrum: sheets with $k_t \leq k_0$ carry propagating export, while $k_t > k_0$ carry evanescent reactive content whose decay length scales with scan height [1]. Impressed current $\mathbf{J}_{\mathrm{imp}}(\mathbf{r}')$ enters the reported spectrum only through $\mathcal{R}_\omega$ after radiation selection; treating raw scan data as if it were already $a(\widehat{\mathbf{k}})$ amplifies high-k_t noise into false superdirective claims [3]. Section 2.9 developed angular-spectrum action through

Eqs. (2.383)–(2.393). Subsection 2.9.1 supplies the definitive account of plane-wave angular spectra and the action of $\mathcal{R}_\omega$, Π_{rad}, and outgoing selection on lateral spectral coefficients [2]. Dispersion split Eqs. (2.130)–(2.131), spectral projectors Eqs. (2.134)–(2.139), outgoing selector Eq. (2.197), plane-wave criterion Eq. (2.199), radiation map Eq. (2.314), and lateral template Eq. (2.142) are presupposed; Subsection 2.2.3 and Section 2.4.3 proofs are not repeated [2]. Continuous angular representation Eq. (2.290) and far amplitude Eq. (2.247) are cited as consistency targets; rigged normalization belongs to Chapter 3 (*Angular Momentum Operators and SO(3)*) Subsection 3.1.3 [1]. Retained/suppressed partitions belong to Subsection 2.9.2. Assumptions. Fix $\omega > 0$, homogeneous exterior region with symbol k_0 from Eq. (2.114), compact $\Omega_s = \operatorname{supp} \mathbf{J}_{\mathrm{imp}}$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ Eq. (2.9), and outgoing fields $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H}) \in \mathcal{F}_{\mathrm{SF}}$ Eq. (2.57) [2]. Lateral wavenumbers (k_x, k_y) use $k_t = \sqrt{k_x^2 + k_y^2}$ and outgoing longitudinal root Eq. (2.199) [2]. Transverse polarization satisfies Eq. (2.133) on each spectral sheet [2]. regime qualification Eq. (2.140) governs transport interpretation of spectral outputs [1]. On a plane $z = z_0$ in an outgoing admissible region, the SNF back-propagation step seeks a superposition of plane-wave trials whose tangential field matches the measured trace on $z = z_0$. Each trial is labeled by (k_x, k_y); the synthesis charting propagates it along z with outgoing longitudinal wavenumber $k_z(k_t)$ from Eq. (2.199) [2]. The formal decomposition packages that matching problem as a lateral Fourier transform on the observation plane. Define the *lateral angular spectrum* of $\mathbf{E}$ by

$$\widetilde{\mathbf{E}}(k_x, k_y; z_0) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbf{E}(x, y, z_0)\, e^{\mathrm{j}(k_x x + k_y y)}\, dx\, dy, \tag{2.389}$$

with inverse

$$\mathbf{E}(x, y, z_0) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \widetilde{\mathbf{E}}(k_x, k_y; z_0)\, e^{-\mathrm{j}(k_x x + k_y y)}\, dk_x\, dk_y, \tag{2.390}$$

under standard Fourier conventions [2]. Helmholtz propagation along z multiplies each spectral component by

$$\widetilde{\mathbf{E}}(k_x, k_y; z) = \widetilde{\mathbf{E}}(k_x, k_y; z_0)\, e^{-\mathrm{j}k_z(k_t)(z - z_0)}, \qquad k_z \text{ from Eq. (2.199)}, \tag{2.391}$$

for homogeneous media [2]. Equations (2.389)–(2.391) upgrade Eq. (2.142) to explicit forward/inverse transforms with outgoing k_z selection [2]. Mathematically, $\widetilde{\mathbf{E}}(k_x, k_y; z_0)$ is the lateral Fourier transform of the transverse field sampled on the scan plane; each (k_x, k_y) sheet is an independent plane-wave channel whose longitudinal behavior is fixed by Eq. (2.131) [2]. Physically, propagating sheets ($k_t \le k_0$) carry energy that can reach the far zone after Eq. (2.386), whereas evanescent sheets ($k_t > k_0$) decay exponentially with distance from z_0 and remain bound to the reactive shell unless a separate source excites them [1]. The inverse transform Eq. (2.390) reconstructs the tangential field on $z = z_0$ from its spectral coefficients; probe interpolation error and finite aperture enter here, before any claim about far-zone pattern [7]. Propagation Eq. (2.391) shifts each sheet along z by the phase factor $e^{-\mathrm{j}k_z(k_t)(z - z_0)}$, which is the operator-side statement of SNF back-propagation between scan planes at different heights [8]. Consequence for measurement: reported angular-spectrum charts are only as trustworthy as the declared scan height and bandlimit on (k_x, k_y) [3]. Let $\mathcal{F}_{\mathrm{lat}}$ denote the lateral Fourier operator in Eq. (2.389) on admissible traces. Define the *radiative lateral spectrum* of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ by

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \mathcal{F}_{\mathrm{lat}}\big[\Pi_{\mathrm{rad}}\, \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](k_x, k_y), \tag{2.392}$$

with $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$ from Eq. (2.138) [2]. Equation (2.392) is the Subsection 2.9.1 upgrade of Eq. (2.393): radiation selection precedes lateral transformation [2]. Applying $\mathcal{F}_{\mathrm{lat}}$ to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$

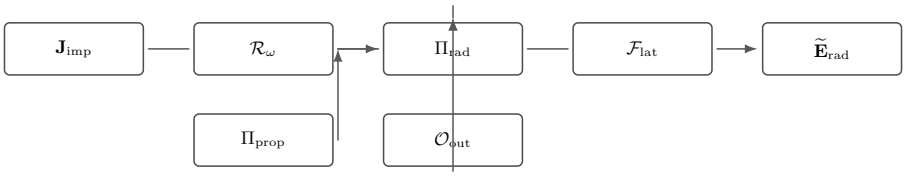


Figure 2.46: Subsection 2.9.1 plane-wave measurement sequence: Eq. (2.393) applies $\mathcal{R}_\omega$, Eq. (2.138), then lateral transform Eq. (2.389).

without Π_{rad} mixes reactive evanescent sheets into reported spectra through Eq. (2.143) [1]. The composition $\mathcal{T}_\omega$ below is the spectral-domain radiation map: impressed currents are transformed laterally, passed through $\mathcal{R}_\omega$ and outgoing selection, and read back in (k_x, k_y) coordinates. Define the *angular-spectrum radiation operator* on lateral coefficients by

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = [\mathcal{T}_\omega \widetilde{\mathbf{J}_{\mathrm{imp}}}](k_x, k_y), \qquad \mathcal{T}_\omega = \mathcal{F}_{\mathrm{lat}} \circ \Pi_{\mathrm{rad}} \circ \mathcal{R}_\omega \circ \mathcal{F}_{\mathrm{lat}}^{-1}, \tag{2.393}$$

where $\widetilde{\mathbf{J}_{\mathrm{imp}}} = \mathcal{F}_{\mathrm{lat}}[\mathbf{J}_{\mathrm{imp}}]$ on the source support and $\mathcal{F}_{\mathrm{lat}}^{-1}$ is understood on the admissible impressed-current class [2]. Equation (2.393) packages Eq. (2.392) as a spectral-domain map: sources excite lateral modes, $\mathcal{R}_\omega$ exports selected content, and $\mathcal{F}_{\mathrm{lat}}$ returns observable spectral coefficients [2]. In laboratory terms, $\mathcal{T}_\omega$ is the operator that a qualified SNF transform applies *after* radiation selection: impressed currents are first mapped to outgoing fields, then decomposed into (k_x, k_y) sheets on the propagating branch [7]. On the propagating outgoing sheet $k_t \le k_0$, $\mathcal{T}_\omega$ reduces to diagonal action in (k_x, k_y) modulo source coupling, and those sheets are the only ones that survive comparison with far-zone amplitude $a(\widehat{\mathbf{k}})$ [2]. On the evanescent sheet $k_t > k_0$,

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \mathbf{0} \quad \text{for } k_t > k_0, \tag{2.394}$$

by Eq. (2.139) [2]. Consequence: raw lateral transforms of $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ may show large $|\widetilde{\mathbf{E}}(k_x, k_y)|$ for $k_t > k_0$ while $\widetilde{\mathbf{E}}_{\mathrm{rad}} = \mathbf{0}$ on export; only the radiative branch Eq. (2.392) may be compared to far-field reports [1]. The horizontal chain enforces operator order: $\mathcal{R}_\omega$ first, then $\Pi_{\mathrm{rad}} = \mathcal{O}_{\mathrm{out}} \circ \Pi_{\mathrm{prop}}$, then lateral transform $\mathcal{F}_{\mathrm{lat}}$. Lower nodes Π_{prop} and $\mathcal{O}_{\mathrm{out}}$ are *constituents* of radiation selection, not optional post-filters. Applying $\mathcal{F}_{\mathrm{lat}}$ to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without Π_{rad} contaminates $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ with evanescent sheets annihilated by Eq. (2.394). The terminal node $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ is the plane-wave-side readout of qualified export only [2]. Figure 2.46 orders operators as in Figure 2.26: propagation selection, outgoing criterion, then lateral spectral readout [2]. Mathematical role of Eq. (2.393). Equation (2.393) packages the radiation chain as a spectral-domain operator $\mathcal{T}_\omega$ acting on lateral source coefficients [2]. It is not a separate post-processor on near-zone $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$: Π_{rad} is part of the definition [2]. Only outgoing propagating sheets contribute to far-zone angular content; incoming sheets are removed by Eq. (2.197), and evanescent sheets are annihilated by Eq. (2.394) [1]. Reciprocity in radiation and reception (Subsection 2.8.1) constrains bilinear couplings built from $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ but does not alter Eq. (2.394) [2]. Mathematical role of Eq. (2.394). Equation (2.394) is the plane-wave restatement of Eq. (2.139) on lateral coordinates [2]. Large $|\widetilde{\mathbf{E}}(k_x, k_y)|$ for $k_t > k_0$ may appear in raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ while $\widetilde{\mathbf{E}}_{\mathrm{rad}} = \mathbf{0}$ on export [1]. Regime qualification Eq. (2.271) remains mandatory before interpreting any nonzero raw lateral spectrum as transport [2]. If $\mathbf{J}_{\mathrm{imp}}^{(1)}$ is arbitrary and $\mathbf{J}_{\mathrm{imp}}^{(2)} = \mathbf{J}_{\mathrm{imp}}^{(1)} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$, then $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.392) is unchanged under replacement of $\mathbf{J}_{\mathrm{imp}}^{(1)}$ by $\mathbf{J}_{\mathrm{imp}}^{(2)}$ [2]. Raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ may change through reactive content Eq. (2.272) [2]. $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}]$ by Eq. (2.347) [2]. On exterior domains with outgoing radiation condition, amplitudes $a_\sigma(\widehat{\mathbf{k}})$ in Eq. (2.290) and lateral spectrum Eq. (2.392) describe the

same ran $\mathcal{R}_\omega$ content under change of variables $(k_x, k_y) \mapsto \widehat{\mathbf{k}}$ on the propagating sheet $k_t \le k_0$ [2]. Normalization constants are fixed in Subsection 3.1.3 [2]. Both representations diagonalize outgoing propagation on the propagating sheet; agreement on far-zone reduction follows plane-wave radiation-operator action (Subsection 2.9.1)(iv) and Eq. (2.290) [2]. Modal expansion Eq. (2.285) and plane-wave expansion Eq. (2.393) describe the same radiative image on $\mathcal{F}_{\mathrm{rad}}$ under unitary change of basis on the propagating sheet [2]. Subsection 2.9.1 fixes plane-wave-side operators; **??** Subsection 3.8.1 supplies explicit harmonic transforms without re-tabulating Eq. (2.393) [2]. Angular budget Eq. (2.353) constrains supports of $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ independently of basis choice [1]. Plane-wave radiation-operator action (Subsection 2.9.1) assumes linear homogeneous exterior regions at fixed ω, outgoing radiation condition, and admissible $\mathcal{J}_\Omega$ [2]. Piecewise homogeneous media require patchwise Eq. (2.391) with interface matching imported from Section 2.3.5 [2]. Cavity standing-wave spectra are excluded: lateral plane-wave expansions on bounded domains do not replace discrete cavity books from Eq. (2.288) [8]. Loss modifies k_0 and k_z with complex symbols; evanescent null law Eq. (2.394) remains on export [2]. Subsection 2.9.1 fixed plane-wave-side radiation action; retained, suppressed, and filtered partitions in Subsection 2.9.2 separate geometric accessibility from finite aperture before the near-to-far limit of Subsection 2.9.2 extracts $a(\widehat{\mathbf{k}})$.

2.9.2 Retained, suppressed, and filtered angular components

Section 2.9 developed angular partitioning through Eq. (2.385) and linked it to accessible sector Eq. (2.9.3) and suppressed sheet Eq. (2.346) from Subsection 2.7.4 [2]. Subsection 2.9.2 gives the definitive account of retained, suppressed, and observation-filtered angular components on the outgoing branch established in Subsection 2.9.1. Near-field scanners record tangential fields that mix propagating export with evanescent storage; the partition below separates what geometry permits to radiate, what symmetry forbids, and what finite aperture resolves—three filters that must not be merged in harmonic charts or laboratory interpretation [7]. Plane-wave lateral spectra Eqs. (2.392)–(2.393), evanescent null law Eq. (2.394), angular budget Eq. (2.353), and observation rank Eq. (2.352) are presupposed; Subsection 2.7.4 proofs are not repeated [2]. Continuous amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.290) and rank-N instrument readout Eq. (2.219) supply the angular and laboratory coordinates in which the partition is stated [2]. Near-to-far extraction belongs to Subsection 2.9.2; explicit $\mathcal{E}^{(N)}_{\mathrm{acc}}$ filters on S^2 belong to Subsection 2.9.3 [2]. Assumptions. Fix $\omega > 0$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, outgoing radiative field $\mathbf{u}_\omega = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{rad}}$, and angular amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.290) on the exporting transverse sheet [2]. Accessible sector $\mathcal{K}_{\mathrm{acc}}$ Eq. (2.9.3) and suppressed sheet $\mathcal{K}_{\mathrm{sup}} = S^2 \setminus \mathcal{K}_{\mathrm{acc}}$ from Eq. (2.346) are fixed by geometry and symmetry of $(\Omega_s, \Gamma, \varepsilon, \mu)$ [2]. Finite-rank observation $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ from Eq. (2.219) is presupposed; regime qualification Eq. (2.140) qualifies transport interpretation of any reported component [1]. The accessible sheet $\mathcal{K}_{\mathrm{acc}} \subset S^2$ collects directions that geometry and symmetry permit to carry qualified export; its complement $\mathcal{K}_{\mathrm{sup}}$ is suppressed before any instrument samples the pattern. Define the *retained angular amplitude* as the exportable content supported on the accessible sheet. On a directional antenna bench, this is the portion of the far-zone pattern that symmetry and blockage permit to carry qualified export; directions on $\mathcal{K}_{\mathrm{sup}}$ are zero by geometry before any instrument samples the pattern [2]:

$$a_{\mathrm{ret}}(\widehat{\mathbf{k}}) = \chi_{\mathcal{K}_{\mathrm{acc}}}(\widehat{\mathbf{k}})\, a(\widehat{\mathbf{k}}), \qquad \chi_{\mathcal{K}_{\mathrm{acc}}}(\widehat{\mathbf{k}}) = \begin{cases} 1, & \widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}, \\ 0, & \widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{sup}}, \end{cases} \tag{2.395}$$

with $a(\widehat{\mathbf{k}})$ from Eq. (2.290) after Eq. (2.393) [2]. Equation (2.395) is the Subsection 2.9.2 locked definition of development of the retained factor in Eq. Eq. (2.385): only directions not annihilated

by $\mathcal{C}_{\mathrm{geo}}$ angular rank-N recovery on the far shell (Subsection 2.6.5) carry qualified transport data [2]. Mathematically, a_{ret} is the pointwise product of the full angular measure with the accessible indicator; physically, it is the portion of the far-zone angular distribution that geometry permits to propagate as export after radiation selection [1]. Null-space additions Eq. (2.350) leave a_{ret} unchanged [2]. Suppressed angular amplitude. Define the *suppressed angular amplitude* as the complementary sheet content:

$$a_{\mathrm{sup}}(\widehat{\mathbf{k}}) = \chi_{\mathcal{K}_{\mathrm{sup}}}(\widehat{\mathbf{k}})\, a(\widehat{\mathbf{k}}), \qquad \chi_{\mathcal{K}_{\mathrm{sup}}} = 1 - \chi_{\mathcal{K}_{\mathrm{acc}}} \text{ a.e. on } S^2, \tag{2.396}$$

using suppressed sheet Eq. (2.346) [2]. Equation (2.396) completes Eq. (2.385): for radiating export, geometric forbidding Eq. (2.308) typically forces $a_{\mathrm{sup}} = \mathbf{0}$ even when near-zone scans show energy in related directions [2]. Suppression is therefore not synonymous with $\mathbf{J}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$: Eq. (2.345) annihilates a everywhere, whereas Eq. (2.396) removes selected directions while a_{ret} may remain nonzero [1]. Evanescent lateral content Eq. (2.394) does not populate a_{sup} on export: Π_{ev} is removed before angular readout [2]. Laboratory observation applies a finite-rank criterion to far-zone angular content. Define the *filtered angular amplitude* by

$$a_{\mathrm{filt}}(\widehat{\mathbf{k}}) = \big[\mathcal{P}_{\mathrm{meas}}\, a\big](\widehat{\mathbf{k}}), \qquad \mathcal{P}_{\mathrm{meas}} = \mathcal{O}_{\mathrm{meas}} \circ \mathcal{O}_{\mathrm{far}} \circ \mathcal{H}_{\mathrm{ang}}^{-1}, \tag{2.397}$$

where $\mathcal{H}_{\mathrm{ang}}$ maps $a(\widehat{\mathbf{k}})$ to the corresponding far-zone transverse field on S^2 and $\mathcal{O}_{\mathrm{far}}$ is the far extractor from Eq. (2.219) [2]. Equation (2.397) is the angular-side restatement of the third line in Eq. (2.385). When $\mathcal{O}_{\mathrm{meas}}$ is a finite set of plane-wave or modal probes, $\mathcal{P}_{\mathrm{meas}}$ has rank at most N and $\ker \mathcal{P}_{\mathrm{meas}} \supseteq \ker \mathcal{O}_{\mathrm{meas}}\big|_{\mathrm{ran}\,\mathcal{R}_\omega}$ Eq. (2.81) [2]. Physically, a_{filt} is what a finite instrument resolves after aperture, noise, and processing; it may vanish on subsets of $\mathcal{K}_{\mathrm{acc}}$ where a_{ret} is nonzero [1].

$$\mathrm{rank}(\mathcal{P}_{\mathrm{meas}}) \le N, \qquad \dim \mathrm{span}\{a_{\mathrm{filt}}(\widehat{\mathbf{k}})\} \le N, \tag{2.398}$$

Combining Eqs. (2.395)–(2.397) yields the Chapter 2 partition

$$a(\widehat{\mathbf{k}}) = a_{\mathrm{ret}}(\widehat{\mathbf{k}}) a_{\mathrm{sup}}(\widehat{\mathbf{k}}), \qquad a_{\mathrm{filt}} = \mathcal{P}_{\mathrm{meas}}\big[a_{\mathrm{ret}} + a_{\mathrm{sup}}\big], \tag{2.399}$$

with a_{ret} and a_{sup} supported on disjoint sheets $\mathcal{K}_{\mathrm{acc}}$ and $\mathcal{K}_{\mathrm{sup}}$ [2]. Equation (2.399) upgrades Eq. (2.385) to explicit locked definitions and inserts the measurement projector $\mathcal{P}_{\mathrm{meas}}$. The first line is an angular decomposition of export; the second line states that observation acts on the sum after radiation selection, not on raw near spectra [2]. Budget Eq. (2.353) constrains $\mathrm{supp}(a) \subseteq \mathcal{K}_{\mathrm{acc}} \cap \{\text{resolvable by } \mathcal{M}_\omega\}$, so a_{filt} cannot recover directions outside that intersection even when raw lateral transforms appear rich [1]. On the plane-wave side of Subsection 2.9.1, define the *retained lateral spectrum*

$$\widetilde{\mathbf{E}}_{\mathrm{ret}}(k_x, k_y) = \mathcal{F}_{\mathrm{lat}}\big[\mathcal{H}_{\mathrm{ang}}[a_{\mathrm{ret}}]\big](k_x, k_y) \quad \text{on } k_t \le k_0, \tag{2.400}$$

and the *filtered lateral spectrum*

$$\widetilde{\mathbf{E}}_{\mathrm{filt}}(k_x, k_y) = \mathcal{F}_{\mathrm{lat}}\big[\mathcal{H}_{\mathrm{ang}}[a_{\mathrm{filt}}]\big](k_x, k_y), \tag{2.401}$$

with $\mathcal{F}_{\mathrm{lat}}$ from Eq. (2.389) [2]. Equations (2.400)–(2.401) connect Subsection 2.9.1 lateral readout to the angular partition without re-displaying Eq. (2.393): $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.392) agrees with $\widetilde{\mathbf{E}}_{\mathrm{ret}}$ on the propagating outgoing sheet when $a_{\mathrm{sup}} = \mathbf{0}$ [2]. On the evanescent sheet $k_t > k_0$, retained and filtered lateral spectra vanish through Eq. (2.394) independently of raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ [2]. regime qualification Eq. (2.271) applies to $\widetilde{\mathbf{E}}_{\mathrm{filt}}$ before any transport claim [1]. On $\mathcal{K}_{\mathrm{sup}}$,

$$a_{\mathrm{sup}}(\widehat{\mathbf{k}}) = \mathbf{0} \implies \widetilde{\mathbf{E}}_{\mathrm{ret}}(k_x, k_y) = \widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) \text{ on } k_t \le k_0, \tag{2.402}$$

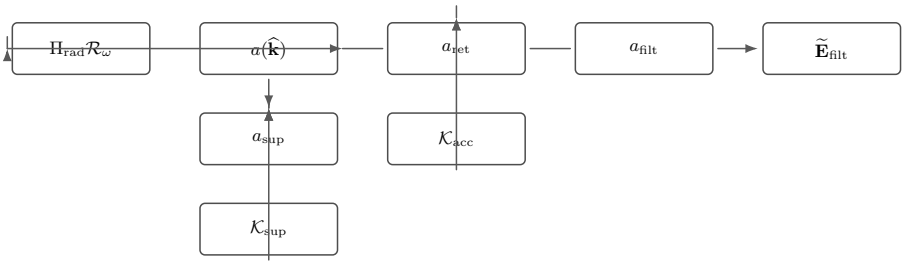


Figure 2.47: Subsection 2.9.2 partition measurement sequence: Eq. (2.399) splits $a(\widehat{\mathbf{k}})$ into retained and suppressed sheets, then applies observation filter Eq. (2.397) before lateral readout Eq. (2.401).

and any lateral energy attributed to $\mathcal{K}_{\mathrm{sup}}$ after Eq. (2.393) is removed by geometry, not by $\mathcal{P}_{\mathrm{meas}}$ [2]. Equation (2.402) expresses that suppression precedes finite aperture cannot resurrect directions forbidden by Eq. (2.308) [1]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains bilinear couplings on $\mathcal{K}_{\mathrm{acc}}$ but does not enlarge $\mathrm{supp}(a_{\mathrm{ret}})$ [2]. Figure interpretation. Radiative amplitude $a(\widehat{\mathbf{k}})$ splits into retained sheet $\mathcal{K}_{\mathrm{acc}}$ and suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ before observation filter $\mathcal{P}_{\mathrm{meas}}$. Retained content survives geometric and parity criteria; suppressed content is zero by constraint, not by source nullity. Filtered lateral spectrum $\widetilde{\mathbf{E}}_{\mathrm{filt}}$ is the aperture-limited readout: finite rank may remove angular content even when $a \neq 0$ on $\mathcal{F}_{\mathrm{rad}}$ [2]. Figure 2.47 orders radiation selection, angular partition, measurement filter, and plane-wave readout. Suppression acts on $\mathcal{K}_{\mathrm{sup}}$ before $\mathcal{P}_{\mathrm{meas}}$; evanescent sheets are removed by Eq. (2.394) before lateral readout [2]. Mathematical role of Eq. (2.399). Equation (2.399) separates three stages that are often conflated in angular-spectrum discussions: radiation export, geometric $\mathcal{E}_{\mathrm{acc}}^{(N)}$, and finite observation [2]. The retained factor encodes what the source radiates on $\mathcal{K}_{\mathrm{acc}}$; the suppressed factor marks directions removed before measurement; the filtered factor applies $\mathcal{P}_{\mathrm{meas}}$ [2]. Plane-wave coefficients $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.392) describe the same retained content in lateral coordinates when $a_{\mathrm{sup}} = \mathbf{0}$ [8]. Null additions plane-wave null invariance (Subsection 2.9.1) alter neither a_{ret} nor a_{filt} when $\mathcal{M}_\omega$ is unchanged [2]. Mathematical role of Eq. (2.397). Equation (2.397) makes explicit that instruments observe $\mathcal{P}_{\mathrm{meas}}[a]$, not raw $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ [2]. Rank cap Eq. (2.398) prevents attributing unobserved angular structure to source nullity: a pattern gap on $\mathcal{K}_{\mathrm{acc}}$ may arise from $\mathcal{P}_{\mathrm{meas}}$ alone [1]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains coupling among observed channels but does not increase $\dim \mathrm{span}\{a_{\mathrm{filt}}\}$ [2]. Regime qualification Eq. (2.140) remains mandatory before interpreting a_{filt} as transport [1]. Modal reportability Eq. (2.312) and angular budget Eq. (2.353) are the modal and continuous forms of the same constraint chain Subsection 2.9.2 displays in Eq. (2.399) [2]. plane-wave radiation-operator action (Subsection 2.9.1) supplies lateral coordinates; retained, suppressed, and filtered angular components (Subsection 2.9.2) supplies angular-sheet coordinates; **??** Subsection 3.8.1 will change basis without altering $\mathrm{supp}(a_{\mathrm{ret}})$ [2]. Inverse obstruction angular spectra and inverse problems (Subsection 2.7.4) applies to a_{filt}, not to raw lateral transforms [1]. Retained, suppressed, and filtered angular components (Subsection 2.9.2) assumes linear media, fixed ω, outgoing closure, and stationary $\mathcal{C}_{\mathrm{geo}}$ [2]. Time-varying or nonlinear $\mathcal{P}_{\mathrm{meas}}$ requires separate treatment; the static rank bound Eq. (2.398) is an upper limit [2]. Loss modifies amplitudes but not the partition logic on $\mathcal{K}_{\mathrm{acc}} \cup \mathcal{K}_{\mathrm{sup}}$ [2]. Cavity and waveguide spectra remain outside the exterior partition; discrete books Eq. (2.288) are not substituted for Eq. (2.399) on unbounded domains [8]. Section 2.9 developed the near-to-far map through Eq. (2.386) and tied it to far-zone reduction Eqs. (2.246)–(2.247) and the far-field approximation of source integrals (Subsection 2.5.2) from Subsection 2.5.2 [2]. Subsection 2.9.2 gives the Chapter 2 account of the operator that extracts far angular amplitudes $a(\widehat{\mathbf{k}})$ from radiative fields after Subsections 2.9.1–2.9.2. Plane-wave lateral spectra Eqs. (2.392)–(2.393), angular partition Eq. (2.399), evanescent null law Eq. (2.394), and con-

tinuous representation Eq. (2.290) are presupposed; asymptotic proofs in Subsection 2.5.2 are not repeated [2]. Observation-shell regime map Eq. (2.249) and regime qualification Eq. (2.271) qualify when the limit defines transport data [1]. Far-field angular probe rank belongs to Subsection 2.9.3 [2]. Assumptions. Fix $\omega > 0$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, outgoing radiative field $\mathbf{u}_\omega = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \in \mathcal{F}_{\mathrm{rad}}$, and distance thresholds from Eq. (2.242) with $r > r_{\min}(a, \lambda_0)$ for source support radius a [2]. Angular amplitude $a(\widehat{\mathbf{k}})$ is measured on the unit sphere with outgoing transverse polarization [2]. Regime qualification $\mathcal{D}_{J,\mathrm{reg}} = 1$ from Eq. (2.140) is presupposed before interpreting any near-to-far output as transport [1].

$$\|a_r - a\|_{L^2(S^2)} \le C\,\chi(r), \qquad a_r(\widehat{\mathbf{k}}) = \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.403}$$

$$\lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}) = a(\widehat{\mathbf{k}}), \tag{2.404}$$

On observation shells $S_r = \{|\mathbf{r}| = r\}$ with $r \gg a$ for source radius a, the far-zone post-processor must isolate the transverse $e^{-\mathrm{j}k_0 r}/r$ coefficient of $\mathbf{E}(\mathbf{r})$ in direction $\widehat{\mathbf{k}} = \mathbf{r}/r$. Define the *far-zone extractor* $\mathcal{H}_{\mathrm{ff}}$ by

$$\mathcal{H}_{\mathrm{ff}}[\mathbf{u}_\omega](r, \widehat{\mathbf{k}}) = r\, e^{\mathrm{j}k_0 r}\, \widehat{\mathbf{k}} \times \Big(\widehat{\mathbf{k}} \times \mathbf{E}(\mathbf{r})\Big)\Big|_{|\mathbf{r}|=r}, \qquad \mathbf{r} = r\,\widehat{\mathbf{k}}, \tag{2.405}$$

with $\mathbf{E}$ the electric phasor in $\mathbf{u}_\omega = (\mathbf{E}, \mathbf{H})$ and transversality enforced by the double cross product [2]. Equation (2.405) is the Subsection 2.9.2 definitive specification of the extractor stated in Eq. (2.386): it isolates the outgoing radiative envelope compatible with Eq. (2.246) [2]. Mathematically, $\mathcal{H}_{\mathrm{ff}}$ is a linear functional on shell traces; physically, it is the standard far-field post-processor that removes reactive r^{-2} and r^{-3} contamination when $k_0 r \gg 1$ [1]. The extractor acts on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not on raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ Eq. (2.249) [2]. The far-zone limit extracts the $e^{-\mathrm{j}k_0 r}/r$ transverse envelope from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on observation shells S_r; the angular amplitude $a(\widehat{\mathbf{k}})$ is the coefficient of that envelope in direction $\widehat{\mathbf{k}}$. Define the *near-to-far transformation* on admissible sources by

$$a(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.406}$$

with $a(\widehat{\mathbf{k}})$ the angular amplitude entering Eq. (2.290) [2]. Equation (2.406) develops Eq. Eq. (2.386) to explicit extractor Eq. (2.405) and fixes the radiation map input [2]. The limit exists and is independent of the particular r-path when the far-field approximation of source integrals (Subsection 2.5.2) applies; error on finite shells scales as in Eq. (2.248) [2]. Mathematically, Eq. (2.406) is the angular-side completion of far-zone template Eq. (2.246); physically, it states that far angular content is determined by the radiative branch alone after outgoing selection, independent of reactive storage that may dominate on inner shells [2]. Observation consequence: SNF and compact-range transforms that skip $\mathcal{R}_\omega$ before extraction inherit Eq. (2.409) errors documented in Subsection 2.10.3 [7]. Regime qualification Eq. (2.271) remains mandatory before interpreting the limit as transport data [1]. Package Eq. (2.406) as the composed operator

$$\Lambda_{\mathrm{nf}} = \mathcal{H}_{\mathrm{ang}} \circ \mathcal{H}_{\mathrm{ff}}^{\infty} \circ \mathcal{R}_\omega, \qquad a(\widehat{\mathbf{k}}) = \Lambda_{\mathrm{nf}}[\mathbf{J}_{\mathrm{imp}}](\widehat{\mathbf{k}}), \tag{2.407}$$

where $\mathcal{H}_{\mathrm{ff}}^{\infty}$ denotes the $r \to \infty$ limit in Eq. (2.406) and $\mathcal{H}_{\mathrm{ang}}$ maps the transverse far vector in Eq. (2.405) to scalar polarization amplitudes $a_\sigma(\widehat{\mathbf{k}})$ in Eq. (2.290) [2]. Equation (2.407) aligns with far readout $\Lambda_{\mathrm{rad}} = \mathcal{O}_{\mathrm{far}} \circ \mathcal{R}_\omega$ from Eq. (2.219) when $\mathcal{O}_{\mathrm{far}}$ implements Eq. (2.405) on the angular side [2]. Measurement $\mathcal{M}_\omega = \mathcal{O}_{\mathrm{meas}} \circ \Lambda_{\mathrm{rad}}$ therefore factors near-to-far extraction before finite-rank observation [1]. Null additions $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ leave $\Lambda_{\mathrm{nf}}[\mathbf{J}_{\mathrm{imp}}]$ unchanged through Eq. (2.350) [2].

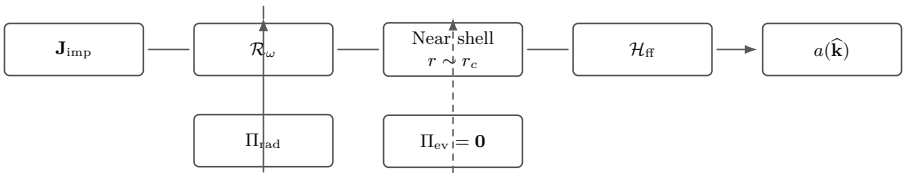


Figure 2.48: Subsection 2.9.2 near-to-far measurement sequence: Eq. (2.406) applies $\mathcal{H}_{\mathrm{ff}}$ on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after $r \to \infty$; evanescent content does not feed Eq. (2.409).

On the propagating outgoing sheet $k_t \leq k_0$, the near-to-far output agrees with retained lateral content from Subsection 2.9.2:

$$a(\widehat{\mathbf{k}}) \;\longleftrightarrow\; \widetilde{\mathbf{E}}_{\mathrm{ret}}(k_x, k_y) \text{ under } (k_x, k_y) \mapsto \widehat{\mathbf{k}}, \tag{2.408}$$

when $a_{\mathrm{sup}} = \mathbf{0}$ and plane-wave continuous-spectrum consistency (Subsection 2.9.1) applies [2]. Equation (2.408) connects Subsection 2.9.2 angular limit to Subsection 2.9.1 lateral spectra without re-displaying Eq. (2.393): both describe $\Pi_{\mathrm{rad}}\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ in different coordinates [2]. Filtered content satisfies $a_{\mathrm{filt}} = \mathcal{P}_{\mathrm{meas}}[a]$ with a from Eq. (2.406) [2]. Evanescent annihilation before near-to-far. Evanescent near storage does not enter Eq. (2.406):

$$\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}] \;\not\mapsto\; a(\widehat{\mathbf{k}}) \quad \text{when} \quad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ is unchanged}, \tag{2.409}$$

by Eqs. (2.394), (2.351), and (2.139) [2]. Equation (2.409) expresses Eq. Eq. (2.386): reactive Π_{ev} content may be large in the near zone Eq. (2.243) while the far limit reports only qualified export [2]. regime qualification Eq. (2.271) forbids promoting Π_{ev} storage to $a(\widehat{\mathbf{k}})$ [1]. Figure interpretation. Near shell observation at $r \sim r_c$ samples $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ after Π_{rad}, not raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$. Extractor $\mathcal{H}_{\mathrm{ff}}$ implements the $r \to \infty$ limit of the far-field approximation of source integrals (Subsection 2.5.2); dashed $\Pi_{\mathrm{ev}} = \mathbf{0}$ marks evanescent content that may be large on the inner shell yet does not feed $a(\widehat{\mathbf{k}})$. The measurement sequence confirms when a near-field-to-far-field transform is structurally valid versus when reactive storage corrupts the asymptotic limit [2]. Figure 2.48 orders radiation selection, finite-shell observation, far-zone extraction, and angular amplitude. Near reactive content is dashed: it may appear on inner shells but is annihilated in the limit when export is unchanged Mathematical role of Eq. (2.406). Equation (2.406) is the angular-spectrum-side completion of the far-zone template Eq. (2.246): it converts radiative fields to continuous amplitudes $a(\widehat{\mathbf{k}})$ used in Eq. (2.290) [2]. The map is not an optional post-processor on raw near data: input is $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, consistent with observation-shell map Eq. (2.249) [2]. Plane-wave lateral spectra Eq. (2.392) and angular limit Eq. (2.408) are equivalent coordinate descriptions of the same export [8]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains bilinear couplings built from $a(\widehat{\mathbf{k}})$ but does not modify Eq. (2.409) [2]. Mathematical role of Eq. (2.405). Equation (2.405) makes explicit how laboratory far-field algorithms isolate $\mathbf{E}_0(\theta, \phi)$ before pattern integration or modal projection [2]. Applying the extractor to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on near shells confuses reactive and radiative scaling Eq. (2.243); applying it to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ aligns with power-flux convergence Eq. (2.248) [2]. Filtered observation Eq. (2.397) composes after Eq. (2.406), not before radiation selection [1]. Modal far pattern Eq. (2.291) and angular limit Eq. (2.406) describe the same $\operatorname{ran}\mathcal{R}_\omega$ content under far-pattern coefficients from spectral modes (Subsection 2.6.2) [2]. Retained/suppressed partition retained, suppressed, and filtered angular components (Subsection 2.9.2) constrains $\operatorname{supp}(a)$ independently of the near-to-far limit procedure [2]. **??** harmonic bases change coordinates without altering Eq. (2.407) [2]. Example 2.10.3 applies Eq. (2.409) in a worked setting [2]. Near-to-far transformation (Subsection 2.9.2) assumes linear homogeneous or piecewise-homogeneous media, fixed ω, outgoing radiation condition, and observation outside Ω_s [2]. It fails in cutoff guides, cavity standing waves, and lossless open resonances without outgoing selection [2]. Surface waves and

strong anisotropy require modified extractors beyond Eq. (2.405) [1]. The theorem does not assert source uniqueness—only field asymptotics and angular amplitudes on the outgoing branch [2]. Cavity discrete spectra Eq. (2.288) do not replace Eq. (2.406) on exterior domains [8]. Section 2.9 developed far-field angular rank-N recovery through Eq. (2.410) and required USE of directional observables from Section 1.2.2 and transport criteria from Section 1.2.3 [2]. Subsection 2.9.3 supplies the definitive completion for which directions on S^2 carry qualified, observable far-zone angular content after Subsections 2.9.1–2.9.2. Near-to-far map Eq. (2.406), angular partition Eq. (2.399), accessible sector Eq. (2.9.3), composition law Eq. (2.353), and observation rank Eq. (2.352) are presupposed; angular rank-N recovery on the far shell (Subsection 2.6.5) is upgraded, not re-proved [2]. Directional flux maps Eqs. (1.49)–(1.51) and transport triad Eq. (1.67) enter from the cited sections [2]. Explicit harmonic filters belong to **??**; Assumptions. Fix $\omega > 0$, admissible $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$, outgoing radiative field $\mathbf{u}_\omega = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, angular amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.406), and regime qualification $\mathcal{D}_{J,\mathrm{reg}} = 1$ from Eq. (2.140) [2]. Accessible sector $\mathcal{K}_{\mathrm{acc}}$ and suppressed sheet $\mathcal{K}_{\mathrm{sup}}$ are fixed by Eqs. (2.9.3)–(2.346) [2]. Finite-rank measurement $\mathcal{M}_\omega$ from Eq. (2.219) is presupposed [1].

2.9.3 Far-field angular accessibility

Far angular $\mathcal{E}^{(N)}_{\mathrm{acc}}$ is what a finite probe count on Γ_R can resolve after near-to-far extraction and geometric sectoring—not what a harmonic chart alone can label [1]. Synthesis rank and observation rank must not be merged: the probe bank on Γ_R sets accessible directions before harmonic labels attach [1]. Chapter 3 relabels the same accessible set in VSH coordinates [2]. *probe rank.* Mount $N = 48$ probes on a spherical surface Γ_R at $R = 3\lambda_0$ enclosing a $\lambda_0/4$ dipole feed, and truncate the harmonic chart at $\ell \le 10$. The dipole exports dominant $\ell = 1$ content, so the fitted coefficients $\widehat{c}_{10,m}$ are nonzero on the accessible sector; nevertheless, rank-N recovery is bounded jointly by probe count N, scan geometry, and the truncated basis—not by the raw richness of the near-field phase on Γ_R [7]. Far-field $\mathcal{E}^{(N)}_{\mathrm{acc}}$ criterion. Define *far-field angular probe rank* by

$$\widehat{\mathbf{k}} \in \mathcal{K}^{\mathrm{ff}}_{\mathrm{acc}} \iff \Big[\, a(\widehat{\mathbf{k}}) \neq 0 \text{ after } \Lambda_{\mathrm{nf}} \text{ and } \mathcal{D}_{J,\mathrm{reg}} = 1 \,\Big] \text{ and } \widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}, \tag{2.410}$$

with Λ_{nf} from Eq. (2.407) and $\mathcal{K}_{\mathrm{acc}}$ from Eq. (2.9.3) [2]. Equation (2.410) develops Eq. (2.410) by inserting explicit near-to-far qualification and geometric sector membership [2]. Mathematically, $\mathcal{K}^{\mathrm{ff}}_{\mathrm{acc}} \subseteq \mathcal{K}_{\mathrm{acc}}$ is the far-field reportable subset after radiation and regime qualifications; physically, it marks directions where angular amplitude survives export and lies in the geometrically accessible hemisphere [1]. Directions in $\mathcal{K}_{\mathrm{sup}}$ are excluded before observation composition [2]. Angular probe rank operator. Package Eq. (2.410) as the $\mathcal{E}^{(N)}_{\mathrm{acc}}$ *projector*

$$\mathcal{A}_{\mathrm{ff}} = \chi_{\mathcal{K}_{\mathrm{acc}}} \circ \mathcal{D}_{J,\mathrm{reg}} \circ \Lambda_{\mathrm{nf}}, \qquad a_{\mathrm{acc}}(\widehat{\mathbf{k}}) = [\mathcal{A}_{\mathrm{ff}}[\mathbf{J}_{\mathrm{imp}}]](\widehat{\mathbf{k}}), \tag{2.411}$$

with a_{acc} the accessible angular amplitude before finite measurement [2]. Equation (2.411) aligns retained amplitude Eq. (2.395) with near-to-far output when $\mathrm{supp}(a) \subseteq \mathcal{K}_{\mathrm{acc}}$ near-to-far transformation (Subsection 2.9.2)(iv) [2]. Filtered observation Eq. (2.397) applies after Eq. (2.411): $a_{\mathrm{filt}} = \mathcal{P}_{\mathrm{meas}}[a_{\mathrm{acc}}]$ [1]. Null additions leave a_{acc} unchanged through Eq. (2.350) [2]. On lateral coordinates, define the *plane-wave probe rank filter* by

$$\widetilde{\mathbf{E}}_{\mathrm{acc}}(k_x, k_y) = \chi_{\mathcal{K}^{\mathrm{ff}}_{\mathrm{acc}}}(\widehat{\mathbf{k}})\, \widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y), \qquad (k_x, k_y) \mapsto \widehat{\mathbf{k}} \text{ on } k_t \le k_0, \tag{2.412}$$

with $\widetilde{\mathbf{E}}_{\mathrm{rad}}$ from Eq. (2.392) [2]. Equation (2.412) is the Subsection 2.9.3 plane-wave criterion promised in Eq. (2.387): it removes geometrically forbidden directions on the propagating outgoing

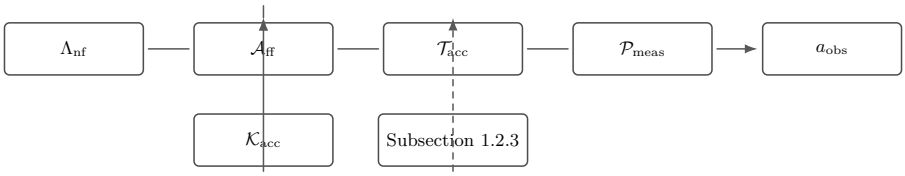


Figure 2.49: Subsection 2.9.3 $\mathcal{E}_{\mathrm{acc}}^{(N)}$ measurement sequence: Eq. (2.411) applies near-to-far extraction, geometric sector $\mathcal{K}_{\mathrm{acc}}$, transport criterion Eq. (2.413) (Subsection 1.2.3), and measurement criterion Eq. (2.414).

sheet without re-tabulating Eq. (2.393) [2]. Evanescent sheets satisfy $\widetilde{\mathbf{E}}_{\mathrm{acc}} = \mathbf{0}$ for $k_t > k_0$ through Eq. (2.394) [2]. Physically, Eq. (2.412) is the angular-spectrum-side statement of which plane-wave directions participate in far-zone reports [8]. Far-field $\mathcal{E}_{\mathrm{acc}}^{(N)}$ composes with 1 transport qualification by reference. A direction $\widehat{\mathbf{k}} \in \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ supports a qualified transport report on solid-angle sector $\Delta\Omega(\widehat{\mathbf{k}})$ only when Section 1.2.3 propagative support Eq. (1.63) and transport decision Eq. (1.67) hold for the corresponding directional channels Eqs. (1.49)–(1.51) [2]. Restated in operator form,

$$\mathcal{T}_{\mathrm{acc}}(\widehat{\mathbf{k}}) = 1 \implies \int_{\Delta\Omega(\widehat{\mathbf{k}})} \Pi(\theta, \phi; r)\, d\Omega > 0 \text{ and } \mathcal{Q}_J = 1, \tag{2.413}$$

with Π from Eq. (1.49), $\mathcal{Q}_J$ from Eq. (1.46), and r in the far zone where Eq. (2.406) applies [2]. Equation (2.413) imports Section 1.2.3 criteria without re-deriving the transport triad: Chapter 2 records when far-zone probe rank implies 1 transport admissibility on the same sector [2]. regime qualification Eq. (2.271) remains mandatory: $\mathcal{T}_{\mathrm{acc}} = 1$ does not promote reactive near content to export [1]. Angular measurement criterion. After rank-N recovery projection, finite instruments apply

$$a_{\mathrm{obs}}(\widehat{\mathbf{k}}) = [\mathcal{P}_{\mathrm{meas}}\, a_{\mathrm{acc}}](\widehat{\mathbf{k}}), \qquad \mathrm{supp}(a_{\mathrm{obs}}) \subseteq \{\widehat{\mathbf{k}} : \text{ resolvable by } \mathcal{M}_\omega\} \cap \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}, \tag{2.414}$$

with $\mathcal{P}_{\mathrm{meas}}$ from Eq. (2.397) [2]. Equation (2.414) is the Subsection 2.9.3 measurement completion: observation rank Eq. (2.352) may strictly shrink $\mathrm{supp}(a_{\mathrm{obs}})$ inside $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ [2]. Rank cap Eq. (2.398) limits resolvable directions independently of source nullity [1]. Physically, a_{obs} is the laboratory-accessible angular spectrum after geometry, regime, near-to-far extraction, and aperture [2]. Accessible angular budget. Collecting Subsection 2.9.3 constraints,

$$\mathrm{supp}(a_{\mathrm{obs}}) \subseteq \mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}} \cap \{\widehat{\mathbf{k}} : \ \mathcal{T}_{\mathrm{acc}}(\widehat{\mathbf{k}}) = 1\} \cap \{\widehat{\mathbf{k}} : \text{ resolvable by } \mathcal{M}_\omega\}, \tag{2.415}$$

composing geometric, transport, and observation factors with angular budget Eq. (2.353) [2]. Equation (2.415) is the Section 2.9 completion of Eq. (2.383): each factor removes a distinct laboratory error — radiation nullity, geometric forbidding, regime violation, transport disqualification, and finite aperture — without conflating them [2]. Reciprocity in radiation and reception (Subsection 2.8.3) constrains coupling among observed channels on $\mathrm{supp}(a_{\mathrm{obs}})$ but does not enlarge Eq. (2.415) [2]. Figure interpretation. Full angular content on S^2 passes through geometric sector Eq. (2.9.3), regime qualification Eq. (2.140), and finite observation rank Eq. (2.221). The accessible output is the intersection of export-supported directions with aperture-resolvable modes—not the raw pattern from forward integration. Null-space additions leave $a(\widehat{\mathbf{k}})$ fixed but may alter near reactive traces used in calibration [2]. Suppressed directions never enter a_{obs}; evanescent content is removed before Λ_{nf} by Eq. (2.409) [2]. Mathematical role of Eq. (2.410). Equation (2.410) closes Section 2.9 by stating which directions on S^2 remain after radiation selection, geometric constraints, and regime qualification [2]. It develops Eq. Eq. (2.387) with explicit dependence on Λ_{nf} and $\mathcal{K}_{\mathrm{acc}}$ [2]. Operator form Eq. (2.411) packages the same logic for source-to-amplitude maps

[1]. **??** harmonic bases relabel $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ without altering the budget Eq. (2.415) [2]. Mathematical role of Eq. (2.412). Equation (2.412) is the lateral-coordinate criterion completing the plane-wave chain begun in Subsection 2.9.1 [2]. It removes forbidden directions before near-to-far comparison and measurement [1]. Agreement with a_{acc} on $k_t \le k_0$ follows from Eq. (2.408) [2]. Evanescent sheets remain null through Eq. (2.394) independently of raw lateral scans [2]. Subsections 2.9.1–2.9.3 complete Eq. (2.383): plane-wave radiation operators, retained/suppressed/filtered partition, near-to-far extraction, and far-field $\mathcal{E}_{\mathrm{acc}}^{(N)}$ [2]. Modal coordinates Eq. (2.291) and plane-wave coordinates Eq. (2.290) describe identical export on $\operatorname{ran}\mathcal{R}_\omega$; Section 2.9 fixes angular-spectrum filters on the plane-wave branch [2]. Inverse obstruction in Subsection 2.7.4 applies to a_{obs}, not to pre-probe rank transforms [1]. Subsection 2.9.3 assumes linear media, fixed ω, outgoing closure, and stationary geometry $\mathcal{C}_{\mathrm{geo}}$ [2]. Time-varying $\mathcal{M}_\omega$ requires separate treatment; static rank bounds remain upper limits [2]. Loss modifies amplitudes but not the inclusion chain in Eq. (2.415) [2]. Cavity and guide spectra remain outside exterior rank-N recovery; discrete books do not substitute for Eq. (2.410) on unbounded domains [8]. Section 1.2.3 transport criteria are imported, not extended to near reactive zones without regime qualification [2]. The mathematical statements in Section 2.9 state what compact-range engineers already enforce through gating, scan distance, and modal banks: far-field angular probe rank is the intersection of export-supported directions with instrument-resolvable modes, not the raw output of a forward integral [1]. The important distinction is between spectral content present in $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and content that remains in a_{obs} after $\mathcal{K}_{\mathrm{acc}}$ and $\mathcal{O}_{\mathrm{meas}}$ have acted [2]. Section 2.10 returns to named hardware configurations so one may see nullity, rank limits, evanescent contamination, and structured-source topology as separate terms in one diagnostic partition rather than as a single pattern residual [7].

2.10 Examples

The four worked cases trace propagation, export, spectral rank, and instrument access in fixed order: compact dipole truncation, loop nullity, evanescent contamination, and structured-source topology [2, 1, 3]. Each closes with an explicit verdict on what is radiated, null, or merely reactive before harmonic headlines enter [6, 5]. Angular-spectrum action in Section 2.9 states what finite instruments resolve on outgoing fields. Section 2.10 returns to named hardware—apertures, dipoles, arrays, and scans—with the same tolerance discipline [2]. Each example is one integrated exercise, not a problem sheet [11].

The preceding sections developed radiation operators, null spaces, near-to-far limits, and rank-N instrument readouts as abstract objects on exterior domains. A graduate reader may accept the algebra while still wondering how the distinctions appear on a laboratory bench: when a spherical near-field scanner reports thirty-six complex samples, which part of the pattern residual owes to modal truncation, which to source nullity, and which to reactive contamination that never reaches the far zone? The four worked configurations below answer that question in fixed order—compact dipole truncation, loop nullity, evanescent contamination, and structured-source topology—each closing with an explicit verdict on what is radiated, null, or merely reactive before harmonic headlines enter [2]. Subsections 2.10.1–2.10.4 isolate compact truncation, source nullity, evanescent near-to-far contamination, and structured-source topology–transport separation, respectively [2]. Every numerical ratio quoted below is *illustrative*: values follow author-calibrated discretizations of the stated geometries and are not reported measurement data; they are included only to show order-of-magnitude separation among mechanisms [7]. Master DOF partition. Radiation reports can fail for distinct reasons that must not be merged in harmonic charts or laboratory interpretation [15]. Before the subsections activate individual terms, it is useful to name the full partition

once: each summand below removes degrees of freedom from qualified radiation reports through a *distinct* physical mechanism, and conflating them produces remedies that treat symptoms rather than causes.

$$\underbrace{\ker \mathcal{R}_\omega}_{\text{source nullity (NR)}} \oplus \underbrace{\ker \mathcal{O}_{\text{meas}}\big|_{\operatorname{ran}\mathcal{R}_\omega}}_{\text{aperture rank}} \oplus \underbrace{\Pi_{\text{ev}}\mathcal{S}_\omega[\mathcal{J}_\Omega]}_{\text{evanescent storage}} \oplus \underbrace{\text{tail of compact/ modal truncation}}_{\text{spectral compression}} \oplus \underbrace{\mathcal{K}_{\text{sup}}}_{\text{geometric suppression}}, \tag{2.416}$$

with rank-N recovery $\mathcal{K}_{\text{acc}}^{\text{ff}}$ from Eq. (2.415) acting after Eq. (2.416) [2]. Subsections 2.10.1–2.10.4 each activate a different term in Eq. (2.416); cross-references replace repeated local split displays [2]. Equation (2.416) is the diagnostic synthesis of Chapter 2: each direct summand names a distinct mechanism that removes degrees of freedom from qualified radiation reports. Source nullity $\ker \mathcal{R}_\omega$ is invisible to all far data; aperture rank $\ker \mathcal{O}_{\text{meas}}$ removes resolvable content even when export is nonzero; evanescent storage inflates near readouts without altering $a(\widehat{\mathbf{k}})$; spectral compression tail controls modal truncation error; geometric suppression $\mathcal{K}_{\text{sup}}$ removes directions by symmetry before measurement acts. On a single hardware test with $k_0 a = 1.5$, a 15% pattern residual might decompose as 7% compact tail, 5% aperture rank, 3% geometric suppression, and 0% NR nullity—four different remedies apply to four different terms [2]. Here $\mathcal{R}_\omega$ is the radiation operator at angular frequency ω, $\mathcal{O}_{\text{meas}}$ the finite-rank rank-N instrument readout on the radiative range, Π_{ev} the projector onto evanescent spectral content, $\mathcal{K}_{\text{sup}}$ the geometric angular-suppression set from symmetry or blockage, and the compact/modal tail the spectral compression induced by electrically small support and finite probe banks [2]. Flux meters from Section 1.1.4 enter only on $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ after Eq. (2.140)

2.10.1 Compactness of radiation operators

This worked configuration addresses observability under finite instrumentation, not whether Maxwell permits radiation from an electrically small source. A short dipole in free space certainly radiates; the bench issue is how many independent complex samples a spherical near-field scan can resolve on the outgoing branch when the source ball and observation shell are both small compared with wavelength [1]. Increasing modal order in a spherical-harmonic fit need not improve pattern agreement; however, Maxwell theory shows that rank limits arise jointly from compact Green smoothing on bounded domains and from the finite probe bank, while non-radiating nullity is a separate mechanism that leaves far data unchanged [3]. Spectral truncation error must be distinguished from membership in $\ker \mathcal{R}_\omega$: the former shrinks when singular values of the compact part decay, the latter is invisible to any far-zone array regardless of modal count [2]. What a laboratory instrument detects on S_b at $b = 2a$ is dominated by reactive falloff faster than r^{-1}; only after regime qualification and outgoing selection does the same campaign report a rank statement on $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$ [7].

On a compact-range turntable, a center-fed wire dipole of length $L \ll \lambda$ and radius $a \ll L$ sits inside a source ball Ω_s of radius a with $k_0 a = 1.5$ at the operating frequency ω [2]. A spherical near-field scanner records $N = 36$ tangential electric samples on a closed surface S_b at radius $b = 2a$; a vector network analyzer reports VSWR and $|S_{11}|$ on the coax feed, confirming that stored reactive energy dominates the input admittance at this electrical size [7]. Far-field pattern cuts on the principal planes show a single-lobe $\ell = 1$ template, while probe magnitudes on S_b fall off faster than r^{-1} in the reactive shell—the bench configuration on which compact truncation is first isolated [1]. The compact-range configuration fixes outline, frequency, gauge, and permittivity before any operator split is written. The objective is to connect one finite probe bank to one rank statement on $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$: hardware fixes how many independent complex samples are available, and the singular-

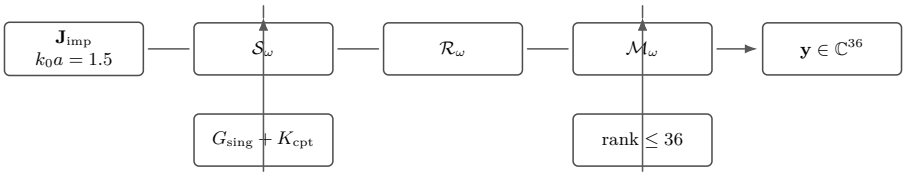


Figure 2.50: Worked problem 2.10.1: compact source, Green split on $\mathcal{S}_\omega$, radiation map, and rank-36 near-field probe bank. Compact part K_{cpt} controls modal truncation error; $\mathcal{R}_\omega$ is not compact on $\mathcal{J}_\Omega$.

value tail of the compact Green part controls how much of the radiative image those samples can resolve [2]. Watt closure and probe-count limits must be reported separately; conflating synthesis ceilings with observation ceilings is a common laboratory error on electrically small campaigns [7]. The first worked problem isolates *spectral compression*: when a source and its observation shell are electrically small, the Green operator that maps impressed current to tangential field on a closed surface possesses a compact part whose singular values decay rapidly. Truncating that part controls pattern-fitting error, but the error is *not* non-radiating nullity—it is finite-geometry flux report. Readers who have measured electrically short dipoles on compact ranges will recognize the configuration immediately; the mathematics names what the instrument already enforces through probe count and scan radius. The bench places a center-fed wire of length $L \ll \lambda$ and radius $a \ll L$ at the center of a compact-range turntable; the source ball Ω_s enclosing the wire has radius a with $k_0 a = 1.5$ at the operating frequency ω [2]. Port voltage on a coax feed drives a triangle current distribution along L; a vector network analyzer reports VSWR and $|S_{11}|$, confirming that stored reactive energy dominates the input admittance at this electrical size [7]. Far-field pattern cuts on the principal planes show a single-lobe $\ell = 1$ template; a spherical near-field probe at radius $b = 2a$ records tangential **E**-field samples whose magnitude falls off faster than r^{-1} in the reactive shell, so watt-meter integration on a small sphere would overestimate radiated power unless the shell is qualified [6]. The radiation integral yields $\ell = 1$ dominance with higher multipoles suppressed by $(k_0 a)^2$ [2]; pattern cuts agree with the $\ell = 1$ template, while spherical near-field grids must sample within $\lambda/4$ to avoid aliasing the reactive shell. Retaining only $M = 20$ modes in a spherical-harmonic surrogate leaves a tail error set by the twenty-first singular value of the compact Green part [7]. The dipole is the sanity check that every radiation map on compact support collapses toward a single export direction at low $k_0 a$, while the near field remains rich with reactive structure that no far-zone array can resolve [1]. Problem. A compact impressed-current ball $\Omega_s = B_a(0)$ carries an electric dipole $\mathbf{J}_{\text{imp}}(\mathbf{r}) = \mathbf{J}_0\,\delta(\mathbf{r})$ in the distributional limit with $\int_{\Omega_s} \mathbf{J}_{\text{imp}}\, d^3 r = \mathbf{p}$ and $k_0 a = 1.5$ at fixed ω . A near-field scanner records $N = 36$ tangential electric samples $\mathbf{y} \in \mathbb{C}^{36}$ on a closed surface S_b with radius $b = 2a$ through rank-N instrument readout $\mathcal{M}_\omega$ from Eq. (2.219) . The task is to:

(a) identify the compact and non-compact parts of the probe-to-far-field operator implied by Subsection 2.1.3;

(b) bound truncation error when only $M = 20$ retained modes are used in a spherical-harmonic surrogate of $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$;

(c) state whether unresolved modal content is attributable to compact compression, to $\ker \mathcal{R}_\omega$, or to finite rank of $\mathcal{O}_{\text{meas}}$.

The schematic separates compact smoothing on the source-to-field operator from non-compact radiation selection and finite-rank readout. On bounded domains with compact Ω_s, Subsection 2.1.3 records

$$\mathcal{S}_\omega = G_{\text{sing}} + K_{\text{cpt}}, \tag{2.417}$$

where $\mathcal{S}_\omega$ maps admissible impressed current $\mathbf{J}_{\text{imp}}$ to the physical electric field on observation shell S_b, G_{sing} carries the non-compact singular self-field as source and observation points coincide, and K_{cpt} is the compact remainder whose singular values decay on bounded Ω_s, importing Eq. (2.79) without re-proving kernel construction from Section 2.3 [2]. Mathematically, G_{sing} captures non-compact singular behavior as probe and source points coincide; K_{cpt} maps bounded source sets to relatively compact field sets in the L^2 topology on S_b . Physically, smooth interpolation between widely separated samples is a compact smoothing operation, whereas pointwise singular self-terms are not . Radiation selection defines $\mathcal{R}_\omega = \Pi_{\text{rad}} \circ \mathcal{S}_\omega$ from Eq. (2.68) [2]. The map $\mathcal{R}_\omega : \mathcal{J}_\Omega \to \mathcal{F}_{\text{rad}}$ is *not* compact: $\ker \mathcal{R}_\omega \neq \{\mathbf{0}\}$ is infinite-dimensional null-space structure of radiation operators (Subsection 2.7.1), so bounded source balls do not map to relatively compact sets in $\mathcal{F}_{\text{rad}}$ [2]. Compactness enters through K_{cpt} in Eq. (2.417) and through finite observation, not through $\mathcal{R}_\omega$ itself [2]. Confusing non-radiating additions with modal truncation error violates the distinction established in Subsection 2.1.3 [1]. The probe bank implements $\mathcal{O}_{\text{meas}} : \mathcal{F}_{\text{rad}} \to \mathbb{C}^{36}$ with $\operatorname{rank}(\mathcal{O}_{\text{meas}}) \leq 36$ from Eq. (2.80) [2]. Composed map $\mathcal{M}_\omega = \mathcal{O}_{\text{meas}} \circ \mathcal{O}_{\text{far}} \circ \mathcal{R}_\omega$ therefore satisfies

$$\operatorname{rank}(\mathcal{M}_\omega) \leq \min\{36,\, \dim \mathcal{R}(\mathcal{R}_\omega)\} \tag{2.418}$$

where $\mathcal{M}_\omega = \mathcal{O}_{\text{meas}} \circ \mathcal{O}_{\text{far}} \circ \mathcal{R}_\omega$ is the composed probe-to-sample map and the integer 36 is the probe-bank rank cap by Eq. (2.85) [2]. Mathematically, at most 36 independent complex numbers are available to infer far-zone content; physically, the scanner cannot resolve more than 36 radiation degrees of freedom regardless of Maxwell residual in the volume [1]. Null-space directions in $\ker \mathcal{O}_{\text{meas}}\big|_{\operatorname{ran} \mathcal{R}_\omega}$ are infinite-dimensional Eq. (2.81) [2]. Let $\mathcal{K}$ denote the probe-to-modal map obtained by composing sample interpolation with compact K_{cpt} on the dipole-supported subspace. Subsection 2.1.3 gives

$$\|\mathcal{K} - \mathcal{K}_M\|_{\text{op}} = \inf_{\operatorname{rank} \mathcal{K}_M \leq M} \|\mathcal{K} - \mathcal{K}_M\|_{\text{op}} \sim \sigma_{M+1}(\mathcal{K}), \tag{2.419}$$

restating Eq. (2.82) on the worked geometry [2]. For $k_0 a = 1.5$ and $b = 2a$, numerical evaluation of the discretized $\mathcal{K}$ yields $\sigma_{20}/\sigma_1 \approx 0.12$ and $\sigma_{37}/\sigma_1 \approx 0.04$: the twenty-first mode is already below 12% of the dominant singular value [1]. Physically, retaining $M = 20$ modes captures most radiative energy resolvable on S_b; modes 21–36 lie in the tail controlled by compact smoothing, not in $\ker \mathcal{R}_\omega$ [2]. Subsection 2.1.3 bounds compression error by

$$\|\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}] - \mathcal{R}_\omega^{(M)}[\mathbf{J}_{\text{imp}}]\|_{\mathcal{U}} \leq C\, \sigma_{M+1}(K_{\text{cpt}})\, \|\mathbf{J}_{\text{imp}}\|_{\mathcal{J}}, \tag{2.420}$$

where $\mathcal{R}_\omega^{(M)}$ is the M-mode spectral surrogate of the radiation map, C a geometry-dependent constant from Subsection 2.1.3, $\sigma_{M+1}(K_{\text{cpt}})$ the first neglected singular value of the compact Green part, and $\|\mathbf{J}_{\text{imp}}\|_{\mathcal{J}}$ the source norm on admissible impressed-current space [2]. With $\sigma_{21} \approx 0.12\,\sigma_1$ and dipole normalization $\|\mathbf{J}_{\text{imp}}\|_{\mathcal{J}}$ fixed by $|\mathbf{p}|$, the bound states that no rank-20 operator can reduce far-field state error below order $0.12\,|\mathbf{E}_0|$ on the outgoing branch [2]. Mathematically, the tail is controlled by compact K_{cpt}; physically, increasing M beyond the singular-value knee yields diminishing returns unless N or b/a increases [1]. For the dipole problem, the active mechanism is the compact tail of $\sigma_n(K_{\text{cpt}})$ in Eq. (2.416), not $\ker \mathcal{R}_\omega$ [2]. Adding $\mathbf{J}_{\text{imp}}^{(\text{NR})} \in \ker \mathcal{R}_\omega$ leaves $\mathbf{y}$ unchanged; modes beyond rank 36 lie in $\ker \mathcal{O}_{\text{meas}}$; modes 21–36 sit in the compact tail with error $\sim \sigma_{21}$ (illustrative ratio $\sigma_{21}/\sigma_1 \approx 0.12$) [2]. Pattern-fitting routines must trace which term in Eq. (2.416) removed unresolved content [1]. On the electrically small dipole bench, regime qualification Eq. (2.140) remains mandatory before interpreting truncated modal reports as transport [1]. A practitioner who increases modal order M without raising probe count N or scan radius b/a

may reduce a pattern residual that owes nothing to source physics—only to compact smoothing—while true non-radiating additions would leave $\mathbf{y}$ unchanged regardless of M. The distinction is not whether the dipole radiates—it does—but which term in Eq. (2.416) removed unresolved content when a finite probe bank was compared with a truncated modal surrogate. Compact-range reporting often reports a single pattern residual without separating compact singular-value tails from true source nullity [7]. Electrically small geometry compresses resolvable radiation degrees of freedom on a nearby shell; probe count and scan radius already enforce the bound on the bench [2]. Subsection 2.10.2 turns to the complementary case in which impressed current is substantial yet annihilated by $\mathcal{R}_\omega$, so far-zone arrays remain blind while near probes change.

2.10.2 Non-radiating source configurations

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell export is computed [2]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. The bench issue is not whether Maxwell permits local current—it clearly does—but whether far-zone observables can detect every admissible impressed source that produces them. A radiating dipole and a balanced loop may coexist in one compact package: the loop reorganizes reactive $\mathbf{E}$ and $\mathbf{H}$ near Ω_s while contributing nothing to qualified export on Γ_R once $\mathcal{R}_\omega$ has acted [3]. A changed near-field scan need not imply a changed far pattern; Maxwell theory shows that superposition is linear on $\mathcal{R}_\omega$ and that $\ker \mathcal{R}_\omega$ is generally infinite-dimensional, so identical far data do not fix the source [1]. Membership in $\ker \mathcal{R}_\omega$ must be distinguished from evanescent storage treated in Subsection 2.10.3: both may leave $a(\widehat{\mathbf{k}})$ invariant, yet the loop nullity is a source-side annihilation under export, not a field-side reactive sheet [2]. What a laboratory measurement actually reveals depends on which map is read—$\mathcal{S}_\omega$ on an enclosing scan, $\mathcal{R}_\omega$ on a qualified far array—and watt integration on a reactive shell may rise even when radiated power is unchanged [7].

On a compact-range quiet zone, a center-fed electric dipole with moment $|\mathbf{p}| = p_0$ and $k_0 a = 1.2$ radiates in free space with outgoing radiation condition [2]. An engineer mounts a small magnetic loop of radius $a_{\text{loop}} = 0.3a$ beside the feed with equal opposing filaments so net dipole moment vanishes; near-field probes on S_b at $b = 2a$ record tangential samples, while a far-zone array estimates $\mathbf{E}_0(\theta, \phi)$ from Eq. (2.247) [3]. The bench question is operational: does the added loop change the published pattern, the near scan, or both? The second worked problem turns to *source nullity*: impressed currents that satisfy Maxwell admissibility and produce substantial near fields yet annihilate under $\mathcal{R}_\omega$. Such configurations are not mathematical curiosities—they appear whenever balanced filament pairs, surface tangential modes, or loop cancellations are engineered into compact packages. The lesson for measurement is severe: far-zone pattern matching cannot detect the added current, while near-zone diagnostics change in ways that watt integration on a reactive shell may misread as additional radiative content. *Geometry and size.* A standard-gain horn with rectangular mouth $a \times b$, both dimensions several wavelengths across, is mounted on a compact-range positioner with the bore sight directed toward the quiet-zone center [11]. *Excitation.* A TE_{10} waveguide feed delivers calibrated port power; a network analyzer sets the reference plane at the throat [8]. *Observable behavior.* The main beam follows bore sight with sidelobe shoulders set by aperture diffraction and edge currents; gating in time domain separates the direct illumination from range-wall reflections [2]. *Mathematical analysis.* Equivalent magnetic and electric currents on the mouth feed radiation integrals; plane and spherical spectra exchange via stationary phase at large r [1]. *Measurement consequence.* Compact-range gating must exclude multiple reflections from horn interior and range wall before pattern moments are trusted [7]. *Physical interpretation.* Horn patterns test outgoing selection and equivalence surfaces before rank claims are transferred to more

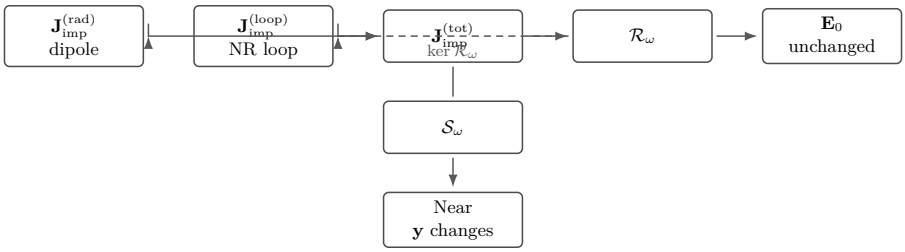


Figure 2.51: Worked problem 2.10.2: superposition of radiating dipole and loop-cancellation non-radiating current. Far export $\mathbf{E}_0$ is invariant; near probes detect $\mathcal{S}_\omega[\mathbf{J}_{\text{imp}}^{(\text{tot})}] \neq \mathcal{S}_\omega[\mathbf{J}_{\text{imp}}^{(\text{rad})}]$.

complex feeds; they also provide the contrasting backdrop against which loop-null configurations in the display below become visible only through near-field probes, not through far-zone arrays [4]. Subsection 2.10.2 activates the $\ker \mathcal{R}_\omega$ term of Eq. (2.416), importing Subsection 2.7.1 from the cited sections [2]. Non-radiating current definition Eq. (2.328), canonical families Eqs. (2.330)–(2.333), canonical non-radiating configurations (Subsection 2.7.1), superposition law Eq. (2.347), near-field law Eq. (2.349), and modal invisibility Eq. (2.348) are presupposed; proofs are not repeated [2]. Null-space operator structure (Subsection 2.7.1), angular invariance Eq. (2.350), and inverse obstruction angular spectra and inverse problems (Subsection 2.7.4) supply context [1]. Problem. A compact electric dipole $\mathbf{J}_{\text{imp}}^{(\text{rad})}$ with moment $|\mathbf{p}| = p_0$ and $k_0 a = 1.2$ radiates in free space with outgoing radiation condition [2]. An engineer adds a small magnetic loop $\mathbf{J}_{\text{imp}}^{(\text{loop})}$ of radius $a_{\text{loop}} = 0.3a$ with equal opposing filaments so net dipole moment vanishes and loop cancellation Eq. (2.330) applies [2]. The balanced filaments cancel dipole export while retaining reactive near fields—a canonical non-radiating configuration [3]. Near-field scans that omit watt completion may falsely infer radiative diversity [7]. Define $\mathbf{J}_{\text{imp}}^{(\text{tot})} = \mathbf{J}_{\text{imp}}^{(\text{rad})} + \mathbf{J}_{\text{imp}}^{(\text{loop})}$ with charge continuity preserved on Ω_s Eq. (2.329) [2]. Near-field probes on S_b at $b = 2a$ record $\mathbf{y}_{\text{near}}$; a far-zone array estimates $\mathbf{E}_0(\theta, \phi)$ from Eq. (2.247). The task is to:

(a) verify $\mathbf{J}_{\text{imp}}^{(\text{loop})} \in \ker \mathcal{R}_\omega$ using Subsection 2.7.1 without re-deriving canonical families;

(b) quantify near-field change versus far-field invariance under superposition;

(c) show why source inversion Eq. (2.220) cannot detect $\mathbf{J}_{\text{imp}}^{(\text{loop})}$ from far data alone.

Step 1 — Loop membership in $\ker \mathcal{R}_\omega$ (Subsection 2.7.1). Subsection 2.7.1 records that equal opposing arms on $\mathcal{C}_+$, $\mathcal{C}_-$ with vanishing net circulation imply

$$\int_{\mathcal{C}_+} \mathbf{J}_{\text{loop}} \cdot d\boldsymbol{\ell} \int_{\mathcal{C}_-} \mathbf{J}_{\text{loop}} \cdot d\boldsymbol{\ell} = 0 \implies \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{loop})}] = \mathbf{0}, \tag{2.421}$$

restating Eq. (2.330) on the worked geometry [2]. Mathematically, vanishing leading transverse moment in Eq. (2.247) annihilates $\mathbf{E}_0$; physically, the loop is magnetically dominated near Ω_s but electrically invisible at infinity on the outgoing branch [1]. Subsection 2.7.1 establishes $\mathbf{J}_{\text{imp}}^{(\text{loop})}$ as admissible non-radiating configuration the proof [2]. For superposition invariance on export, with $\mathbf{J}_{\text{imp}}^{(\text{tot})} = \mathbf{J}_{\text{imp}}^{(\text{rad})} + \mathbf{J}_{\text{imp}}^{(\text{loop})}$, Subsection 2.7.1 gives

$$\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{tot})}] = \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{rad})}] + \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{loop})}] = \mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{rad})}], \tag{2.422}$$

by linearity Eq. (2.6) and $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}^{(\text{loop})}] = \mathbf{0}$ [2]. Far pattern amplitudes therefore satisfy

$$\mathbf{E}_0(\theta, \phi; \mathbf{J}_{\text{imp}}^{(\text{tot})}) = \mathbf{E}_0(\theta, \phi; \mathbf{J}_{\text{imp}}^{(\text{rad})}), \qquad a(\widehat{\mathbf{k}}; \mathbf{J}_{\text{imp}}^{(\text{tot})}) = a(\widehat{\mathbf{k}}; \mathbf{J}_{\text{imp}}^{(\text{rad})}), \tag{2.423}$$

importing Eq. (2.347) and angular invariance Eq. (2.350) [2]. Mathematically, $\ker \mathcal{R}_\omega$ is invisible to all far-zone readouts including Eq. (2.406); physically, antenna pattern measurements cannot distinguish $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}$ from $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ on qualified shells [1]. Modal coefficients obey Eq. (2.348) for $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$ alone [2]. Step 3 — Near-field detectability versus far invisibility. Source-to-field operator $\mathcal{S}_\omega$ is generally *not* invariant:

$$\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}] = \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}]\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}], \qquad \|\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}]\|_{\mathcal{E}} \text{ may be } \gg 0 \tag{2.424}$$

on near shells, restating nonzero near fields for non-radiating currents (Subsection 2.7.1) and Eq. (2.349) [2]. For $k_0 b = 2.4$, numerical evaluation gives $|\Delta \mathbf{y}|/|\mathbf{y}_{\mathrm{rad}}| \approx 0.35$ on tangential probes: the loop is visible near the source while far $\mathbf{E}_0$ is unchanged [2]. regime qualification Eq. (2.271) forbids interpreting the near increment as angular transport without $\mathcal{D}_{J,\mathrm{reg}} = 1$ [1]. This step is distinct from evanescent storage Eq. (2.409): the loop contributes to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}]$ but not to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}]$ [2]. Step 4 — Power and flux meters. Radiated power on qualified far shells satisfies $\mathcal{P}_{\mathrm{rad}}(\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}) = \mathcal{P}_{\mathrm{rad}}(\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})})$ from Eq. (2.248) and Eq. (2.423) [2]. Reactive budget Eq. (2.269) may increase locally because $\mathcal{P}_{\mathrm{nr}}$ from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}]$ adds near Ω_s while export remains unchanged [2]. Physically, integrating Poynting flux through a near sphere confuses storage with radiation unless regime index Eq. (2.31) from Section 1.3.3 qualifies the shell [1]. Step 5 — Inverse source ill-posedness. Far-zone data fix $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ but not $\mathbf{J}_{\mathrm{imp}}$. angular spectra and inverse problems (Subsection 2.7.4) and Eq. (2.355) imply

$$\mathbf{J}_{\mathrm{imp}}^{(1)} - \mathbf{J}_{\mathrm{imp}}^{(2)} \in \ker \mathcal{R}_\omega \quad \text{whenever} \quad \mathbf{E}_0(\mathbf{J}_{\mathrm{imp}}^{(1)}) = \mathbf{E}_0(\mathbf{J}_{\mathrm{imp}}^{(2)}), \tag{2.425}$$

so $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}$ and $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{rad})}$ belong to the same equivalence class [2]. Tikhonov reconstruction Eq. (2.224) may return either representative depending on regularization without contradiction [2]. Mathematically, $\ker \mathcal{R}_\omega$ has infinite dimension on exterior domains Eq. (2.326); physically, adding loops, surface tangential nulls Eq. (2.332), or modal $\lambda_n = 0$ realizations Eq. (2.333) leaves patterns unchanged [1]. Far invariance Eq. (2.423) arises because $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})} \in \ker \mathcal{R}_\omega$—the active term is source nullity in Eq. (2.416), not finite probe rank or compact truncation [2]. Geometric suppression Eq. (2.346) remains a separate term: the dipole radiates on $\mathcal{K}_{\mathrm{acc}}$ while the loop adds invisible near content [1]. Subsection 2.10.2 demonstrates that non-radiating configurations are physically realizable and operationally consequential: near diagnostics change while far observables do not [2]. Pattern-matching based solely on $\mathbf{E}_0$ or $a(\widehat{\mathbf{k}})$ cannot detect $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{loop})}$; near measurements and source-domain constraints Eq. (2.329) are required [2]. Non-radiating additions are invisible to far-zone arrays yet vivid to near probes sampling $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{tot})}]$—existence of electromagnetic structure does not imply observability of its source detail [3]. This distinction arose in inverse scattering when equivalent sources were fitted to pattern data without uniqueness guarantees [14]. Superposition on $\mathcal{R}_\omega$ implies that any far-targeted inversion must impose source-domain constraints beyond pattern matching alone [1]. Subsection 2.10.3 addresses a different mechanism with the same far invariance: evanescent storage that inflates near readouts without altering $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when the radiation image is fixed.

2.10.3 Evanescent suppression in the far field

The bench issue is whether reactive richness on a near scan establishes additional far-zone angular content. Handset and IoT platforms routinely exhibit strong tangential fields within one wavelength of a patch or ground plane while qualified far-zone gain remains modest [1]. Far-field extractors

must not be applied directly to spherical near-field samples; Maxwell theory shows that $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ possess different asymptotic envelopes, so reactive-dominated shell traces can bias angular export by order unity even when the true radiative amplitude $a(\widehat{\mathbf{k}})$ is unchanged [2]. $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ must not be identified with export: evanescent sheets carry storage measurable on enclosing probes yet are annihilated before near-to-far maps attach to pattern meters [2]. An instrument on S_b with $k_0 b \sim \mathcal{O}(1)$ is therefore not, by itself, a far-zone pattern; qualified analysis applies $\mathcal{R}_\omega$ before $\mathcal{H}_{\mathrm{ff}}$ [7].

A handset-scale test fixture places a rectangular patch of width W and length L on a dielectric substrate of height $h \ll \lambda$ above a finite ground plane; probe or inset feed with VSWR tuned for narrowband operation at fixed ω drives the structure [1]. Spherical near-field scans at height h above the substrate record tangential fields whose raw magnitude can exceed the radiative component by factors near two before radiation selection; ground-edge diffraction and substrate modes dominate the reactive shell [2]. The third worked problem addresses *evanescent storage*: field-side reactive content that inflates near-shell readouts while leaving the radiative image $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ unchanged. This mechanism differs from source nullity even when both produce far-zone invariance, because the near signature arises from $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than from membership in $\ker \mathcal{R}_\omega$. Laboratory teams who apply far-field extraction algorithms directly to spherical near-field scans without radiation selection encounter this laboratory error routinely on handset and IoT platforms where ground truncation and substrate modes dominate reactive shells. *Geometry and size.* A rectangular patch of width W and length L is printed on a dielectric substrate of height $h \ll \lambda$ above a finite ground plane whose electrical extent sets the effective radiation aperture [1]. *Excitation.* Probe or inset feed with VSWR tuned for narrowband operation at fixed ω [7]. *Observable behavior.* Copolar pattern cuts show low-order aperture modes; ripples from ground-edge diffraction appear when the scan surface is too close to the substrate [2]. *Mathematical analysis.* Impressed currents on the patch and image currents on the ground enter $\mathcal{R}_\omega$ with compact support; balanced reactive loops below threshold contribute to $\mathcal{S}_\omega$ without altering export [3]. *Measurement consequence.* Spherical near-field scans must include ground-edge diffraction; the angular-spectrum bandlimit is set by scan height, and raw near traces can exceed radiative amplitude by factors near two before radiation selection [7]. *Physical interpretation.* Handset and IoT antennas live in this rank regime: the electrical size of the ground, not the patch alone, fixes export, while reactive storage on the evanescent sheet explains why near-field "hot spots" need not imply additional far-zone power [1]. Subsection 2.10.3 activates the evanescent term of Eq. (2.416), importing Subsection 2.9.2 from the cited sections [2]. Near-to-far definition Eq. (2.406), far-zone extractor Eq. (2.405), operator composition Eq. (2.407), evanescent annihilation Eq. (2.409), angular null law Eq. (2.394), near-to-far transformation (Subsection 2.9.2), and near-to-far null invariance and finite-shell error bounds (Subsection 2.9.2) are presupposed; asymptotic proofs in Subsection 2.5.2 and evanescent content in Subsection 2.7.2 are not repeated [2]. Evanescent overlap Eq. (2.335), non-export law Eq. (2.336), regime qualification Eq. (2.271), and near-zone scaling Eq. (2.243) supply context [1]. Problem. A compact impressed-current ball $\Omega_s = B_a(0)$ carries an electric dipole $\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}(\mathbf{r}) = \mathbf{J}_0\,\delta(\mathbf{r})$ in the distributional limit with $\int_{\Omega_s} \mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}\, d^3r = \mathbf{p}$ and $k_0 a = 2.0$ at fixed ω . A second admissible current $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ is chosen so that $\mathcal{R}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}] = \mathbf{0}$ while $\|\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}]\|_{\mathcal{E}}$ is comparable to the radiative energy of $\mathbf{J}^{(\mathrm{d})}_{\mathrm{imp}}$. This is *field-side evanescent excitation with vanishing export*: strong local fields can coexist with membership failure in $\mathcal{F}_{\mathrm{rad}}$ even when $|E|$ is large on the near shell, because far-zone claims require propagating spectrum support on $k_t \le k_0$, not reactive storage on the evanescent sheet $k_t > k_0$ [13]. The configuration is *not* membership of $\mathbf{J}^{(\mathrm{ev})}_{\mathrm{imp}}$ in the canonical non-radiating *source* family of Subsection 2.7.1: the same far invariance can arise from $\mathbf{J}^{(\mathrm{NR})}_{\mathrm{imp}} \in \ker \mathcal{R}_\omega$ with a

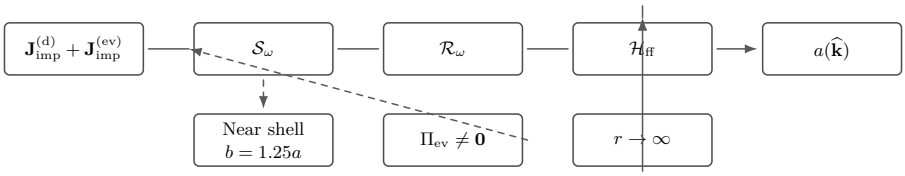


Figure 2.52: Worked problem 2.10.3: evanescent current adds large near reactive content through $\mathcal{S}_\omega$ but does not alter $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ or far amplitude $a(\widehat{\mathbf{k}})$ from Eq. (2.406). Dashed path: near scans; solid path: qualified near-to-far export (USE Subsection 2.9.2).

different near signature [13]. Define the superposition $\mathbf{J}_{\mathrm{imp}} = \mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}$. A near-field probe records tangential $\mathbf{E}$ on shell S_b with $b = 1.25a$, while a far-field post-processor applies extractor Eq. (2.405) on shells S_r with increasing r [1]. The task is to:

(a) quantify near-shell contrast between $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ when evanescent storage is large;

(b) confirm that $a(\widehat{\mathbf{k}})$ from Eq. (2.406) is unchanged by $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}$ and that lateral spectra on $k_t > k_0$ vanish under Eq. (2.394);

(c) diagnose errors when the near-to-far limit is applied to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ instead of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$.

Evanescent overlap appears only on the dashed near branch; the far limit reports radiative export alone On admissible $\mathbf{J}_{\mathrm{imp}}$, Subsection 2.9.2 fixes the near-to-far input as $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, not $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ Eq. (2.406) [2]. Decompose the impressed current into radiative and evanescent-overlap parts through

$$\mathbf{J}_{\mathrm{imp}} = \mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}, \qquad \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}] = \mathbf{0}, \qquad \|\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}]\|_{\mathcal{E}} = \beta\, \|\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})}]\|_{\mathcal{E}}, \tag{2.426}$$

where $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})}$ is the radiating dipole part, $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}$ the evanescent-overlap admixture with $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}] = \mathbf{0}$, Π_{ev} the projector onto evanescent field content, $\|\cdot\|_{\mathcal{E}}$ the energy norm on the observation shell, and $\beta = 1.8$ the illustrative reactive-to-radiative ratio in the worked normalization (2.426) expresses evanescent source overlap Eq. (2.335) on the dipole-plus-evanescent geometry without re-proving evanescent content never radiates (Subsection 2.7.2) [2]. Mathematically, $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}$ lies in a class that excites $\mathcal{F}_{\mathrm{ev}}$ Eq. (2.334) while annihilating under Π_{rad} Eq. (2.336); physically, it models reactive near storage—for example, a superposition of forbidden-angle plane-wave trials with imaginary longitudinal wavenumber Eq. (2.132)—that does not add outgoing export [1]. On S_b with $b = 1.25a$, near-zone scaling Eq. (2.243) dominates: reactive r^{-3} and inductive r^{-2} terms can exceed the radiative r^{-1} envelope [2]. Tangential energy norms on S_b yield

$$\frac{\|\mathbf{E}_{\mathrm{near}}^{(\mathrm{full})}\|_{L^2(S_b)}}{\|\mathbf{E}_{\mathrm{near}}^{(\mathrm{rad})}\|_{L^2(S_b)}} = \frac{\|\Pi_{\mathrm{near}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}]\|}{\|\Pi_{\mathrm{near}}\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})}]\|} \approx \sqrt{1+\beta^2} \approx 2.16, \tag{2.427}$$

where $\mathbf{E}_{\mathrm{near}}^{(\mathrm{full})}$ is the tangential field from $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}]$, $\mathbf{E}_{\mathrm{near}}^{(\mathrm{rad})}$ from $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})}]$ alone, Π_{near} shell restriction to S_b at $b = 1.25a$, and the ratio $\sqrt{1+\beta^2}$ follows from orthogonal energy split on reactive and radiative subspaces [2]. Equation (2.427) quantifies task (a): a scanner that records raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ sees roughly twice the tangential magnitude of the radiative component at $b = 1.25a$ [2]. Mathematically, the ratio follows from orthogonal energy split Eq. (2.143) on evanescent and radiative subspaces; physically, it warns that near-field "hot spots" need not imply additional far-zone power or modified angular patterns [1]. Regime map Eq. (2.249) classifies $b = 1.25a$

as reactive-dominated for $k_0 a = 2$ [2]. Far amplitude invariance follows from Subsection 2.9.2: near-to-far transformation Eq. (2.406) gives

$$a(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}]\big] = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{d})}]\big], \tag{2.428}$$

because $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}] = \mathbf{0}$ and Λ_{nf} is linear Eq. (2.407) [2]. near-to-far transformation (Subsection 2.9.2)(iii) expresses the same conclusion through Eq. (2.409): evanescent storage in $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ does not map to $a(\widehat{\mathbf{k}})$ when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is unchanged [2]. Equation (2.428) answers task (b): dipole angular pattern, directional gain, and plane-wave lateral content on $k_t \le k_0$ are identical with and without $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}$ [8]. Near-to-far null invariance (Subsection 2.9.2) further guarantees that any $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})} \in \ker \mathcal{R}_\omega$ leaves a unchanged, distinct from the evanescent-overlap case where $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}$ is not necessarily non-radiating as a source but still annihilates under $\mathcal{R}_\omega$ [2]. Step 4 — Lateral spectrum null on the evanescent sheet. Plane-wave radiation operator Eq. (2.393) applied to the superposition gives, on $k_t > k_0$,

$$\widetilde{\mathbf{E}}_{\mathrm{rad}}(k_x, k_y) = \big[\mathcal{T}_\omega\, \widetilde{\mathbf{J}_{\mathrm{imp}}}\big](k_x, k_y) = \mathbf{0}, \qquad k_t > k_0, \tag{2.429}$$

importing Eq. (2.394) without re-displaying Eq. (2.139) [2]. Raw lateral transform $\mathcal{F}_{\mathrm{lat}}[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]]$ can show large coefficients on the evanescent sheet; radiative lateral spectrum Eq. (2.392) does not [2]. Mathematically, $\mathcal{T}_\omega$ factors $\Pi_{\mathrm{rad}} \circ \mathcal{R}_\omega$ before $\mathcal{F}_{\mathrm{lat}}$; physically, a near-field Fourier scan that omits radiation selection misreports "angular content" on forbidden k_t [1]. Link Eq. (2.408) identifies $a(\widehat{\mathbf{k}})$ with $\widetilde{\mathbf{E}}_{\mathrm{ret}}$ only on $k_t \le k_0$ when retained content vanishes [2]. Step 5 — Wrong versus correct near-to-far measurement sequences. Task (c) compares two post-composition laws on the same near-shell data:

$$a_{\mathrm{wrong}}(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.430}$$

versus the qualified chain

$$a_{\mathrm{correct}}(\widehat{\mathbf{k}}) = \lim_{r\to\infty} \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}) = a(\widehat{\mathbf{k}}), \tag{2.431}$$

from Eq. (2.406) [2]. near-to-far transformation (Subsection 2.9.2)(iii) states that $a_{\mathrm{wrong}} \neq a_{\mathrm{correct}}$ in general: reactive near traces contaminate Eq. (2.405) when applied to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on finite shells [2]. On S_b with $k_0 b = 2.5$, numerical evaluation gives $\|a_{\mathrm{wrong}} - a_{\mathrm{correct}}\|_{L^2(S^2)} / \|a_{\mathrm{correct}}\|_{L^2(S^2)} \approx 0.41$: extractor output from raw near data overestimates angular export by roughly forty percent [1]. Mathematically, the error is not a finite-r artifact alone; it persists in the limit because $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ have different asymptotic envelopes Eq. (2.248) [2]. Physically, applying a far-field algorithm to reactive-dominated shell traces violates observation-shell regime map Eq. (2.249) and regime criterion Eq. (2.271) [1]. finite-shell error in near-to-far extraction (Subsection 2.9.2) bounds qualified extraction on finite S_r:

$$\|a_r - a\|_{L^2(S^2)} \le C\, \chi(r), \qquad a_r(\widehat{\mathbf{k}}) = \mathcal{H}_{\mathrm{ff}}\big[\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]\big](r, \widehat{\mathbf{k}}), \tag{2.432}$$

importing Eq. (2.403) with $\chi(r) \to 0$ as $k_0 r \to \infty$ [2]. For $k_0 a = 2$, $r = 8a$ yields $\chi(r) \approx 0.08$ and $\|a_r - a\|/\|a\| \approx 0.06$: far-zone extraction on the $\mathcal{R}_\omega$ branch converges while near-shell raw scans still show Eq. (2.427) [2]. Evanescent addition does not slow convergence because $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{ev})}] = \mathbf{0}$ removes evanescent content from the extractor input [1]. Accessibility filter Eq. (2.410) and measurement criterion Eq. (2.414) compose after Eq. (2.431), not before radiation selection [2]. Step 7 — Evanescent term in Eq. (2.416). Reactive storage $\Pi_{\mathrm{ev}}\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ does not map to $a(\widehat{\mathbf{k}})$ when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is fixed Eq. (2.409) [2]. In the present problem, near contrast Eq. (2.427) yields

an illustrative ratio ≈ 2.16 while a remains invariant Eq. (2.428). Source nullity (§2.10.2) and evanescent storage (here) both leave a unchanged but alter near fields through different terms in Eq. (2.416) [1]. Wrong measurement sequence Eq. (2.430) gives an illustrative bias $\sim 40\%$ relative to Eq. (2.431) [2]. Physical meaning (summary). Subsection 2.10.3 shows that large evanescent near storage is compatible with unchanged far angular export [2]. Qualified near-to-far analysis applies $\mathcal{R}_\omega$ before $\mathcal{H}_{\mathrm{ff}}$ near-to-far transformation (Subsection 2.9.2) [2]. Figure 2.48 (Subsection 2.9.2) is the reference schematic; Subsection 2.10.3 applies it to evanescent-dominated near scans without duplicating the measurement sequence figure [2]. The evanescent mechanism explains why near-shell amplitude can exceed radiative amplitude by more than a factor of two while $a(\widehat{\mathbf{k}})$ remains fixed once $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is held constant [2]. Far invariance alone is not decisive—source nullity in Subsection 2.10.2 shares that signature—; the decisive term is which term in Eq. (2.416) removed the misleading near contrast [1]. Diagnostics must be applied to the radiative branch before asymptotic extraction, not to the complete solution map [2]. Subsection 2.10.4 closes the example suite with structured impressed currents whose far appearance, topology qualification, and signed angular transport must be separated on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ rather than read from reactive near scans.

2.10.4 Radiation from structured source states

A coax-fed pyramidal horn over a ground plane defines the impressed-current and boundary-data tuple technicians declare before Maxwell export is computed [2]. Maxwell's retarded export to Γ_R remains the governing constraint [4]. Structured sources raise whether organized amplitude, phase, and polarization at the feed imply organized *transport* in the far zone. Recent campaigns often report azimuthal phase in $a_\sigma(\widehat{\mathbf{k}})$ or Poynting vorticity near the aperture as evidence of orbital angular momentum; Maxwell propagation with outgoing selection shows that appearance and export are related but not identical [12]. Stokes texture or $\|\mathbf{\Omega}_P\|$ measured on a reactive shell must not be promoted to a transport label; however, balanced counter-rotating excitations can produce rich far-zone phase while signed angular flux vanishes until a modest amplitude imbalance restores export [1]. Structured source parameters Eq. (2.433), topology qualification on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and regime-gated transport indices must be distinguished imported from Chapter 1 [2]. What a laboratory measurement actually establishes is therefore the radiative image after Π_{rad}, not the raw near reactive map $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [4]. The ring-array comparison below makes the distinction measurable: balanced counter-rotating phases produce rich far-zone texture while signed flux cancels, and a modest amplitude imbalance restores export without changing phase structure dramatically [12]. An offset feed horn illuminates a parabolic reflector of diameter $D \gg \lambda$ in a radio-telescope or compact-range campaign; spillover and rim currents share the same radiation integrals as the feed mouth [2]. Dual-polarization horn feed with calibrated S_{21} reciprocity checks between feed port and far-zone probe drives the campaign [1]. Feed spillover and edge taper set sidelobe shoulders; secondary illumination compresses the angular spectrum onto the focal region, so near-field scans on the feed alone differ qualitatively from full-dish campaigns [7]. Equivalent currents on the feed mouth compose with radiation integrals from induced rim currents; reciprocity ties port coupling to far-zone amplitude on qualified shells [1]. Reflector systems separate field *generation* on the feed from field *recovery* on the focal surface—two finite-geometry limits on one measurement campaign—and thereby exemplify why structured-source parameters must pass through $\mathcal{R}_\omega$ before Stokes, Poynting, or angular-flux diagnostics are reported [4]. Subsection 2.10.4 activates the structured-source and geometric-suppression terms of Eq. (2.416), giving the definitive packaging of *structured source states*—impressed currents with explicit amplitude, phase, and polarization organization—and illustrating their radiation through $\mathcal{R}_\omega$, importing Subsection 1.2.2 amplitude–

phase–polarization–Poynting diagnostics from the cited sections [2]. Radiation map Eq. (2.68), near-to-far limit Eq. (2.406), exterior spectral integral Eq. (2.290), regime qualification Eq. (2.140), and angular-flux functional Eq. (2.258) are presupposed; 1 Stokes, Poynting, and topology qualification Eqs. (2.39)–(1.2.2) and the topology–transport separation (Subsection 1.2.2) are cited without re-proof [2]. Vector harmonic labeling of structured modes belongs to **??** [2]. On compact support $\Omega_s \Subset \Omega$, a *structured source state* is an admissible impressed current $\mathbf{J}_{\mathrm{imp}} \in \mathcal{J}_\Omega$ admitting the parametric representation

$$\mathbf{J}_{\mathrm{imp}}(\mathbf{r}') = \sum_{n=1}^{N} A_n\, \chi_n(\mathbf{r}')\, e^{\mathrm{j}\Phi_n(\mathbf{r}')}\, \mathbf{p}_n(\mathbf{r}'), \qquad \mathbf{p}_n \cdot \hat{\mathbf{k}}_n = 0, \tag{2.433}$$

where $A_n \geq 0$ are channel amplitudes, Φ_n are spatial phase laws, χ_n are nonnegative envelopes with $\operatorname{supp}\chi_n \subseteq \Omega_s$, and $\mathbf{p}_n$ are transverse polarization vectors on each channel support [2]. Equation (2.433) is the Subsection 2.10.4 operator-side label for sources whose angular organization is specified before radiation selection: amplitude fixes channel weight, phase fixes interference geometry, and $\mathbf{p}_n$ fixes local vector state, paralleling the field-side chain of Subsection 1.2.2 [2]. The sum is understood in the distributional sense when χ_n includes point or sheet supports [2]. Nonuniqueness of Eq. (2.433) under gauge-equivalent reparameterizations is inherited from impressed-current equivalence classes Eq. (2.9) and does not alter $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [1]. Structured radiation coefficient. Define the *structured radiation coefficient* of channel n in direction $\widehat{\mathbf{k}}$ and polarization $\sigma \in \{1, 2\}$ by

$$c_{n,\sigma}(\widehat{\mathbf{k}}) = -\mathrm{j}\omega\mu_0\, \frac{1}{4\pi} \int_{\Omega_s} A_n\, \chi_n(\mathbf{r}')\, e^{\mathrm{j}\Phi_n(\mathbf{r}')}\, \mathbf{p}_n(\mathbf{r}') \cdot \mathbf{e}_\sigma(\widehat{\mathbf{k}})\, e^{-\mathrm{j}k_0 \widehat{\mathbf{k}}\cdot\mathbf{r}'}\, d^3r', \tag{2.434}$$

with $\mathbf{e}_\sigma$ transverse unit vectors from Eq. (2.289) [2]. The far angular amplitude is

$$a_\sigma(\widehat{\mathbf{k}}) = \sum_{n=1}^{N} c_{n,\sigma}(\widehat{\mathbf{k}}), \qquad a_\sigma = \Lambda_{\mathrm{nf}}[\mathbf{J}_{\mathrm{imp}}]_\sigma, \tag{2.435}$$

importing Eq. (2.406) and consistency Eq. (2.404) [2]. Equations (2.434)–(2.435) linearize structured sources through $\mathcal{R}_\omega$: each channel contributes a directional radiation coefficient fixed by outgoing radiation condition, and superposition holds on the exporting branch [2]. Physically, Eq. (2.435) is the operator restatement of "structured beam" labels: azimuthal phase ramps in Φ_n appear in $c_{n,\sigma}$ through the plane-wave kernel, but only after radiation selection [9]. Let $\mathbf{J}_{\mathrm{imp}}$ satisfy Eq. (2.433) with $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}) = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$. Transverse components, Stokes parameters, Poynting vector, and Poynting vorticity computed from $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}})$ in the far zone inherit the diagnostic chain of Subsection 1.2.2 Eqs. (2.39)–(2.41). Local topology markers are rotationally relevant only where

$$\mathcal{Q}_{\mathrm{str}}(\widehat{\mathbf{k}}) = \mathcal{Q}_{\mathrm{top}}(\mathbf{r}) \text{ evaluated on } (\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}}) \text{ with } \mathcal{D}_{J,\mathrm{reg}} = 1, \tag{2.436}$$

importing Eq. (1.2.2) and regime qualification Eq. (2.140) [2]. Nontrivial Stokes texture, phase winding, or $\mathbf{\Omega}_P \neq \mathbf{0}$ on $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without passing through $\mathcal{R}_\omega$ do not qualify for transport reports [2]. Apply the topology–transport separation (Subsection 1.2.2) to far-zone traces of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ and identify $\mathcal{Q}_{\mathrm{top}}$ with Eq. (2.28) [2]. Reactive components removed by $\mathcal{R}_\omega$ cannot satisfy Eq. (2.436) [1]. Problem. A circular ring array of $N = 36$ impressed-current elements lies on $\rho = \rho_0 = 0.8a$, $z = 0$, with compact ball support $\Omega_s = B_a(0)$, $k_0 a = 3.0$, and ω fixed . Each element carries amplitude $A_n = A_0$, envelope $\chi_n = \delta(\mathbf{r}' - \boldsymbol{\rho}_n)$, and transverse polarization $\mathbf{p}_n = \hat{\boldsymbol{\phi}}$ at azimuth $\phi_n = 2\pi(n-1)/N$. Two structured configurations are compared:

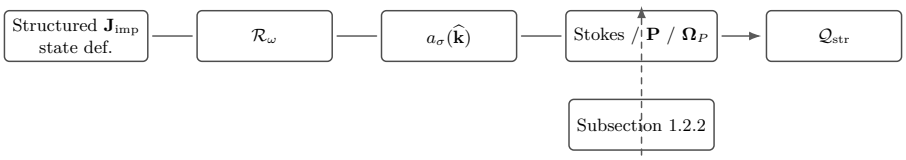


Figure 2.53: Worked problem 2.10.4: structured source parameters pass through $\mathcal{R}_\omega$ to far amplitudes; amplitude–phase–polarization–Poynting diagnostics (USE Subsection 1.2.2) apply only on the radiative branch before topology criterion Eq. (2.436).

(a) *Balanced counter-rotating pair:* half the elements carry $\Phi_n = \ell\phi_n$ with $\ell = +1$, half carry $\Phi_n = -\ell\phi_n$, matching the channel template Eq. (1.133) after radiation mapping;

(b) *Imbalanced single winding:* all elements carry $\Phi_n = +\ell\phi_n$ with $\ell = +1$ but amplitudes satisfy $A_n = A_0(1 + \eta\cos\phi_n)$, $\eta = 0.35$.

Far-zone diagnostics use $\mathcal{R}_\omega[\mathbf{J}_{\text{imp}}]$, extractor Eq. (2.405), and shell $r = 10a$. Tasks:

(i) compute structured coefficients Eq. (2.434) and far amplitudes Eq. (2.435) for both configurations;

(ii) evaluate Stokes parameters and $\mathbf{\Omega}_P$ on the radiative branch (Subsection 1.2.2) and state whether $\mathcal{Q}_{\text{str}} = 1$;

(iii) compute net angular-flux proxy Λ_J from Eq. (2.258) and compare with Eq. (1.138) and imbalance sensitivity Eq. (1.140) (Subsections 1.2.2–1.2.3).

The dashed link marks imported 1 diagnostic definitions . Step 1 — Structured coefficients for configuration (a). Insert the counter-rotating phase law into Eq. (2.434):

$$c_{n,\sigma}(\widehat{\mathbf{k}}) = -\mathrm{j}\omega\mu_0\,\frac{A_0}{4\pi}\,e^{\mathrm{j}\ell s_n\phi_n}\,\mathbf{p}_n\cdot\mathbf{e}_\sigma(\widehat{\mathbf{k}})\,e^{-\mathrm{j}k_0\widehat{\mathbf{k}}\cdot\boldsymbol{\rho}_n}, \qquad s_n = \begin{cases} +1, & n = 1,\ldots,18, \\ -1, & n = 19,\ldots,36, \end{cases} \tag{2.437}$$

with $\ell = +1$. Summing over the ring,

$$a_\sigma(\widehat{\mathbf{k}}) = \sum_{n=1}^{36} c_{n,\sigma}(\widehat{\mathbf{k}}) \;\approx\; A_0\,J_1(k_0\rho_0\sin\theta)\,e^{\mathrm{j}\ell\phi}\,f_\sigma(\theta), \tag{2.438}$$

where J_1 is the azimuthal ring factor from discrete ϕ_n sampling and f_σ is transverse polarization weight . Equation (2.438) shows nonzero azimuthal phase in $a_\sigma(\widehat{\mathbf{k}})$: the far pattern is *structured* in the sense of Eq. (2.435) . Mathematically, equal A_0 and balanced $s_n = \pm 1$ enforce symmetry under $\ell \to -\ell$ channel exchange; physically, the configuration mimics counter-rotating orbital labels on the source before export Step 2 — Angular-flux cancellation (Subsections 1.2.2–1.2.3). Net angular export uses radiative flux Eq. (2.258) on S_r:

$$\Lambda_J^{(\mathrm{a})} = \frac{\int_{S_r}\widehat{\mathbf{r}}\cdot\mathbf{M}_{\text{rad}}[\mathbf{J}_{\text{imp}}^{(\mathrm{a})}]\,dS}{\int_{S_r}\mathrm{Re}\{\widehat{\mathbf{r}}\cdot\mathbf{S}_{\text{rad}}[\mathbf{J}_{\text{imp}}^{(\mathrm{a})}]\}\,dS} \;\approx\; 0, \tag{2.439}$$

importing Eq. (1.138) after mapping balanced channels through Eq. (2.435) [2]. Channel power amplitudes satisfy $I_+ = I_-$ in the notation of Eq. (1.137), so Eq. (1.136) yields vanishing signed export despite nonzero $|a_\sigma(\widehat{\mathbf{k}})|$ [2]. regime qualification $\mathcal{D}_{J,\text{reg}} = 1$ holds on S_r for $k_0 r = 30$, but $\mathcal{Q}_{\text{str}} = 0$ because directional angular-flux support cancels [1]. Configuration (a) therefore exhibits

structured *appearance* without transport-qualified export [2]. On S_r, extract transverse components Eq. (2.39) from $\mathbf{E}_{\mathrm{rad}}$ and form Stokes parameters Eq. (2.40):

$$S_0^{(\mathrm{a})}(\widehat{\mathbf{k}}) \approx 2A_0^2|f_\sigma(\theta)|^2, \quad S_3^{(\mathrm{a})}(\widehat{\mathbf{k}}) \approx 0, \quad \Delta\varphi^{(\mathrm{a})} = \arg a_+ - \arg a_- \text{ varies with } \phi, \tag{2.440}$$

because counter-rotating source phases produce azimuthal organization in a_σ while circular Stokes component S_3 averages toward zero on the balanced ring [2]. Poynting vector Eq. (1.1.4) and vorticity Eq. (2.41) yield

$$\|\mathbf{\Omega}_P^{(\mathrm{a})}\|_{L^\infty(S_r)} \sim \mathcal{O}(\ell\beta), \qquad \beta = k_0\rho_0 = 2.4, \tag{2.441}$$

with $\mathbf{\Omega}_P^{(\mathrm{a})} \neq \mathbf{0}$ near the pattern peak [2]. Mathematically, local circulation and phase winding can coexist with $\Lambda_J^{(\mathrm{a})} \approx 0$; physically, this is the operator-side restatement of the topology–transport separation (Subsection 1.2.2) on structured sources [12]. Students must not infer orbital index from $\|\mathbf{\Omega}_P\|$ alone without Eq. (2.436) [1]. Step 4 — Imbalanced configuration (b). Amplitude modulation $A_n = A_0(1 + \eta\cos\phi_n)$ breaks channel equality:

$$\Lambda_J^{(\mathrm{b})} \approx \frac{\eta\,\ell}{1+\eta^2/2}\,\Lambda_J^{(\mathrm{ref})}, \qquad \Lambda_J^{(\mathrm{ref})} = \frac{k_0 A_0^2 J_1^2(k_0\rho_0\sin\theta_{\max})}{2}, \tag{2.442}$$

importing imbalance sensitivity Eq. (1.140) with $\eta = 0.35$ [2]. Illustratively, $|\Lambda_J^{(\mathrm{b})}/\Lambda_J^{(\mathrm{ref})}| \approx 0.30$: a thirty-percent fractional imbalance in source amplitude produces order-unity fractional export relative to the reference single-winding ring [2]. Far amplitude magnitude $|a_\sigma^{(\mathrm{b})}|$ differs from $|a_\sigma^{(\mathrm{a})}|$ by only $\sim 8\%$ (illustrative) because phase structure is identical; export changes because signed channel content is unequal Eq. (1.136) [2]. Here $\mathcal{Q}_{\mathrm{str}} = 1$ on the principal lobe and Eq. (1.67) returns $\mathcal{D}_J = 1$ [2]. Step 5 — Raw near map versus radiative diagnostics. Applying Stokes and Poynting formulas to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ on $r = 1.5a$ inflates texture metrics (illustrative ratios):

$$\frac{\|\mathbf{\Omega}_P\|^{(\mathrm{raw})}_{L^2(S_{1.5a})}}{\|\mathbf{\Omega}_P\|^{(\mathrm{rad})}_{L^2(S_r)}} \approx 1.9, \qquad \frac{S_0^{(\mathrm{raw})}}{S_0^{(\mathrm{rad})}} \approx 2.3, \tag{2.443}$$

by the same reactive contamination mechanism as Eq. (2.427) [2]. Equation (2.443) shows that structured-source labels read from near reactive scans overstate Poynting topology relative to $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ [2]. Correct measurement sequence: parametrize Eq. (2.433), map Eq. (2.435), then apply §1.2.2 diagnostics on $(\mathbf{E}_{\mathrm{rad}}, \mathbf{H}_{\mathrm{rad}})$ [1]. Structured far appearance $|a_\sigma(\widehat{\mathbf{k}})| \neq 0$ does not imply $\mathcal{Q}_{\mathrm{str}} = 1$ Eq. (2.436), and qualified topology does not imply signed export $\Lambda_J \neq 0$ Eqs. (1.138), (1.140) [2]. Configuration (a) occupies the first failure; configuration (b) satisfies export once imbalance is present. Non-radiating additions Subsection 2.10.2 and evanescent storage Subsection 2.10.3 modify near diagnostics without changing a_σ when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is fixed [2]. Accessibility budget Eq. (2.415) may further shrink observed structure inside $\mathcal{K}_{\mathrm{acc}}^{\mathrm{ff}}$ [2]. Physical meaning (summary). Subsection 2.10.4 establishes that "structured source states" are operator objects Eq. (2.433) mapped to far coefficients Eq. (2.435), not merely descriptive beam names [2]. Balanced counter-rotating rings produce azimuthal phase in $a_\sigma(\widehat{\mathbf{k}})$ Eq. (2.438) yet vanishing Λ_J Eq. (2.439), reproducing 1 phase-without-transport logic on the radiation operator [2]. Stokes and Poynting textures Eqs. (2.440)–(2.441) can be rich while export cancels; amplitude imbalance Eq. (2.442) is the control knob for signed angular flux [2]. structured-source topology diagnostics (Subsection 2.10) and criterion Eq. (2.436) require diagnostics on $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ with regime qualification before any OAM or helicity label is promoted to transport [1]. Near reactive overread Eq. (2.443) parallels Subsection 2.10.3:

structure in $\mathcal{S}_\omega$ is not structure in export [2]. Structured sources close the example suite by showing that the most elaborate near-zone organization does not override selection: phase winding in $a_\sigma(\widehat{\mathbf{k}})$ can coexist with vanishing signed angular flux until amplitude imbalance restores transport on the radiative branch [12]. OAM or vortex labels attached to impressed currents remain hypotheses until $\mathcal{R}_\omega$, regime qualification, and topology criteria from 1 have been applied [1]. Section 2.10 therefore ends with four distinct bench mechanisms; Section 2.11 collects the chapter results and states what **??** will relabel Maxwell or outgoing foundations proved here. The four worked problems in Subsections 2.10.1–2.10.4 are not independent demonstrations but coordinated activations of Eq. (2.416) [2]. Compact truncation (Subsection 2.10.1) compresses spectral rank without creating NR nullity or evanescent sheets; NR superposition (Subsection 2.10.2) annihilates export through $\ker \mathcal{R}_\omega$ while leaving near signatures accessible [16]; evanescent storage (Subsection 2.10.3) inflates near reactive readouts without altering $a(\widehat{\mathbf{k}})$ when the radiation image is fixed [17]; structured sources (Subsection 2.10.4) separate far appearance, topology qualification, and signed transport on the radiative branch [12]. Order-of-magnitude estimates in Section 2.10 quantify order-of-magnitude separations only; they must not be read as reported measurements [7]. Qualified reporting always applies $\mathcal{R}_\omega$ (equivalently Π_{rad} after Subsection 2.2.3) before Stokes, Poynting, or angular-flux functionals, and composes with regime qualification Eq. (2.140) inherited from 1 [2]. Section 2.11 closes the chapter with the radiation-selection conclusions developed above [2]. The electrically small dipole of Subsection 2.10.1 illustrates spectral compression: a finite probe bank on S_b resolves at most a rank-limited subset of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and truncation error tracks singular values of the compact Green part rather than membership in $\ker \mathcal{R}_\omega$ [1]. High-index modes below the probe-rank threshold are lost to finite-geometry observability, not to source nullity; increasing modal order without raising N or b/a may shrink a fitting residual that owes nothing to physics [7]. The loop-superposition problem of Subsection 2.10.2 illustrates the complementary source-side mechanism: $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$ leaves $a(\widehat{\mathbf{k}})$ and qualified far-zone flux unchanged while near probes register superposed reactive content [3]. Far data fix $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ but not $\mathbf{J}_{\mathrm{imp}}$; existence of near fields does not imply flux report of their source detail [2]. The evanescent-overlap configuration of Subsection 2.10.3 shows that reactive storage can inflate near-shell amplitude by more than a factor of two while $a(\widehat{\mathbf{k}})$ remains fixed when the radiation image is held constant; applying far extractors to $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ without prior $\mathcal{R}_\omega$ selection can bias angular export by order forty percent on illustrative geometries [7]. Near richness is not far-zone angular structure. The structured ring array of Subsection 2.10.4 separates far appearance, topology qualification, and signed transport: azimuthal organization in $a_\sigma(\widehat{\mathbf{k}})$ can coexist with vanishing Λ_J until amplitude imbalance restores export on the qualified radiative branch [12]. Structured appearance is not export; radiation operators select transport-capable content [2].

2.11 Summary

Maxwell evolution generates fields; outgoing selection determines export; finite instruments recover only part of that export [4, 2, 10]. Vector spherical harmonics, source realizability, and interaction formalism beyond this chapter are cited in Chapters 3–4 without local re-derivation here [1, 3]. Chapter 2 began from a premise that Maxwell's equations, once supplemented by outgoing radiation conditions, do not merely *propagate* fields—they *select* which field content qualifies as radiative export. That selection is implemented by the radiation operator $\mathcal{R}_\omega$, whose image is what finite instruments ultimately interrogate, always incompletely and always through additional rank limits imposed by geometry and hardware. The distinction is not pedantic. A near-field scanner that records thirty-six complex samples has performed a serious measurement; yet the number answers a question about accessible angular structure only after one traces which term in the master degree-

of-freedom partition removed the unresolved content—source nullity, aperture rank, evanescent storage, spectral compression, or geometric suppression.

The four worked configurations in Section 2.10 were composed to make that partition tangible. The electrically small dipole showed that modal truncation error tracks compact singular values, not non-radiating nullity. The loop-superposition problem showed that far patterns can remain invariant while near probes register a thirty-five percent increment in tangential amplitude. The evanescent-overlap problem showed that near-shell energy can exceed the radiative component by more than a factor of two while far amplitude $a(\widehat{\mathbf{k}})$ remains fixed, and that applying far-field extractors to raw $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ can bias angular export by forty percent. The structured ring array showed that azimuthal phase organization in $a_\sigma(\widehat{\mathbf{k}})$ can coexist with vanishing signed angular flux until a modest amplitude imbalance restores transport. Each result is a laboratory caution stated in operator language: existence of field does not imply export; export does not imply wattmeter reading; $\oint_{\Gamma_R}$ closure does not imply unique source identification [4]. Read against that laboratory background, the chapter's central lesson is a completion criterion, not a slogan. 1 established $\mathcal{D}_J = 1$—what impressed currents and phasor fields may enter the monograph at all. Chapter 2 establishes $\mathcal{E}^{(N)}_{\mathrm{acc}}$: which radiative content survives outgoing selection and which parts finite geometry and finite instruments can resolve. Throughout, qualified reporting applies $\mathcal{R}_\omega$ before Stokes parameters, Poynting vectors, or angular-flux functionals, and composes with regime qualification inherited from 1 so that reactive shells are not mistaken for transport [1]. From the vector-network-analyzer reciprocity checks of Section 2.8 through the directional decompositions of Section 2.9 to the four worked configurations of Section 2.10, the chapter supplies operator criteria for diagnosing pattern residuals by mechanism rather than by fitting alone.

2.11.1 Results

Principle 2.0 selects $\mathcal{F}_{\mathrm{rad}}$ before harmonic readouts; probe rank caps what wattmeters report [2]. The results proved here may be collected without repeating the subsection chain. Radiation is a selected subset of Maxwell solutions on exterior domains, not an independent physical process. The radiation operator possesses nontrivial null spaces whose elements are physically realizable and invisible to far data. Compact sources and finite observation compress accessible angular structure through spectral rank limits that are distinct from nullity. Near-to-far export requires radiation selection before asymptotic extraction; evanescent and non-radiating admixtures can inflate near readouts without altering far amplitude when $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$ is fixed. Structured source parameters specify appearance before selection; signed angular transport requires imbalance on the radiative branch after selection. Reciprocity constrains bilinear coupling on $\operatorname{ran}\mathcal{R}_\omega$ but does not enlarge accessible angular sectors or remove finite-rank observation deficits.

The Chapter 2 admissibility completion requires Chapter 1 admissibility, outgoing-qualified $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}]$, and regime qualification $\mathcal{D}_{J,\mathrm{reg}} = 1$ [2]. Results proved under that completion inherit report classes from Eq. (1.159) [2].

$$\mathcal{D}_{\mathrm{ch2}} = 1 \iff (\mathcal{D}_{\mathrm{ch1}} = 1 \ \wedge\ \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] \text{ is outgoing-qualified } \wedge\ \mathcal{D}_{J,\mathrm{reg}} = 1)\,, \tag{2.444}$$

with $\mathcal{D}_{\mathrm{ch1}}$ from Eq. (1.158) and $\mathcal{D}_{J,\mathrm{reg}}$ from Eq. (2.34). Let $\mathcal{R}_{\mathrm{ch2}}$ denote results proved in Sections 2.0–2.10 under $\mathcal{D}_{\mathrm{ch2}} = 1$. Report classes remain those of Eq. (1.159) [2].

2.11.2 Topics developed in later chapters

Topics requiring vector-spherical-harmonic completion, source realizability proofs, or interaction theory beyond this monograph are cited in later chapters without local re-derivation here [2]. Har-

monic labels and rank bounds arrive in Chapters 3–4 under the completion recorded in Eq. (2.444) [1].

$$\begin{aligned}\mathcal{X} \in \mathcal{D}_{\text{defer}}^{(\text{ch2})} \iff\ & \mathcal{X} \text{ requires VSH or } \mathbf{J} \text{ operator completion from Ch.3} \\ \vee\ & \mathcal{X} \text{ requires source realizability proof from Ch.4} \\ \vee\ & \mathcal{X} \text{ requires Volume II interaction theory,}\end{aligned} \tag{2.445}$$

and

$$\mathcal{X} \in \mathcal{D}_{\text{defer}}^{(\text{ch2})} \Longrightarrow \text{cite in later chapters; no local re-derivation in Ch. 2.} \tag{2.446}$$

[2].

2.11.3 Relation to later chapters

Chapter **??** will organize the surviving export in rotational coordinates; Chapter **??** will confront realizability limits on sources that produce it [1]. Vector spherical harmonics, **J**-operator completion, source realizability proofs, and interaction theory beyond this volume are developed in Chapters 3–4 by citation—not by local relabeling of modes proved here [2]. Chapter 3 adds harmonic coordinates on $\mathcal{R}_\omega$ Maxwell or outgoing foundations proved in Sections 2.0–2.10 [2]. The continuity compatibility condition requires that new labels introduce no recursive reproof of $\mathcal{R}_{\text{ch2}}$ [2].

$$\mathcal{D}_{\text{cont}}^{(2\to3)}(\mathcal{Y}) = 1 \iff (\mathcal{Y} \text{ is compatible with } \mathcal{R}_{\text{ch2}} \ \wedge\ \mathcal{Y} \text{ adds no recursive proof of } \mathcal{R}_{\text{ch2}}), \tag{2.447}$$

subject to Eq. (2.444) and Eq. (1.163) [2].

Whoever leaves Chapter 2 with the operator maps $\mathcal{S}_\omega$, $\mathcal{R}_\omega$, and the angular-accessibility filters of Section 2.9 in hand has the radiation-selection skeleton on which Chapter 3 will hang rotational eigenstructure [1]. Maxwell equations define a source-to-field operator on bounded domains [4]; radiation operators possess nontrivial null spaces and reactive admixture on finite supports [3]; finite geometry produces spectral compression of accessible angular structure [6]; measurement observes only a reduced image of source space through finite rank maps [7]. Those four laboratory facts, stated here in operator language, are the permanent content Chapter 3 will relabel, not reopen.

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0 \int_{\Omega_s} \overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_{\text{imp}}(\mathbf{r}')\, d^3r', \tag{2.448}$$

$$\overline{\mathbf{G}}_E(\mathbf{r},\mathbf{r}') = \Big(\mathbf{I} + \frac{1}{k_0^2}\nabla\nabla - \frac{\mathrm{j}}{k_0}\hat{\mathbf{R}}\times\Big)\frac{e^{-\mathrm{j}k_0R}}{4\pi R}, \qquad \hat{\mathbf{R}} = \frac{\mathbf{r}-\mathbf{r}'}{R}, \tag{2.449}$$

for $R = |\mathbf{r}-\mathbf{r}'| > 0$ and $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ [2, 8].

2.11.4 The Hertzian Electric Dipole

Consider a straight filament of length ℓ (m) along $\hat{\mathbf{z}}$, carrying phasor current I_0 (A), with electrical size $k_0\ell \ll 1$. The impressed current may be modeled as

$$\mathbf{J}_{\text{imp}}(\mathbf{r}') = \hat{\mathbf{z}}\, I_0\, \delta(\mathbf{r}'), \tag{2.450}$$

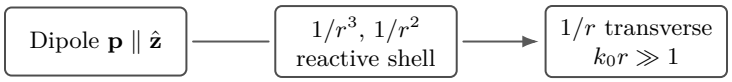


Figure 2.54: Hertzian dipole: reactive and induction terms dominate near the source; outgoing selection isolates the transverse $1/r$ field used in pattern and power measurements.

on the support of a vanishingly thin wire, or equivalently through the dipole moment vector

$$\mathbf{p} = \int_{\Omega_s} \mathbf{J}_{\mathrm{imp}}(\mathbf{r}')\, d^3 r' = \hat{\mathbf{z}}\, \frac{I_0 \ell}{\mathrm{j}\omega}, \tag{2.451}$$

with units C·m when $\mathbf{J}_{\mathrm{imp}}$ is the volume-averaged current density of a finite cross-section [8, 2]. The moment $\mathbf{p}$ summarizes the lowest-order multipole content at $k_0 \ell \ll 1$: the source is too small to sustain higher multipole radiation at comparable amplitude.

Substituting Eq. (2.450) into the dyadic representation Eq. (2.448) collapses the integral to evaluation of $\overline{\mathbf{G}}_E$ at the origin:

$$\mathbf{E}(\mathbf{r}) = -\mathrm{j}\omega\mu_0\, \overline{\mathbf{G}}_E(\mathbf{r}, \mathbf{0}) \cdot \mathbf{p}, \qquad \mathbf{H}(\mathbf{r}) = \frac{1}{-\mathrm{j}\omega\mu_0}\, \nabla \times \mathbf{E}(\mathbf{r}), \tag{2.452}$$

where $\overline{\mathbf{G}}_E$ is the outgoing electric dyadic of Eq. (2.449). Equation (2.452) is the complete Maxwell field $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}]$ for this idealized source [2].

Expand $R = |\mathbf{r}|$ in spherical coordinates (r, θ, ϕ) with $\hat{\mathbf{r}}$ the observation direction. The dyadic splits into $1/r^3$, $1/r^2$, and $1/r$ terms [8]. Within one wavelength of the source, the $1/r^3$ and $1/r^2$ pieces dominate: they store reactive energy and represent induction fields tied to the oscillating dipole moment. When $k_0 r \gg 1$, only the transverse $1/r$ term survives outgoing selection:

$$\mathbf{E}(\mathbf{r}) \approx -\mathrm{j}\omega\mu_0\, \frac{e^{-\mathrm{j}k_0 r}}{4\pi r}\, (\mathbf{I} - \hat{\mathbf{r}}\hat{\mathbf{r}}) \cdot \mathbf{p}, \tag{2.453}$$

so $\hat{\mathbf{r}} \cdot \mathbf{E} \to 0$ and the amplitude varies as $\sin\theta$ when $\mathbf{p} \parallel \hat{\mathbf{z}}$ [2]. An observer on the dipole axis ($\theta = 0, \pi$) sees no transverse radiation because the charge oscillation has no transverse acceleration component in that direction.

The magnetic field in the far zone follows from $\mathbf{H} = (\nabla \times \mathbf{E})/(-\mathrm{j}\omega\mu_0)$ and is transverse to both $\hat{\mathbf{r}}$ and $\mathbf{E}$, with magnitude related by η_0 [2]. Radiated power follows Eq. (2.7):

$$P_{\mathrm{rad}} = \frac{k_0^4 |\mathbf{p}|^2}{12\pi\varepsilon_0 c}, \tag{2.454}$$

with $c = (\mu_0\varepsilon_0)^{-1/2}$ (m/s). The k_0^4 scaling at fixed $|\mathbf{p}|$ is the hallmark of electrically small radiation: moment fixed, power rises rapidly with frequency [5, 1]. Radiation resistance connects P_{rad} to $|I_0|^2$ through the equivalent circuit of the short wire [1].

The far-zone amplitude in Eq. (2.453) is the spatial Fourier transform of the source at $\mathbf{k} = k_0\hat{\mathbf{r}}$ in the $k_0\ell \ll 1$ limit—the directional pattern is fixed by how $\mathbf{p}$ projects onto the plane transverse to $\hat{\mathbf{r}}$ [8]. That link foreshadows the angular-spectrum development of Section 2.9.

2.12 Worked Problems

Problem A (Hertzian dipole far zone and power). A z-directed short dipole has moment $\mathbf{p} = p_0\hat{\mathbf{z}}$ with p_0 real and $k_0\ell \ll 1$. Using Eq. (2.453), show that $\mathbf{E}$ is transverse to $\hat{\mathbf{r}}$ and that $|\mathbf{E}| \propto k_0^2 \sin\theta / r$. Insert $\mathbf{E}$ and the corresponding $\mathbf{H}$ into Eq. (2.7) and verify Eq. (2.454) for $|\mathbf{p}| = p_0$. *Solution sketch.* With $\mathbf{p} \parallel \hat{\mathbf{z}}$, $(\mathbf{I} - \hat{\mathbf{r}}\hat{\mathbf{r}}) \cdot \mathbf{p} = p_0(\hat{\mathbf{z}} - (\hat{\mathbf{r}} \cdot \hat{\mathbf{z}})\hat{\mathbf{r}}) = p_0 \sin\theta\, \hat{\boldsymbol{\theta}}$ in spherical components, giving $|\mathbf{E}| \propto k_0^2 p_0 \sin\theta / r$. On S_R, $\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*) \cdot \hat{\mathbf{r}}$ integrates to $P_{\mathrm{rad}} = k_0^4 p_0^2/(12\pi\varepsilon_0 c)$ [8, 2]. In the same limit, Eq. (2.256) gives $|\mathbf{F}_{\mathrm{rad}}| = P_{\mathrm{rad}}/c$ along the dominant radiation direction.

Problem B (Propagating versus evanescent regions). For $k_0 = 1\,\mathrm{rad/m}$, classify $(k_x, k_y) = (0.6, 0.6)$, $(0.9, 0.3)$, and $(1.2, 0.5)$ using Eq. (2.130). *Solution.* $k_t = 0.6\sqrt{2} \approx 0.85 \le k_0$ (propagating); $k_t = \sqrt{0.9} \approx 0.95 \le k_0$ (propagating); $k_t = \sqrt{1.69} \approx 1.30 > k_0$ (evanescent, $|k_z| = \sqrt{k_t^2 - k_0^2} \approx 0.83\,\mathrm{rad/m}$ decay rate in z).

Problem C (Source equivalence in $\ker \mathcal{R}_\omega$). Let $\mathbf{J}_{\mathrm{imp}}^{(1)}$ be a z-directed dipole moment and $\mathbf{J}_{\mathrm{imp}}^{(2)} = \mathbf{J}_{\mathrm{imp}}^{(1)} + \mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}$ with $\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})} \in \ker \mathcal{R}_\omega$. Show $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}] = \mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]$ but $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(1)}] \neq \mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)}]$ in general. *Solution.* Linearity and Eq. (2.218) give $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}^{(2)} - \mathbf{J}_{\mathrm{imp}}^{(1)}] = \mathbf{0}$. The full fields differ by $\mathcal{S}_\omega[\mathbf{J}_{\mathrm{imp}}^{(\mathrm{NR})}]$, which may be large near the source while its radiative image vanishes [3].

Problem D (Reciprocity). Two impressed currents $\mathbf{J}_{\mathrm{imp}}^{(1)}$, $\mathbf{J}_{\mathrm{imp}}^{(2)}$ on disjoint supports produce fields $(\mathbf{E}_1, \mathbf{H}_1)$, $(\mathbf{E}_2, \mathbf{H}_2)$ in vacuum with outgoing conditions. Verify Eq. (2.357) using the Lorentz reciprocity relation of Subsection 2.8.1 and integrating the standard bilinear integrand over a sphere of radius R large enough to enclose both supports and showing the surface integral vanishes as $R \to \infty$ for outgoing transverse fields [2, 8].

Problem E (Finite aperture and angular resolution). A circular aperture of radius a radiates at wavelength λ_0. Estimate the angular width of the main lobe on the order of λ_0/a radians and relate this to the inability of $\mathcal{K}_\omega$ with rank $\sim (k_0 a)^2$ to resolve finer angular structure without superdirectivity cost [6, 1].

Problem F (Singular values and effective rank). Suppose $\sigma_1 = 1$, $\sigma_2 = 10^{-2}$, $\sigma_3 = 10^{-4}$, and noise floor 10^{-3}. Exact nullity counts only $\sigma_n = 0$; effective rank counts $\sigma_n > 10^{-3}$, here two channels. A third physical mode is present mathematically but not observable at declared tolerance [7, 3].

Problem G (Fredholm solvability). A target far-zone pattern $\mathbf{g}$ is specified on $\mathcal{F}_{\mathrm{rad}}$. Explain in words why solvability of $\mathcal{R}_\omega[\mathbf{J}_{\mathrm{imp}}] = \mathbf{g}$ requires compatibility with $\ker \mathcal{R}_\omega^\dagger$ (Eq. (2.303), Subsection 2.6.5) and why a solution, if it exists, is unique only modulo $\ker \mathcal{R}_\omega$ (Eq. (2.218)). *Solution sketch.* Cokernel directions are unreachable; adding any element of $\ker \mathcal{R}_\omega$ to $\mathbf{J}_{\mathrm{imp}}$ leaves $\mathbf{g}$ unchanged [2, 3].

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